Teacher Edition

Science Focus 3

second edition

PDF Version

Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

This grid contains information related to book three only. A final version will be supplied online which includes information across all four books once they are all released.

PRESCRIBED FOCUS AREAS OUTCOMES

STAGE 4

Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.1 identifies historical examples of how scientific knowledge has changed people's understanding of the world																		
4.2 uses examples to illustrate how models, theories and laws contribute to an understanding of phenomena																		
4.3 identifies areas of everyday life that have been affected by scientific developments																		
4.4 identifies choices made by people with regard to scientific developments																		
4.5 describes areas of current scientific research																		

STAGE 5

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.1 explains how social factors influence the development and acceptance of scientific ideas		●																
5.2 describes the processes that are applied to test and validate models, theories and laws			●	●														
5.3 evaluates the impact of applications of science on society and the environment	●				●		●											
5.4 discusses evidence supporting different viewpoints						●			●									
5.5 analyses how current research might affect people's lives					●			●										

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

PRESCRIBED FOCUS AREA – CORE CONTENT

STAGE 4

Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.1a) identify some of the scientific ideas that different cultures have contributed to science throughout history																		
4.1b) describe (using examples including those developed by Aboriginal peoples) ideas developed by different cultures to explain the world around them																		
4.1c) describe some models and theories that have been considered in science and then been modified or rejected as a result of available evidence																		
4.1d) discuss examples where societal, religious or ethical values have had an impact on scientific developments																		
4.1e) describe historical cases where developments in science have led to the development of new technologies																		
4.1f) describe historical cases where developments or improvements in technology have transformed science																		
4.2a) evaluate the role of creativity, curiosity, objectivity and logical reasoning in describing phenomena, carrying out investigations and in the devising and testing of hypotheses																		
4.2b) distinguish between scientific argument and economic or legal argument																		
4.2c) apply scientific processes to test the validity of ideas and theories																		
4.2d) describe how an idea can gain acceptance in the scientific community as either theory or law																		
4.2e) use examples which show that scientists isolate a set of observations, identify trends and patterns and construct hypotheses or models to explain these																		
4.2f) give examples that demonstrate the benefits and limitations of using models																		
4.2g) identify that the nature of observations made depends upon the understanding that the observer brings to the situation																		

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Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.3a) identify and describe examples of scientific concepts and principles that have been used in technological developments (including Australian examples)																		
4.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science																		
4.3c) identify and describe examples where technological advances have impacted on science																		
4.3d) give reasons why society should support scientific research																		
4.4a) discuss viewpoints about some issues with a major scientific component																		
4.4b) give examples to show that different societal groups may use or weight criteria differently to make a decision about an issue involving a major scientific component																		
4.4c) identify choices that need to be or have been made when considering whether to use particular scientific advances																		
4.4d) analyse reasons why different cultures or groups within a society, including Aboriginal people, may have different views in relation to scientific issues																		
4.4e) discuss the place of social and ethical considerations in scientific practice and in applications of science																		
4.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions																		
4.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding																		
4.5c) identify scientific skills that can be useful in a broad range of careers																		
4.5d) identify possible career paths in science																		

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

PRESCRIBED FOCUS AREA – CORE CONTENT

STAGE 5

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.1a) identify some of the ideas from different cultures (including those of Aboriginal and other Indigenous people) that different cultures have contributed to science throughout history		●																
5.1b) describe some models and theories that have been considered in science and then been modified or rejected as a result of available evidence		●																
5.1c) discuss examples where societal, religious or ethical values have had an impact on scientific developments		●																
5.1d) describe historical cases where developments in science have led to the development of new technologies		●																
5.1e) describe historical cases where developments or improvements in technology have transformed science		●																
5.2a) evaluate the role of creativity, curiosity, objectivity and logical reasoning in describing phenomena, carrying out investigations and in the devising and testing of hypotheses			●	●														
5.2b) distinguish between scientific argument and economic or legal argument			●	●														
5.2c) apply scientific processes to test the validity of ideas and theories			●	●														
5.2d) describe how an idea can gain acceptance in the scientific community as either theory or law			●	●														
5.2e) use examples which show that scientists isolate a set of observations, identify trends and patterns and construct hypotheses or models to explain these			●	●														
5.2f) give examples that demonstrate the benefits and limitations of using models	●		●	●														
5.2g) identify that the nature of observations made depends upon the understanding that the observer brings to the situation	●		●	●														
5.3a) identify and describe examples of scientific concepts and principles that have been used in technological developments (including Australian examples)	●				●		●											

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science	●				●		●											
5.3c) identify and describe examples where technological advances have impacted on science	●				●		●											
5.3d) give reasons why society should support scientific research	●				●		●											
5.4a) discuss viewpoints about some issues with a major scientific component						●			●									
5.4b) give examples to show that different cultures or groups within a society (including Aboriginal and other Indigenous people) may use or weight criteria differently to make a decision about an issue involving a major scientific component						●	●		●									
5.4c) identify choices that need to be or have been made when considering whether to use particular scientific advances	●					●			●									
5.4d) discuss the place of social and ethical considerations in scientific practice and in applications of science	●					●	●		●									
5.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions					●			●										
5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding					●			●										
5.5c) identify scientific skills that can be useful in a broad range of careers					●			●										
5.5d) identify possible career paths in science	●				●			●										

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

DOMAIN – SKILL OUTCOMES

STAGE 4

Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem																		
4.14 follows a sequence of instructions to undertake a first-hand investigation																		
4.15 uses given criteria to gather first-hand data																		
4.16 accesses information from identified secondary sources																		
4.17 evaluates the relevance of data and information																		
4.18 with guidance, presents information to an audience to achieve a particular outcome																		
4.19 draws conclusions based on the information available																		
4.20 uses an identified strategy to solve problems																		
4.21 uses creativity and imagination to suggest plausible solutions to familiar problems																		
4.22 undertakes a variety of individual and team tasks with guidance																		

DOMAIN – SKILL OUTCOMES

STAGE 5

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch 1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.13 identifies a problem and independently produces an appropriate investigation plan	●	●	●	●	●	●	●	●	●									
5.14 undertakes first-hand investigations independently with safety and competence	●	●	●	●		●	●	●	●									

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch 1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.15 gathers first-hand data accurately	●	●	●	●		●	●	●	●									
5.16 accesses information from a wide variety of secondary sources	●	●	●		●	●		●	●									
5.17 explains trends, patterns and relationships in data and/or information from a variety of sources	●	●	●	●		●	●	●	●									
5.18 selects and uses appropriate forms of communication to present information to an audience	●	●		●		●	●	●	●									
5.19 uses critical thinking skills in evaluating information and drawing conclusions	●	●	●	●	●	●	●	●	●									
5.20 selects and uses appropriate strategies to solve problems	●	●	●	●		●		●	●									
5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions	●		●	●		●	●	●	●									
5.22 independently plans, implements and evaluates the effectiveness of a variety of tasks as an individual and as a team member		●		●		●	●	●	●									

SKILLS – CORE CONTENT

STAGE 4

Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.13.1a) describe a problem, hypothesis or question that can be tested or researched																		
4.13.1b) propose possible sources of data and/or information relevant to the investigation																		
4.13.1c) identify what type of information or data needs to be collected																		
4.13.1d) justify why particular types of data or information are to be collected																		
4.13.1e) identify the appropriate units to be used in collecting data																		

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Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information																		
4.13.1g) formulate a means of recording the data to be gathered or the information to be collected																		
4.13.2a) identify variables that need to be kept the same if first-hand data is to be collected																		
4.13.2b) specify the dependent and independent variables when planning controlled experiments																		
4.13.2c) describe a logical procedure for undertaking a simple or controlled experiment																		
4.13.2d) establish an appropriate timeline for an investigation																		
4.13.3a) identify advantages and limitations of using particular laboratory equipment for a specific task																		
4.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task																		
4.13.3c) describe ways to reduce the risk to themselves and others when working in the laboratory or field																		
4.14a) follow the planned procedure when performing an investigation																		
4.14b) use time and resources effectively																		
4.14c) safely and efficiently construct, assemble and manipulate identified equipment																		
4.14d) record data using the appropriate units																		
4.14e) evaluate and modify experimental procedures																		
4.14f) demonstrate the use of safe and hygienic work practices including the correct use of safety equipment																		
4.15a) make and record observations and measurements accurately over a number of trials																		
4.15b) use independently a range of data collection strategies and technologies such as data loggers																		
4.16a) use a range of sources, including CD-ROMs and the Internet, to access information																		

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Outcomes		*Science Focus 1*									*Science Focus 2*								
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.16b)	use a variety of techniques, such as key words, skimming and scanning, to identify appropriate information																		
4.16c)	extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio/visual resources																		
4.16d)	summarise information from identified oral and written secondary sources																		
4.17a)	collate information from a number of sources																		
4.17b)	distinguish between relevant and irrelevant information																		
4.17c)	check reliability of gathered data and information by comparing them with observations or information from other sources																		
4.17d)	organise data using a variety of methods including diagrams, tables, spreadsheets and databases																		
4.17e)	critically analyse the accuracy of scientific information presented in mass media																		
4.17f)	identify trends, patterns, relationships and contradictions in data and information																		
4.17g)	apply mathematical concepts and computer-based technologies to assist analysis of data and information																		
4.18a)	select, and use appropriately, types of texts for different purposes and contexts including a discussion, explanation, procedure, exposition, recount, report, response or experimental records for oral or written presentation																		
4.18b)	select and use an appropriate medium to present data and information																		
4.18c)	select and use an appropriate method to acknowledge sources of information																		
4.18d)	use symbols to express relationships, including mathematical ones, and appropriate units for physical quantities																		
4.18e)	use drawings, diagrams, graphs, tables, databases, spreadsheets and flow charts to show relationships and present information clearly and/or succinctly																		

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Outcomes		*Science Focus 1*									*Science Focus 2*								
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.18f)	select and draw the appropriate type of graph (from column graph, histogram, divided bar, sector or line graph) or diagram to convey information and relationships clearly and accurately																		
4.19a)	justify inferences in light of gathered information																		
4.19b)	identify data which support or discount a hypothesis, a question being investigated or a proposed solution to a problem																		
4.19c)	predict outcomes and generate plausible explanations directly related to observations made																		
4.19d)	make generalisations in relation to a relevant set of observations or experimental results																		
4.19e)	anticipate and/or respond to problems as they arise in practical situations																		
4.19f)	use models, including mathematical ones, to explain phenomena or make predictions																		
4.19g)	use cause-and-effect relationships to explain ideas																		
4.20a)	identify the nature of a presented problem																		
4.20b)	describe different strategies that could be employed to solve an identified problem																		
4.20c)	use identified strategies to develop a range of possible solutions to a particular problem																		
4.20d)	evaluate the appropriateness of different strategies for solving an identified problem																		
4.21a)	seek evidence to support claims																		
4.21b)	evaluate evidence for reliability and validity																		
4.21c)	produce creative solutions for problems																		
4.21d)	propose ideas that demonstrate coherence and logical progression																		
4.21e)	apply critical thinking in the consideration of proposals																		
4.21f)	formulate cause-and-effect relationships																		
4.22.1a)	independently plan and conduct investigations, communicate information and understandings and solve problems																		

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Outcomes	*Science Focus 1*									*Science Focus 2*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
4.22.1b) set and work to realistic timelines and goals																		
4.22.1c) accept responsibility for maintenance of a safe working environment for themselves and others																		
4.22.1d) evaluate the effectiveness of their performance in completing the task																		
4.22.2a) identify the specific roles needed when working in a team																		
4.22.2b) match the tasks to the team members according to the requirements of the task and the skills of the individual																		
4.22.2c) negotiate and allocate individual roles to members of the team																		
4.22.2d) accept specific roles in a team while planning and conducting investigations, communicating information and understandings and solving problems																		
4.22.2e) set and work to realistic timelines and goals as a group																		
4.22.2f) accept personal responsibility for maintenance of a safe working environment for the team																		
4.22.2g) monitor progress of the team towards completion of a task																		
4.22.2h) evaluate the process used by the team and effectiveness of the team in completing the task																		

SKILLS – CORE CONTENT

STAGE 5

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.13.1a) describe a problem, hypothesis or question that can be tested or researched				**4.1, 4.5**	**5.1**	**6.1**		**SF**	**9.3**									
5.13.1b) propose possible sources of data and/or information relevant to the investigation		**2.4**		**4.2, 4.5**					**9.3**									

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.13.1c) identify what type of information or data needs to be collected		**SF**		**4.1, 4.2, 4.5**		**6.1**												
5.13.1d) justify why particular types of data or information are to be collected			**3.2**	**4.1, 4.5**														
5.13.1e) identify the appropriate units to be used in collecting data				**4.1, 4.5**		**6.2**												
5.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information			**3.2**	**4.1, 4.5**		**6.1**												
5.13.1g) formulate a means of recording the data to be gathered or the information to be collected		**2.4**		**4.5**														
5.13.2a) identify variables that need to be kept the same if first-hand data is to be collected		**2.3**	**3.3**	**4.4, 4.5**	**5.1**	**6.2**												
5.13.2b) specify the dependent and independent variables when planning controlled experiments		**2.3**	**3.3**	**4.1, 4.4, 4.5**	**5.1**													
5.13.2c) describes a logical procedure for undertaking a simple or controlled experiment to collect valid first-hand data	**1.1, 1.2**	**2.3**		**4.4, 4.5**	**5.1**	**6.2**	**7.1**		**9.6**									
5.13.2d) establish an appropriate timeline for an investigation				**4.4, 4.5**	**5.1**	**6.2**			**9.6**									
5.13.3a) identify advantages and limitations of using particular laboratory equipment for a specific task							**7.1**											
5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task						**6.2**	**7.1, 7.2**											
5.13.3c) describe ways to reduce the risk to themselves and others when working in the laboratory or field																		
5.14a) follow the planned procedure when performing an investigation	**1.1, 1.3**	**2.1**	**3.1, 3.4**	**4.1, 4.2, 4.3, 4.4, 4.5, 4.6**			**7.1, 7.2, 7.3**		**9.2, 9.4, 9.5, 9.6**									
5.14b) use time and resources effectively		**2.1**	**3.4**	**4.4, 4.5, 4.6**					**9.2**									

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Outcomes		*Science Focus 3*									*Science Focus 4*								
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.14c)	safely and efficiently construct, assemble and manipulate identified equipment	**1.1, 1.3**	**2.1**	**3.1**	**4.1, 4.2, 4.4, 4.5**			**7.1, 7.2, 7.3**		**9.5**									
5.14d)	record data using the appropriate units			**3.1**				**7.2**											
5.14e)	evaluate and modify experimental procedures	**1.1, 1.3**	**2.1**	**3.1**	**4.1**			**7.1, 7.2**		**9.2, 9.5**									
5.14f)	demonstrate the use of safe and hygienic work practices including the correct use of safety equipment			**3.1, 3.4**	**4.1, 4.3, 4.5**														
5.15a)	make and record observations and measurements accurately over a number of trials	**1.1, 1.2, 1.3**	**2.1, 2.3**	**3.4**	**4.1, 4.2, 4.3, 4.4**		**6.2**	**7.1, 7.2, 7.3**		**9.5**									
5.15b)	use independently a range of data collection strategies and technologies such as data loggers	**1.1, 1.2, 1.3**		**3.4**	**4.2**		**6.2**	**7.2**											
5.16a)	use a range of sources, including CD-ROMs and the Internet, to access information	**SF**	**2.1**	**3.4**		**5.2, 5.3, 5.3**	**6.1**		**8.3, 8.4, 8.5**	**9.3, 9.4**									
5.16b)	use a variety of techniques, such as key words, skimming and scanning, to identify appropriate information	**1.3**	**2.2**			**5,4**			**8.4, 8.5**	**9.4**									
5.16c)	extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio/visual resources		**2.2, SF**		**4.6**	**5.2, 5.3**				**9.3**									
5.16d)	summarise information from identified oral and written secondary sources		**2.2**	**3.4**	**4.6**	**5.2, 5.3**	**6.1**		**8.3, 8.4, 8.5**	**9.4**									
5.17a)	collate information from a number of sources	**1.2**	**2.3**		**4.6**				**8.3, 8.4, 8.5**										
5.17b)	distinguish between relevant and irrelevant information	**1.2, 1.3**	**2.3**		**4.6**				**8.4, 8.5**										

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Outcomes		*Science Focus 3*									*Science Focus 4*								
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.17c)	check reliability of gathered data and information by comparing them with observations or information from other sources	**1.1, 1.2, 1.3, SF**	**2.4**		**4.6**			**7.2**											
5.17d)	organise data using a variety of methods including diagrams, tables, spreadsheets and databases				**4.6**			**7.2**	**8.1**										
5.17e)	critically analyse the accuracy of scientific information presented in mass media	**SF**		**3.4,**		**5.4**		**7.2**	**8.1, 8.4, 8.5**										
5.17f)	identify trends, patterns, relationships and contradictions in data and information			**3.1, 3.4**			**6.2**	**7.2**											
5.17g)	apply mathematical concepts and computer-based technologies to assist analysis of data and information		**2.1, 2.3**		**4.1**			**7.2**		**9.5**									
5.18a)	select, and use appropriately, types of texts for different purposes and contexts including a discussion, explanation, procedure, exposition, recount, report, response or experimental records for oral or written presentation						**6.1**		**8.4, 8.5**	**9.2, 9.4**									
5.18b)	select and use an appropriate medium to present data and information		**2.3**		**4.4**				**8.2**	**9.5**									
5.18c)	select and use an appropriate method to acknowledge sources of information								**8.2**	**9.5**									
5.18d)	use symbols to express relationships, including mathematical ones, and appropriate units for physical quantities		**2.2**							**9.3**									
5.18e)	use drawings, diagrams, graphs, tables, databases, spreadsheets and flow charts to show relationships and present information clearly and/or succinctly							**7.2, 7.3**		**9.3**									
5.18f)	select and draw the appropriate type of graph (from column graph, histogram, divided bar, sector or line graph) or diagram to convey information and relationships clearly and accurately							**7.2, 7.3**											
5.19a)	justify inferences in light of gathered information				**4.3, 4.4**	**5.1**		**7.3**	**8.3**	**9.3, 9.5, 9.6**									
5.19b)	identify data which support or discount a hypothesis, a question being investigated or a proposed solution to a problem			**3.2**	**4.3, 4.6**														
5.19c)	predict outcomes and generate plausible explanations directly related to observations made		**2.4**	**3.2, 3.4**	**4.1**					**9.1, 9.5**									

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

Outcomes		*Science Focus 3*									*Science Focus 4*								
		Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.19d)	make generalisations in relation to a relevant set of observations or experimental results		**2.4**	**3.4**	**4.3, 4.6**			**7.3**		**9.1**									
5.19e)	anticipate and/or respond to problems as they arise in practical situations			**3,4**															
5.19f)	use models, including mathematical ones, to explain phenomena or make predictions								**8.1**										
5.19g)	use cause-and-effect relationships to explain ideas			**3.2**	**4.4, 4.6**				**8.1**	**9.1, 9.3**									
5.20a)	identify the nature of a presented problem			**3.2**			**6.1**			**9.1**									
5.20b)	describe different strategies that could be employed to solve an identified problem			**3.2**					**SF**										
5.20c)	use identified strategies to develop a range of possible solutions to a particular problem			**3.2**															
5.20d)	evaluate the appropriateness of different strategies for solving an identified problem			**3.2**															
5.21a)	seek evidence to support claims			**3.2**	**4.3**			**7.2**		**9.3**									
5.21b)	evaluate evidence for reliability and validity							**7.2**											
5.21c)	produce creative solutions for problems								**8.2, 8.3**	**9.2**									
5.21d)	propose ideas that demonstrate coherence and logical progression			**3.4**															
5.21e)	apply critical thinking in the consideration of proposals		**2.4**							**9.2**									
5.21f)	formulate cause-and-effect relationships		**2.4**	**3.4**			**6.1**												
5.22.1a)	independently plan and conduct investigations, communicate information and understandings and solve problems				**4.1**			**7.3**											
5.22.1b)	set and work to realistic timelines and goals				**4.1**														
5.22.1c)	accept responsibility for maintenance of a safe working environment for themselves and others																		
5.22.1d)	evaluate the effectiveness of their performance in completing the task				**4.1**														

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Science Focus 3 Curriculum Correlation Grid for *PFA* and *Skills* outcomes

Outcomes	*Science Focus 3*									*Science Focus 4*								
	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9	Ch1	Ch2	Ch3	Ch4	Ch5	Ch6	Ch7	Ch8	Ch9
5.22.2a) identify the specific roles needed when working in a team				**4.3, 4.4**	**5.1**		**7.3**	**8.2**	**9.3**									
5.22.2b) match the tasks to the team members according to the requirements of the task and the skills of the individual							**7.3**		**9.3**									
5.22.2c) negotiate and allocate individual roles to members of the team				**4.2, 4.4**	**5.1**		**7.3**		**9.3**									
5.22.2d) accept specific roles in a team while planning and conducting investigations, communicating information and understandings and solving problems								**8.2**										
5.22.2e) set and work to realistic timelines and goals as a group				**4.2**														
5.22.2f) accept personal responsibility for maintenance of a safe working environment for the team				**4.2, 4.3**			**7.3**	**8.2**										
5.22.2g) monitor progress of the team towards completion of a task				**4.4**			**7.3**	**8.2**										
5.22.2h) evaluate the process used by the team and effectiveness of the team in completing the task					**5.1**													

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Stage 4

Science Focus 1 and Science Focus 2

Outcome 4.6 identifies and describes energy changes and the action of forces in common situations			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.6.1 the law of conservation of energy	a) identify situations or phenomena in which different forms of energy are evident b) use models to describe different forms of energy c) identify objects that possess energy because of their motion (kinetic) or because of other properties (potential) d) qualitatively account for the total energy involved in energy transfers and transformations		
4.6.2 forces	a) identify changes that take place when particular forces are acting b) use the term 'field' in describing forces acting at a distance		
4.6.3 electrical energy	a) associate electricity with energy transfer in a simple circuit b) construct and draw circuits to show transfer of energy		
4.6.4 sound energy	a) describe sound as a form of energy requiring a medium for propagation		
4.6.5 light energy	a) describe light as a form of energy not requiring a medium for propagation		
4.6.6 heat energy	a) identify processes of heat transfer by conduction, convection and radiation		

Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.6.7 frictional force	a) describe friction as a contact force which opposes motion b) identify everyday situations where friction acts		
4.6.8 electrostatic force	a) describe ways in which objects acquire an electrostatic charge b) identify everyday situations where the effects of electrostatic forces can be		

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	observed c) describe the behaviour of charges when they are brought close to each other		
4.6.9 magnetic force	a) describe the behaviour of magnetic poles when they are brought close to each other b) identify everyday situations in which magnets and electromagnets are used		
4.6.10 gravitational force	a) identify that all objects exert a force of gravity on all other objects in the universe		

Outcome 4.7 describes observed properties of substances using scientific models and theories			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.7.1 the particle model of matter	a) describe the behaviour of matter in terms of particles that are continuously moving and interacting b) describe expansion and contraction of materials in terms of a simple particle model c) relate an increase or decrease in the amount of energy possessed by particles to changes in particle movement		
4.7.2 properties of solids, liquids and gases	a) relate properties of solids, liquids and gases to the particle model of matter b) describe the physical changes that occur during observations of evaporation, condensation, boiling, melting and freezing c) explain density in terms of a simple particle model d) explain changes in pressure in terms of increases or decreases in frequency of particle collisions		
4.7.3 change of state	a) relate changes of state to the motion of particles as energy is removed or added b) relate energy transfers in melting and freezing, condensation, evaporation and boiling to the particle model		

Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.7.4 elements	a) classify elements as metals or non-metals according to their common		

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	characteristics b) identify internationally recognised symbols for common elements		
4.7.5 mixtures	a) identify some common mixtures b) identify, using examples, the importance of water as a solvent c) describe aqueous mixtures in terms of solute, solvent and solution d) identify situations where the processes of filtration, sedimentation, sieving, distillation, chromatography, evaporation, condensation, crystallisation and magnetic attraction are appropriate to separate components of a mixture		

Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.7.6 compounds and reactions	a) distinguish between elements and compounds b) identify when a chemical reaction is taking place by observing changes in temperature, the appearance of a new substance or the disappearance of an original substance c) distinguish between compounds and mixtures		

Outcome 4.8 describes features of living things			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.8.1 cell theory	a) identify that living things are made of cells b) identify and describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, chloroplast c) identify that substances move into and out of cells d) distinguish between unicellular and multicellular organisms		

Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.8.2 classification	a) classify living things according to structural features and identify that they have patterns of similarities and differences b) identify a range of plants and animals using simple keys		

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Students learn about:	Students learn to:	Science Focus 1	Science Focus 2
4.8.3 unicellular organisms	a) identify the beneficial and harmful effects that microorganisms have on living things and the environment b) explain that reproduction in unicellular organisms takes place by cell division		

Students learn about:	Students learn to:	Science Focus 1	Science Focus 2
4.8.4 multicellular organisms	a) identify that there is a wide range of multicellular organisms b) identify that tissues, organs and organ systems in multicellular organisms consist of different types of cells c) explain why multicellular organisms require specialised organs and systems d) identify the materials required by multicellular organisms for the processes of respiration and photosynthesis e) describe the role of the root, stem and leaf in maintaining flowering plants as functioning organisms		

Students learn about:	Students learn to:	Science Focus 1	Science Focus 2
4.8.5 humans	a) describe the role of the digestive, circulatory, excretory, skeletal and respiratory systems in maintaining humans as functioning organisms		

Outcome 4.9 describes the dynamic structure of Earth and its relationship to other parts of our solar system and universe			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.9.1 the Newtonian model of the solar system	a) describe qualitatively relative sizes, distances and movements of components of our solar system b) describe relative movements of the planets, moons and Sun c) explain night and day in terms of Earth's rotation d) explain the seasons in terms of the tilt of Earth's axis and its revolution around the Sun		

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4.9.2 components of the universe	a) describe some major features of the universe, including galaxies, stars, nebulae and solar systems b) use appropriate scales to describe differences in sizes of, and distances between, structures making up the universe		

Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.9.3 the structure of Earth	a) describe the inner structure of Earth in terms of core, mantle, crust and lithosphere		
4.9.4 the atmosphere	a) identify gases that comprise the greater percentage of air and explain the difference between Earth's atmosphere and space b) describe the importance of atmospheric gases, including ozone and greenhouse gases, to life on Earth		
4.9.5 the hydrosphere	a) describe the water cycle in terms of the physical processes involved b) describe the effect of the forces of the Sun and Moon on the hydrosphere		
4.9.6 the lithosphere	a) identify that rocks are composed of minerals b) explain the breaking down of rocks in terms of physical and/or chemical changes c) relate the formation of landforms to weathering, erosion and deposition d) describe the origins of sedimentary, igneous and metamorphic rocks		

Outcome 4.10 identifies the factors affecting survival of organisms in an ecosystem			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.10 ecosystems	a) describe some adaptations of living things to factors in their environment b) describe, using examples of food chains and food webs from Australian ecosystems, how producers, consumers and decomposers are related c) describe the roles of photosynthesis and respiration in ecosystems d) discuss some effects of bushfires, drought and flood on Australian ecosystems		

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Outcome 4.11 identifies resources used by humans and where they are found, and describes ways in which they are exploited			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.11 natural resources	a) distinguish between natural and made resources b) give examples of resources from living things and resources extracted from the air, earth and oceans c) identify fossil fuels and describe some of their uses d) identify renewable and non-renewable sources of energy		

Outcome 4.12 identifies, using examples, common simple devices and explains why they are used			
Students learn about:	**Students learn to:**	**Science Focus 1**	**Science Focus 2**
4.12 technology	a) identify that technologies make tasks easier or more convenient b) identify a variety of energy transformations in everyday devices involving either electrical, sound, light and/or heat energy		

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Stage 5

Science Focus 3 and Science Focus 4

Outcome 5.6 applies basic physical models, theories and laws to situations involving energy, force and motion			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.6.1 the wave model	a) identify waves as carriers of energy b) qualitatively describe features of waves including frequency, wavelength and speed c) give examples of different types of radiation that make up the electromagnetic spectrum and identify some of their uses		

Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.6.2 Newton's Laws – motion	a) describe qualitatively the relationship between force, mass and acceleration b) explain qualitatively the relationship between distance, speed and time c) relate qualitatively acceleration to a change in speed and/or direction as a result of a net force d) analyse qualitatively common situations involving motion in terms of Newton's Laws		

Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.6.3 electrical energy	a) design, construct and draw circuits containing a number of components b) describe voltage, resistance and current using analogies c) describe qualitatively the relationship between voltage, resistance and current d) compare the characteristics and applications of series and parallel circuits		

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.6.4 light energy	a) distinguish between the absorption, reflection and refraction of light and identify everyday situations where each occurs	7.1, 7.2, 7.3	

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.6.5 nuclear energy	a) identify that energy and particles may be released from the nuclei of atoms		

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.6.6 gravitational force	a) distinguish between the terms 'mass' and 'weight'		

Outcome 5.7 relates properties of elements, compounds and mixtures to scientific models, theories and laws			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.7.1 atomic theory	a) describe features of and the location of protons, neutrons and electrons in the atom	2.1, 2.2	
	b) distinguish between elements, using information about the numbers of protons, neutrons and electrons	2.1, 2.2	
	c) describe an appropriate model that has been developed to describe atomic structure	2.1, 2.2	
5.7.2 elements	a) identify the atom as the smallest unit of an element and distinguish between atoms and molecules	2.1	
	b) describe some relationships between elements using the periodic table	2.2, 2.3, 2.4	

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.7.3 compounds and reactions	a) identify that a new compound is formed by rearranging atoms rather than by creating matter	3.1, 3.3	
	b) classify compounds into groups based on common chemical characteristics	3.3, 3.4	
	c) construct word equations from observations and written descriptions of a range of chemical reactions	3.1	

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	d) identify a range of common compounds using their common names and chemical formulae	3.3	
	e) qualitatively describe reactants and products in the following chemical reactions:		
	i) combustion	3.2, 3.3	
	ii) corrosion		
	iii) precipitation		
	iv) acids on metals and carbonates		
	v) neutralisation		
	vi) decomposition		
	f) describe the role of indicators	3.4	

Outcome 5.8 relates the structure and function of living things to models, theories and laws			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.8.1 cell theory	a) explain that systems in multicellular organisms serve the needs of cells	5.1	
	b) identify the role of cell division in growth, repair and reproduction in multicellular organisms	5.1, 5.3	
5.8.2 the Watson-Crick model of DNA	a) explain the advantages of DNA replicating exactly		
	b) explain the advantages and disadvantages of DNA mutating		
	c) identify that information is transferred as DNA on chromosomes when cells reproduce themselves		
	d) identify that genes are part of DNA	1.1	
	e) identify the role of genes and environmental factors in determining the features of an organism		

Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.8.3 the theory of evolution and natural selection	a) discuss evidence that present-day organisms have developed from organisms in the distant past		
	b) relate natural selection to the theory of evolution		

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Syllabus map for *Science Focus* series NSW Stage 4 and 5

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.8.4 humans	a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms	4.1,4.2, 4.3, 4.4, 4.5, 4.6	
	b) describe some responses of body systems to infectious and non-infectious diseases	5.4	
	c) relate the organs involved in human reproductive systems to their function	5.2, 5.3, 5.4	

Outcome 5.9 relates the development of the universe and the dynamic structure of Earth to models, theories and laws and the influence of time			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.9.1 the Big Bang theory	a) discuss current scientific thinking about the origin of the universe	8.1, 8.2, 8.3	
	b) identify that some types of electromagnetic radiation are used to provide information about the universe	8.1, 8.2	
	c) describe some of the difficulties in obtaining information about the universe	8.4, 8.5, 8.SF	
5.9.3 components of the universe	a) relate some major features of the universe to theories about the formation of the universe	8.2, 8.3	
	b) describe some changes that are likely to take place during the life of a star	8.2, 8.3	

Students learn about:	Students learn to:	Science Focus 3	Science Focus 4
5.9.2 the theory of plate tectonics	a) discuss evidence that suggests crustal plates move over time	9.1	
5.9.4 natural events	a) identify that geological history can be interpreted from the formation, by sediments, of horizontal layers in which the oldest are at the base and the youngest at the top	9.6	
	b) describe conditions under which fossils form	9.6	
	c) relate the fossil record to the age of Earth and the time over which life has been evolving	9.6	
	d) relate movements of Earth's plates to convection currents in the mantle and to	9.1	

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	gravitational forces e) explain how interactions at plate boundaries may result in earthquakes, volcanic activity and new landforms f) explain some impacts of natural events including cyclones, volcanic eruptions and earthquakes on the atmosphere, hydrosphere, lithosphere and/or biosphere	9.1, 9.2, 9.3, 9.4, 9.5 9.3, 9.4	

Outcome 5.10 assesses human impacts on the interaction of biotic and abiotic features of the environment			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.10 ecosystems	a) distinguish between biotic and abiotic features of the local environment b) describe the importance of cycles of materials in ecosystems c) describe some impacts of human activities on ecosystems	6.2, 6.3 6.3, 6.SF	

Outcome 5.11 analyses the impact of human resource use on the biosphere to evaluate methods of conserving, protecting and maintaining Earth's resources			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.11.1 energy resources	a) discuss the importance of energy as a resource b) identify properties that make some natural resources economically important and describe their uses	6.1 6.3	
5.11.2 waste from resource use	a) relate pollution to contamination by unwanted substances b) identify excessive use of fossil fuels as a contributing factor to a greenhouse effect c) discuss strategies used to balance human activities and needs in ecosystems with conserving, protecting and maintaining the quality of the environment	6.3 6.3 6.3, 6.SF	

Outcome 5.12 describes scientific principles underlying some common technologies			
Students learn about:	**Students learn to:**	**Science Focus 3**	**Science Focus 4**
5.12 technology	a) describe some everyday uses and effects of electromagnetic radiation, including applications in communication technology		

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	b) discuss the benefits and problems associated with medical and industrial uses of nuclear energy c) describe some benefits and problems of using biotechnology d) describe ways in which technology has increased the variety of made resources	1.1	

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Science Focus Teaching Program Introduction

The main teaching features of each unit were highlighted in these teaching programs. The skills from the syllabus that are identified in each unit are those that form a major component of skill development for that unit. Where possible, additional content has been provided in each chapter.

It is suggested that these programs be used only as a scaffold for developing individual school-based programs.
When modifying the programs please include:

- relevant guidelines and directives from your educational sector
- school-based curriculum development policies
- occupational health and safety standards
- chemical safety in schools
- animal welfare guidelines.

When modifying the programs consider:

- what teaching and learning approach your faculty uses
- students' learning styles, cultural backgrounds and specific learning needs
- school resources and resource priorities.

The Teacher's Edition wrap around notes provide extra activities that are useful in teaching the program. Choose which suit your school's teaching practices and your students' learning styles according to the Multiple Intelligences.

Refer to the Science Year 7–10 syllabus support document, *Advice on Programming and Assessment* and the *Quality Science* CD for information and guidance on program development and evaluation.

The document *Advice on Programming and Assessment* is an invaluable resource. It provides advice about constructing a program, maps a process for planning and sequencing of units and offers advice on how to develop teaching and learning activities.
It provides scaffolds for:

- stage scope and sequence plans
- sample lesson sequences
- building a teaching and learning program
- developing units of work to ensure coverage of the scope of the syllabus.

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1: Forensics

Prescribed focus area	**The application and uses of science** 5.3a) identify and describe examples of scientific concepts and principles that have been used in technological development (including Australian examples). 5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.3c) identify and describe examples where technological advances have impacted on science. 5.3d) give reasons why society should support scientific research.

Domain outcomes	
Knowledge and understanding	5. 3 evaluate the impact of applications of science on society and the environment. 5.12 relate the interactions involved in using some common technologies to their underlying scientific principles. 5.15 gather first-hand data accurately. 5.16 access information from a wide variety of secondary sources. 5.17 explain trends, patterns and relationships in data and/or information from a variety of sources.
Skills	5.13 identifies a problem and independently produces an appropriate investigation plan. 5.14 undertakes first-hand investigations independently with safety and competence. 5.15 gathers first-hand data accurately. 5.16 accesses information from a wide variety of secondary sources. 5.17 explains trends, patterns and relationships in data and/or information from a variety of sources. 5.18 selects and uses appropriate forms of communication to present information to an audience. 5.19 uses critical thinking skills in evaluating information and drawing conclusions. 5.20 selects and uses appropriate strategies to solve problems.

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	5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 plans, implements and evaluates the effectiveness of a variety of tasks independently and as a team member.
Values and attitudes	5.23 demonstrates confidence and willingness to make decisions and to take responsible actions. 5.24 respects differing viewpoints on science issues and is honest, fair and ethical. 5.26 recognises the role of science in providing information about issues being considered and in increasing their understanding of the world around them.

Unit 1.1 Forensics and identification

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.15 gathering data	4.12a) identify that technologies make tasks easier or more convenient. 5.8.2d) identify that genes are part of DNA. 5.12d) describe ways in which technology has increased the variety of manufactured resources. 5.13.2c) describe a logical procedure for undertaking a simple or controlled experiment to control valid first-hand data. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified	**Quick Quiz**: Chapter 1 (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Forensic science/ Forensic methods of identification** Discuss the basics involved with forensic science and the people who work in forensics. **Facial recognition** Look at photographic identification, Identikit drawings and biometric facial recognition using computer technology. **Prac 1**: Make your own Identikit **Drag and drop**: Crime time (*Science Focus 3 Second Edition LiveText*) **Identification using the body** Look at the Bertillon system, fingerprints and identification using the eye.	Quick Quiz Prac 1 Drag and drop	

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	equipment. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately. 5.15b) independently use a range of data collection strategies and technologies such as data loggers. 5.17c) check the reliability of gathered data and information by comparing.	**Prac 2**: Fingerprints **Worksheet 1.1**: Fingerprints **Genetic identification** Look at DNA profiling and the use of DNA samples. Also consider where DNA can go wrong. **Identifying a body** Consider evidence to identify dead bodies; sometimes these are known and other times these are unknown. **Worksheet 1.2**: Time of death **Prac 3**: Collecting teeth impressions **Prac 4**: Forensic anthropometry **Case study**: The 'Pyjama Girl Murder Case' **Unit 1.1 questions**	Prac 2 Worksheet 1.1 Worksheet 1.2 Prac 3 Prac 4 Case study Unit questions	

Unit 1.2 Is it real?

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.17 processing information	5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.15a) make and record observations and measurements accurately. 5.15b) independently use a range of data collection strategies and technologies. 5.17a) collate information from a number of sources.	**Analysis of writing and print** Writing can be analysed no matter what form it is in: handwriting, matching typewriters, printer matching, ink analysis. **Prac 1**: Testing handwriting **Prac 2**: Chromatography catches a criminal **Forgery** Forgery can be used to counterfeit money. There are methods used by governments to avoid this.	Prac 1 Prac 2	

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	5.17b) distinguish between relevant and irrelevant information. 5.17c) check the reliability of gathered data and information by comparing.	**Prac 3**: Examining money **Prac 4**: Forging a photograph **Drag and drop**: Money protection (*Science Focus 3 Second Edition LiveText*) **Case study**: The medicine man **Unit 1.2 questions** **Web destination**: Who makes Australia's banknotes? (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: How are coins minted? (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 3 Prac 4 Drag and drop Case study Unit questions Web destination Web destination	

Unit 1.3 Evidence

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.17 processing information	5.2f) give examples that demonstrate the benefits and limitations of using models. 5.2g) identify that the nature of observations made depends upon the understanding that the observer brings to the situation. 5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.4c) identify choices that need to be	**What is evidence?** Define evidence **Collecting fingerprints** Using fingerprints is useful; however, these need to be collected at the crime scene. **Prac 1**: Comparing fingerprints **Collecting body products and fibres** It is possible to use fibres and products on the body to help in evidence collection. Fibres can be investigated but can sometimes be classed as circumstantial evidence and be potentially inconclusive.	Prac 1	

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	made or have been made when considering whether to use particular scientific advances. 5.5d) identify possible career paths in science. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately. 5.15b) independently use a range of data collection strategies and technologies. 5.16b) use a variety of techniques, such as keywords, skimming and scanning to identify appropriate information. 5.17b) distinguish between relevant and irrelevant information. 5.17c) check the reliability of gathered data and information by comparing.	**Prac 2**: Fibre analysis **Video**: How elements can help solve crimes (*Science Focus 3 Second Edition LiveText*) **Collecting impressions** Discuss the use of tool marks or track impressions in collecting evidence. **Prac 3**: Collecting foot impressions **Collecting biological evidence** Water organisms and insects can be helpful in evidence collection. **Collecting electronic evidence** Discuss how security cameras and timed frames can provide useful information about crimes. **What does a bloody mess tell us?/What do wounds tell us?/What do gunshot injuries tell us?** Investigate blood, wounds and gunshots and what information they provide **Case study**: Eaten twice by shark! **Prac 4**: Investigating stride length **Prac 5**: Blood drips **Unit 1.3 Questions** **Worksheet 1.3**: Careers **Drag and drop**: Forensic jobs (*Science Focus 3 Second Edition LiveText*)	Prac 2 Video Prac 3 Case study Prac 4 Prac 5 Unit questions Worksheet 1.3 Drag and drop	

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1: Forensics

Science Focus: Investigating the death of Azaria Chamberlain

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.16 gathering information from secondary sources	5.4e) discuss the place of social and ethical considerations in scientific practice and in applications of science. 5.17c) check the reliability of gathered data and information by comparing. 5.17e) critically analyse the accuracy of scientific information presented in the mass media. 5.16a) use a range of sources, including databases, CD-ROMs and the Internet, to access information.	Consider the evidence presented and how it was used to incorrectly convict Lindy Chamberlain. Student activities: Investigate and present what occurred. Review *Evil Angels*. Discuss why this case is still so renowned.		

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 1.4**: Crossword	Worksheet 1.4	
Complete **Worksheet 1.5**: Sci-words	Worksheet 1.5	
Interactive: Forensics interactive crossword (*Science Focus 3 Second Edition LiveText*)	Interactive	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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1: Forensics

Program review and evaluation

Topic: Forensics
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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2: The periodic table

Prescribed focus area	The history of science
	5.la) identify some of the ideas from different cultures (including those of Aboriginal and Indigenous peoples) that have contributed to science throughout history. 5.1b) describe some models and theories that have been considered in science and have then been modified or rejected as a result of available evidence. 5.1c) discuss examples of where societal religious or ethical values have had an impact on scientific developments. 5.1d) describe historical cases where developments in science have led to the development of new technologies. 5.1e) describe historical cases where developments or improvements in technology have transformed science.

Domain outcomes	
Knowledge and understanding	5.7.1a) describe features and the location of protons, neutrons and electrons in the atom. 5.7.1b) distinguish between elements using information about the number of protons, neutrons and electrons. 5.7.1c) describe an appropriate model that has been developed to describe atomic structure. 5.7.2a) identify the atom as the smallest unit of an element and distinguish between atoms and molecules. 5.7.2b) describe some relationships between elements using the periodic table.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.14 follows a sequence of instructions to undertake a first-hand investigation. 5.15 uses given criteria to gather first-hand data. 5.16 accesses information from identified secondary sources. 5.17 evaluates the relevance of data and information.

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	5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.26 recognises the role of science in providing information about issues being considered and in increasing understanding of the world around them.

Unit 2.1 Atoms and elements

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.7.1 atomic theory	5.7.1a) describe features and the location of protons, neutrons and electrons in the atom. 5.7.1b) distinguish between elements using information about the number of protons, neutrons and electrons. 5.7.1c) describe an appropriate model that has been developed to describe atomic structure. 5.7.2a) identify the atom as the smallest unit of an element and distinguish between atoms and molecules.	**Quick Quiz**: Chapter 2 (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Atoms** What are atoms made of? Identify the three particles that make up an atom: neutron, proton and electron. **Atomic and mass numbers** Define each term and carry out an example of fluorine (F). **Interactive**: Atom X (*Science Focus 3 Second Edition LiveText*) **Elements** Define as the building blocks of all matter. Relate the history of elements to today's understanding of atoms and	Quick Quiz Interactive	

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	5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources effectively. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.17g) apply mathematical concepts and computer-based technologies to assist analysis of data and information.	elements. **Worksheet 2.1**: Elements **Prac 1**: Making elements **Compounds** Brief introduction to compounds **Molecules and lattices** Distinguish between molecules and a lattice, giving examples. **Prac 2**: Making a compound **Mixtures** Recall the definition of mixtures **Prac 3**: Compounds in soft drinks **Drag and drop**: What's the matter? (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Video**: Mixtures and compounds (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Unit 2.1 Questions** **Web destination**: Periodic table of comic books (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 2.1 Prac 1 Prac 2 Prac 3 Drag and drop Video Unit questions Web destination	

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Unit 2.2 Structure of the periodic table

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.7.2 elements	5.7.1a) describe features and the location of protons, neutrons and electrons in the atom. 5.7.1b) distinguish between elements, using information about the number of protons, neutrons and electrons. 5.7.1c) describe an appropriate model that has been developed to describe atomic structure. 5.7.2b) describe some relationships between elements using the periodic table. 5.15a) make and record observations and measurements accurately over a number of trials. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to assess information. 5.16b) use a variety of techniques, such as key words, skimming and scanning to identify appropriate information. 5.16c) extract information from column graphs, histograms, divided bar and	**The periodic table** Summary of the periodic table. **Features of the periodic table** Discuss the periodic table as shown in Fig 2.2.2. In particular, point out the metals, non-metals and semi-metals. Also identify groups and periods. **Prac 1**: Investigating a physical property **Prac 2**: Comparing elements **Worksheet 2.2**: Who am I? **Interactive**: Periodic table (*Science Focus 3 Second Edition LiveText*) **The role of electrons** Explain where electrons are in an atom, and the concept of electron shells or levels. Examine the electron arrangement in the first 20 elements. **Video**: Electronic structure (*Science Focus 3 Second Edition LiveText*) **Video**: Periodic trends: electronegativity (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Periods, groups and electrons** Discuss the relationship between the periods and groups of the periodic table and electrons.	Prac 1 Prac 2 Worksheet 2.2 Interactive Video Video	

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	sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources. 5.16d) summarise information from identified oral and written secondary sources. 5.18b) select and use an appropriate medium to present data and information.	**Unit 2.2 questions** **Web destination**: Chemical elements (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Pictorial periodic table (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Royal Society of Chemistry (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Web elements (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Unit questions Web destination Web destination Web destination Web destination	

Unit 2.3 Using the periodic table

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.7.1 atomic theory	5.7.2b) describe some relationships between elements using the periodic table. 5.13.2a) identify variables that need to be kept held constant if reliable first-hand data is to be collected. 5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task.	**Atoms that react and atoms that don't** Examine what a full outer shell has to do with reactivity. **Ions** Define ions as atoms that have gained or lost electrons. In particular, with positive ions, make sure students don't think there is a gain of protons. **Prac 1**: Firework colours **Prac 2**: Ions get together! **Worksheet 2.3**: Ions	Prac 1 Prac 2 Worksheet 2.3	

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	5.13.3c) describe ways to reduce the risk to themselves and others when working in the laboratory or field. 5.15a) make and record observations and measurements accurately. 5.17a) collate information from a number of sources. 5.17b) distinguish between relevant and irrelevant information. 5.17g) apply mathematical concepts and computer-based technologies to assist analysis of data and information.	**The odd couple: H and He** Compare hydrogen and helium, in a league of their own, with only one and two electrons, respectively **Drag and drop**: Ions and atoms (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Those electrons again!** Discuss the difference between how electrons react in metals and non-metals **Metals, non-metals and semi-metals** What is a metal and what is not a metal? Have students compare features between the three types. **Prac 3**: Metal crystals **Prac 4**: More crystals **Prac 5**: Changing the properties of metals **Prac 6**: Using metals to make non-metals **Worksheet 2.4**: Periodic table properties **Worksheet 2.5**: The periodic table **Unit 2.3 questions**	Drag and drop Prac 3 Prac 4 Prac 5 Prac 6 Worksheet 2.4 Worksheet 2.5 Unit questions	

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Unit 2.4 Families of elements

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.7.2 elements	5.7.2b) describe some relationships between elements using the periodic table. 5.13.1b) propose possible sources of data and/or information relevant to the investigation. 5.13.1g) formulate a means of recording the data to be gathered or the information to be collected. 5.17c) check reliability of gathered data and information by comparing with observations or information from other sources. 5.19c) predict outcomes and generate plausible explanations directly related to observations made. 5.l9d) make generalisations in relation to a relevant set of observations or experimental results. 5.21e) apply critical thinking in the consideration of proposals. 5.21f) formulate cause and effect relationships.	**Group I: the alkali metals** Compare with group VII and note their properties. **Group II: the alkaline earths** **Prac 1**: The alkaline earths **Group IV** Discuss their chemical characteristics and practical applications of their properties. **Prac 2**: Group IV **Group VII: the halogens** Discuss differences with previous group. **Group VIII: the noble gases** Discuss their chemical characteristics. **The transition metals** Discuss their chemical characteristics. **Interactive**: Interactive periodic table (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Drag and drop**: Families of elements (*Science Focus 3 Second Edition LiveText*) **Unit 2.3 questions**	Prac 1 Prac 2 Interactive Drag and drop Unit questions	

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2: The periodic table

Science Focus: Development of the periodic table

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.1 the history of science	5.1a) identify some of the ideas from different cultures (including those of Aboriginal and Indigenous peoples) that have contributed to science throughout history. 5.1b) describe some models and theories that have been considered in science and have then been modified or rejected as a result of available evidence. 5.13.1c) identify what type of information or data needs to be collected. 5.16c) extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources.	Discuss the early development of elements from Aristotle. Examine the work of Dalton, Dobereiner, Newlands, Mendeleev, Lother Meyer and Moseley. Discuss the formation of artificial elements.	Various student activities including research and investigations	

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2: The periodic table

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 2.6**: Crossword	Worksheet 2.6	
Complete **Worksheet 2.7**: Sci-words	Worksheet 2.7	
Interactive: Periodic table interactive crossword (*Science Focus 3 Second Edition LiveText*)	Interactive	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: The periodic table
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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3: Chemical change

Prescribed focus area	**The nature and practice of science** 5.2a) evaluate the importance of using creativity, curiosity, objectivity and logical reasoning in describing phenomena in their surroundings, stimulating investigations about phenomena and devising and testing hypotheses. 5.2c) apply scientific processes to test the validity of ideas and theories. 5.2d) describe how an idea can gain acceptance in the scientific community as either theory or law. 5.2e) use examples which show that scientists isolate a set of observations, identify trends and patterns and construct hypotheses or models to explain these. 5.2f) give examples that demonstrate benefits and limitations of using models. 5.2g) identify that the nature of observations made depends upon the understanding that the observer brings to the situation.

Domain outcomes	
Knowledge and outcomes	5.7.3a) identify that a new compound is formed by rearranging atoms rather than by creating matter. 5.7.3b) classify compounds into groups based on common chemical characteristics. 5.7.3c) construct word equations from observations and written descriptions of a range of chemical reactions. 5.7.3d) identify a range of common compounds using their common names and chemical formulae. 5.7.3e) qualitatively describe reactants and products in the following chemical reactions: i) combustion ii) corrosion iii) precipitation iv) acids on metals and carbonates v) neutralisation

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	vi) decomposition. 5.7.3f) describe the role of indicators.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.14 follows a sequence of instructions to undertake a first-hand investigation. 5.15 uses given criteria to gather first-hand data. 5.16 accesses information from identified secondary sources. 5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.25 recognises the relevance and importance of lifelong learning and acknowledges the continued impact of science in many aspects of everyday life. 5.26 recognises the role of science in providing information about issues being considered and in increasing their understanding of the world around them.

Unit 3.1 Chemical reactions

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
		Quick Quiz: Chapter 3 (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Quick Quiz	

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5.7.3 compounds and reactions	5.7.3a) identify that a new compound is formed by rearranging atoms rather than by creating matter. 5.7.3c) construct word equations from observations and written descriptions of a range of chemical reactions. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14d) record data using the appropriate units. 5.14e) evaluate and modify experimental procedures. 5.14f) demonstrate the use of safe and hygienic work practices, including the correct use of safety equipment. 5.17f) identify trends, patterns, relationships and contradictions in data and information.	**Physical change/Chemical change** Describe physical and chemical changes and compare each. **Prac 1**: Chemistry in the kitchen **Animation**: Physical *vs* chemical change (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Chemical equations** Introduction to chemical equations, formulae and the state of matter. **Animation**: Representation of a chemical equation (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Signs of chemical change** Describe the chemical changes to look for in a chemical reaction, such as colour changes or gas production. **Prac 2**: Signs of chemical change **Prac 3**: Light sticks: chemiluminescence **Video**: Exothermic reactions (*Science Focus 3 Second Edition LiveText*) **Interactive**: Burning magnesium (*Science Focus 3 Second Edition LiveText*) **Drag and drop**: Chemical reactions (*Science Focus 3 Second Edition LiveText*, *Science*	Prac 1 Animation Animation Prac 2 Prac 3 Video Interactive Drag and drop	

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		Focus 3 Second Edition Student Lounge) **Why do chemical reactions occur?** Discuss chemical bonding and diatomic gases. **Unit 3.1 questions** **Web destination**: Acid rain (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: The chemistry of acid rain (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	 Unit questions Web destination Web destination	

Unit 3.2 Combination, combustion and decomposition

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.7.3 compounds and reactions	5.7.3e) qualitatively describe reactants and products in the following chemical reactions: i) combustion ii) corrosion iii) precipitation iv) acids on metals and carbonates	**Combination reactions** Define reaction and give an example. **Video**: Formation of sodium chloride (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Combustion reactions** Discuss as a burning reaction in oxygen. May also be called oxidation.	 Video	

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	v) neutralisation vi) decomposition. 5.13.1d) justify why particular types of data or information are to be collected. 5.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information. 5.19b) identify data which supports or discounts an hypothesis, question being investigated or the proposed solution to a problem. 5.19c) predict outcomes and generate plausible explanations directly related to observations made. 5.19g) use cause and effect relationships to explain ideas. 5.20a) identify the nature of a presented problem. 5.20b) describe different strategies that could be employed to solve an identified problem. 5.20c) use identified strategies to develop a range of possible solutions to a particular problem. 5.20d) evaluate the appropriateness of different strategies for solving an identified problem. 5.21a) seek evidence to support claims.	**Prac 1**: Burning magnesium ribbon (teacher demonstration) **Prac 2**: Stopping combustion **Decomposition reactions** Define as the opposite to combination reaction. **Video**: Airbags (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Thermal decomposition** Describe the special type of decomposition reaction where heat breaks down two or more components. **Prac 3**: Decomposition of a metal carbonate **Unit 3.2 questions**	Prac 1 Prac 2 Video Prac 3 Unit questions	

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Unit 3.3 Precipitation reactions

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.7.3 compounds and reactions	5.7.3a) identify that a new compound is formed by rearranging atoms rather than by creating matter. 5.7.3b) classify compounds into groups based on common chemical characteristics. 5.7.3d) identify a range of common compounds using their common names and chemical formulae. 5.7.3e) qualitatively describe reactants and products in the following chemical reaction: iii) precipitation. 5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task.	**Precipitation reactions** Define and discuss the rules governing how a precipitate forms. **Video**: Precipitation reactions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Ions and salts** Reconsider electron configuration to discuss ions and salts. **Worksheet 3.1**: Ionic compounds **Animation**: Naming ionic compounds (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Predicting precipitation reactions** Discuss solubility rules. **Prac 1**: Precipitation reactions **Prac 2**: Precipitation of unknowns **Interactive**: Bonding (*Science Focus 3 Second Edition LiveText*) **Unit 3.3 questions** **Web destination**: Chemical reactions quiz (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Chembalancer (*Science*	Video Worksheet 3.1 Animation Prac 1 Prac 2 Interactive Unit questions Web destination Web destination	

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		Focus 3 Second Edition LiveText, *Science Focus 3 Second Edition Student Lounge*)		

Unit 3.4 Acids and bases

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.7.3 compounds and reactions	5.7.3b) classify compounds into groups based on common chemical characteristics. 5.7.3f) describe the role of indicators. 5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources effectively. 5.14f) demonstrate the use of safe and hygienic work practices including the correct use of safety equipment. 5.15a) make and record observations and measurements accurately. 5.15b) use independently a range of data collection strategies and technologies such as data loggers. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16d) summarise information from identified oral and written secondary sources.	**Acids** Define an acid and list some of their properties. Distinguish between strong and weak acids, giving examples that may be in everyday use. Contrast concentrated and dilute acids and demonstrate the correct procedure for dilution of an acid. Always add acid to water. **Drag and drop**: Common names for acids (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Bases** Define a base and an alkali. Contrast strong and weak bases and some common uses. **Drag and drop**: About acids and bases (*Science Focus 3 Second Edition LiveText*) **Concentration** Discuss the concentration of acids and bases. **The pH scale** Define this scale as one that allows the classification of the strength of acids and bases. Describe indicators as chemicals that can show the pH of a solution.	Drag and drop Drag and drop	

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	5.17e) critically analyse the accuracy of scientific information presented in popular media. 5.17f) identify trends, patterns, relationships and contradictions in data and information. 5.19c) predict outcomes and generate plausible explanations directly related to observations made. 5.19d) make generalisations in relation to a relevant set of observations or experimental results. 5.19e) anticipate and/or respond to problems as they arise in practical situations. 5.21d) propose ideas that demonstrate coherence and logical progression. 5.21f) formulate cause and effect relationships.	**Interactive**: pH scale (*Science Focus 3 Second Edition LiveText*) **Interactive**: Indicators (*Science Focus 3 Second Edition LiveText*) **Drag and drop**: The pH scale (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Worksheet 3.3**: pH levels **Prac 1**: Common indicators **Prac 2**: Universal indicator **Prac 3**: Natural indicators **Video**: Natural indicators (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 4**: Testing household solutions **Neutralisation** Define and give examples. **Video**: Sodium and potassium in water (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Worksheet 3.2**: Neutralisation **Prac 5**: Acids and hydroxides **Prac 6**: Acids and carbonates **Acids and metals** Discuss the reactions	Interactive Interactive Drag and drop Worksheet 3.3 Prac 1 Prac 2 Prac 3 Video Prac 4 Video Worksheet 3.2 Prac 5 Prac 6	

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		between acids and metals. **Prac 7**: Acids and metals **Unit 3.4 questions**	 Prac 7 Unit questions	

Chapter review

Suggested teaching and learning strategies	***Science Focus 3* Resources**	**Register**
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 3.4**: Crossword	Worksheet 3.4	
Complete **Worksheet 3.5**: Sci-words	Worksheet 3.5	
Interactive: Chemical change interactive crossword (*Science Focus 3 Second Edition LiveText*)	Interactive	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Chemical reactions
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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4: Sense and control

Prescribed focus area	The nature and practice of science
	5.2c) apply scientific processes to test the validity of ideas and theories. 5.2d) describe how an idea can gain acceptance in the scientific community as either theory or law. 5.2e) use examples which show that scientists isolate a set of observations, identify trends and patterns and construct hypotheses or models to explain these. 5.2f) give examples that demonstrate the benefits and limitations of using models. 5.2g) identify that the nature of observations made depends upon the understanding that the observer brings to the situation.

Domain outcomes	
Knowledge and understanding	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.14 follows a sequence of instructions to undertake a first-hand investigation. 5.15 uses given criteria to gather first-hand data. 5.16 accesses information from identified secondary sources. 5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.21 uses creativity and imagination in the analysis of problems and in the development of possible solutions.

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	5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.23 demonstrates confidence and willingness to make decisions and to take responsible actions. 5.25 recognises the relevance and importance of lifelong learning and acknowledges the continued impact of science in many aspects of everyday life.

Unit 4.1 Sight

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.12a) describe some everyday uses and effects of electromagnetic radiation, including applications in communication technology. 5.13.1a) describe a problem, hypothesis or question that can be tested or researched. 5.13.1c) identify what type of information or data needs to be collected. 5.13.1d) justify why particular types of data or information are to be	**Quick Quiz**: Chapter 4 (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **The eye** Investigate the structure of the eye and compare it to a simple camera. Discuss the importance of eye protection. Investigate the advantages having two eyes. **Prac 1**: Eye tests **Prac 2**: Persistence of vision **Worksheet 4.1**: The eye **Video**: The structure of the eye (*Science Focus 3 Second Edition LiveText*) **Prac 3**: Eye dissection **Colour vision** Describe how eyes can see colours. Define the role of rods and cones. Investigate colour blindness. **Animal eyes** Discuss why eyes are situated in various	Quick Quiz Prac 1 Prac 2 Worksheet 4.1 Video Prac 3	

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	collected. 5.13.1e) identify the appropriate units to be used in collecting data. 5.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information. 5.13.1g) formulate a means of recording the data to be gathered or the information to be collected. 5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.13.2b) specify the dependent and independent variables when planning controlled experiments. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14e) evaluate and modify experimental procedures. 5.14f) demonstrate the use of safe and hygienic work practices, including the correct use of safety equipment. 5.17g) apply mathematical concepts and computer-based technologies to	locations on different animals. Investigate whether animals see colours. **Eye defects** Discuss why people wear glasses. Describe the two different defects and the methods used to overcome them. Recent advances in contact lenses allow for longer wearing times. Investigate the application of laser technology to correct defects. **Interactive**: The eye (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 4**: Lens and retina **Unit 4.1 questions** **Web destination**: Online dissection of a cow's eye (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: The eye (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Label the eye (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Interactive Prac 4 Unit questions Web destination Web destination Web destination	

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	assist analysis of data and information. 5.19c) predict outcomes and generate plausible explanations directly related to observations made. 5.22.1a) independently plan and conduct investigations, communicate information and understandings and solve problems. 5.22.1b) set and work to realistic timelines and goals. 5.22.1d) evaluate the effectiveness of their performance in completing the task.			

Unit 4.2 Hearing

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.13.1b) propose possible sources of data and/or information relevant to the investigation. 5.13.1c) identify what type of information or data needs to be	**Sound** Recall from Book 1 that sound waves transfer energy. Describe how the ear converts this sound energy to electrical energy. **The ear** Describe the ear as having three main sections. Identify each section and its structure. Define each part within each section. Discuss with students why they think two ears are an advantage. Compare with eyes.		

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	collected. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14d) record data using the appropriate units. 5.15a) make and record observations and measurements accurately. 5.15b) independently use a range of data collection strategies and technologies, such as data loggers. 5.22.2c) negotiate and allocate individual roles to members of the team. 5.22.2e) set and work to realistic timelines and goals as a group. 5.22.2f) accept personal responsibility for maintenance of a safe working environment for the team.	**Prac 1**: Hearing tests 1 **Ear problems** Describe how excessive sound energy may damage the inner structures of the ear. This damage can be either temporary or permanent. Discuss the Australian invented and developed bionic ear. **Worksheet 4.2**: The ear **Worksheet 4.3**: Hearing **Ear protection** Discuss the importance of protecting your ears. Investigate the noise level that causes damage. **Prac 2**: Modelling hearing difficulties **Prac 3**: Measuring decibels **Unit 4.2 questions** **Web destination**: The ear (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 1 Worksheet 4.2 Worksheet 4.3 Prac 2 Prac 3 Unit questions Web destination	

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Unit 4.3 Smell, taste and touch

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.14a) follow the planned procedure when performing an investigation. 5.14f) demonstrate the use of safe and hygienic work practices, including the correct use of safety equipment. 5.15a) make and record observations and measurements accurately. 5.19a) justify inferences in light of gathered information. 5.19b) identify data which supports or discounts an hypothesis, questions being investigated or the proposed solution to a problem. 5.l9d) make generalisations in relation to a relevant set of observations or experimental results. 5.21a) seek evidence to support claims. 5.22.2a) identify the specific roles needed when working in a team.	**Smell** Describe how we smell. Brainstorm why freshly cut grass and BBQ smells travel long distances. Why do we sneeze? **Taste** Describe the method of taste and the five basic tastes. Recent research appears to disprove specific taste areas on the tongue. Investigate the interaction of taste and smell. **Prac 1**: Taste regions **Prac 2**: Taste trickery **Touch** Describe the role of the various layers of the skin and touch. **Prac 3**: Skin receptors **Worksheet 4.4**: The skin **Skin conditions** Describe various skin conditions, with emphasis on skin cancers. Perhaps, invite a skin cancer specialist to talk to students about being sun safe. **Worksheet 4.5**: Codes **Unit 4.3 questions**	Prac 1 Prac 2 Prac 3 Worksheet 4.4 Worksheet 4.5 Unit questions	

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	5.22.2f) accept personal responsibility for maintenance of a safe working environment for the team.			

Unit 4.4 Responding

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.13.2b) specify the dependent and independent variables when planning controlled experiments. 5.13.2c) describe a logical procedure for undertaking a simple or controlled experiment to collect valid first-hand data. 5.13.2d) establish an appropriate timeline for an investigation. 5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources	**Responding to stimuli** Recall that responding is one of the characteristics of all living things. **Drag and drop**: Receptors and stimuli (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Keeping things constant** Define how senses are detected by humans. Describe the location of the sense receptors. Describe a simple stimulus-response model. Have students find two more examples. **Prac 1**: Sweet and salty **Prac 2**: Sound threshold **Feedback and coordination** Describe the coordination required for a functioning human. Investigate what occurs when the messages are mixed up. **Worksheet 4.6**: Responding **Unit 4.6 questions**	Drag and drop Prac 1 Prac 2 Worksheet 4.6 Unit questions	

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	effectively. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.15a) make and record observations and measurements accurately. 5.18b) select and use an appropriate medium to present data and information. 5.19a) justify inferences in light of gathered information. 5.19g) use cause and effect relationships to explain ideas. 5.22.2a) identify the specific roles needed when working in a team. 5.22.2c) negotiate and allocate individual roles to members of the team. 5.22.2g) monitor progress of the group towards completion of a task.			

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Unit 4.5 Nervous control

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.13.1a) describe a problem, hypothesis or question that can be tested or researched. 5.13.1b) propose possible sources of data and/or information relevant to the investigation. 5.13.1c) identify what type of information or data needs to be collected. 5.13.1d) justify why particular types of data or information are to be collected. 5.13.1e) identify the appropriate units to be used in collecting data. 5.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information. 5.13.1g) formulate a means of	**The nervous system** Describe the nervous system as consisting of two parts, the CNS and PNS. Define the main method of transferring messages, the neuron. Identify the functions of parts of a neuron. Define the role of the synapse in the nervous system. **Drag and drop**: Neuron structure (*Science Focus 3 Second Edition LiveText*) **Neurons and nerves** Nerve cells called neurons pass messages through the system. Discuss the three types of neurons: connecting, motor and sensory neurons. **Interactive**: Neurons (*Science Focus 3 Second Edition LiveText*) **Worksheet 4.7**: The nervous system **The brain** Define the role of the brain and the function of the main parts. **Interactive**: The brain (*Science Focus 3 Second Edition LiveText*) **Damage to the nervous system** Explore some diseases, conditions or injuries that affect the nervous system such as spinal cord injuries, spina bifida and multiple sclerosis. Have students research one of these in more detail. **Prac 1**: Brain dissection	Drag and drop Interactive Worksheet 4.7 Interactive Prac 1	

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	recording the data to be gathered or the information to be collected. 5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.13.2b) specify the dependent and independent variables when planning controlled experiments. 5.13.2c) describe a logical procedure for undertaking a simple or a controlled experiment. 5.13.2d) establish an appropriate timeline for an investigation. 5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources effectively. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14f) demonstrate the use of safe and hygienic work practices, including the correct use of safety equipment.	**Prac 2**: A model brain **Prac 3**: Brain wars: the Stroop effect **Web destination**: Interactive Stroop effect experiment (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Using the nervous system** Describe the function of the reflex arc in reflex actions. Contrast with what happens for more complex actions. **Interactive**: Brain wars–the Stroop effect (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 4**: Response time **Video**: Reflex actions (*Science Focus 3 Second Edition LiveText*) **Drag and drop**: Revising the reflex arc (*Science Focus 3 Second Edition LiveText*) **Unit 4.5 questions** **Web destination**: Nervous system–reflexes (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 2 Prac 3 Web destination Interactive Prac 4 Video Drag and drop Unit questions Web destination	

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4: Sense and control

Science Focus: Understanding memory

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.2c) apply scientific processes to test the validity of ideas and theories. 5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.13.2b) specify the dependent and independent variables when planning controlled experiments . 5.14a) follow the planned procedure when performing an investigation. 5.19b) identify data which supports or discounts an hypothesis, question being investigated or the proposed solution to a problem. 5.19d) make generalisations in relation to a relevant set of observations or experimental results. 5.21a) seek evidence to support claims. 5.22.1d) evaluate the effectiveness of their performance in completing the task.	The passage describes memory. However, use the passage to test students' short-term and sensory memories. Have them read the section and then ask them to close their books. Ask students questions about what they have just read, such as what was in the figures and what the titles were, as well as what they remember of the textual information. Discuss how their memories hold up. **Prac 5**: What's that smell? **Prac 6**: Chunking	Prac 5 Prac 6	

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Unit 4.6 Chemical control

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4a) describe the role of, and interaction between, coordination systems in maintaining humans as functioning organisms. 5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources effectively. 5.16c) extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources. 5.16d) summarise information from identified oral and written secondary sources. 5.17a) collate information from a number of sources. 5.17b) distinguish between relevant and irrelevant information. 5.17c) check reliability of gathered data and information by comparing with observations or information from	**Hormones** Describe the function of hormones and where they are produced. Illustrate the position of the major endocrine glands in the human body. Identify which glands produce which hormone and for what reason. **Drag and drop**: Major human endocrine glands (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Systems working together** Describe how the nervous and endocrine systems work together. **Controlling growth** Discuss the role of hormones in growth, reproduction and development. Examine the effect of too much and too little of various hormones. **Controlling glucose levels/ Diabetes** Investigate the interaction of glucose and insulin. Define type I and II diabetes **Worksheet 4.8**: Chemical control **Worksheet 4.9**: Hormonal control **Pheromones** Define the roles and effect of pheromones in animals.	Drag and drop Worksheet 4.8 Worksheet 4.9	

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	other sources. 5.17d) organise data using a variety of methods including graphs, diagrams, tables, spreadsheets and databases. 5.19b) identify data which supports or discounts a hypothesis, a question being investigated or the proposed solution. 5.l9d) make generalisations in relation to a relevant set of observations or experimental results. 5.19g) use cause and effect relationships to explain ideas.	**Prac 1**: Plant hormones **Prac 2**: Plants and gravity **Unit 4.6 questions** **Interactive**: Hormone action (*Science Focus 3 Second Edition LiveText*) **Web destination**: Human anatomy online (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 1 Prac 2 Unit questions Interactive Web destination	

Chapter review

Suggested teaching and learning strategies	***Science Focus 3* Resources**	**Register**
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 4.10**: Wordfind	Worksheet 4.10	
Complete **Worksheet 4.11**: Crossword	Worksheet 4.11	
Complete **Worksheet 4.12**: Sci-words	Worksheet 4.12	
Interactive: Sense and control interactive crossword (*Science Focus 3 Second Edition LiveText*)	Interactive	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Sense and control
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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5: Reproduction

Prescribed focus area	**The applications and uses of science** 5.3a) identify and describe examples of scientific concepts and principles that have been used in technological development (including Australian examples). 5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.3c) identify and describe examples where technological advances have impacted on science. 5.3d) give reasons why society should support scientific research. **Current issues in research and development in science** 5.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions. 5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding. 5.5c) identify scientific skills that can be useful in a broad range of careers. 5.5d) identify possible career paths in science.

Domain outcomes	
Knowledge and understanding	5.8.1a) explain that systems in multicellular organisms serve the needs of cells. 5.8.1b) identify the role of cell division in growth, repair and reproduction in multicellular organisms. 5.8.4b) describe some responses of body systems to infectious and non-infectious diseases. 5.8.4c) relate the organs involved in human reproductive systems to their function.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.16 accesses information from identified secondary sources.

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	5.19 draws conclusions based on the information available. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.23 demonstrates confidence and willingness to make decisions and to take responsible actions. 5.24 respects differing viewpoints on science issues and is honest, fair and ethical. 5.25 recognises the relevance and importance of lifelong learning and acknowledges the continued impact of science in many aspects of everyday life. 5.26 recognises the role of science in providing information about issues being considered and in increasing their understanding of the world around them.

Unit 5.1 Types of reproduction

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
		Quick Quiz: Chapter 5 (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*)	Quick Quiz	
5.8.1 cell theory	5.8.1a) explain that systems in multicellular organisms serve the needs of cells.	**Cells and reproduction** Revise understanding of cells and focus on chromosomes and genes to introduce reproduction.		
	5.8.1b) identify the role of cell division in growth, repair and reproduction in multicellular organisms.	**Asexual reproduction** Define asexual reproduction, clones and mutation. Indicate the advantages of asexual reproduction.		
	5.13.1a) describe a problem, hypothesis or question that can be tested or researched.	**Prac 1**: Asexual reproduction in plants **Prac 2**: Examination of spores	Prac 1 Prac 2	
	5.13.2a) identify variables that need to be	Define the main types of asexual reproduction: fusion, budding, spores and fragmentation. Indicate examples of each.		

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	held constant if reliable first-hand data is to be collected. 5.13.2b) specify the dependent and independent variables when planning controlled experiments. 5.13.2c) describes a logical procedure for undertaking a simple or controlled experiment to collect valid first-hand data. 5.13.2d) establish an appropriate timeline for an investigation. 5.19a) justify inferences in light of gathered information. 5.22.2a) identify the specific roles needed when working in a team. 5.22.2c) negotiate and allocate individual roles to members of the team. 5.22.2h) evaluate the process used by the group and effectiveness of the team in completing the task.	**Sexual reproduction** Define sexual reproduction and contrast with asexual reproduction. Define the terms 'gamete' and 'zygote'. Explain and investigate sexual reproduction in flowering plants. **Worksheet 5.1**: Plant reproduction **Prac 3**: Flower dissection **Drag and drop**: Parts of a flower (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Unit 5.1 questions** **Web destination**: Reproduction in plants (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 5.1 Prac 3 Drag and drop Unit questions Web destination	

Unit 5.2 Human reproductive systems

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4c) relate the organs involved in human reproductive systems to their function.	**The male reproductive system** Describe the various parts and function of the male reproductive system. **Drag and drop**: The male reproductive system (*Science Focus*	Drag and drop	

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	5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16c) extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources. 5.16d) summarise information from identified oral and written secondary sources.	*3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **The female reproductive system** Describe the various parts and function of the female reproductive system. **Prac 1**: Model reproductive systems **Worksheet 5.3**: Ovulation **Worksheet 5.2**: Human reproductive organs **Drag and drop**: The female reproductive system (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Puberty** Compare and contrast changes in puberty for males and females. **Drag and drop**: Human reproductive systems (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Unit 5.2 questions** **Web destination**: Puberty (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Male and female genitals (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 1 Worksheet 5.3 Worksheet 5.2 Drag and drop Drag and drop Unit questions Web destination Web destination	

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Unit 5.3 Human reproduction

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.1 cell theory 5.8.4 humans	5.8.1b) identify the role of cell division in growth, repair and reproduction in multicellular organisms. 5.8.4c) relate the organs involved in human reproductive systems to their function. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16c) extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources. 5.16d) summarise information from identified oral and written secondary sources.	**Becoming pregnant** Define the terms and relate to the function of the relevant organs. Examine the various methods of contraception, in particular, the advantages and disadvantages. Compare this information with what may be presented in the media. **Drag and drop**: Conception (*Science Focus 3 Second Edition LiveText*) **The next nine months** Explain the changes that occur from fertilisation to implantation to birth. **Worksheet 5.4**: Pregnancy **Video**: 3-D baby scans (*Science Focus 3 Second Edition LiveText*) **Drag and drop**: Biotechnology (*Science Focus 3 Second Edition LiveText*) **Twins** Discuss how twins are conceived. What is the difference between fraternal, identical and conjoined twins? **Prac 1**: Survey of twins **Unit 5.3 questions** **Web destination**: Pregnancy–week by week (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Drag and drop Worksheet 5.4 Video Drag and drop Prac 1 Unit questions Web destination	

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Unit 5.4 Reproductive health

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.8.4 humans	5.8.4b) describe some responses of body systems to infectious and non-infectious diseases. 5.8.4c) relate the organs involved in human reproductive systems to their function. 5.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions. 5.5d) identify possible career paths in science. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16b) use a variety of techniques, such as key words, skimming and scanning to identify appropriate information. 5.17e) critically analyse the accuracy of scientific information presented in popular media.	**Contraception** Explain the terminology of contraception and getting pregnant. **Prac 1**: Methods of contraception **Sexually transmitted infections** Describe some of the more common sexually transmitted infections. Students may investigate a disease that is current such as AIDS, which occurs through HIV or pelvic inflammatory disease, which may occur from chlamydia. **Cancers and cysts** Describe the cancers that can occur in reproductive systems. **A healthy pregnancy** Describe how science technology has assisted the health of the mother during pregnancy. Outline the tests that may be carried out. Discuss the effects of taking drugs during pregnancy. **Infertility** Discuss how recent scientific advancements have helped infertile families. Research the work of Australians in this field. **Career Profile**: Medical practitioner **Unit 5.4 questions**	Prac 1 Career profile Unit questions	

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5: Reproduction

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 5.5**: Crossword	Worksheet 5.5	
Complete **Worksheet 5.6**: Sci-words	Worksheet 5.6	
Interactive: Reproduction interactive crossword (*Science Focus 3 Second Edition LiveText*)	Interactive	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Reproduction
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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6: Ecosystems

Prescribed focus area	**The implications of science for society and the environment** 5.4a) discuss viewpoints about some issues with a major scientific component. 5.4b) give examples to show that different cultures or groups within a society (including Aboriginal and other Indigenous people) may use or weight criteria differently to make a decision about an issue involving a major scientific component. 5.4c) identify choices that need to be or have been made when considering whether to use particular scientific advances. 5.4d) discuss the place of social and ethical considerations in scientific practice and in applications of science.

Domain outcomes	
Knowledge and understanding	5.10a) distinguish between biotic and abiotic features of the local environment. 5.10b) describe the importance of cycles of materials in ecosystems. 5.10c) describe some impacts of human activities on ecosystems. 5.11.1a) discuss the importance of energy as a resource. 5.11.1b) identify properties that make some natural resources economically important and describe their uses. 5.11.2a) relate pollution to contamination by unwanted substances. 5.11.2c) discuss strategies used to balance human activities and needs in ecosystems with conserving, protecting and maintaining the quality and sustainability of the environment.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.14 follows a sequence of instructions to undertake a first-hand investigation. 5.15 uses given criteria to gather first-hand data. 5.16 accesses information from identified secondary sources.

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	5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.23 demonstrates confidence and a willingness to make decisions and to take responsible actions. 5.24 respects differing viewpoints on science issues and is honest, fair and ethical. 5.26 recognises the role of science in providing information about issues being considered and in increasing understanding of the world around them. 5.27 acknowledges their responsibility to conserve, protect and maintain the environment for the future.

Unit 6.1 Energy for life

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.11.1 energy resources	5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.11.1a) discuss the importance of energy as a resource. 5.13.1a) describe a problem, hypothesis or question that	**Quick Quiz**: Chapter 6 (*Science Focus 3 Second Edition* LiveText, *Science Focus 3 Second Edition Student Lounge*) **Conservation of energy in an ecosystem** Recall the Law of Conservation of Energy. Define producers and consumers. **Interactive**: Photosynthesis (*Science Focus 3 Second Edition LiveText*) **Plants in an ecosystem** Discuss photosynthesis.	Quick Quiz Interactive	

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	can be tested or researched. 5.13.1c) identify what type of information or data needs to be collected. 5.13.1f) recommend the use of an appropriate technology or strategy for collecting data or gathering information. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16d) summarise information from identified oral and written secondary sources. 5.18a) select and use appropriately, types of texts for different purposes and contexts, including a discussion, explanation, procedure, exposition, recount, report, response or experimental records for oral or written presentation. 5.20a) identify the nature of a presented problem. 5.21f) formulate cause and effect relationships.	**Interactive**: Photosynthesis (*Science Focus 3 Second Edition LiveText*) **Animals in an ecosystem** Talk about the different types of animals based on what they eat: herbivores, carnivores, omnivores, consumers etc. **A simple food chain/Energy flow** State that energy only flows in one direction in an ecosystem. Follow a simple food chain's energy flow. **Conservation of matter in an ecosystem** Define the Law of Conservation of Mass. **Prac 1**: Biomass **Food pyramids** Present a pyramid showing the change in dry weight of organisms and energy changes at each level. **Worksheet 6.1**: Energy for life **Prac 2**: Food pyramid **Unit 6.1 questions**	Interactive Prac 1 Worksheet 6.1 Prac 2 Unit questions	

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Unit 6.2 Recycling in nature

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.10 ecosystems	5.10a) distinguish between biotic and abiotic features of the local environment 5.10b) describe the importance of cycles of materials in ecosystems. 5.13.1e) identify the appropriate units to be used in collecting data. 5.13.2a) identify variables that need to be held constant if reliable first-hand data is to be collected. 5.13.2c) describe a logical procedure for undertaking a simple or controlled experiment. 5.13.2d) establish an appropriate timeline for an investigation. 5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task. 5.15a) make and record	**Two types of matter** Define the two types of matter that occur in an ecosystem. **Drag and drop**: Energy for life (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Organic matter** (containing carbon). **Inorganic matter** (no carbon). Define the terms 'biotic' and 'abiotic'. **Prac 1**: Testing for water **Prac 2**: Measuring the boiling point of water **The water cycle** Discuss the importance of the water cycle in an ecosystem. **Interactive**: The water cycle (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **The carbon cycle** Define as a simple cycle that moves carbon atoms between abiotic and biotic environments. **Video**: The carbon cycle (*Science Focus 3 Second Edition LiveText*) **The nitrogen cycle** Discuss this complex cycle and note that all cycles are operating together all the time. Describe how nitrogen	Drag and drop Prac 1 Prac 2 Interactive Video	

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	observations and measurements accurately over a number of trials. 5.15b) use independently a range of data collection strategies and technologies such as data loggers. 5.17f) identify trends, patterns, relationships and contradictions in data and information.	flows from the atmosphere into the soil, into plants and animals, and back to the atmosphere. **Video**: The nitrogen cycle (*Science Focus 3 Second Edition LiveText*) **Worksheet 6.2**: Recycling in nature **Unit 6.2 questions** **Web destination**: The carbon cycle (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: The water cycle (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Video Worksheet 6.2 Unit questions Web destination Web destination	

Unit 6.3 Human impact on ecosystems

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.10 ecosystems	5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.3c) identify and describe examples where technological advances have impacted on science. 5.4a) discuss viewpoints about some issues with a major scientific component. 5.10c) describe some impacts of human activities on	**Environmental revolution** Discuss how history has changed the way environments exist. **Pollution** Consider different types of pollution. **Prac 1**: Water pollution **Worksheet 6.3**: Waste **Prac 2**: Climate in a beaker **Global warming** Talk about the greenhouse effect, the enhanced greenhouse effect and how global warming affects the environment. **Prac 3**: Simulating global warming	Prac 1 Worksheet 6.3 Prac 2 Prac 3	

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	ecosystems. 5.11.1b) identify properties that make some natural resources economically important and describe their uses. 5.11.2a) relate pollution to contamination by unwanted substances. 5.11.2b) identify excessive use of fossil fuels as a contributing factor to a greenhouse effect. 5.11.2c) discuss strategies used to balance human activities and needs in ecosystems with conserving, protecting and maintaining the quality and sustainability of the environment.	**Drag and drop**: Global warming (*Science Focus 3 Second Edition LiveText*) **Introduced species** Consider foxes, rabbits and other species that affect the environment and the existence of Australian animals and plants. **Worksheet 6.4**: Rabbit advance **Conservation** Discuss aspects of conservation. **Worksheet 6.5**: Australian plants **Case study**: The hairy-nosed wombat Consider other threatened or endangered species. **Unit 6.3 questions**	Drag and drop Worksheet 6.4 Worksheet 6.5 Case study Unit questions	

Science Focus The right balance – a human problem

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
	5.3a) identify and describe examples of scientific concepts and principles that have been used in technological development (including Australian examples).	The implications of science for society and the environment Environmental impacts before white settlement The early settlers: a struggle to survive		

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	5.4b) give examples to show that different societal groups may use or weight criteria differently to make a decision about an issue involving a major scientific component. 5.4d) analyse reasons why different cultures or groups within a society, including Aboriginal people, may have different views in relation to scientific issues. 5.4e) discuss the place of social and ethical considerations in scientific practice and in applications of science. 5.10c) describe some impacts of human activities on ecosystems. 5.11.2c) discuss strategies used to balance human activities and needs in ecosystems with conserving, protecting and maintaining the quality and sustainability of the environment.	From then until now Additional activities: Research how Aboriginal belief systems relate to environmental management. Describe how the land management practices and techniques of Aboriginal and non-Aboriginal peoples have changed the environment.		

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6: Ecosystems

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 6.6**: Crossword	Worksheet 6.6	
Complete **Worksheet 6.7**: Sci-words	Worksheet 6.7	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Energy in ecosystems
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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7: Light

Prescribed focus area	**The application and uses of science** 5.3a) identify and describe examples of scientific concepts and principles that have been used in technological development (including Australian examples). 5.3b) discuss, using examples, the positive and negative impacts of applications of recent developments in science. 5.3c) identify and describe examples where technological advances have had an impact on science. 5.3d) give reasons why society should support scientific research.

Domain outcomes	
Knowledge and understanding	5.6.4a) distinguish between absorption, reflection and refraction, and identify everyday situations where each occurs.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.14 follows a sequence of instructions to undertake a first-hand investigation. 5.15 uses given criteria to gather first-hand data. 5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.23 demonstrates confidence and willingness to make decisions and to take responsible actions. 5.25 recognises the relevance and importance of lifelong learning and acknowledges the continued impact of science in many aspects of everyday life

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	5.26 recognises the role of science in providing information about issues being considered and in increasing their understanding of the world around them.

Unit 7.1 Bending light

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.6.4 light energy	5.6.4a) distinguish between absorption, reflection and refraction, and identify everyday situations where each occurs. 5.13.2c) describe a logical procedure for undertaking a simple or controlled experiment to collect valid first-hand data. 5.13.3a) identify advantages and limitations of using particular laboratory equipment for a specific task. 5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the task. 5.14a) follow the planned procedure when performing an	**Quick Quiz**: Chapter 7 (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*) **Reflection** Revise reflection. **Interactive**: Reflect (*Science Focus 3 Second Edition LiveText*) **Interactive**: Mirror (*Science Focus 3 Second Edition LiveText*) **Interactive**: Light snooker (*Science Focus 3 Second Edition LiveText*) **Worksheet 7.1:** Reflection **Refraction** Describe as the bending of light. Recall how to draw ray diagrams. Define the normal – explain that it is only a construction line and should be shown as a dotted line. **Interactive**: Refraction (*Science Focus 3 Second Edition LiveText*) **Video**: Refraction (*Science Focus 3 Second Edition LiveText*) **Prac 1**: Measuring angles of refraction **Worksheet 7.2**: Snell's law	Quick Quiz Interactive Interactive Interactive Worksheet 7.1 Interactive Video Prac 1 Worksheet 7.2	

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	investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately.	**Total internal reflection** Present as a special case. Define the critical angle. Discuss the applications of this property of light. **Prac 2**: Observing light **Optical fibres** Describe how optical fibres trap light using total internal reflection. **Drag and drop**: Bending light (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Refraction illusions** Illustrate this with a pencil in water and discuss students' experiences with home swimming pools and apparent depth. Discuss the danger it presents to young children. **Prac 3**: Refraction illusions **Prac 4**: Apparent depth **Interactive**: Refract (*Science Focus 3 Second Edition LiveText*) **Interactive**: Rainbows (*Science Focus 3 Second Edition LiveText*) **Unit 7.1 questions** **Web destination**: Refraction (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Prac 2 Drag and drop Prac 3 Prac 4 Interactive Interactive Unit questions Web destination	

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Unit 7.2 Focusing devices: lenses and curved mirrors

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.6.4 light energy	5.6.4a) distinguish between absorption, reflection and refraction, and identify everyday situations where each occurs. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14d) record data using the appropriate units. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately. 5.15b) use independently a range of data collection strategies and technologies such as data	**The language of optics** Define the terms 'real', 'virtual', 'inverted', 'upright', 'enlarged', 'diminished' and 'magnification'. **Lenses** Define the two types of lenses: concave and convex. Discuss their features and define 'principle focus', 'focal length' and the method of drawing light rays through lenses. Describe ray tracing diagrams and the importance of scale drawing. **Prac 1**: Water lenses **Drag and drop**: Various lens types (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 2**: Lenses and a light box **Worksheet 7.3**: Lenses **Prac 3**: Images in a convex lens **Focusing devices** Describe how the eye works in two stages. Discuss long and short sightedness and how the lenses work in each case. **Prac 4**: Simulating imperfect vision **Video**: A model lens (*Science Focus 3 Second Edition*	Prac 1 Drag and drop Prac 2 Worksheet 7.3 Prac 3 Prac 4 Video	

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	loggers. 5.17c) check reliability of gathered data and information by comparing with observations or information from other sources. 5.17d) organise data using a variety of methods including graphs, diagrams, tables, spreadsheets and databases. 5.17e) critically analyse the accuracy of scientific information presented in the popular media. 5.17f) identify trends, patterns, relationships and contradictions in data and information. 5.17g) apply mathematical concepts and computer-based technologies to assist analysis of data and information. 5.18e) use drawings, diagrams, graphs, tables, databases, spreadsheets and flow	*LiveText*) **Curved mirrors** Define concave and convex mirrors and examine applications. **Prac 5**: Telescopes and microscopes **Prac 6**: Exploring curved surfaces: spoons **Prac 7**: Forming images with curved mirrors **Worksheet 7.4**: Mirrors **Unit 7.2 questions** **Web destination**: Optical instruments (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	 Prac 5 Prac 6 Prac 7 Worksheet 7.4 Unit questions Web destination	

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	charts to show relationships and present information clearly and/or succinctly. 5.18f) select and draw the appropriate type of graph or diagram to convey information and relationships clearly and accurately. 5.21a) seek evidence to support claims. 5.21b) evaluate evidence for reliability and validity.			

Unit 7.3 Colour

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.6.4 light energy	5.6.4a) distinguish between absorption, reflection and refraction, and identify everyday situations where each occurs. 5.13.3b) select appropriate equipment (including appropriate safety equipment) and/or resources to perform the	**Dispersion** Recall the colours of the rainbow. How do they occur? Define 'dispersion'. **Prac 1**: Dispersion: splitting white light **Rainbows** Describe how rainbows form using dispersion of light in raindrops. **Blue skies and red sunsets** Define the term 'scattering' and show how this gives the changing colours in the sky.	Prac 1	

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	task. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.15a) make and record observations and measurements accurately. 5.18e) use drawings, diagrams, graphs, tables, databases, spreadsheets and flow charts to show relationships and present information clearly and/or succinctly. 5.18f) select and draw the appropriate type of graph or diagram to convey information and relationships clearly and accurately. 5.19a) justify inferences in light of gathered information. 5.19d) make generalisations in relation to a relevant set	**Colour addition** Describe primary, secondary and complementary colours. **Prac 2**: Mixing coloured light **Colour subtraction** Describe how we see the colour of objects and the use of filters. **Prac 3**: Seeing things in a different light **Prac 4**: Measuring light absorption **Mixing pigments** Use the diagrams to show how pigments are seen when mixed. **Prac 5**: Mixing pigments **Interactive**: Mixing pigments (*Science Focus 3 Second Edition LiveText*) **Printing** Describe how this book was printed. **Unit 7.3 questions**	Prac 2 Prac 3 Prac 4 Prac 5 Interactive Unit questions	

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	of observations or experimental results. 5.22.1a) independently plan and conduct investigations, communicate information and understandings and solve problems. 5.22.2a) identify the specific roles needed when working in a team. 5.22.2b) match the tasks to the team members according to the requirements of the task and the skills of the individual. 5.22.2c) negotiate and allocate individual roles to members of the team. 5.22.2f) accept personal responsibility for maintenance of a safe working environment for the team. 5.22.2h) evaluate the process used by the group and effectiveness of the team in completing the task.			

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7: Light

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 7.5**: Crossword	Worksheet 7.5	
Complete **Worksheet 7.6**: Sci-words	Worksheet 7.6	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Light
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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8: The universe

Prescribed focus area	**Current issues, research and development in science** 5.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions. 5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding. 5.5c) identify scientific skills that can be useful in a broad range of careers. 5.5d) identify possible career paths in science.
Domain outcomes	
Knowledge and understanding	5.9.1a) discuss current scientific thinking about the origin of the universe. 5.9.1b) identify that some types of electromagnetic radiation are used to provide information about the universe. 5.9.1c) describe some of the difficulties in obtaining information about the universe. 5.9.3a) relate some major features of the universe to theories about the formation of the universe. 5.9.3b) describe some changes that are likely to take place during the life of a star.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem. 5.16 accesses information from identified secondary sources. 5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 undertakes a variety of team and individual tasks with guidance.

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8: The universe

Values and attitudes	5.25 recognises the relevance and importance of lifelong learning and acknowledges the continued impact of science in many aspects of everyday life. 5.26 recognises the role of science in providing information about issues being considered and in increasing understanding of the world around them.

Unit 8.1 The expanding universe

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.1 the Big Bang theory	5.9.1a) discuss current scientific thinking about the origin of the universe. 5.9.1b) identify that some types of electromagnetic radiation are used to provide information about the universe. 5.17d) organise data using a variety of methods including graphs, diagrams, tables, spreadsheets and databases. 5.17e) critically analyse the accuracy of scientific information presented in the popular media. 5.19f) use models, including mathematical ones, to explain phenomena or make predictions.	**Quick Quiz**: Chapter 8 (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*) **The Doppler effect and sound** Discuss the Doppler effect in relation to the sounds of an approaching ambulance's siren. **The Doppler effect and light** Recall that sound is in the form of waves, as is light. Light can do the same. **Prac 1**: Using a spectroscope **Video**: Red shift and the expanding universe (*Science Focus 3 Second Edition LiveText*) **Drag and drop**: The Doppler effect (*Science Focus 3 Second Edition LiveText*) **The universe is expanding** Define 'red shift', 'Hubble's law' and the 'expanding universe'. **Prac 2**: A balloon universe	Quick Quiz Prac 1 Video Drag and drop Prac 2	

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	5.19g) use cause and effect relationships to explain ideas.	**Going back in time** Introduce the Big Bang theory. **Drag and drop**: The expanding universe (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Unit 8.1 questions** **Web destination**: Doppler effect (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Drag and drop Unit questions Web destination	

Unit 8.2 The Big Bang

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.1 the Big Bang theory	5.9.1a) discuss current scientific thinking about the origin of the universe. 5.9.1b) identify that some types of electromagnetic radiation are used to provide information about the universe. 5.9.3a) relate some major features of the universe to theories about the formation of the universe. 5.9.3b) describe some changes that are likely to take place during the life of a star. 5.18b) select and use an appropriate	**The Big Bang theory** Introduce the Big Bang theory. **Drag and drop**: The Big Bang (*Science Focus 3 Second Edition LiveText*) **The first few minutes** Discuss with students their understanding of the Big Bang. Introduce the terms 'singularity', 'matter' and 'antimatter'. **The fog clears** Examine the formation of atoms and elements. **Messages from the birth of time** Look at evidence of matter clumping together to form galaxies.	Drag and drop	

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	medium to present data and information. 5.18c) select and use an appropriate method to acknowledge sources of information. 5.21c) produce creative solutions for problems. 5.22.2a) identify the specific roles needed when working in a team. 5.22.2d) accept specific roles in a team while planning and conducting investigations, communicating information and understandings and solving problems. 5.22.2f) accept personal responsibility for maintenance of a safe working environment for the team. 5.22.2g) monitor progress of the group towards completion of a task.	**Galaxies, stars and planets** Examine the formation of young planets or planetesimals. **Worksheet 8.1**: Origin of the universe **Drag and drop**: The universe (*Science Focus 3 Second Edition LiveText*) **The future** Present some current theories of what will occur in the future for the universe. **Unit 8.2 questions** **Web destination**: History of the universe (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 8.1 Drag and drop Unit questions Web destination	

Unit 8.3 The life of a star

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.1 the Big Bang theory	5.9.1a) discuss current scientific thinking about the origin of the universe. 5.9.3a) relate some major features of the	**Birth of a star** Define the terms 'nebula' and 'protostar' in terms of star formation. As the protostar changes with age it forms a main sequence star.		

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	universe to theories about the formation of the universe. 5.9.3b) describe some changes that are likely to take place during the life of a star. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16d) summarise information from identified oral and written secondary sources. 5.17a) collate information from a number of sources. 5.19a) justify inferences in light of gathered information. 5.21c) produce creative solutions for problems.	**Death of a star** Discuss the aging process for varying sized stars. Explain how red giants, white dwarf and red supergiants form. Look at the formation of a supernova and a neutron star. Define the characteristics of each. **Drag and drop**: Life and death of a star (*Science Focus 3 Second Edition LiveText*) **Black holes** Define as a rapidly rotating neutron star emitting radio waves. **Worksheet 8.2**: Stars **Unit 8.2 questions** **Web destination**: Images of the universe (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Drag and drop Worksheet 8.2 Unit questions Web destination	

Unit 8.4 Are we alone?

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.5 current issues, research and developments in science	5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding. 5.9.1c) describe some of the difficulties	**Long distance space travel** The closest stars are 4.3 light years away! This makes travel very difficult. **Trying to make contact** Popular media presents many images of life from outer space. Scientists are searching for life in the		

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	in obtaining information about the universe. 5.17e) critically analyse the accuracy of scientific information presented in the popular media. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16b) use a variety of techniques, such as key words, skimming and scanning to identify appropriate information. 5.16d) summarise information from identified oral and written secondary sources. 5.17a) collate information from a number of sources. 5.17b) distinguish between relevant and irrelevant information. 5.18a) select and use appropriately, types of texts for different purposes and contexts, including a discussion, explanation, procedure, exposition, recount, report, response or experimental records for oral or written presentation.	universe using spacecraft with messages and radio waves. **SETI** Discuss what the letters stand for and Australia's involvement in the project. **For and against** An excellent area to start a classroom discussion or debate. **Worksheet 8.3**: Are we alone? **Drag and drop**: Searching for life (*Science Focus 3 Second Edition LiveText*) **Unit 8.4 questions** **Web destination**: The SETI institute (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 8.3 Drag and drop Unit questions Web destination	

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Unit 8.5 Using space

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.5 current issues, research and developments in science	5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding. 5.9.1c) describe some of the difficulties in obtaining information about the universe. 5.17e) critically analyse the accuracy of scientific information presented in the popular media. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16b) use a variety of techniques, such as key words, skimming and scanning to identify appropriate information. 5.16d) summarise information from identified oral and written secondary sources. 5.17a) collate information from a number of sources. 5.17b) distinguish between relevant and irrelevant information. 5.18a) select and use appropriately, types	**Artificial satellites** Define artificial and geostationary satellites. **Satellite remote sensing** Discuss remote sensing through passive and active sensors. **International Space Station: A global effort** Consider the countries that have been involved in the International Space Station (ISS). **Space tourism** Ask students if they would like to travel into space for a 'tour'. **Worksheet 8.4**: Space travel **Unit 8.5 questions** **Web destination**: NASA space travel (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Time travel (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 8.4 Unit questions Web destination Web destination	

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	of texts for different purposes and contexts, including a discussion, explanation, procedure, exposition, recount, report, response or experimental records for oral or written presentation.			

Science Focus Long-distance space travel

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.5 current issues, research and developments in science	5.5a) describe some recent scientific contributions made by male and female scientists, including Australians, and discuss the effect of their contributions. 5.5b) evaluate the potential impact of some issues raised in the mass media that require some scientific understanding. 5.9.1c) describe some of the difficulties in obtaining information about the universe. 5.13.1a) describe a problem, hypothesis or question that can be tested or researched. 5.20b) describe different strategies that could be employed to solve an identified problem.	Discuss the distances involved in space travel. Recall the term 'light year'. Discuss what would need to be taken on such a trip. Most students only consider a one-way trip. Present some new forms of space travel. Discuss how each one may work. This should appeal to the *Star Trek* and *Star Wars* fans. Discuss warp drive engines, wormholes and gravity as methods of space travel. **Worksheet 8.5**: *Mir*'s plunge **Worksheet 8.6**: Global positioning	 Worksheet 8.5 Worksheet 8.6	

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8: The universe

Chapter review

Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 8.7**: Crossword	Worksheet 8.7	
Complete **Worksheet 8.8**: Sci-words	Worksheet 8.8	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: The universe
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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9: Earth's fragile crust

Prescribed focus area	The implications of science for society and the environment
	5.4a) discuss viewpoints about some issues with a major scientific component.
	5.4b) give examples to show that different cultures or groups within a society (including Aboriginal and other Indigenous people) may use or weight criteria differently to make a decision about an issue involving a major scientific component.
	5.4c) identify choices that need to be or have been made when considering whether to use particular scientific advances.
	5.4d) discuss the place of ethical considerations in scientific practice and in applications of science.

Domain outcomes	
Knowledge and understanding	5.9.2a) discuss evidence that suggests crustal plates move over time.
	5.9.4a) identify that geological history can be interpreted from the formation, by sediments, of horizontal layers in which the oldest are at the base and the youngest at the top.
	5.9.4b) describe conditions under which fossils form.
	5.9.4c) relate the fossil record to the age of the Earth and the time over which life has been evolving.
	5.9.4d) relate movements of the Earth's plates to convection currants in the mantle and to gravitational forces.
	5.9.4e) explain how interactions at plate boundaries may result in earthquakes, volcanic activity and new landforms.
	5.9.4f) explain some impacts of natural events, including cyclones, volcanic eruptions and earthquakes, on the atmosphere, hydrosphere, lithosphere and/or biosphere.
Skills	5.13 clarifies the purpose of an investigation and, with guidance, produces a plan to investigate a problem.
	5.14 follows a sequence of instructions to undertake a first-hand investigation.
	5.15 uses given criteria to gather first-hand data.
	5.16 accesses information from identified secondary sources.

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	5.17 evaluates the relevance of data and information. 5.18 with guidance, presents information to an audience to achieve a particular outcome. 5.19 draws conclusions based on the information available. 5.20 uses an identified strategy to solve problems. 5.21 uses creativity and imagination in the analysis of problems and the development of possible solutions. 5.22 undertakes a variety of team and individual tasks with guidance.
Values and attitudes	5.26 recognises the role of science in providing information about issues being considered and in increasing understanding of the world around them. 5.27 acknowledges their responsibility to conserve, protect and maintain the environment for the future.

Unit 9.1 Plate tectonics

Students learn about	**Students learn to**	**Suggested teaching and learning strategies**	***Science Focus 3* Resources**	**Register**
5.9.2 the theory of plate tectonics	5.9.2a) discuss evidence that suggests crustal plates move over time. 5.9.4d) relate movements of the Earth's plates to convection currents in the mantle and to gravitational forces. 5.1c) describe some models and theories that have been considered in	**Quick Quiz**: Chapter 9 (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*) **Continents that move** Describe evidence that the continents were once one supercontinent. The development of new scientific equipment provided more evidence. Examine this new evidence for the theory. **Prac 1**: The planet Splatter **Worksheet 9.1**: The planet Splatter **Floating plates** Describe the layers of the Earth and how the	Quick Quiz Prac 1 Worksheet 9.1	

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	science and then been modified or rejected as a result of available evidence. 5.19c) predict outcomes and generate plausible explanations directly related to observations made. 5.19d) make generalisations in relation to a relevant set of observations or experimental results. 5.19g) use cause and effect relationships to explain ideas. 5.20a) identify the nature of a presented problem.	lithosphere floats on the asthenosphere. The ridged lithosphere plates are moving. Define the theory of plate tectonics. Present some alternative theories, particularly that by Australian S. Warren Carey. **Drag and drop**: Plate tectonics (*Science Focus 3 Second Edition LiveText, Science Focus 3 Second Edition Student Lounge*) **Prac 2**: Convection currents **Prac 3**: Future Earth **Worksheet 9.2**: Future Earth **Unit 9.1 questions** **Web destination**: Maps through time (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Drag and drop Prac 2 Prac 3 Worksheet 9.2 Unit questions Web destination	

Unit 9.2 At the edges

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.4 natural events	5.9.4e) explain how interactions at plate boundaries may result in earthquakes, volcanic activity and new landforms. 5.14a) follow the planned procedure when performing an investigation. 5.14b) use time and resources	**Plate boundaries** Define the three types of boundaries. **Spreading boundaries** Describe this as a construction zone that makes new lithosphere. **Prac 1**: Plates that separate **Collision boundaries**	Prac 1	

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	effectively. 5.14e) evaluate and modify experimental procedures. 5.18a) select, and use appropriately, types of texts for different purposes and contexts including a discussion, explanation, procedure, exposition, recount, report, response or experimental record for oral or written presentation. 5.21c) produce creative solutions for problems. 5.21e) apply critical thinking in the consideration of proposals.	Describe as the area where old lithosphere is remelted. **Worksheet 9.3**: Ocean trenches Describe the method by which islands and mountain ranges are formed. **Drag and drop**: The ocean floor (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 2**: Colliding plates **Drag and drop**: Plate boundaries (*Science Focus 3 Second Edition LiveText*) **Scraping boundaries** Describe that when plates scrape along each other earthquakes can occur. **Worksheet 9.4**: Earthquakes **Worksheet 9.5**: Volcanoes **Unit 9.2 questions** **Web destination**: Slip, slide and collide (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 9.3 Drag and drop Prac 2 Drag and drop Worksheet 9.4 Worksheet 9.5 Unit questions Web destination	

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Unit 9.3 Earthquakes

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.4 natural events	5.9.4e) explain how interactions at plate boundaries may result in earthquakes, volcanic activity and new landforms . 5.9.4f) explain some impacts of natural events, including cyclones, volcanic eruptions and earthquakes, on the atmosphere, hydrosphere, lithosphere and/or biosphere. 5.13.1a) describe a problem, hypothesis or question that can be tested or researched. 5.13.1b) propose possible sources of data and/or information relevant to the investigation. 5.16a) use a range of sources including databases, CD-ROMs and the Internet, to access information. 5.16c) extract information from column graphs, histograms, divided bar and sector graphs, line graphs, composite graphs, flow diagrams, other texts and audio-visual resources. 5.18d) use symbols to express	**Why earthquakes happen** Summarise why earthquakes occur. **Video**: What caused the earthquake? (*Science Focus 3 Second Edition LiveText*) **Focus and epicentre** Define each term and indicate where each is located. **Prac 1**: Slinky springs **Seismic waves** Describe each type and how they are recorded. **Prac 2**: Build a seismometer **Body waves** Define and describe their effects on the Earth. Describe earthquake shadows. **Finding the epicentre** Describe and model how to determine the location of the epicentre of an earthquake. **Worksheet 9.6**: Earthquake epicentres **Prac 3**: Locating the epicentre **Worksheet 9.11**: Locating the epicentre **Surface waves** Illustrate how these waves cause the greatest damage. **Worksheet 9.7**: Waves in earthquakes	Video Prac 1 Prac 2 Worksheet 9.6 Prac 3 Worksheet 9.11 Worksheet 9.7	

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	relationships, including mathematical ones, and appropriate units for physical quantities. 5.18e) use drawings, diagrams, graphs, tables, databases, spreadsheets and flow charts to show relationships and present information clearly and/or succinctly. 5.19a) justify inferences in light of gathered information. 5.19g) use cause and effect relationships to explain ideas. 5.21a) seek evidence to support claims. 5.22.2a) identify the specific roles needed when working in a team. 5.22.2b) match the tasks to the team members according to the requirements of the task and the skills of the individual. 5.22.2c) negotiate and allocate individual roles to members of the team.	**Measuring earthquakes** Compare the two scales – Richter and Mercalli. Students need to understand that a magnitude change of 1 in the Richter scale is related to 10 times the size and 30 times the energy release of an earthquake. **Worksheet 9.8**: Earthquake statistics **Other earthquake nasties** Examine the impact of earthquakes on the living population of the area affected. **Worksheet 9.9**: Tsunami statistics **Unit 9.3 questions** **Web destination**: Seismic waves (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Understanding earthquakes (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: National Earthquake Information Centre (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Earthquake zones (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Worksheet 9.8 Worksheet 9.9 Unit questions Web destination Web destination Web destination Web destination	

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Unit 9.4 Volcanoes

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.4 natural events	5.9.4e) explain how interactions at plate boundaries may result in earthquake, volcanic activity and new landforms. 5.9.4f) explain some impacts of natural events, including cyclones, volcanic eruptions and earthquakes, on the atmosphere, hydrosphere, lithosphere and/or biosphere. 5.14a) follow the planned procedure when performing an investigation. 5.16a) use a range of sources including databases, CD-ROMs and the Internet to access information 5.16b) use a variety of techniques such as key words, skimming and scanning to identify appropriate information. 5.16d) summarise information from identified oral and written secondary sources. 5.18a) select, and use appropriately, types of texts for different purposes and contexts including a discussion, explanation, procedure,	**Volcanoes everywhere** Define the status of volcanoes. Discuss historically volcanoes or volcanic activity in the local region. **Just like a pimple!** Examine a cross-section of a volcano, noting its major features. **Animation**: Moving volcanoes (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Volcanic material** Describe the material that is ejected from a volcano and its effects on local communities. **Drag and drop**: Erupting volcanoes (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 1**: Volcanic clouds **Prac 2**: Build a 3D model of a volcano **Worksheet 9.10**: Labelling a volcano **Unit 9.4 questions** **Web destination**: Active volcanoes (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Web destination**: Volcano world (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Animation Drag and drop Prac 1 Prac 2 Worksheet 9.10 Unit questions Web destination Web destination	

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	exposition, recount, report, response or experimental record for oral or written presentation.			

Unit 9.5 Landscaping the crust

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.4 natural events	5.9.4e) explain how interactions at plate boundaries may result in earthquakes, volcanic activity and new landforms. 5.14a) follow the planned procedure when performing an investigation. 5.14c) safely and efficiently construct, assemble and manipulate identified equipment. 5.14e) evaluate and modify experimental procedures. 5.15a) make and record observations and measurements accurately over a number of trials. 5.17g) apply mathematical concepts and computer-based technologies to assist analysis of data and information. 5.18b) select and use an appropriate	**Faulty landscaping** Describe the formation of normal, reverse and transcurrent faults. **Folding** Describe the formation of folds in the Earth's crust. **Prac 1**: Faults and folds **Volcanic landscapes** Describe the various volcanic cones and how they are formed. Give examples of each. **Plugs and other intrusions** Describe dykes, sills, batholiths and laccoliths. Locate any in the local area. **Prac 2**: Shaping volcanoes **Moving volcanoes** Examine the formation of volcanic islands. **Treasure from below** Discuss the economic importance of the moving plates. **Unit 9.5 questions**	Prac 1 Prac 2 Unit questions	

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	medium to present data and information. 5.18c) select and use an appropriate method to acknowledge sources of information. 5.19a) justify inferences in light of gathered information. 5.19g) use cause and effect relationships to explain ideas.	**Web destination**: Fault motion (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Web destination	

Unit 9.6 Geological time

Students learn about	Students learn to	Suggested teaching and learning strategies	*Science Focus 3* Resources	Register
5.9.4 natural events	5.9.4a) identify that geological history can be interpreted from the formation, by sediments, of horizontal layers in which the oldest are at the base and the youngest at the top. 5.9.4b) describe conditions under which fossils form. 5.9.4c) relate the fossil record to the age of the Earth and the time over which life has been evolving. 5.13.2c) describe a logical procedure for undertaking a simple or controlled experiment to	**Fossils/How fossils form/Dating fossils** Describe how fossils are formed and that they form in horizontal sedimentary layers in which the oldest are at the base and the youngest at the top. Define an index fossil. **Prac 1**: Inspecting fossils **Drag and drop**: Fossil formation (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Prac 2**: Dinosaur fossils **Radioactive dating** Describe how chemical elements are used to date rocks.	Prac 1 Drag and drop Prac 2	

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	collect valid first-hand data. 5.13.2d) establish an appropriate timeline for an investigation. 5.14a) follow the planned procedure when performing an investigation. 5.19a) justify inferences in light of gathered information.	**Prac 3**: 'Radioactive' cubes **The geological time scale** Relate the fossil record to the age of the Earth and the time over which life has been evolving. **Drag and drop**: The geological time scale (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*) **Unit 9.6 questions**	Prac 3 Drag and drop Unit questions	

Chapter review

Suggested teaching and learning strategies	***Science Focus 3* Resources**	**Register**
Complete Chapter review questions at the end of the chapter	Chapter Review Questions	
Complete **Worksheet 9.12**: Crossword	Worksheet 9.12	
Complete **Worksheet 9.13**: Sci-words	Worksheet 9.13	
Chapter Review Questions (*Science Focus 3 Second Edition LiveText*, *Science Focus 3 Second Edition Student Lounge*)	Chapter Review Questions	
Chapter Test	Chapter Test	

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Program review and evaluation

Topic: Earth's fragile crust
Modifications required in program
Additional practicals and skill development
Assessment for learning strategies
Projects undertaken
Websites

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1.1 Answers

Remembering

1 **a** Forensic scientist: helps investigators collect scientific evidence.

b Crime scene unit: is made up of police who are specially trained to collect, bag and label all types of evidence.

c Forensic pathologist: carries out autopsies to find signs of injury that may determine the likely time and cause of death.

2 driver's licence, credit cards

3 **a** Advantages: reliable method of recording, identifying and proving identity.

Disadvantages: difficult to classify, may require sifting through hundreds of photographs, searching through photos may alter the memory of eye witnesses, a person's appearance may differ from his or her photograph as they get older.

b Advantages: allows witnesses to create a composite drawing without the need for an artist. Computers can create composite images quickly.

Disadvantages: witness' impressions are not always accurate.

c Advantages: extremely fast, can be used in crowds.

Disadvantages: lower percentage of accuracy, privacy.

4 **a** Advantage: simple.

Disadvantages: assumes that no two people have exactly the same measurements, it is time-consuming and laborious.

b Advantages: fingerprints are unique, so less chance of mistaken identity.

Disadvantages: difficult to sort through, requires expert human analysis.

c Advantages: don't change over time, less chance for error, more detailed than a fingerprint.

Disadvantages: expensive, difficult to obtain from unwilling subjects.

5 dental records, X-rays of the bones to look for previous breaks etc.

6 measurement of femur, pelvis and skull, teeth

Understanding

7

Method	How it is used	Effectiveness	Comments
Wanted posters and Identikit	A drawing of a criminal is used to warn and inform the public in the hope of finding more eye witnesses.	Poor (2%)	The drawing or witness' memories may be inaccurate. The criminal can change the way he or she looks.

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Method	How it is used	Effectiveness	Comments
Photographs	Photos carried on a person can help in their identification if they are dead or arrested. A database of photos can be used to help a witness identify a criminal from a list of known offenders.	Personal identification, such as a driver's licence and passport, are very effective. Databases of photos for criminal identification are a poor means of identification.	Searching through a database of photos may change a witness's idea of what the criminal looked like. Photos do not prevent criminals from changing the way they look.
Facial recognition	Software that very quickly compares the structure of a person's face with pictures of known criminals. This can be used to identify known criminals in a crowd.	Not yet fool-proof, as the percentages of false positives and false negatives are high.	The technique is sensitive to differences in angle and image quality. The technique raises privacy issues.
Anthropometry	An older technique that identifies bodies based on measurements of their body parts.	Poor	Based on the idea that no two people have the same measurements. This is not always correct and the process of taking measurements is time-consuming.
Fingerprints	Used to identify corpses and can be used to identify who was present at a crime scene. They can also be used as passwords for computers or for entry into a building.	Very good	With the advent of pattern-recognition software, fingerprint databases have become easier to search.
Iris and retina scans	Iris and retina scans can be used to provide security access to important facilities.	Extremely effective form of identification.	Difficult to obtain.

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Method	How it is used	Effectiveness	Comments
DNA	DNA samples collected from a crime scene can be compared with DNA samples from likely suspects for identification. DNA can be used to determine the identity of an unknown body.	Very effective, although this varies depending on sample type: blood 90% effective, saliva 67%, fallen hair 25%, sweat 17%. DNA can provide identification even if there is only a small sample of blood, skin etc.	DNA is very useful for eliminating suspects, as well as determining who is guilty.

8 In the past, there was no photographic record of people.

9 All primates: apes, monkeys and humans. They help to provide grip.

10 DNA is a molecule that carries all of our genetic information. It is stored in the nuclei of our cells.

11 Negative DNA matches can allow suspects of a crime to be eliminated from an investigation.

12 The teeth impressions may have come from animals attacking the corpse after death.

Applying

13 **a** Forensic scientist: this is crucial evidence.

b To stop cross-contamination of evidence; e.g. DNA from the forensic scientist contaminating the sample.

Evaluating

14 **a** Fillings may fall out, teeth can become chipped or knocked out.

b Your teeth may have more cavities, or grow in different directions.

c X-ray technology was relatively new at the time, so was likely to be inaccessible to forensic scientists.

15 The delay would not occur today. DNA testing would soon show a match between the victim and her relatives.

16
- There were huge numbers of bodies.
- Entire villages were wiped out, so few family members or relatives were available to help identify bodies.
- Unlikely to be fingerprint records of all the victims.
- Any records would probably not be in a computerised database.
- Water-logged or dismembered bodies may be difficult to identify from photographs.

17 Teeth are very hard, they do not decompose over time and are unlikely to be eaten by animals. They can also survive fire.

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18 Computers can identify similarities but cannot identify slight differences that might mean the difference between guilt or innocence. This is especially true in cases where a fingerprint from the crime scene is imperfect.

19 **a** TV shows would become too complicated if there was a different character doing each job.

b Every job in a crime scene investigation requires a specialist with specific skills. It is not practical for a different person to play each of these roles in a TV show. Also, in real life, the crime will be solved much quicker with many people working on different jobs simultaneously.

20 Blind tests ensure that a scientist's own beliefs about who is innocent and who is guilty do not affect the results.

1.1 Practical activities

Prac 1: Make your own Identikit

Common mistakes

Ensure that the photographs of the subjects are taken in similar light conditions and that the subjects are standing in exactly the same position. A tripod may assist in taking a clearer photo.

Possible results

Features from different people are used to construct different and imaginary faces.

Suggested answers

1 *Depends on student results.*

2 *Probably not, a real database would include a huge range of images with which to compile faces.*

3 *Depends on student results.*

4 *A computer could blend the edges of each body part to make an image that is not disjointed.*

Prac 2: Fingerprints

Common mistakes

Rolling your thumb or forefinger back and forward will smudge the print. Allow students to practice with ink and a blank piece of paper.

Possible results

By collecting and examining the patterns of fingerprints, students should find that all fingerprints contain one of three basic patterns: loops, whorls and arches.

Suggested answers

1 *Depends on student results.*

2 *Students should find that loops are the most common, followed by whorls, arches and composites.*

3 *Composites are the least common; they are a combination of all three patterns.*

4 *Some fingerprints are hard to distinguish because they do not follow the common patterns well.*

5 *You could tell them apart by examining small details and differences in the fingerprints.*

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Prac 3: Collecting teeth impressions

Common mistakes

Students will tend to bite down too hard on the lolly, causing it to tear.

Possible results

By examining each set of teeth impressions, students should find that the impressions will vary depending on the different numbers of teeth and the different sizes of the skull and jaw.

Suggested answers

1 *Depends on student results.*

2 *Each skull and jaw is a different size, meaning that different numbers of teeth will make different numbers of impressions in the jelly.*

3 *Each curved length is probably a little different.*

4 *Each skull and jaw is a different size, meaning that the curved length of the teeth and the jaw in the impressions will vary.*

5 *The investigator would look for oddities in the tooth pattern; e.g. missing teeth, larger than normal gaps etc.*

6 *Saliva and, possibly, blood would be taken for DNA analysis.*

Prac 4: Forensic anthropometry

Common mistakes

Incorrect measurements due to not knowing the beginning or end points of the bones.

Possible results

The height of a person can be determined by measuring the length of their femur.

Suggested answers

1 *Answers will vary. Encourage students to give a numerical answer either in centimetres or as a percentage error: (observed – expected)/expected.*

2 *incorrect measurements due to not knowing the beginning or end points of the bones, errors in recording data, mistakes in calculations*

1.2 Answers

Remembering

1 banknotes, passport, driver's licence, wills, medical records, historical documents

2 the type of paper they were printed on was too new and the diaries contained historical mistakes

3 shape or font of the letters and microscopic differences in the type-bar or hammer

4 much more difficult to forge, more durable, recyclable

5 identical serial numbers, fine details are of a poorer standard, printed on a different material

Understanding

6 The same person may have a different slant or spacing in their writing from day to day and these characteristics of handwriting are easy to disguise.

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7 The computer printer makes unique marks on the paper due to the transport mechanism that moves the paper through the machine.

8 **a** Inks that look the same under normal white light may fluoresce differently under UV light. This allows different inks to be distinguished.

b Inks that look to be the same colour may be made up of different colours. Chromatography splits the ink into its different components for a more detailed analysis.

9 **a** Intaglio printing: The image is first etched onto a plate. The plate is then inked and stamped onto a piece of paper.

b Microprinting: Very small text printed on a banknote or document to prevent forgery. It may be invisible so that criminals don't know that it's there or be so small that simplistic forgery techniques (e.g. printing from a normal computer printer) cannot copy it.

c Watermark: An image overlaid on top of a banknote or document made up of lighter or darker regions depending on the thickness of the paper. Sometimes, a watermark cannot be seen unless viewed under special conditions, such as holding the document up to light.

10 Early $100 bills were mainly black and white and shades of grey which could easily be photocopied and still look very similar to the original.

11 Black ink is actually made up of many different coloured inks mixed together. When the ink is passed through a chromatographic filter, some colours pass through faster than others. As a result, the colours in the ink separate.

12 Money would become essentially worthless.

Applying

13 Practical activity.

Analysing

14 Evidence that helped convict Dr Harold Shipman: forged will, morphine in the bodies of his patients/victims, computer records showing that many other files were modified on the day of death of the patients.

Direct evidence: forged will, morphine in the bodies of his patients/victims.

Circumstantial evidence: computer records showing that many other files were modified on the day of death of the patients.

15 Practical activity.

Evaluating

16 Although certain aspects of a person's personality may be deduced from their handwriting, it will also depend on the person's mood and state of mind at the time.

17 Identity theft is when a criminal pretends to be someone else in order to gain access to the victim's money. Identity theft has become more common because of computerisation of accounts. People are identified using only an account number, username(s) and password(s), which are easy to imitate.

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18 The note should be verified, which is done by looking for distinguishing features, such as fine details, watermarks or microprinting; checking that the note is printed on similar paper to a real note; noticing whether the note fluoresces under UV light and checking whether other notes from the same batch have different serial numbers.

1.2 Practical activities

Prac 1: Testing handwriting

Common mistakes

Advise students to press hard when writing to leave a significant imprint. A darkened room will also help with the oblique lighting.

Possible results

By tracing over an impression made on a piece of paper, the students should be able to legibly determine what was written.

Suggested answers

1 *Results will vary. Depends on how hard the person wrote on the top paper, and the surface that was underneath both sheets.*

2 *This technique could be used to find evidence on a chequebook cover or to prove that a note was written on a certain notepad.*

Prac 2: Chromatography catches a criminal

Common mistakes

Do not allow the letter to get wet. The letter must be placed just above the ethanol or methylated spirits. Be careful not to let the chromatography paper touch the sides of the gas jar.

Possible results

Using chromatography, the students will compare the migration rate and the different patterns of colours that have separated from the black ink of each pen. The pattern on one of the strips will match that of another strip, enabling identification of the pen used to write the ransom note.

Suggested answers

1 *Answers will vary. If there are any, it means that they are not soluble in the solvent used.*

2 *Answers will vary.*

3 *Answers will vary.*

4 *To absorb all the different colours to make a black colour.*

5 *If the ink is water soluble, there should be different colours. If the pigments are separated according to their molecular weight, they may be the same colour and in the same order.*

Prac 3: Examining money

Common mistakes

A well-lit room will assist this exercise.

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Possible results

Students identify the different features and hidden marks on banknotes that make forgery of the banknote difficult.

Suggested answers

1 *The other optical features in Australian banknotes make metal bands and watermarks superfluous. Also, the plastic material makes it difficult to insert bands or print with watermarks.*

2 *Answers will depend on the banknote chosen.*

3 *The poem is an early version of Advance Australia Fair.*

4 *Answers will vary.*

Prac 4: Forging a photograph

Common mistakes

A tripod may assist in taking a clearer photo.

Possible results

Using image manipulation software, students will produce a forged photograph.

Suggested answers

1 *Answers will vary depending on computer skills.*

2 *Answers will vary.*

3 *Nowadays, people are more aware of doctored photos. If the photo was plausible (i.e. not far-fetched), then they might accept it without question.*

4 *Students may see an edge or a pixilation, showing it is a fake.*

5 *Answers will vary.*

1.3 Answers

Remembering

1 **a** Evidence is information that has legal value in court. Rumours and gossip are not evidence.
b physical evidence and eye witness accounts or direct and circumstantial evidence
c DNA left at a scene, weapons, fingerprints, hair, bullets etc.

2 blood, saliva, skin, semen, hair, clothing

3 • imprinted in a soft surface like mud, snow etc.
• imprinted in a liquid, such as blood or grease, on a hard surface
• left behind from mud, dust or liquid on dirty shoes

4 within 100 m

5 the height from which the blood dropped, the direction of movement, the type of wound and the weapon, the position of the attacker, the time of the attack, the identity of the victim or the killer based on DNA testing

6 the type of weapon, whether the attacker was left- or right-handed, the shape of the weapon

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Answers to *Science Focus 3 second edition* Student Book questions

Understanding

7 **a** Dusting the surface, the dust sticks to the oil in the latent fingerprints.

b Poly lights shone on the surface cause the latent fingerprint to fluoresce.

8 It is difficult to prove that the fibre definitely came from the attacker.

9 **a** Insects will inhabit a corpse after a certain amount of time and have a well-defined life cycle. By studying what stage their life cycle is at, a forensic scientist can determine how long the body has been dead.

b Different types of insects inhabit different areas. Therefore, the origin of the drug can be determined by studying the insects found in a drug haul and testing their DNA.

10 No, deleting a file just means that that part of the disk is allowed to be overwritten. It may still remain on the hard disk unless the disk is completely wiped or formatted.

11 Bullets usually have a small entry wound and a large exit wound. Death can be instantaneous if the bullet pierces vital organs, or may be prolonged if the bullet misses vital organs but causes severe blood loss.

12 Bullets will generally travel in a straight line. Therefore, investigators can look at the path that the bullet passed through the body and determine which direction it came from.

Applying

13 **a** screw driver or crow bar marks on windows and doors

b marks from scissors or a shredder

c knife wounds on bone and in the flesh of the victim

d heavy tool marks on the tracks or bolts

e markings from a blade on the brake cable

14 **a** the pattern of the tread on the bottom of the shoe

b where the pattern of the shoe had been worn down the most, where the most weight was placed on the shoe, the size of the shoe and the length of the stride

15 insects in the body and lungs

16 **a** closed circuit television, street cameras, ATM cameras

b mobile phone, use of debit and credit cards, passports

c telephone conversations, SMS recordings, emails

17 Although fingerprints are good for identification purposes, they cannot be used to prove a murder in their own right, as they do not show how or even if a person was murdered. They can only place a person at the scene.

Analysing

18 No, shoe prints are not unique, as many people can own the same type and size of shoe. However, if the pattern of wear is similar between the shoe print and the suspect's shoes, then it provides very strong circumstantial evidence.

19 If the audio-visual is unclear, of poor quality or obscured from view, then a clear representation of the events cannot be obtained.

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Creating

20 Creative exercise, various responses are possible.

21 Creative exercise, various responses are possible.

22 The timeline needs to be ordered as follows.
eggs → small maggots → large maggots → pupa → small blowfly → large blowfly

1.3 Practical activities

Prac 1: Comparing fingerprints

Common mistakes

Students tend to apply too much powder or wipe too vigorously.

Possible results

Students dust for fingerprints, then compare them with the prints they obtained in Unit 1.1 Prac 2.

Suggested answers

1 *Most students will find this difficult, as they tend to apply too much powder or wipe too vigorously.*

2 *commonly applied too much powder, brushed away the print*

3 *Answers will vary.*

4 *Students may correctly identify only one or two.*

5 *Answers will vary.*

Prac 2: Fibre analysis

Common mistakes

Some fibre samples and animal fur can look very similar. Labelled samples will assist students in drawing and identifying each sample. Some samples may be too thick to observe under a compound microscope.

Possible results

Students observe the different surface structures that make up different fibres.

Suggested answers

Part A

1 *Answers will vary, although most students will be able to identify that the fibres look radically different under the microscope.*

2 *Some materials will need the use of a monocular microscope at ×40 magnification.*

3 *Answers will vary.*

4 *Students should write about general appearances (i.e. animal hairs have inner and outer layers, synthetic fibres are very smooth with no layers etc.).*

Part B

5 *Answers will vary, although ×40 magnification is good for most tasks.*

6 *should be able to unless the hair is very dark*

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7 *This chart will depend on the fibres and the microscopes used.*

Prac 3: Collecting foot impressions

Common mistakes

To obtain a clean solid impression, the sand or soil needs to be damp and free from sticks, stones and other debris.

Possible results

Students will make plaster casts of footprints and make comparisons between the different soles of shoes.

Suggested answers

1 *Answers will vary.*

2 *to make a thicker section of plaster so that it doesn't break*

3 *Reasons for a bad mould may be shoe imprint wasn't deep enough, bubbles in the mixture, or mixture too thick or too runny.*

4 *make deeper impressions with your feet, get the plaster exactly the right consistency, or buy professional plaster used by the police*

Prac 4: Investigating stride length

Common mistakes

Ensure that students are consistent in their measurements (i.e. measure each stride using the same parameters—toe to toe, heel to heel, toe to heel).

Possible results

The height of a person is determined by measuring the length of their stride.

Suggested answers

1 *Joining dot-to-dot implies that there are no errors in the experiment and that these are the only stride lengths possible. This is not true. There are many possible stride lengths for people of the same height and so a line of best fit is best.*

2 *Various answers are possible.*

3 *Various answers are possible.*

Prac 5: Blood drips

Common mistakes

If conducting this experiment outside, ensure that there is no wind.

Possible results

The greater the height from which a drop of blood falls the greater the diameter of the splat that will form.

Suggested answers

1 *Not necessarily, as the blood would be dripping onto different surfaces from that tested here.*

2 *Eventually the blood drops will fall at their terminal velocity. Once they do this, all splatters will be the same size. Various answers are possible for this height.*

3 *The splatters would elongate.*

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Chapter answers

Remembering

1 identifying dead bodies, identifying criminals, as passwords to computers, allowing people who cannot write to sign their name

2 Forensic pathologists carry out autopsies to find the time and cause of death.

3 temperature, exposure to the environment, exposure to insects, fire

4 Digital photographs, video footage and audio recordings: Record crimes in action.

Computer files: Deleted files may include incriminating evidence.

Internet and mobile phone records: Record communications between criminals.

Understanding

5 Pattern matching is an automated computer technique for comparing complex patterns such as fingerprints.

6 Bite marks may be made by animals after the time of death.

7 **a** Take a DNA sample from the body and compare it with the grandfather's family, check dental records, examine the bones and compare with medical records.

b Study the inside of the lungs to determine whether any smoke from the fire had been inhaled before death.

c They might look for the area that looks most burnt. This is the place the fire started. They would also test for accelerants, such as fuel, kerosene etc.

8 Examine each wound individually and distinguish which marks appear to be from animals and which may have been caused by an implement.

Analysing

9
- Clear bystanders away from the building.
- Call for back-up.
- Make sure no robbers are still present at the crime scene.
- Carefully enter the building, checking for danger.
- Get everyone to sit down.
- Ask everyone to write down everything that they saw and heard.
- Get the names and addresses of all people present.
- Allow people to leave the building.
- Get the security camera footage.

10 what the robber is wearing (not very helpful), approximate height and build (helpful), the type of magazines stolen (not very helpful), any distinguishing marks such as tattoos (very helpful), skin colour (helpful)

11 the fact that she was seen at the scene of the crime, the fact that the jewellery matches the jewellery that was taken, the fact that the jewellery was hidden under the bed

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Evaluating

12 **a** This may be a domestic violence or rape case where the person was held face down with their arms held behind their back.

b Doctors can look for signs of forced sexual activity and semen, and look for skin under the fingernails.

13 Scan both photos into a computer and remove the distortion, try facial recognition software.

14 **a** The fact that he's carrying eggs and probably didn't declare them. The fact that he crushed the eggs to prevent identification.

b Check the egg shell and compare them with egg shells of galahs and pigeons. Take a DNA sample from the eggs and compare them to eggs of galahs and pigeons.

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2.1 Answers

Remembering

1 an atom

2 **a** 1 carbon and 2 oxygen

b 2 hydrogen and 1 sulfur

c 12 carbon, 22 hydrogen and 11 oxygen

d 2 hydrogen, 1 sulfur and 4 oxygen

e 2 carbon, 4 hydrogen and 2 oxygen

Understanding

3 An atom is made up of a small, positively charged nucleus, surrounded by a cloud of negative electrons. The nucleus is made up of positively charged protons and neutral neutrons.

4 The atom is neutral, so the number of positively charged protons is equal to the number of negatively charged electrons. If there is not an equal number of protons and electrons, the atom develops a charge and becomes an ion.

5 **a** Atomic number: the number of protons in the nucleus.

b Mass number: the total number of protons and neutrons in the nucleus.

c Nucleus: a positively charged cluster of protons and neutrons that sits at the centre of an atom.

6 Compounds are formed when different types of atoms chemically bind to form molecules or crystal lattices.

7 A substance can be identified as a mixture if its components can be separated by physical processes. For example, a solution of table salt in water is a mixture because evaporation can separate the salt and water. Similarly, sand in water is a mixture because they can be separated by decantation or filtration.

8 Cordial has no chemical formula because it is made up of a mixture of different types of molecules.

Applying

9 **a** electron

b protons and neutrons have essentially the same mass; however, technically, neutrons are slightly heavier.

c proton

d electron

e neutron

f electron

g protons and neutrons

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10
- $^{56}_{26}Fe$: 26 prtons, 26 electrons, 30 neutrons
- $^{59}_{28}Ni$: 28 protons, 28 electrons and 31 neutrons
- $^{64}_{29}Cu$: 29 protons, 29 electrons and 35 neutrons
- $^{197}_{79}Au$: 79 protons, 79 electrons and 118 neutrons

11 **a** carbon dioxide, water, glucose

b Carbon dioxide: found in the air and is in particularly high concentrations in the air we breathe out.

Water: air, our bodies, the oceans, rivers.

Glucose: ripe fruit, leaves, blood.

12 **a** Element: contains only one type of atom.

b Compound: contains only one type of molecule but different types of atoms.

c Mixture: contains two compounds, salt (NaCl) and water (H_2O) that can be separated by evaporation.

d Compound: contains only one type of molecule but different types of atoms.

e Mixture: contains many compounds that can be separated physically.

13 **a** compound

b lattice

c atom

d molecule

e mixture

Analysing

14 The atomic number (i.e. the number of protons in the nucleus) determines what type of element it is.

15 Molecules are groups of atoms that are bonded chemically to each other. A single oxygen atom by itself is just an atom, but two oxygen atoms chemically bonded together form an oxygen molecule.

16 **a** An element contains only one type of atom, whereas a compound contains more than one type of atom.

b The element iron is a substance made up of iron atoms.

c The compound water is a substance made up of many water molecules.

d A compound is made up of different elements but only one type of molecule. However, a mixture is made up of more than one type of molecule.

e A molecule is made up of several atoms bonded chemically to each other.

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17

Atom	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons	Symbol of the atom
Carbon	6	12	6	6	6	$^{12}_{6}C$
Sulfur	16	32	16	16	16	$^{32}_{16}S$
Sodium	11	23	11	12	11	$^{23}_{11}Na$
Oxygen	8	16	8	8	8	$^{16}_{8}O$
Fluorine	9	19	9	10	9	$^{19}_{9}F$
Iodine	53	127	53	74	53	$^{127}_{53}I$

Evaluating

18 **a** True: the only exception is hydrogen, which can have a mass number equal to its atomic number.

b False: the chemical symbol for iron is Fe.

c False: table salt is the compound NaCl.

d True

e False: a molecule is made up of groups of atoms and each molecule can move independently. In a lattice, all atoms are bonded together into a single crystal.

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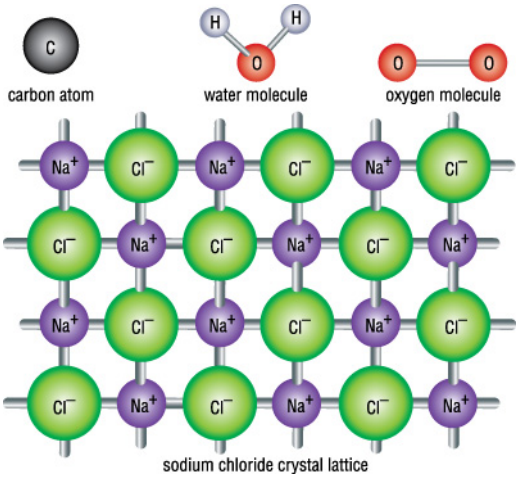
Creating

19



20 Creative writing exercise.

2.1 Practical activities

Prac 1: Making elements

Common mistakes

A common error is that the electrodes will sometimes touch, causing the power supply to short circuit. This can be avoided by placing a rubber stopper into the beaker to separate the electrodes.

Possible results

Hydrogen is produced at the negative electrode (cathode) and oxygen is produced at the positive electrode (anode) at a ratio of 2:1, respectively; therefore, more bubbles should be observed at the hydrogen electrode.

Suggested answers

1 *Elements cannot decompose, as they are made up of one type of atom only. A compound such as water, which consists of two or more atoms, can decompose when electricity is passed through it. In this experiment, the water decomposed into oxygen and hydrogen.*

2 *There were more gas bubbles at the electrode connected to the negative terminal.*

Prac 2: Making a compound

Common mistakes

The more limewater used, the greater the amount of carbon dioxide required for the limewater to turn milky. Using small amounts of limewater will assist students in achieving positive results.

Possible results

The compound carbon dioxide is produced, turning the limewater milky.

Suggested answers

1 *CO_2 (carbon dioxide)*

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2 *They both turned the limewater milky, so the gases made may be the same gas.*

3 *Carbon dioxide is a compound of carbon and oxygen. Each carbon dioxide molecule contains one carbon atom bonded to two oxygen atoms.*

Prac 3: Compounds in soft drinks

Common mistakes

Soft drinks will release more CO_2 at room temperature than at a refrigerated temperature. To assist students in obtaining measurable results, ensure that each soft drink tested is at room temperature.

Possible results

The varying amount of CO_2 in different brands and types of soft drink is measured.

Suggested answers

1 *Answers will vary.*

2 *Answers will vary.*

3 *Answers will vary.*

4 *'Lite' soft drinks are usually less dense because they contain less mass of dissolved sugar.*

5 *'Lite' should refer to low energy content (measured in calories or kilojoules). Sometimes, however, 'lite' is a misleading term. Sometimes, it refers to low fat content in food or drink (but high sugar content; e.g. many breakfast cereals) or light in flavour (e.g. recently, 'lite' cigarettes were sold in Australia. Although consumers thought they were light in nicotine, tobacco companies later claimed that the light meant a 'lighter' flavour).*

2.2 Answers

Remembering

1 boron, silicon, germanium, arsenic, antimony, tellurium, polonium, astatine

2 hydrogen, helium, carbon, nitrogen, oxygen, fluorine, neon, phosphorus, sulfur, chlorine, argon, selenium, bromine, krypton, iodine, xenon, radon

3 iron, cobalt, nickel, zinc, copper etc.

4 lithium, sodium, potassium, rubidium etc.

5 lithium, beryllium, boron, carbon, nitrogen, oxygen, fluorine, neon

6 The first shell needs two electrons, the second needs eight, the third shell can hold 18 but is stable with eight and the fourth shell can hold 32 but is stable with eight electrons.

Understanding

7 **a** The first four horizontal rows between Groups II and III are all transition metals.

b Vertical columns are called 'groups'.

c The most reactive metallic element would be francium.

d True, if you follow the trends of the periodic table.

e True

8 Energy levels are shells around an atom that can hold electrons. Electrons can only sit in an energy level and cannot exist between energy levels.

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9 **a** Atoms in the same group have the same number of electrons in their outer-most electron shell. These elements will share many chemical properties.

b Atoms in the same period have all their outer-most electrons in the same shell. This means that the atoms will be similar in size.

10 **a** 4

b 37

c 58

Applying

11 **a** V

b IV

c VII

d II

12 **a** mercury

b sodium, magnesium, aluminium

c tin, lead

13 **a** solids: 5

b liquids: 1

c gases: 11

14 **a** hydrogen, helium, nitrogen etc.

b bromine is the only one

c nitrogen, phosphorus and arsenic (semi-metallic)

d carbon, nitrogen, oxygen

e fluorine, bromine, iodine

f phosphorus, sulfur, chlorine

15 Most metals are found in Groups I, II or III, or as transition metals, or in the Lanthanides and Actinides series. Most non-metals are in Groups V, VI, VII and VIII.

16 Any five of:

- state at room temperature; i.e. solid, liquid or gas
- melting or boiling points
- colour
- density
- metal, semi-metal or non-metal
- thermal conductivity
- electrical conductivity.

17 **a** oxygen, sulfur, selenium etc.

b sodium, magnesium, aluminium

c lithium, sodium, potassium etc.

d fluorine, chlorine, bromine

e helium, neon, argon

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Answers to *Science Focus 3 second edition* Student Book questions

Analysing

18 Cl: chlorine, non-metal
Na: sodium, metal
Ar: argon, non-metal
Si: silicon, semi-metal
Cu: copper, metal
Ge: germanium, semi-metal

19

Atomic number	Element (name and symbol)	Number of protons	Number of electrons
7	Nitrogen (N)	7	7
8	Oxygen (O)	8	8
18	Argon (Ar)	18	18
3	Lithium (Li)	3	3
19	Potassium (K)	19	19
9	Fluorine (F)	9	9
12	Magnesium (Mg)	12	12
15	Phosphorus (P)	15	15
13	Aluminium (Al)	13	13

Evaluating

20 **a** cuprum: copper
b aurum: gold
c plumbum: lead
d wolfram: tungsten
e bromos: bromine

21 These words come from the Latin word for lead: plumbum.

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Creating

22

Scientist	Dates	Main contribution
John Dalton	1808	Proposed the atomic theory and created a table of the 20 substances thought to be elements.
John Newlands	1864	Arranged the 60 known elements in order of increasing atomic mass. Arranged elements in columns of seven and noticed that similar elements tended to be in the same rows but this was not always the case.
Dmitri Ivanovich Mendeleev	1869	Considered the father of the periodic table. Also arranged the known elements in order of atomic mass but also grouped elements into families of similar elements by leaving gaps in the table. From these gaps, he was able to predict the properties of undiscovered elements.
Julius Lother Meyer	1868–69	He constructed a table similar to Mendeleev but did not leave gaps for undiscovered elements.
Henry Moseley	1913	Suggested that the physical and chemical properties of elements was determined by their atomic number, not their atomic mass. Created a periodic table much like the one we use today.

2.2 Practical activities

Prac 1: Investigating a physical property

Common mistakes

Errors in measuring either mass or volume will cause errors in the measured and calculated values for density.

Possible results

Different types of matter with densities less than 1.0 g/cm^3 will float, whereas those with densities greater than 1.0 g/cm^3 will sink.

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Suggested answers

1 *Substances with densities less than 1.0 g/cm^3 will float, whereas those with densities greater than 1.0 g/cm^3 will sink.*

2 *1 mL = 1 cm^3*

3 *Errors in measuring either the mass or the volume will cause errors in the measured and calculated values for the density of water.*

4 *A tonne of lead has the same mass as a tonne of feathers (i.e. neither is heavier than the other). Because feathers are much less dense than lead, a tonne of feathers would take up far more volume than a tonne of lead.*

Prac 2: Comparing elements

Common mistakes

Students will often connect the ammeter to the circuit using the lowest scale, causing the ammeter's fuse to melt. To avoid damage, instruct students to first connect their circuit to the 5 amp scale, then down to the 1 amp scale if required.

Possible results

The different samples of the various elements will vary in their chemical and physical properties.

Suggested answers

1 *The metals to be inspected by the students should be solid, shiny (when polished), malleable, dense (sink in water, although some might float) and be able to conduct electricity.*

2 *The non-metals to be inspected by the students should be solid, crumbly powders, non-shiny, non-malleable and be insulators (not able to conduct electricity).*

2.3 Answers

Remembering

1 2,8,4

2 lattices

3 Oxygen: oxide (O^{2-})
Fluorine: fluoride (F^-)
Chlorine: chloride (Cl^-)
Bromine: bromide (Br^-)

4 non-metals

5 **a** metalloids
b brittle, conduct electricity to some extent
c silicon, germanium

6 Positive ions: potassium K^+; calcium Ca^{2+}; aluminium Al^{3+}
Negative ions: hydride H^-; fluoride F^-; oxide O^{2-}

7 First shell: maximum of 2 electrons
Second shell: maximum of 8 electrons

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Third shell: maximum of 18 electrons

Fourth shell: maximum of 32 electrons

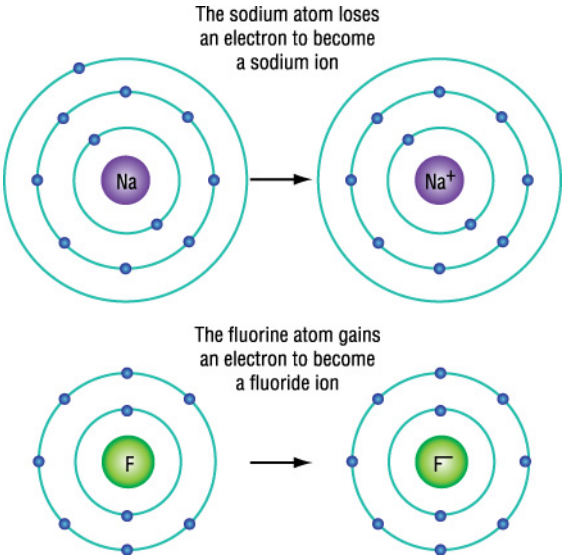
Understanding

8 The electronic configuration shows how many electrons are in each electron shell.

9 Magnesium has two electrons in its innermost shell, eight electrons in its second shell and two electrons in its outer-most shell.

10 A sodium ion is formed when the sodium atom loses an electron from its outer shell.

11 A positive ion is formed when an atom loses one or more electrons, a negative ion is formed when an atom gains one or more electrons.



12 The sodium ion has a positive charge (1+) and the chloride ion has a negative charge (1−), but in sodium chloride there is an equal number of sodium and chloride ions so these charges balance.

13 **a** Hydrogen has only one electron in its outer-most electron shell, so might be included in Group I.

b Hydrogen is missing an electron from a complete outer shell so could be put in Group VII. Also, many of its physical properties are similar to other elements in Group VII.

c Hydrogen shares physical and chemical properties from Groups I and VII, so might be better placed on its own.

14 Helium has only two electrons in its outer shell but this gives it a complete outer shell, making it an inert gas. All other inert gases are in Group VIII so helium is also placed in this group.

15 **a** lustrous: has a metallic shine

b malleable: can be bent into different shapes without breaking

c ductile: can be drawn into wires

d brittle: breaks or crumbles easily

e semi-metal: has many of the properties of a non-metal but conducts electricity.

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16

Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group VIII
+1	+2	+3	Uncertain	–3	–2	–1	0

Applying

17 Group VIII

18 **a** Period 2 Group IV
b Period 3 Group VI
c Period 2 Group V
d Period 3 Group V

19 **a** 2,6
b 2,8,8
c 2
d 2,8,2
e 2,8,6

20

Element	Atomic number	Number of electrons	Electron configuration	The atom could lose	The atom could gain	Most likely scenario	Most likely ion formed
H	1	1	1	$1e^-$	$1e^-$	Uncertain	H^+ or H^-
He	2	2	2	Unreactive		No ion formed	
Li	3	3	2,1	$1e^-$	$7e^-$	Lose $1e^-$	Li^+
Be	4	4	2,2	$2e^-$	$6e^-$	Lose $2e^-$	Be^{2+}
B	5	5	2,3	$3e^-$	$5e^-$	Lose $3e^-$	B^{3+}
C	6	6	2,4	$4e^-$	$4e^-$	Uncertain	
N	7	7	2,5	$5e^-$	$3e^-$	Gain $3e^-$	N^{3-}
O	8	8	2,6	$6e^-$	$2e^-$	Gain $2e^-$	O^{2-}

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Element	Atomic number	Number of electrons	Electron configuration	The atom could lose	The atom could gain	Most likely scenario	Most likely ion formed
F	9	9	2,7	$7e^-$	$1e^-$	Gain $1e^-$	F^-
Ne	10	10	2,8	Unreactive		No ion formed	
Na	11	11	2,8,1	$1e^-$	$7e^-$	Lose $1e^-$	Na^+
Mg	12	12	2,8,2	$2e^-$	$6e^-$	Lose $2e^-$	Mg^{2+}
Al	13	13	2,8,3	$3e^-$	$5e^-$	Lose $3e^-$	Al^{3+}
Si	14	14	2,8,4	$4e^-$	$4e^-$	Uncertain	
P	15	15	2,8,5	$5e^-$	$3e^-$	Gain $3e^-$	P^{3-}
S	16	16	2,8,6	$6e^-$	$2e^-$	Gain $2e^-$	S^{2-}
Cl	17	17	2,8,7	$7e^-$	$1e^-$	Gain $1e^-$	Cl^-
Ar	18	18	2,8,8	Unreactive		No ion formed	
K	19	19	2,8,8,1	$1e^-$	$7e^-$	Lose $1e^-$	K^+
Ca	20	20	2,8,8,2	$2e^-$	$6e^-$	Lose $2e^-$	Ca^{2+}

21 **a** Na^+
b S^{2-}
c I^-

Analysing

22 Atoms that can react have an incomplete outer shell of electrons and so react with other atoms to either gain or lose electrons in order to stabilise their outer electron shell. Atoms with a stable outer shell (Group VIII) do not react because they already have a stable electron configuration.

23 A chloride ion is a chlorine atom with one extra electron.

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24

Number of protons	Number of neutrons	Number of electrons	Overall charge	Is it an atom or an ion?	Symbol
8	6	8	0	Atom	O
10	10	10	0	Atom	Ne
11	10	10	+1	Ion	Na^+
17	16	18	–1	Ion	Cl^-
15	15	18	–3	Ion	P^{3-}
19	18	18	+1	Ion	K^+
20	19	18	+2	Ion	Ca^{2+}
8	7	10	–2	Ion	O^{2-}

Evaluating

25 The neutrons are charge neutral, so do not contribute to the overall charge of an ion.

26 Metallic atoms do not hold onto their outer electrons tightly. Metals are malleable and ductile because as the atoms are arranged in different ways the electrons are free to move along with them. On the contrary, in non-metals the electrons are held tightly to each atom. Therefore, the atoms cannot be rearranged easily without breaking the lattice.

Creating

27 Practical and creative task.

28

Metals	Non-metals
Ductile, dense, malleable, lustrous, excellent conductors, normally solid	Normally gas or liquid, brittle, dull, poor conductors

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2.3 Practical activities

Prac 1: Firework colours

Common mistakes

A well-lit room is not ideal for this practical, as it may not allow students to clearly see the initial colour of the flame. A darkened room will assist this exercise. Sodium chloride should be left until last, as it may contaminate everything else.

Possible results

Students will identify the element by the colour its compound produces when placed in a flame.

- barium chloride: green
- copper chloride: blue-green
- potassium chloride: violet
- sodium chloride: yellow
- strontium chloride: red

Suggested answers

1 *It served as the 'control' so we can see what the burning stick looked like with no chemical.*

2 *Tap water contains impurities that can change the colour of the flame.*

3 ***a*** *red: strontium*

b *green: barium*

c *blue-green: copper*

4 ***a*** *blue-green then purple: copper salt layer covering core of potassium salt (potassium gives a light pink-purple colour)*

b *red, then green: barium salt inside covered with a layer of strontium salt*

5 *In this experiment, the electrons gained their energy from the heat of the Bunsen burner. In a firework, the electrons gain their energy from the burning explosive black powder.*

Prac 2: Ions get together!

Common mistakes

If the jigsaw shapes are cut incorrectly, then the pieces may not interlock.

Possible results

The overall charge of the compound made should have an equal number of positive and negative charges of ions in order to be neutral.

Suggested answers

1 *The overall charge of a compound that has an equal number of positive and negative charges of ions is neutral. The overall charge of a compound that has more positive charges of ions than negative charges of ions is positive, whereas a compound with fewer positive charges of ions than negative charges of ions is negative.*

2 *The amount of positive charge carried by the cations must be balanced exactly by the amount of negative charge carried by the anions.*

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Prac 3: Metal crystals

Common mistakes

Instruct students to make sure that the agar is completely dissolved before adding the required chemical. This will ensure that the agar sets.

Possible results

Clearly visible silver crystals will grow as branches from the zinc strip.

Suggested answers

1 *Being a jelly, agar does not easily move. It allows a stable medium for the crystals to grow, allowing them to grow larger. If grown in liquid, every movement would break the crystals formed.*

2 *These metal crystals formed are a lattice because they are solid silver metal and solid metals have a lattice structure.*

3 *will be brown in colour*

4 *Used to culture fungi and bacteria. The agar provides a stable, surface medium for the bacteria to grow. The bacterial colonies can then be observed through the clear cover of the Petri dish without exposing the fungi or bacteria to the air and pose a health risk to the observer.*

Prac 4: More crystals

Common mistakes

Soiled or coated copper wire may inhibit crystal growth. If soiled or coated, clean with a piece of sandpaper or steel wool.

Possible results

Silver crystals will grow on the copper wire while the copper is being dissolved. The colourless solution becomes blue.

Suggested answer

1 *The copper dissolves and the clear silver nitrate becomes blue as the silver ions in the solution are replaced by the copper ions.*

Prac 5: Changing the properties of metals

Common mistakes

To ensure that the desired amount of heat is obtained, instruct students to use a blue flame and to place the pin at the top of the blue cone inside the flame.

Possible results

Quenching the pin caused the metal to become more brittle. Annealing caused the metal to become more malleable.

Suggested answers

1
- *Normalising: heating until red, cooling naturally.*
- *Quenching: heating until red, cooling quickly in cold water.*
- *Tempering: heating until red, cooling quickly in cold water, polishing, reheating until the metal itself has a blue colour, then cooling naturally.*

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2 *Various answers are possible depending on students' results, but the following results should have been obtained:*

- ***a*** *Quenching caused the metal to become more brittle.*
- ***b*** *Normalising caused the metal to become more malleable.*

3 *Normalising had a slow cooling stage and so would produce the largest crystals.*

4 *Larger crystals mean fewer bonds and fewer linkages between the bonds.*

5 *Steel is an alloy of iron, containing small amounts of carbon to make it stronger.*

Prac 6: Using metals to make non-metals

Common mistakes

Instruct students to stopper the test tube to ensure a sufficient collection of gas.

Possible results

When metals are added to a hydrochloric acid solution, they form metal chloride and displace hydrogen from the acid, forming bubbles of hydrogen gas. The rate of bubbling is dependent on the reactivity of the metal; the more bubbles produced the more reactive the metal is.

In this experiment, magnesium is the most reactive with hydrogen gas produced; iron is less reactive with hydrogen gas produced; and copper is the least reactive with no gas produced at all.

Suggested answers

1 ***a*** *Popping: hydrogen.*

- ***b*** *Flares brightly: oxygen.*
- ***c*** *Extinguishes flame: carbon dioxide.*

2 *Hydrogen and oxygen gases are elements, whereas carbon dioxide is a compound.*

3 *(Some) metals react with acids to release hydrogen gas.*

2.4 Answers

Remembering

1 lithium, sodium, potassium, rubidium, cesium or francium

2 helium, neon, argon, krypton, xenon or radon

3 silicon and germanium

Understanding

4 Hydrogen is explosive and helium is inert.

5 They increase in mass and density.

6 The melting and boiling points increase.

7 They react violently with acids and water, producing an alkaline (basic) solution and hydrogen gas.

8 **a** Diamond: jewellery or cutting tools, such as saws and drills.

- **b** Graphite: lubricant or in 'lead' pencils.
- **c** Silicon: the basis of computer chips and other electronics.

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d Germanium: as the catalyst in fluorescent lights; its oxides are used in the production of lenses for optical instruments, such as microscopes.

9 When carbon is in the form of graphite, it is brittle but conducts electricity and heat like a semi-metal.

Applying

10 **a** iodine and chlorine
b fluorine
c iodine
d chlorine
e fluorine

11 **a** sodium
b sodium
c lithium
d francium
e lithium

12 **a** calcium
b barium
c calcium
d magnesium
e beryllium

13 **a** yttrium, zirconium, niobium, molybdenum etc.
b iron, cobalt, nickel
c gold, silver, platinum
d silver, platinum, mercury
e gold, mercury, silver, tungsten, iron

14

	20°C	**100°C**	**–199°C**	**150°C**
Fluorine	Gas	Gas	Liquid	Gas
Chlorine	Gas	Gas	Solid	Gas
Bromine	Liquid	Gas	Solid	Gas
Iodine	Solid	Solid	Solid	Liquid

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Evaluating

15 **a** True
b True
c False; they are forms of carbon.
d False; carbon is in all things that live or have lived.

16 Most non-metals are gaseous and therefore were more difficult to discover. Moreover, carbon is extremely common, being a part of all living things and is easily formed simply by burning wood or other plant matter.

Creating

17 Research task.

2.4 Practical activities

Prac 1: The alkaline earths

Common mistakes

Instruct students to stopper the test tube to ensure a sufficient collection of gas.

Possible results

Group II metals are less reactive than Group I metals.

Suggested answers

1 *Calcium is more reactive than magnesium.*

2 *Reactivity increases as we move down Group II.*

3 *Group II metals are less reactive than Group I metals.*

Prac 2: Group IV

Common mistakes

Ensure the light globe selected has a suitable voltage tolerance.

Possible results

The properties become more metallic as we move down Group IV.

Suggested answers

1
- *metals: lead and tin*
- *non-metals: carbon and silicon*
- *semi-metals: silicon and germanium are metalloids*

2 *The properties become more metallic as we move down Group IV.*

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Chapter answers

Remembering

1
- **a** False; it is the number or protons and neutrons.
- **b** False; it is a liquid.
- **c** False; there are approximately 118 known atoms.
- **d** True
- **e** True
- **f** False; Period 3 Group V.
- **g** False; it is a compound.
- **h** False; it is a mixture.
- **i** True
- **j** True
- **k** True
- **l** False; lose or gain electrons.
- **m** False; it becomes a positive ion.
- **n** True

Understanding

2 Electrons exist in shells or energy levels around the nucleus. The innermost shell can hold only two electrons, the second shell holds eight, the third shell can hold 18 electrons and the fourth shell holds 32 electrons.

3 The period number indicates how many of the atom's electron shells contain electrons. The group number indicates how many electrons are in the outer-most shell.

4 Elements in the same family have similar chemical properties. The chemical properties are largely determined by the number of electrons in the outer-most shell and, therefore, elements in the same group are also in the same family.

5 Hydrogen has one electron in its outer-most shell. It can, therefore, be placed in Group I with the reactive metals sodium, potassium etc. However, it is also missing just one electron from a full outer shell. This makes it a non-metallic gas and means it could also be placed in Group VII with the reactive non-metals, such as fluorine and chlorine. Similarly, helium has two electrons in its outer-most shell so could be placed in Group II, but it also has a complete outer shell, which means that its physical and chemical properties are more like those of Group VIII.

6 The atoms get bigger and heavier.

7

Scientist	Dates	Main contribution
John Dalton	1808	Proposed the atomic theory and created a table of the 20 substances thought to be elements.

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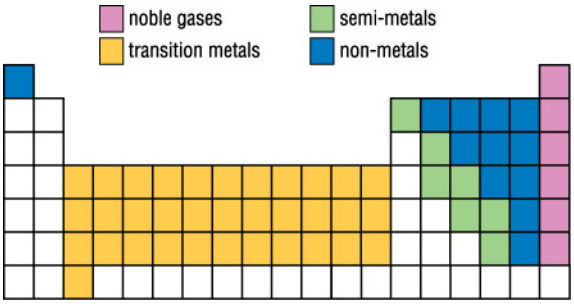
Scientist	Dates	Main contribution
John Newlands	1864	Arranged the 60 known elements in order of increasing atomic mass. Arranged elements in columns of seven and noticed that similar elements tended to be in the same rows but this was not always the case.
Dmitri Ivanovich Mendeleev	1869	Considered the father of the periodic table. Also arranged the known elements in order of atomic mass but also grouped elements into families of similar elements by leaving gaps in the table. From these gaps, he was able to predict the properties of undiscovered elements.
Julius Lother Meyer	1868–69	He constructed a table similar to Mendeleev but did not leave gaps for undiscovered elements.
Henry Moseley	1913	Suggested that the physical and chemical properties of elements was determined by their atomic number, not their atomic mass. Created a periodic table much like the one we use today.

8 The potassium atoms would lose an electron to the fluorine atoms, creating potassium ions (K^+) and fluoride ions (F^-). The positive and negative ions would attract to form a crystal lattice of potassium chloride.

9 $SiCl_4$, $GeCl_4$, $SnCl_4$, $TlCl_4$

Applying

10



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11 **a** Group VII
b Group VIII
c Group II

12 **a** 3^+
b 1^-
c 2^-
d 0
e 2^+
f 3^-

13

Atom	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons	Atomic symbol
Sulfur	16	32	16	16	16	$^{32}_{16}S$
Hydrogen	1	1	1	0	1	$^{1}_{1}H$
Beryllium	4	9	4	5	4	$^{9}_{4}Be$
Iodine	53	127	53	74	53	$^{127}_{53}I$
Nickel	28	59	28	31	28	$^{59}_{28}Ni$

Analysing

14 Metals conduct electricity, are malleable, are ductile, have a metallic lustre and conduct heat.

15 A chloride ion is a chlorine atom that has gained one electron.

16 **a** 17 protons, 17 electrons and 18 neutrons
b 1 proton, 1 electron and 2 neutrons
c 79 protons, 79 electrons and 118 neutrons

17 When a chemical reaction takes place the atoms exchange or share electrons. The outer-most electrons are less tightly bound to the nucleus, so it is these electrons that rearrange themselves in a chemical reaction.

Evaluating

18 The outer-most electrons in francium are in the seventh electron shell and, therefore, much further away from the nucleus. As a result, it is much easier for francium to lose its outer-most electron, making it more chemically reactive.

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19 Fluorine and iodine both tend to gain electrons during a chemical reaction. However, the outermost shell of fluorine is much closer to the nucleus, making it more efficient at 'stealing' electrons from other atoms and, hence, more chemically reactive.

Creating

20 Practical task.

21 Creative task.

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3.1 Answers

Remembering

1 melting ice, boiling water, breaking glass, dissolving sugar etc.

2 During a chemical change, new substances form. During a physical change, no new substance is formed.

3
- permanent colour change
- a gas is produced
- a precipitate is formed
- heat energy is produced or absorbed

4
- The matchstick turns black.
- Energy is given off as light and heat.
- You can smell new substances.

5
a energy is given off
b energy is absorbed
c The reaction can gain enough energy from its environment to proceed.
d The reaction requires a constant input of energy to proceed.

6
a burning magnesium ribbon
b photosynthesis
c rusting
d electrolysis of water

Understanding

7 Solution: a mixture formed when a substance is dissolved in a liquid.

8 Solutions do not scatter light and are never cloudy or murky. However, they can absorb light of different colours.

9 A precipitate forms when two solutions of soluble substances are mixed, forming an insoluble precipitate.

10 Burning methane produces enough heat energy to maintain the reaction. The lit match is needed to initiate the reaction.

Applying

12 Chemical changes: burning toast, baking a cake, striking a match, fireworks, rusting etc.
Physical changes: melting, boiling, freezing, condensation, dissolution, crystallisation, breaking, mixing etc.

13 **i**
a water
b hydrogen, oxygen
c endothermic
d products

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ii **a** methane, oxygen
b carbon dioxide, water
c exothermic
d reactants

14 **a** Yes, a precipitate has formed.
b No, no new substance has formed.
c Yes, a new gas has been produced.
d Yes, heat energy is being given off.
e No, no new substance has formed.
f Yes, a permanent colour change.

Analysing

15 **a** physical change
b chemical change
c chemical change
d physical change
e physical change
f physical change
g chemical change. Although the aspirin is dissolving, CO_2 is also being produced.

Evaluating

16 Once the reaction has started, the sparkler produces enough energy to allow the reaction to continue. Other spontaneous reactions include burning a match, burning natural gas on a stovetop, rusting, rotting of fruit.

Creating

17 **a** $copper_{(s)} + nitric\ acid_{(aq)} \rightarrow nitrogen\ monoxide_{(g)} + water_{(l)} + copper\ nitrate_{(aq)}$
b $sulfuric\ acid_{(aq)} + sodium\ carbonate_{(s)} \rightarrow carbon\ dioxide_{(g)} + water_{(l)} + sodium\ sulfate_{(aq)}$
c $magnesium_{(s)} + oxygen_{(g)} \rightarrow magnesium\ oxide_{(s)}$
d $carbon\ dioxide_{(g)} + water_{(l)} \rightarrow glucose_{(s)} + oxygen_{(g)}$
e $iron_{(s)} + water_{(l)} + oxygen_{(g)} \rightarrow hydrated\ iron\ oxide_{(s)}$
f $lead\ nitrate_{(aq)} + sodium\ iodide_{(aq)} \rightarrow lead\ iodide_{(s)} + sodium\ nitrate_{(aq)}$

18 Creative task. Various solutions are possible.

3.1 Practical activities

Prac 1: Chemistry in the kitchen

Common mistakes

Ensure that the oven is set to the correct temperature. Instruct students not to open the oven during baking as the fluctuation of the temperature will hinder the process of baking.

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Possible results

Baking of cookie/muffin/cake involves both physical and chemical changes. During the baking, a gas is produced and causes the cookie/muffin/cake to rise as the gas (carbon dioxide) gets trapped within it.

Physical changes: melting butter, evaporation of water.

Suggested answers

1 *Thick liquid becomes a solid, permanent colour change. Gas forms small bubbles in the cake mixture that are 'set in place' as the cake bakes, increasing the volume and making it lighter to eat. Heat causes a change and makes the cake firm. These are chemical changes that cannot be reversed.*

2 *Before baking: thick liquid without any air bubbles, heavy, dense. After baking: solid, lighter colour, lighter, firm.*

Prac 2: Signs of chemical change

Common mistakes

When magnesium is reacted with dilute nitric acid, the hydrogen gas produced may not be sufficient to make the 'pop' sound. A slightly higher concentration of nitric acid should produce better results.

Possible results

Copper carbonate: green powder turns to black powder.

Nitric acid and magnesium: hydrogen gas is produced.

Dilute barium nitrate and sodium sulfate: white precipitate is formed.

Dilute nitric acid and sodium hydroxide: heat released.

Zinc and dilute copper sulfate: copper deposits on zinc and the blue colour of copper sulfate fades away.

Suggested answers

1 *hydrogen gas*

2 *It burns with a 'pop' sound, which confirms hydrogen gas. It is an explosive gas.*

3 *Barium sulfate, as sodium nitrate is soluble.*

4 *exothermic*

5 *No reaction would occur because silver is less reactive than copper.*

Prac 3: Light sticks: chemiluminescence

Common mistakes

The light sticks can sometimes break when activating them, causing the contents to leak. Instruct the students to gently bend the light sticks when activating them.

Possible results

The light stick in the hot water will glow more brightly because the extra heat energy speeds up the chemical reaction.

Suggested answers

1 *Answers will vary, but it is probable that the light stick in the hot water will be glowing more brightly because the extra heat energy speeds up the chemical reaction.*

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2 *It would get dimmer but last longer.*

3 *Light sticks are used for providing light while camping or out at night away from electricity; for signalling; and for car breakdowns and similar emergencies.*

3.2 Answers

Remembering

1
- Combination: combining carbon and oxygen to form carbon dioxide; burning magnesium ribbon in oxygen to form magnesium oxide.
- Combustion: burning methane in oxygen to form carbon dioxide and water; burning magnesium ribbon in air to form magnesium oxide.
- Decomposition: decomposition of hydrogen peroxide to form oxygen and water; decomposition of calcium carbonate to form calcium oxide and carbon dioxide.

2 Combination: $x + y \rightarrow xy$
Decomposition: $xy \rightarrow x + y$

3 carbon dioxide

4 $2Mg_{(s)} + O_{2(g)} \rightarrow 2MgO_{(s)}$

Understanding

5 Oxygen is made up of molecules, each with two oxygen atoms. When a reaction occurs, both atoms will take part in the reaction. Therefore, it is important to write O_2 to ensure that the equation can be balanced.

6 In combination reactions, two or more substances come together to form one substance. In decomposition reactions, one substance splits apart to form two or more substances.

Applying

7 **a** decomposition
b combustion
c combination

8 **a** $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$
b $CuCO_{3(s)} \rightarrow CuO_{(s)} + CO_{2(g)}$
c $C_3H_{8(g)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$

9 **a** magnesium + oxygen → magnesium oxide
b water → hydrogen + oxygen
c calcium oxide + water → calcium hydroxide

Analysing

10 To heat our homes, to cook food on a gas stove, to power cars by burning fuels and to produce electricity.

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11 When humans first discovered fire, they were using a combustion reaction. These reactions not only helped humans to heat themselves and cook, but were used to create tools and smelt metals. Later, humans used combustion reactions as fuels to power trains, cars and other modes of transport and to generate electricity.

Evaluating

12 The candle would soon die out. The process of combustion consumes all the oxygen and the reaction stops.

13 Hydrogen peroxide spontaneously decomposes into oxygen and water. Keeping the hydrogen peroxide cool slows this reaction down.

14 **a** combustion

b oxygen

c The reaction is spontaneous because once it has started it will continue by itself.

d **i** For a fire to start, the reactants must first gain enough energy from their environment to react. In summer there is more energy and so it is easier for the fire to get started.

ii There is less heat energy in the environment in winter, so the reaction is easier to control.

e Burnoffs reduce the fuel for fires.

Creating

15 Creative task. Various responses are possible.

3.2 Practical activities

Prac 1: Burning magnesium ribbon (teacher demonstration)

Common mistakes

Magnesium ribbon should be cleaned to remove the oxide coating to facilitate rapid burning.

Possible results

When pure magnesium burns in oxygen, a combination reaction occurs, forming the chemical compound magnesium oxide.

Suggested answers

1 *magnesium + oxygen → magnesium oxide*

2 *This reaction involves the 'combining' of two elements (magnesium and oxygen) to form a chemical compound (magnesium oxide). A combustion reaction is a chemical reaction that involves oxygen and produces energy (heat) rapidly.*

3 *Exothermic reaction, as all combustion reactions release heat energy and so are exothermic.*

4 *Non-spontaneous reaction. Heating is needed for the reaction.*

Prac 2: Stopping combustion

Common mistakes

To assist with the collection of CO_2, ensure that the plastic tube in the conical flask is well above the solution level.

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Possible results

Combustion is stopped when oxygen is removed.

Suggested answers

1 *Heat is produced.*

2 *Spontaneous, as there is no external source of ignition.*

3 *CO_2 gas does not support combustion and it is denser than air, so if CO_2 can be directed downwards onto the burning material it covers the burning substance, depriving it of O_2 so combustion stops.*

Prac 3: Decomposition of a metal carbonate

Common mistakes

Lift the delivery tube from the limewater as soon as the heating has stopped to avoid the liquid being sucked back up the tube.

Possible results

When heated, copper carbonate decomposes into copper oxide and gives off carbon dioxide gas.

Suggested answers

1 *Green powder turns to black powder. A colour change has taken place and therefore a chemical reaction has occurred.*

2 *copper carbonate + oxygen → copper oxide + carbon dioxide*

3 *It would be less than the mass of copper carbonate, as carbon and some oxygen is removed as carbon dioxide.*

4 *Carbon dioxide turns limewater milky.*
carbon dioxide + calcium hydroxide → calcium carbonate + water

3.3 Answers

Remembering

1 (s): solid, (l): liquid, (g): gas, (aq): aqueous. A precipitate is a solid (s).

2 hydrogen ion (H^+), ammonium (NH_4^+)

3 ammonium (NH_4^+), carbonate (CO_3^{2-}), nitrate (NO_3^-), sulfate (SO_4^{2-}), phosphate (PO_4^{3-})

4 soluble

Understanding

5 A salt is any substance made up of a lattice of positive and negative ions.

6 when two clear solutions are mixed to form a solid or the solutions become cloudy

Applying

7 Nitrates: no exceptions; chlorides except silver chloride AgCl, lead chloride $PbCl_2$ and mercury chloride $HgCl_2$.

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8 **a** silver chloride AgCl

b mercury(I) iodide (HgI)

c calcium carbonate ($CaCO_3$)

d barium sulfate ($BaSO_4$)

9 **a** silver nitrate$_{(aq)}$ + sodium chloride$_{(aq)}$ → silver chloride$_{(s)}$ + sodium nitrate$_{(aq)}$
$AgNO_{3(aq)} + NaCl_{(aq)} \rightarrow AgCl_{(s)} + NaNO_{3(aq)}$

b mercury(I) nitrate$_{(aq)}$ + potassium iodide$_{(aq)}$ → mercury(I) iodide$_{(s)}$ + potassium nitrate$_{(aq)}$
$HgNO_{3(aq)} + KI_{(aq)} \rightarrow HgI_{(s)} + KNO_{3(aq)}$

c calcium nitrate$_{(aq)}$ + lithium carbonate$_{(aq)}$ → calcium carbonate$_{(s)}$ + lithium nitrate$_{(aq)}$
$Ca(NO_3)_{2(aq)} + Li_2CO_{3(aq)} \rightarrow CaCO_{3(s)} + 2LiNO_{3(aq)}$

d barium nitrate$_{(aq)}$ + sodium sulfate$_{(aq)}$ → barium sulfate$_{(s)}$ + sodium nitrate$_{(aq)}$
$Ba(NO_3)_{2(aq)} + Na_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + 2NaNO_{3(aq)}$

Analysing

10 Classifying reactions allows scientists to predict the likely products of a reaction based on the type of reaction. Also, understanding how one reaction works allows scientists to infer how other reactions of the same type work.

11 **a** 4+

b 32+

c 9–

12 **a** 2 nitrogen, 8 hydrogen, 1 sulfur, 4 oxygen

b 2 potassium, 2 chromium, 7 oxygen

c 1 calcium, 2 oxygen, 2 hydrogen

Evaluating

13 **a** no

b yes

c no

d yes

14 **a** NaBr

b MgS

c CaF_2

d Li_3N

e Al_4C_3

15 **a** rubidium bromide

b potassium sulfide

c beryllium oxide

d sodium nitride

e ammonium chloride

f lithium hydroxide

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g silver carbonate
h zinc sulfate

Creating

16 Creative and investigation task.
17 Creative task.

3.3 Practical activities

Prac 1: Precipitation reactions

Common mistakes

Barium sulfate precipitate is white, although it appears light blue in a copper solution. Students may identify it as a blue precipitate. Allow precipitates to settle before making a conclusion about the precipitation.

Possible results

Following the rules of solubility, students will identify which precipitate is formed from the reactions between the ions of two different substances.

Suggested answers

1 *All nitrates are soluble. Therefore, silver chloride, silver oxide and barium sulfate are precipitates.*

2 *silver nitrate$_{(aq)}$ + sodium chloride$_{(aq)}$ → silver chloride$_{(s)}$ + sodium nitrate$_{(aq)}$*
silver nitrate$_{(aq)}$ + sodium hydroxide$_{(aq)}$ → silver oxide$_{(s)}$ + sodium nitrate$_{(aq)}$ + water
barium nitrate$_{(aq)}$ + copper(II) sulfate → barium sulfate$_{(s)}$ + copper nitrate$_{(aq)}$

Prac 2: Precipitation of unknowns

Common mistakes

Some solutions may be too dilute to form a precipitate and students may make wrong conclusions regarding the solubility of some compounds. Consider more concentrated solutions when reactions are needed to form silver sulfate, calcium sulfate and lead chloride precipitates.

Possible results

Following the rules of solubility, students will identify the precipitates formed from a series of reactions between the ions of two unknown substances.

Suggested answers

1 *If any results were inconclusive, it could be due to products being slightly soluble, leading to small precipitates.*

2 *Answers will vary, but could include adding sodium iodide, sodium carbonate or sodium sulfate.*

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3.4 Answers

Remembering

1 Acids are substances that produce hydrogen ions (H^+) when dissolved in water, whereas bases produce hydroxide ions (OH^-) when dissolved in water.

2 Acids:

- have a sour taste
- can be corrosive
- turn blue litmus red
- neutralise bases.

Bases:

- taste bitter
- have a soapy feel
- can be caustic
- turn red litmus blue
- neutralise acids.

3 **a** hydrochloric acid, sulfuric acid

b carbonic acid, phosphoric acid

c sodium hydroxide, lithium hydroxide

d ammonia, carbonate ion

4 **a** as low as 0, depending on concentration

b below 7

c 7 exactly

d above 7

e as high as 14, depending on concentration

5 **a** green

b blue

c blue

6 acid + carbonate → salt + carbon dioxide + water
acid + metal → salt + hydrogen gas

Understanding

7 using an acid to dissolve lime deposits, using an antacid to relieve heartburn

8 **a** Pop test: glowing splint at the top of a test tube will ignite the hydrogen and create a popping sound.

b Limewater test: bubbling carbon dioxide gas through limewater will cause the limewater to become cloudy.

9 **a** bees, wasps, jellyfish

b applying a weak acid or a weak base to neutralise the sting, depending on the type of sting

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c Vinegar and the chemical in bee stings are both acids and therefore no neutralisation will occur.

10 The burning sensation is due to the concentrated acid in the stomach burning the lining of the oesophagus.

11 **a** aluminium + nitric acid → hydrogen gas + aluminium nitrate

b zinc + nitric acid → hydrogen gas + zinc nitrate

c iron + nitric acid → hydrogen gas + iron nitrate

d lithium + nitric acid → hydrogen gas + lithium nitrate

12 **a** hydrochloric acid + iron(II) hydrogen carbonate → iron(II) chloride + carbon dioxide + water

b nitric acid + silver hydroxide → silver nitrate + water

c sulfuric acid + barium oxide → barium sulfate + water

d **a** $2HCl_{(aq)} + Fe(HCO_3)_{2(s)} \rightarrow FeCl_{2(aq)} + 2CO_{2(g)} + 2H_2O_{(l)}$

b $HNO_{3(aq)} + AgOH_{(aq)} \rightarrow AgNO_{3(aq)} + H_2O_{(l)}$

c $H_2SO_{4(aq)} + BaO_{(s)} \rightarrow BaSO_{4(s)} + H_2O_{(l)}$

Applying

13 acetic acid, also known as ethanoic acid

14 Indicators are often used to test the pH of water in swimming pools to minimise the levels of bacteria in the water.

15 orange, lemon, mandarin

16 **a** strontium nitrate

b copper sulfate

c silver chloride

d magnesium nitrate

17 **a** hydrochloric acid and barium hydroxide

b nitric acid and calcium hydroxide

c sulfuric acid and iron(III) hydroxide

Analysing

18 A dilute solution of nitric acid has less nitric acid and more water than a concentrated solution.

19 Acids and bases are both very important in our everyday lives. We have acids in our stomach to help us digest food. Also, many of the foods we eat are acidic, such as citrus fruits and vinegar. Bases are most commonly found in cleaning products, such as toothpaste, soap or caustic soda.

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20

Reaction type	Example (word equation and chemical formulae)	Reactant(s)	Product(s)
acid + hydroxide	hydrochloric acid + sodium hydroxide → sodium chloride + water; $HCl + NaOH \rightarrow NaCl + H_2O$	hydrochloric acid + sodium hydroxide	sodium chloride + water
acid + oxide	nitric acid + barium oxide → barium nitrate + water; $2HNO_3 + BaO \rightarrow Ba(NO_3)_2 + H_2O$	nitric acid + barium oxide	barium nitrate + water
acid + carbonate	sulfuric acid + calcium carbonate → calcium sulfate + carbon dioxide + water; $H_2SO_4 + CaCO_3 \rightarrow CaSO_4 + CO_2 + H_2O$	sulfuric acid + calcium carbonate	calcium sulfate + carbon dioxide + water
acid + metal	hydrochloric acid + lithium → lithium chloride + hydrogen gas; $2HCl + 2Li \rightarrow 2LiCl + H_2$	hydrochloric acid + lithium	lithium chloride + hydrogen gas

21 990 mL

Evaluating

22 **a** acid

b Both acid and water will lower the pH. However, water lowers the pH by changing the concentration, so as the water evaporates, the pH will increase again. However, the acid will neutralise the base, permanently lowering the pH. Also, water can never reduce the pH below 7.

23 It is most likely to be acetic acid, as acetic acid is a weak acid whereas hydrochloric acid is a strong acid. Also, acetic acid is vinegar, so is more likely to be in food.

3.4 Practical activities

Prac 1: Common indicators

Common mistakes

To avoid contamination and false results, instruct students to thoroughly rinse each test tube between tests.

Possible results

Students test the effects of acids and bases on a variety of commonly used indicators. Results should be comparably similar to the results shown in Figure 3.4.14.

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Suggested answers

1 *Tap water contains impurities and may not have a pH of exactly 7.0.*

2 *Although answers will vary, they should be comparable with the colours seen in Figure 3.4.14.*

Prac 2: Universal indicator

Common mistakes

A good dilution technique will produce a varied colour change with each dilution.

Possible results

Students dilute the solutions by a factor of 10 to change from one pH value to the next, then test the effects of the increasing pH on universal indicator. Results should be comparably similar to the results shown in Figure 3.4.9.

Suggested answers

1 *Tubes 1 to 6 all contain HCl, a strong acid. HCl is concentrated in tubes 1 and 2 and dilute in tubes 3 to 6. Tubes 8 to 14 all contain NaOH, a strong base. NaOH is concentrated in tubes 13 to 14 and dilute in tubes 8 to 12.*

2 *Dilution factor = ratio of final volume/aliquot volume (final volume = aliquot + diluent)*
10 mL/1 mL = 1:10

a *1:100*

b *1:10*

Prac 3: Natural indicators

Common mistakes

Observations of the colours will vary among the students due to the concentrations of the solutions prepared.

Possible results

Many natural substances of vegetable origin have the property to change colour because of the acidity of the environment. Results will vary depending on the flowers used. Hydrangea petals are white or blue in an acidic environment and pink in a basic environment. Black tea is dark red in an acidic environment and yellow in a basic environment. Beetroot juice is light brown in an acidic environment and orange brown in a basic environment.

Suggested answers

1 *Many natural pigments are indicators. Many, but not all, are red in acidic conditions. Flower petals work, but plant leaves do not.*

2 *Answers will vary. Many flower petals have more distinct colour changes than black tea or beetroot juice.*

3 *The colours using the paper are less clear and less dramatic.*

4 *Answers will vary. One example is red cabbage juice. It has a dark purple pigment called anthocyanin. The cabbage juice changes from purple to bright pink in acids to blue or yellow-green in bases.*

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Prac 4: Testing household solutions

Common mistakes

Dark-coloured household solutions are not ideal for testing, as the colour change of the indicator will not be distinct.

Possible results

Household substances such as vinegar, fruit juice and baking powder will be acidic, whereas detergents, soap and bleach will be basic.

Suggested answers

1 *Answers will vary depending on the household liquids chosen. Vinegar and citrus fruits will be acidic. Detergents and soaps might be basic.*

2 *at the 'most acidic' end, with a lower pH value than anything tested here*

3 *Milk becomes sour because of the production of lactic acid, which makes sour milk more acidic than fresh milk (acids taste sour).*

Prac 5: Acids and hydroxides

Common mistakes

Students may measure volumes incorrectly.
A burette will assist in achieving more accurate results.

Possible results

The pH reading should start from a lower range (pH 1) and should increase slowly to about 2. Thereafter, it should increase rapidly to pH 7 when 25 mL of sodium hydroxide is added to the hydrochloric acid. After this point, pH will increase slowly to 12 with the addition of more sodium hydroxide. If the concentrations of the solutions are not accurate, slight variations may occur.

Suggested answers

1 *The solution is neutral when 25 mL of 0.1 M sodium hydroxide is added. The pH meter should read 7 and the colour of the universal indicator should indicate neutral.*

2 *hydrochloric acid + sodium hydroxide → sodium chloride + water*

Prac 6: Acids and carbonates

Common mistakes

The reaction in the test tube may bubble over if too much solid is added. Instruct students to use less rather than more.

Possible results

Metal carbonate compounds react with acids to produce salt, water and carbon dioxide gas.

Suggested answers

1, 2 *Word equations (salts formed shown in bold).*

sodium hydrogen carbonate + nitric acid → ***sodium nitrate*** *+ water + carbon dioxide*

lithium carbonate + nitric acid → ***lithium nitrate*** *+ water + carbon dioxide*

sodium carbonate + nitric acid → ***sodium nitrate*** *+ water + carbon dioxide*

ammonium carbonate + nitric acid → ***ammonium nitrate*** *+ water + carbon dioxide*

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*sodium hydrogen carbonate + hydrochloric acid → **sodium chloride** + water + carbon dioxide*
*lithium carbonate + hydrochloric acid → **lithium chloride** + water + carbon dioxide*

3 *Similar to Figure 2.1.9 on page 40.*

Prac 7: Acids and metals

Common mistakes

Sometimes, the concentration of acids may not be strong enough to produce the desired amount of hydrogen gas. If this occurs, consider a more concentrated acid solution of 1–2 M.

Possible results

Metals react with acids to produce salt and hydrogen gas. Hydrochloric acid produces chloride salts, sulfuric acid produces sulfates and acetic acid produces acetates.

Suggested answers

1 *Five equations of the form:*
metal + acid → salt + hydrogen

2 *Hydrochloric acid produces chloride salts, sulfuric acid produces sulfates and acetic acid produces acetates.*

3 *Most active to least active: magnesium, aluminium, zinc, iron, tin.*

Chapter answers

Remembering

1 During a chemical change a new substance is formed; e.g. rusting iron, burning a match. During a physical change, no new substances are formed; e.g. melting of ice, dissolving salt in water.

2
- Permanent colour change: e.g. burnt toast.
- A gas is given off: e.g. reacting acid with a carbonate.
- Precipitate forms: e.g. bubbling carbon dioxide through limewater.
- Energy is given off or absorbed: e.g. fireworks exploding.

3 ammonium (NH_4^+), carbonate (CO_3^{2-}), nitrate (NO_3^-), sulfate (SO_4^{2-})

4 **a** LiOH
b $BaSO_4$
c $AlBr_3$

5 soluble: nitrates, chlorides
insoluble: carbonates, hydroxides

6 **a** hydrogen carbonate
b iodide
c sulfide
d ammonium

7 Acids contain hydrogen ions combined with a negative non-metal ion. Bases contain hydroxide or oxide ions combined with a positive metal ion.

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8 reactants: acid + metal
products: salt + hydrogen gas
e.g. hydrochloric acid + sodium → sodium chloride + hydrogen gas

Understanding

9 **a** Using classifications, scientists can study one reaction and generalise their understanding to other reactions of the same type. For example, by understanding how silver chloride precipitates from a solution of silver nitrate and sodium chloride, scientists can understand how all other precipitation reactions work.

b Models provide a simplified picture that allows scientists to make predictions and explain how things work. For example, the atomic model can be used to explain the properties of solids, liquids and gases.

c By applying rules, scientists can determine how something works with no direct experience of it themselves. For example, by following the rules of balancing equations, scientists can write balanced equations for any reaction.

10 An exothermic reaction gives off heat energy; e.g. fireworks. An endothermic reaction absorbs heat energy; e.g. cold pack.

11 As a match burns it produces enough energy to sustain the reaction.

12 permanent colour change, production of gas, a precipitate may form, energy may be given off

13 A salt is a substance made up of a lattice of positive and negative ions.

14 If the ions from two solutions combine to produce an insoluble substance, it can be expected that a precipitate will form.

15 A weak base steals a hydrogen ion from a water molecule to form a hydroxide ion.

Applying

16 **a** combination

b neutralisation

17 Various answers possible: aluminium hydroxide, magnesium hydroxide, aluminium carbonate.

18 **a** red

b red

c red-orange

d orange

Analysing

19 **a** insoluble

b soluble

c soluble

d insoluble

e soluble

f insoluble

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g insoluble
h insoluble

20 a base
b base
c acid
d base
e acid

21 90 mL

Evaluating

22 **a** +1
b +2
c +3

23 **a** sodium chloride + carbon dioxide + water
b calcium nitrate + water

24 Put 10 mL of a solution in a test tube and add the indicator to determine the pH. If the pH is less than 7, then it is hydrochloric acid; if the pH is 7 then it is water; and if the pH is greater than 7 then the solution is sodium hydroxide. Add 90 mL of another solution to the first solution. If the pH changes by 1 then the second solution is water. If the pH goes up by more than 1, then the second solution is sodium hydroxide and if the pH goes down by more than 1, then the second solution is hydrochloric acid. The third solution can then be deduced.

Creating

25 Investigation task. You would expect to see the pH increase by 1 every time the solution is diluted by a factor of 10.

26 Research and creative task. Students should demonstrate that they understand that a carbonate would provide the safest neutralisation. Strong bases can be just as dangerous, and strong acids and metals produce hydrogen gas, which is explosive.

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4.1 Answers

Remembering

1 **a** cornea and lens
b iris and pupil
c retina

2 **a** rods
b cones

3 If you only had one eye, you would not be able to judge the distance of objects.

4 Eyebrows and eye lashes protect from dust and particles, tear ducts wash away particles, orbits stop the eye from being knocked.

5 hyperopia: long-sightedness; myopia: short-sightedness

Understanding

6

Part	Description/function
Conjunctiva	Thin, clear layer covering front of eye
Iris	Colourful part of the eye that changes the size of the pupil to regulate the amount of light that enters the eye
Pupil	The hole at the centre of the iris that allows light to enter the eye
Cornea	Transparent convex layer in front of the pupil that focuses light rays as they enter the eye
Lens	A jelly-like convex lens that can be stretched or compressed to focus light rays on the back of the eye
Retina	A layer of cells on the back of the eyeball that detects light and sends the image to the brain
Vitreous humour	The liquid inside the eye that gives it shape
Aqueous humour	Liquid between the cornea and the iris that maintains the shape of the cornea
Sclerotic layer	Tough outer layer of the eyeball

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Part	Description/function
Choroid	A layer of blood vessels between the retina and the sclerotic layer that nourishes the back of the eye. It is black and prevents light from reflecting around inside the eyeball.
Suspensory ligaments	Ligaments that hold the lens in place
Ciliary muscles	Muscles that adjust the shape of the lens to allow focusing

7 Colour blindness is caused by a lack of differentiation between the cones that detect red, green or blue light.

8 The blind spot does not have any light-sensitive cells because the blind spot is where blood vessels and the optic nerve join the eyeball.

9 Blinking moistens the eye, stimulates the tear ducts to produce tears and removes foreign particles.

Applying

10 **a** humans, monkeys, dogs, cats, owls

b rabbits, mice, fish, chickens

11 long-sightedness (hyperopia)

12 presbyopia

Analysing

13 Predators: monkeys, dogs, cats, owls.

Prey: rabbits, mice, fish, chickens.

Humans (apes) can be predator or prey but have developed binocular vision to help jump between trees etc. (in the case of apes).

14 Both the eye and a camera have a series of lenses that focuses light from an image onto a surface. In the case of the eye, the surface is the retina, whereas for a camera it is film. However, eyes focus the image by changing the shape of the lens, whereas a camera changes the relative distance between the lenses.

15 Lasik surgery involves the cornea being lifted up in order for a laser to reshape the eye. It results in less discomfort and healing time. PRK involves completely removing a layer of cells from the surface of the cornea and remodelling its shape.

Evaluating

16 **a** In a bright room, your pupils are very small, so when you walk into a dark room, your eye does not receive enough light. It takes some time before your pupils adjust to the new reduced light levels.

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b When you are suddenly exposed to bright light your pupils need time to contract. You squint to reduce the amount of light initially going in to your eye.

c The choroid is black to absorb light and stop it being reflected around your eye. Without it, the inside of your eyeball would appear bright and wash out the images projected onto your retina.

d Flies have many eyes that allow them to see in all directions at the same time so they can always see you coming.

17 Not very good. An owl is mostly nocturnal; therefore, there is very little need for an owl to see colour. It is much more important that they can detect great detail in dim lighting.

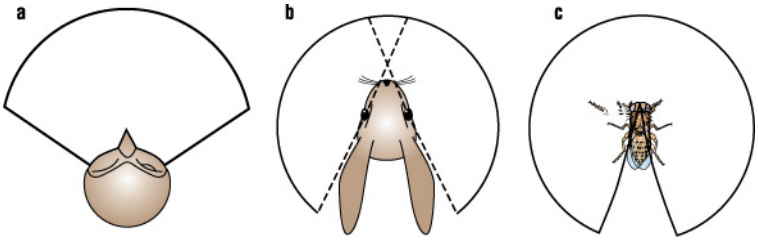
18 Humans, like all apes, need two forward-facing eyes to judge distances. This is very important when driving a car, but our ancestors would have needed two eyes to jump between trees etc.

19 Insects and animals need to distinguish colours in order to determine which flowers and plants are good to eat, to find a mate (e.g. the bower bird) and to determine which animals might be dangerous. Certain animals display colours to show that they are poisonous and shouldn't be eaten. These colours are usually red, black and yellow. Other animals mimic these colours so they won't be eaten themselves.

Creating

20 Creative and practical task.

21



4.1 Practical activities

Prac 1: Eye tests

Common mistakes

Measurement of the blind spot can be easily missed if the object of focus is moved too quickly.

Possible results

Students work through a series of tests to investigate their sense of sight, blind spot and discover whether they have a dominant eye. All results will vary.

Suggested answers

1 *Students' measurements.*

2 *Students' measurements*

3 *We would not be able to judge distance as well if we had only one eye.*

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Prac 2: Persistence of vision

Common mistakes

A brighter drawing will assist the brain in retaining the image longer.

Possible results

The bird appears to be in the cage. When you see the image of the bird, your brain holds onto the image for a short time even when the image appears and disappears quickly. The same thing happens with the image of the cage. The two images overlap in your brain so the bird appears to be in the cage.

Suggested answers

1 *Persistence of vision refers to the length of time the retina retains an image.*

When you look at an image, your brain remembers it for a fraction of a second after it has disappeared. This happens because our eyes cannot keep up with the speed of change between each image.

2 *Slow rotation: the bird is out of the cage. Fast rotation: the bird appears to be inside the cage.*

3 *When the human eye sees more than 10 images within a second, our brain assembles them together as a series of moving images. Movies use the same technology, except they show 24 frames per second, so the effect is much smoother and more natural.*

Prac 3: Eye dissection

Common mistakes

Students may cut too deep and accidentally cut through the lens.

Possible results

Through dissecting a cow's eye, students gain an understanding of the structure and function of the human eye.

Suggested answers

1 *Answers will vary.*

2 *upside-down*

3 *The lens is flexible because it needs to change shape according to the distance of vision.*

Prac 4: Lens and retina

Common mistakes

A well-lit room will assist this exercise.

Possible results

The paper behaves as the retina, and the convex lens behaves as the lens of the eye. The convex nature of the lens of the human eye turns an image upside-down onto the retina.

Suggested answers

1 *upside-down*

2 **a** *lens: convex lens*

b *retina: sheet of paper*

3 *The thicker the lens, the clearer and further away the image appeared to be.*

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4.2 Answers

Remembering

1 Sound travels at ~340 m/s and is transmitted by vibrating waves of air particles

2 **a** inner ear

b ossicles (hammer, anvil and stirrup)

c cochlea

3 middle ear

4 decibel

5 100–110 dB

Understanding

6 Cotton buds are small enough to reach and damage the eardrum.

7 **a** protects dust and bacteria entering the ear

b may stop the eardrum from vibrating correctly, causing temporary deafness.

8 Prolonged exposure to loud noise can cause the hairs in the cochlea to flatten or cause tinnitus. A blow to the head can cause the tympanic membrane to tear. Infection can cause a build-up of wax that can cause temporary deafness or a jamming of the ossicles.

9 You may experience temporary partial deafness and a ringing in the ears caused by damage to the hairs lining the cochlea.

Applying

10 The sound came from the right.

11 when working in situations with prolonged loud noise; e.g. ground crew for aircraft; when experiencing short but very loud sounds such as on a gun-firing range; protection from infection when swimming

12 **a** train, lawn mower

b jumbo jet taking off, thunder clap

c whisper

Analysing

13 A horizontal Eustachian tube makes it easier for bacteria to enter deeply into the ear canal where infection can easily take hold.

Evaluating

14 **a** The inner ear contains fluid in the semi-circular canals that help you determine whether you are standing up, lying down or at an angle. An infection affecting the inner ear can confuse this mechanism.

b Your body can be tilted in three directions—forwards and back, left and right, up and down, so you need three semi-circular canals to determine your position.

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c The act of swallowing can help open the Eustachian tube, allowing the pressure in the inner and outer ears to be adjusted.

d Large ears allow animals to collect more of the sound waves to aid hearing and give them pre-warning of approaching predators.

15 Initially from the intensity of the sound. Our ears are forward facing, so sounds in front will appear clearer and louder. However, we will turn our heads slightly to determine which direction the sound is coming from.

16 Caleb has a problem with how sound is collected and amplified; i.e. in the outer and middle ears. However, Sarah may have a problem with how the sound is processed; i.e. in the inner ear.

17 Yes, two ears allow prey to determine the direction of a predator attack, and two ears allow predators to pin-point prey.

Creating

18 Creative and practical task.

19 Creative and practical task.

4.2 Practical activities

Prac 1: Hearing tests 1

Common mistakes

Ensure that the blindfold is not too loose or too tight.

Possible results

Students discover the directional ability our ears have to detect sounds.

Suggested answers

1 *We can detect distance from the sounds we hear, because the loudness is less and the sound is often muffled the further away the sound.*

2 *The direction of sound changes.*

3 *To distinguish whether a sound is coming from the left or the right. This facilitates our sense of distance, direction and balance.*

Prac 2: Modelling hearing difficulties

Common mistakes

The length of the paragraphs selected should be long enough to contain a wide array of hissing sounds.

Possible results

Students discover how sounds and speech can be different for people with hearing difficulties.

Suggested answers

1 *Students' responses.*

2 *Speech is often learned by hearing it and repeating it. People with hearing difficulties do not have the ability to hear their own voices so cannot correct the pitch of their voice.*

3 *Men's voices have a lower frequency.*

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Prac 3: Measuring decibels

Common mistakes

Check that the batteries of the sound meters are charged before proceeding outside, or take spare ones with you.

Possible results

Using a sound meter, students measure various sounds in decibels and classify each sound as harmful, dangerous, loud, normal or quiet.

Suggested answers

1, 2 *Students' results.*

4.3 Answers

Remembering

1 **a** sweet, sour, salty, bitter, umami

b

Sense	Sense organ
Sight	Eye
Hearing	Ear
Smell	Nose
Taste	Tongue
Touch	Skin

2 **a** True

b True

c True

d False

3 **a** dead skin layer, epidermis, dermis, fatty layer

b pain receptor, light contact receptor, heat receptor, cold receptor, pressure receptor

c basal cell carcinoma, squamous cell carcinoma, melanoma

Understanding

4 Molecules from the toilet are dispersed into the air and taken into your nose where they dissolve, triggering the olfactory cells to send a message to the brain.

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5 If there is a gas leak, the gas can be smelt.

6 Hold your nose so you can't smell it.

7 The fatty layer allows energy to be stored in the form of fat and also provides thermal insulation.

8 Sweat requires energy to change from a liquid to a gas; i.e. evaporate. When sweat evaporates, it absorbs heat energy from your skin, making you feel cooler.

9 Wear sunscreen, stay out of the Sun during the middle of the day, wear a hat, wear long-sleeved clothing and, when outdoors, try and stay in the shade.

Applying

10 touch

11 sweet: sugar, sour: lemon, bitter: vegemite, salty: salt, umami: cheese

12 chef, cooking

13 burned: burning paper, foul: sewage, fragrant: flowers, fruity: fresh apples, resinous: paint stripper, spicy: cinnamon

14 to help chew, language

15 dermatitis

Analysing

16 Papillae and taste buds are different. Papillae are raised protrusions on the tongue. Taste buds sit on top and in between the papillae and are the taste receptors.

Evaluating

17 Although both kissing and stubbing a toe rely on the sense of touch, stubbing a toe sends a very strong message to the pain receptor. However, kissing sends a very slight message via the pain receptors and a stronger message from the pressure receptors, which the brain interprets as pleasurable.

18 The high skin cancer rate in Australia is both cultural and geographical. As a nation, Australia receives a large amount of sunlight during the year. Also, Australia is very close to the hole in the ozone layer, meaning that Australia is more exposed to UV radiation from the Sun than other countries. However, Australia also has a 'sun-worshipping' culture where being tanned is considered beautiful and healthy. Australians tend to spend a lot of time in the sun, at the beach or playing sports etc.

Creating

19 Creative task.

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4.3 Practical activities

Prac 1: Taste regions

Common mistakes

Ensure that the blindfold is not too loose or too tight.

Possible results

Students determine whether some regions of the tongue are more receptive to certain tastes. Results should be comparably similar to the results in Figure 4.3.5.

Suggested answers

1 *A: sour, B: bitter, C: sour, D: sweet, E: sweet and salty.*

2 *Students' results.*

3 *Students' results.*

4 *Students' results.*

Prac 2: Taste trickery

Common mistakes

The experiments designed by the students should be assessed by the teacher before carrying them out.

Possible results

Students' design.

Suggested answers

Students' design.

Prac 3: Skin receptors

Common mistakes

Ensure that the blindfold is not too loose or too tight.

Possible results

The palm of the hand, back of the neck and fingertips are more sensitive to touch because the two toothpicks could be felt on the skin, concluding that there are more touch receptors in these areas.

Suggested answers

1 *Students' results.*

2 *Students' results.*

3 *There are more touch receptors in the areas of the skin where sensitivity is important; e.g. lips to detect temperature before we swallow; fingertips to enable fine motor skills.*

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4.4 Answers

Remembering

1 Foot is lifted up rapidly, muscles in the leg become tense, might vocalise (ouch), breathing might be held or become shallower.

2 The most usual response is to move to a more favourable position.

3 **a** 37°C

b 7.38

4

Receptor	Stimulus
a Cells of the retina	**iv** light
b Cells of the inner ear	**v** sound
c Taste buds	**ii** chemicals
d Osmoreceptors in the brain	**vi** water levels
e Semicircular canals in the ears	**i** gravity
f Thermoreceptors in the skin	**iii** heat

Understanding

5 **a** Stimulus: anything that triggers a change in the way an organism behaves.

b Response: how an organism changes after receiving certain stimuli.

c Homeostasis: maintenance of a constant internal environment of an organism.

d Feedback: information given back to the organism after a stimulus and response in order to allow the organism to make an adjustment.

e Coordination: the organisation of a chain of responses set off by stimuli.

6 A driver affected by alcohol is not able to respond to stimuli as quickly. Thus, in the event of an emergency, where every millisecond counts, the driver may not be able to react in time.

7 Answers will vary depending on personal experience.

Applying

8 water, glucose and hydrogen ions; i.e. pH

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9

Stimulus	Response
A very cold wind	Shivering
Bad smell from a drain	Emily pinches her nose
A bus sounds its horn	John covers his ears
Temperature rises	Reuben takes off his shirt
The smell of sausages on the BBQ	Helen salivates
Susan yawns	Patrick yawns
A salty meal	Kristy drinks more water
Scary scene in a movie	Lan screams

10 **a** barks back, cowers, runs away

b more fierce barking and growling, stands his ground, relaxes

c engage in a fight, the second dog accepts the first dog's authority/dominance, second dog never returns to the area

11 **a** main arteries

b brain

c lungs and heart

d breathing rate would increase and heart beats faster

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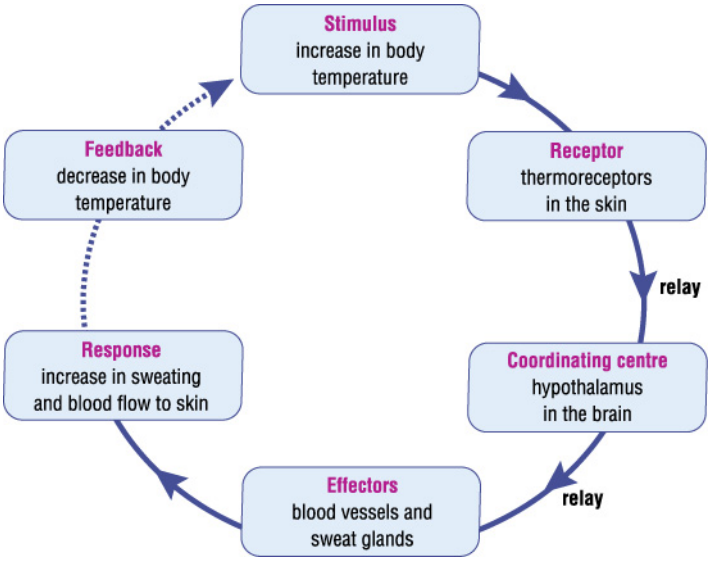
12



13 **a** Feeling cold, shivers and puts on a coat. She may also have rubbed her hands together or jumped on the spot to get warm, moved to a warmer location.

b She is now feeling hot, so she starts sweating.

c Removes the coat and puts on lighter clothes that will not cause her to get so hot.

Evaluating

14 The child might start screaming, which the dog may interpret as aggressive behaviour and so continues to bark and growl. The child may show fear and retreat and the dog may be happy that its territory is no longer threatened, so it stops barking.

15 Various creative responses possible.

a Student is holding a pen in the flame of a Bunsen burner.

b shouts at the student, bangs a ruler on the desk, turns off the gas to the Bunsen burner

c surprise, shock, embarrassment

d puts the pen down

e continues to burn the pen, shouts back, leaves the classroom

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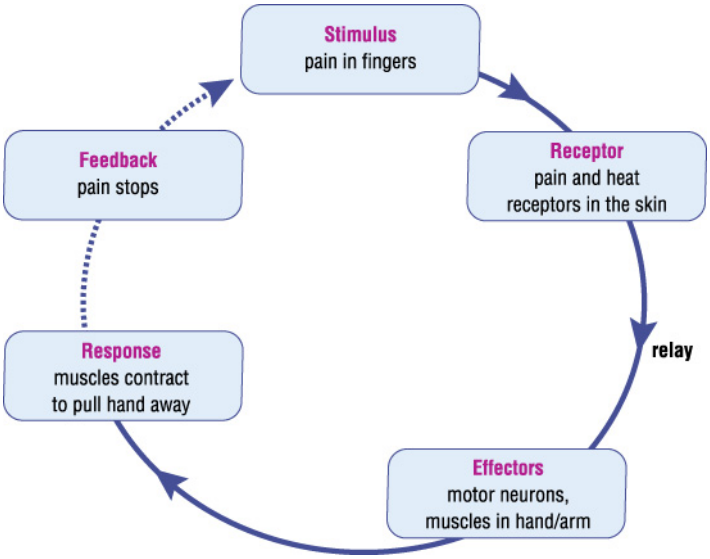
Creating

16



4.4 Practical activities

Prac 1: Sweet and salty

Common mistakes

Students rinsing their mouth thoroughly between sweet and salty samples will produce better results.

Possible results

Results will vary depending on the class.

Suggested answers

1 *Various answers. The minimum concentration at which sweetness was tasted will vary for each student.*

2 *Various answers. The minimum concentration at which saltiness was tasted will vary for each student.*

3 *Results will vary depending on the class.*

4 *Thresholds are important for stimuli that are able to cause harm.*

5 ***a*** *Expectations may alter senses and the interpretation of them.*

b *Blindfolds or 'blind testing' (where students do not know what they are tasting) could be used to avoid this possibility.*

Prac 2: Sound threshold

Common mistakes

Any experiments designed by students should be assessed by the teacher before carrying them out.

Possible results

Results will vary depending on the class.

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Suggested answers

Students' results.

4.5 Answers

Remembering

1 **a** central nervous system and peripheral nervous system

b CNS: brain and spinal cord. Acts as a control centre, receiving, processing and sending messages.

PNS: sensory receptors and nerves. Continuously informs the CNS of changing conditions.

2 **a** skull, cerebrospinal fluid, meninges

b blow to the head, deprived of oxygen, deprived of blood (stroke)

c spinal cord may be broken by a fall, multiple sclerosis, spina bifida

3 **a** blink, squint

b cough

c salivate

d tan, sweat

4 eating, walking, talking

Understanding

5 Neurotransmitters carry chemical messages between synapses.

6 Synapses are like switches turning neural pathways on and off. Without synapses all the neurons would be connected all the time.

7 A wrinkled cerebrum increases the surface area of the brain, allowing more neurons to be crammed into a smaller volume. This allows humans to have a very sophisticated brain without the need for a huge head.

8 A bright light is shone in the eye.

Receptors detect a change in light intensity.

An impulse is sent along a motor neuron to iris muscles.

Iris muscles contract, causing the pupil to narrow.

An impulse is sent along a sensory neuron to the brain (simultaneously).

Applying

9 **a** motor neuron

b **i** cell body

ii nucleus

iii dendrite

iv myelin

v axon

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10 **a** cerebrum
b cerebellum
c medulla
d spinal cord

11 **a** cerebrum
b medulla
c cerebellum
d cerebrum

12 coughing, sneezing, blinking

13 left hemisphere

14 **a** light
b sound
c chemical
d heat

Analysing

15 **a** A neuron has all the main features of cells. It has a nucleus, cytoplasm, cell membrane.
b A neuron has additional specialised features; i.e. dendrites, axon. Neurons may be specialised; i.e. interneurons, motor neurons, sensory neurons.

16 A nerve is made up of many neurons bundled together.

17 Sensory neurons have specialised endings that respond to stimuli, such as heat, light, pressure etc. Motor neurons transmit information from the central nervous system to organs such as muscles to cause movement in response to stimuli. Interneurons allow messages to be sent within the central nervous system for thinking, sensing, creativity and memory.

4.5 Practical activities

Prac 1: Brain dissection

Common mistakes

Light cuts rather than deep cuts into the tissue of the brain will assist in obtaining a thin layer of the membrane.

Possible results

Through dissecting a lamb's brain, students gain an understanding of the structure and function of the human brain.

Suggested answers

1 *The brain is easier to dissect when partially frozen, as an unfrozen brain has a 'mushy' consistency that makes it difficult to cut cleanly.*

2 *The cerebrum is waxy and pale pink/grey, with a folded surface.*

3 *The cerebellum indicates the back of the brain. This allows easy identification of the left and right hemispheres of the brain.*

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4 *The cerebellum is a darker pink/red, has a firmer consistency and has finer (but shallower) folds than the cerebrum.*

5 *The medulla is a creamy-white tough, thick cord.*

Prac 2: A model brain

Common mistakes

Students may accidentally cut right through the orange, separating the two hemispheres. A demonstration on how to cut the orange beforehand may be helpful.

Possible results

Students construct a model of the human brain.

Suggested answers

1 *The orange peel represented the skull in this model.*

2 *The flesh of the orange represented the cerebrum.*

3

Function	Left hemisphere	Right hemisphere
Artistic and musical ability	no	yes
Intuition	no	yes
Perception	no	yes
Language	yes	no
Mathematics	yes	no
Logical thinking	yes	no

Prac 3: Brain wars: the Stroop effect

Common mistakes

Time measurements may vary depending on the reflexes of the time keeper.

Possible results

Students are presented with the names of different colours, written in coloured inks, and then asked to either read the word or name the colour of the print. Students will find that word reading is quicker than colour naming, the interference occurs because words are read faster than colours are named.

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Suggested answers

1 *The left hemisphere of the brain is associated with reading and speaking language.*

2 *The right hemisphere is most associated with identifying colours.*

3 *Two contradictory pieces of information (colours and names of colours) are coming into the different hemispheres of the brain. The brain must then rapidly communicate between both hemispheres to solve the problem.*

4 *The left hemisphere is reading the words. The right side is detecting the actual colours. Both sides are in contradiction.*

5 *If the names of the colours were written in another language, it is unlikely that you would experience the same problem. The left hemisphere would not be 'reading' the words (as they are a nonsensical jumble of letters). Hence, only the right hemisphere would be providing information about what is seen (i.e. the actual colours).*

Prac 4: Response time

Common mistakes

Any experiments designed by students should be assessed by the teacher before carrying them out.

Suggested answers

Students' results.

Science Focus investigations

Prac 5: What's that smell?

Common mistakes

Make sure that the lids of the containers are tight fitting, ensuring smells are not weakened by loss of vapour.

Possible results

Students will identify a variety of common smells from memory.

Suggested answers

1 *Various scores will be possible.*

2 *Different people will detect different parts of the scent and will relate them to different sources. Discussion will allow a pooling of sensations and memories.*

3 *Wine experts discuss the taste and smell of a wine to pool their sensations and memories.*

4 *If the vapours are allowed to escape, then the sample will become progressively 'weaker' and harder to smell.*

5 *Various answers are possible. Perception is likely to change once its source is known.*

6 *Various answers are possible.*

Prac 6: Chunking

Common mistakes

Instruct students not to look at the cards before the activity begins. Thick, non-see-through paper will assist this exercise.

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Possible results

Students test whether memory can be improved by chunking information.

Suggested answers

1 *Various answers are possible. The easiest to remember will be the ones that most resemble real words.*

2 *Some things are easier to remember than others because they bear a memorable pattern (e.g. they form actual words, letters are in alphabetical order etc.) or form 'natural' chunks of easy-to-remember information.*

3 *Various answers are possible. Some factors that might be given are the form of the chunks, the number of times this sort of task has been attempted in the past, distractions, size of print etc.*

4 *No answer required.*

4.6 Answers

Remembering

1 Not all responses need to be immediate. Hormones provide a longer-lasting response, and hormones do not need to be supplied continuously.

2

Gland	Hormone
a Adrenal	**ii** Adrenalin
b Pancreas	**iv** Insulin
c Pituitary	**v** ADH
d Thyroid	**vi** Thyroxin
e Ovaries	**i** Oestrogen
f Testes	**v** Testosterone

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Understanding

3 **a** Hormones are chemicals produced by the body to stimulate a change in the body's biochemistry at the cellular level.

b Hormones are produced by specialised glands located in different areas of the body; e.g. brain, pancreas, reproductive organs.

c Hormones are released into the bloodstream and carried to target cells by the circulatory system.

4 Like a piece of a jigsaw puzzle, hormones have a specific shape and will only react with receptors of the correct shape.

5 The pituitary gland because it links the nervous and endocrine systems. Like the conductor of an orchestra, it stimulates other glands to release hormones.

6 Increased heart rate: Causes blood to be pumped around the body quicker so that more oxygen is sent to the muscles and brain for increased physical and mental alertness.

Dilation of bronchioles: Allows large volumes of air to be taken into the lungs to provide more oxygen for the circulatory system.

Glucose release from the liver: Glucose provides a very quick energy source for the muscles.

Increased breathing rate: Increases oxygen to the circulatory system to increase respiration and so release energy.

7 In emergency or panic situations (e.g. predator attack), the nervous system provides a very quick and instinctual response to the danger while the endocrine system provides a longer-term response to fight the attacker or flee the danger.

8 Thyroid-stimulating hormone (TSH): causes the thyroid to release thyroxin.

Thyroxin: regulates the speed of cell reactions.

Human growth hormone (HGH): causes all tissues and organs in the body to grow.

Testosterone and oestrogen: simulate growth of the reproductive organs and maturation of the body; i.e. puberty.

9 No, by the time the person is an adult the changes will be permanent. Giving thyroxin only prevents the changes, it cannot reverse them.

10 The urine and faeces of dogs contain hormones called pheromones that provide information about the dog which deposited them. This is a form of chemical communication between dogs and can be used to mark territory or indicate when a dog is ready to mate.

Applying

11 **a** M

b O

c L

d K

e N

f P

12 Type I: 15%, Type II: 85%

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13 Ants use pheromones to mark food trails, dogs use pheromones to mark territory, many animals use pheromones to attract a mate.

Analysing

14 A hormone is a chemical that is released inside the body to stimulate a change in an organism's own biochemistry. A pheromone is released outside the body into the environment to stimulate a response from other organisms.

15

	Nervous system	**Endocrine system**
Nature of message	Stimuli are normally detected by small receptors and the response sent to the brain via electrical impulses.	Stimuli are normally detected by receptors in large organs and the response is sent to the brain by hormonal relay.
Distribution of message	The electrical impulses are sent along neurons (nerve cells).	The hormones are released into the bloodstream and carried to the relevant organs by the circulatory system.
Speed of delivery	Almost instantaneous.	Can take several minutes or even days to create the desired change.
Length of response	Normally only as long as the message is being sent.	Changes may exist for minutes, days or even a lifetime.

Evaluating

16 Pheromones are often used to stimulate a pleasurable, sexual response. Manufacturers of perfumes and aftershaves would like to simulate this response with their products to make their customers appear and feel more attractive to others. It would also promote higher product sales.

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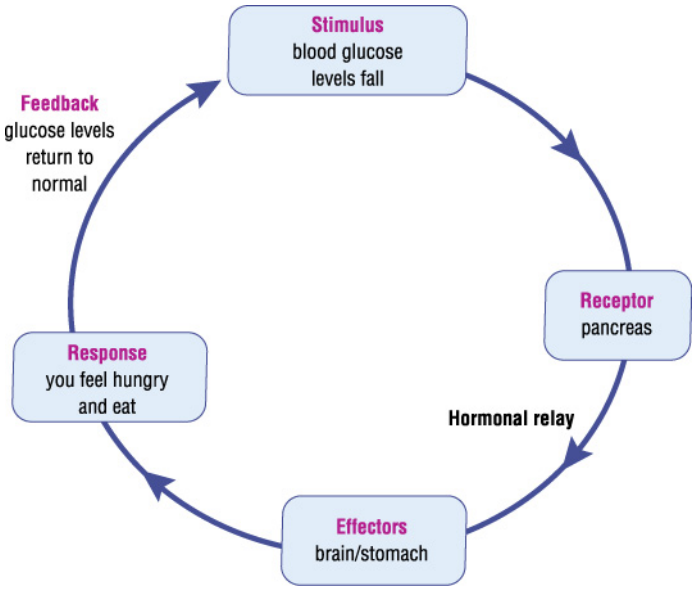
Creating

17



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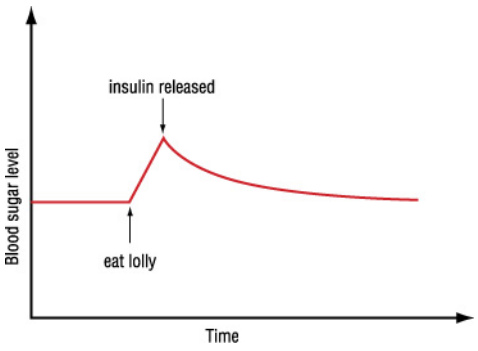

4.6 Practical activities

Prac 1: Plant hormones

Common mistakes

Make sure that the test tubes are tightly sealed to avoid water escaping, causing the plant to dry out.

Possible results

The plants will bend towards the light.

Suggested answers

1 *Diagrammatic answer required.*

2 *This experiment tests the response of plants to the stimulus of light. This tropism is called phototropism.*

3 *Placing the tubes both in the dark and in the light allowed a comparison to ensure that the only variable tested was light.*

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Prac 2: Plants and gravity

Students' design.

Chapter answers

Remembering

1
- **a** sight, smell, hearing, taste, touch
- **b** outer, middle, inner
- **c** sweet, sour, salty, bitter, umami
- **d** brain, spinal cord
- **e** cerebrum, cerebellum, medulla
- **f** testosterone, oestrogen, thyroxin

2

Part	Function
a Cerebellum	**ii** Centre for sight, hearing and speech
b Medulla	**i** Controls involuntary actions such as breathing
c Meninges	**iv** Protect the brain from injury
d Cerebrum	**iii** Controls muscle movements while you are cycling

3
- **a** growth, puberty, fight or flight response
- **b** seed germination and growth, production of flowers etc.

4

Functions	Hormones
a Blood glucose levels	**iii** Insulin
b Female reproductive functions	**iv** Oestrogen
c The rate of chemical reactions in cells	**v** Thyroxin
d Water levels within the body	**i** ADH

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Functions	Hormones
e Readiness of the body for action	**vi** Adrenalin
f Deepening of the male voice at puberty	**ii** Testosterone

Understanding

5 Iris: regulates the amount of light entering the eye.

Lens: focuses light rays onto the retina.

Retina: senses light.

Choroid: provides oxygen and nourishment to the back of the eye. Also stops light from being reflected.

6 The pupil dilates in dim lighting. This allows more light to enter the eye, which helps vision in low light conditions.

7 Eardrum: converts sound waves in the air into vibrations of the ossicles.

Ossicles: transmit vibrations of the eardrum to the liquid-filled cochlea, which sends the signals to the brain.

Semicircular canals: sense when you are tilted and provide balance.

8 Molecules from the substance are dissolving in our nostrils and are sensed by receptors.

9 You could reduce the sensation of taking an unpleasant medicine by blocking your nose.

10 **a** Stimulus: something that causes a change or response; e.g. pricking your finger with a needle.

b An effector: something that creates a change; e.g. muscles contracting when you burn yourself.

c A receptor: special organelles designed to sense particular stimuli; e.g. pressure receptors in the skin.

d A response: a change that occurs as a result of a stimulus; e.g. a plant turning towards sunlight.

11 **a** synapses

b When the electrical signal travelling along a neuron reaches a synapse, neurotransmitters are released and quickly move across the gap, restarting the electrical impulse.

12 **a** A reflex action is a response that does not require any brain control.

b sneezing, coughing, blinking

c Reflex actions need to be fast because they are often your first line of defence; e.g. if you touch a hot iron. High speed is achieved by sending the message from the receptors straight to the motor neurons and bypassing the brain.

Applying

13 The right and left edges of the tongue detect sour taste.

14 **a** to respond to possible attack from a predator, to respond to changes in the environment

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b Often a response to a single stimulus requires a change to many body parts and functions.

c nervous and endocrine systems

15 **a, b** light: rods and cones in the retina on the back of the eyeball
flavour: taste buds on the tongue
heat: thermoreceptors in the skin
sound: eardrum in the middle ear, hairs of the cochlea

16 **a** cerebrum

b medulla

c spinal cord

d cerebellum

17 **a** homeostasis

b Blood sugar levels that are too low result in hypoglycaemia, which can result in unconsciousness. Sugar levels that are too high result in hyperglycaemia, which may lead to fatigue or coma.

c endocrine system

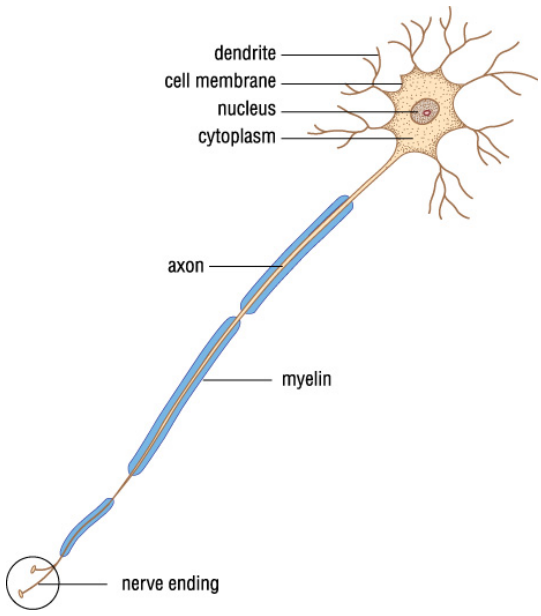
d diabetes

Evaluating

18 Our senses are all very important to our survival, although some more so than others. Sight and hearing are probably our most important senses. Touch is also very important to prevent people from hurting themselves accidentally. People are sometimes forced to live without one of their senses; for example, in the cases of blindness or deafness. Living without two or more senses makes life very challenging and would require significant care and supervision.

Creating

19



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5.1 Answers

Remembering

1 **a** sperm
b egg or ova (plural), ovum (singular)
2 **a** humans, mealworm beetles, elephants
b fish, amphibians
3 24 from each parent

Understanding

4 **a** Transfer of pollen grains, by wind, water or insects, from the anther of a flower to the stigma of a flower.
b The fusing/joining of sperm and egg in animals, and pollen and egg in plants.
c An organism that possesses both male and female reproductive organs.
d When a parent cell splits, a fault can occur in the DNA. If this happens, then the genes of the daughter cells will be slightly different to the parent cells.

5 During growth and repair, our cells form new cells when parent cells divide to form two identical daughter cells.

6 Bacteria reproduce by fission. Therefore, when food is not kept below 4°C, bacteria found on the food reproduce very quickly, causing the food to go 'off'.

7 The small size of spores allows a light breeze to disperse thousands of spores great distances. Rainwater can also disperse spores and transport them across the soil. Spores, like pollen, have the ability to stick to the fur or feathers of animals, which will carry them until they drop off.

8 Sexual reproduction produces offspring that share similarities with both their parents but are nonetheless different, leading to variation within a population. If such variation exists, the organism with the most desirable characteristics will have a better chance of survival in a changing environment.

9 Kangaroos have one offspring at a time because the mother kangaroo needs to nurse her young for up to one year and because the joey stays in the pouch for six to eight months.

10 The environment influences the expression of the genes of each tree.

11 Fish lay lots of eggs at one time to increase the chances of some offspring surviving to reach adulthood.

12 Overuse of insecticides will reduce the number of flowers pollinated. If these flowers don't reproduce, their numbers will reduce and eventually the plants could die out.

13 As birds, mammals or insects feed on the nectar, the pollen on the anthers will drop onto their bodies. As the bird, mammal or insect moves on to the next flower, it will introduce pollen from the previous flower, thereby pollinating it.

14 The different brightly coloured flowers attract birds and insects. The nectar is food reward. The birds and insects will rub up against the pollen and transfer it to the stigma of other flowers as they move from flower to flower, resulting in pollination.

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Applying

15 **a** iii **b** ii **c** iv **d** i

16 asexual (fragmentation)

17 **a** 4 **c** 4096

b 64 **d** $4.97\,323\,236 \times 10^{86}$

Analysing

18 **a** mitosis **b** meiosis **c** mitosis

19 **a**

Similarities (found in both animal and plant cells)	Differences (found only in plant cells)
Cytoplasm	Cell wall
Mitochondria	Chloroplast
Vacuoles	
Nucleus	
Chromosomes	
Cell membrane	

b Similarities: In fission and budding, a parent cell splits into two. Algae can reproduce by fission and spores. Fungi can reproduce by fission, spores and fragmentation.

Differences: Fragmentation is for regeneration and propagation, not 'reproduction'. In fission, a parent cell splits evenly into two identical daughter cells, whereas in budding, the parent cell splits into two uneven cells.

Evaluating

20 Regeneration of body parts happens only at certain sections or body regions, so if earthworms or mushrooms are cut in body regions where regeneration doesn't occur they will die.

21 37°C

22 Cells have specialised functions.

Creating

23 Student responses will vary.

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5.1 Practical activities

Prac 1: Asexual reproduction in plants

Common mistakes

Students may accidentally cut the onion horizontally. A demonstration beforehand on how to cut the onion may be helpful. A flattened stem may be identified as the base of the root.

Possible results

Non-flowering plants generally reproduce asexually. The potato and onion are examples of non-flowering plants.

Suggested answers

1 *The onion is reproducing without a male and a female and, hence, without sex. This is an example of asexual reproduction.*

2 *Although the 'eye' of a potato is a bud and should be able to grow by itself, the rest of the potato is a food supply for its growth. Hence, the eye is more likely to grow if it remains attached to the potato.*

Prac 2: Examination of spores

Common mistakes

Fern spore vessels are very hard to open, encourage students to be patient.

Possible results

Students examine the structure of spores using a stereomicroscope.

Suggested answers

1 *Various answers are possible. It is likely to be very difficult due to the resilient nature of spores. Many spores are usually in each spore vessel.*

2 *Fern spores should be easy to spread due to their small size, as they are light enough to be carried by even a gentle breeze.*

3 *Spores are transported by water, wind or other living organisms.*

Prac 3: Flower dissection

Possible results

Students dissect a flower to examine and identify the male (stamens) and female (stigmas) reproductive parts of the flower.

Suggested answers

1 *Various answers are possible. Students should at least be able to identify petals, stigma, stamen (anther and filament) and sepals, which are easy to identify.*

2 *A hermaphrodite contains both male and female parts.*

3 *The plants in Figure 5.1.13 should all be able to reproduce sexually, as all seem to contain both male (stamens) and female (stigmas etc.) parts in their flowers.*

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5.2 Answers

Remembering

1 **a** testis or testicle
b epididymis
c prostate gland, Cowper's gland

2 **a** ovary
b fallopian tubes
c uterus (womb)

3 fluid and sperm

4 when no more eggs are released from the ovaries

5 **a** testosterone
b oestrogen and progesterone

Understanding

6 Puberty is when the reproductive organs that have been present since birth become functional; that is, when males and females reach sexual maturity.

7 Sperm needs the right temperature to form and survive. For this reason, the testes are located outside the male body in a sac called the scrotum, at a temperature cooler than 37°C. The scrotum is able to stretch or wrinkle, allowing it to adjust in different temperatures to keep the sperm alive.

8 Every time an egg is released from the follicles on the surface of the ovary, it leaves a scar.

Applying

9 **a** prostate gland, Cowper's gland and seminal vesicle
b urethra
c epididymis
d testicle
e scrotum

10 **a** fallopian tube
b ovary
c uterus
d cervix
e vagina

11 Primary sexual characteristics are the anatomical parts of the body involved in sexual reproduction.

Female: cervix, fallopian tubes, ovaries, uterus, vagina.

Male: epididymis, penis, prostate, scrotum, seminal vesicles, testicles.

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Answers to *Science Focus 3 second edition* Student Book questions

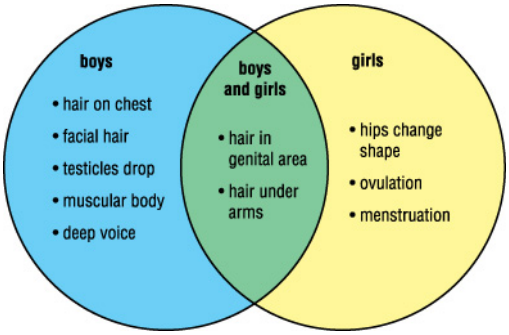
Analysing

12 Similarities: reproductive organs are present from birth but become fully developed and functional after puberty, hair appears in genital area and under arms.

Differences: men grow chest and facial hair, women's hips change and prepare for childbirth, women start menstruation.

Evaluating

13



14 **a** 125 cm

b 140 cm

c 12–14 years for girls and 13–15 years for boys

d Until puberty occurs, boys and girls have the same average height. Once puberty occurs in girls they are slightly taller, on average, than boys. However, once boys go through puberty, they are taller, on average, than girls.

15 **a** develops a more muscular body

b If sportspeople take testosterone they will have greater strength due to the extra musculature, which will give them an edge over those who do not take testosterone.

c extra muscle development, early baldness and/or extra body and facial hair

d Students' responses will vary.

16 **a** Students' responses will vary.

b Students' responses will vary.

Creating

17 The flow chart needs to list the following points in the order shown.

Millions of sperm form in seminiferous tubules.

Cilia line the tubule walls and beat back and forth, moving the sperm along to the epididymis.

Sperm matures and is stored in the epididymis.

On orgasm, the sperm leaves the epididymis and passes through the prostate, seminal vesicle and Cowper's gland.

As sperm passes through these structures more fluids are added, forming semen.

The semen travels to the penis via the sperm duct and urethra into the vagina during sexual intercourse.

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Sperm travels up the uterus and into the fallopian tubes to fertilise an egg.

18 Students' responses will vary.

5.2 Practical activities

Prac 1: Model reproductive systems

Common mistakes

If the jigsaw shapes are cut incorrectly, then the pieces may not interlock.

Possible results

Students construct a jigsaw or mobile of the male or female reproductive systems.

Suggested answer

1 *Answers will vary.*

5.3 Answers

Remembering

1 **a** several hundred million

b one

Understanding

2 **a** Copulation is sexual intercourse during which sperm comes into contact with the egg in the female.

b Gestation is the period during which an embryo develops into a full-term baby.

3 **a** iv

b iii

c i

d ii

4
- Muscular contractions pump sperm cells out of the penis and into the vagina.
- Once inside the vagina the sperm use their tails (flagella) to swim towards the fallopian tubes.

5 Once the sperm reaches the fallopian tubes it can live for three to five days. If ovulation happens two days after copulation, sperm still has the ability to surround and fertilise the egg.

6 The foetus is protected by a pool of amniotic fluid, surrounded by the amniotic membrane.

7 The foetus doesn't drown because it doesn't need to use lungs, as it gets its oxygen through the placenta.

8 At approximately 280 days after fertilisation the cervix dilates (relaxes) to let the baby through. The amniotic membrane splits and the amniotic fluid rushes out of the woman's vagina. This process is known as 'breaking of the waters'.

9 The crying of a baby indicates the lungs are clear of amniotic fluid.

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Applying

10 embryonic development

Analysing

11 Similarities:

- Identical twins and conjoined twins form when one egg is fertilised by one sperm and the egg then splits.
- Identical twins and conjoined twins are genetically identical.

Differences:

- Fraternal twins occur when two separate eggs are fertilised by two separate sperm.
- Identical twins and conjoined twins form when one egg is fertilised by one sperm, but the split of the egg in conjoined twins is incomplete.

12 sperm, egg, zygote, morula, blastocyst

13 blastocyst, embryo, foetus, baby

Evaluating

14 **a** 36.25 weeks

b 33 weeks

15 **a** It may affect the development of the external genitalia, ears, palate and central nervous system of the foetus.

b It may affect the development of the eyes, limbs (arms and legs) and heart of the embryo.

16 During birth, the uterus contracts strongly to help squeeze out the baby's head first. As the bones of the skull are not yet joined, this allows for easier movement in relation to each other, allowing the head to change shape and to fit the shape and size of the pelvic opening and the uterus during childbirth.

Creating

17 The flow chart should include the following points in the order shown.

- Sperm and egg fuse to form a zygote.
- Zygote divides to form a clump of 80 cells called the morula.
- Further division occurs to form a blastocyst.
- Blastocyst implants itself into the uterus.
- Cells start to differentiate into major body systems.
- Week 3: Central nervous system and heart start to form.
- Week 4: Eyes, ears and limbs start to develop.
- Week 6: Teeth and palate start to develop.
- Week 7: External genitalia start to develop.
- Week 9: Completion of palate and heart development.
- Week 16: Limbs, ears and teeth are fully developed.
- Week 40: CNS, eyes and external genitalia are completely developed.

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5.3 Practical activities

Prac 1: Survey of twins

Common mistakes

Ensure that the sample size is large enough to obtain significant results.

Possible results

Students construct a survey to determine the percentage of twins in a population.

Suggested answers

1 *Identical twins are formed when one fertilised egg (ovum) splits and develops into two embryos with exactly the same genetic information.*

Fraternal twins are formed when two eggs (ova) are fertilised by two sperm and produce two genetically unique embryos, just like brothers and sisters born at different times.

2 *Fraternal twins form when two separate eggs are fertilised by two separate sperm, each of which can produce a male or a female offspring.*

Identical twins form when one egg is fertilised and divides into two while it is still a tiny collection of cells. The two halves then develop into two identical embryos with exactly the same genetic information, and so are of the same sex.

5.4 Answers

Remembering

1 **a** intra-uterine device

b sexually transmitted infections

c in-vitro fertilisation

2

Contraception	Used by	Advantages	Disadvantages
Condom	Males	Few to no side-effects and protects against STIs	Reduces sensation and spontaneity
The Pill	Females	Easy to use, has ability to protect against cervical cancer and makes periods lighter	No protection against STIs and some pills can't be used by smokers or people with circulatory problems

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Contraception	Used by	Advantages	Disadvantages
Cap and diaphragm	Females	Few side-effects	Reduced spontaneity, no protection against STIs and increased risk of bladder infections
IUD	Females	Once inserted, no further maintenance required	Can lead to infections and heavier periods and no protection against STIs

3 condoms

4 Curable STIs: gonorrhoea, chlamydia and pubic lice.
Non-curable STIs: final stages of syphilis, herpes and HIV/AIDS.

5 testicular and breast

6 Legal drugs: alcohol, cigarettes, cortisone, antibiotics.
Illegal drugs: LSD, hallucinogens, heroin, thalidomide.

Understanding

7 **a** asymptomatic: when there is an infection without visible symptoms
b chancre: an ulcer-like structure formed in the early stages of syphilis
c cyst: non-cancerous growth that can be found in the reproductive systems of men and women

8 Sexual contact is the touching of any genitalia, internally or externally.

9 Students' responses will vary.
Sample answer: Syphilis (the pox). The pathogen enters through any break in the skin and can be transmitted through oral, genital and anal sex. If syphilis is detected in the early stages it can be treated with antibiotics; however, there is no treatment once individuals enter the final stages.

10 Women, not men, suffer from cervical cancer, so to reduce the number of cases diagnosed each year, the vaccine is now being given to all girls 12 years and older. Rubella can cause serious birth deformities if contracted during a woman's first trimester of pregnancy.

11 During the first eight weeks, the body systems of the embryo first develop.

12 In the first three months of pregnancy, folate provides protection against chromosomal abnormalities in an unborn child.

13 **a** In IVF, a woman's ovaries are stimulated to release multiple eggs. These eggs are fertilised by sperm and often more than one of the fertilised eggs are implanted into the uterus, increasing the chances of multiple pregnancies.
b Women's bodies are not built to carry more than one child at a time.

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Applying

14 Restricted blood vessels reduce the supply of nutrients and oxygen required by the embryo for development.

15 infection and radiation

Analysing

16 Reproductive technologies such as IVF are now being used more frequently than in the past for couples and individuals to have their own child rather than adopt.

17 Men are more likely to notice symptoms of sexually transmitted infections because their reproductive organs are external, unlike women who require an internal examination to detect symptoms of many STIs.

Evaluating

18 pubic lice, chlamydia, gonorrhoea, herpes, syphilis, HIV/AIDS

19 Although people are using contraception to prevent pregnancy, they are not using condoms as contraception, which prevent the spread of STIs.

20 In the first trimester, a woman is still getting used to the changes her body is experiencing; for example, altered blood volume, which affects blood pressure. If a woman exercises heavily in the first trimester she could become short of breath, limiting oxygen supply to the developing embryo.

21 Students' responses will vary.

22 Regular check ups will help treat STIs, reduce the spread of STIs and allow for early detection of cancers linked to the reproductive system.

23 **a** Students' responses will vary.

b It is unknown what harm a few drinks over a short period of time, particularly early in a pregnancy, can do to a developing embryo. Therefore, the experts caution to avoid alcohol altogether during pregnancy.

c It is not realistic to assume that a pregnant woman will not drink during her pregnancy because there may be external factors, such as family and social life, which play a part in determining the amount of alcohol women are exposed to during their pregnancy.

d not drinking any alcohol during their partner's pregnancy

Creating

24 The flow chart should include the following:

- woman's ovaries stimulated to release multiple eggs
- eggs retrieved from ovary
- eggs placed into a salt solution at body temperature
- fresh sperm fertilise eggs
- fertilised eggs incubated overnight
- fertilised eggs develop in the laboratory for several days
- one or more embryos are transferred to the uterus.

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25 Students' responses will vary.

5.4 Practical activity

Prac 1: Methods of contraception

Common mistakes

Accuracy in the collection and recording of data.

Possible results

Students inspect and rank different forms of contraception.

Suggested answer

1 *Students' results.*

Chapter answers

Remembering

1 asexual

2

Asexual reproduction	Example
Fission	Bacteria
Spores	Ferns
Budding	Yeast
Fragmentation	Starfish

3 muscular development, deeper voice and hair growth on body and face

4 fallopian tubes

5 280 days = 9 months

6 **a** uterus wall
b amniotic fluid
c cervical canal
d umbilical cord
e foetus

7 take drugs, smoke or drink alcohol

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Answers to *Science Focus 3 second edition* Student Book questions

Understanding

8 gametes, fertilisation, implantation, zygote, embryo, foetus

9

Term	Definition
a Ova	**viii** The female sex cell
b Ovaries	**xiv** Female sex organ that produces ova
c Ovulation	**vi** The process of releasing an ovum once a month
d Oviduct	**xi** Tube that 'catches' the ovum
e Uterus	**vii** Area where embryo implants and grows
f Cervix	**i** Ring of muscle separating uterus and vagina
g Vagina	**x** The passageway for menstrual flow and birth of a baby. Sperm is also deposited here.
h Sperm	**iii** The male sex cell
i Testes	**ii** Male sex organs that make sperm cells
j Scrotum	**xiii** Pocket of skin that holds testes
k Epididymis	**iv** Area where sperm cells are stored
l Seminal fluid	**ix** Nourishes and activates sperm cells
m Semen	**v** Consists of seminal fluid and sperm
n Urethra	**xii** Passes through the penis. Urine and sperm pass through this tube.

10 In humans, the developing baby is protected and has a good chance of surviving. When the embryo develops on the outside, as in fish, there is a greater chance of not surviving until the eggs hatch because they are exposed to predation and the elements.

11 Sperm production is ensured by keeping the testes outside the body at temperatures cooler than normal body temperature.

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12 Avoid coming into contact with common cold sores and use condoms if sores are present to avoid spreading the virus.

Applying

13 Make observations, apply logic and scientific knowledge to solve problems and draw conclusions from information presented.

14 obstetrician, gynaecologist

Analysing

15 Asexual reproduction does not require sex; therefore, only one parent is required. In asexual reproduction, new cells (daughter cells) form from old cells (parent cells) and are identical to the parent cell.

Sexual reproduction requires a male and a female sex gamete to fuse to form a new cell called a zygote.

16 Similarities: Parent cells have a complete set of chromosomes that double before each process starts.

Differences: Meiosis produces cells with half the number of chromosomes compared to mitosis. In mitosis, the daughter cells have the same number of chromosomes as the parent cell; i.e. the daughter cells are identical to the parent cell.

In meiosis, the procedure involves two cell division stages, whereas in mitosis there is only one cell division stage.

17

Flower part	Human part	Comparable function
Anther	Testes	Production of male sex cells
Ovary	Ovary	Produces and releases female sex cells
Pollen	Sperm	Male sex cell
Ovule	Ovum	Female sex cell
Stigma	Vagina	Male sex cells deposited here

Evaluating

18 Occasionally, an error can occur when parent cells split. If this happens, the gene of the daughter cell will be slightly different from that of the parent cell, causing the offspring to exhibit a new characteristic, which can then be passed on to new generations. Sometimes, this new characteristic is favourable, leading to improvements in a species.

19 fertilising and implanting fewer eggs

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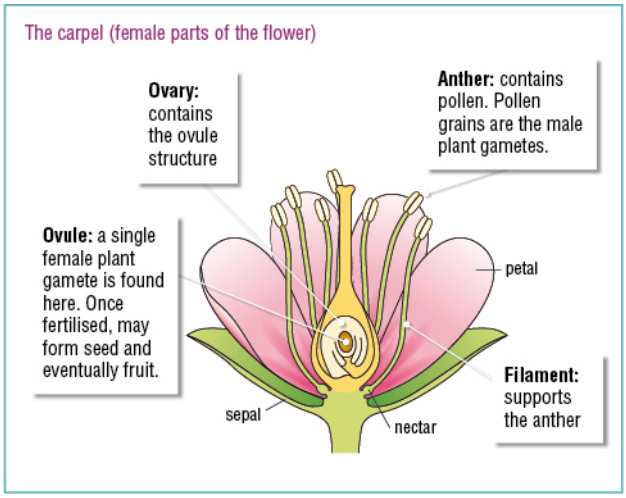
20



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6.1 Answers

Remembering

1 joules (J)

2 carbon dioxide (CO_2), water (H_2O), sunlight

3 carbon dioxide + water + sunlight → glucose + oxygen gas

$6CO_2 + 6H_2O$ + sunlight → $C_6H_{12}O_6 + 6O_2$

4 in volcanic vents on the sea floor

5 dogs, cats (any mammal or bird)

6 **a** fewer carnivores, more herbivores

b more plants, fewer carnivores

Understanding

7 **a** A specific area in which different organisms live and depend on one another (interact).

b Plants are producers because they make their own food by converting light energy into chemical energy.

c Organisms that eat other organisms.

d An organism that eats only plants.

e An organism that eats only meat.

f An organism that eats both meat and plants.

8 The arrows point towards the organism ingesting the food source. The arrows show the flow of energy.

9 Energy is lost through production of waste products, life processes, movement, heat and sound.

10 **a** Energy is neither created nor destroyed but can only be changed from one form to another.

b In chemical reactions, atoms are never destroyed but are recycled.

11 Yes, because they use the process of photosynthesis to make their own food as they convert light energy into chemical energy.

12 If the kookaburra can't find one food source or if one species of food source dies out, the kookaburra will still have a food source from which to obtain energy for life processes.

Applying

13 **a** plants (producers)

b animals (consumers)

14 cricket: because it directly eats the plant (producer)

15 **a** algae

b water snail

c large fish and kingfisher

16 **a** nuclear energy → light energy and heat energy

b light energy → chemical energy and heat energy

c chemical energy → chemical energy and heat energy

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17 Line A represents the kookaburra because it is an endothermic organism, which means it maintains a constant internal body temperature throughout the day and night; i.e. it regulates its internal temperature from the inside, using the energy it obtains from the food it eats. Line B represents the snake because it is an ectothermic organism, which means it uses the external environment to regulate its temperature; i.e. at 6 a.m. the snake's body temperature is lower than at 6 p.m. because at 6 p.m. the snake's body has been warmed all day by the Sun.

Analysing

18 **a** First order consumers eat producers; these organisms are known as herbivores.

Second order consumers eat herbivores; these organisms can be either carnivores or omnivores.

b Endothermic organisms regulate their internal temperature from the inside using the energy they obtain from the food they eat. Ectothermic organisms use the external environment to regulate their temperature.

19 Energy flows in one direction in an ecosystem. Energy flows from the Sun to producers and then to consumers. Matter is also transferred through the food chain. Not all the energy is transferred to the next level by organisms, as some is used at each level, as in the case of matter. Most of the material (matter) is used by the organism; however, some is returned to the environment as waste. Matter is never lost; it simply changes its form so that the amount of matter in an ecosystem will always be the same.

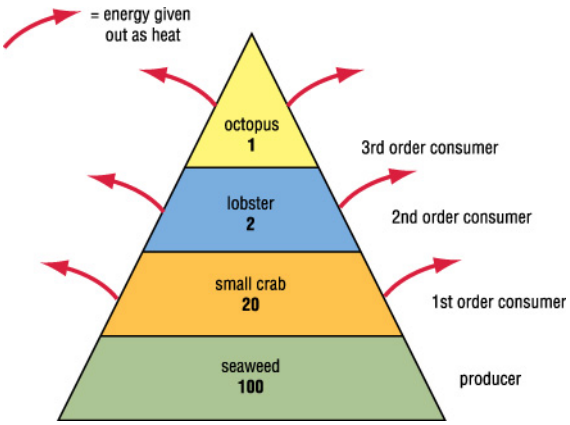
Evaluating

20 **a** To sustain the herbivores there needs to be more plants than herbivores. If there is no light for one week there will be little or no effect on the herbivores because plants store energy to sustain life processes for a short period.

b To sustain the herbivores there needs to be more plants than herbivores. If there is no light for one year the herbivores will reduce in number because there will not be enough plants to meet the dietary and energy needs of the herbivores in an ecosystem.

Creating

21



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22 Students' food chains will vary. An example is shown below.
algae → prawn → large fish → human
a consumer
b yes
c shark

23 Students' responses will vary. An example is shown below.
grass → grasshopper → sparrow → eagle

24 Students' responses will vary.

25 Students' responses will vary.

6.1 Practical activities

Prac 1: Biomass

Common mistakes

Ensure that the plant sample is completely dry before reweighing.

Possible results

Biomass is determined by removing the water from the plant material, then measuring the mass.

Suggested answers

1 *Green plant matter is fresh and green in colour, whereas dry plant matter is wrinkled and brown-green or brown.*

2 *Need to ensure that water is completely removed from the plant matter.*

3 *All organisms at each level of the pyramid are made up of roughly the same organic molecules in similar proportions; a measure of their dry weight is a rough measure of the energy they contain.*

Prac 2: Food pyramid

Common mistakes

Arithmetic error.

Possible results

(top of pyramid) kookaburra → grasshopper → grass (bottom of pyramid)

Suggested answers

1 *because we are investigating energy flow through trophic levels for a particular area*

2 *biomass of each level*

3 *grass → grasshopper → kookaburra*

4 *At each trophic level along a food chain, there is always a 'loss' of matter and energy in the forms of waste (e.g. carbon dioxide, faeces), heat and kinetic energy. Only 10% of the energy (chemicals used for growth) is passed on to the next level.*

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6.2 Answers

Remembering

1 **a** skin and muscles
b air and water

2 **a** C
b N
c O

3 **a** H_2O
b CO_2
c NH_3

4 glucose + oxygen → carbon dioxide + water + energy
$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy

5 moist surface, transpiration in plants, sweat, lakes/oceans

6 respiration, animal droppings, decomposition of dead matter

7 **a** colourless, odourless gas N_2
b nitrates
c urine: as urea $(NH_2)2CO$
faeces: as ammonia (NH_3)

Understanding

8 **a** land based
b water based
c cooling of water vapour to form liquid water
d condensed water vapour that falls as either rain or snow

9 Matter that was created during the Big Bang has been recycled many times because matter is not destroyed in biological reactions. It is rearranged. Hence, matter found in dinosaurs millions of years ago could be found within us today.

10 Condensation: cooling of water vapour in clouds to form rain and snow.
Precipitation: falling of rain and snow.
Evaporation: heating of water molecules from moist areas and surfaces by the Sun.

11 Most organisms cannot use nitrogen when it is in its atmospheric form so when lightning strikes, N_2 undergoes the process of nitrogen fixation. N_2 is converted to nitrates and is dissolved in raindrops.

12 There will be a build-up of dead animals, plants and wastes and less CO_2 will be returned to the atmosphere.

Applying

13 carbon

14 Carbon enters plants through the stomata in the leaves of plants, in the form of CO_2.

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15 **a** photosynthesis
b the Sun
c oxygen
d Oxygen is required by all living cells for the process of respiration.

16 Amino acids and nucleic acids. They are vital because amino acids are the building blocks of protein and nucleic acids are the building blocks of DNA.

Analysing

17 Abiotic: Water, soil, air, rock, cloud.
Biotic: Leaf, algae, bacteria, grass, human.

18 Nitrifying bacteria: convert ammonia into nitrates.
Denitrifying bacteria: convert nitrates into atmospheric nitrogen gas.
Bacteria decompose dead organisms and return nitrogen back into the soil as nitrates.

Evaluating

19 Organic because carbon is the basic element of organic compounds and the basic atoms within organic matter are carbon and hydrogen.

20 **a** True
b True
c True
d True

Creating

21 Students' responses will vary; however, they need to include the following:

- release of carbon into the atmosphere from burning fossil fuels
- removal of carbon dioxide from the atmosphere due to photosynthesis
- animals gain carbon by eating other organisms
- respiration from all organisms releases carbon dioxide into the atmosphere
- waste products formed when dead organisms decompose and release carbon dioxide back into the atmosphere
- carbon forms part of fossil fuels when organisms die.

22 Students' responses will vary.

6.2 Practical activities

Prac 1: Testing for water

Common mistakes

The watch-glass must be clean and dry to ensure the results are entirely due to the substance being tested.

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Possible results

Liquid tested	Copper(II) sulfate	Cobalt chloride paper
Tap water	Turns blue	Turns pink
Methylated spirits	No colour change	No colour change
Salt water	Turns blue	Turns pink
Sucrose solution	Turns blue	Turns pink
Acetic acid	Turns blue	Turns pink
Unknown X	Students' results	Students' results
Unknown Y	Students' results	Students' results
Unknown Z	Students' results	Students' results

Suggested answers

1 *Copper(II) sulfate dissolves and heat is produced (exothermic reaction).*

2 *all except the methylated spirits*

3 *Answer depends on the liquid being tested.*

4 *to remove any traces of water*

Prac 2: Measuring the boiling point of water

Common mistakes

Factors such as differences in the measured amounts of salt by tablespoon and the accuracy of each thermometer may result in varying temperature readings for the boiling point of water.

Possible results

Liquid	Boiling temperature
Beaker 1 (distilled water)	100°C

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Liquid	Boiling temperature
Beaker 2 (1 tablespoon salt)	101.5°C
Beaker 3 (2 tablespoons salt)	102.5°C
Beaker 4 (3 tablespoons salt)	103.5°C

Suggested answers

1 *Students' results.*

2 *The more salt that is added, the higher the boiling temperature.*

3 *As antifreeze and the water mixture don't freeze as readily as just water in the radiator, the antifreeze must be lowering the melting point of ice/freezing point of water. Similarly, the addition of antifreeze seems to increase the boiling point of water, so the water in the radiator is less likely to boil in hot weather.*

4 *Impurities in water increase its boiling point because more energy is required to separate the water molecules from the impurity*

6.3 Answers

Remembering

1 any three of sulfur dioxide, nitrogen dioxide, carbon monoxide, carbon dioxide, hydrogen sulfide, dust, smoke

2 any three of carbon dioxide, methane, nitric oxide, ozone

3 **a** HNO_3
b CO_2
c CH_4

4 **a** nitrogen dioxide
b carbon monoxide
c ozone

5 wild pig, rabbit, European carp, prickly pear, tamarisk tree

Understanding

6 **a** A pollutant is anything that makes the environment unhealthy for the organisms that live in it.
b Sewage is the waste water that goes into the pipes leading away from a house or factory. It is a pollutant that contains cleaning agents and human wastes.
c Desertification is the process of turning good land into desert.

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7 Frogs are usually the first organisms to be affected by pollutants because they breathe using their lungs and skin and spend a large amount of their lives in water. Thus, the fewer pollutants in water and air, the healthier and more populous the frogs in an ecosystem.

8 Blue-green algae thrive in water polluted by fertiliser run-off. They are dangerous because they can cause severe illness, as they are capable of producing several different toxins. Blue-green algae also grow uncontrollably until they clog up waterways.

9 **a** When trees are removed, the water table deep under the surface rises, bringing with it dissolved salts.

b Irrigation also raises the water table, bringing with it dissolved salts.

10 Overgrazing: allowing more animals to graze in an area that the area can sustain.

Deforestation: large-scale tree removal.

11 Greenhouse gases in the atmosphere reflect the Sun's heat back to the surface and keep the atmosphere warm enough to sustain life. The enhanced greenhouse effect is due to the increased industrialised gases in the atmosphere which trap more heat than usual, leading to global warming.

12 Pigs were introduced in 1788 as domestic animals. They became feral after escaping. They dug up large areas, damaging native plants and destroying the nesting grounds of birds and native mammals.

13 The hydrosphere, atmosphere and lithosphere in country areas may have fewer pollutants to damage the tree and cause it to die.

14
- Burn less fossil fuel (e.g. petrol) so that less carbon dioxide is released into the atmosphere.
- Cut down fewer trees so that more carbon dioxide is taken in by plants during photosynthesis.

Applying

15 8000–12 000 million

16 Melting of the glaciers in the polar regions, which has caused sea levels to rise.

17 cleaning agents, human waste, agricultural run-off, sediment pollution, inorganic chemicals

18 Electricity (controlled by power stations) is needed for the daily running of factories, to heat houses and to provide lighting. Electricity is generated by burning fossil fuels, which adds extra carbon dioxide to the Earth's atmosphere.

Analysing

19 **a** Endangered species are still found in their habitats or in zoos, albeit in low numbers, whereas extinct species have completely disappeared and are no longer in existence.

b Lithosphere is the land part (rocks, mountains and soil) of the Earth, whereas the hydrosphere consists of all the water components of the Earth (rivers, lakes, rain and oceans).

Evaluating

20 Cane toads, because they have no natural predators in Australia and the skin of the cane toad is highly poisonous to other animals that try to kill or eat it.

21 **a** Captive breeding program refers to animals in zoos or 'captive' human-operated facilities that breed animals to sustain the species.

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b Zoos and captive breeding programs prevent and/or slow down the rate at which endangered species become extinct.

c Students' responses will vary.

Creating

22 Students' responses will vary.

6.3 Practical activities

Prac 1: Water pollution

Possible results

Simulated waste water is taken through a series of filtration methods to produce clean water.

Suggested answers

1 *murky water*

2 *Step 1: large particles are separated*
Step 2: visible layers are created
Step 3: medium particles are separated
Step 4: minute particles are separated

3 *To show that the clean water is now ecologically sound and is able to sustain plant life.*

Prac 2: Climate in a beaker

Common mistakes

Avoid the cheap brands of aluminium foil for this prac, as they tend to tear very easily and will not support the weight of the ice blocks.

Possible results

A mini-climate is simulated inside a beaker, forming smog to represent a typical city climate.

Suggested answers

1 *Smog forms when a mixture of moisture and pollutants from cars rises into a cooler layer of the atmosphere, reacting there with sunlight. The air above rivers is likely to be more humid, containing more moisture than the air above solid land. The pollutants released in car exhausts in cities with rivers will therefore be more likely to form into smog than in cities without rivers.*

2 *In winter, the air is cooler high in the atmosphere, trapping the smog before it can rise and disperse.*

3 *Wetting the inside of the bowl provided moisture to combine with the smoke in the air to form smog.*

4 *The model does accurately show how smog is formed, causing pollution, but in a simplistic way, as each part of the model is only a crude representation of each aspect of the geography of a city.*

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Prac 3: Simulating global warming

Common mistakes

Ensure the hole in the juice carton is completely sealed around the thermometer to prevent heat loss.

Possible results

The juice container with the glass microscope slides represents the layer of greenhouse gases surrounding the Earth. They allow warmth in but not out, a simulation of global warming.

Suggested answers

1 *Popper A should increase in temperature more significantly than popper B.*

2 *Students will give various answers.*

3 *The glass microscope slides represent the layer of greenhouse gases surrounding the Earth. They allow warmth in but not out. Therefore, this model does show what happens on Earth, but in a simplistic way.*

Chapter answers

Remembering

1 direction of energy flow

2 predators

3 **a** carbon dioxide + water + sunlight → oxygen + glucose

b oxygen + glucose → carbon dioxide + water + energy

4 • Split the species colony into several groups and relocate each group.

• Monitor the population.

• Assist during birth to prevent unnecessary deaths.

• Release the offspring into suitable and protected habitats where introduced species have been removed.

Understanding

5 Organisms have specific needs which are met in localised areas.

6 10%

7 Energy cannot be created or destroyed; it can only be converted into different forms.

8 **a** The unit with which we measure energy is the joule.

b Endothermic animals use some of their energy to maintain a constant body temperature.

c Bacteria and fungi are decomposers that have a major role in returning atoms in the biotic environment back to the abiotic environment of the ecosystem.

9 When a first order consumer (herbivore) eats a plant, the herbivore obtains the energy produced by the plant through photosynthesis and respiration.

10 A pollutant is anything that makes the environment unfit or unhealthy for the organisms that live there.

11 Desertification is the process of turning good land into desert.

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12 If we don't maintain our environment, we will impact on the survival of species, their habitats (deforestation, rising water tables, salinity, desertification), the diet requirements of all organisms (alter food chains and webs) and the abiotic factors (water, light and air) that interact with the biotic factors.

13 Worms act as decomposers, whereby they return atoms in the biotic environment back to the abiotic environment, in this case, the soil.

Applying

14 Energy flows in one direction in an ecosystem. Matter is also transferred through the food chain. Not all of the energy is transferred by organisms to the next level, as some is used at each level, as is the case of matter. Most of the material (matter) is used by the organism; however, some is returned to the environment as waste. Matter is never lost to an ecosystem; the amount of matter in an ecosystem will always be the same.

15 nitrogen and phosphorus

16 deforestation and irrigation

17 Students' responses will vary.

Evaluating

18 Lightning is a necessary danger because most organisms can't use nitrogen when it is in its atmospheric form. During lightning strikes, N_2 undergoes the process of nitrogen fixation. N_2 is converted to nitrates and dissolved in raindrops so that it can be recycled and reused by the biotic components of the ecosystem.

19 When trees are removed the water table deep under the surface rises, bringing with it dissolved salts which increase salinity of the soil, resulting in loss of vegetation and, hence, erosion.

20 Although they are small, if bacteria did not exist there would be a build-up of dead animals, dead plants and wastes, and fewer atoms would be returned to the abiotic components of the ecosystem for use by all the elements of the ecosystem.

21 Having an omnivorous diet provides the bilby with an advantage in obtaining energy for survival because if the bilby is unable to find or rely on one food source due to extinction or unavailability, it can hunt or feed on another food source.

Creating

22 Generic example:
producer (plant) → herbivore (first order consumer) → carnivore (second order consumer) → omnivore/carnivore (third order consumer)

23 light energy → chemical and heat energy → chemical and heat energy
The Sun is the original energy source

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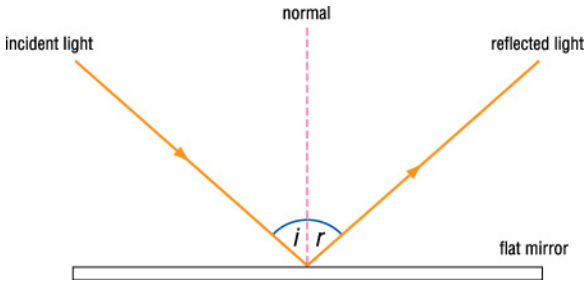
7.1 Answers

Remembering

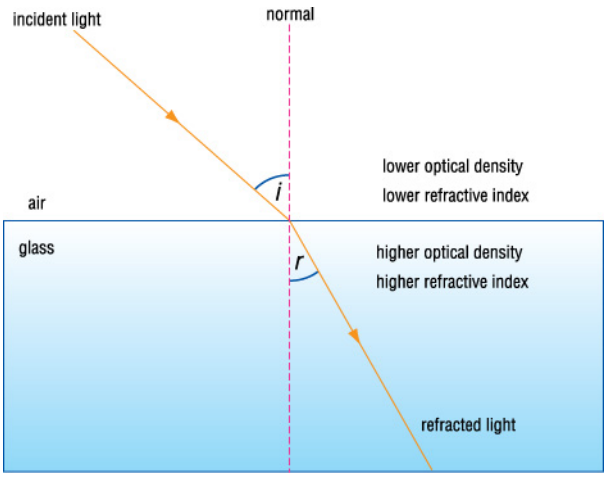
1 Law of reflection: Angle of incidence is equal to the angle of reflection, $i = r$.



2 **a** A transparent substance allows light to pass through it without absorbing the light.

b Possible answers include glass, water, Perspex, cling wrap.

3 Refraction: When a light ray bends as it passes from a substance of one optical density into a substance of a different optical density.



4
- the direction of the light ray
- the speed of the light ray

5 thinner, cheaper, more durable, faster information transfer, carries more information, higher data security, not affected by interference from electromagnetic radiation

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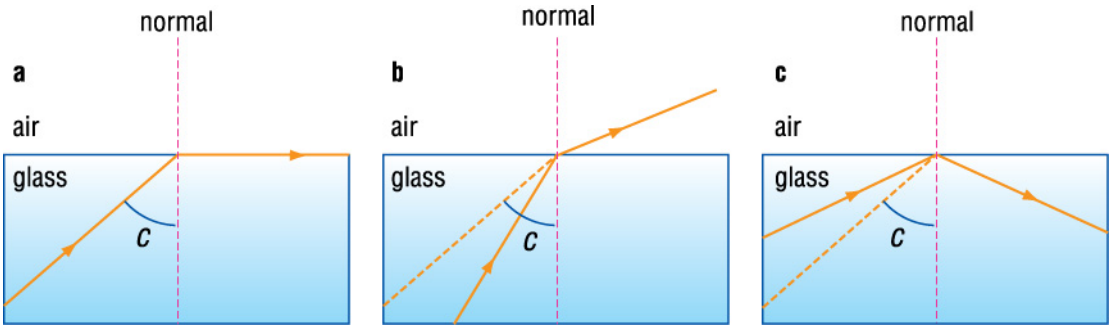
6 **a** The light ray is refracted along the surface.

b The light ray passes out of the glass and into the air, bending away from the normal.

c The light ray undergoes total internal reflection, reflecting back into the glass.



Understanding

7 **a** Refraction: When a light ray bends as it passes from a substance of one optical density into a substance of a different optical density.

b Total internal reflection: When a light ray travelling in an optically dense substance attempts to pass into a substance that is less optically dense but, instead, is reflected back into the original substance.

c Critical angle: The angle of an incident light ray that will cause the light to refract exactly 90° to the normal. Light incident at angles greater than the critical angle will undergo total internal reflection. Light incident at angles less than the critical angle will pass through the interface and undergo refraction.

8 Bike reflectors are shaped so that the angle of the incoming light is always greater than the critical angle. This allows the light from the car behind to be reflected back, making the reflector appear to be shining bright red. There is no need for a mirror (see Figure 7.1.8).

9 **a** The light will be refracted along the diamond and air boundary.

b The light will undergo total internal reflection, being reflected back into the diamond.

c The light will pass out of the diamond and into the air and be refracted away from the normal.

10 Refraction occurs in air when there are layers of air at different temperatures. Layers of air at different temperatures have different optical densities.

11 any solar-powered device; e.g. solar-powered calculator, solar-powered lamps, emergency telephones

Applying

12 **a** towards the normal

b towards the normal

c towards the normal

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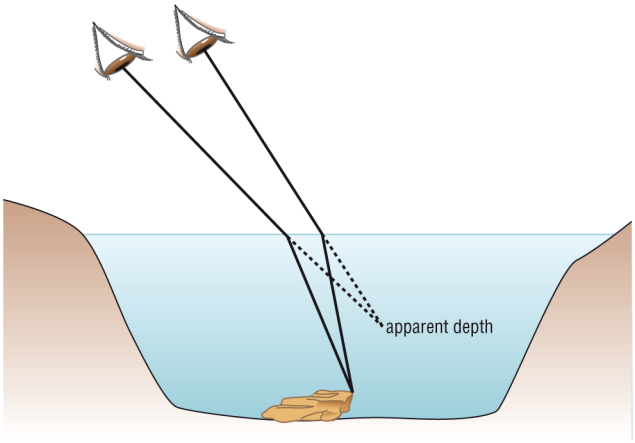
13



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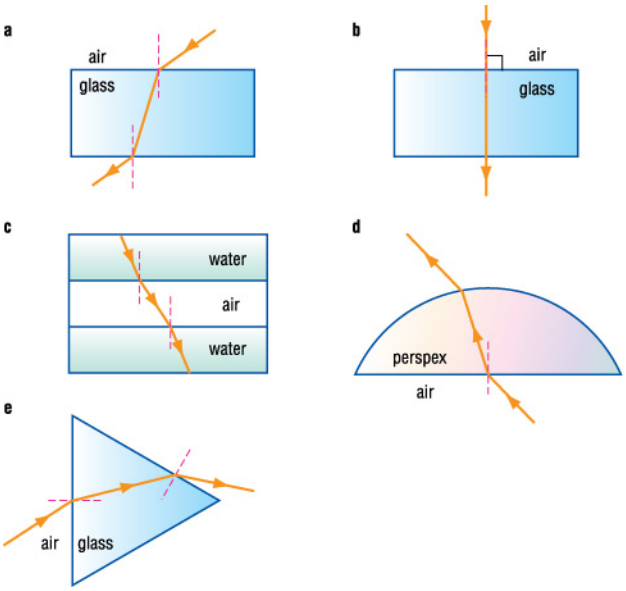

Analysing

15 Similarities:

- Reflection and refraction both change the direction of light.
- Neither refraction nor reflection absorb light.

Differences:

- Reflected light always travels in the same substance, whereas refracted light passes through different substances.
- Reflected light always travels at the same speed, whereas refracted light changes speed.
- Reflected light is always at the same angle from the normal, whereas refracted light changes its angle from the normal.

16 C

17 **a** The car is travelling fastest on the bitumen.

b Sand has the higher 'refractive index'.

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Evaluating

18 No, reflection must obey the law that the angle of reflection is equal to the angle of incidence, whereas this is never true with refraction. Also, refraction requires that the light passes into a new substance, whereas reflection implies that the light ray stays in the original substance.

19 **a** Scratches on optical fibres will cause the light travelling through the optical fibre to travel less efficiently. This means that data sent through scratched optical fibres may be noisy, corrupted or not transmitted at all.

b Optical fibres work by passing light along the fibre through totally internal reflection. This requires that the surface of the fibre be very smooth. Scratches on the surface of the optical fibre would prevent the light from being reflected. Instead, the light might be scattered or escape from the fibre.

20 **a** Optical fibres have only a small diameter so can detect light only from a small area. As a result, fibres need to be bundled in endoscopes with each fibre carrying a small part of the image, together making up the whole image.

b In an endoscope, other optical fibres are used to shine light onto what the doctor needs to see; e.g. the inside of the stomach. The light then reflects off the stomach and into the bundle of optical fibres, which feeds the light out and forms an image.

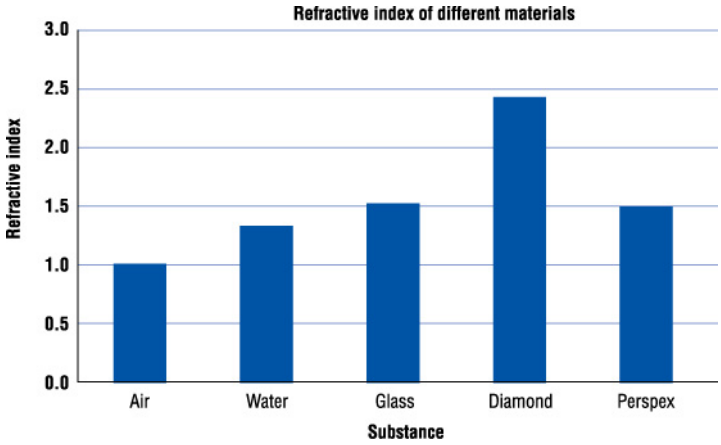
21 **a** Optical fibres can be used in endoscopes to look inside the human body without the need for surgery. They can also be used to shine high-powered laser light into parts of the body to remove blockages in arteries, damaged cells, tumours etc.

b Optical fibres can replace some surgical techniques, such as investigative surgery to look inside the stomach, intestines, lungs etc. They can also replace some open heart surgery procedures by using lasers to remove damaged cells in the heart. They may even replace cancer treatment by applying radiation directly to a tumour inside the body.

c Optical fibre technology means that many medical procedures can be done more quickly, more cheaply, less invasively and with less discomfort and shorter recovery time for the patient.

Creating

22 **a**



b air, water, Perspex, glass, diamond

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7.1 Practical activities

Prac 1: Measuring angles of refraction

Common mistakes

A common error is that students will measure the angle of incidence as being between the incident ray and the surface of the slab. A well-lit room is not ideal for this prac, as it may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

- The greater the angle of incidence used, the greater the angle of refraction produced.
- For any non-zero angle of incidence, the angle of incidence will be greater than the angle of refraction.

Suggested answers

1 *The semicircular shape ensures that refraction only occurs at one boundary (the straight edge).*

2 *As the light ray enters the block it is bent towards the normal.*

3 *The angle of incidence should be represented on the x-axis and the angle of refraction on the y-axis.*

4 *The graph of the diamond slab will have a smaller gradient than the Perspex slab, as diamond is more optically dense and thus produces a greater bending of the light ray.*

Prac 2: Observing light

Common mistakes

Adding too much milk to the water will cause the beam to be lost.

Possible results

At small angles of incidence, the light ray will emerge from the water to the air. As the angle of incidence increases, it will reach an angle called the critical angle at which the light will not emerge from the fish tank but will be reflected back along the surface of the water at the critical angle.

Suggested answers

1 *For small angles of incidence you get refraction occurring when the light ray emerges from the fish tank but, for large angles of incidence (i.e. greater than the critical angle) you get total internal reflection.*

2 *The angle of incidence equals the angle of reflection.*

3 *If the liquid has a higher refractive index, then the critical index will be smaller.*

Prac 3: Refraction illusions

Common mistakes

Add water to the bowl slowly, taking care not to displace the coin.

Possible results

Objects in water are always deeper than they appear to be.

The refraction of light causes the ruler to appear to bend under water.

The image of the coin will appear to be above the position of the actual coin.

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Suggested answers

1 *The greater the angle of the ruler, the more the ruler appears to bend.*

2 *The ruler appears to be larger under water because of the light rays bending as they pass through one medium into another.*

3 *The more water that is in the bowl, the higher the coin appears to be, allowing a clearer image of the coin.*

Prac 4: Apparent depth

Common mistakes

Students may take measurements from the image reflected on the side of the glass block.

Possible results

The apparent depth will be significantly shallower than the actual depth.

Suggested answers

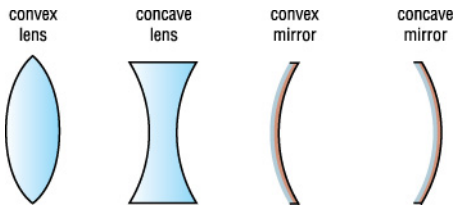
1 *Students' results.*

2 *The apparent depth will increase.*

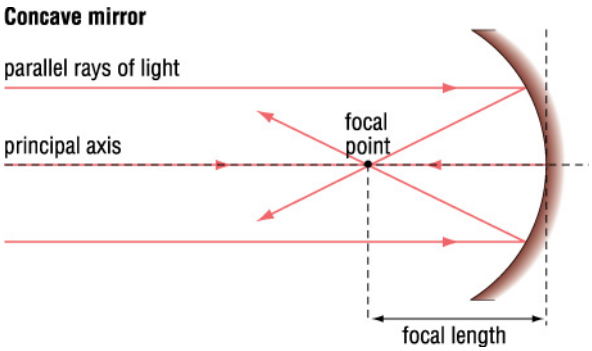
7.2 Answers

Remembering

1 concave and convex.



2 **a**



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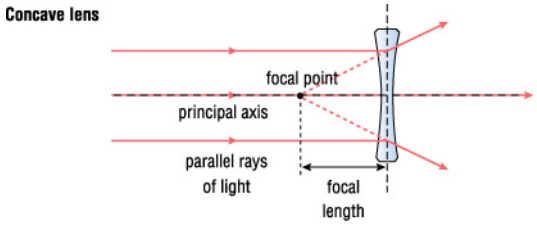
b



3 cornea and lens

Understanding

4 **a** Focal length: The distance from the lens to the focus; i.e. the point through which all parallel light rays pass or appear to pass in the case of a convex lens.

b Principal axis: The line that passes through the centre of a lens.

c Myopic: Short-sighted; i.e. when the focal length of the lens of the eye is too short and the image forms in front of the retina.

d Hyperopic: Long-sighted; i.e. when the focal length of the lens of the eye is too long and the image forms beyond the retina.

5 Unlike real images, a virtual image cannot be projected onto a screen and, therefore, cannot be seen at the movies.

6 **a** Real images formed by convex lenses are always smaller than the original object.

b Virtual images formed by convex lenses are always bigger than the original object.

c Concave lenses can form only virtual images.

d Images in a concave lens are always the right way up.

e Virtual images in a convex lens are always the right way up.

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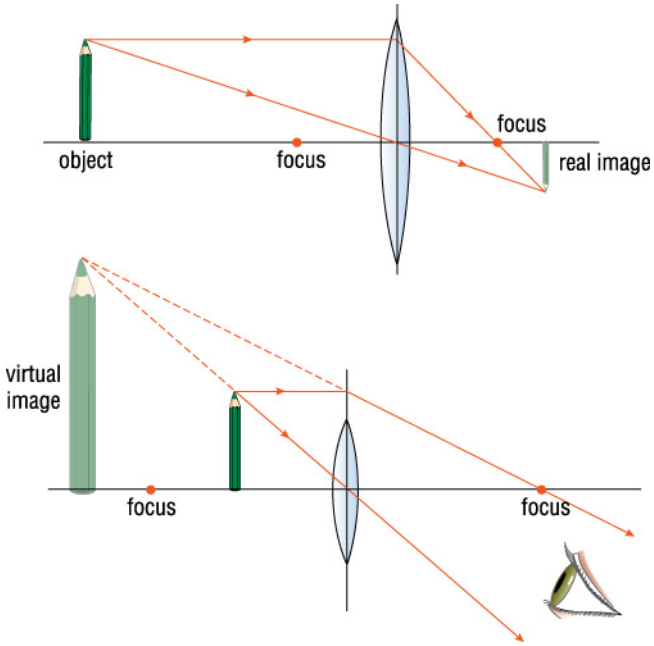
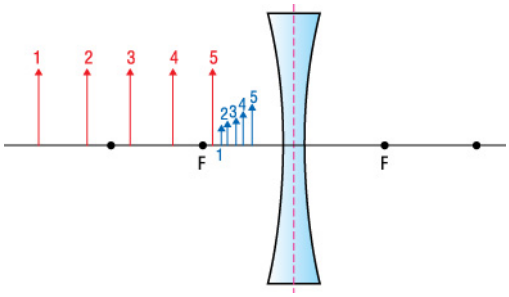
7 **a** When an object is further than the focal point of a convex lens, the image is real, diminished and inverted and appears on the opposite side of the lens. When the object is brought closer, the image blurs as the object passes through the focal point and then becomes virtual, magnified, upright and is formed behind the object.



b For an object beyond the focal length of a concave lens, its image is virtual, diminished and upright. As the object is brought closer, the image gets bigger but these properties remain the same.



8 A lens or a mirror can focus sunlight on a specific spot. Therefore, by angling the lens towards the Sun and focusing the light rays into a small area you can intensify the heat coming from the Sun and start a fire.

9 **a** A myopic eye focuses the image in front of the retina, so concave glasses are used to diverge the light rays before entering the eye so that they take longer to converge and are therefore focused on the retina.

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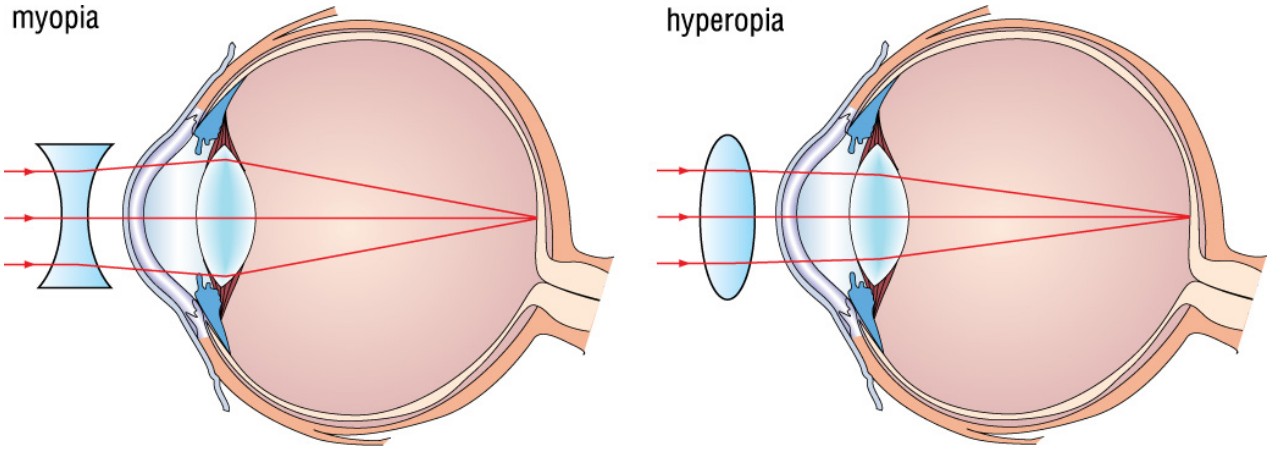
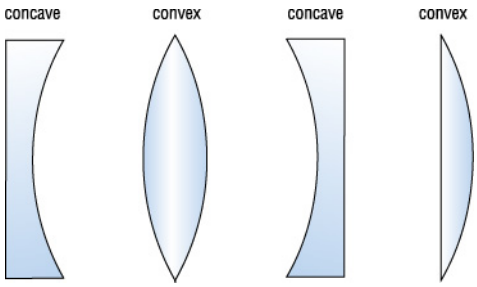
b A hyperopic eye focuses the image beyond the retina, so convex glasses are used to converge the light rays before entering the eye so that the image is focused on the retina.



Applying

10



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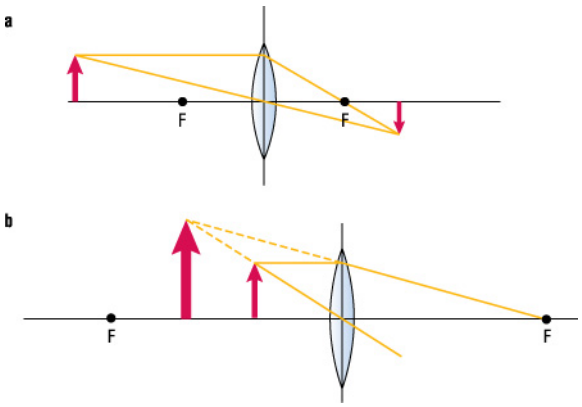

12 **a** Convex lenses are converging; i.e. the light rays passing through them converge to a single point.

b Concave lenses are diverging; i.e. the light rays passing through them diverge.

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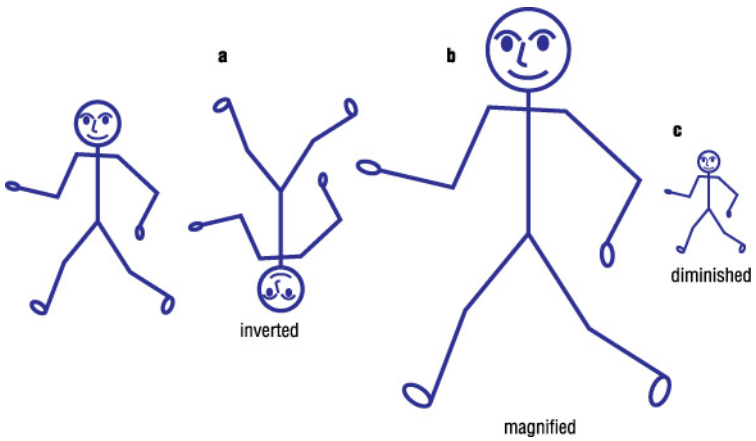
13 Real image: object further than the focal length.
Virtual image: object within the focal length.
Blurred image: object exactly at the focal length.

14 The lens is a convex lens and the hot spot is the focal point of the lens where the Sun's rays meet.

15



16 **a** You can't touch the image, only the ant itself.
b The image is virtual.

17

Object height	Image height	Magnification
2 cm	6 cm	3
5 cm	20 cm	4
25 mm	5 mm	0.2
16 mm	4 mm	0.25
8 mm	160 mm	20

18 **a** A convex mirror allows drivers to see as much of the intersection as possible.
b A concave mirror magnifies the teeth.

19 concave mirror

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Analysing

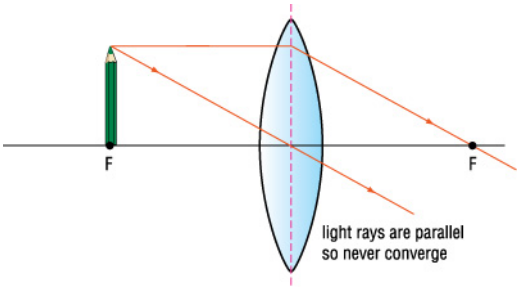
20 A convex lens produces enlarged images only when the object is within one focal length. A lens will never produce an enlarged image of the Moon because the Moon is so far away. It will always be further than one focal length away from the lens.

Evaluating

21 The objective lens should be thin because the rays coming from a distant object are almost parallel, so only need to be converged slightly in order to produce an image. However, the eyepiece should be thick in order to converge the light rays rapidly from the image to the eye.

Creating

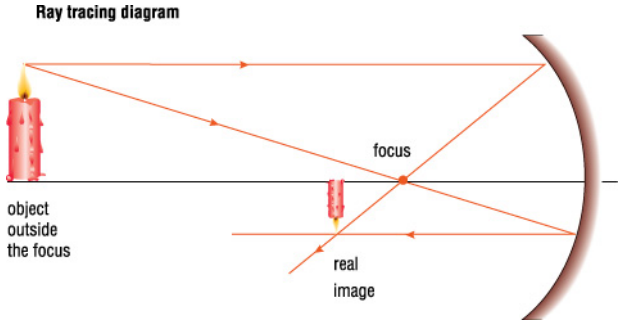
22



23 **a** concave mirror

b The image is real, diminished and inverted.

c



7.2 Practical activities

Prac 1: Water lenses

Common mistakes

Glossy paper works best. The drops of water may soak into matt paper.

Possible results

Drops of water behave as a convex lens.

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Suggested answers

1 *convex*

2 *Smaller drops are more curved.*

Prac 2: Lenses and a light box

Common mistakes

Instruct students on the correct use of light boxes. A common error is that students may place the multi-slit slides into the wrong end of the light box, making it difficult to obtain parallel beams.

Possible results

For convex lenses, the focal length is the distance between the centre of the lens and the focal point (where the light rays intercept).

For concave lenses, the focal length is the distance between the centre of the lens and where the light rays would intercept if you traced them back.

Suggested answers

1 **a** *A convex lens causes the light rays to converge.*

b *A concave lens causes the light rays to diverge.*

2 *A thick convex lens causes greater convergence than a thin convex lens.*

3 *A light ray passing through the centre of the lens will remain unbent as it enters and exits at 90°.*

4 *Results will depend on the lenses used.*

Prac 3: Images in a convex lens

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

If the candle is too close to the lens, no real image will be formed.

Possible results

As the object gets closer to the lens, the real image on the opposite side of the lens gets further away from the lens and becomes larger. When the object is at the focal point, the image is at an infinite distance away and is infinitely large.

As the object moves closer than one focal length, the image becomes a virtual image, upright, enlarged and on the same side as the object.

Suggested answers

1 *The image of the candle becomes larger and further away from the lens.*

2 **a** *A real image can be formed only with a convex lens and the object is greater than one focal length away.*

b *A virtual image can be obtained only when the object is closer than one focal length to the convex lens. Concave lenses always form virtual images.*

c *No image is obtained when the candle is at the focal point of a convex lens.*

3 *It is not possible.*

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4 *As the object gets closer to the lens, the real image on the opposite side of the lens gets further away from the lens and becomes larger. When the object is at the focal point, the image is at an infinite distance away and is infinitely large. With a concave lens, the image is always virtual and becomes larger as the object approaches the lens.*

Prac 4: Simulating imperfect vision

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

The smaller the exposed area of the lens, the clearer the image will be.

Suggested answers

1 *Students' measurements.*

2 *The image becomes duller and clearer.*

3 *The lens forms the image by combining multiple images (the blurred image) into a single clear image.*

Prac 5: Telescopes and microscopes

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

The arrangement of lenses will produce an image that appears larger than the object.

Suggested answers

1 *A telescope is used to view large objects at great distances, whereas a microscope is used to view small objects at short distances.*

2 *The image will remain visible through the eyepiece.*

Prac 6: Exploring curved surfaces: spoons

Common mistakes

When looking at the concave side of the spoon, students may not hold the spoon close enough to their eyes to produce an upright image.

Possible results

When close up, the concave side of the spoon will produce an upright real image. When the spoon is moved further away, the image will become inverted.

The concave side of the spoon will produce an upright virtual image.

Suggested answers

1 *Real image: When the concave side of the spoon is greater than one focal length.*

Virtual image: When the concave side of the spoon is closer than one focal length.

The convex side of the spoon will always produce a virtual image.

2 *The light rays cross over.*

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Prac 7: Forming images with curved mirrors

Common mistakes

A well-lit room is not ideal for this prac, as it may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

A concave mirror will produce a real, inverted and smaller image or a virtual, upright and enlarged image, depending on the distance between the object and the mirror.

A convex mirror will always produce a virtual and upright image. The size of the image is dependent on the distance between the object and the mirror.

Suggested answers

1 ***a*** *Concave mirror: The image is enlarged and upright.*

b *Convex mirror: The image is enlarged and upright.*

2 ***a*** *Concave mirror: Real, inverted and smaller or virtual, upright and enlarged.*

b *Convex mirror: Virtual, upright and smaller.*

7.3 Answers

Remembering

1 Sun or light globe

2 triangular prism

3 dispersion

4 red, orange, yellow, green, blue, indigo and violet

5 scattering

6 **a** red, green and blue

b cyan, magenta and yellow

7 **a** magenta

b red

8 **a** red, green and blue

b cyan, magenta, yellow and black

9 magenta and yellow

Understanding

10 Sunsets and sunrises are caused by scattering in which molecules of gas and dust particles in the atmosphere alter the direction of light rays. The dust and gas is better at scattering blue light than red light, which is why the sky usually appears blue. The red light escapes into space. At sunrise and sunset, when sunlight travels further through the atmosphere, almost all of the blue rays are scattered and the light that reaches us is mainly red or orange.

11 A primary rainbow has red at the top and blue at the bottom. A secondary rainbow has the reverse effect on the colours, so that the bottom band is red.

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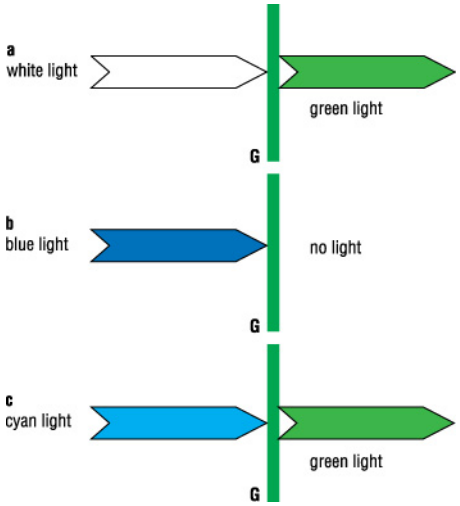
12 A colour television screen can simulate all colours by using the three primary colours red, blue and green. The screen is coated in patterns of dots called pixels and each pixel contains cells of each of the primary colours. When an electron strikes each cell they become excited and glow. Your eyes then combine the colours from each of the glowing cells in each pixel to construct the different colours.

Applying

13 C

14



15 **a** blue

b green

c black

16 Black will heat up the quickest because it absorbs all the colours of light unlike white, which reflects all the colours of light, or the green car, which only absorbs blue and red light.

Evaluating

17 The light energy absorbed by the filter is converted to heat energy.

18 Water droplets behave like tiny prisms in the sky and as each drop catches the Sun's rays, light refracts (bends), breaking it up into its various wavelengths, creating a rainbow.

19 Rainbows are formed by light entering a water droplet and undergoing total internal reflection re-emerging from the droplet. Therefore, rainbows can be seen only when the Sun is behind you and only from light rays that strike the back of the water droplet at angles greater than the critical angle of 42°.

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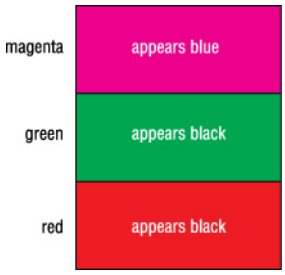
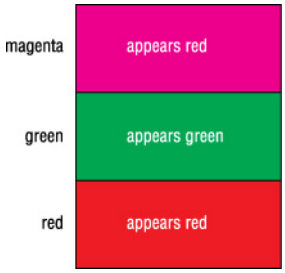
Creating

20 **a**



b



21 Practical activity: Different responses possible depending on observations.

7.3 Practical activities

Prac 1: Dispersion: splitting white light

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

Colours are dispersed using a triangular prism, with violet being refracted the most and red the least.

Suggested answers

1 ***a*** *Violet refracts the most.*

b *Red refracts the least.*

2 *Another triangular prism could be used to bend the coloured light back into the same path.*

Prac 2: Mixing coloured light

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

Complementary colours combine to form white light.

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Suggested answers

1 *green and magenta, red and cyan, blue and yellow. These combinations are called complementary colours.*

2 *the difference in the intensity of colours in the filters*

Prac 3: Seeing things in a different light

Common mistakes

A well-lit room may not allow students to see the behaviour of the light ray clearly. A darkened room will assist this exercise.

Possible results

Coloured cards look a particular colour because they reflect that colour. They absorb other colours.

Suggested answers

1 **a** *any combination of colours that contain red light*

b *any coloured light except red*

2 *When a coloured light which contains none of the colour of the object is shone on that object, the object will appear black. Various specific examples are possible.*

3 *Labelling the cards makes it easier to recall their colour while different coloured lights are shining on them.*

4 *Cards that reflected grey did not absorb all the coloured light being shone on them.*

Prac 4: Measuring light absorption

Common mistakes

Ensure the hole in the plastic lid is sealed around the thermometer to prevent heat loss.

Possible results

Black will absorb the most heat and white will absorb the least. The amount of heat the other colours will absorb will be within the two extremes.

Suggested answers

1 *Different colours absorb and reflect different amounts of light. The less reflection and more absorption, the more the can will heat up.*

2 *Black: reflects nothing, absorbs everything.*

White: reflects everything, absorbs nothing.

Red: reflects red, absorbs all other colours.

Blue: reflects blue, absorbs all other colours.

Green: reflects green, absorbs all other colours.

3 *Students' measurements.*

4 *The more colours absorbed, the greater the temperature rise.*

Prac 5: Mixing pigments

Common mistakes

Use a white-coloured mixing surface to avoid false results.

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Possible results

Students will achieve various colour combinations depending on which pigments they mix.

Suggested answers

1 *Different combinations of pigments will produce different combinations of colours. Answers will vary.*

2 *There are no more colours left to reflect, as all colours have been absorbed.*

Chapter answers

Remembering

1 When a light ray travelling in air strikes a glass boundary, it is refracted towards the normal. The speed of the ray in the glass is slower than it is in air.

2 **a** False, not if it strikes the boundary at 90°.

b True

c True, but only if the substance has different optical densities, such as warm and cold air.

d False

e True

3 transmission of data, endoscopes

4 **a** cyan

b green

5 **a** magnified virtual image

b diminished virtual image

6 **a** movie projector

b magnifying glass

7 **a** one

b two

Understanding

8 **a** Two uses for optical fibres are data transfer and medical applications, such as endoscopy.

b Data transfer: Optical fibres may be used to replace copper wires for the transmission of information. Optical fibres use light that is transmitted along the fibre. Optical fibres are cheaper, lighter, faster and more secure than copper wires.

Medical applications: Optical fibres can be used to replace some surgical procedures. In endoscopes, optical fibres allow doctors to look at organs inside the body without surgery. Doctors can also use optical fibres to perform laser surgery on blockages in blood vessels or tumours.

9 **a** black

b black

c green

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10 Planar mirrors are used in bathrooms so people can see what they look like without any distortion.

Concave mirrors are used in shaving mirrors to magnify the image.

Convex mirrors are used on sharp, dangerous street corners so drivers can see what is coming around the corner.

11 **a** Security mirrors in a shop are convex mirrors that allow the entire room to be viewed from any angle.

b A slide projector uses lenses to form a magnified real image of the picture that's on the slide. This real image can be projected onto a screen for everyone to see.

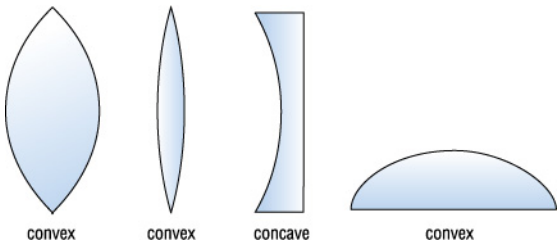
12 **a** orange

b blue

c violet

Applying

13



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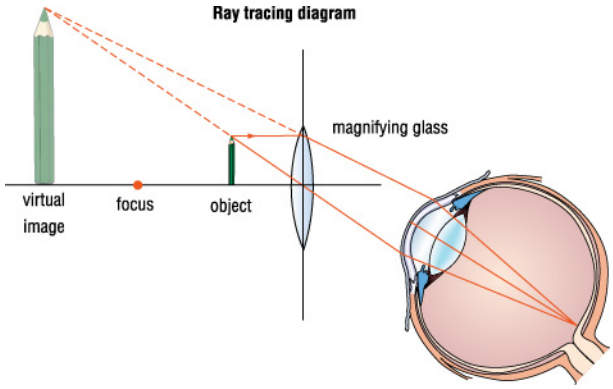

Analysing

15 With more dust or pollution in the air, more blue light is scattered, making sunsets red.

16 **a** real, inverted and diminished

b real, inverted and diminished

c virtual, upright and magnified

d virtual, upright and diminished

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Evaluating

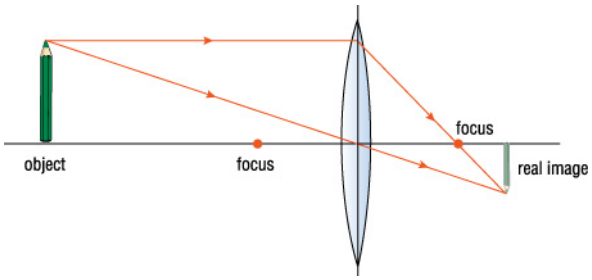
17 Yellow dots absorb blue light but reflect red and green. Cyan dots absorb red light but reflect blue and green. Therefore, the patch should appear green.

18 The lemons are yellow, so reflect green and red light. Therefore, the lemons should appear red.

Creating

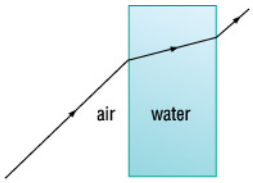
19 **a** Concave lenses do not produce real images.

b



20

a



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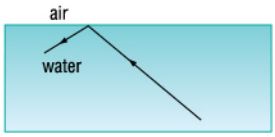

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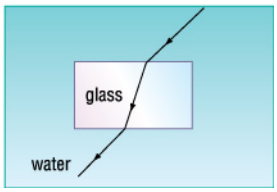

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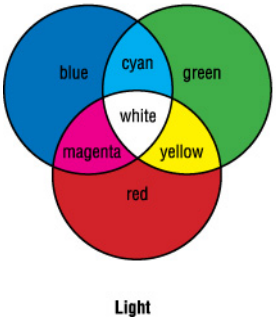

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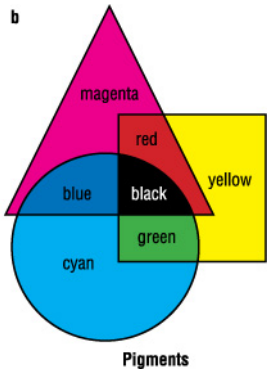

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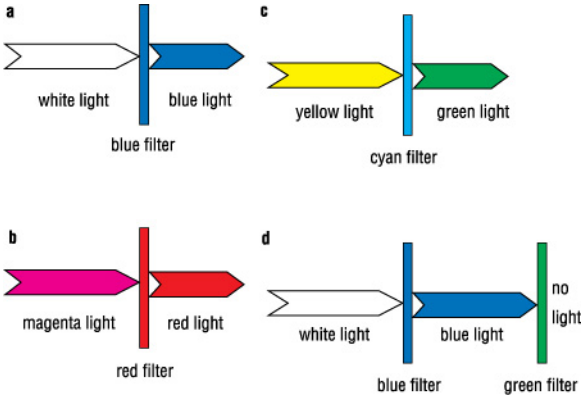

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8.1 Answers

Remembering

1 **a** short
b long

2 **a** spectrometer
b A prism refracts light and separates it into the visible spectrum.

3 Atoms in the atmosphere of stars absorb some of the light of a galaxy's spectrum and this produces dark lines within them.

4 red shift

5 Using the dark lines emitted from different stars and galaxies, astronomers can determine:
- whether distant stars and galaxies are moving away from Earth
- the further away they are, the faster they are going.

Understanding

6 If the car is moving towards you, it catches up with the sound waves in front of you, bringing the sound waves closer. This shortens the wavelength and, in turn, increases the frequency that produces a high-pitched sound.

7 A stationary ambulance's siren spreads the sound waves evenly in all directions; therefore, the wavelength, frequency and pitch will be exactly the same in all directions.

8 Astronomers have evidence that the universe is still expanding, so they imagine this expansion process in reverse to a starting point.

9 **a** Christian Doppler and Edwin Hubble
b Christian Doppler described how the wavelength and pitch changes as objects approach and move away. Edwin Hubble proved that the universe has many galaxies and that it is expanding.

10 The pitch is higher as they fly towards you and the pitch lowers as they fly away from you.

Applying

11 police siren, ice-cream truck music

12 **a** lower pitch
b higher pitch
c lower pitch

13 **a** females
b males
c males
d females

14 **a** red
b blue

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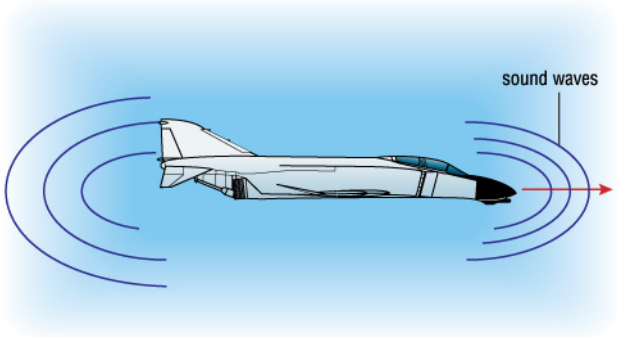
15



16 **a** a
b h, b and a
c e
d f and d

Analysing

17 The faster plane will produce a deeper sound compared to the slower plane as they move away from you.

18 Astronomers believe that before the Big Bang all matter was packed into a point smaller than a proton, suggesting the density within that point was enormous. Just after the Big Bang, the universe had a high density because all the matter was packed into a small space, whereas now, the universe is still expanding and all the matter that was created has spread apart, forming a less dense universe.

Evaluating

19 **a** green
b red (the fastest)
c blue

20 The red shift colour a star emits tells astronomers how fast the star is moving away from us.

Creating

21 Students' responses will vary. An example: What was the temperature at the time of the Big Bang?

8.1 Practical activities

Prac 1: Using a spectroscope

Common mistakes

A well-lit room may not allow students to see the colours of the spectrum clearly. A darkened room will assist this exercise.

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Possible results

Many colours can be observed when looking at white light through a spectroscope. Only one colour can be observed when looking at a coloured filter through a spectroscope.

Suggested answers

1 *The spectrum of white light is made up of all the colours. The range extends from red through to violet, with orange, yellow, green and blue in between.*

The coloured light is one of the colours within the visible spectrum.

2 *Diagrammatic answer is required.*

Prac 2: A balloon universe

Common mistakes

To minimise the risk of puncturing the balloon, use a felt tip pen rather than a ball point pen to draw on the balloon.

Possible results

The balloon will expand in such a way as to show that all points on the surface of the balloon are moving away from a single point at the centre of the balloon.

Suggested answers

1 *The other galaxies moved further away from the Milky Way.*

2 *All galaxies should move farther apart, regardless of which one is circled.*

3 *This suggests that the universe is expanding.*

4 *Yes, the balloon has a centre from which everything is moving.*

8.2 Answers

Remembering

1 13.7 billion years

2 3:1

3 cosmic background explorer

4 300 000 years after the Big Bang

5 heat, radio waves, microwaves

Understanding

6 Particles of matter bumped into antimatter, annihilating each other. When all the matter–antimatter annihilations were finished, a relatively small amount of excess matter was left, which became the building blocks of the universe.

7 It is the simplest atom and was the first atom formed.

8 one proton

9 It contained only energy in the form of radiation, such as X-rays and light.

10 As electrons slowed down, they were captured by hydrogen and helium to form new types of atoms and elements. As more and more elements formed the fog cleared.

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11 They detected microwaves formed from the first photon of light.

12 Invisible material, called dark matter, drew gas clouds together until they compacted enough to form galaxies. Smaller clouds of gas collapsed even more to form the first stars.

13 When rings of gas orbiting stars condense they form young planets. As they attract more and more matter, they increase in size until they become planets.

14 galaxies

Applying

15 within a fraction of a second (one ten million trillion trillion trillion trillionths of a second)

16 quarks

Analysing

17
- The open universe theory holds that if the mass of the universe is too low, then gravity will be too weak to stop it expanding. Eventually, the stars will cease to shine and the universe will become a dark and cold place.
- The closed universe theory holds that the expansion of the universe will eventually stop if it has sufficient mass and strong enough gravity. The gravitational pull in this situation will cause a reversal of the Big Bang.
- The flat universe model states that the universe will eventually stop expanding, but will never reverse.
- The accelerating universe theory is the only theory that does not accept gravitational attraction as slowing or stopping the expansion of the universe.

Evaluating

18 Students' responses will vary.

19 This theory is less likely because there is evidence to support that the universe is expanding. Therefore, if the universe is still growing, there must have been a starting point.

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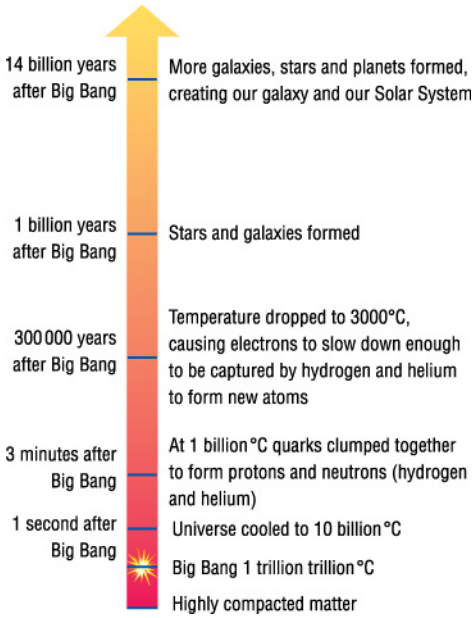
Creating

20



8.3 Answers

Remembering

1 Stars begin as dense clouds of gas and dust known as a nebula. The denser regions within the nebula collapse under the pull of their own gravity. The nebula's gas and dust come closer together, forming a protostar. Eventually, nuclear reactions begin, during which hydrogen atoms fuse together to form helium.

2 hydrogen

3 burns hydrogen, burns helium, red giant, white dwarf

4 blue super giant, red super giant, supernova, neutron star

Understanding

5 Nebula is a dense cloud of gas and dust.

6 Pulsar is a rapidly rotating neutron star with a strong magnetic field.

7 If a massive star collapses more than a neutron star, it forms a black hole. A black hole is an object so dense that anything in close proximity will be drawn into it by its own gravity. Black holes emit no light and distort the space around them.

8 No, because one teaspoon of a neutron star has a mass of around 1 billion tonnes.

9 The strong gravitational force of the black hole would draw in all the planets and life forms in our galaxy, including Earth.

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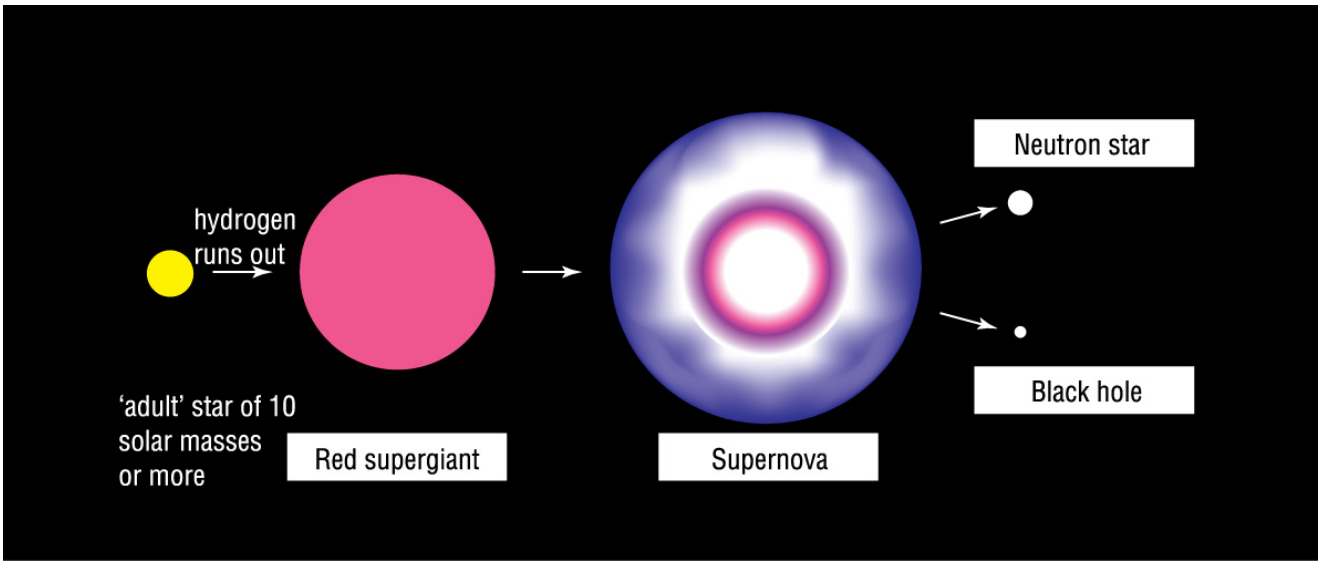
Answers to *Science Focus 3 second edition* Student Book questions

Applying

10



Analysing

11 All stars are born as nebulae, whereby gravitational forces collapse the cloud of gas and dust to form a protostar. As stars such as our Sun grow older, they run out of hydrogen and helium. The outer layers then expand, forming a red giant. However, in a star ten times the mass of our Sun, the fuel burns faster and the star expands to become a blue super giant. In a star the size of our Sun, the centre then collapses to form a white dwarf; this is the final stage for stars like our Sun. In stars ten times the mass, the centre collapses, which results in a massive supernova (explosion) and then a neutron star.

Evaluating

12 All atoms originated after the Big Bang. As the temperature changed, quarks formed hydrogen and helium. These were the first simplest atoms. As electrons slowed down, they joined with helium and hydrogen to form atoms such as carbon. As stars are born and die, their carbon is used in different components of the universe and we are simply one of these components. Hence, elements within our bodies would have come from stars.

13 Black holes are dark and emit no light; however, we know they exist because they emit X-rays. Astronomers study how nearby stars behave—if they 'wobble' there is a black hole nearby.

Creating

14 Flow chart should show the following stages in the order shown here.

- Star swells up to burn hydrogen in its atmosphere, expanding 100 times to become a red giant.
- Helium fuel used to produce carbon.
- Outer layers and carbon are blown away in space to form new stars and planets.
- The remaining centre collapses under its own gravity to form a white dwarf.

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8.4 Answers

Remembering

1 The closest star to Earth is the Sun and we believe we are the only planet around it that has life. Our closest star is 4.3 light years away and it would take 80 000 years to get there.

2 laser and radio waves

3 Search for Extra Terrestrial Intelligence, NASA

4 Arecibo and Parkes

Understanding

5 **a** the distance light travels in one year (9.46×10^{12} kilometres)

b relates to Earth and its inhabitants

c a being or life form occurring outside Earth

d Search for Extra Terrestrial Intelligence: search for life within the universe other than on Earth.

6 The heading implies that we may not be alone and questions whether we are the only life form in the universe.

7 there is less background interference

8 The radiating lines show where Earth is located in relation to the closest pulsars. The hydrogen molecule symbolises the most abundant element in the universe and the humans display the generic appearance of intelligent life forms on Earth (the place of the plaque's origin).

9 Many politicians believe that SETI programs are ridiculous and expensive.

10 Any signal we may receive would have to have been sent a long time ago because it would take a long time to reach Earth, sometimes light years. In the time the signal took to arrive, the civilisation that sent it could have been wiped out.

Applying

11 Determine how many seconds in a year, then multiply by the speed of light, then divide by 1000 to convert from metres to kilometres as follows.

$60 \times 60 = 3600$ seconds in an hour

$3600 \times 24 = 86\,400$ seconds in a day

$86\,400 \times 365 = 31\,536\,000$ seconds in a year

$31\,536\,000 \times (3 \times 10^8) = 9.46 \times 10^{15}$ metres in one year

$9.46 \times 10^{15} \times 1000 = 9.46 \times 10^{12}$ kilometres

12 $1/50 \times 2000\,000\,000 = 40\,000\,000$

Analysing

13 The electromagnetic spectrum is made of electrical and magnetic fields. Magnetic fields can still be detected even after switching off transmitters.

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Evaluating

14 Students' responses will vary.
- Humans are depicted naked.
- Humans are depicted as Caucasian only.
- No other life forms are depicted.
- Some people may be offended by the suggestion that there are possible life forms outside of Earth.

15 Students' responses will vary.

Creating

16 Against: It could be dangerous because we don't know what weapons, knowledge or diseases they may carry.

For: They could bring knowledge to advance our technologies that will improve our quality of life.

17 Students' responses will vary.

18 Students' responses will vary.

8.5 Answers

Remembering

1 **a** MISR: multi-angle imaging spectroradiometer
b ISS: International Space Station
c EVA: extra-vehicular activity
d GMT: Greenwich Mean Time

2 **a** Venus and Mars (any two planets)
b geostationary and remote sensing

3 weather patterns, temperature of the Earth and oceans, shape of land surface, details of the sea floor, natural phenomena, vegetation in agriculture and forestry, ozone hole, tracking of animals, monitoring pollution and navigation (Students to list any five.)

4 Dennis Tito, A$27 million

Understanding

5 an object in space that travels around a larger object

6 satellites placed at a fixed position over the Equator travelling at the same speed at which the Earth rotates, making them appear stationary and, hence, useful for communication

7 because the geostationary satellite has a fixed position

8 Microgravity is the feeling of weightless-ness when riding a rollercoaster as you lift out of your seat and then back down into your seat. Here, you feel weightless but gravity pushes you back down.

9 The Russian Elektron generator makes oxygen by splitting water into hydrogen and oxygen.

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Answers to *Science Focus 3 second edition* Student Book questions

Applying

10 cables

11 rollercoaster ride

Analysing

12 Passive sensors detect and record the amount of electromagnetic radiation reflected from natural sources, whereas active sensors make their own electromagnetic radiation. Both sensors code signals as binary numbers that are transmitted to ground stations as electromagnetic signals.

13 ISS provides research in various scientific fields, such as engineering, tissue culture and materials science. However, a disadvantage is the extremely high cost (A$200 billion).

Evaluating

14 Students' responses will vary.

15 to provide financial support to maintain the station and experts in all fields and to provide the research that benefits all nations

16 for spectacular views of Earth and other celestial bodies, also the possibility to partake in microgravity sporting activities

17 Students' responses will vary. Some examples are given below.

Human race: Less competition for resources on Earth.

World government: Research the positive and negative impacts of living in space for extended periods.

Private industry: Financial benefits from ticket sales to fly to space and from property sales.

Creating

18 Students' responses will vary.

19 The flow chart should include the following points.

- Release from mothership at 15.2 km.
- Launch to Mach 3 and ascent into space.
- Space entry at 54.86 km.
- Suborbital tour at 109.7 km.
- Re-entry at 54.8 km.
- Spaceship defeathers into glider mode at 30.5 km.

Chapter answers

Remembering

1 The fog cleared about 300 000 years after the Big Bang.

2 **a** True

b False

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c False

d True

e False

Understanding

3 Initially, all the matter of the universe was packed together in a point smaller than a proton. It then exploded out; this is what we know as the Big Bang. It is believed that it was the moment that time and space was created.

4 The current temperature of the universe is much cooler than that of the early universe, which was approximately 3000°C.

5 Astronomers detect black holes by observing X-ray emissions and whether nearby stars wobble.

6 **a** effects of microgravity on humans

b rocket technology

7 Christian Doppler: described the relationship between wavelength and pitch. This concept is used to observe
fast-moving light sources such as stars.

Edwin Hubble: worked out whether stars or galaxies are moving towards Earth or away from Earth and at what speed.

Applying

8 hydrogen and helium

9 **a** 8.6 years

b 160 000 years

10 areas where thousands of stars are forming

11 When the electromagnetic wave used in making the phone call crosses paths with another radio wave.

Analysing

12 Dark matter is the leftover subatomic particles from the Big Bang. This matter drew clouds of gases together until they compacted sufficiently to form galaxies. Therefore, if dark matter did not exist, galaxies could not be currently forming.

13 In an open universe, if the mass of the universe is too low then the weak gravity will decrease the rate at which the universe expands. Eventually, the stars and galaxies will cease to shine and the universe will become a dark and cold place. In a closed universe, scientists believe that the expansion will stop if it has a high enough mass and strong enough gravity. The gravitational pull in this situation will cause reversal of the Big Bang. The flat universe theory also states that the universe will stop expanding but it will remain as it is. The accelerating universe theory is the only theory that does not accept gravitational attraction as slowing or stopping the growth of the universe. It states that the universe is expanding at an accelerated rate due to a cosmological force called ‘dark energy’.

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Evaluating

14 Students' responses will vary.

World space stations provide the following advantages and disadvantages.

Advantages: provide research vital for humans in various fields.

Disadvantages: the A$200 billion cost of running such a program.

Overall, space stations provide humans with valuable research and information.

Creating

15 Students' responses will vary.

For: extraterrestrial life could bring knowledge and technologies far more advanced than ours on Earth.

Against: the unknown dangers that extraterrestrial life may bring.

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9.1 Answers

Remembering

1
- Underwater volcanic mountain ranges were found down the centre of the ocean.
- Deep ocean trenches were found.
- The age of the rocks of the ocean floor vary from recent to 200 million years old.
- The rock of the continent is less dense than that of the ocean floor and it seems to float.
- Rocks of the ocean floor have magnetic stripes.

2
a 8–64 km
b 2800 km
c 2300 km
d 1400 km

Understanding

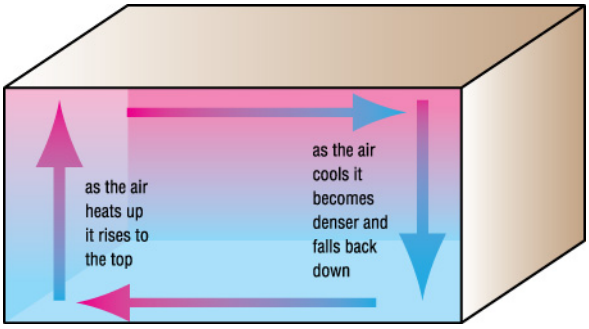
3
- There is a spread of identical fossils across the southern continents.
- The structure, age and rock composition of mountains across different continents can be matched.
- Ancient glaciers have left valleys and debris across many continents.
- Coal has been found above the Arctic Circle. Coal comes from decomposed plants and the Artic Circle is far too cold now for plants to grow, so this region must have moved from a warmer climate.
- As lava from a volcano cools, it adopts the current magnetism of the Earth. Three hundred million-year-old magnetic rocks in South and North America have been found with their poles pointed in different directions. If the continents are joined the magnetic poles will all point to the same spot.

4
a Tectonic plates: cracks (fault lines) in the lithosphere form slabs called tectonic plates.
b Mantle: the next layer after the crust. The crust and mantle together form the lithosphere.
c Crust: the layer of Earth we live on. The crust varies in thickness.

5 The tectonic plates in the lithosphere move by convection currents in the Earth's mantle.

6 Heat deep within the Earth causes molten rock of the mantle to move upwards. When this hot molten rock comes into contact with the colder crust it cools and sinks.

7



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8 Rock deep in the mantle is constantly being heated by radioactive decay.

9 **a** Triassic reptiles could not have swum the distance required to populate different continents.

b There are similar mountain ranges in the USA and Europe, and also in Africa and South America.

c Continents that do not have glaciers now have not always been too warm to have them.

d Coal deposits above the Arctic Circle suggest that the land floated there from warmer climates.

e The rock of the ocean floor and that of the continents are not necessarily the same age.

f Continental rock is less dense than the rock of the ocean floor.

g Magnetic stripes on the ocean floor suggest that new rock is made along mid-ocean ridges.

Applying

10 **a** Australia, Antarctica, South America, Africa and India

b North America, Europe and most of Asia

11 The oldest rocks are along trenches and the newest are located along ridges.

Analysing

12 75 × 5 = 375 cm (3.75 m)

Evaluating

13 The temperature of the rising hot molten rock is hotter than that of the hardened surrounding surfaces.

14 The plates would stop floating and moving around.

15 The temperature of the entire Earth could have been cooler, whereby glaciers covered the Earth. This is in contrast to Earth's current warmer temperature.

9.1 Practical activities

Prac 1: The planet Splatter

Common mistakes

If the continent shapes are cut incorrectly, then the pieces may not fit together.

Possible results

The shapes of each continent are cut out and rebuilt, joining each complementary shape to form a supercontinent.

Suggested answers

1 *The ways in which the edges of the continents line up show that the continents could have fitted together. The magnetic lines show that the continents were once oriented in different directions that are consistent with them fitting together. The fossil remains are also consistent with the continents having once been joined together.*

2 *No, there are no other ways. If all the information is taken into account, there is only one solution that fulfils all of them.*

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Prac 2: Convection currents

Common mistakes

Placing more than one crystal of potassium permanganate into the beaker of water will make it difficult for students to trace the movement of the current.

Possible results

The potassium permanganate stain moves up with the current in warm water.

The potassium permanganate stain moves down with the current in cold water.

Suggested answers

1 *the fact that hotter material is less dense than cooler material*

2 *hot currents rise; diagrammatic answer required*

3 *cold currents sink*

Prac 3: Future Earth

Common mistakes

Accuracy in measurement.

Possible results

Possible future changes to the position of the Earth's continents.

Suggested answers

1 *No, different plates move at different rates.*

2 ***a*** *The Mediterranean Sea will close up.*

b *The Atlantic Ocean will eventually close up.*

c *The Red Sea will get much wider.*

3 ***a*** *further north, possibly 'crashed into' Asia*

b *further north*

c *further north*

d *further west*

e *much further east*

4 *Due to the shift towards the Equator for the Indo–Australia continental plate, New South Wales' climate would be significantly warmer.*

5 *Answers will vary.*

9.2 Answers

Remembering

1 20 cm a year on average

2 divergent or spreading boundaries; convergent or collision boundaries; transform or scraping boundaries

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3 **a** fast
b heavy
c oceanic plate

Understanding

4 In areas of spreading boundaries, rocks crack and new magma fills the cracks. This is like a scab because as a wound repairs itself it can crack due to stress and the blood oozes again to fill in the cracks.

5 Oceanic plates are denser than continental plates so they are forced under the continental plate. The angle (20–60°) at which the plate dives is called the subduction zone.

6 When heat from a collision of two continental plates melts some rock, this forms an area called a 'mountain root' that resists weathering.

7 At transform boundaries, plates scrape alongside each other, but they do not create or destroy rocks, so they are known as conservative boundaries. However, they produce many earthquakes.

8 **a** Land is not created or destroyed; these boundaries cause earthquakes.
b Land is destroyed and returned to the mantle for recycling; this happens at subduction zones.
c Constructive zones are also known as convergent (collision) boundaries, whereby new rock is made; for example, along ocean ridges.
d Subduction zones form when plates collide and one is pushed under the other; for example, in areas where ocean trenches are formed.
e Rift valleys form along spreading boundaries and are found mostly under water.
f These are collision boundaries that form subduction zones by pushing oceanic plates under the continental plate into the mantle.
g These form at collision (convergent) boundaries where the oceanic plate drops below the continental plate, as in the Peru–Chile Trench.
h When two continental plates of similar densities collide (convergent boundaries) and are unable to push one under the other, instead the plates crumble, fold and push up.
i Transform boundaries, such as the western coast of North America where fault lines are formed.
j These form at converging boundaries, where an oceanic plate collides with another oceanic plate and the magma rises from volcanoes to form island chains.
k At converging boundaries, such as in Asia where continental plates of similar densities collide to form mountain ranges such as the Himalayas.

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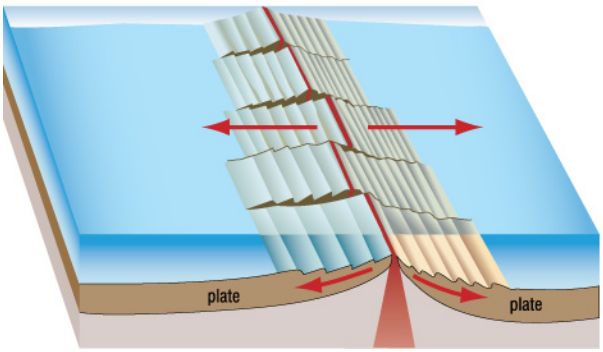
9 a



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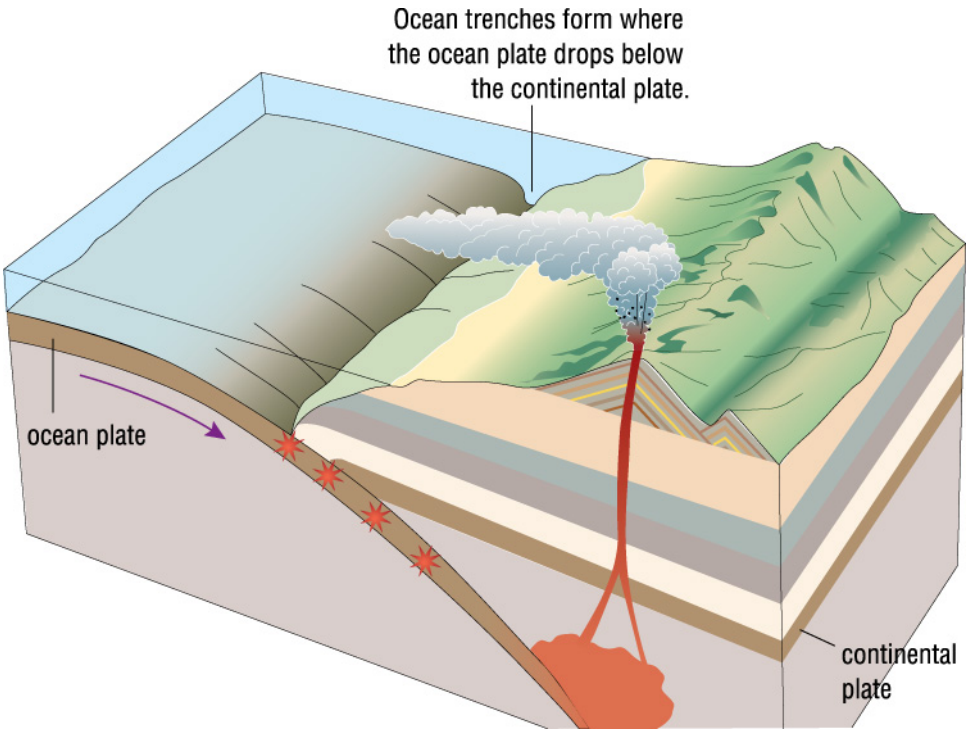

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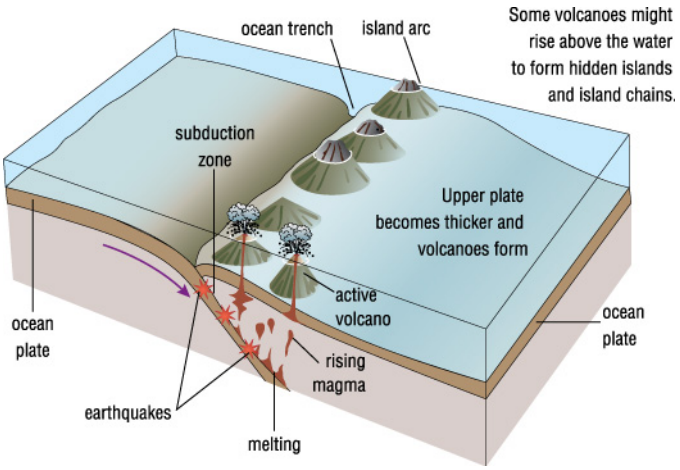
c



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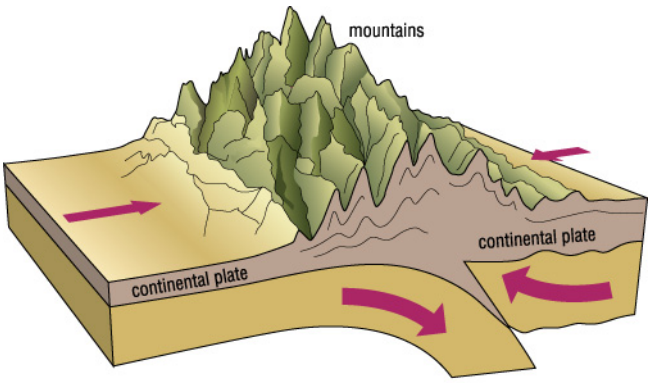
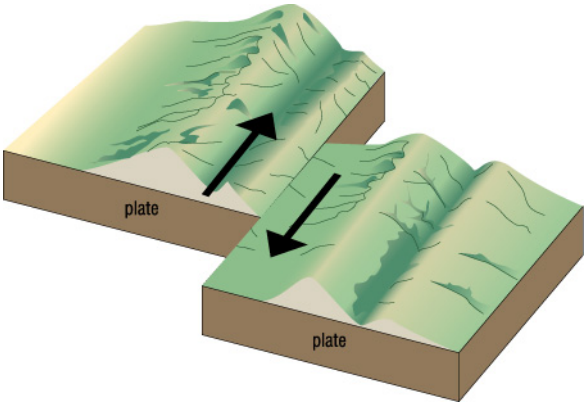


Both plates have similar densities and neither can push the other underneath. Plates crumple, fold and push up to form mountains.

e



10 The upper plate gets crushed and thickens as it folds, forming mountains. Also, if any magma seeps through cracks it solidifies causing further thickening of the upper plate.

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Answers to *Science Focus 3 second edition* Student Book questions

Applying

11 convergent; divergent; transform

12 **a** divergent
b transform
c convergent
d divergent
e divergent
f divergent

13 **a** Indo–Australian and Eurasian plates
b Nazca and South American plates
c North American and Eurasian plates
d Cocos and Caribbean plates
e Eurasian and Philippine plates
f Pacific and Philippine plates
g North American and Pacific plates
h Eurasian and Indian plates

14 **a** l
b e
c j

Analysing

15 **a** 1 × 75 = 75 cm
b **i** 1000 years
ii 10 000 years
iii 100 000 years

16 **a** 500 – 240 = 260 km
260 × 1000 = 260 000 m
260 000 × 100 = 26 000 000 cm
26 000 000 ÷ 20 = 1 300 000 years
b (6100 – 240) = 5860 km
5860 × 1000 = 5 860 000 m
5 860 000 × 100 = 586 000 000 cm
586 000 000 ÷ 20 = 29 300 000 years
c 14 000 – 240 = 13 760 km
13 760 × 1000 = 13 760 000 m
13 760 000 × 100 = 1 376 000 000 cm
1 376 000 000 ÷ 20 = 68 800 000 years

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Creating

17 Students' responses will vary.

18 Students' responses will vary but students need to consider the following:

- cities away from coastlines
- cities away from any convergent, divergent and transform boundary
- cities with large populations
- cities not on islands.

9.2 Practical activities

Prac 1: Plates that separate

Common mistakes

Accuracy in measurement.

Possible results

The movement of the paper is similar to the creation of rock at a mid-ocean rift.

Suggested answers

1 *It models the spreading process that occurs.*

2 *The paper rises either side of the join.*

3 ***a*** *the paper lying on the desks*

b *the air above the paper*

c *the emerging new paper*

d *gravity*

e *the gap between the desks*

f *the colouring of the different strips of paper as they emerge*

4 *Those coloured first are the oldest rock and those coloured last the youngest.*

5 *the oldest, first-coloured strips*

Prac 2: Colliding plates

Common mistakes

Pushing the paper too hard will cause it to crumple.

Possible results

The paper simulates plate collisions.

Suggested answers

1 *They model collision boundaries.*

2 ***a*** *two stacks of paper*

b *two stacks of paper*

c *a stack of paper with a book*

3 *Himalayas, Peru–Chile Trench, Japan*

4 *Yes, some rocks have layers because they have been laid down over time as sedimentary rocks.*

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9.3 Answers

Remembering

1 They are too small or too far away from human activity to be felt.

2 Body waves: primary (P), secondary (S).
Surface waves: Rayleigh (R), Love (L).

3 **a** R and L (R more than L)
b S
c P and S
d P and S
e P
f R
g R
h L
i S

4 Richter scale 3 and Mercalli scale 11

Understanding

5 As tectonic plates separate, collide and scrape past each other, they build up pressure to overcome the friction that causes the plates to 'stick'.

6 When the pressure build-up of overcoming friction is released an earthquake will occur.

7 S waves are not recorded directly opposite; this suggests the core is liquid because S waves cannot travel through liquids.

8 200 km, as this is the depth at which an oceanic plate will melt into the asthenosphere.

9 7–8, because buildings collapse, dams break, leading to floods, and deep cracks appear in the ground.

10 Aftershocks are capable of moving large slabs of crust and rock and have the ability to destroy buildings made unstable by the original earthquake.

11 A tsunami is an earthquake with its epicentre under the ocean floor, forming a wave. As this wave travels it gathers incredible speed and height. Tsunamis end when they hit coastlines, causing great devastation and death.

12 Papua New Guinea and New Zealand have a plate boundary running through their land mass. In contrast, Australia is located further away from the plate boundary. This means that Australia is less likely to experience earthquakes.

13 The epicentre, because it is the shortest distance from the focus; therefore, the time delay is insignificant.

14 A seismogram displays the time delay between the arrival of each different wave type and its energy.

15 **a** Aftershocks are smaller powerful movements occurring after earthquakes.
b An earthquake of strength 5 on the Richter scale is ten times the strength of a 4.

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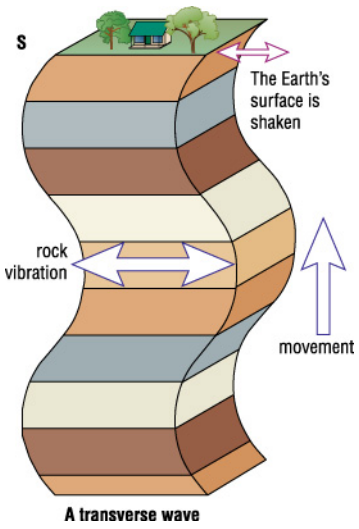
c Earthquakes are caused by plates moving over each other.

d The focus of an earthquake is the exact point at which an earthquake starts.

e Tsunamis are huge when in shallow water.

Applying

16



17 **a** sound

b water waves

Analysing

18 The epicentre is the point on the Earth's surface that is directly above the focus of an earthquake.

19 **a** R waves have a rolling motion whereas L waves move side to side like a snake. R waves move slower compared to L waves but they cause more damage.

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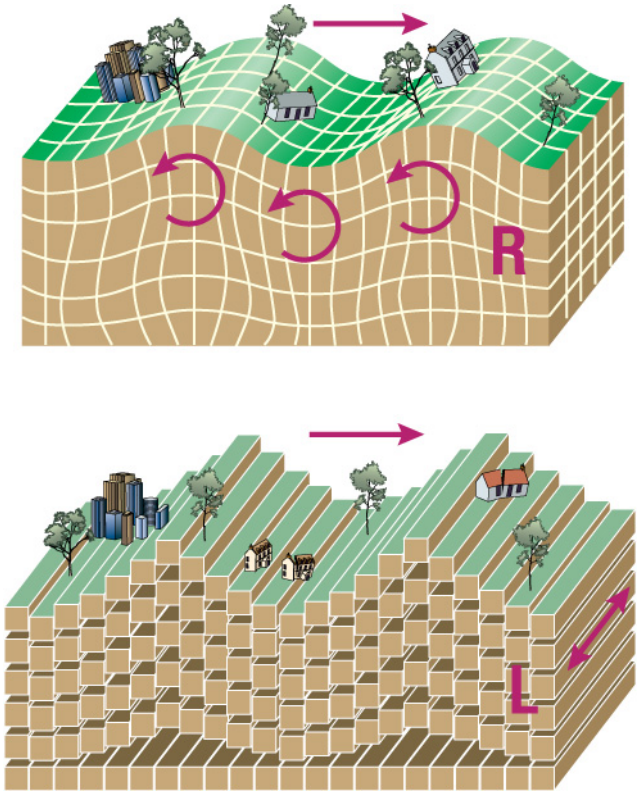
b



20 **a** Body waves travel through the Earth whereas surface waves travel around the Earth. Surface waves cause more damage than body waves.

b Longitudinal wave particles vibrate sideways and push and pull the materials they travel through, whereas transverse wave particles move up and down like water waves.

21 Distance = (time ÷ 8.6) × 7000

a (4 ÷ 8.6) × 7000 = 3256 km

b 1791 km

c 2849 km

d 6593 km

22 **a** 7.4 minutes

b 1.8 minutes

c 4.1 minutes

d 1.1 minutes

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23

Arrival time of P waves (h:min:s)	Arrival time of S waves (h:min:s)	Time difference (min:s)	Time (min)	Distance of epicentre (km)
10:24:00	10:32:00	8:00	8.0	6512
04:48:20	04:52:50	4:30	4.5	3663
02:55:21	03:01:21	6:00	6.0	4884
07:37:03	07:42:33	5:30	5.5	4477
14:08:34	14:11:46	3:12	3.2	2605
20:21:02	20:25:50	4:48	4.8	3907
05:45:10	05:50:10	5:00	5.0	4070
11:34:30	11:41:00	6:30	6.5	5291
08:12:56	08:16:02	3:06	3.1	2500
15:21:04	15:28:10	7:06	7.1	5800

Evaluating

24 People and equipment in the eye of the tsunami would not survive or would be damaged, respectively.

25 **a** Most buildings will collapse, causing great destruction and death. Landslides due to the earthquake will cause valleys to be filled in. Floods will also occur because river boundaries may change and dam walls will break. Transport systems will cease to function because roads and railway lines will break and twist.

b zero to ten times

9.3 Practical activities

Prac 1: Slinky springs

Common mistakes

If the slinky is overstretched it will not return to its original form.

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Possible results

Using a slinky to model the movement of P and S waves.

Suggested answers

1 ***a*** *sideways*

b *back and forth*

2 *An ant in a transverse wave may become twisted, whereas in a longitudinal wave it would get squashed.*

3 ***a*** *Part B*

b *Part A*

c *Part A, except up and down instead of sideways*

Prac 2: Build a seismometer

Common mistakes

Ensure that the retort stand is securely clamped to the bench.

If the spring is overstretched it will not return to its original form.

Possible results

Creating a working model of a seismometer.

Suggested answers

1 *Answers will vary.*

2 *Energy stored in the spring and heavy weight keeps it moving for some time.*

Prac 3: Locating the epicentre

Common mistakes

Convert times in minutes and seconds into a decimal value of the minute before reading the graph; e.g. 4 minutes and 36 seconds is 4.6 minutes. If this step is missed, the circles will not intercept.

Possible results

The epicentre of an earthquake is located.

Suggested answers

1 *on the Flinders river, central Queensland*

2 *under the ground directly below the epicentre*

3 *The distance is approximately 2250 km to Broome from the epicentre. Using the graph in Figure 9.3.6 on page 299, this gives a time of 3.5 minutes, which equals 3 minutes and 30 seconds.*

4 *Cloncurry*

9.4 Answers

Remembering

1 1500 active volcanoes

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2 Krakatoa was a small volcanic island that destroyed itself. It is steadily building a new island. It is famous because in 1883 it caused the largest explosion ever experienced in the recorded history of humans.

3 Mt Gambier in South Australia

Understanding

4 Most volcanoes occur at the edges of tectonic plates and are sometimes located underwater.

5 Most volcanoes are found along the edges of plates because as plates move, the pressure from gases in the mantle squeezes molten rock upwards.

6 **a** The steam, gas and smoke burst out due to pressure and travel as a fast-moving cloud that reaches incredible heights because the ash is carried by winds.

b
- may crush a city
- may suffocate and bury a city
- can turn into lahar (river of mud) if it rains
- may block the Sun due to jet-stream winds

7 **a** Lava is the matter that erupts from volcanoes.

b The magma chamber is where magma forms deep under the Earth's surface.

c Rain turns ash into lahar, which is a river of mud.

d Fume is when a cloud of volcanic ash travels the Earth up to 30 km in height.

e Jet-stream winds exist 30 km above the Earth.

8 **a** Lava is when magma erupts from a volcano.

b A dormant volcano is a sleeping volcano.

c Volcanic ash moves faster than lava.

d White lava is hotter than red.

e Ash clouds can travel, up to 500 km.

Applying

9 steam, gas (hydrogen sulfide), magma, smoke, ash

10 Volcanic bombs are explosions from volcanoes, which form in three ways:
- gas explosions with large pieces blown out as solid rock
- hot lava thrown into the air
- rocks block the vent until gases build enough pressure, leading to an explosion.

11 hydrogen sulfide (rotten egg gas)

Analysing

12 Vents are tubes by which lava erupts. A volcano has a central vent and side vents. Fissures, however, are long cracks by which lava erupts.

13 1250 km/h

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Evaluating

14 The Pacific Ocean is a plate; therefore, along the edges, volcanoes constantly erupt due to plate movement.

15 Volcanic eruptions occur along plate boundaries, as do earthquakes. When earthquakes occur they can disturb and agitate molten lava (magma), which could lead to volcanic eruptions.

Creating

16 Students' responses will vary. Student responses need to include:

- a correct scale
- dates
- names and locations of volcanoes
- volcanoes that erupted in the 1900s.

17 Students' responses will vary.

18 The diagram will be scaled as follows.

a 3.05 mm

b 8.85 cm

c 80 cm

d 10 cm

9.4 Practical activities

Prac 1: Volcanic clouds

Common mistakes

To obtain better results, ensure that the temperature of the water in the flask is very hot.

Possible results

The hot coloured water rises to represent volcanic gas. The clear cold water represents the surrounding air.

Suggested answers

1 *Diagrammatic answer required.*

2 *hot water rises*

3 *Coloured water represents volcanic gas and dust. Clear water represents the surrounding air.*

4 *hot ash would rise*

5 *The ash is hotter and less dense than the surrounding cold air.*

Prac 2: Build a 3D model of a volcano

Common mistakes

Experiments designed by the students should be assessed by the teacher before students carry them out.

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Possible results

Students build a scale model of a volcano.

Suggested answers

1 *Answers will vary.*

2 *Answers will vary.*

9.5 Answers

Remembering

1 normal, reverse, transcurrent

2 buckling and folding without breaking

Understanding

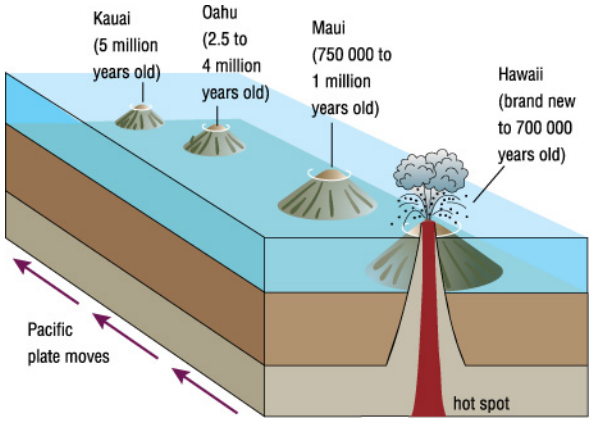
3 The movement along normal and reverse faults is up–down, so it leaves a rounded edge. However, if weathering had occurred, it would form a cliff.

4 Faults are fractures in the Earth's crust caused by extreme forces of divergent boundaries. As divergent plates spread, new crust (land) forms, and when these new blocks of crust are formed near an area with two fault lines the blocks shift, thrusting up (horst) or sinking down (graben).

5 Rocks need to be subjected to extreme pressure both horizontally and vertically. Under these conditions rock acts like modelling clay and starts to buckle and fold without breaking.

6 Volcanoes usually lie at the weak edges of tectonic plates or over hot spots. The islands of Hawaii lie 3200 km from the nearest plate boundary. Hot spots never change positions but the plate above the hot spots do. Hawaii is currently over a hot spot but it will eventually move west with the plate. A new volcano is already forming east of the chain, suggesting the islands have moved.

7



8 With extreme pressure, both horizontally and vertically, the layers of sedimentary rock folded at 80° to the horizontal layers to build Uluru. The rocks formed an arch called an anticline.

9 D, C, B, A from newest to oldest

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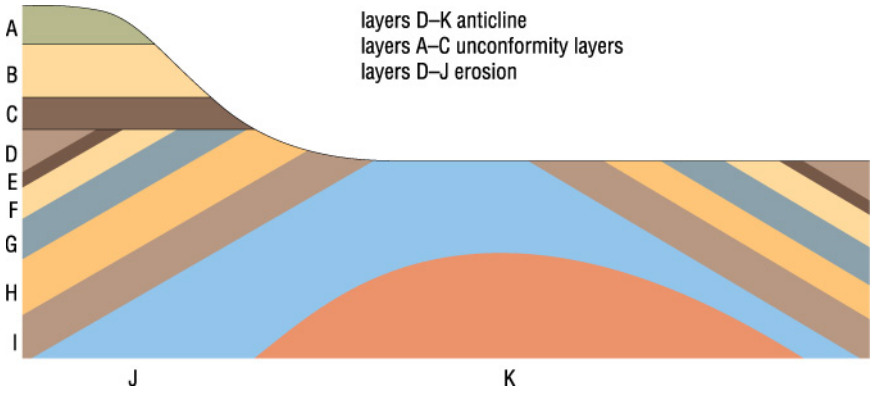
Applying

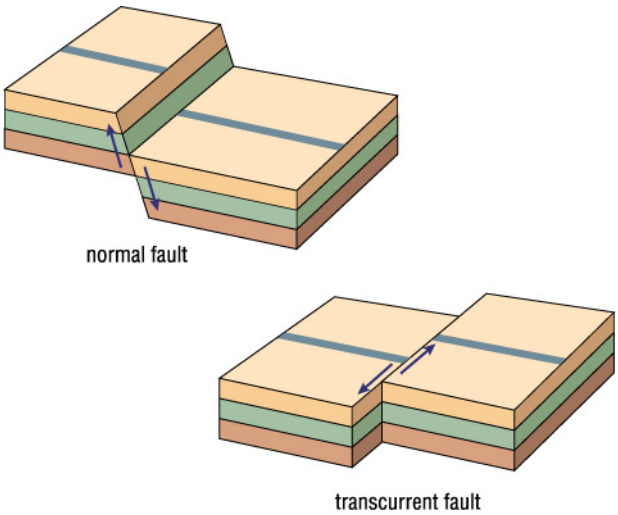
10 Bass Strait was once over a hot spot.

11



Analysing

12 a



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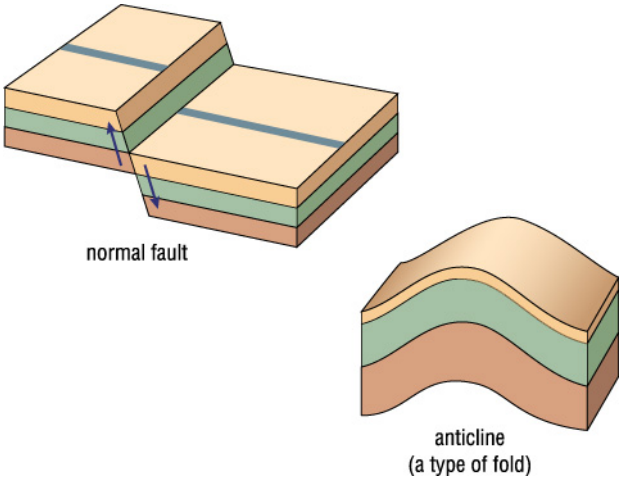
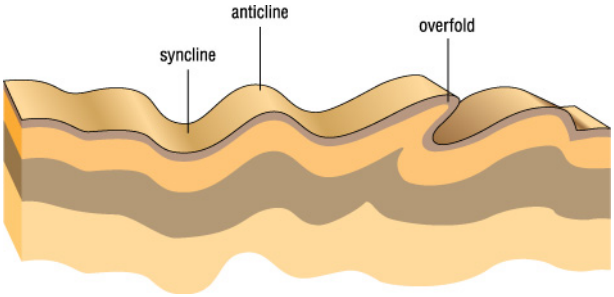
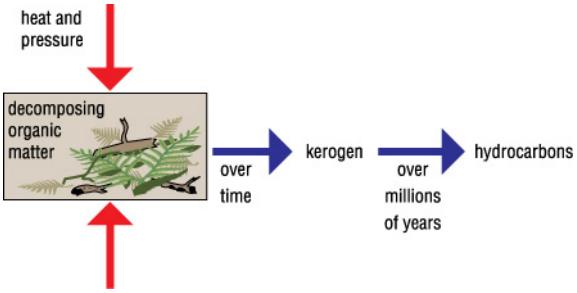


c



d



13 **a** A plug is the vent of a volcano that has been exposed due to the walls of the volcano eroding way. A dyke is an example of an igneous intrusion where the magma cools below the surface.

b Shield cones form away from boundaries and grow larger with every eruption. Lava flows slowly down the gentle slopes. Cinder cones are small and consist of piles of rock and cinder (ash).

c When two faults allow a block of rock to thrust up it forms a horst, but if it sinks down it forms a graben.

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Evaluating

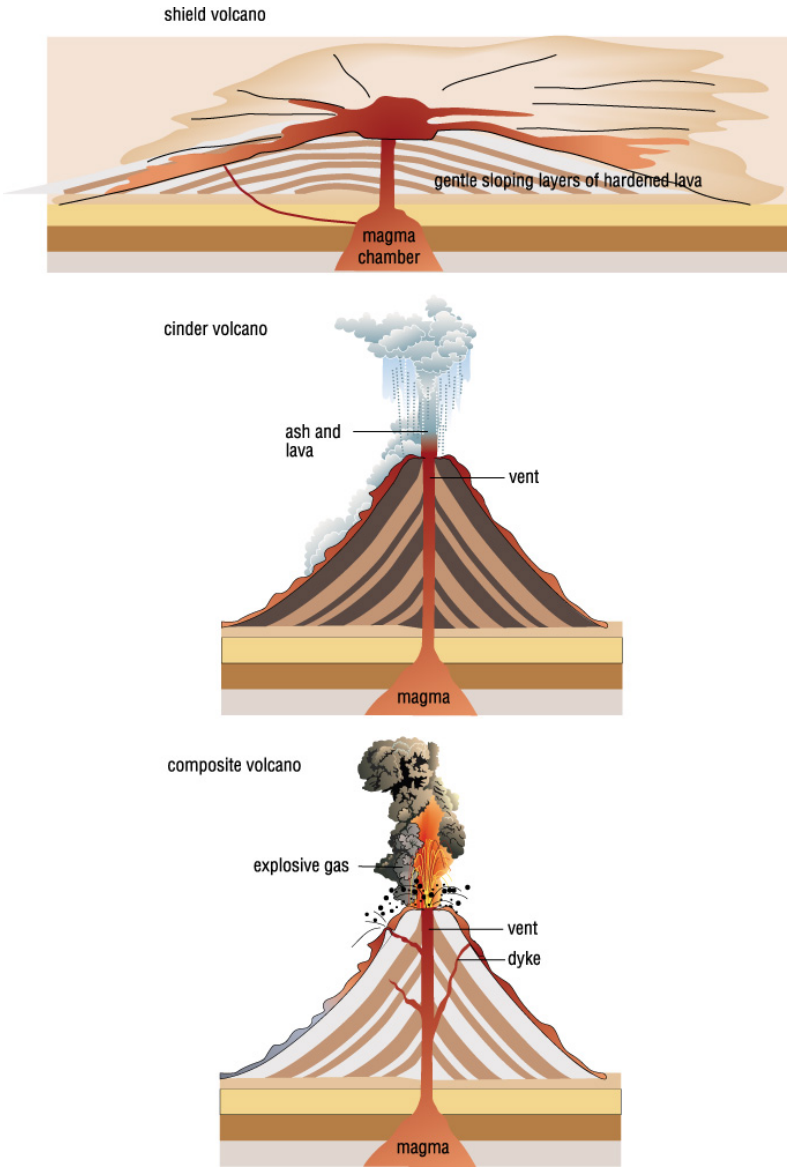
14 The intense pressure and heat at weak spots may provide the right conditions to convert decomposed plant and animal matter into fossil fuels. Also, along weak spots, sponge-like rocks form into which oil can seep.

15 For volcanoes to form away from plate boundaries they must be directly over hot spots or plumes. There must also be intense pressure for the magma to force its way through.

16 The North Island is located over hot spots and the plates are separating; therefore, volcanoes will form and erupt. The South Island, however, is on a section of the plate that is colliding with another plate, forming large mountains in the process.

Creating

17



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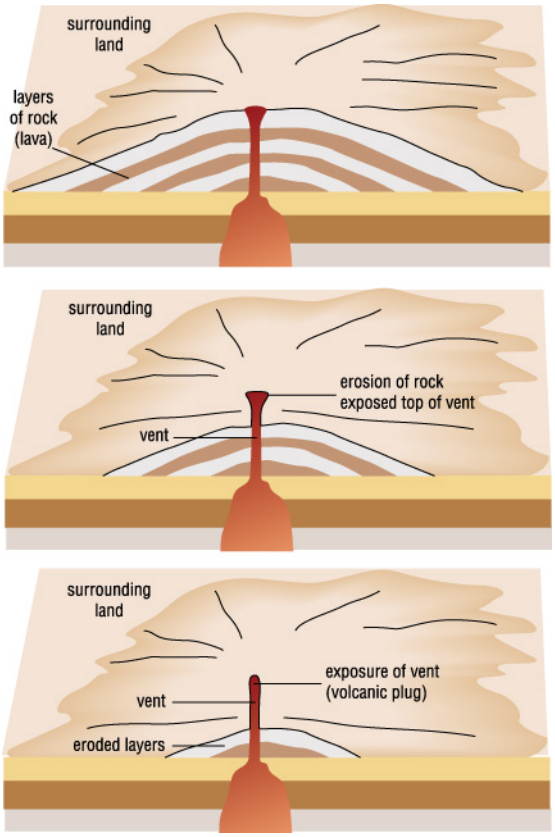
18



19 Students' responses will vary. Their presentations need to include the following information.

- tectonic theory due to convection currents
- climate will get warmer and more humid
- initially, an increase in earthquake, volcano and tsunami activity
- once a collision occurs, mountains will form and the destruction of plates will occur
- Sydney, Brisbane, Adelaide, Melbourne, Alice Springs and Perth are some cities that will not be impacted.

9.5 Practical activities

Prac 1: Faults and folds

Common mistakes

Soft malleable plasticine will assist this exercise. If the plasticine is too hard, place on a paper towel and microwave for a very brief time (too long and it melts).

Possible results

Students model faults in the land.

Suggested answer

1 *could be an anticline after erosion*

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Prac 2: Shaping volcanoes

Common mistakes

To maximise the shape of the volcano, students must accurately drop the wax onto the peak of the volcano.

Possible results

The candle wax volcano will have a larger angle of repose.

Suggested answers

1 *Part A should closely resemble the structure of a shield volcano due to the gently sloping sides. Part B should have steeper sides and resemble the structure of a composite cone or a cinder cone.*

2 *Various answers are possible, depending on the available materials.*

3 *Different substances have different properties that influence their ability to form steep slopes without assistance.*

4 *Bigger volcanoes will most likely have smaller angles of repose.*

5 *Various answers are possible.*

9.6 Answers

Remembering

1 4.5–4.7 billion years old

2 Cenozoic, Mesozoic, Palaeozoic, Proterozoic, Archaeozoic, Azoic

3 Triassic, Jurassic, Cretaceous

Understanding

4

Period	Span (millions of years)
Quaternary	2
Tertiary	63
Cretaceous	79
Jurassic	64
Triassic	40
Permian	42

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Period	Span (millions of years)
Carboniferous	72
Devonian	46
Silurian	30
Ordovician	67
Cambrian	65
Proterozoic	2130
Archaeozoic	800
Azoic	1100

5 Evidence of past life in a rock. It can be part of or a complete preserved animal, bones, teeth or shell, footprints or petrified minerals of bone, shell or wood.

6 **a** A dinosaur footprint is a fossil.
b Minerals may replace the shell or bones of an animal to make a fossil.
c Soft-bodied animals are less likely to form fossils than animals with shells or skeletons.
d Fossils are found under layers of sediments.
e Generally speaking, lower layers of rock in a region contain older fossils.
f Fossils of complete animals do exist.

Applying

7 Ancient insects have been found trapped in tree sap called amber.

8 lead

9 carbon-14

10 ammonite

11 **a** Carboniferous
b Cretaceous
c Silurian
d Archaeozoic
e Jurassic
f Jurassic

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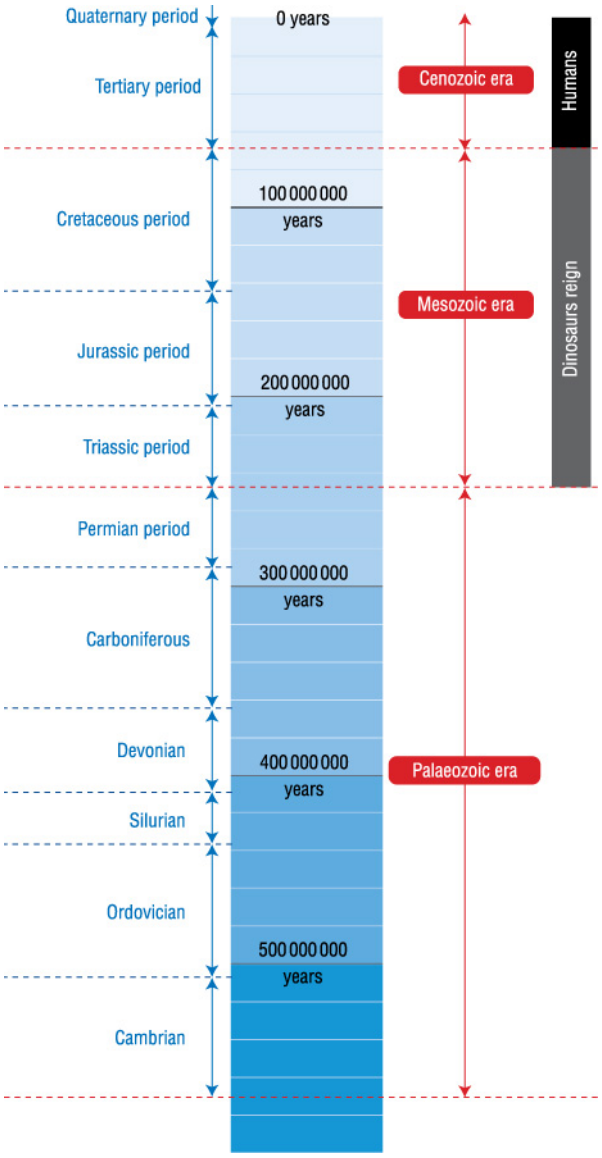
g Ordovician
h Proterozoic
i Jurassic
j Cretaceous

12 **a**,**b**



c The year range in the Proterozoic period alone is over 2000 years and it would be difficult to design a scale that would fit the whole geological timeline because the maximum range of years for the periods in the first three eras is approximately 60 years.

Evaluating

13 Yes it can, because radioactive dating can determine the ages of fossils as old as 100 million years and as young as 70 000 years.

14 environmental pressures, disease and meteorites

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15 During a geological event, such as a fault, movement in the Earth's crust may thrust layers of sedimentary rock containing older fossils upwards, over layers containing more recent fossils.

16 Fossils of marine biota can be found in areas far away from the ocean because of geological events. Over millions of years, divergent plates will move, causing sea floor spreading, whereby new land is formed and the older land is pushed further away. If the older blocks of crust contain fossils, after millions of years, the older crust can become exposed if it comes into contact with a convergent plate forming mountain ranges above the sea, or if it is thrust up by a fault or a fold.

17 plaster cast

18 Two smaller animals were walking together when one of the smaller animals was in a scuffle with a larger animal. The smaller animal was killed or eaten because its footprints do not continue. However, the other smaller animal walked away during the fight, as did the larger animal after the fight.

Creating

19 Students' responses will vary.

9.6 Practical activities

Prac 1: Inspecting fossils

Common mistakes

Some fossils may be fragile. Handle with care.

Possible results

Students make detailed drawings of fossils.

Suggested answers

1 *Animals decay on the surface, animal remains are eaten or scattered by scavengers, remains are crushed by sedimentary layers, remains are in the wrong type of rock, or fossils are destroyed by later folding of rock.*

2 *plaster cast*

3 *Jelly fish, as they do not have a skeleton; all herbaceous plants (non-woody plants) which will decompose before they can mineralise.*

Prac 2: Dinosaur fossils

Common mistakes

Ensure that the plaster mixture is of a thick consistency before pouring into the mould.

Possible results

Students will create impressions in the clay that will produce a recognisable dinosaur cast.

Suggested answers

1 *sedimentary rock*

2 *mould: clay; cast: plaster*

3 *footprint: mould; bone: cast*

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Prac 3: 'Radioactive' cubes

Common mistakes

Rolling the cubes on a tray with sides will contain the cubes to a small area and prevent them from rolling onto the floor.

Possible results

Students model the half-life radioactive decay of uranium.

Suggested answers

1 *The curve should look similar to the one shown in Figure 9.6.13.*

2 *Answer will vary depending on chance, and will be different for different groups.*

3 *Answer will vary depending on chance, and will be different for different groups.*

4 *The half-life of uranium-235 is 700 million years, and that for uranium-238 is 4.5 billion years.*

Chapter answers

Remembering

1 Alfred Wegener

2 See bullet points on pages 281–282 of this chapter for evidence to support Alfred Wegener's theory.

3 The Earth's crust is divided into plates that move by convection currents in the Earth's mantle.

4 200 kilometres

5 radioactive substances in rocks

Understanding

6 The soup creates convection currents where the hot liquid rises to the top and, as the liquid touches the cooler toast, it cools down and then falls back down to the bottom of the bowl; this is a continual process. In this movement, the toast on top of the soup moves in the direction of the convection current.

7 **a** As lava from a volcano cools, it adopts the current magnetism of the Earth. Three hundred million-year-old magnetic rocks in South and North America have been found with their poles pointed in different directions. If the continents are joined the magnetic poles will all point to the same spot.

b The rocks on the ocean floor have magnetic stripes parallel with underwater ridges, which is the location of sea floor spreading.

8 The hot molten rock rises and when it comes into contact with the relatively cold crust it cools and sinks. This process results in convection currents in the mantle. The convection currents carry the tectonic plates with them.

9 As tectonic plates separate, collide and scrape over each other, the plates build up pressure to overcome the friction forces that cause them to stick. The pressure release is sudden and explosive, resulting in an earthquake.

10 • Two plates can press against each other until the land is lifted and folded over itself.

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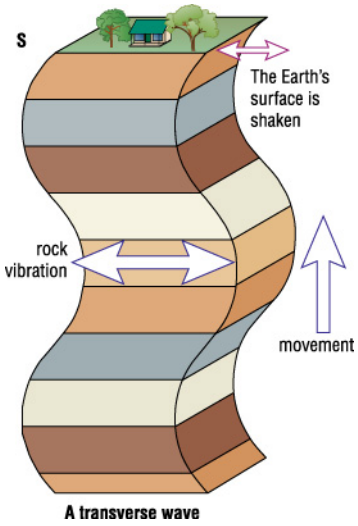
- Severe folding of rock layers, as in the case of Uluru.
- In the ocean, underwater mountains can form when plates spread away from one another, and melted rock pushes up through the gap between them.

11 magma, clouds of steam, gas (fumes: hydrogen sulfide), smoke

12 **a** When tectonic plates slip past each other, they build pressure to overcome the friction that causes the plates to 'stick'. The release of this pressure leads to an earthquake.

b Slipping plates expose cracks in the crust and if these cracks are directly above hot spots, volcanoes will appear.

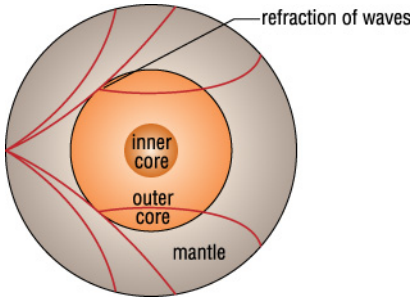
13



14 Magma is molten rock that forms in a magma chamber deep under the surface of the Earth. Magma sometimes squeezes out through cracks in the plates as they move and, at other times, intense pressure builds up, so the magma explodes out through vents.

15 When liquids and gases gain heat they rise. As a room is filled with air (gas), the hot air rises to the ceiling to be cooled. Once it has cooled it falls back down to the floor.

16 As P and S waves pass through rocks of different densities, they change speed and direction; this is known as refraction.



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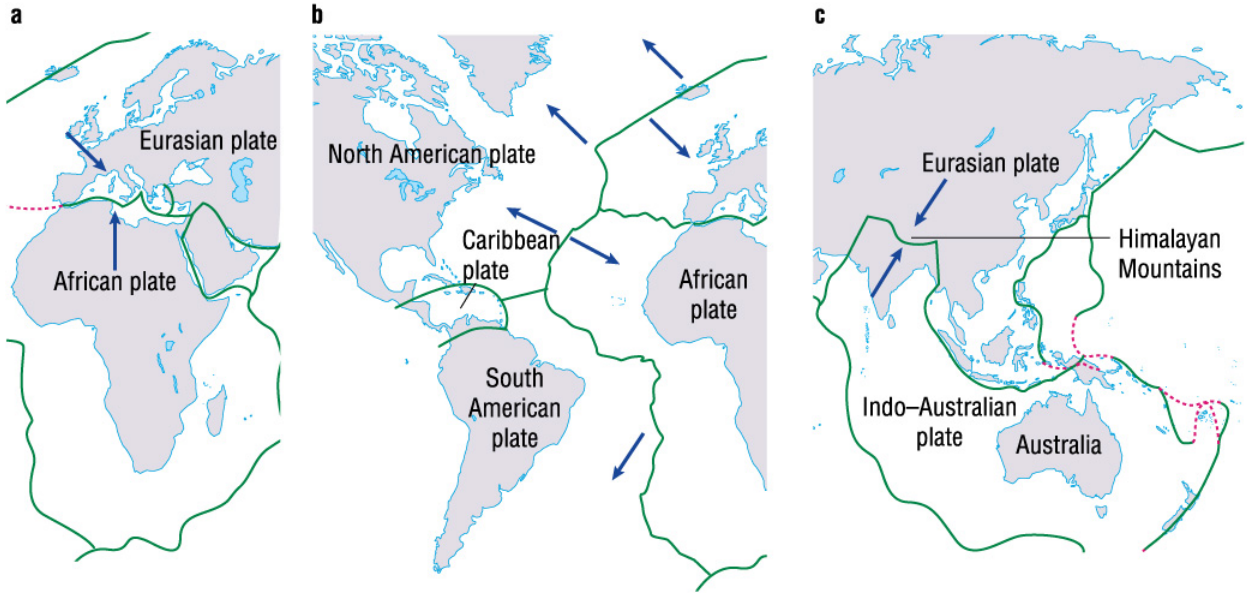
17



18 Due to the extreme pressure and heat, volcanic ash forms a fast-moving cloud that gets carried by winds and is spread across the atmosphere.

19 The Pacific Ocean will widen because the Pacific and Nazca plates are moving apart. The Indo–Australian plate will be further north, possibly having destroyed some of Asia because the Indo–Australian and Eurasian plates are colliding further into each other. North and South America could separate because their plates are also moving apart.

Applying

20 P and S waves are detected by a seismometer and the P waves will be detected first.

21 **a** S waves
b P waves
c R waves

22 **a** True
b True
c True
d False
e True
f False
g True

23 The longest mountain ridge: underwater along divergent boundaries.
The highest mountain ridge: Himalaya.

24 **a** Animals with hard exoskeletons or shells, especially marine organisms.
b In order for fossils to form they need a quick burial in sediments that will not be exposed to weathering.

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25 **a** Palaeozoic
b Palaeozoic
c Cenozoic
d Mesozoic

26 Any one of the following for each.
a fish, first land animals, amphibians, insects, corals, molluscs, fungi and tree-sized land plants
b algae and bacteria
c any animal currently on Earth

Analysing

27 The theory of continental drift proposes that the world was made up of a single continent through most of geological time. Eventually, over time, these continents drifted apart, forming into the seven continents we currently know today. According to the plate tectonics theory, scientists believe that the Earth's surface is broken into a number of plates, which move relative to one another due to convection currents.

28 Faults are fractures in the Earth's crust caused by extreme forces from the slow movement of rock in the asthenosphere. Folds, in contrast, form when continental plates collide, subjecting the crust to extreme pressure horizontally and vertically and causing the rocks to act like clay as they bend without snapping under the pressure.

29 **a** The Earth's mantle is solid, but it can still flow very slowly because the convection currents within the lower mantle cause the solid components to move.
b tea leaves in a cup of tea

30 The denser oceanic plates are forced under the lighter continental plates when the two collide. The plate with the greater density (oceanic) sinks into a subduction zone within the mantle, leaving lighter and less dense plates to float. When plates with similar densities collide, they push up together until they crumble and fold to form mountains.

Evaluating

31 Pangaea is a good name for the original supercontinent because 'all earth' refers to all the continents ('land') that were joined together. Pangaea split into its 'offspring or children', which are called Gondwana (Australia, Antarctica, South America, Africa and India) and Laurasia (North America, Europe and most of Asia).

32 The mysteries of the ocean floor weren't discovered until the late twentieth century because technologies such as submarines that could reach the depths of trenches were not yet developed.

33 A conveyer belt moves objects along; similarly, the ocean floor spreads to move and, in the process, pushes blocks of crust further away.

34 Student responses will vary. An example is shown below.

'P' is for primary, which means first, so they come first. 'S' is for secondary and these waves come second. 'R' is for Rayleigh and these are rolling waves and 'L' (Love waves) is for 'like a snake'.

35 Magma is less dense than the surrounding rock and full of gas. The pressure from the gases and its lower density allows the magma to rise.

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36 In a shield volcano, the lava slowly slides down the sides all the way to the bottom until it cools and hardens, forming gentle slopes. In the case of cinder volcanoes, rocks and ash spew out; these are usually deposited closer to the top, forming steeper slopes.

Creating

37 It is thought that the Earth cannot be 6000 years old because scientists have found and have used radioactive dating to determine that tiny sand grains in Western Australia are 4.25 billion years old. Over the years, scientists have used radioactive dating to determine the ages of various living and non-living components of the Earth.

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1 Refer to Fig 1.1.8 in *Science Focus 3* Second Edition. This is a diagram of each fingerprint type.

2 Students should collect three fingerprints.

3 The prints are stuck onto a black surface because the dust is white, allowing us to better see the patterns.

4 It is easier to see the prints on some surfaces than on others. White powder used on a white surface can be very hard to see. It is easier to see prints of white powder on dark-coloured surfaces. Students should give examples that they found.

5 Carbon powder is black, so would be used for dusting light-coloured or white surfaces. Using carbon powder makes it easier to see prints in situations when it is not easy to see them with white powder.

6 A latent print is caused by the oils and sweat from your skin, which contain salts, water and amino acids, remaining on anything you touch.

Extension

7 Aluminium powder is used for dark and light surfaces and glass. Lanconide powder is used for smooth dark-coloured surfaces.

8 Place the piece of paper that you suspect has a print on it into a jar with iodine. If there is a print the iodine vapour will stick to the print and make it visible. (There are other techniques that may produce the same outcome.)

9 Minutiae are the points at which the ridges in the fingerprint split or end. These features are unique and help to identify each person's fingerprints. The two main types of minutiae are ridge endings and bifurcations, where a ridge splits.

10 Fingerprints begin to form when a foetus is approximately 10–12 weeks old. This occurs when the deeper layers of the skin grow faster than the skin on the surface of the fingers. This difference in growth rate causes the skin on top to buckle and form the ridges and valleys that make up the fingerprints. The actual pattern is partly inherited, but the patterns also vary as the foetus moves and presses on the womb or on itself while fingerprint formation is occurring. This pressure on the fingerprints changes the pattern slightly, meaning that even identical twins do not have exactly the same fingerprints.

Answers 1.2 Time of death

1 **a** most in summer, least in winter

b A dead body would contain a large number of species 4 and a small number of species 3 in winter.

2 **a** 16 days

b 8 days

3 flies

4 **a** bed sores = 3 days, eyes/ears = 1 day

b Eggs are laid in the sores before the person died.

c It is possible, as the flies developed in the last 3 days.

5 **a** The minimum number of days would have been 2 to 3.

b The minimum number of days could be 7.

6 The body was only recently placed in the freezer, as species 1 and 2 are most active in late spring/early summer or the body had been in the freezer since last summer.

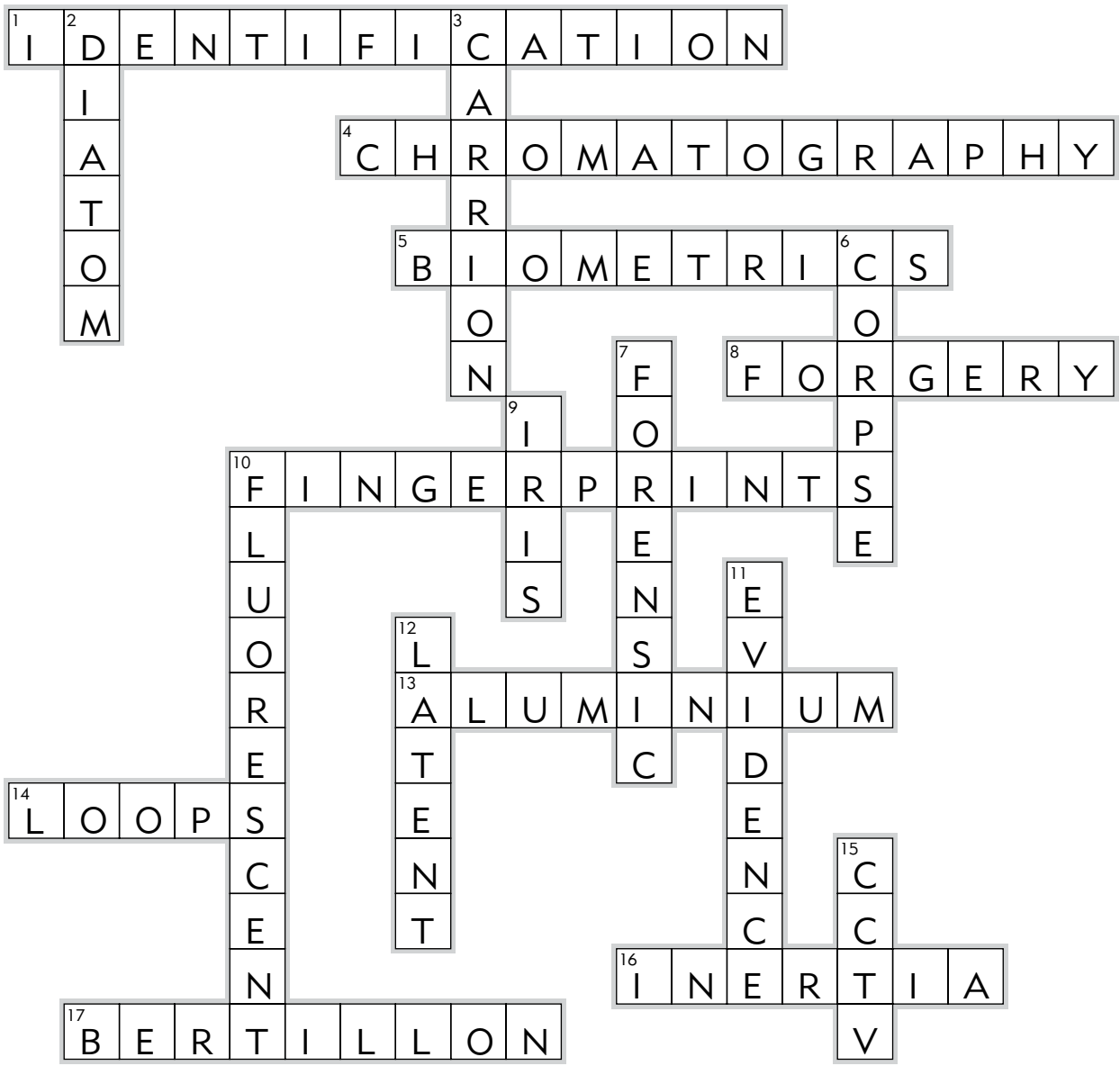
7 **a** blowflies, house flies, carrion beetles, rove beetles

b The body could have been in a location where insects are unable to access it, such as in a locked room, or in a very dry location where fewer insects are found. In each case, insects would not be able to successfully colonise the body.

Answers 1.3 Careers

Student responses will vary.

Answers 1.4 Crossword



Unit 1.1: Forensics and identification

Clue	Word
1 A chemical that contains your genetic code	DNA
2 A specialist in the study of disease and the damage it does to organs and tissues	pathologist
3 Information that has legal value in a court of law	evidence
4 Identification by measurements of body parts	anthropometry
5 The science of measuring biological features	biometrics
6 Interference that confuses the memory	retroactive
7 A 'cut-and-paste' method of identification	identikit
8 A type of fingerprint	whorl
9 Part of the eye that can be scanned for identification	retina
10 Set of tests that measures the DNA code	DNA profiling

Unit 1.2: Is it real?

Clue	Word
1 Angled lighting	oblique
2 Glow	fluoresce
3 Raised printing	intaglio
4 Marks that can only be seen when held up to the light	water marks
5 A way of identifying inks	chromatography

Unit 1.3: Evidence

Clue	Word
1 Absorbent	porous
2 A type of microscope used to compare bullets and fibres	comparison
3 Evidence that is not definite	circumstantial
4 A type of tool mark	striation
5 Made by shoes or tyres	impressions
6 Single-celled organisms found in water	diatoms
7 Many individual images	frames

1 **a** B inside a circle
b K inside a circle
2 **a** can include hydrogen, oxygen, nitrogen, sulfur, iodine etc.
b can include helium, lithium, neon, aluminium etc.
3 **a** They consist of the first letter and the first consonant of their names.
b They consist of the first letter and the first consonant not in common.
4 **a** Cp
b Po
c So
5 Their symbols are derived from either their old name or their Latin name.
6 **a** The first two letters of its old name, Natrium, were used.
b The non-metal, nitrogen, according to the rules, would be N.
7 helium (He) – Greek helios for Sun; uranium (U) – Uranus; neptunium (Np) – Neptune.
8 mendelevium (Md)
9 ununoctium (Uuo)

Answers 2.2 Who am I?

Name		Symbol	Description
1	Hydrogen	H	I am the first element in the periodic table.
2	Iron	Fe	You also do this to clothes to take the creases out.
3	Arsenic	As	I am a very old poison.
4	Neon	Ne	Tubes used for light signs are often filled with me.
5	Calcium	Ca	Dairy products such as milk, cheese and yoghurt contain lots of me and I am needed for bone strength.
6	Krypton	Kr	Superman comes from this planet.
7	Copper	Cu	My name is also an old-fashioned nickname for a police officer.
8	Californium	Cf	I am named after a state of the USA.
9	Curium	Cm	I am named after an important female scientist.
10	Lead	Pb	My ancient Latin name was *plumbum*.
11	Neptunium	Np	I am named after another planet in the Solar System.
12	Fluorine	F	Compounds of me are added to water to help prevent tooth decay.
13	Francium	Fr	I am the most metallic of the elements.
14	Fluorine	F	I am the most non-metallic of the elements.
15	Bromine	Br	I am a halogen that is a red liquid at room temperature.
16	Chlorine	Cl	I was first successfully separated by Scheele in 1774.
17	Beryllium	Be	I am the alkaline earth with a boiling point of over 2000°C.
18	Rubidium	Rb	I am the alkali metal that would be most likely to melt on a very hot day.
19	Carbon	C	Graphite and diamond are two of my allotropes.
20	Helium	He	I was discovered on the Sun before I was discovered on Earth.
21	Bromine	Br	The element that reacts to form bromide ions.
22	Oxygen	O	The element that reacts to form oxide ions.
23	Lithium	Li	The first of the alkali metals.
24	Thorium	Th	The first of the actinides.
25	Scandium	Sc	The first transition element.
26	Germanium	Ge	The metalloid element in Group IV.
27	Xenon	Xe	The heaviest of the inert gases.
28	Calcium	Ca	The third element in the alkaline earths.
29	Iodine	I	A halogen that is sometimes used as an antiseptic on wounds.
30	Chlorine	Cl	The halogen in table salt.
31	Sodium	Na	The alkali metal in table salt.
32	Fluorine	F	The smallest of the halogens.
33	Silicon	Si	The element that would show chemical properties most similar to carbon.
34	Sulfur	S	An atom that would show very similar chemical properties to oxygen.
35	Calcium	Ca	I have an atomic number of 20.
36	Boron	B	My atomic number is 3.
37	Antinomy	Sb	I am the first of the metals in Group V.
38	Radium	Ra	The biggest of the atoms in Group II.
39	Arsenic	As	I am in Period 4, Group V.

Answers 2.4 Periodic table properties

1

Period	Group	Atomic number	Symbol	Element	Melting point (°C)	Boiling point (°C)	Diameter of atom (Angstrom)
1	I	1	H	Hydrogen	–	–	0.60
	II	2	He	Helium	–	–	1.86
2	I	3	Li	Lithium	181	1342	3.04
	II	4	Be	Beryllium	1278	2970	2.22
	III	5	B	Boron	2200	3927	1.76
	IV	6	C	Carbon	3500	3800	1.54
	V	7	N	Nitrogen	–211	–196	1.40
	VI	8	O	Oxygen	–219	–183	1.32
	VII	9	F	Fluorine	–220	–188	1.28
	VIII	10	Ne	Neon	–249	–246	2.24
3	I	11	Na	Sodium	98	883	3.72
	II	12	Mg	Magnesium	649	1107	3.20
	III	13	Al	Aluminium	660	2467	2.86
	IV	14	Si	Silicon	1410	2355	2.34
	V	15	P	Phosphorus	44	287	2.20
	VI	16	S	Sulfur	119	445	2.08
	VII	17	Cl	Chlorine	–101	–35	1.98
	VIII	18	Ar	Argon	–190	–186	3.08
4	I	19	K	Potassium	63	760	4.62
	II	20	Ca	Calcium	839	1484	3.94

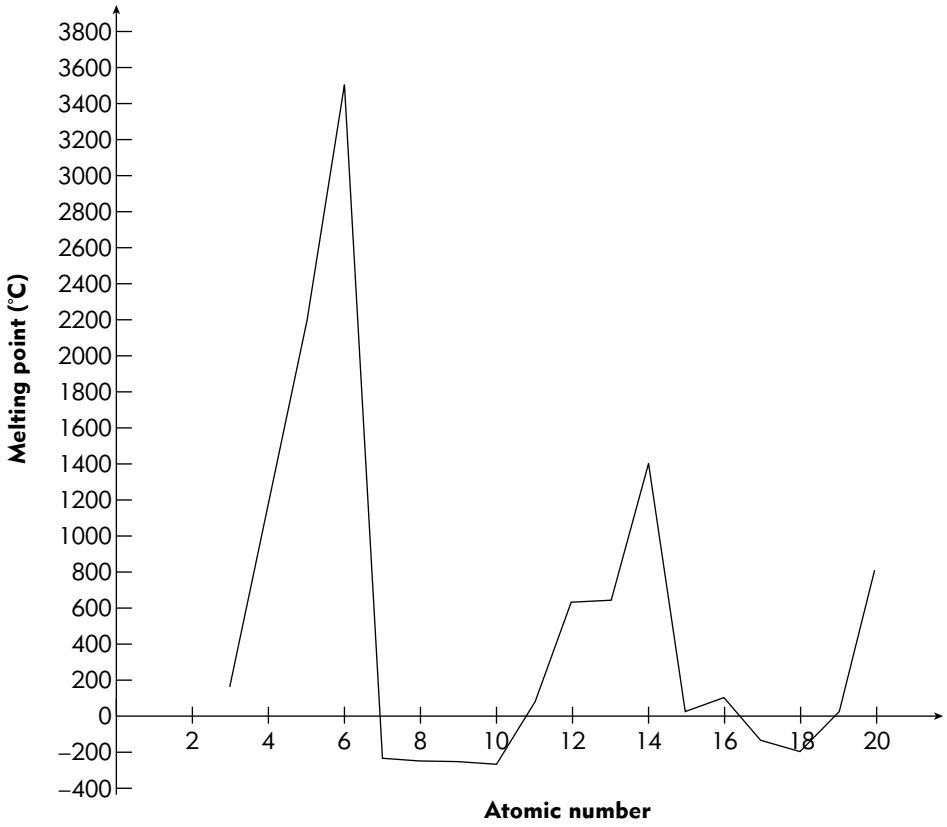
2 **a** alkali metals
b alkaline earths
c halogens
d inert gases or noble gases

3 Atomic number is the number of protons in the nucleus of an atom.

Answers 2.3 Ions

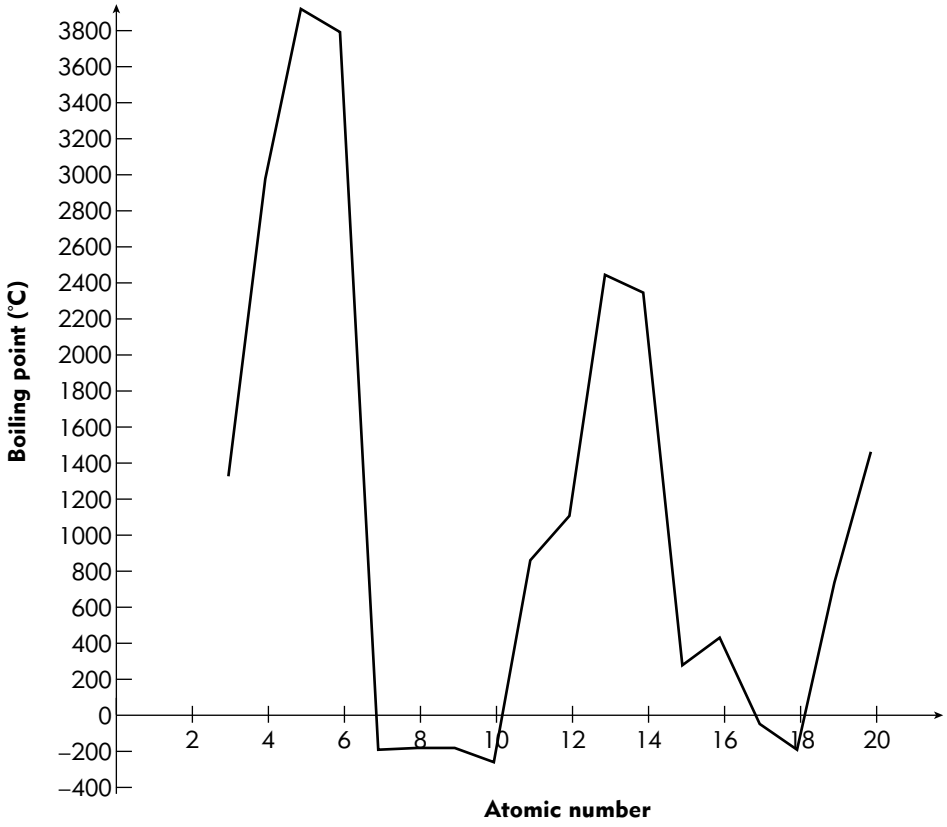
Worksheet supports Prac 2, Unit 2.3 in *Science Focus 3* Second Edition.

4



5 The graph shows a series of peaks and troughs. The bottoms of the troughs coincide with the noble gas elements (atomic numbers 2, 10 and 18). The peaks coincide with the Group IV elements (C and Si).

6



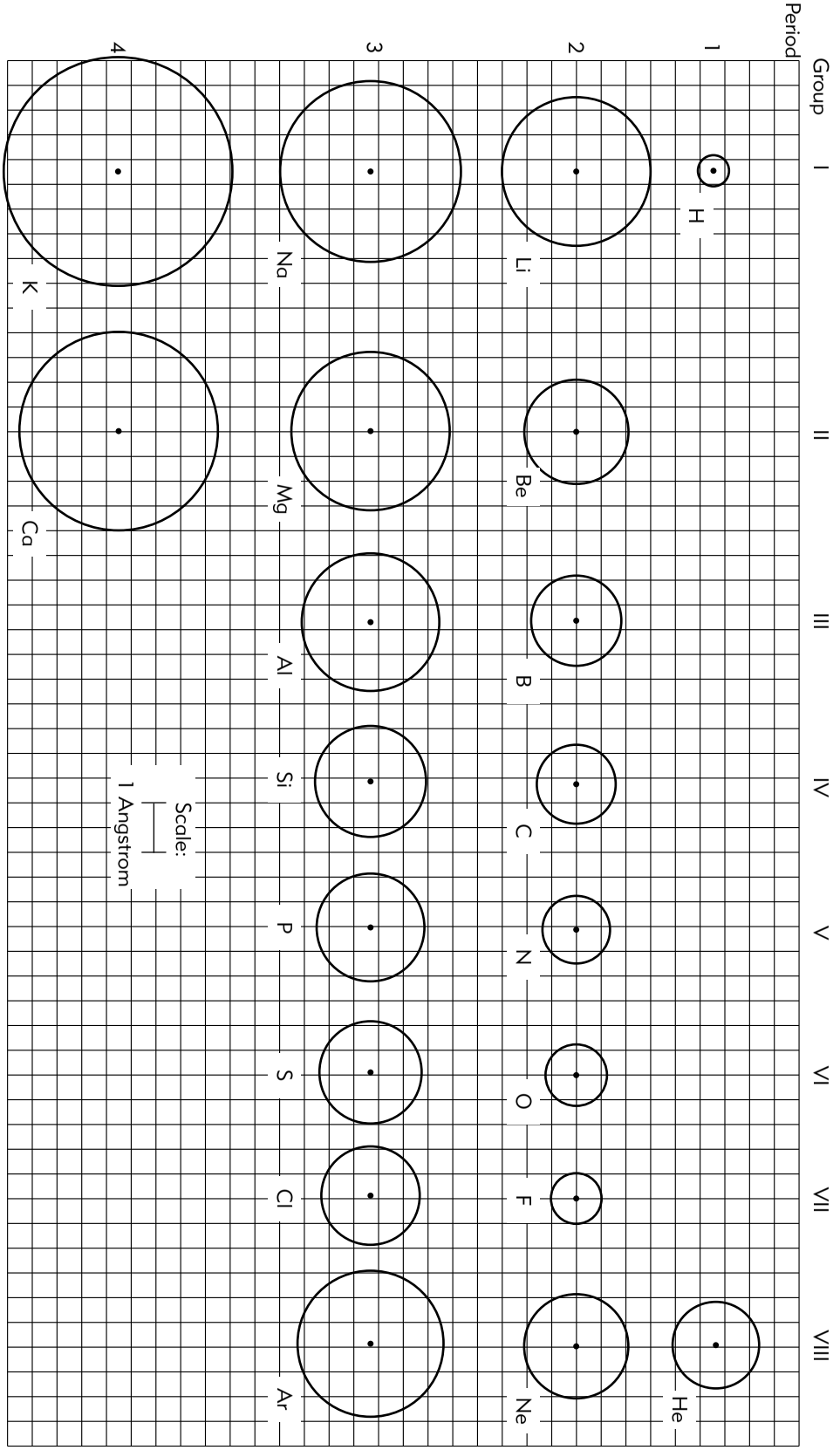
7 Once again, the graph shows a series of peaks and troughs, the lowest point of the graph coinciding with the noble gases. The highest point of the peaks is now in Group III, only marginally higher than the points for the Group IV elements.

8 Atoms get smaller as we move across the first seven groups of Period 2. The Group VII element, Ne, reverses this trend, however, as it is bigger than its neighbour.

9 The same pattern is shown in the sizes of the elements in Period 3—atoms get smaller until we get to Group VIII, which is bigger than its neighbour.

10 The sizes of the atoms should increase as we move down a group, because more shells are being added to the atoms.

Extension


Period
Group
I
II
III
IV
V
VI
VII
VIII
1
2
3
4
H
He
Li
Be
B
C
N
O
F
Ne
Na
Mg
Al
Si
P
S
Cl
Ar
K
Ca
Scale:
1 Angstrom

Answers 2.5 The periodic table

1 Diagrammatic answer is required. Refer to Fig 2.2.2 in *Science Focus 3* Second Edition for support.

2 Group VIII

3 halogens

4
a	Platinum	78
b	Gold	79
c	Oxygen	8
d	Magnesium	12
e	Californium	98
f	Bromine	35
g	Krypton	36

5
a	6	Carbon
b	3	Lithium
c	40	Zirconium
d	13	Aluminium
e	92	Uranium
f	56	Barium
g	1	Hydrogen

6 **a** any three of carbon, chlorine, cobalt, copper, chromium, calcium, cesium, cerium, californium

b C, Cl, Co, Cu, Cr, Ca, Cs, Ce, Cf

c All elements must have a unique symbol. Therefore, only one 'c' element can start with one letter.

7 Elements in the same group share common properties.

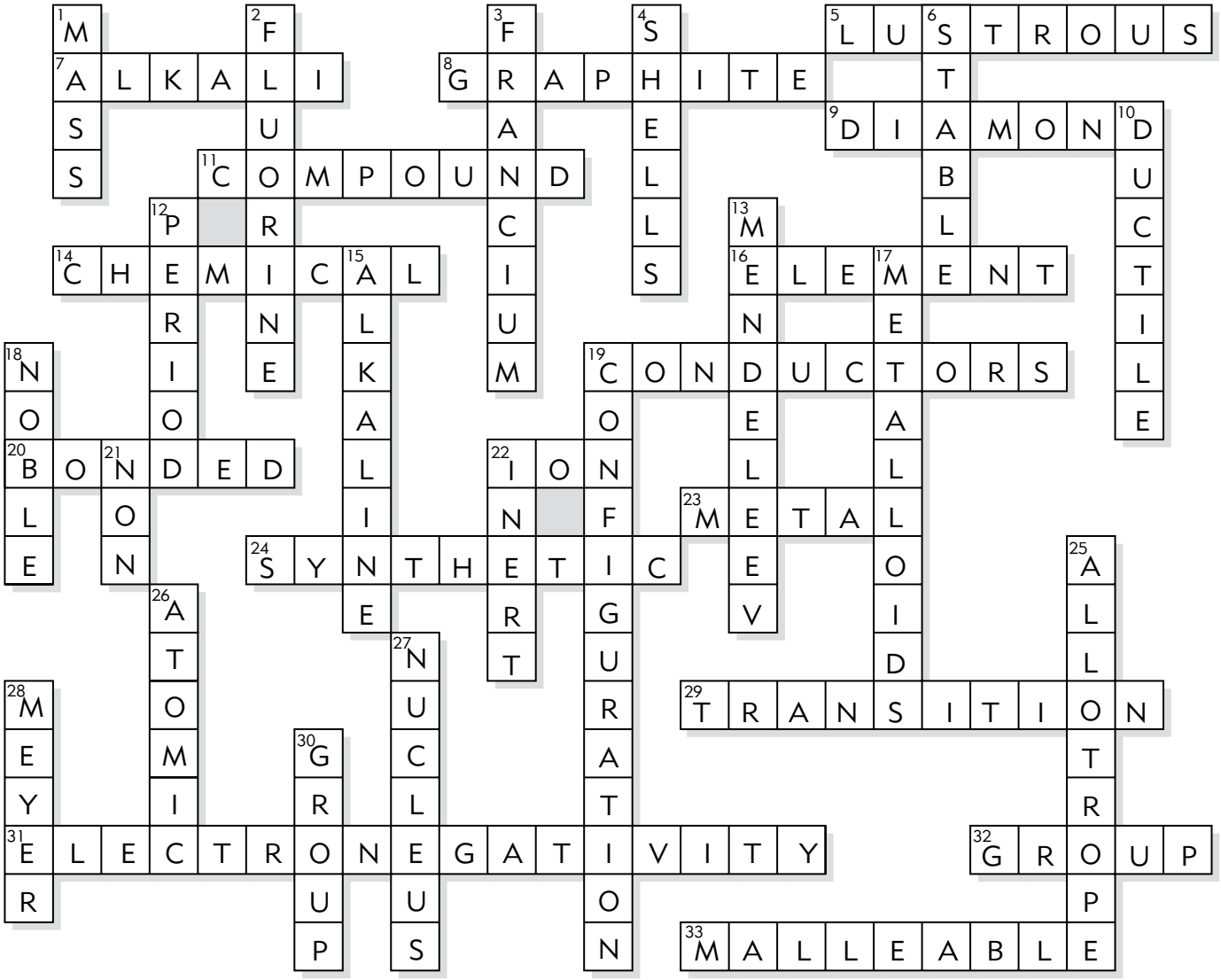
8 Atomic number tells us the number of protons that are in the nucleus of an element.

9 Any three of rutherfordium—Rutherford; nielsbohrium—Niels Bohr; curium—Curie; einsteinium—Einstein; fermium—Fermi; mendelevium—Mendeleev; nobelium—Nobel; seaborgium—Seaborg; lawrencium—Ernest Lawrence (inventor of the cyclotron).

Extension

Student responses will vary.

Answers 2.6 Crossword



Unit 2.1: Atoms and elements

Clue	Word
1 The Greek word for indivisible	atomos
2 The number of protons	atomic number
3 The number of protons and neutrons	mass number
4 Only made of one type of atom	element
5 Two or more elements combined	compound
6 English scientist who gave us the atomic theory	Dalton
7 Atoms joined together are said to be _____	bonded
8 Different atoms joined together	molecule
9 Contains protons and neutrons	nucleus

Unit 2.2: Structure of the periodic table

Clue	Word
1 Horizontal row of the periodic table	period
2 Vertical column of the periodic table	group
3 Made in the laboratory	synthetic
4 Melting point, boiling point, colour etc.	physical properties
5 How an element reacts	chemical properties
6 Energy levels	shells
7 The way electrons are arranged	electron configuration
8 Number of electrons in the outer shell	group number
9 Number of shells	period number
10 Noble gases	inert gases
11 Rarely react	stable

Unit 2.3: Using the periodic table

Clue	Word
1 An atom that has lost or gained an electron	ion
2 Shiny	lustrous
3 Can be bent	malleable
4 Can be drawn into thin wires	ductile
5 These can pass heat or electricity	conductors
6 Holding power of an atom for electrons	electronegativity
7 80% of the elements	metal
8 Mainly gases and liquids	non-metals
9 Another name for semi-metals	metalloids

Unit 2.4: Families of elements

Clue	Word
1 Group VIII	noble gases
2 Group VII	halogens
3 Group I	alkali metals
4 Group II	alkaline earths
5 Some of our most useful metals	transition metals
6 Different forms of the same element	allotropes
7 A slippery form of carbon	graphite
8 A hard form of carbon	diamond

Answers 3.1 Chapter 3: Chemical Change
Ionic compounds

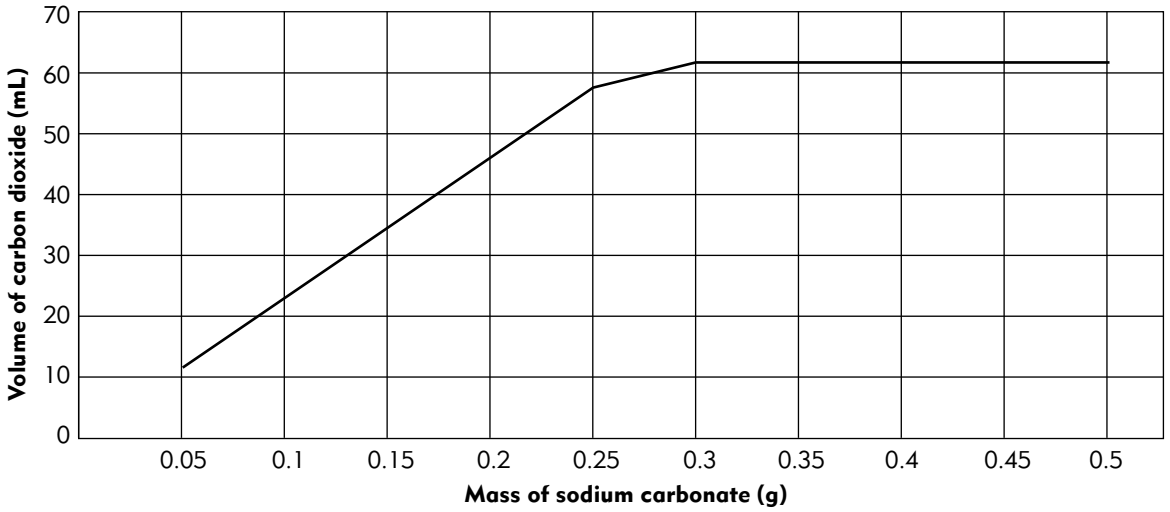
1 **a** magnesium chloride
b ammonium oxide
c calcium nitride
d copper(II) carbonate
e chromium(III) sulfate
f chromium(III) nitrate
g aluminium hydroxide

2 **a** Li_2CO_3
b $AlBr_3$
c $Cu(NO_3)_2$
d NaCl
e $CaBr_2$
f NH_4OH
g Li_3N

3 **a** CrO_4^{2-}
b **i** $MgCrO_4$
ii $Al_2(CrO_4)_3$

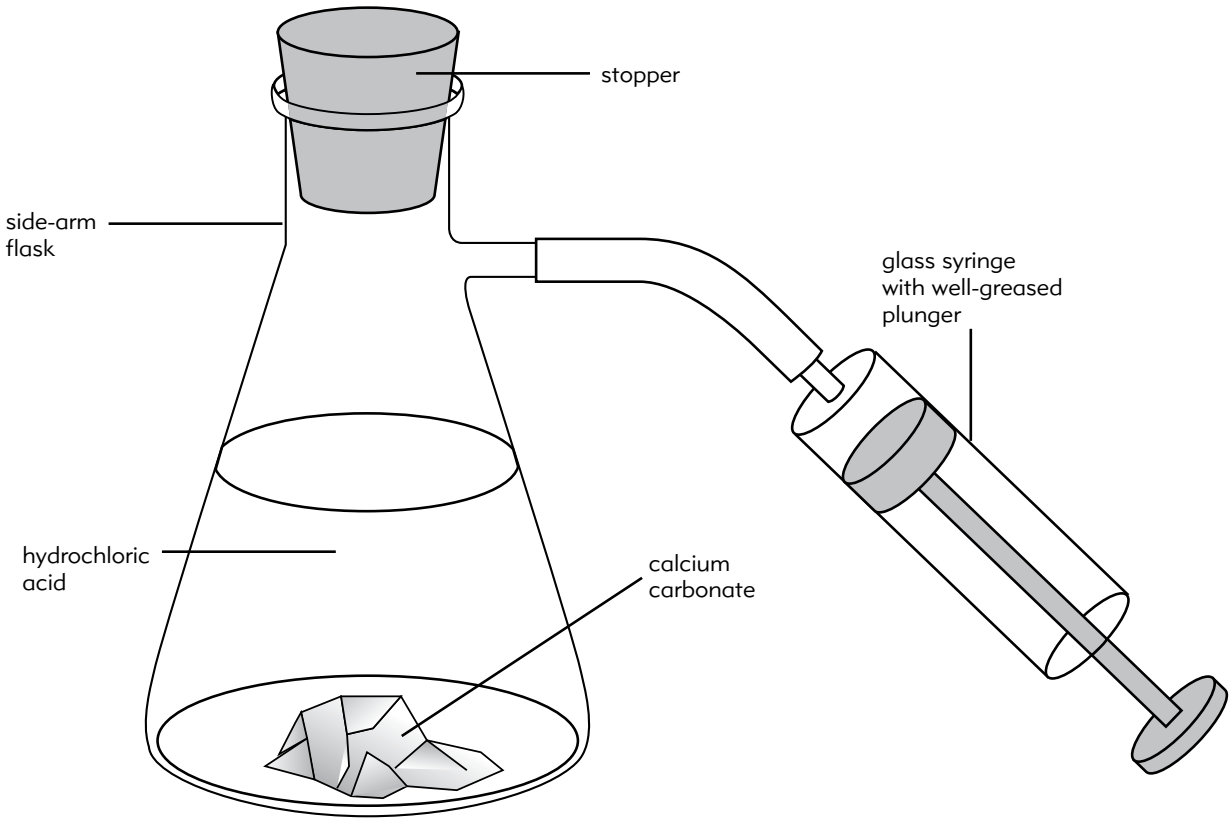
Answers 3.2
Neutralisation

1



2 From the graph, 0.27 g of sodium carbonate.

3



Answers 3.3 pH levels

1 **a** cola

b milk

2 any three of water, milk, coffee, tea

3 **a** The probe was washed after each test to remove any drink from the probe that might have contaminated the next test.

b The water was re-tested after washing the probe because if the previous drink was not fully washed off the probe it would record a pH for water that was less or greater than 6.49.

4 Any two of the following:

- the probe still had acid on it
- the probe was faulty
- the water sample was contaminated.

5 Any two of the following:

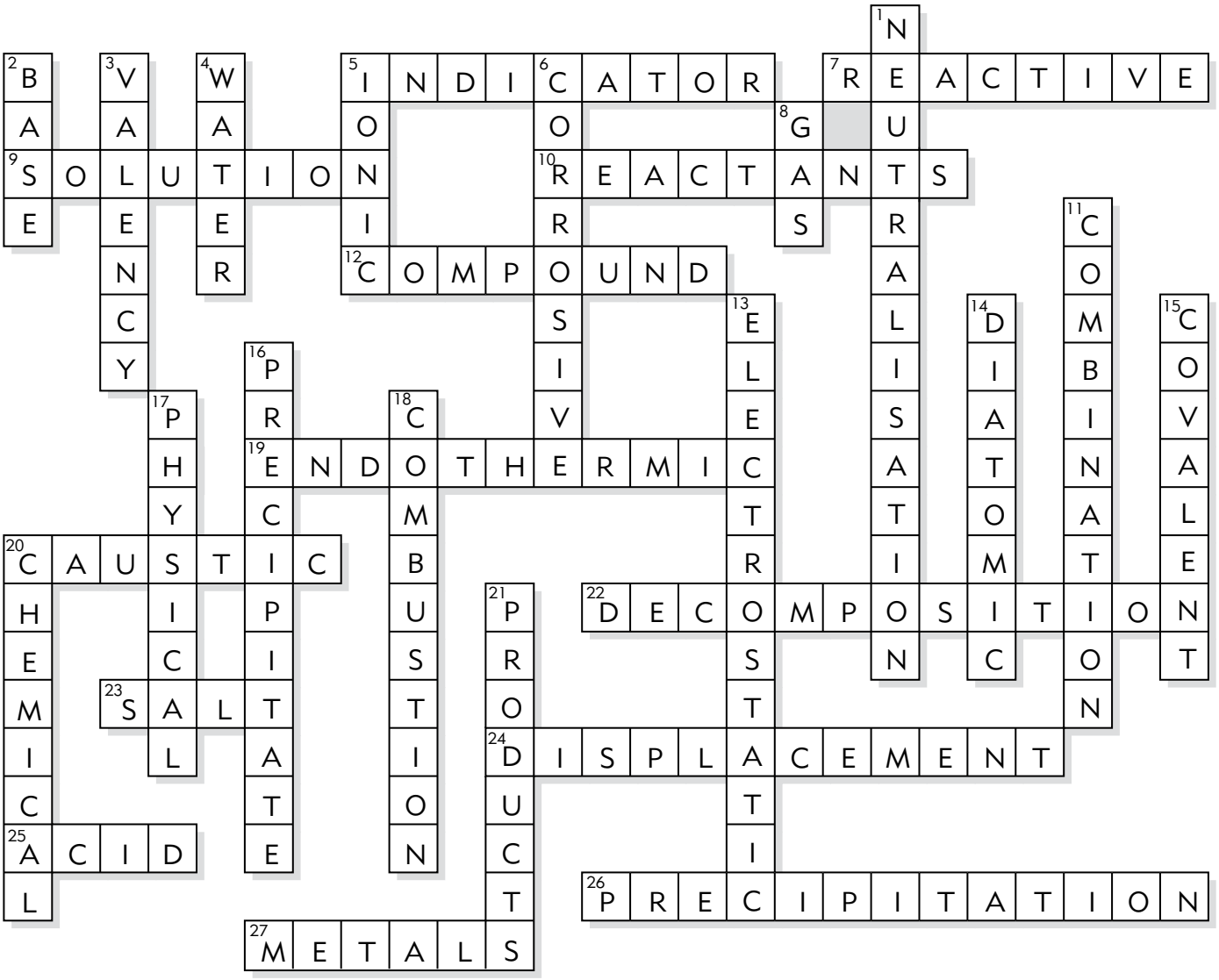
- more than one drink should have been sampled
- the temperature should have been kept constant
- distilled water should have been used for the control.

6 Students' answers will vary.

Extension

Student responses will vary.

Answers 3.4 Crossword



Unit 3.1: Chemical reactions

Clue	Word
1 Sign of chemical change	gas
2 Substances present at the end of a reaction	products
3 Substances present at the start of a reaction	reactants
4 Results in at least one new substance	chemical reaction
5 Does not result in a new substance being formed	physical change
6 Reaction that absorbs energy	endothermic
7 Solute + solvent = _____	solution

Unit 3.2: Combination, combustion and decomposition

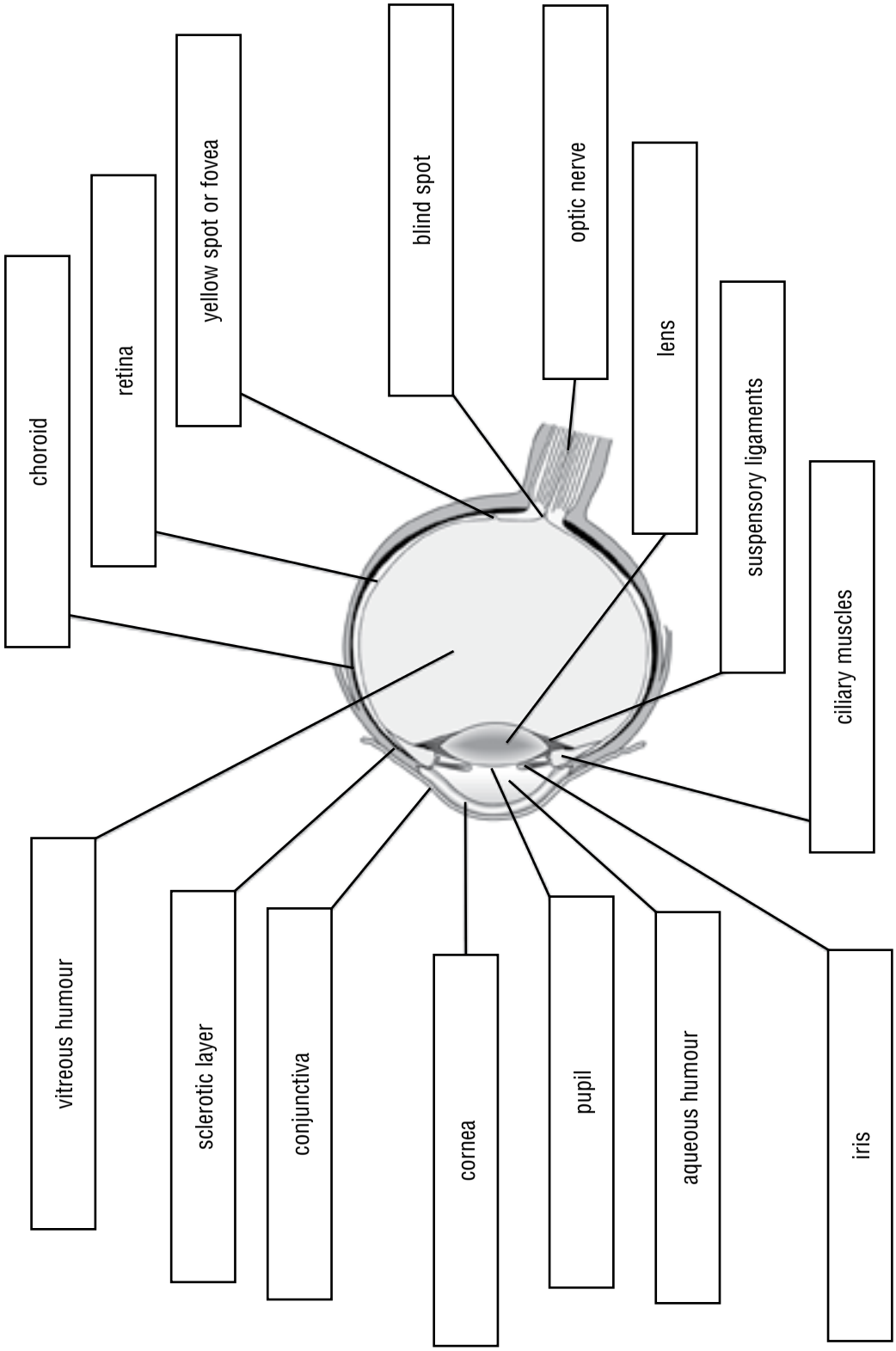
Clue	Word
1 Reaction that always gives one product	combination
2 Molecule made up of two atoms	diatomic
3 Reaction in which one substance breaks down	decomposition
4 Oxygen is always a reactant in this reaction	combustion

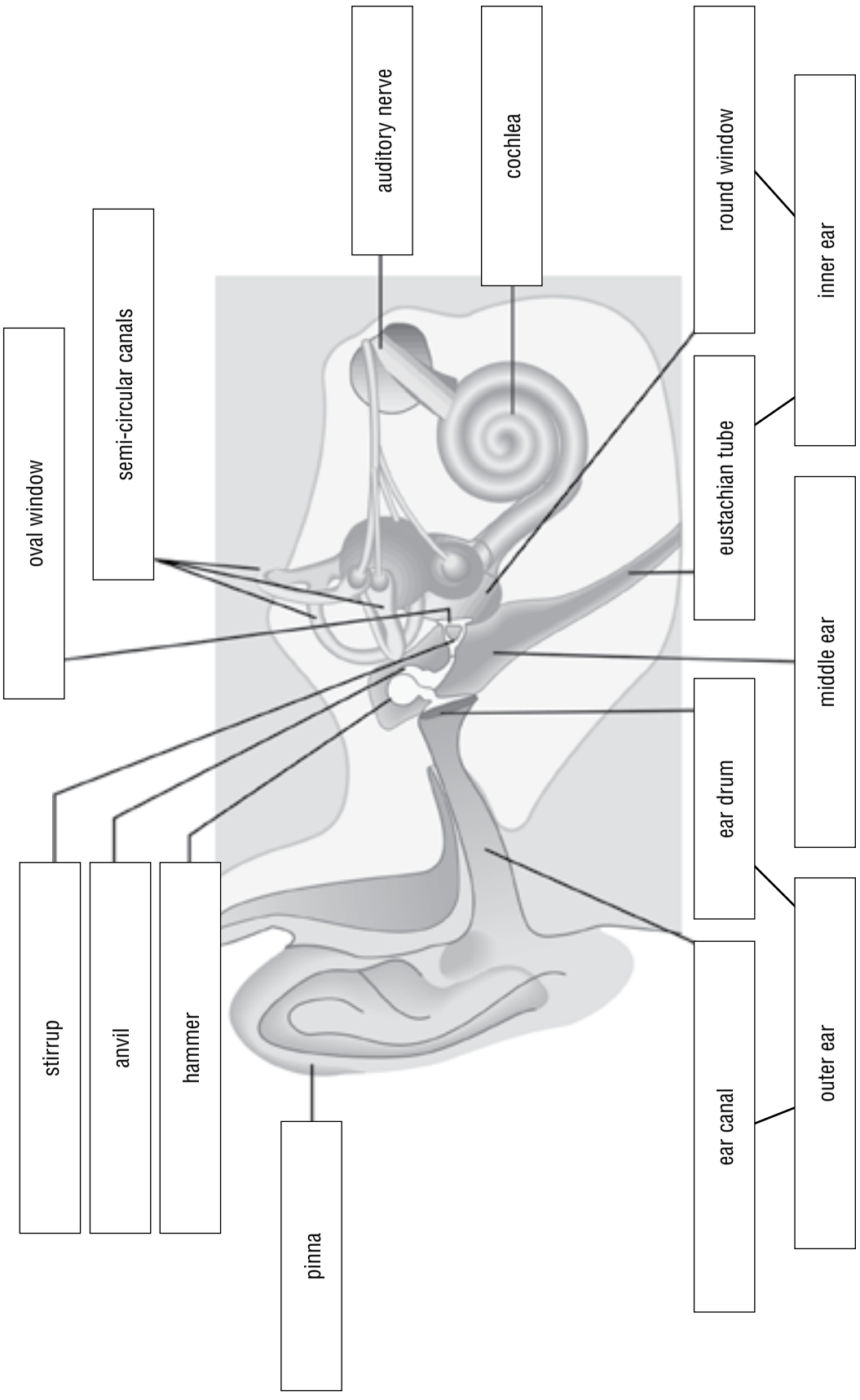
Unit 3.3: Precipitation reactions

Clue	Word
1 Insoluble salt forming in solution	precipitation
2 Another name for an ionic compound	salt
3 Made up of positive and negative ions	ionic compound
4 Contain only one atom that has lost or gained electrons	simple ions
5 Always form negative ions	non-metals
6 Bonding in which electrons are shared	covalent
7 Distribution of electrons around the nucleus	electronic configuration

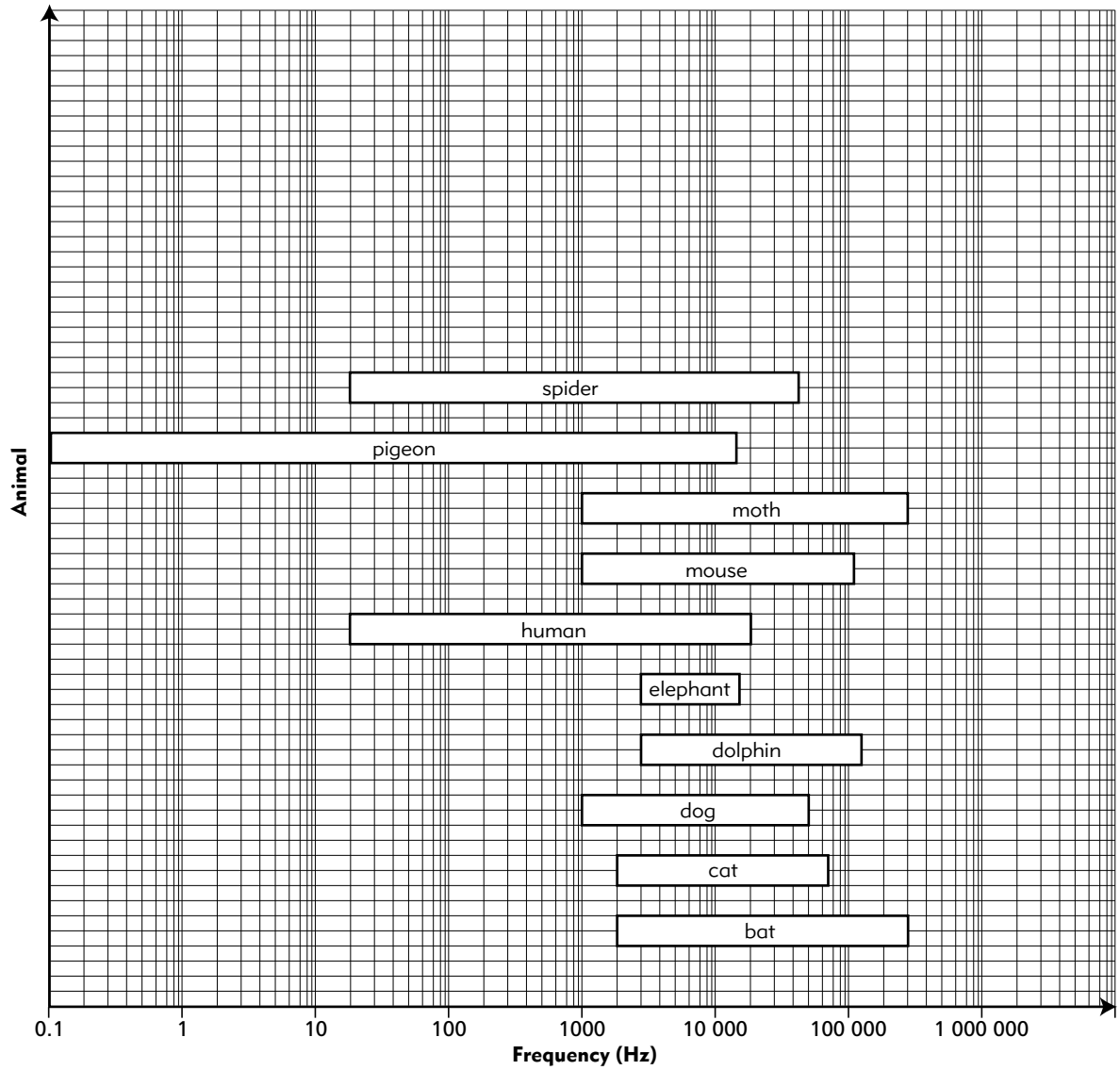
Unit 3.4: Acids and bases

Clue	Word
1 Produces hydrogen ions in solution	acid
2 Neutralises an acid	base
3 Strong acids have this property	corrosive
4 Strong bases have this property	caustic
5 Shows the pH of a substance	indicator
6 Water is always one product of this reaction	neutralisation
7 Always a product in neutralisation reactions	water


choroid
retina
yellow spot or fovea
blind spot
optic nerve
lens
suspensory ligaments
ciliary muscles
vitreous humour
sclerotic layer
conjunctiva
cornea
pupil
aqueous humour
iris

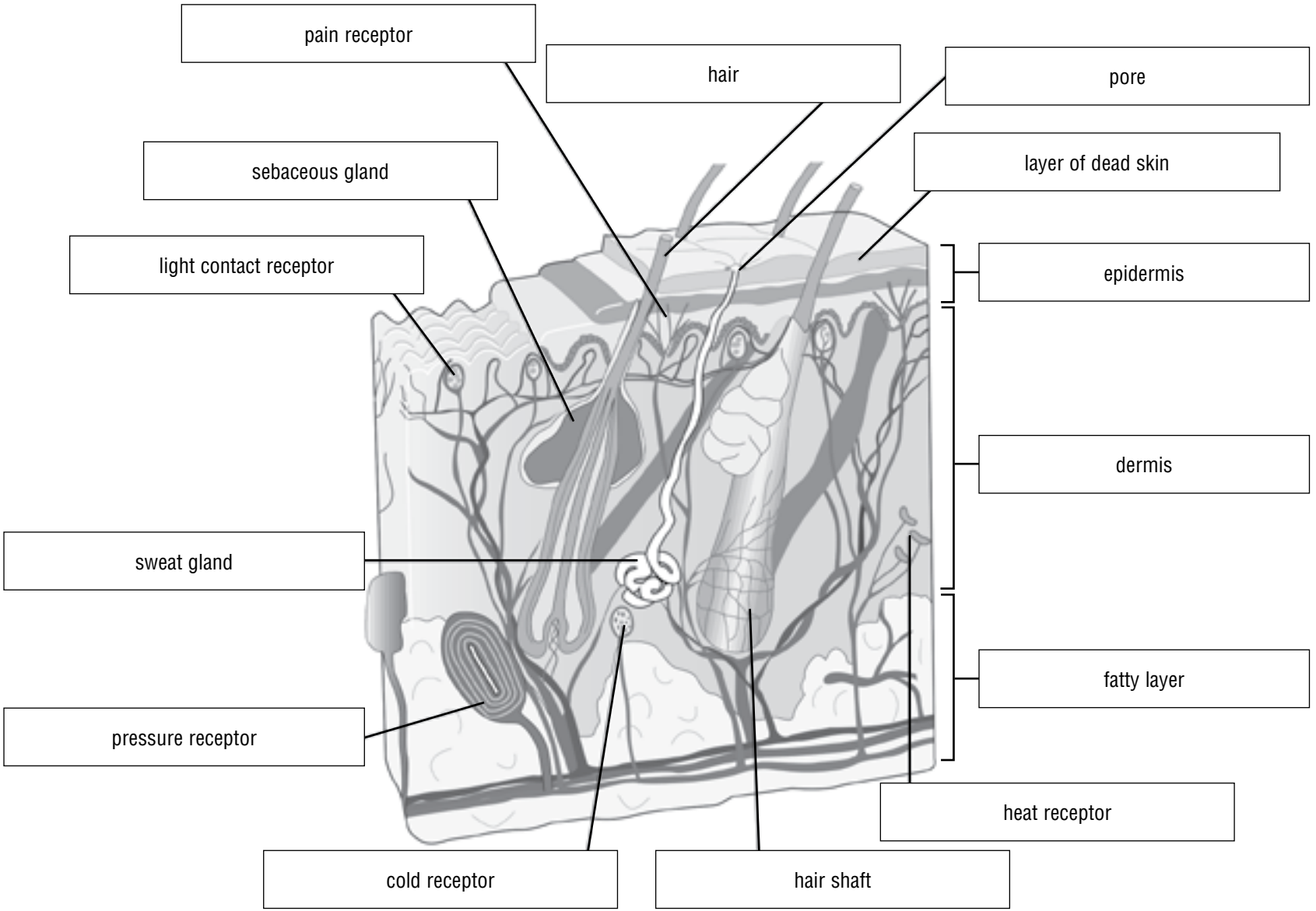

auditory nerve
cochlea
round window
inner ear
semi-circular canals
oval window
eustachian tube
middle ear
ear drum
stirrup
anvil
hammer
ear canal
outer ear
pinna

1



2 **a** pigeon

b elephant


pain receptor
hair
pore
sebaceous gland
layer of dead skin
light contact receptor
epidermis
dermis
sweat gland
fatty layer
pressure receptor
heat receptor
cold receptor
hair shaft

Answers 4.5 Codes

Student responses will vary.

Answers 4.6 Responding

1 Concussion is when the body's nervous function shuts down following a mild injury to the brain.

2 amnesia

3 at least seven days

4 It reduces the brain's capacity to process information at its normal rate.

5 It may be due to the neurons producing less energy to transmit messages. The brain swells and doesn't function properly.

6 **a** two players colliding head-on (head clash)

b a player's head being struck by a ball that was kicked from close range; a player's head hitting the ground

7 E = electro, E = encephalo, G = gram

8 Goalies could be considered non-headers and may have a lower number of neurological problems than a forward who heads the ball more often.

Extension

Student responses will vary.

Answers 4.7 The nervous system

1 The nervous system allows our bodies to detect stimuli, process information and react accordingly.

2 **a** the neuron

b The neuron consists of the cell body, an axon and dendrites.

c sensory and motor

d Any three of heat, pain, cold, touch, light, sound etc.

e muscle or gland

3 A synapse is the gap between two neurons.

4 **a** fastest: 1.8/180 = 0.01 seconds

b slowest: 1.8/0.4 = 4.5 seconds

5 **a** a

b up

c e

d d

Answers 4.8 Chemical control

1 Juvenile diabetes. It has this name because it usually occurs in childhood.

2 An auto-immune reaction destroys approximately 90% of the cells that produce insulin in the pancreas.

3 A number of daily injections of the hormone are required.

4 The body's cells cannot process glucose, which it needs for energy. To obtain energy, the body begins to use its own fat as a substitute and this causes dangerous chemical substances to accumulate in the blood.

5 The percentage of Australians who have type 1 diabetes is 15% of 7%; i.e. $0.15 \times 0.07 = 0.0105 = 1.05\%$.

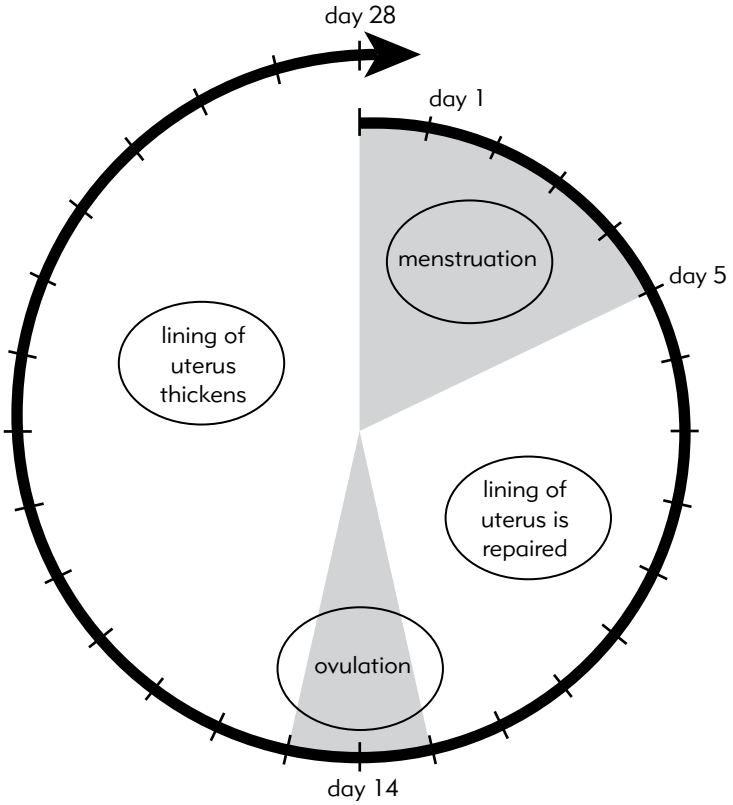
6 The types of insulin are classified by the duration of their effects (i.e. how long they act within the body). This classification may be ultra short-acting, short-acting, intermediate-acting, long-acting or ultra long-acting.

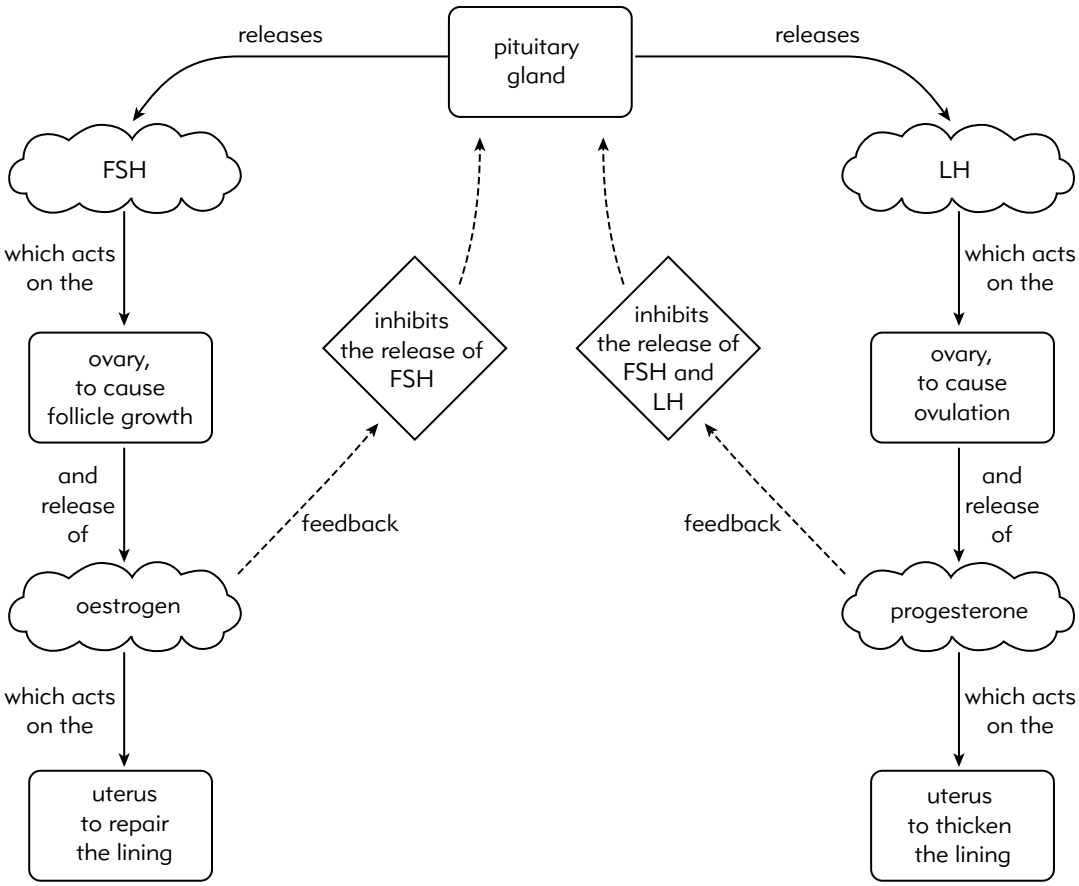
7 onset – when the insulin starts to work; peak – when it is working the hardest; duration – how long it works for

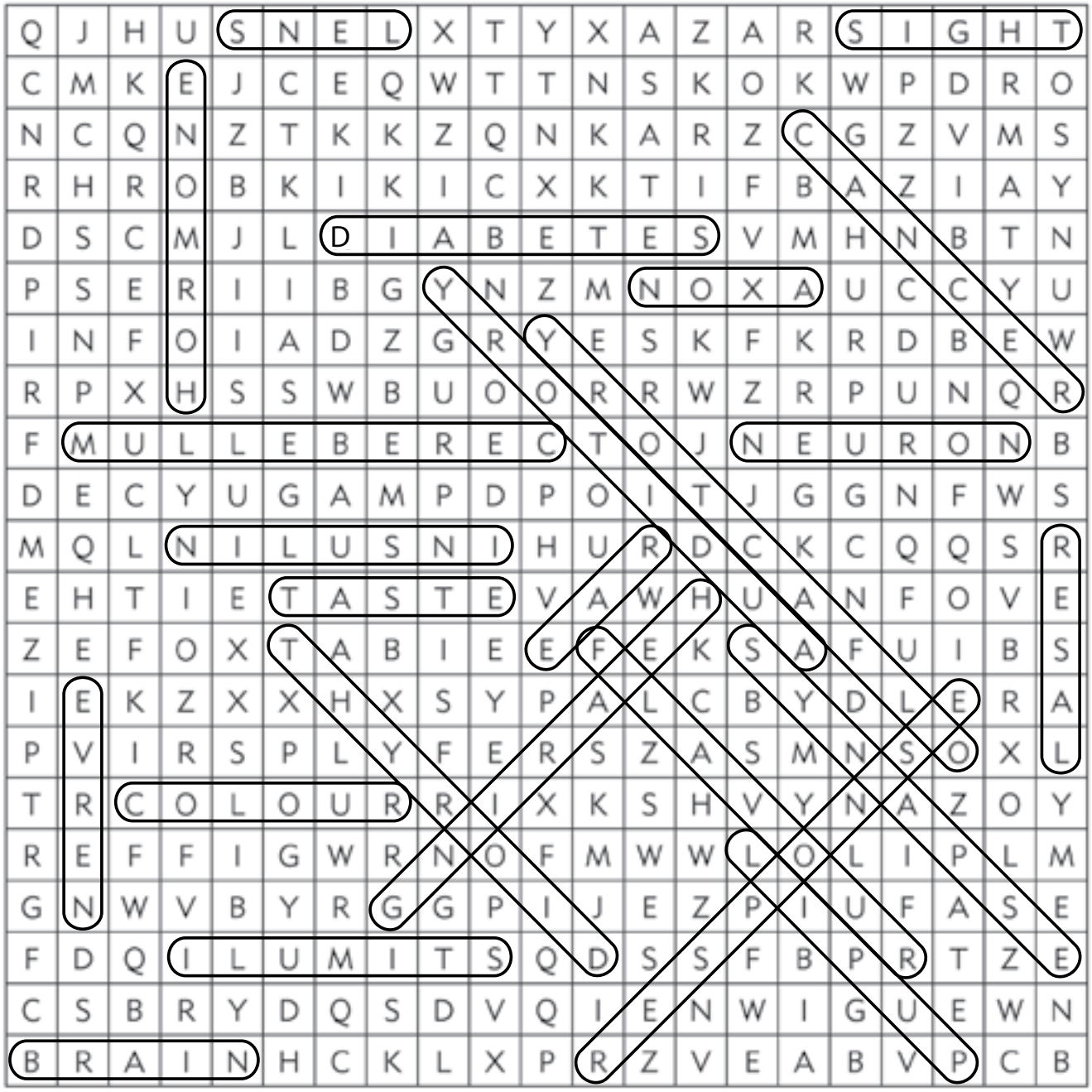
8 ultra short-acting (Humalog), because of its fast onset and short peak

Extension

Student responses will vary.


day 28
day 1
menstruation
day 5
lining of uterus thickens
lining of uterus is repaired
ovulation
day 14

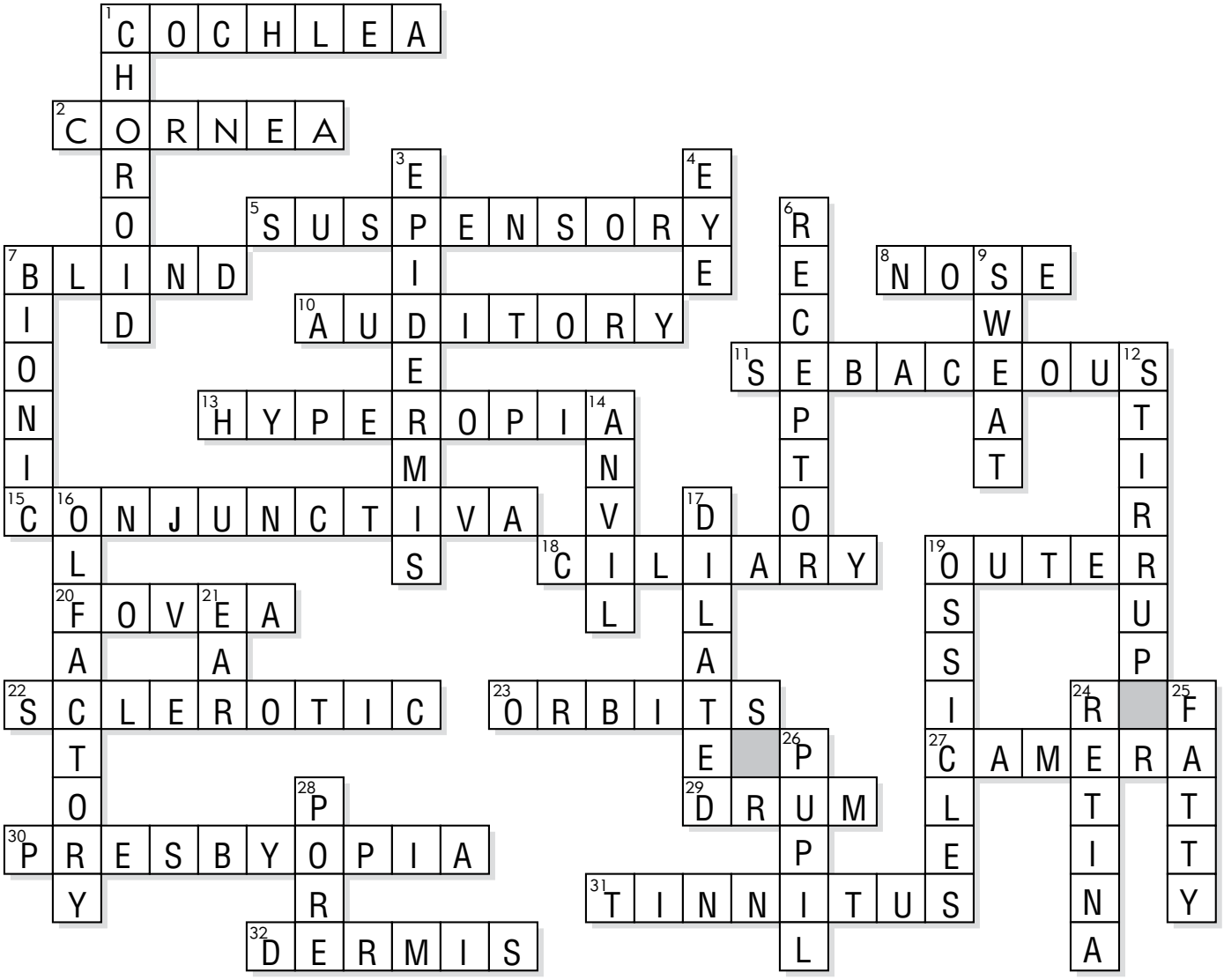

releases
pituitary gland
releases
FSH
LH
which acts on the
ovary, to cause follicle growth
and release of
oestrogen
which acts on the
uterus to repair the lining
inhibits the release of FSH
feedback
inhibits the release of FSH and LH
feedback
which acts on the
ovary, to cause ovulation
and release of
progesterone
which acts on the
uterus to thicken the lining



Word List

AUDITORY
AXON
BRAIN
CANCER
CEREBELLUM
COLOUR
DIABETES
EAR
FLAVOUR
HEARING
HORMONE
INSULIN
LASER
LENS
NERVE
NEURON
OLFACTORY
PUPIL
RESPONSE
SIGHT
STIMULI
SYNAPSE
TASTE
THYROID

Answers 4.11 Crossword



Unit 4.1: Sight

Clue	Word
1 Sense organ concerned with sight	eye
2 Clear, thin layer covering the eye	conjunctiva
3 Clear window that allows light to enter the eye	cornea
4 The lens is filled with this watery liquid	vitreous humour
5 Changes size to control how much light enters the eye	iris
6 Hole in the centre of the iris	pupil
7 Increased in size	dilated
8 Helps focus an image	lens
9 Ligaments that hold the lens in place	suspensory
10 Muscles that change the shape of the lens	ciliary
11 Layer of light-sensitive cells at the back of the eye	retina
12 Where blood vessels and nerves join the eyeball	blind spot
13 Ability of your eyes to focus at different distances	accommodation

Unit 4.2: Hearing

Clue	Word
1 Sense organ for detecting sound	ear
2 The three sections of the ear are outer, middle and _____	inner
3 Thin membrane that vibrates	eardrum
4 Name given to three small bones in the ear	ossicles
5 Where the stirrup joins the cochlea	oval window
6 Snail-shell-shaped part of the ear containing nerve cells that signal the brain	cochlea
7 Nerves that carry messages to the brain from the ear	auditory
8 Help us balance	semicircular canals

Unit 4.3: Smell, taste and touch

Clue	Word
1 Nerve cells in nasal cavity	olfactory cells
2 Taste organ	tongue
3 Bumps on the surface of the tongue	papillae
4 Our largest organ, responsible for sense of touch	skin
5 Outer layer of skin	epidermis
6 Middle layer of skin	dermis
7 Bottom layer of skin	fatty layer
8 Detects touch, heat, pressure etc.	receptor
9 Small hole in the skin	pore
10 Produces sweat	sweat gland
11 Gland that produces oil	sebaceous
12 Australia has a high incidence of this	skin cancer

Unit 4.4: Responding

Clue	Word
1 The maintenance of a constant internal environment	homeostasis
2 Something that acts on an organism to bring about a change in activity	stimulus
3 An action brought about by an effector	response
4 Something that detects a stimulus	receptor
5 Something that brings about a response to a stimulus	effector
6 The nervous and endocrine systems carry out this function	coordination

Unit 4.5: Nervous control

Clue	Word
1 A type of action carried out without thinking	reflex
2 Neurotransmitters transmit a message across a _____	synapse
3 Neurons that carry messages from the central nervous system to muscles and glands	motor
4 The part of the nervous system made up of sensory receptors and nerves	peripheral
5 A neuron with endings sensitive to environmental stimuli	sensory
6 This structure and the brain make up the central nervous system	spinal cord
7 Carries information away from a nerve cell body	axon
8 Carries information towards a nerve cell body	dendrite
9 A bundle of neurons linked together	nerve
10 Part of the brain that controls breathing and heart beat	medulla
11 Part of the brain controlling complex muscle movements like running	cerebellum
12 Layers of connecting tissue that protect the brain	meninges
13 The part of the brain responsible for complex thoughts	cerebrum

Unit 4.6: Chemical control

Clue	Word
1 A chemical messenger	hormone
2 The ‘master’ gland	pituitary
3 Known as the ‘fight or flight’ hormone	adrenalin
4 Part of the brain that sends messages to the pituitary gland	hypothalamus
5 The major male hormone	testosterone
6 A deficiency of this hormone is one cause of dwarfism	HGH
7 A deficiency of this hormone can cause diabetes	insulin
8 The organ that produces insulin	pancreas
9 Chemicals used by many insects to attract mates	pheromones

Answers 5.1

Chapter 5: Reproduction

Plant reproduction

1 Cycads are considered unusual because they have separate male and female plants.

2 **a** weevil

b The ripening cones release a strong odour that attracts the weevil to the male plant.

3 The mating weevils are covered in pollen, the male sex cell.

4 The female cycad emits a similar odour to the male cycad. This draws some of the pollen-coated weevils from the male plant to the female plant.

5 cross-pollination

6 **a** the anther and filament, which are called the stamen

b the stigma, style, ovary and ovule

c The male sex cell is the pollen and the female sex cell is the ovule.

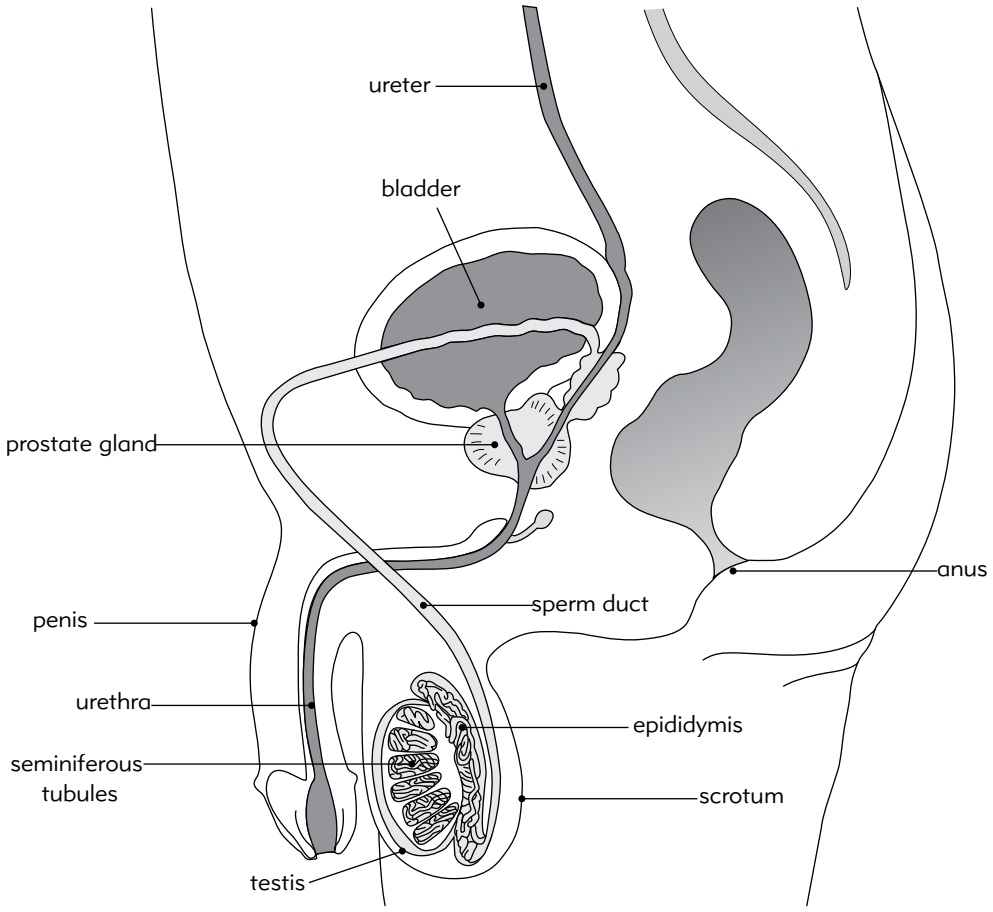
7 Pollination is the transfer of pollen from anther to stigma. Fertilisation is when the pollen and egg fuse together.

8 Any two of wind, insects such as bees or ants, or animals may act to transfer pollen from an anther to a stigma, resulting in pollination; e.g. an animal may carry pollen on its fur; the wind is able to blow the pollen from an anther to a stigma.

Answers 5.2

Human reproductive organs

1



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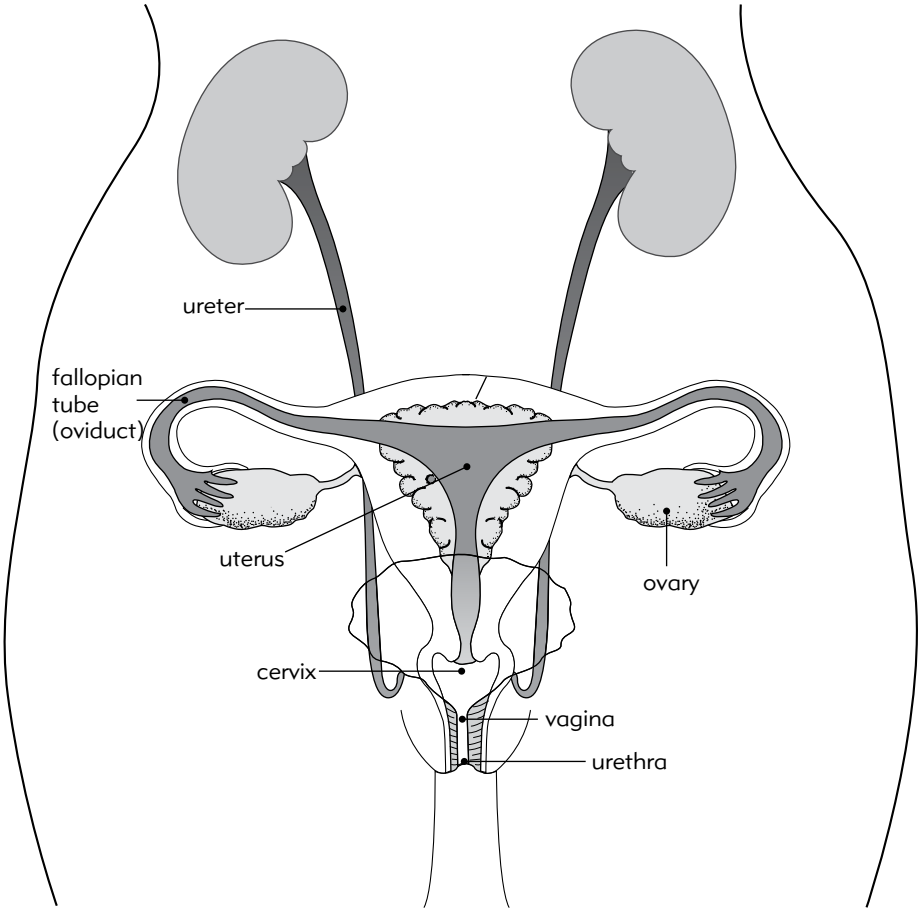


3 **a** Ovaries produce eggs and sex hormones.

b The epididymis is a coiled tube in which sperm matures.

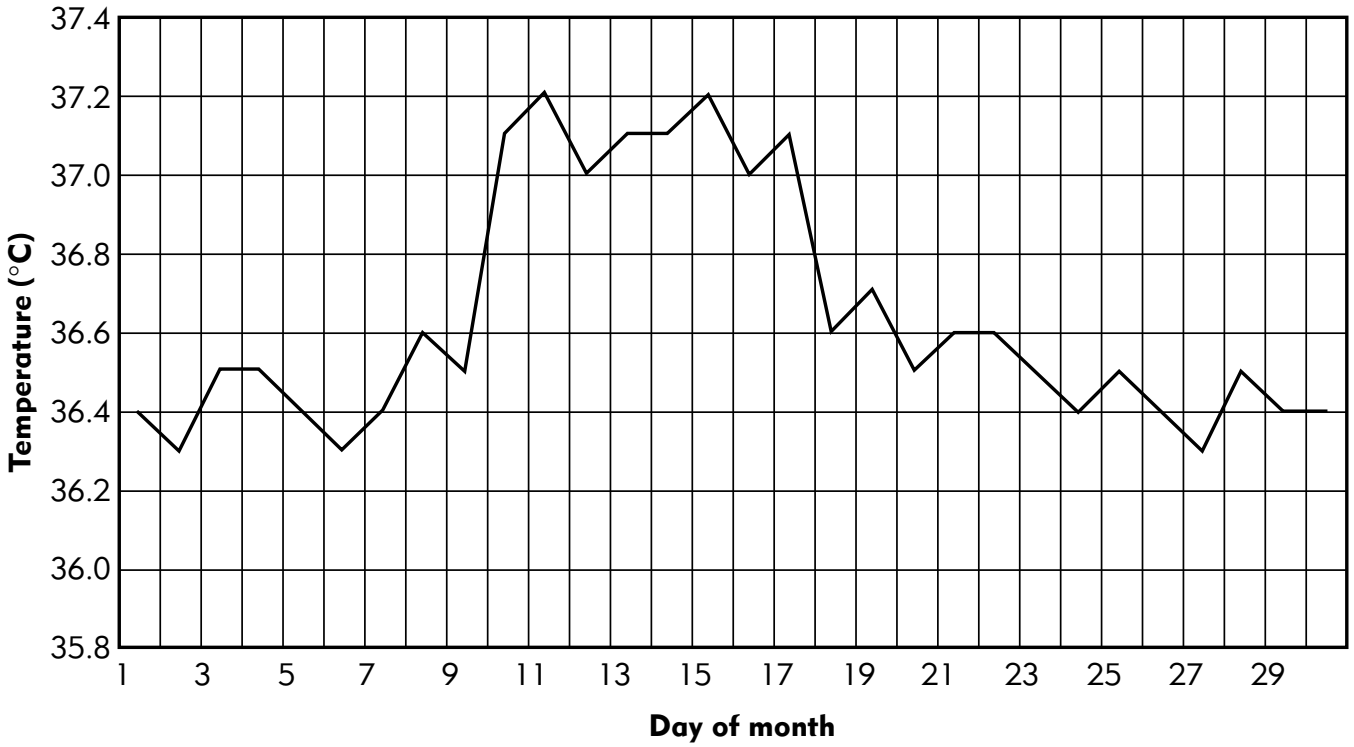
c The fallopian tubes are a pair of very narrow tubes that carry eggs from the ovary to the uterus. Fertilisation occurs in one of the fallopian tubes.

d The seminiferous tubules produce sperm.

e The uterus is where the foetus develops during pregnancy.

Answers 5.3 Ovulation

1



2 The woman ovulated on day 10.

3 days 1 to 9, and days 20 to 30

Extension

a Developing countries. Many answers are possible; e.g. poor medical care, poor nutrition etc.

b Same as answer to part **a**, because of a lack of contraception.

c Developed democratic countries; e.g. Australia, because of education and the availability of contraception.

d same as answer to part **c**

Answers 5.4 Pregnancy

1 Male sex cell is sperm. Female sex cell is ovum or egg.

2 Fertilisation is the union of male sperm and female egg.

3 A morula is a clump of up to 80 cells that forms in the uterus.

4 **a** e: fallopian tube

b zygote

c b

5 a, e, d, c, b

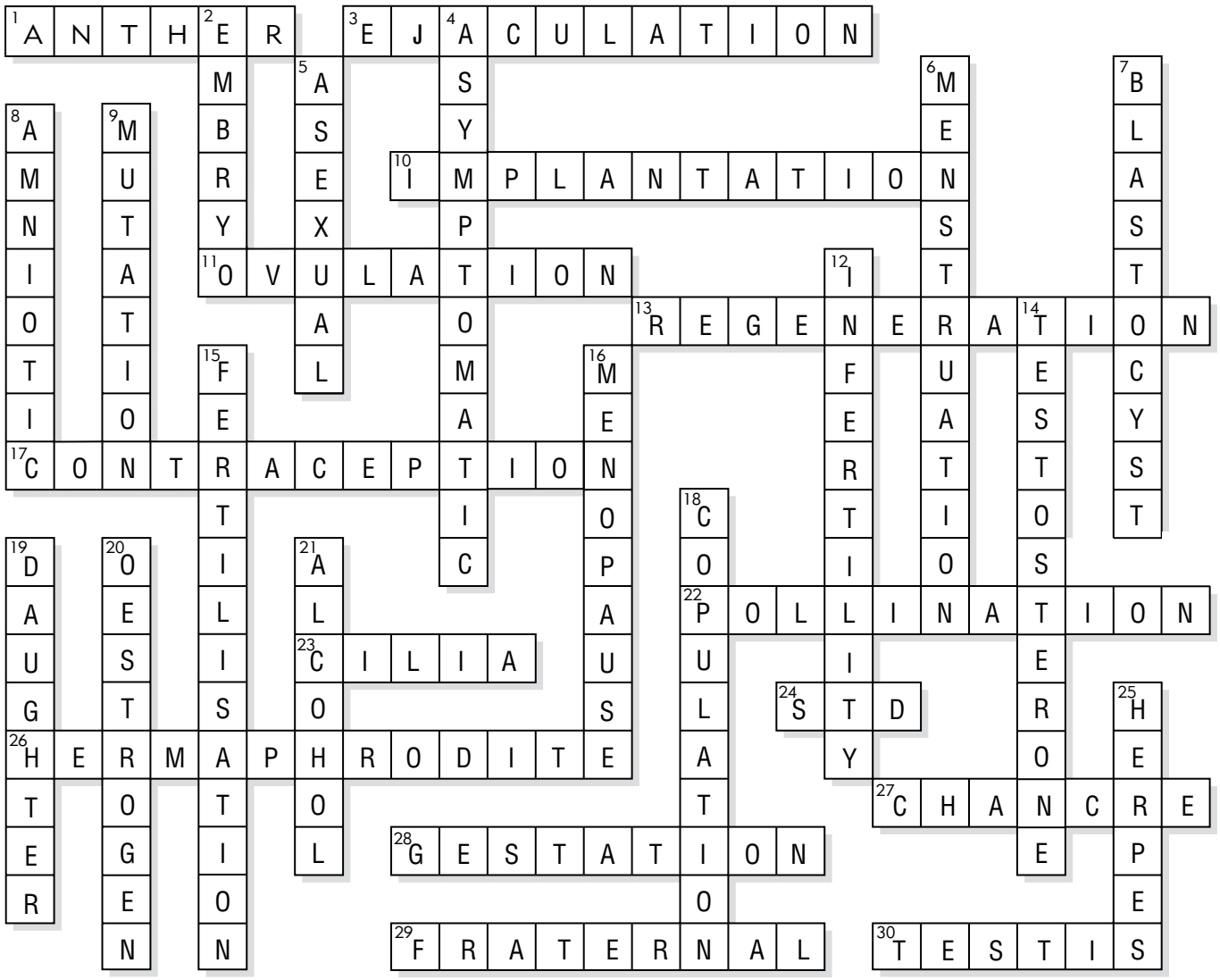
6 **a** The zygote is identified as a foetus after eight weeks.

b During the time in the womb, the baby is protected by the amniotic fluid.

7 a: umbilical cord; b: amniotic fluid

8 36 weeks

Answers 5.5 Crossword



Unit 5.1: Types of reproduction

Clue	Word
1 Individual genetically identical to another	clone
2 Reproduction type requiring only one cell	asexual
3 Change in genetic information	mutation
4 Where male plant gametes are produced	anther
5 Where female plant gametes are produced	ovary
6 Occurs when one plant fertilises another	cross-pollination
7 Successful formation of a zygote	fertilisation
8 Where ova (eggs) are produced	ovum
9 Sex cell	gamete
10 Reproduction type requiring two different cells	sexual
11 Tail on sperm	flagellum
12 Special chemicals that control many body processes	hormones
13 Release of an egg by the female	ovulation
14 Early stage of development	embryo

Unit 5.2: Human reproductive systems

Clue	Word
1 Sac in which testes are located	scrotum
2 Hair-like structures	cilia
3 Male sex organ	penis
4 Tube connecting testis to urethra	sperm duct
5 Mixture of fluid and sperm	semen
6 When no more eggs are released	menopause
7 Small sac of cells	follicle
8 Tube into which egg is released	oviduct
9 Shedding of the uterus lining	menstruation
10 Time when sexual maturity is reached	puberty
11 Female hormone	oestrogen
12 Male hormone	testosterone

Unit 5.3: Human reproduction

Clue	Word
1 Sexual intercourse	copulation
2 Release of semen	ejaculation
3 Birth control devices	contraception
4 Fluid-filled ball	blastocyst
5 Blastocyst burying itself in the uterus lining	implantation
6 New individual in the time from implantation to approximately eight weeks	embryo
7 Fluid surrounding the foetus	amnioticn fluid

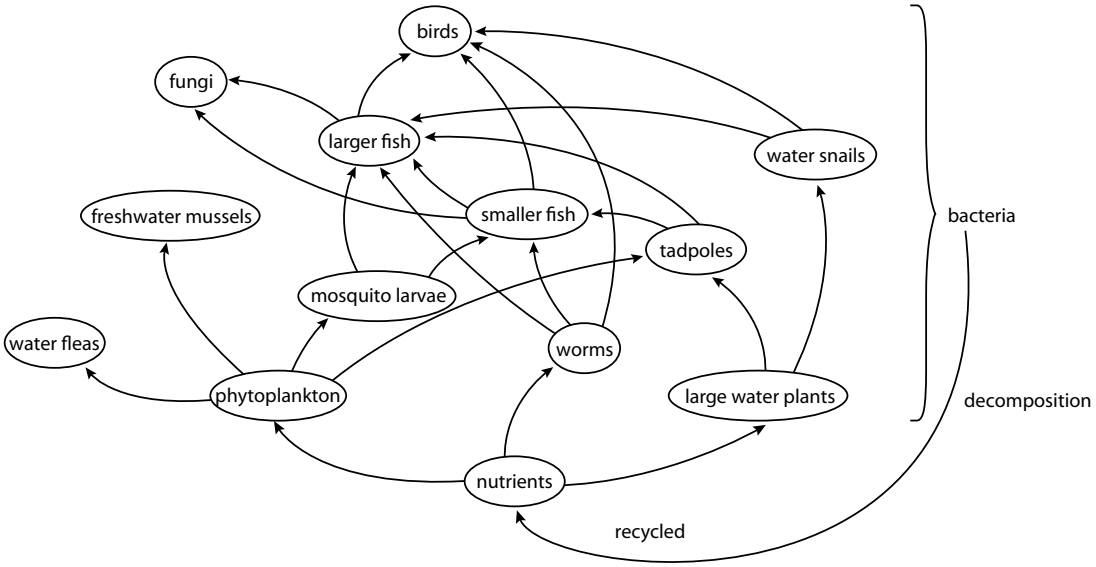
Unit 5.4: Reproductive health

Clue	Word
1 Disease spread by sexual contact	sexually transmitted disease
2 Recommended nutrient in the first three months of pregnancy	folate
3 Disease spread by contact with open sore	herpes
4 In-vitro fertilisation	IVF

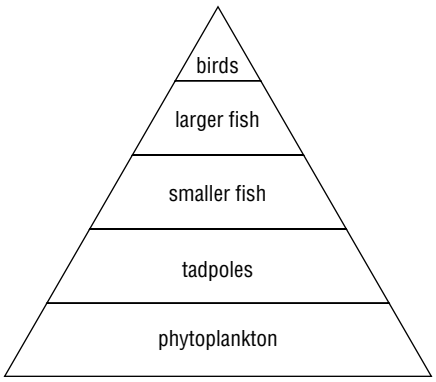
1 There are many possible food chains that can be created. Three examples are shown below.

4th order consumers	→	birds		birds
		↑		↑
3rd order consumers	→	larger fish		larger fish
		↑		↑
2nd order consumers	→	smaller fish	birds	smaller fish
		↑	↑	↑
1st order consumers	→	tadpoles	small fish	mosquito larvae
		↑	↑	↑
Producers	→	phytoplankton	larger water plants	phytoplankton

2 A food web for the food chains shown in Question 1 is shown below.



3 A food pyramid for this ecosystem is shown below. However, many examples are possible.



4 The pyramid follows a food chain from bottom to top. There are always more organisms at the bottom of a food chain, and this is shown in a pyramid for that chain.

Answers 6.2 Recycling in nature

1 Nature's cycles provide us with all our resources such as food, fuel, shelter and minerals. They also recycle our wastes, including carbon dioxide.

2 **a** Plants absorb water and use it in photosynthesis, combining it with carbon dioxide.

b Carbon dioxide enters plants through leaves and then it is used in photosynthesis.

3 **a** Through the process of photosynthesis, plants absorb carbon from the air as carbon dioxide and incorporate it into their biomass (leaves, wood, roots, flowers, fruits). This organic matter provides food for heterotrophic organisms (consumers). Hence, the carbon cycle provides food in the form of plants.

The second way in which the carbon cycle is essential to life on Earth is that plants recycle carbon dioxide from the air and make oxygen.

b Nature—all our resources are provided by natural cycles.

c The resources that support our lives are very distant and therefore our impact on them is hidden. Also, people tend to act only when a problem affects them personally.

4 Eco-footprint is the land area required to sustain our level of resource consumption and waste disposal. Eco-footprint unit = 1 hectare or 10 000 sq m.

5 10/2.5 = 4. NSW's average footprint is currently four times greater than the world average.

6 We are currently using four times the world's average footprint—our foot is too big for the world's average boot. We use far too many resources in an unsustainable way.

Extension

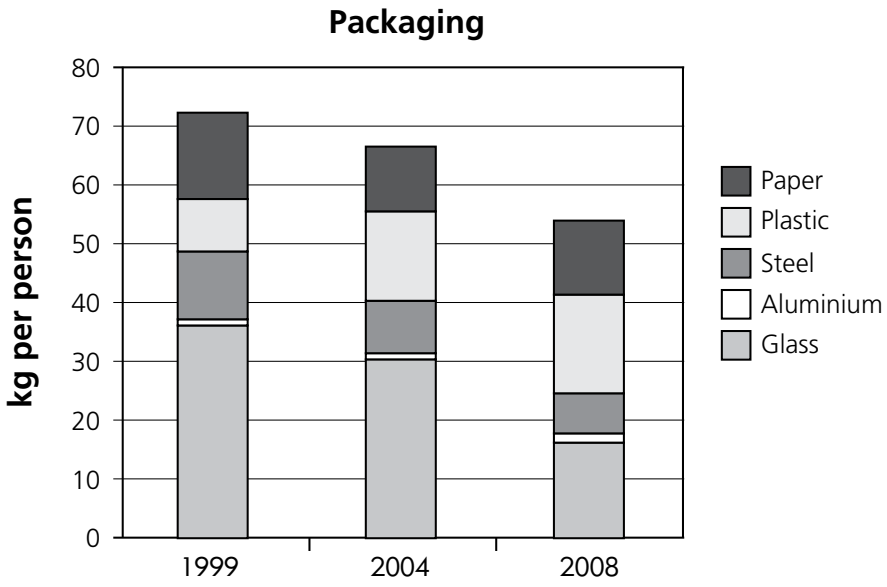
7 Student responses will vary.

8 Student responses will vary.

9 **a** You could personally reduce your eco-footprint by planting trees, composting kitchen waste, walking/cycling to school, growing your own vegetables etc.

b Society could reduce its eco-footprint by improving the efficiency of our use of resources, reducing the amount of waste produced and increasing productivity.

1



2 **a** 71.76 kg

b 66.04 kg

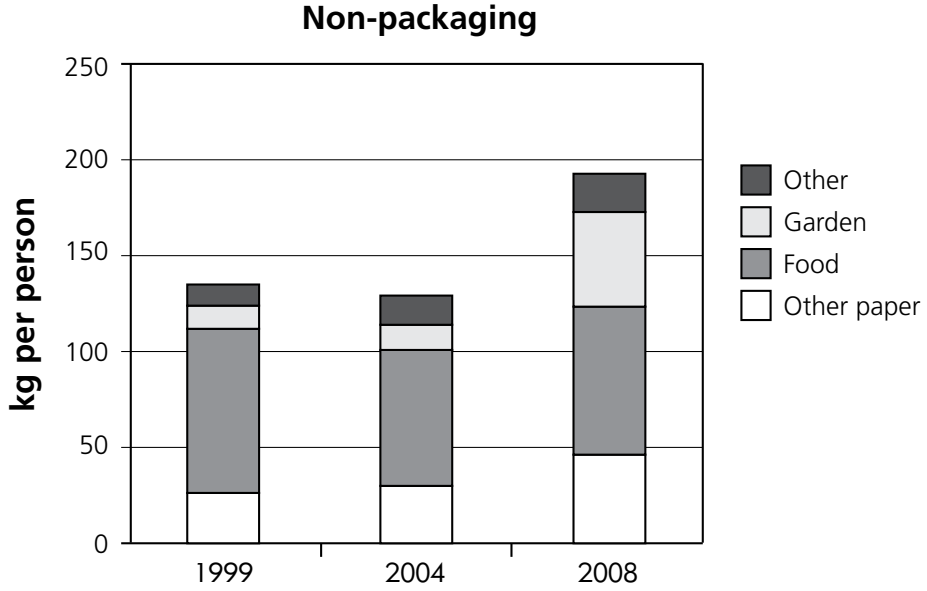
c 53.56 kg

3 **a** steady decrease

b initial increase, then steady

4 It is possible that plastic is being used to replace glass containers.

5



6 **a** 133.64 kg

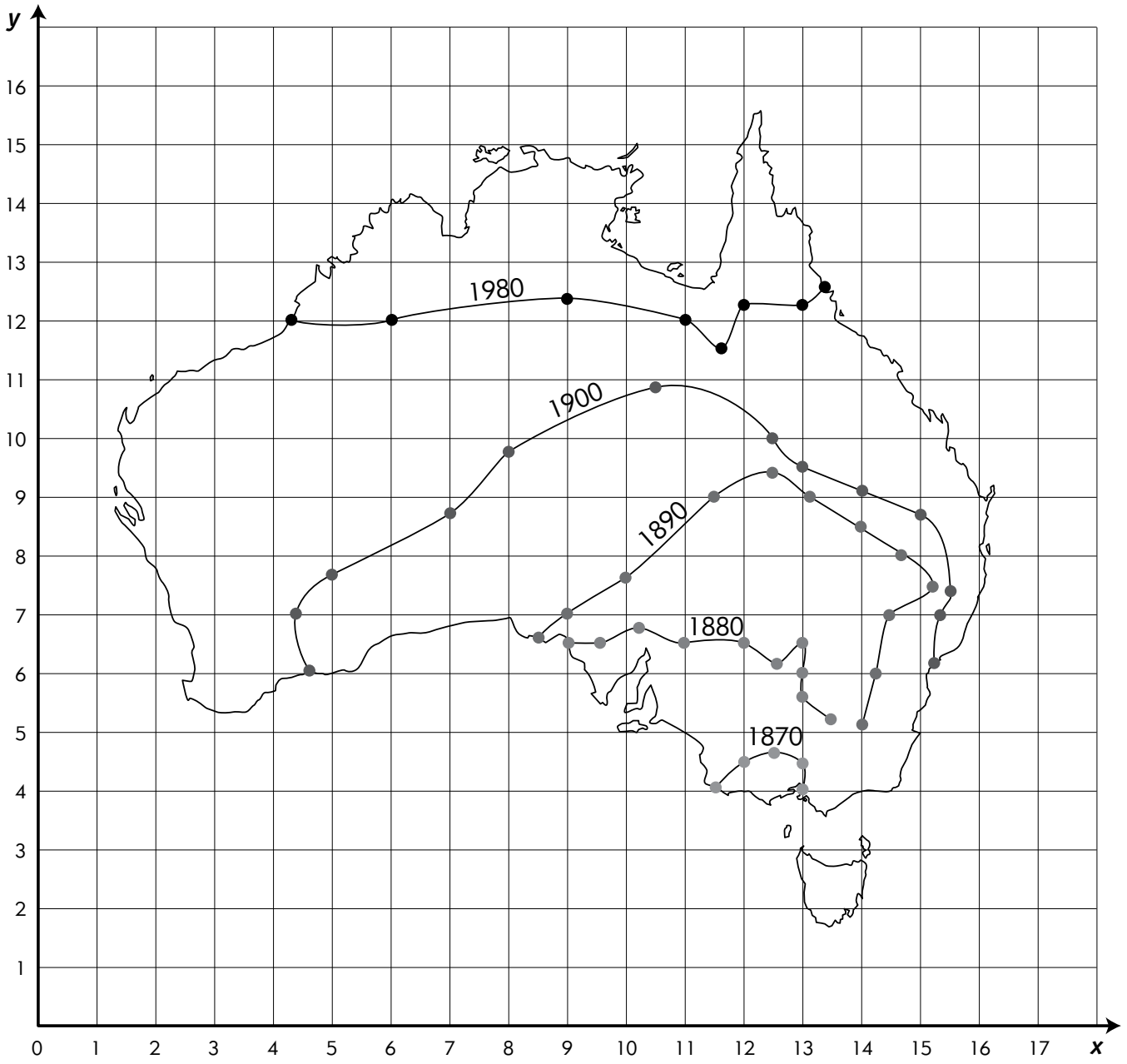
b 127.92 kg

c 190.84 kg

7 Various answers are possible; e.g. the amount of paper used in non-packaging steadily increases.

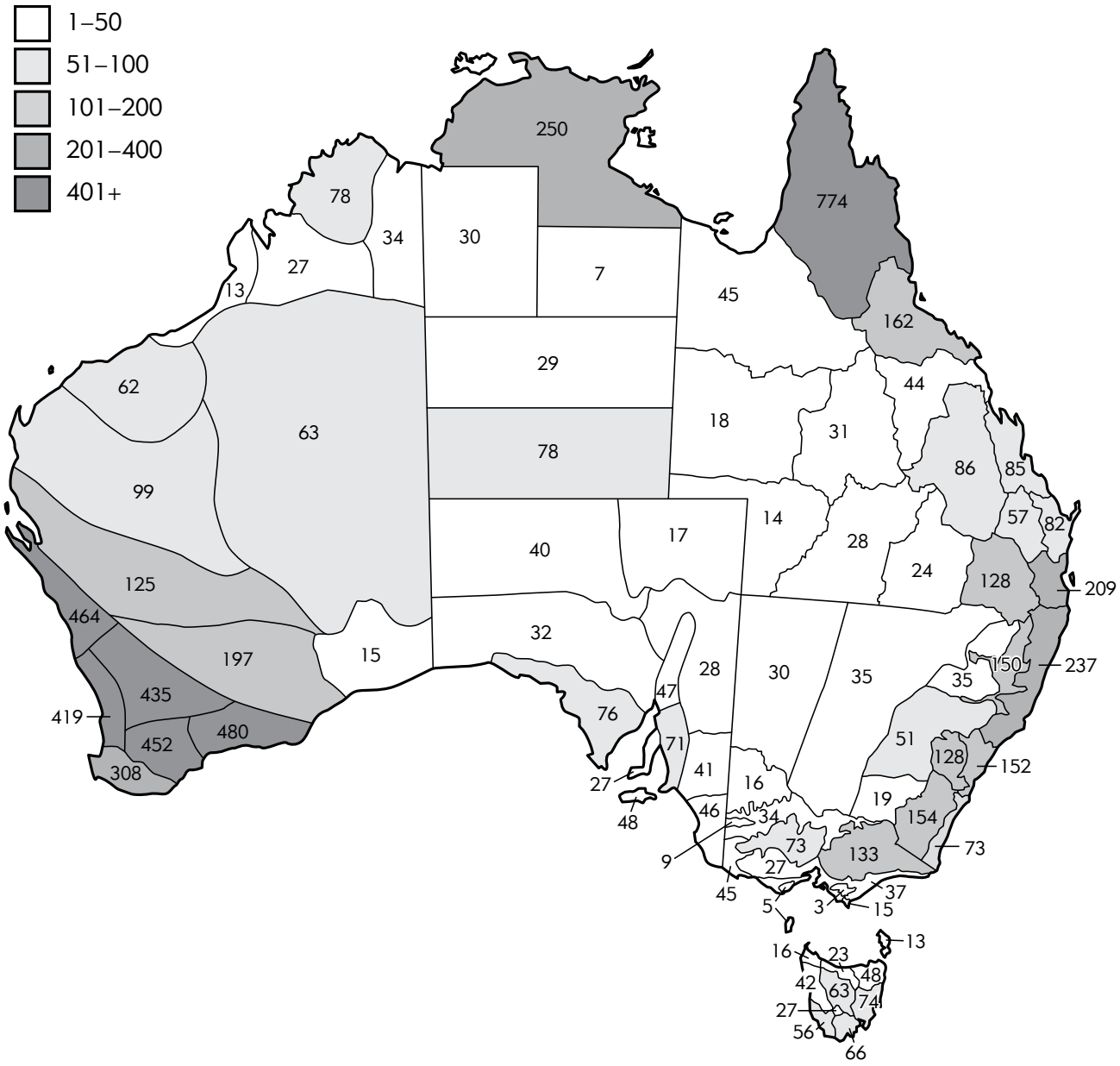
8 1999: 205.40 kg
2004: 193.96 kg
2008: 244.40 kg

9 Packaging and non-packaging totals show different trends. The vertical scale for the Non-packaging graph uses values approximately three times larger than those for the Packaging graph.


y
x
1980
1900
1890
1880
1870

1

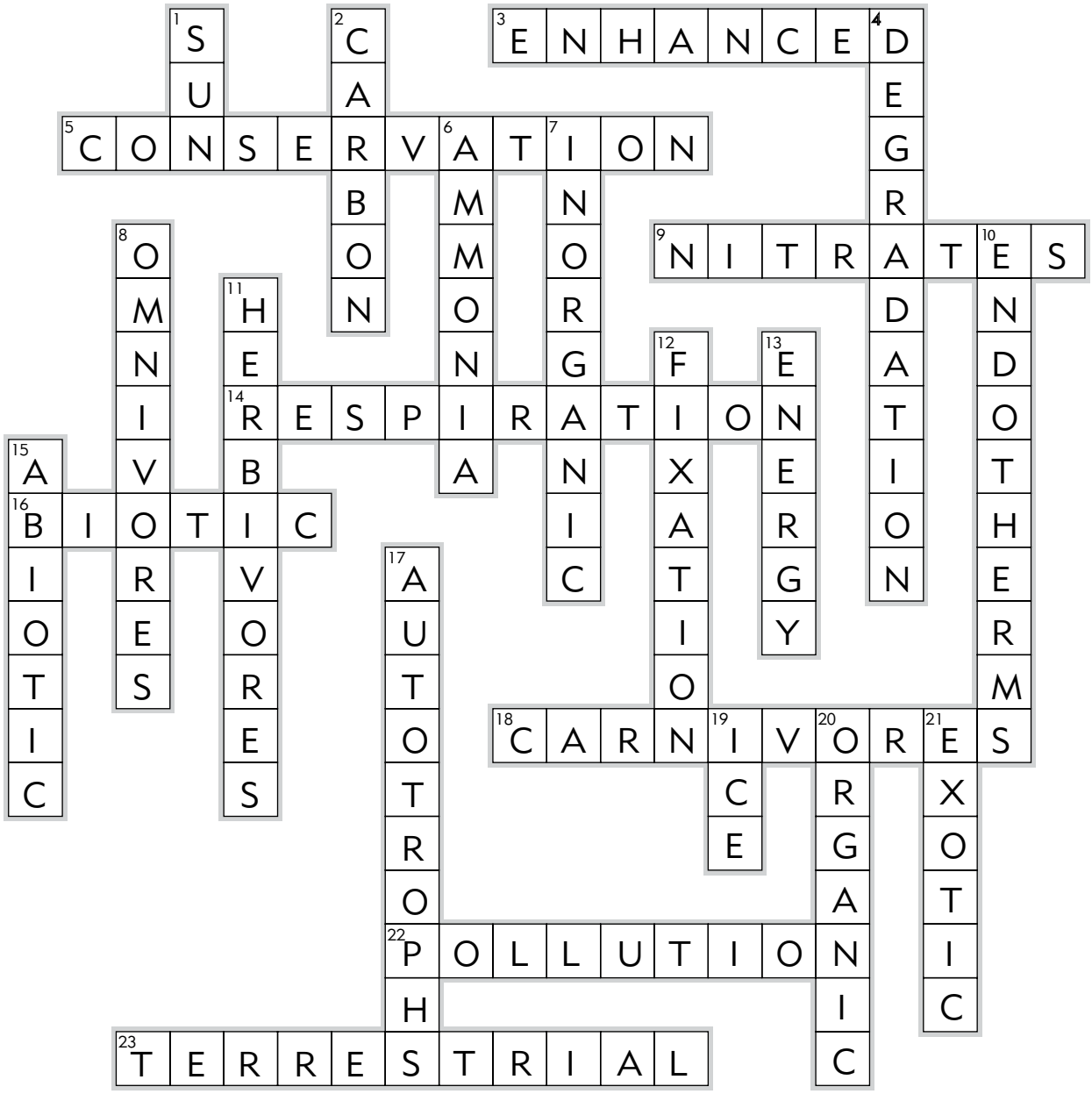
Total numbers of rare or threatened Australian plants in all conservation categories



2 Various answers are possible; e.g. cattle grazing, land clearing, plants are eaten by rabbits and other introduced species.

Extension

Student responses will vary.


1 SUSTAINABLE
2 CARBON
3 ENHANCED
4 DEGRADATION
5 CONSERVATION
6 AMMONIA
7 INORGANIC
8 OMNIVORES
9 NITRATES
10 ENDOTHERMS
11 HERBIVORES
12 FIXATION
13 ENERGY
14 RESPIRATION
15 ABIOTIC
16 BIOTIC
17 AUTOTROPHS
18 CARNIVORE
19 ICE
20 ORGANIC
21 EXOTIC
22 POLLUTION
23 TERRESTRIAL

Unit 6.1: Energy for life

Clue	Word
1 The unit by which we measure energy	joule
2 Organisms that can make their own food	producers
3 Organisms that cannot make their own food	consumers
4 Organisms that do not have a constant internal temperature	ectothermic

Unit 6.2: Recycling in nature

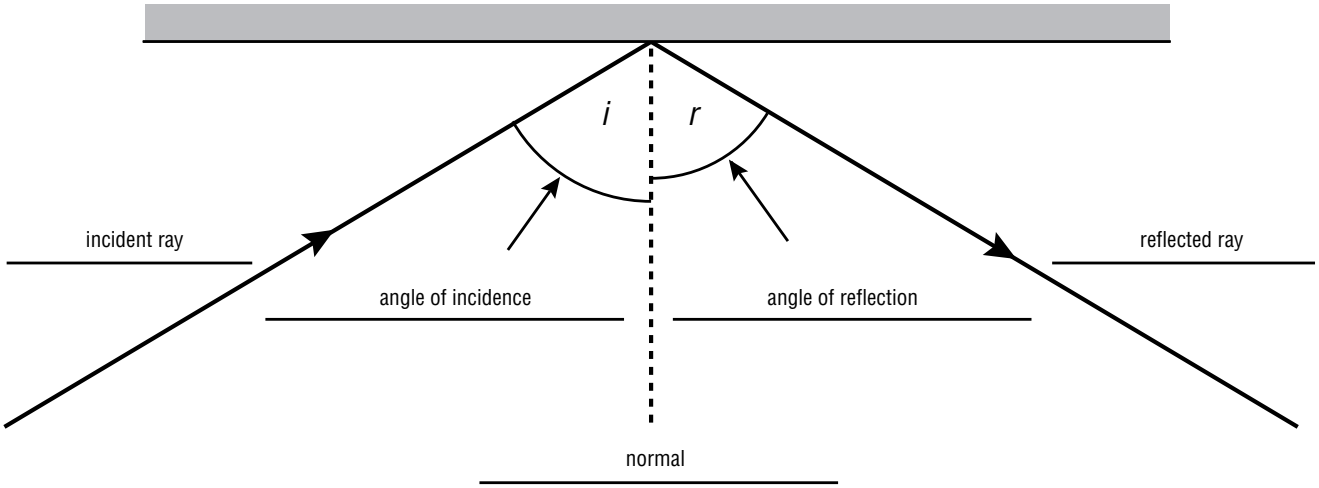
Clue	Word
1 The non-living factors that influence an ecosystem	abiotic
2 The living factors that influence an ecosystem	biotic
3 Matter that is derived from living material and contains the carbon atom	organic matter
4 Matter that does not contain the carbon atom	inorganic matter
5 Pores in the under-surface of a plant leaf that allow gases to enter and exit	stomata
6 The process whereby plants use the Sun's energy to power their food production	photosynthesis
7 A process of living organisms that returns carbon dioxide to the atmosphere	respiration
8 The breakdown of dead organic matter	decomposition
9 The process of converting gaseous nitrogen into a form that can be used by living organisms	nitrogen fixation

Unit 6.3: Human impact on ecosystems

Clue	Word
1 Something that makes the environment unhealthy	pollutant
2 The term used to describe the Earth's rising temperature	global warming
3 Rain that is polluted with chemicals, making it acidic	acid rain
4 Too many animals grazing in one area, causing the area to be destroyed	overgrazing
5 Large-scale removal of trees from an area	deforestation
6 Organisms that are introduced into an area	exotic species
7 Process of doing what is necessary to keep plant and animal species alive	conservation

1 Student responses will vary.

2



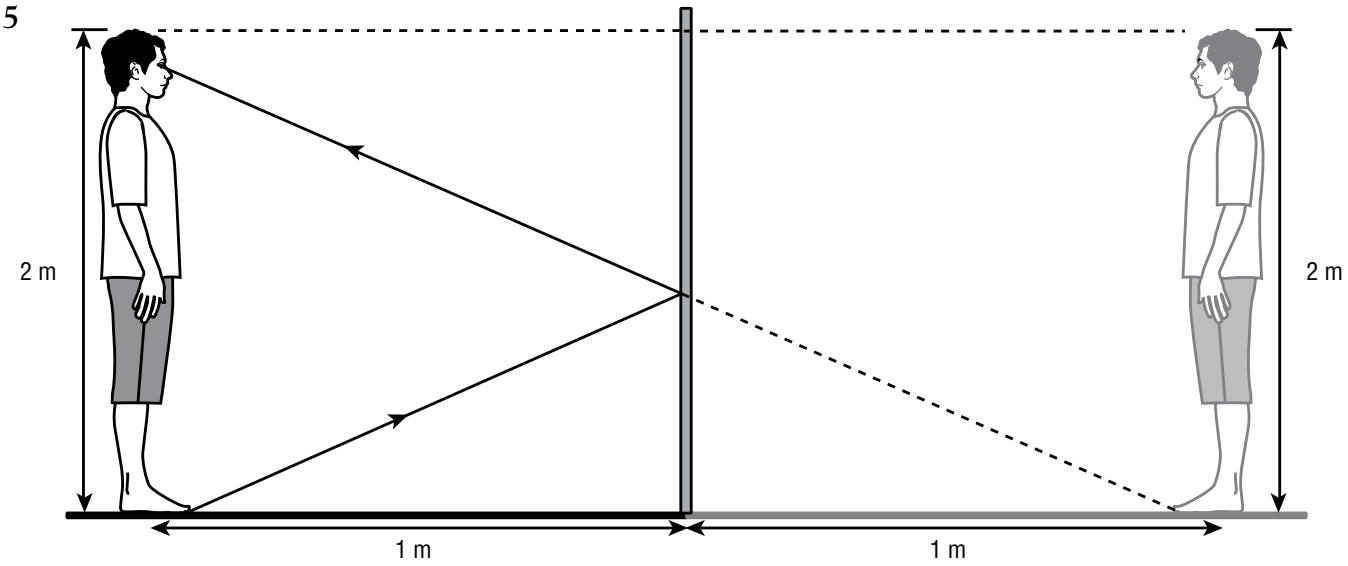
3 **a** $a = 60°$, $b = 60°$, $c = 30°$

b $d = 70°$, $e = 20°$, $f = 70°$

c $g = 50°$, $h = 50°$, $i = 40°$, $j = 50°$, $k = 40°$, $l = 40°$, $m = 50°$

4 The incoming and final outgoing light rays in part c in Question 3 are parallel.

5



Answers 7.2 Snell's law

1 $n_1 = 1.0, \theta_1 = 70°$
$n_2 = 1.33, \theta_2 = ?$
$n_1 \sin \theta = n_2 \sin \theta_2$
$1.0 \sin 70° = 1.33 \sin \theta_2$
$\theta_2 = 1.0 \sin 70°/1.33$
$\theta_2 = 45°$

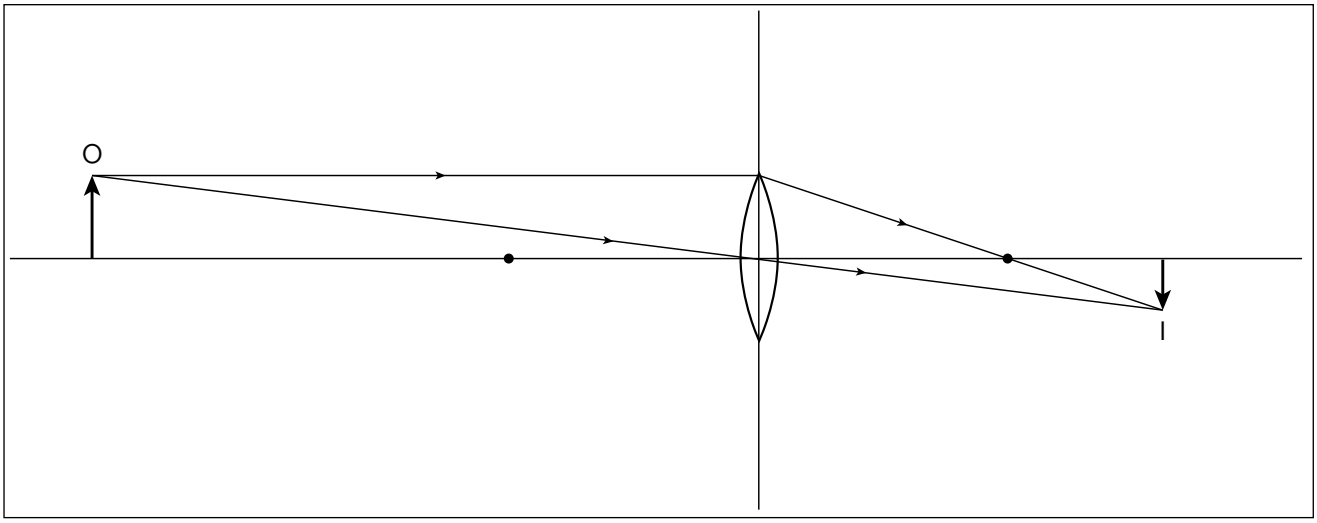
2 $n_1 = ?, \theta_1 = 40°$
$n_2 = 1.0, \theta_2 = 57°$
$n_1 \sin \theta_1 = n_2 \sin \theta_2$
$n_1 \sin 40° = 1.0 \sin 57°$
$n_1 = 1.0 \sin 57°/\sin 40°$
$n_1 = 1.3$

3 $n_1 = 1.33, \theta_1 = ?$
$n_2 = 1.0, \theta_2 = 36$
$n_1 \sin \theta_1 = n_2 \sin \theta_2$
$1.33 \sin \theta_1 = 1.0 \sin 36°$
$\sin \theta_1 = 1.0 \sin 36°/1.33$
$\theta_1 = 26.2°$

4 $n_1 = 1.33, \theta_1 = 15°$
$n_2 = ?, \theta_2 = 17°$
$n_1 \sin \theta_1 = n_2 \sin \theta_2$
$1.33 \sin 15° = n_2 \sin 17°$
$n_2 = 1.33 \sin 15°/\sin 17°$
$n_2 = 1.18°$

Convex lenses

1



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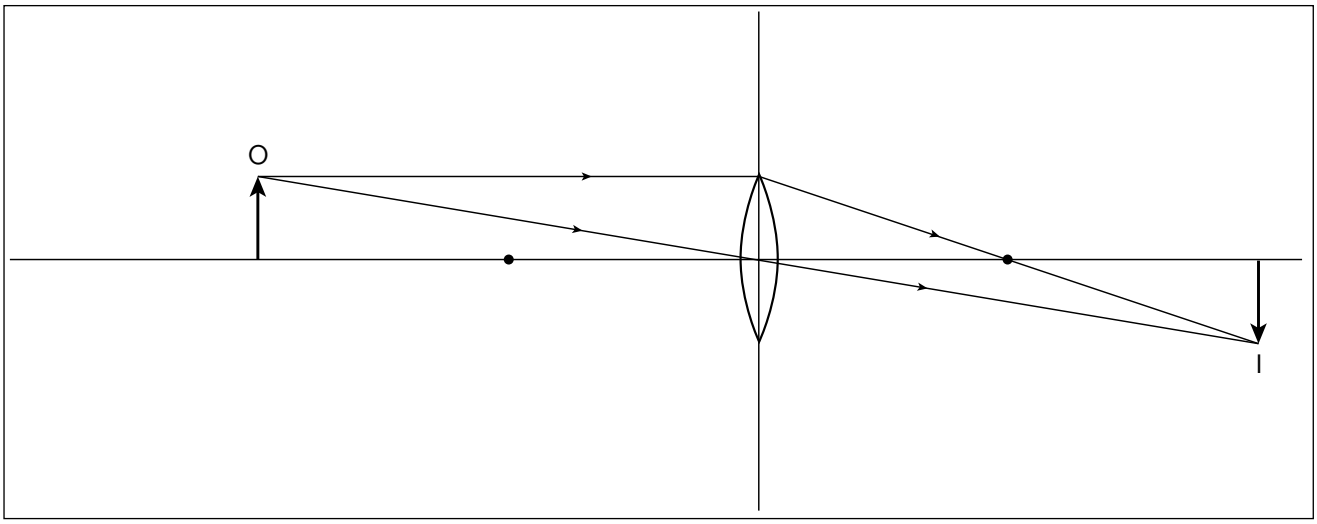

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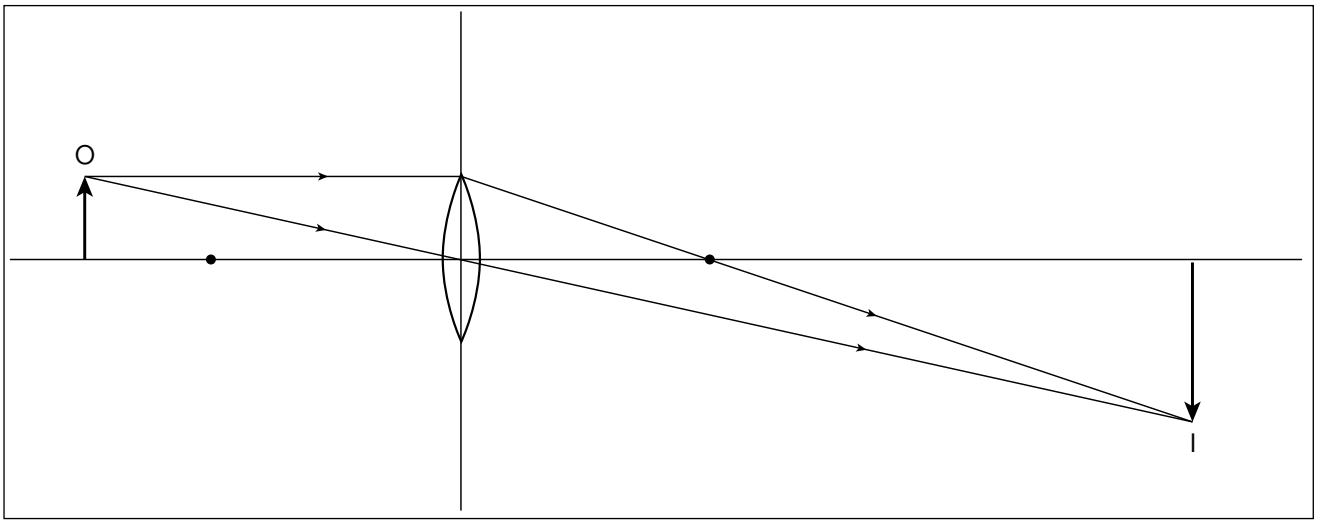

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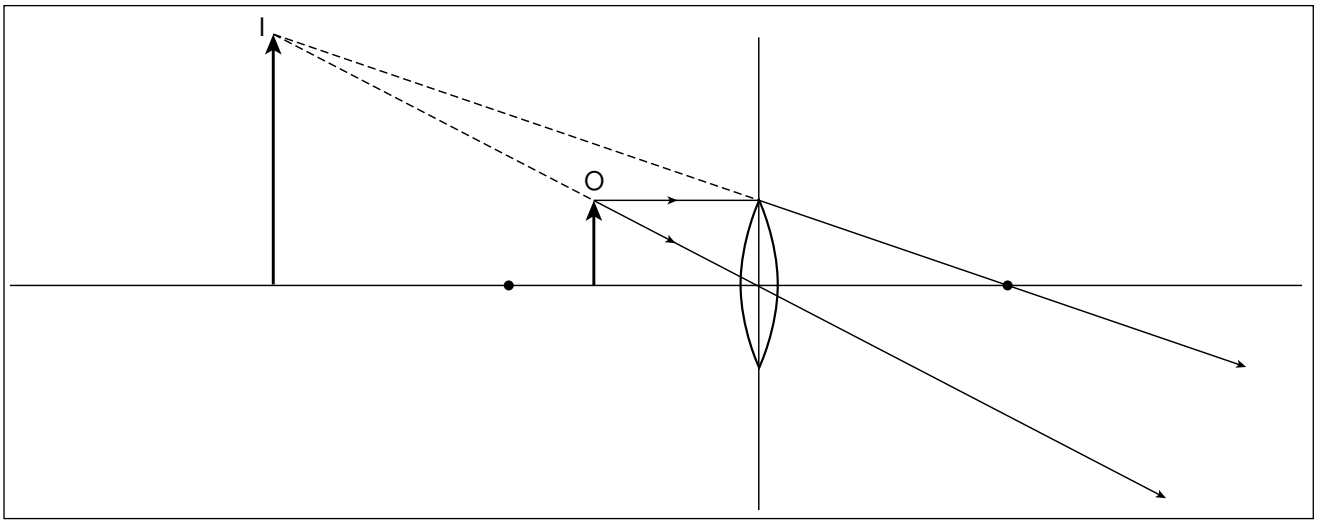

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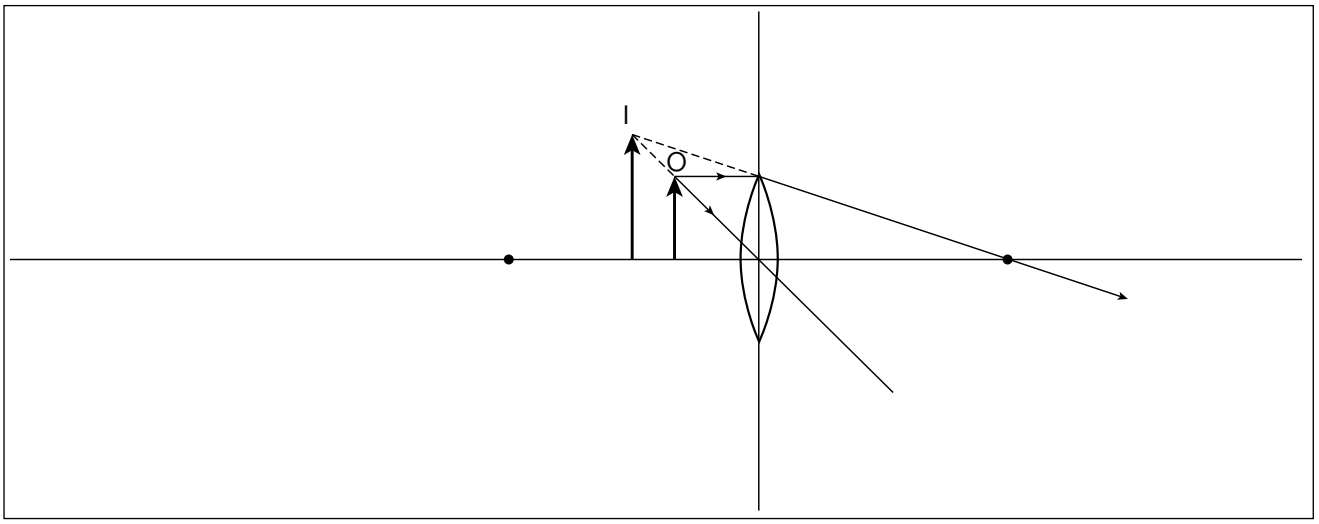
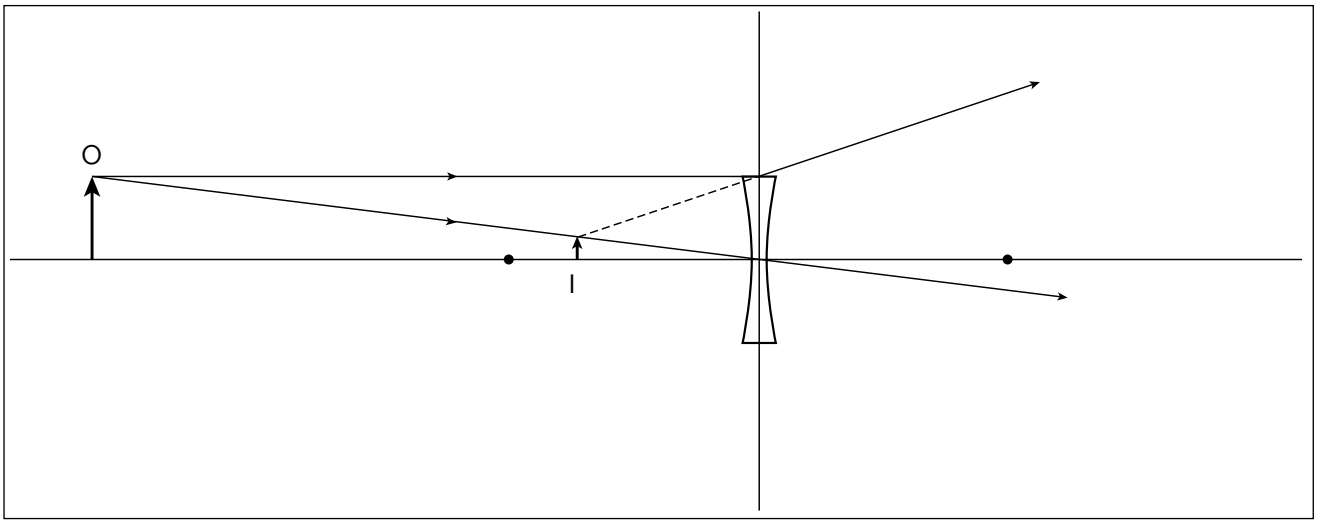


Concave lenses

6



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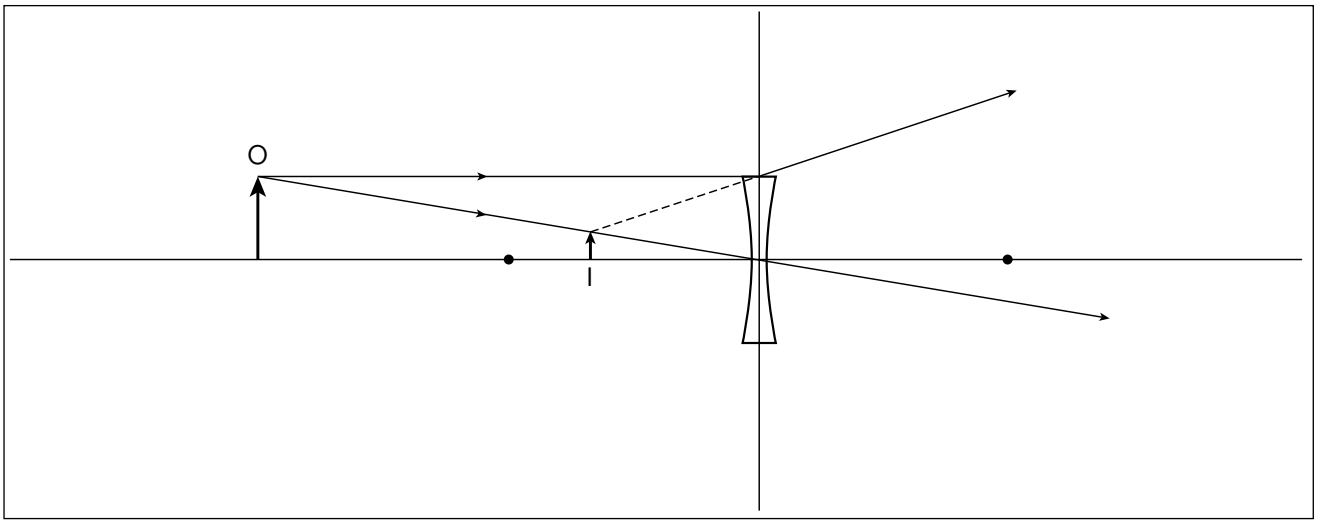

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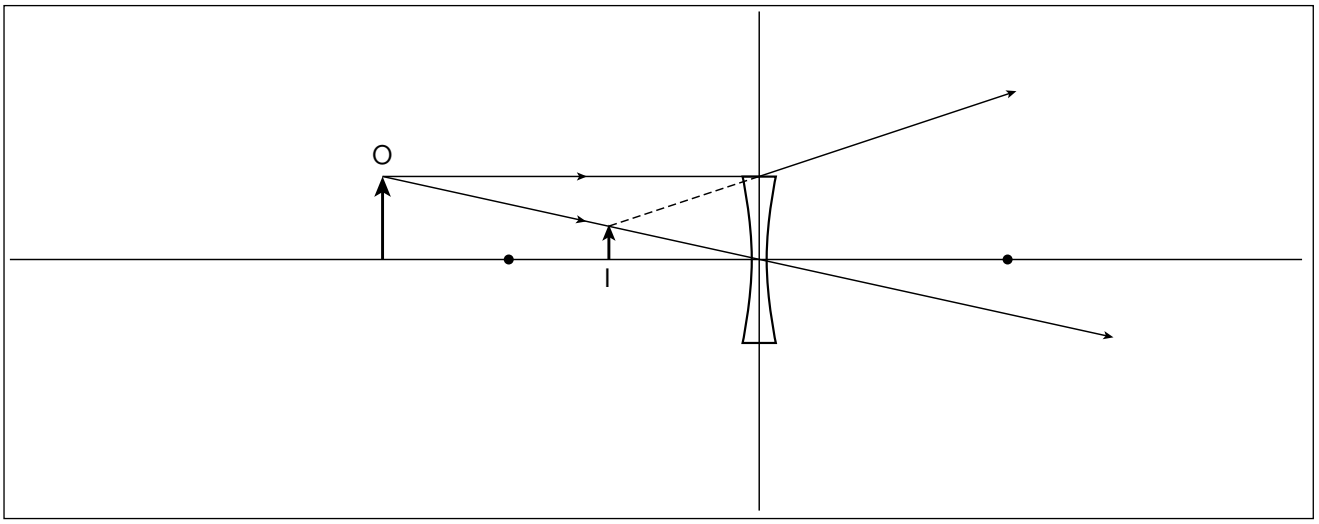

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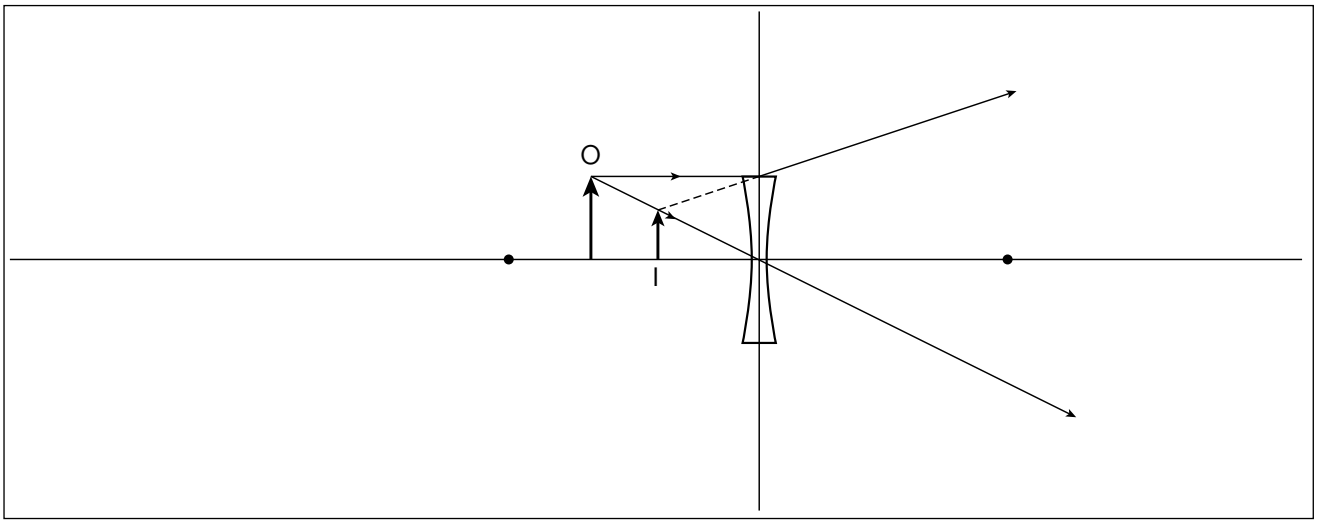

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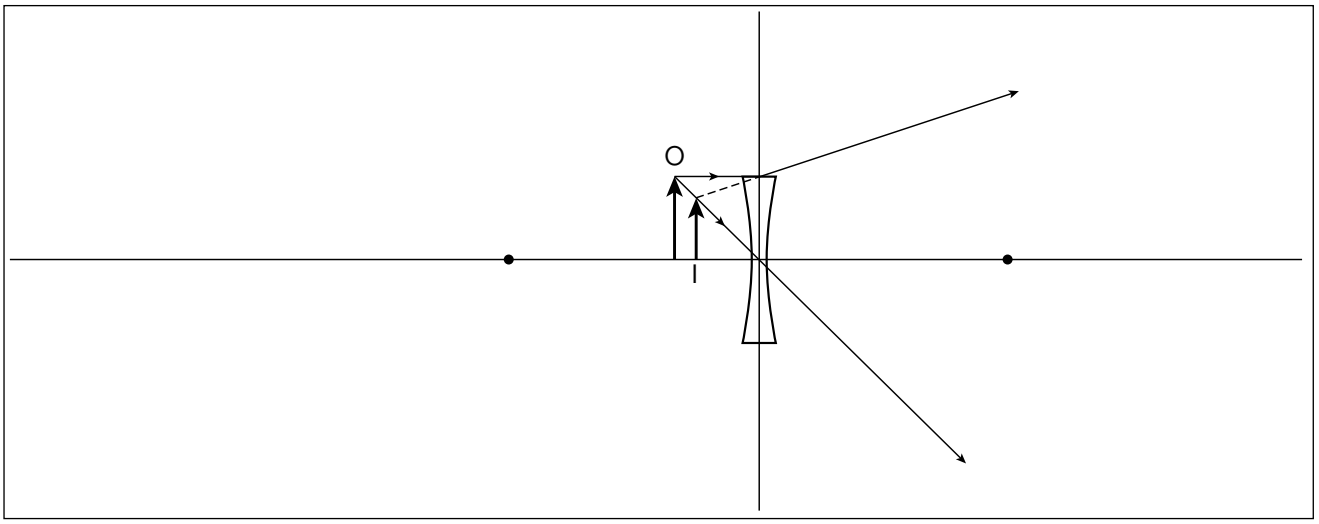


In a convex lens

11 **a** The image moves further away on the other side of the lens and increases in size.

b The image moves closer to the lens on the same side and decreases in size.

12 **a** A virtual image occurs when the object is closer than one focal length to the lens.

b A real image occurs when the object is further than one focal length from the lens.

In a concave lens

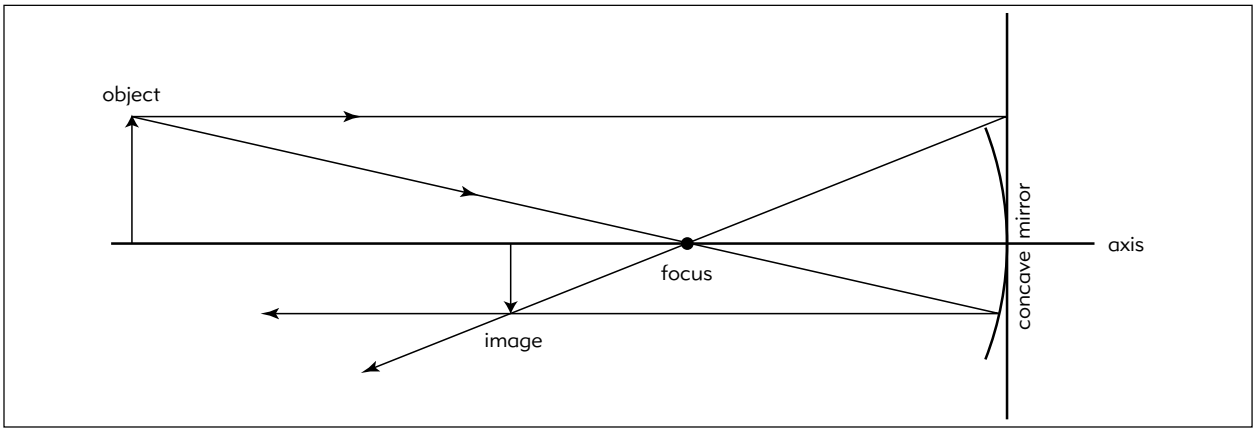
13 **a** The image moves closer to the lens on the same side and increases in size.

b The image moves closer to the lens on the same side and increases in size.

14 **a** A virtual image occurs for any object position.

b A real image never occurs.

1



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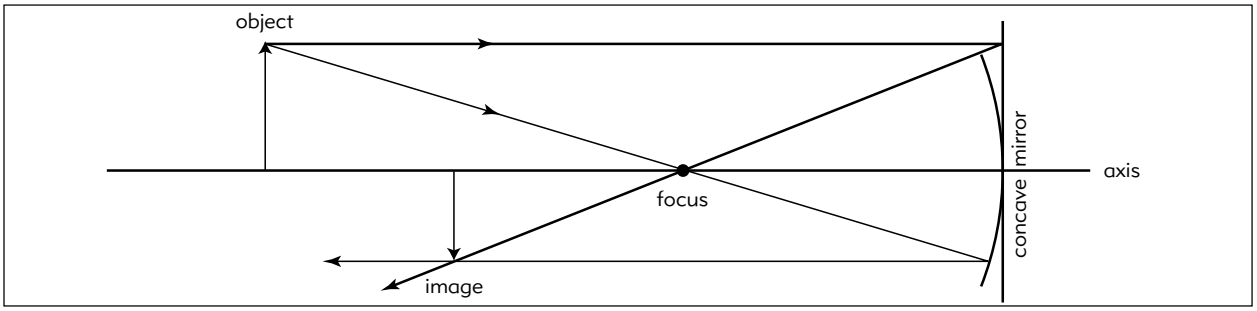

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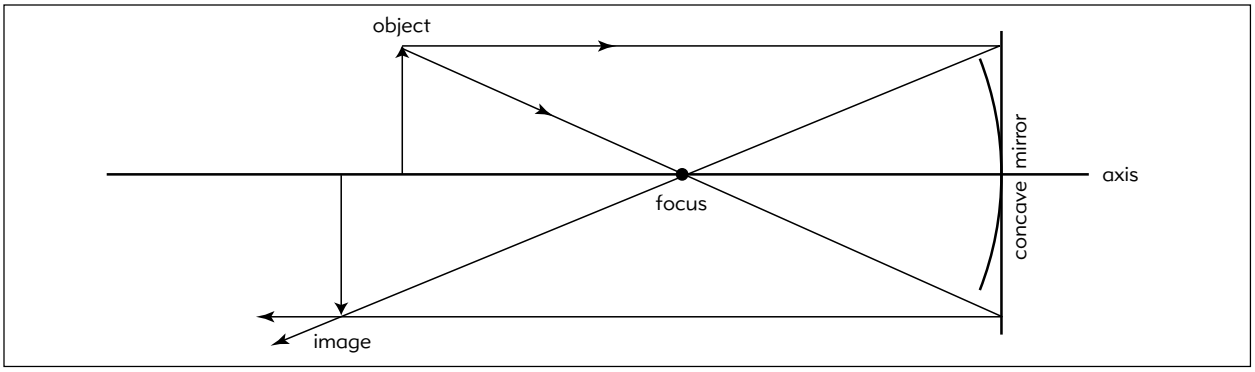

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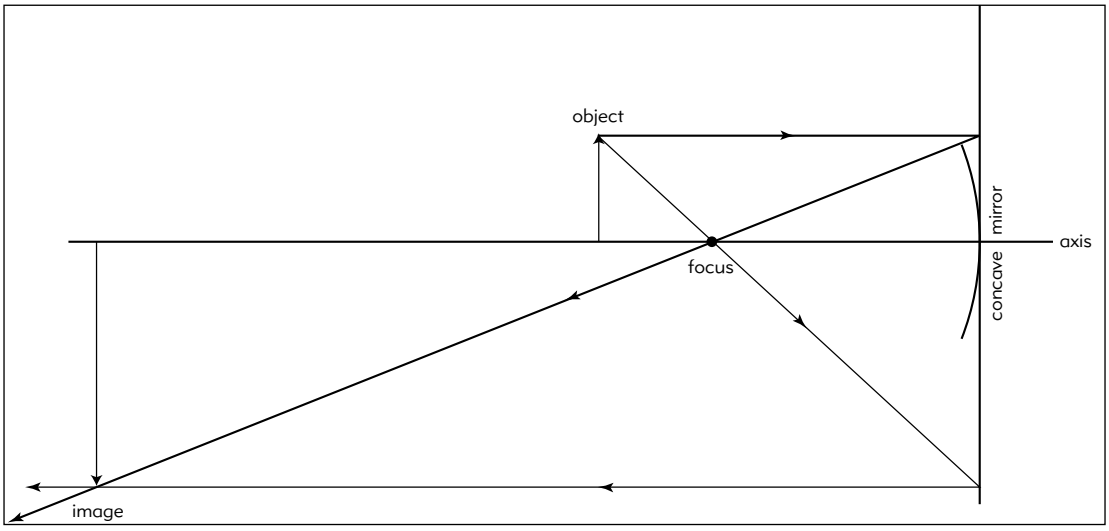
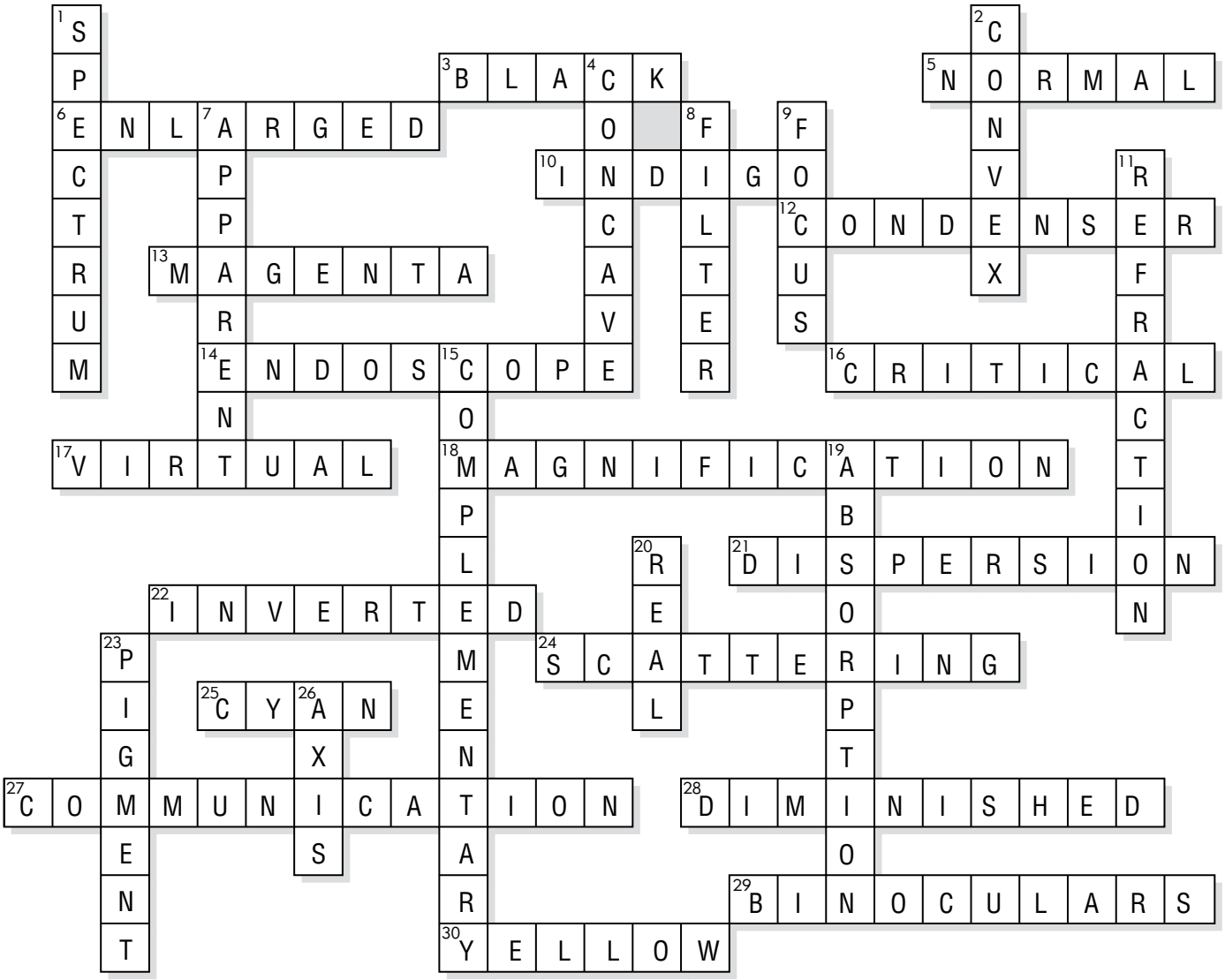


5 The image increases in size and moves further away from the mirror.

Extension

Student responses will vary.

Answers 7.5 Crossword



Unit 7.1: Bending light

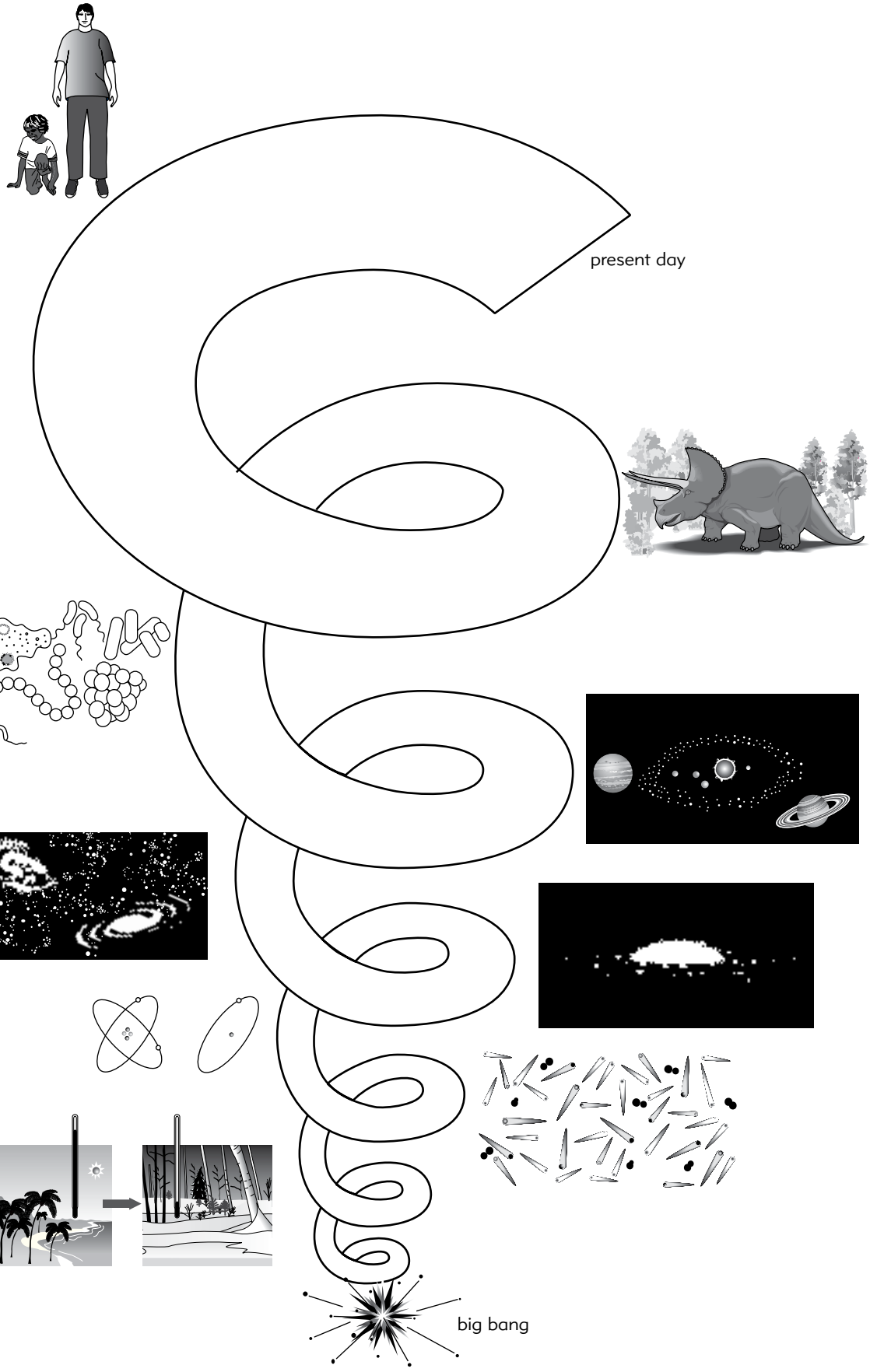
Clue	Word
1 A line at right angles to a boundary between two substances	normal
2 Bending of light that occurs with a change in speed	refraction
3 Optical _____ describes how much a substance bends light	density
4 _____ depth is how deep a pool appears to be	apparent
5 Angles of incidence greater than this result in total internal reflection	critical
6 Medical instrument incorporating optical fibres	end _____ _____ cop _____

Unit 7.2: Focusing devices: lenses and curved mirrors

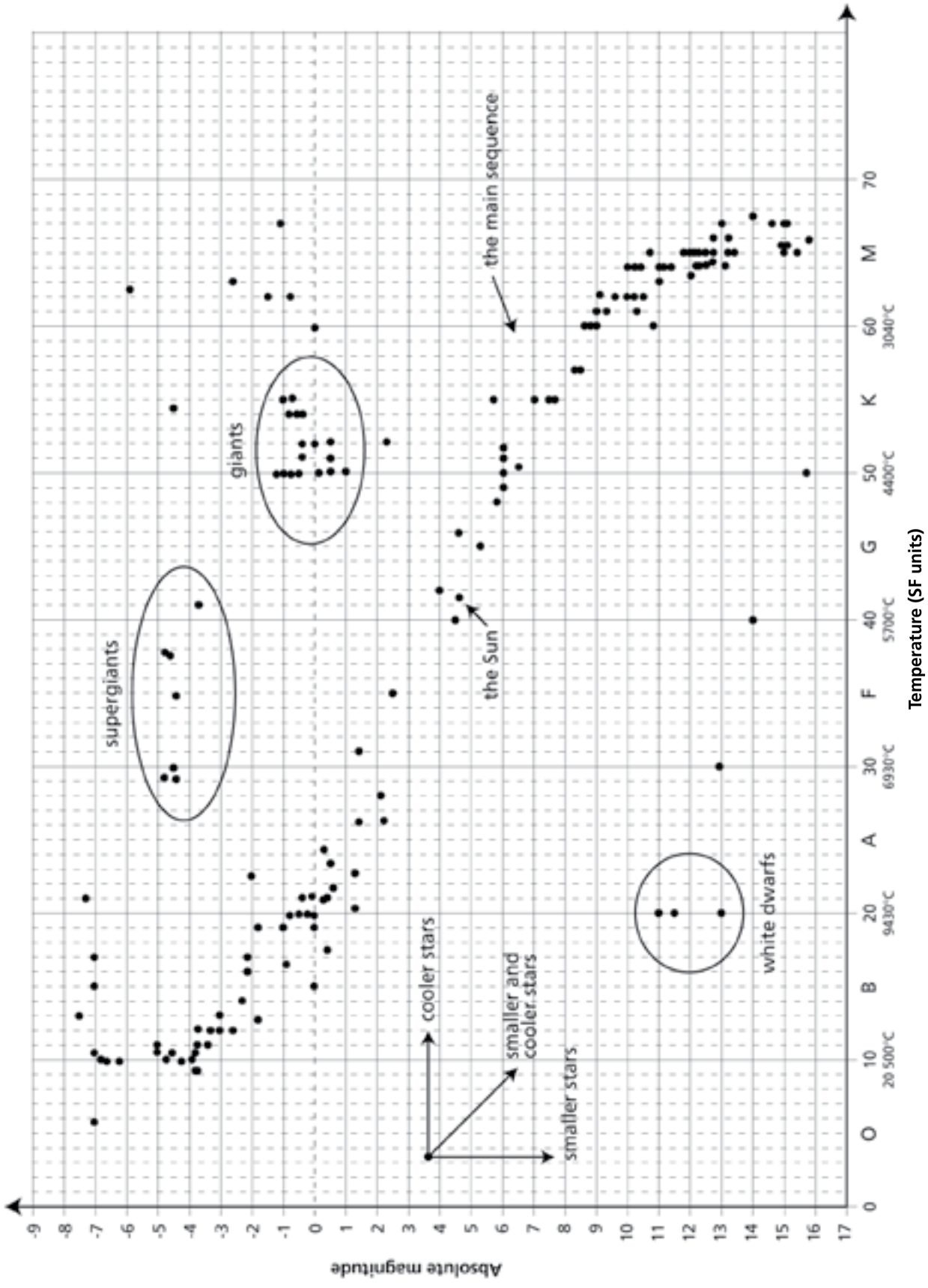
Clue	Word
1 Type of lens that bulges in the middle	convex
2 Opposite of convex	concave
3 Where parallel rays of light meet after passing through a lens	focus
4 Image size divided by object size	magnification
5 The principal _____ is a line through the middle of a lens that contains the focus	axis
6 Type of image that can be projected onto a screen	real
7 Image that cannot be projected onto a screen	virtual
8 Upside down	inverted
9 Smaller	diminished
10 Bigger	enlarged
11 Optical devices used for birdwatching that contain lenses and prisms	binoculars
12 Mirror that curves inwards	concave
13 Mirror that curves outwards	convex

Unit 7.3: Colour

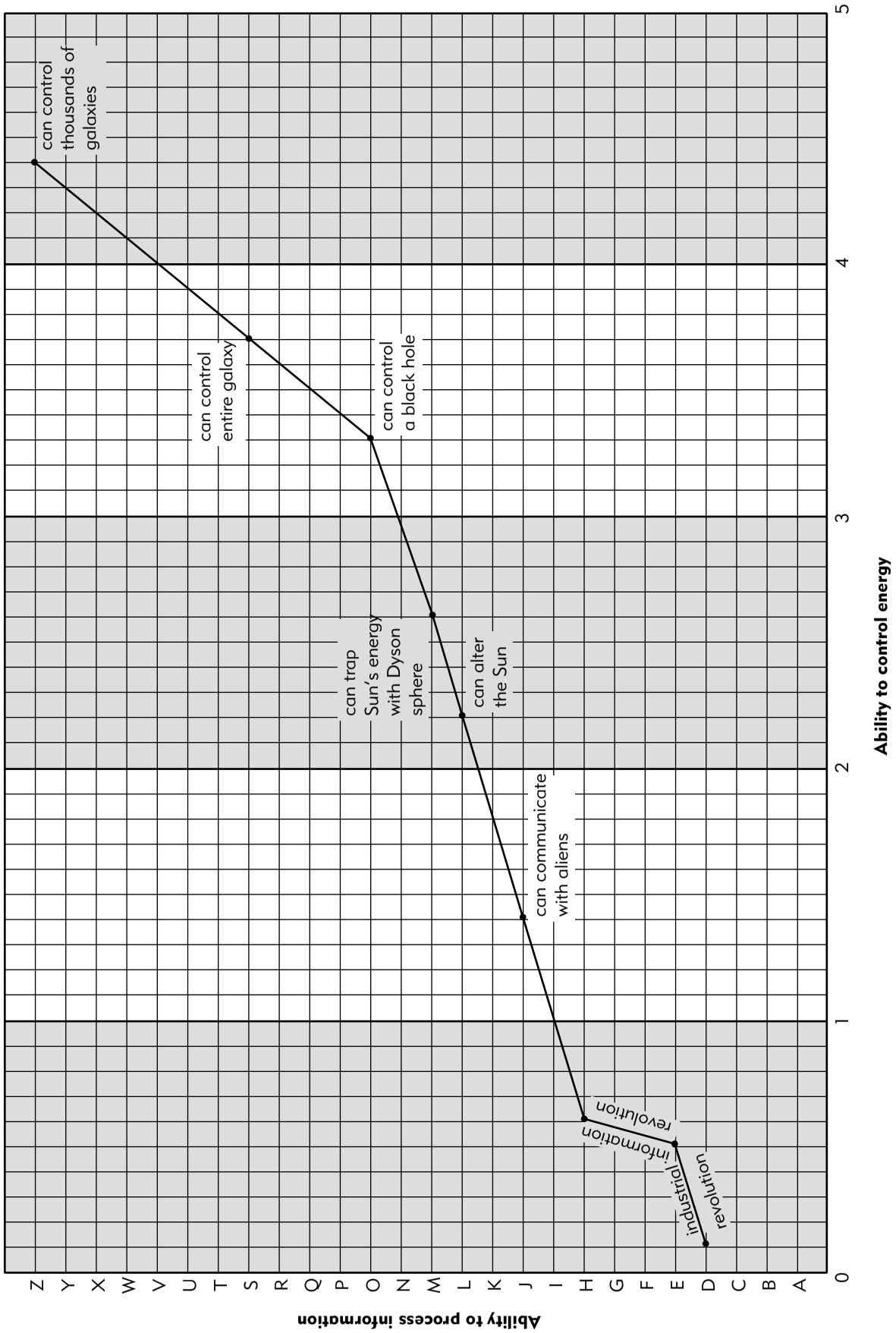
Clue	Word
1 Red, blue and green are the _____ colours for light	primary
2 Set of colours	spectrum
3 The splitting of white light into colours	dispersion
4 Causes blue sky and red sunsets	scattering
5 Caused by light totally internally reflecting in raindrops	rainbow
6 The 'I' in ROYGBIV	indigo
7 A secondary colour of light	magenta
8 Allows only light of a particular colour to pass through	filter
9 Occurs when light strikes a coloured object	absorption
10 Colours that add to form white light	complementary
11 Finely spread out solid particles of colour found in substances such as paint	pigment
12 What we see when cyan, magenta and yellow dyes are combined	black
13 A pigment that absorbs blue light	yellow


present day
big bang

1–4


Absolute magnitude
-9
-8
-7
-6
-5
-4
-3
-2
-1
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
supergiants
giants
the main sequence
the Sun
cooler stars
smaller and cooler stars
smaller stars
white dwarfs
O
B
A
F
G
K
M
20 500°C
9430°C
6930°C
5700°C
4400°C
3040°C
Temperature (SF units)

1–3



Extension

Student responses will vary.

Answers 8.4 Space travel

1 a Mercury (92)
$92 \times 10^6 / 3 \times 10^5$
= 306.7 seconds
= 306.7/60
= 5.11 minutes

b Venus (42)
$42 \times 10^6 / 3 \times 10^5$
= 140 seconds
= 140/60
= 2.33 minutes

c Mars (78)
$78 \times 10^6 / 3 \times 10^5$
= 260 seconds
= 260/60
= 4.33 minutes

d Jupiter (628)
$628 \times 10^6 / 3 \times 10^5$
= 2093.3 seconds
= 2093.3/60
= 34.89 minutes

e Saturn (1250)
$1250 \times 10^6 / 3 \times 10^5$
= 4166.7 seconds
= 4166.7/60
= 69.44 minutes
= 1.16 hours

f Uranus (2725)
$2725 \times 10^6 / 3 \times 10^5$
= 9083.3 seconds
= 9083.3/60
= 151.38 minutes
= 2.52 hours

g Neptune (4350)
$4350 \times 10^6 / 3 \times 10^5$
= 14 500 seconds
= 14 500/60
= 241.67 minutes
= 4.03 hours

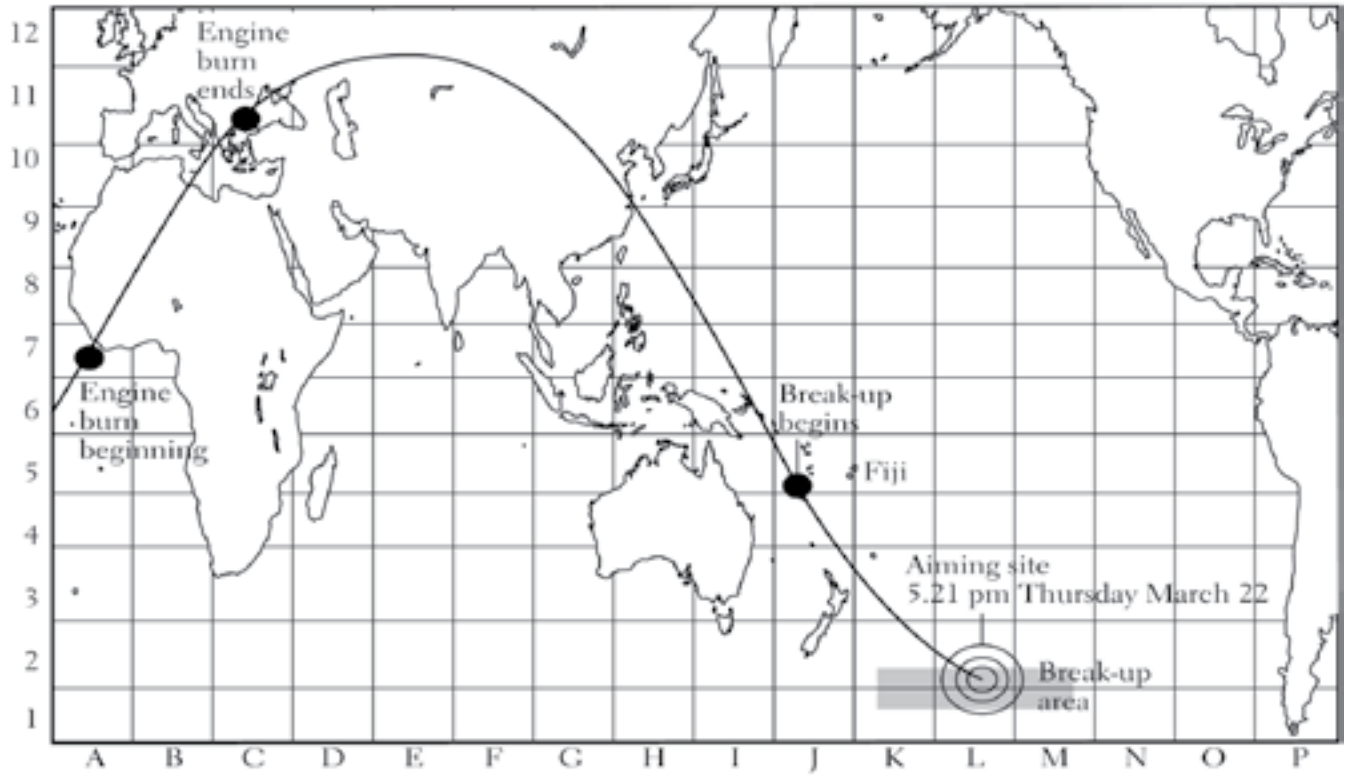
h Alpha Centauri (4.27)
4.2/2.5
= 1.71 years

i Betelgeuse (520)
520/2.5
= 208 years

j Crab Nebula (4000)
4000/2.5
= 1600 years

Answers 8.5 *Mir's* plunge

1–3



4

Answers 8.6 Global positioning

1 **a** Celestial navigation is the technique of finding position by observing the Sun, Moon, stars and planets.

b Objects in the sky are so far from the Earth that they appear to be located on the inside surface of the celestial sphere.

c G = Global, P = Positioning, S = System

2 A worldwide radio-navigation system formed from a constellation of 24 satellites or 'stars' that can be used for navigation.

3 NAVSTAR could mean navigational stars. Actual meaning is **NAV**igation **S**ystem by **T**iming **A**nd **R**anging.

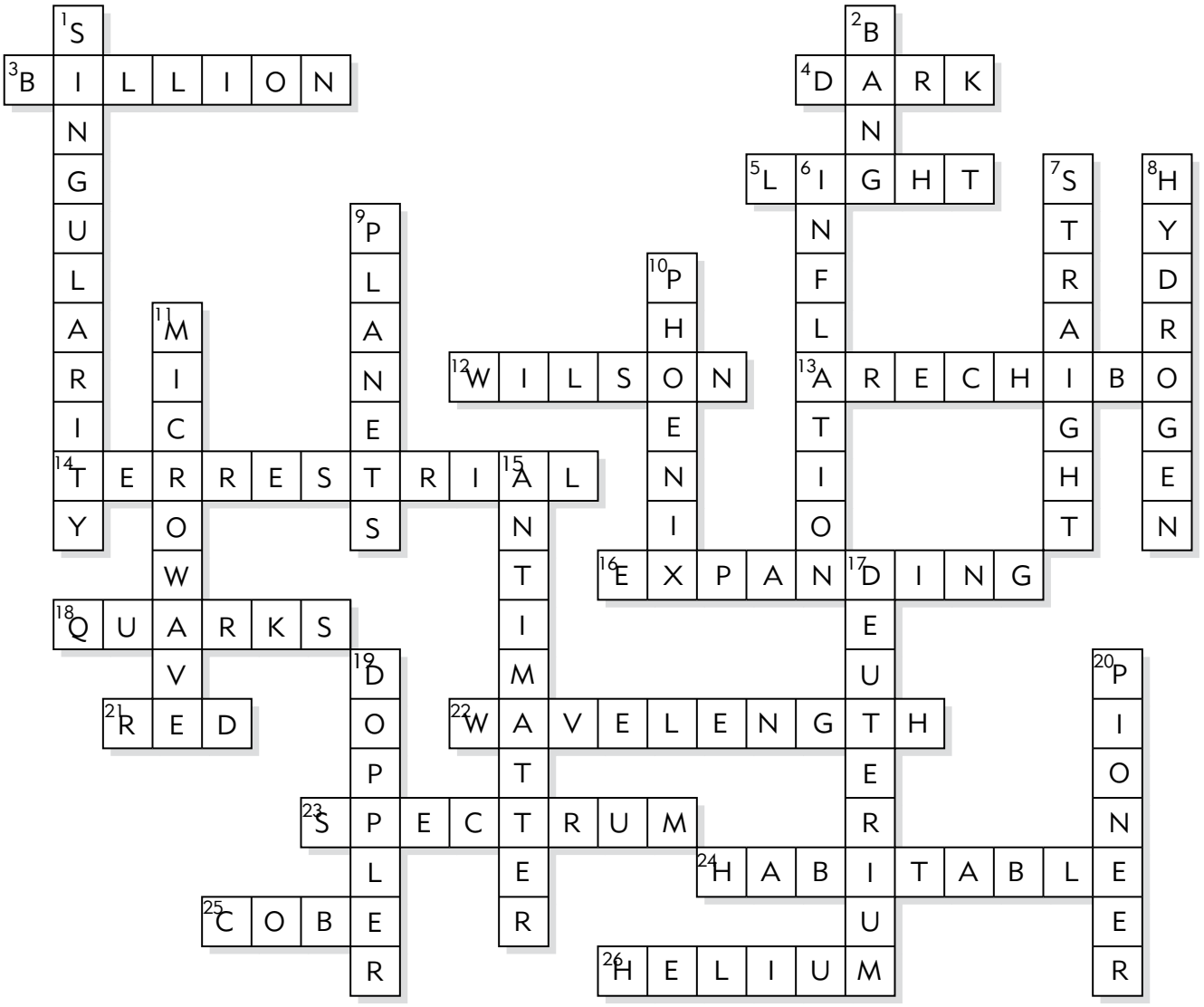
4 positional accuracy to within metres

5 Three satellites will give a latitude/longitude position. This is called a 2D position fix. With four or more satellites, you get a 3D position which includes latitude, longitude and altitude.

Extension

Student responses will vary.

Answers 8.7 Crossword



Answers 8.8 Sci-words

Unit 8.1: The expanding universe

Clue	Word
1 A spectrum from a star moving away from us may exhibit _____ shift	red
2 Effect involving shorter wavelengths produced in front of a moving object, and longer wavelengths behind	Doppler
3 Range of colours	spectrum
4 Distance between waves	wavelength
5 Scientists believe the universe started with the Big _____	Bang

Unit 8.2: The Big Bang

Clue	Word
1 The point from which the universe exploded	singularity
2 Matter and _____ destroy each other	antimatter
3 The building blocks of protons and neutrons	quarks
4 One of the first elements formed in the early universe	hydrogen
5 The second element formed in the early universe	helium
6 Penzias and _____ discovered the 'afterglow' of the young universe	Wilson
7 Invisible matter that may explain why particles in the universe compacted to form the first galaxies	dark
8 Formed by gas and dust orbiting the Sun	planetesimals

Unit 8.3: The life of a star

Clue	Word
1 Massive cloud of dust and gas in space	nebula
2 Causes stars to form from clouds of dust and gas	gravity
3 Our Sun will eventually become a _____ giant	red
4 A _____ dwarf is the final stage of an old star	white
5 Exploding star	supernova
6 Super dense star formed after a supernova	neutron
7 A rapidly rotating star that sends out a beam of radio waves	pulsar
8 The remains of a collapsed star whose gravity is so strong that not even light can escape	black hole

Unit 8.4: Are we alone?

Clue	Word
1 _____ Centauri is our nearest star	Proxima
2 Zone around a sun where life may be possible	habitable
3 Radio waves travel at the speed of _____	light
4 Range of frequencies in which background interference is less	microwave
5 The T in SETI	terrestrial
6 A project involving the Parkes radio telescope in New South Wales	Phoenix

Unit 8.5: Using space

Clue	Word
1 Object that orbits another	satellite
2 Use of satellites to gather information	remote sensing
3 Extravehicular activity	EVA
4 Feeling of weightlessness	microgravity

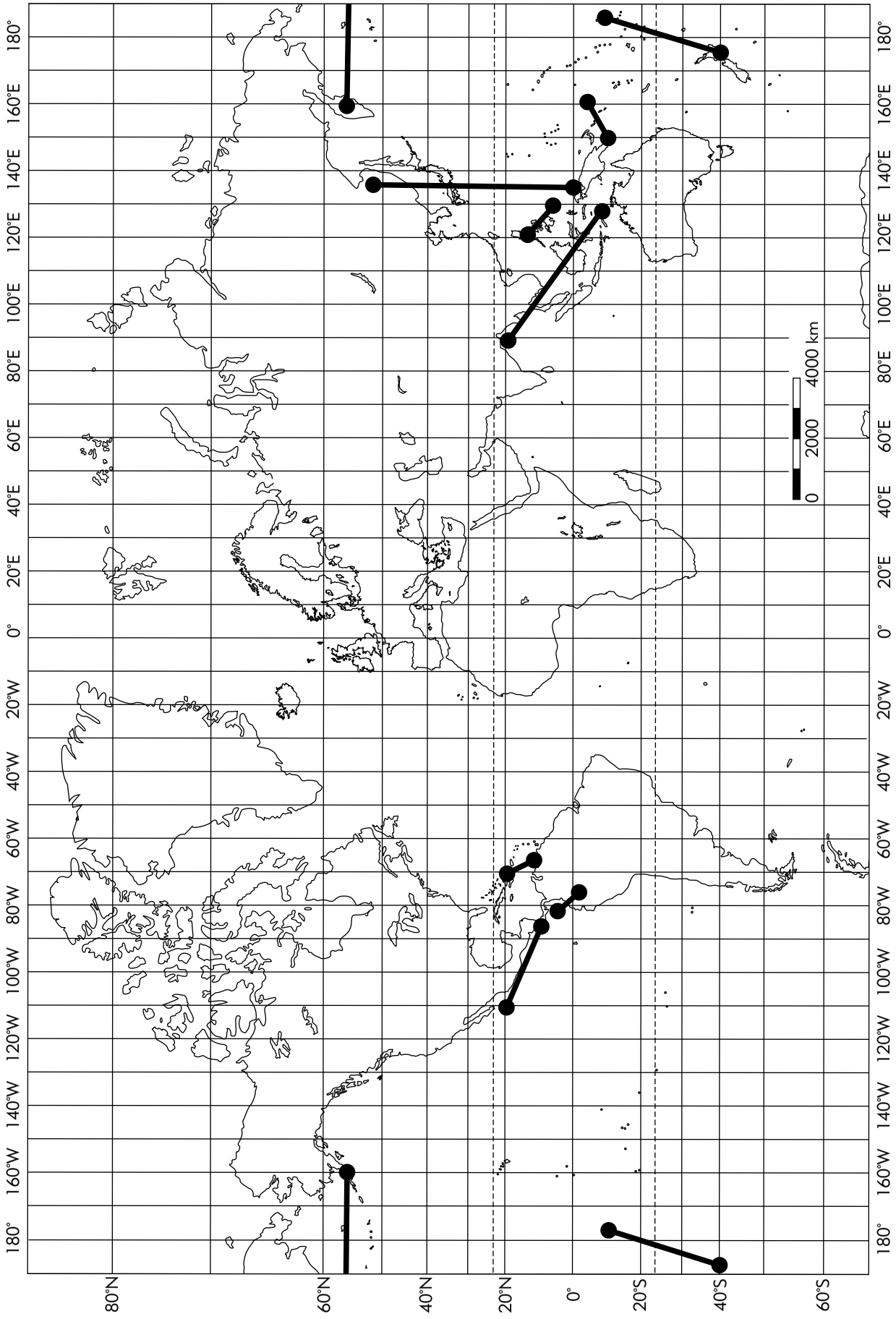
Answers 9.1 The planet splatter

Worksheet supports Prac 1, Unit 9.1 of *Science Focus 3* Second Edition.

Answers 9.2 Future Earth

Worksheet supports Prac 3, Unit 9.1 of *Science Focus 3* Second Edition.

1,2 Refer to the map below showing the positions of trenches.



3 **a** Ocean trenches occur at boundaries where an ocean plate collides with a continental plate and sinks under it, the 'sink' forming the trench.

b The molten rock from the subduction zone is full of dissolved gases, making it lighter than the surrounding rock. It thus pushes upwards, forming volcanoes or a chain of volcanic islands. The upper continental plate also thickens on collision, often forming a folded mountain range.

4 As the ocean plate sinks, it begins to melt and return to the mantle below. The ocean plate is thus gradually destroyed as it sinks beneath the continental plate.

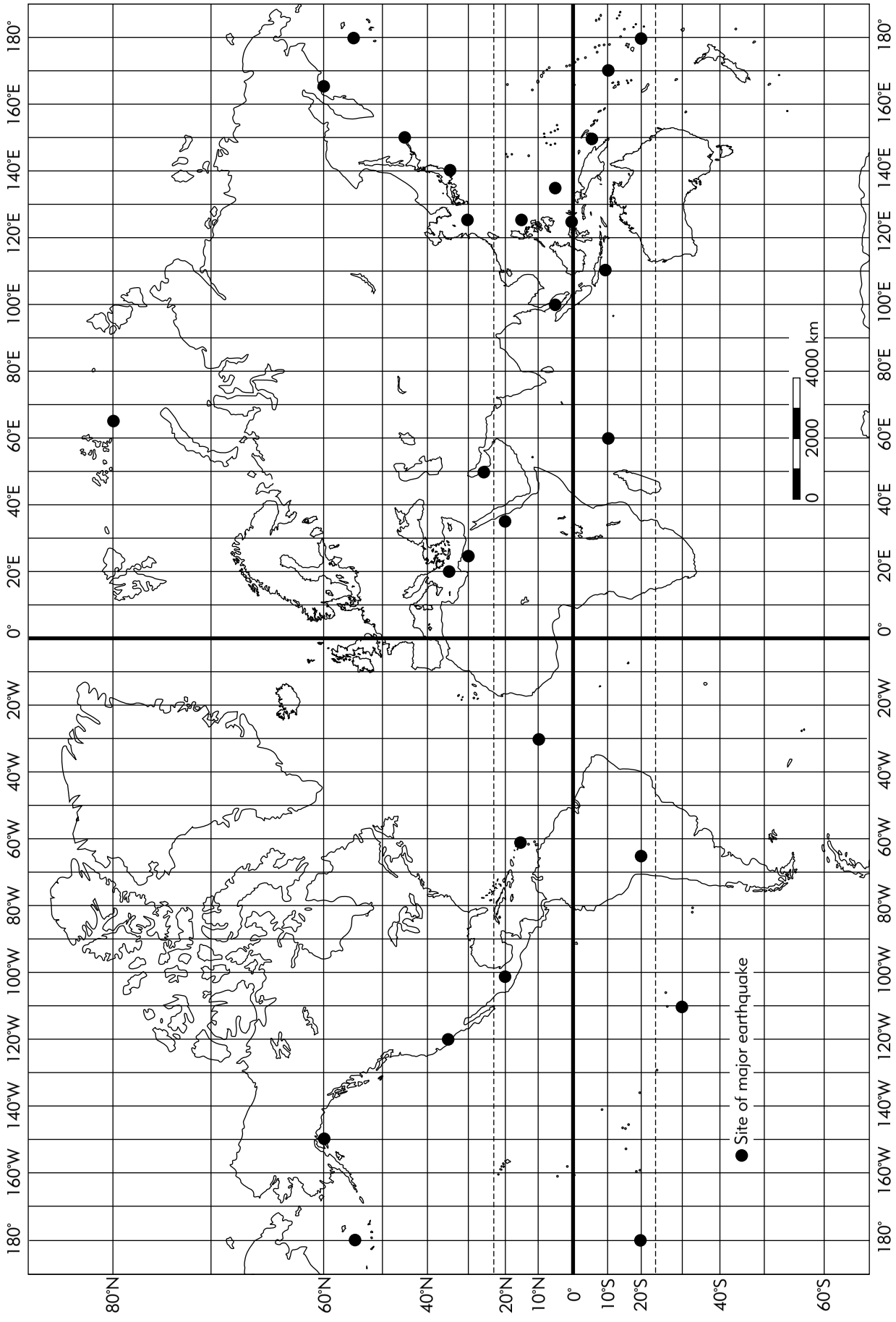
5 The subduction zone is a region of molten rock caused by the friction between the two plates.

6 Trenches are located at the boundary where a continental shelf meets an ocean plate. The edge of the plate will often follow the coast of the continent itself.

7 Eroded matter flows down the rivers of the continents and into the oceans. This matter will settle into the deepest sections of the ocean, partly filling the trenches.

Answers 9.4 Earthquakes

1 Refer to the map below showing the locations of major earthquakes.



2 From the plotted earthquakes, the stable continents are mainland Asia, Europe and Australia.

3 **a** Any four of Australia, Britain, Russia, France, Greenland, Antarctica, Ireland and Finland.

b Any four of USA (west coast), Mexico, Malaysia, Indonesia (Java), the Philippines, Japan, Papua New Guinea, the countries along the west coast of South America (Peru, Chile and Bolivia), the countries of North Africa (Libya, Egypt and Sudan), the island countries of the Caribbean (West Indies) and some Pacific Island countries.

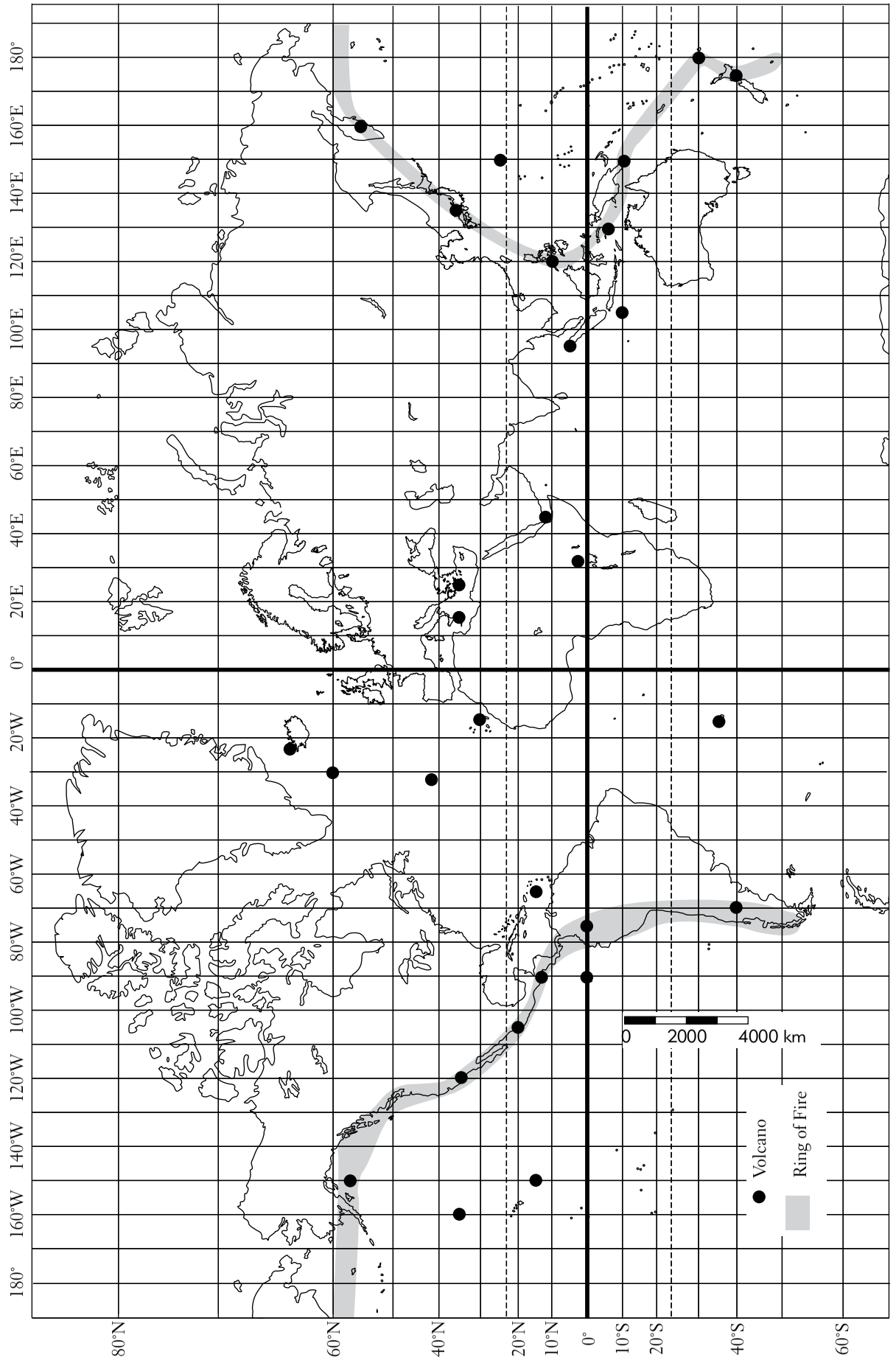
4 **a** The locations of the plotted quakes follow the boundaries of the tectonic plates.

b Plate boundaries are where any movement of the plates will become obvious.

5 Underwater earthquakes will often cause a tsunami that can pound and drown lengths of coastline, villages and towns.

Answers 9.5 Volcanoes

1 Refer to the map below showing the locations of major volcanoes.



Answers 9.5 Volcanoes

2 Volcanoes are often in areas prone to earthquakes because both occur at plate boundaries.

3 See map on previous page.

4 Kauai is in the far west and so is the oldest, having formed and moved west first. The most volcanic would be Hawaii, being the most easterly and thus relatively new.

Answers 9.6 Earthquake epicentres

Earthquake 1: J8

Earthquake 2: G5–H5

Earthquake 3: N15

Earthquake 4: I8

Answers 9.7 Waves in earthquakes

1 **a** epicentre

b focus

2 A body wave travels through the Earth. Surface waves travel around the globe, not through it.

3 **a** P and S are body waves.

b L is a surface wave.

c S and L are transverse waves.

d P is a longitudinal wave.

4 P stands for primary, S for secondary and L for Love.

5 **a** 2.5 minutes

b 11.9 minutes

c 3.4 minutes

6 L is the slowest, followed by S, then P. The graph shows that S, and then L, take longer times to travel the same distance as the P wave.

7 P: average speed is 780 km/min
S: average speed is 420 km/min
L: average speed is 300 km/min

8 **a** 4 minutes

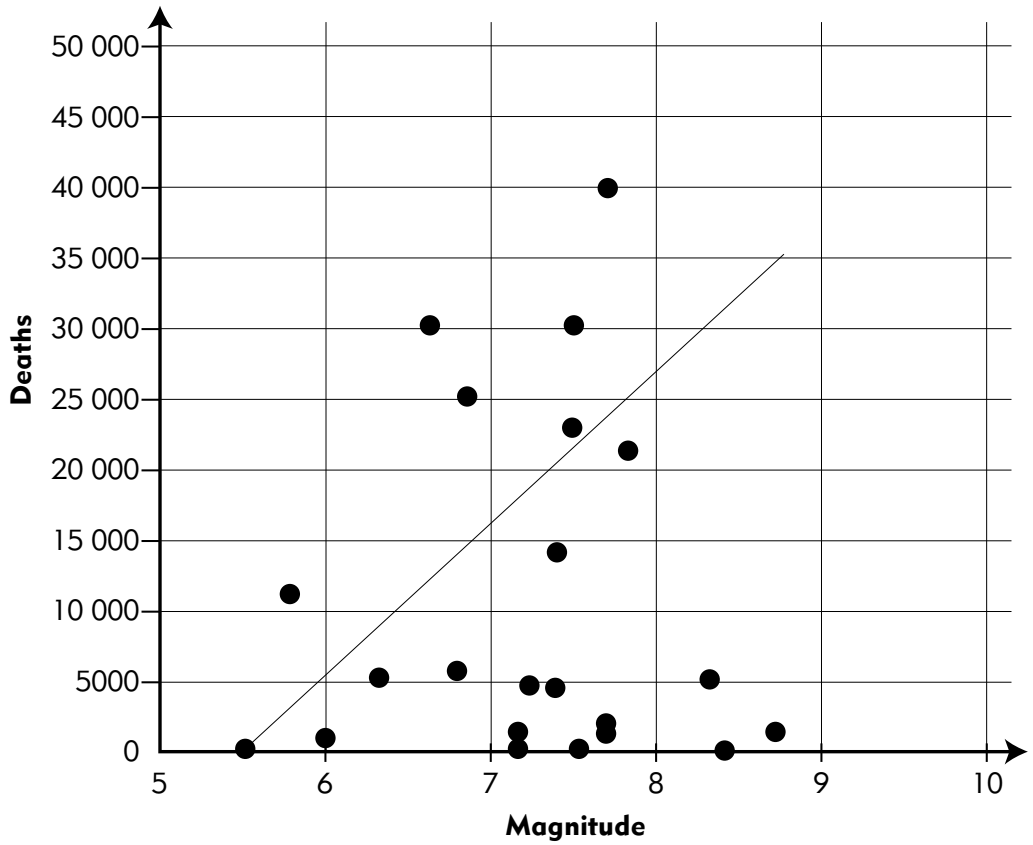
b 7.1 minutes

c 10 minutes

9 A seismometer is a device that measures the energy of a quake. A seismograph is the graph it produces.

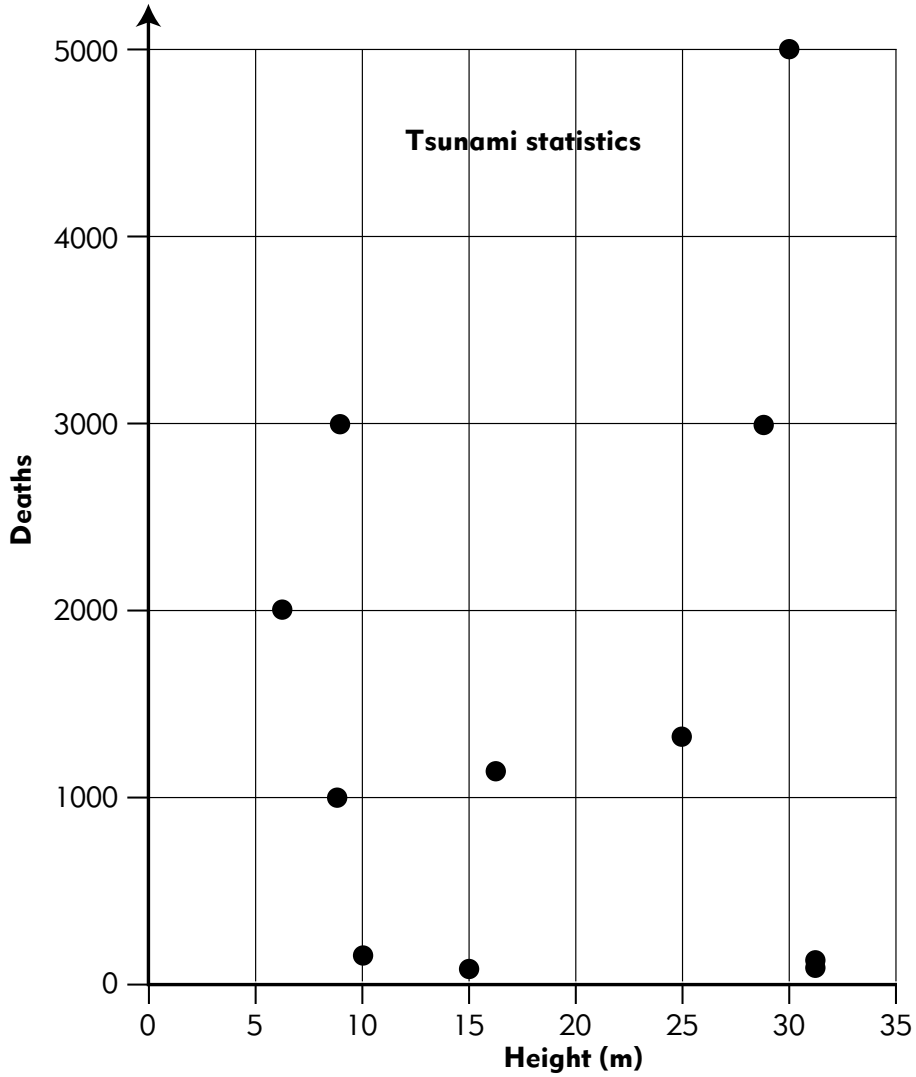
10 The quake occurred at 10.34 am, P waves arrived at 10.41 am, S waves arrived at 10.46 am, and L waves arrived at 10.51 am.

1



2 In general, the greater the magnitude of an earthquake, the greater the number of deaths. However, factors such as population density, quality of buildings, time of day etc. would affect the number of fatalities.

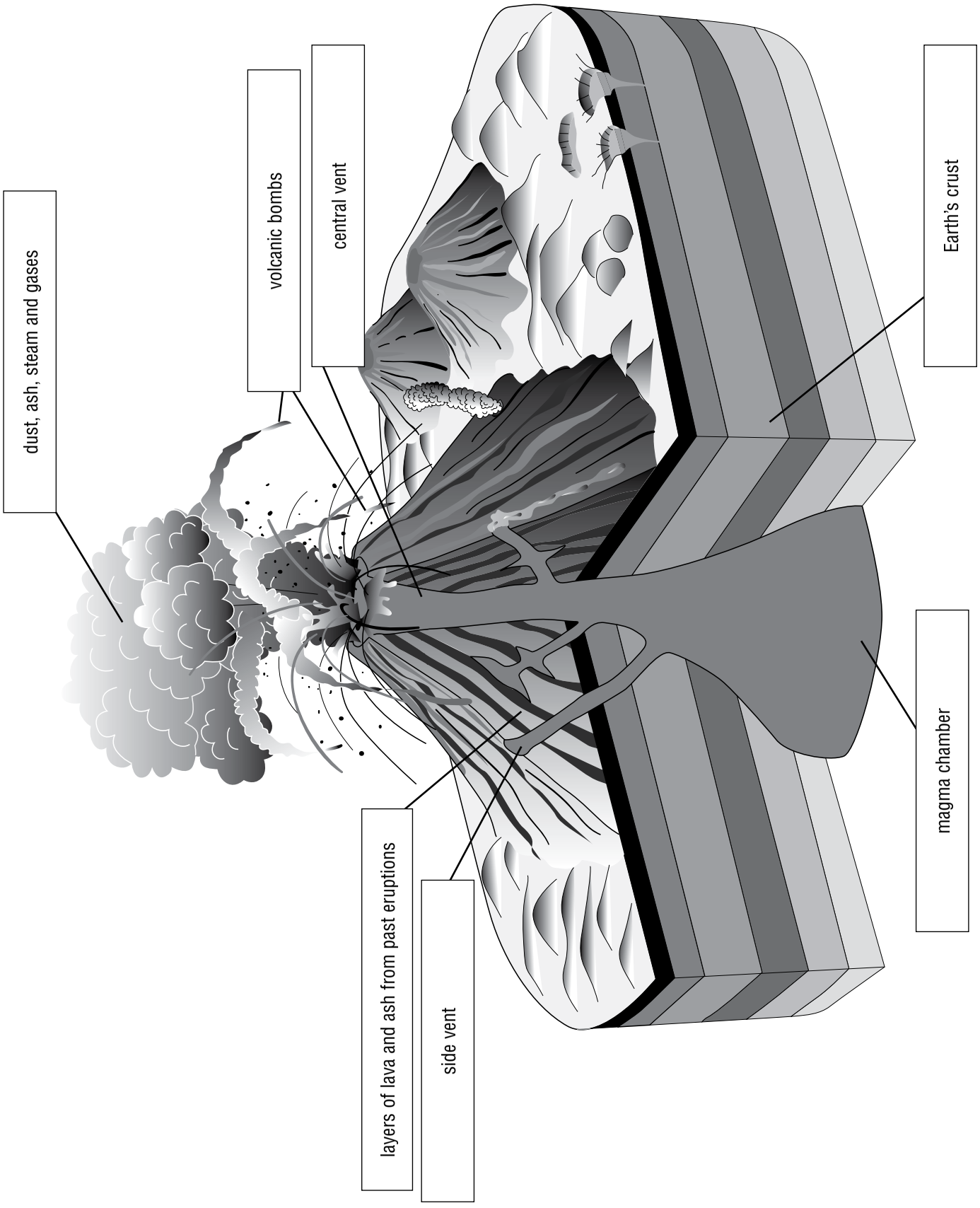
1



2 **a** There is no definite trend. Bigger waves don't necessarily mean more deaths (and, conversely, smaller waves can still cause a large number of deaths).

b Factors that might affect the number of deaths include location, population density, local geography and amount of warning.

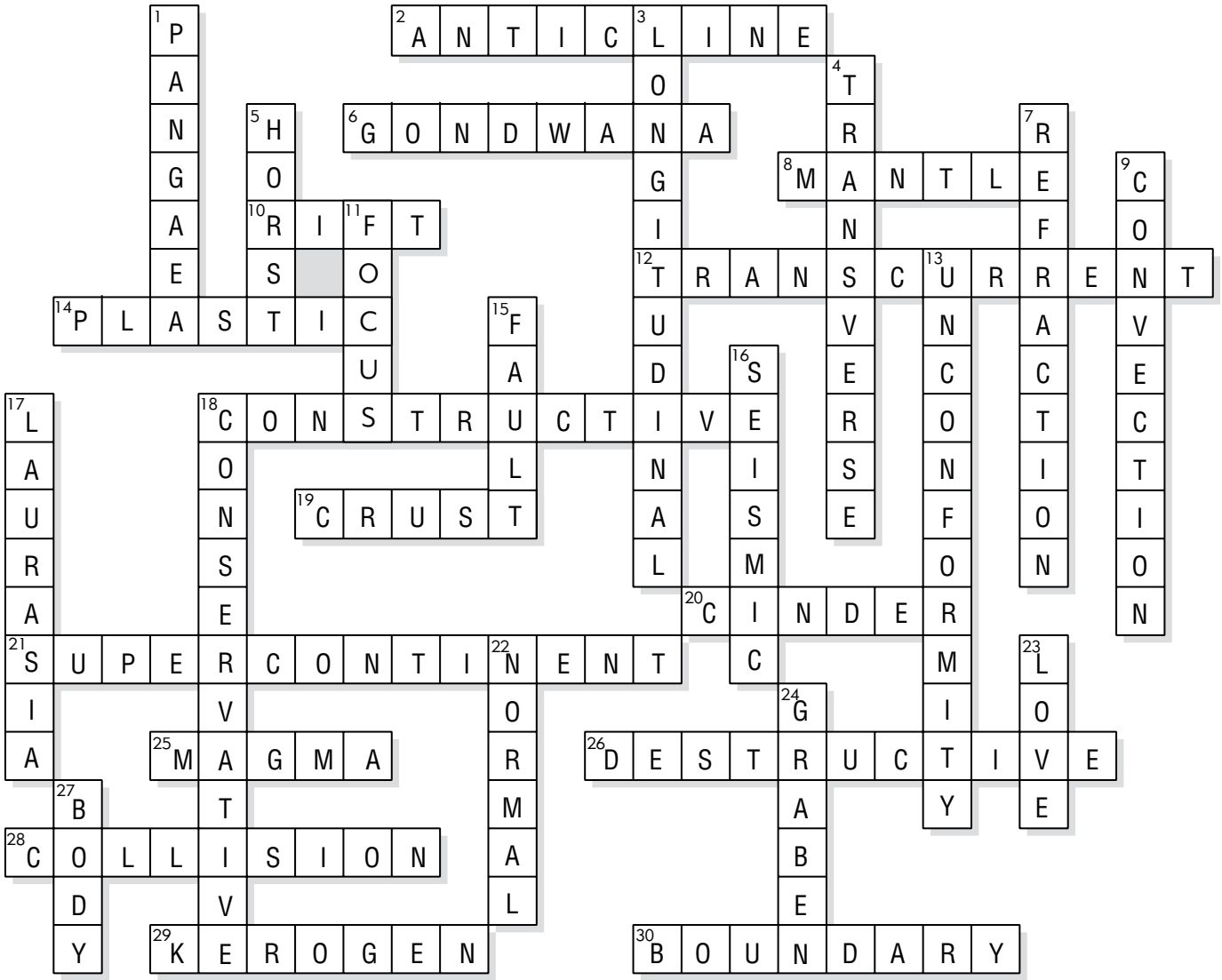
Answers 9.10 Labelling a volcano



Answers 9.11 Locating the epicentre

Worksheet supports Prac 3, Unit 9.3 of *Science Focus 3* Second Edition.

Answers 9.12 Crossword



Unit 9.1: Plate tectonics

Clue	Word
1 Wegener thought this split into the continents of today	Pangaea
2 A continent that split into others	supercontinent
3 Contained Australia, Antarctica, South America, Africa and India	Gondwana
4 Contained North America, Europe and Asia	Laurasia
5 Thin layer of rock that we live on	crust
6 The second layer of Earth	mantle
7 The movement of hot and cold rock, liquid or air	convection current

Unit 9.2: At the edges

Clue	Word
1 Edge of a plate	boundary
2 Where plates are separating	spreading boundaries
3 Subduction zones occur here	collision boundaries
4 Scraping boundaries	transform boundaries
5 Boundary where new rock is made	constructive
6 Molten gassy rock	magma
7 Angled dive where plates collide	subduction zone

Unit 9.3: Earthquakes

Clue	Word
1 Where a quake starts	focus
2 P, S, R and L waves	seismic waves
3 These travel through the Earth	body waves
4 These travel around the Earth	surface waves
5 Bending of waves as their speed changes	refraction
6 A scientist who studies earthquakes	seismologist
7 An instrument that measures an earthquake	seismometer
8 A graph showing the size of an earthquake	seismograph
9 The spot on the Earth's surface directly above the focus	epicentre
10 Measured by the Richter scale	energy
11 A scale from I to XII that rates a quake	Mercalli
12 Smaller earthquakes that happen after the main quake	aftershock
13 A 'tidal' wave that occurs when the epicentre is below the ocean	tsunami

Unit 9.4: Volcanoes

Clue	Word
1 When a volcano lets out material	eruption
2 A hole that acts as an exit for volcanic material	vent
3 Molten rock and gases that form underground	magma
4 Molten rock that has emerged in an eruption	lava
5 Volcanic dust made into mud by rainfall	lahar
6 Fragments of the walls of a volcano that are thrown out in an eruption	volcanic bombs
7 A volcano that will not erupt again	dead or extinct
8 A live but 'sleeping' volcano	dormant
9 A volcano that is erupting or could erupt soon	active

Unit 9.5: Landscaping the crust

Clue	Word
1 Nearly vertical fault	normal fault
2 A sideways fault	transcurrent fault
3 A cliff formed at a normal fault	fault scarp
4 An upthrust block	horst
5 Rift valley	graben
6 Downfold	syncline
7 Upfold	anticline
8 Erosion flattens a fold but new material is laid on top	unconformity
9 Rock formed from eroded material	sedimentary
10 Flattish volcano formed from lava	shield cone
11 Sharper volcano made from falling rock	cinder cone
12 Volcano made from lava and falling rock	composite cone
13 Magma cooled in the vent of a volcano	volcanic plug
14 Rock made by a volcano	igneous rock
15 Magma that cooled before reaching the surface	igneous intrusion
16 Weak point nowhere near a plate boundary	hot spot
17 Tar-like substance	kerogen
18 Actual energy source in fossil fuel	hydrocarbon

Unit 9.6: Geological time

Clue	Word
1 Evidence of past life	fossil
2 Widespread fossil that evolved quickly	index
3 Element used in radioactive dating of rocks	uranium
4 Element used in radioactive dating of plants and animals	carbon
5 The Earth is thought to be about 4.5 _____ years old	billion
6 Example includes Mesozoic or Palaeozoic	era
7 The period of the 'rise of the mammals'	tertiary
8 Era in which there was no life on Earth	azoic
9 Sea fossil	trilobite

Science Focus 3 TEST

1: Forensics

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 78 marks

Section A—Multiple choice (17 marks)

1	The Bertillon system of identification involved: **A** taking fingerprints **B** making up a composite drawing **C** measuring and recording the dimensions of a series of bony body parts **D** retroactive interference.		1
2	Choose the best answer. Using photographs to identify people is difficult because: **A** they might be wearing different clothes **B** the photographs are unrealistic **C** drawings are better **D** the photographs themselves can distort a person's memory.		1
3	Who performs an autopsy? **A** the victim **B** a police officer **C** a forensic pathologist **D** a forensic scientist		1
4	The Identikit system of identification involved: **A** taking fingerprints **B** making up a composite drawing **C** measuring and recording the dimensions of a series of bony body parts **D** retroactive interference.		1
5	There are four types of fingerprints. Which of these is not one of them? **A** whirl **B** loop **C** composite **D** arch		1

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6	Which of the following statements is true regarding iris and retina identification? **A** False eyes can be used to 'bluff' the iris and retina identification. **B** The chance of incorrect identification is very high. **C** Iris identification is less accurate than using fingerprints. **D** Scans are sometimes difficult to obtain from uncooperative people.		1
7	Which of the following contains your DNA and could be used to identify you? **A** your dandruff **B** your blood **C** your saliva **D** all of the above		1
8	Chromatography is: **A** a technique used to get inks to fluoresce **B** when oblique lighting is used to identify writing **C** a method used to counterfeit money **D** a technique that separates the colours in inks and dyes, allowing their identification.		1
9	Which of the following is incorrect? Good track impressions that can be used in forensics are created when you walk: **A** onto mud, clay or snow **B** out from somewhere wet onto a dry path **C** through something spilt (such as blood or grease) over a hard surface **D** through mud or dirt and then onto a clean surface.		1
10	The shape of blood drips on the ground can be used to determine: **A** the direction of movement of the victim **B** the time the crime was committed **C** the blood group of the victim **D** the DNA of the victim.		1
11	Latent fingerprints are usually left on non-porous materials. This means they would easily be found on: **A** newspaper **B** unpolished wood **C** polished wood **D** any animal without paws.		1

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12	Forensic scientists usually: **A** help investigators collect evidence **B** make arrests **C** are pathologists **D** all of the above.		1
13	Evidence can be gathered from: **A** biological remains **B** computers **C** recording devices **D** all of the above.		1
14	Which of the following statements are true (there may be more than one)? **A** Blood sprays are not very useful in a crime investigation. **B** Wounds provide information about the type of attack, such as whether a blunt or a sharp object was used. **C** Gunshot wounds are more likely to kill near organs. **D** Gunshot wounds always kill.		2
15	Where can fingerprints be *easily* found? **A** glass, plastic and mirrors **B** unpolished wood, mirrors and stone **C** in water or other liquids **D** all of the above		1
16	A cold case is: **A** a bag used to store your clothes when you go snow skiing **B** a box that is used to store fruit in a cold-store **C** an unsolved crime that has no more leads for police to investigate **D** a person who has the flu.		1

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Section B—Written answers (61 marks)

1	**Define** biometric facial recognition.		2
2	**State** whether the following statement is true, false or a bit of both. **Justify** your answer. *Every person's DNA is different.*		3
3	Draw sketches of the following types of fingerprints, showing their main identifying characteristics. *loop*, *arch*		4
4	**Present** four reasons why iris identification is very reliable, providing almost 100% correct identification of an individual.		4

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5	**Describe** the methods of analysing writing and print? **Explain** each using one sentence.		8
6	A student has collapsed on the way home from school. **State** four ways that the police might use to identify him/her.		4
7	Briefly **describe** two techniques that can be used to identify the ink used to write or print a document.		2

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8	**List** five methods that make it difficult to forge Australian banknotes.		5
9	The first $100 bills in Australia were printed on paper and were predominantly black, white and grey in colour. **Present** three reasons why these bills were easily forged when first introduced.		3
10	**Explain** how you could tell whether a threatening document has been typed on an old-fashioned typewriter or printed from a computer printer.		2
11	Identity theft is an increasing crime. **Propose** what you think the term 'identity theft' means.		2

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12	**Explain** why rumour and gossip have no value in court as evidence. Second, **explain** how rumours can be dangerous in court.		3
13	**List** an example of a fibre that is: **a** animal in origin **b** vegetable in origin **c** mineral or synthetic in origin.		3
14	**List** three features of shoes that can lead to the identification of a criminal.		3
15	A decaying body has been found in the bush. It is covered in different types of insects, insect eggs, worms and maggots. **Explain** how you could use these to determine how long the body has been there.		2

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16	**Demonstrate** how information can be gained and used as evidence from each of a computer, a mobile phone or video footage.		3
17	**Explain** four ways in which blood splatters and drips can tell a forensic scientist what happened at a crime scene.		8

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Science Focus 3 TEST

2: The periodic table

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 71 marks

Section A—Multiple choice (21 marks)

1	Hydrogen peroxide: **A** is water **B** has one more hydrogen atom than water **C** has one more oxygen atom than water **D** is none of the above.		1
2	The atomic number of an element is equal to the number of: **A** electrons and neutrons **B** protons **C** neutrons **D** protons, neutrons and electrons.		1
3	Which one of the following is *not* a property of non-metals? Non-metals are: **A** normally gases or liquids at room temperature **B** either poor electrical conductors or non-conductors **C** able to be hammered into sheets **D** dull, with little or no shine.		1
4	The atomic number of zinc (Zn) is 30. Therefore, the ion represented by the symbol $^{65}Zn^{2+}$ has: **A** 30 protons, 35 neutrons and 28 electrons **B** 65 protons, 30 neutrons and 32 electrons **C** 30 protons, 65 neutrons and 28 electrons **D** 30 protons, 35 neutrons and 32 electrons.		1
5	An atom of chlorine (Cl) has 17 protons, 18 neutrons and 17 electrons. Which of the following statements is *incorrect* about the atom of chlorine? **A** It has an atomic number of 17. **B** The atom would have no overall electrical charge. **C** It would be in Period VII. **D** It would be in Period 3.		1

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2: The periodic table

6	Use the periodic table to determine which of the following statements is *incorrect* about chlorine atoms. **A** They form the ion Cl^-. **B** Chlorine atoms are smaller than fluorine atoms. **C** They are 'related' to F, Br and I. **D** Chlorine belongs to the halogen family.		1
7	A compound forms when: **A** two or more elements chemically combine with each other **B** two or more elements are physically mixed together **C** a large number of identical atoms join together **D** a mixture is separated into its components.		1
8	The vertical columns in the periodic table are known as: **A** periods, and they contain elements with the same number of outer-shell electrons **B** periods, and they contain elements with the same number of used electron shells **C** groups, and they contain elements with the same number of outer-shell electrons **D** groups, and they contain elements with the same number of used electron shells.		1
9	An atom of calcium has 20 electrons. Its electron configuration would best be written as: **A** $^{42}_{20}Ca$ **B** 20 **C** 2,8,8,2 **D** 2,8,10.		1
10	The electron configuration of an element in Period 3, Group IV of the periodic table is: **A** 2,8,6 **B** 2,8,8,3 **C** 2,8,18,3 **D** 2,8,4.		1
11	In the flame colour experiment, different metallic salts gave out different colours when placed in a Bunsen burner flame. This was because: **A** the heat from the Bunsen burner caused the electrons to jump to another shell. Coloured light emerged when they jumped back **B** the heat from the Bunsen burner caused the electrons to make colours **C** the electrons moved very fast around the nucleus **D** the light came from the Bunsen burner.		1

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2: The periodic table

12	A particular element is shiny, malleable and a good conductor of electricity. Which section of the periodic table would this element *not* be found in? **A** transition elements **B** Group I **C** Group III **D** Group VII		1
13	Which of the following is *not* a physical property of an element? **A** melting point **B** colour **C** reactivity with oxygen **D** density		1
14	Which of the following lists contains only compounds? **A** water (H_2O), glucose ($C_6H_{12}O_6$), carbon dioxide (CO_2) **B** tungsten (W), phosphorus (P), chlorine (Cl_2) **C** Vegemite, limewater ($Ca(OH)_2$), plutonium (Pu) **D** salt (NaCl), air, iron (Fe)		1
15	Moving down Group VII in the periodic table, which property would be expected to decrease? **A** the reactivity **B** the number of outer-shell electrons in each atom **C** the size of atoms **D** the melting point		1
16	In 1869, Mendeleev devised a periodic table. Which of the following did Mendeleev *not* do in his periodic table? **A** Predict properties of undiscovered elements. **B** List elements with atomic numbers greater than 100. **C** Arrange elements in order of increasing atomic mass. **D** Place elements in vertical columns based on chemical properties.		1
17	Which of the following elements would be expected to conduct electricity? **A** helium (element number 2) **B** chlorine (element number 17) **C** scandium (element number 21) **D** krypton (element number 36)		1

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2: The periodic table

18	The mass of one electron is approximately equal to the mass of: **A** 1800 protons **B** 1/1800 of a proton **C** 1 proton **D** 1 neutron.		1
19	Semi-metals have properties similar to those of: **A** metals **B** gases **C** non-metals **D** both metals and non-metals.		1
20	The most likely formula for magnesium iodide is: **A** MgI **B** Mg_2I **C** Mg2I **D** MgI_2		1

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2: The periodic table

Section B—Written answers (50 marks)

1	In a neutral atom, **list** which two particles have: **a** equal but opposite charges **b** approximately the same mass.		2
2	Write the electron shell configuration for each of the following elements: **a** an atom of phosphorus (atomic number 15) **b** an atom of potassium (atomic number 19) **c** an atom with atomic number 7.		3
3	Write the chemical symbol for any element: **a** in the same group as fluorine (F) **b** in the same period as sodium (Na) **c** in the group known as the alkali metals.		3
4	**List** three properties of non-metals.		3

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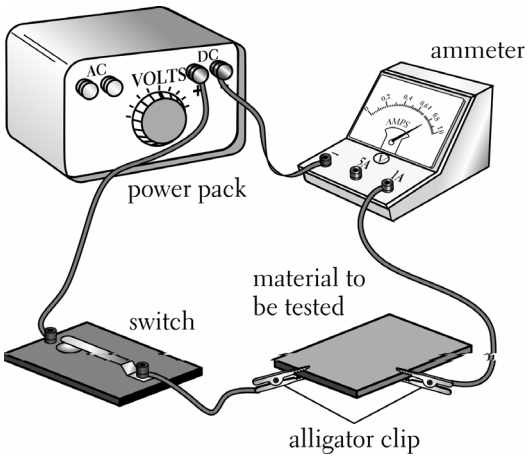
5	**List** three properties of metals.		3
6	The apparatus shown below may be used to test a property of several materials.  **Identify:** **a** the property of materials being tested **b** which of the materials listed below would be expected to cause the ammeter needle to move when the switch is closed: *sulfur, lead, copper, phosphorus.* **Justify** your answer to part **b**.		4

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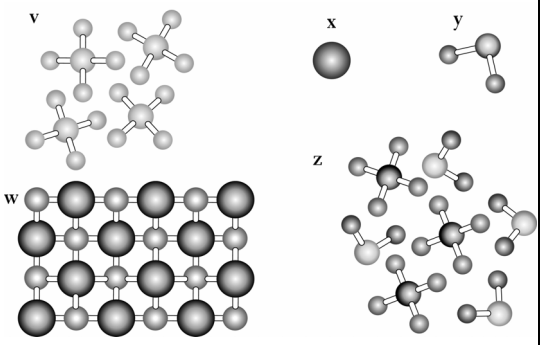
7 Five diagrams, labelled *v* to *z*, are shown below. Use the letters *v* to *z* to answer the questions that follow.



Identify the diagram that could represent:

a an atom of sulfur

b a water molecule (H_2O)

c a mixture of two compounds

d the potassium chloride lattice (KCl).

4

8 Complete the following table.

Atom and change	**Name of ion formed**	**Symbol of ion formed**
Magnesium (Mg) atom loses two electrons		
Nitrogen (N) atom gains three electrons		

4

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<table>
<tr>
<td>9</td>
<td>Write the chemical formulas for:
a calcium oxide
b barium iodide
c beryllium nitride
d aluminium oxide.</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>10</td>
<td>The flame colours of metal salts are given in the table below.
<table>
<tr><th>Salt</th><th>Flame colour</th></tr>
<tr><td>strontium chloride</td><td>red</td></tr>
<tr><td>sodium chloride</td><td>orange</td></tr>
<tr><td>potassium chloride</td><td>purple</td></tr>
<tr><td>barium chloride</td><td>green</td></tr>
</table>
Produce evidence that shows that the colours came from the metals and not from the non-metal parts of the salts?</td>
<td></td>
<td>2</td>
</tr>
<tr>
<td>11</td>
<td>Identify the group of the periodic table described in each statement below:
a reactive metals whose atoms form ions with a charge of +2
b reactive non-metals whose atoms form ions with a charge of –1
c colourless gases whose atoms are very unreactive.</td>
<td></td>
<td>3</td>
</tr>
</table>

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12	Listed below are five atoms using symbols that are not the usual symbols of the elements. **Use** the letters *V* to *Z* to answer the following questions. $^{40}_{18}V$ $^{20}_{10}W$ $^{40}_{19}X$ $^{39}_{19}Y$ $^{40}_{20}Z$ **Identify:** **a** which two atoms have the same number of neutrons **b** which atom has the smallest mass number **c** which two atoms belong to the same element **d** which two symbols represent noble gases.		4
13	Write the symbol of a: **a** Period 2 alkali metal **b** member of the lanthanide series **c** Period 5 transition element.		3
14	**State** one way in which helium (He) is: **a** like all other gases in Group VIII **b** different from all other gases in Group VIII.		2
15	**State** one way in which hydrogen (H) is: **a** similar to the Group VII elements **b** similar to the Group I elements.		2

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16	**Identify** which element in each pair (atomic numbers are given in parentheses) is more reactive. **a** Li (3) and Cs (55) **b** F (9) and Cl (17) **c** Na (11) and Mg (12) **d** O (8) and Ne (10)		4

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Science Focus 3 TEST

3: Chemical change

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 80 marks

Section A—Multiple choice (18 marks)

1	Which of the following involves a chemical change? **A** A piece of chocolate becomes soft when warmed. **B** Dry ice (solid CO_2) is warmed to form gaseous CO_2. **C** Fireworks explode. **D** Water boils to become steam.		1
2	Most reactions need an input of energy. These reactions are called: **A** spontaneous **B** non-spontaneous **C** explosive **D** energy needing.		1
3	The equation for the overall reaction in photosynthesis is shown below. $energy + 6CO_{2(g)} + 6H_2O_{(l)} \rightarrow C_6H_{12}O_{6(aq)} + 6O_{2(g)}$ Photosynthesis is: **A** an endothermic reaction **B** an exothermic process **C** a combustion reaction **D** an acid-base reaction.		1
4	$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$ This is an example of a: **A** neutralisation reaction **B** displacement reaction **C** precipitation reaction **D** decomposition reaction.		1

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5	A precipitate is a: **A** clear and colourless solution **B** gas **C** solid lump **D** fine solid powder that forms when two solutions are mixed.		1
6	A precipitate: **A** may form when two clear solutions are mixed **B** is a soluble salt **C** forms because the ions in a solution repel each other **D** collects at the surface when two solutions are mixed.		1
7	A weak acid is an acid which: **A** easily donates its hydrogen ion to a base **B** does not easily donate its hydrogen ion to a base **C** has few solute particles dissolved in the solvent **D** is highly corrosive.		1
8	When solutions of sodium chloride (NaCl) and silver nitrate ($AgNO_3$) are mixed, solid silver chloride (AgCl) forms. Which ions remain dissolved in the solution? **A** Na^+ and Cl^- **B** Ag^+ and Cl^- **C** Na^+ and NO_3^- **D** Ag^+ and NO_3^-		1
9	An acid reacting with a metal hydroxide to produce salt and water is an example of a: **A** combustion reaction **B** displacement reaction **C** neutralisation reaction **D** decomposition reaction.		1
10	Reduction occurs when: **A** a metal atom forms a positively charged ion **B** a non-metal atom forms a negatively charged ion **C** two ions combine to form a precipitate **D** steam condenses to form liquid water.		1

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11	A combustion reaction: **A** always has carbon dioxide as a product **B** always has water as a product **C** always has oxygen as a reactant **D** is always endothermic.		1
12	Which of the following is a property of bases? **A** taste sour **B** turn blue litmus red **C** react with acids to produce a salt and hydrogen gas **D** feel soapy		1
13	Which test would show that a sample of gas is carbon dioxide? The gas: **A** produces a popping noise when a match is placed in it **B** causes a flame to flare up **C** is odourless and yellow **D** produces a precipitate when it is bubbled through a limewater solution.		1
14	When hydrochloric acid reacts with magnesium, the products are: **A** salt and hydrogen gas **B** salt and water **C** salt, carbon dioxide and water **D** salt and oxygen gas.		1
15	A change from pH 3 to pH 5 means that the hydrogen ion present in the solution: **A** decreases by a factor of 2 **B** increases by a factor of 2 **C** decreases by a factor of 100 **D** increases by a factor of 100.		1
16	What is the pH of pure or distilled water? **A** 0 **B** 5 **C** 7 **D** 10		1
17	A solution of pH 8 is: **A** slightly basic **B** slightly acidic **C** strongly basic **D** strongly acidic.		1

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18	100 mL of a solution with pH 10 is added slowly to 900 mL of cold water. What will be the pH of the diluted solution? **A** 8 **B** 9 **C** 11 **D** 12		1

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Section B—Written answers (62 marks)

1	Write a chemical equation for a reaction where the products are dissolved $CaCl_2$, gaseous CO_2 and liquid H_2O. The reactants are solid $CaCO_3$ and dissolved HCl.		2
2	**Identify** three signs that a chemical reaction has occurred.		3
3	**Propose** two reasons why some chemical reactions do not occur spontaneously.		2

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4 The table below shows the symbol or formula of several ions.

Name of ion	Symbol or formula
magnesium	Mg^{2+}
carbonate	CO_3^{2-}
hydroxide	OH^-
nitride	N^{3-}
ammonium	NH_4^+

Construct the chemical formula for:

a magnesium nitride

b ammonium hydroxide

c magnesium hydroxide.

4

5 **Use** the information in the previous question to name these chemicals.

a $Al_2(SO_4)_3$

b $MgCO_3$

c $Ca(OH)_2$

3

6 **Identify** how many atoms of each element there are in the compounds below.

a $Zn(NO_3)_2$

b $(NH_4)_2SO_4$

2

7 **Predict** the missing chemicals in the equations below.

a $HCl + NaOH \rightarrow NaCl + \ldots$

b $2HNO_3 + Zn \rightarrow Zn(NO_3)_2 + \ldots$

c $2HCl + \ldots \rightarrow CaCl_2 + H_2O$

3

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8	Complete the following word equation. acid + metal oxide →		1
9	**Name** the salt produced when: **a** hydrochloric acid (HCl) reacts with magnesium (Mg) **b** nitric acid (HNO_3) reacts with barium carbonate ($BaCO_3$)		2
10	Two clear aqueous solutions, $BaCl_2$ and Na_2SO_4, are mixed. A white solid forms. **Identify:** **a** the solvent for this reaction **b** the name given to a solid that forms when the two solutions are mixed **c** the chemical formula of the solid that is formed in this reaction.		3
11	Write a chemical equation for the reaction between solid zinc (Zn) and an aqueous solution of hydrochloric acid (HCl). One of the products of the reaction is a soluble salt, zinc chloride ($ZnCl_2$).		2

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12 The table below shows the solubility of several ions.

Negative ion	Positive ion	Solubility of compounds
NO_3^-	all	soluble
SO_4^{2-}	Ba^{2+} Pb^{2+} others	insoluble soluble
Cl^-	Ag^+ Pb^{2+} others	insoluble soluble

Use this table to help you write equations for the following reactions. You should include states (aq, s, g) in your equations.

a Solutions of sodium sulfate (Na_2SO_4) and barium chloride ($BaCl_2$) are mixed.

b Solutions of lead(II) nitrate ($Pb(NO_3)_2$) and potassium chloride (KCl) are mixed.

4

13 **Define** the terms 'thermolysis reaction' and 'decomposition temperature'.

2

14 **Propose** whether water can decompose.

2

15 **Explain** what salts are made up of.

1

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16	**Explain** why it is necessary to have solubility rules.		2
17	Write the chemical formula for: **a** dinitrogen pentoxide **b** phosphorus trichloride **c** sulfur hexafluoride.		3
18	In a solution of pH 7, **state** the colour of: **a** red litmus **b** blue litmus.		2
19	You are provided with 10 mL of a solution of pH 4. **Describe** how you would prepare a solution of pH 5.		2
20	**List** three properties of acids.		3
21	**Use** any of the terms in the list below to **name** each of the reactions shown. *decomposition, precipitation, displacement, combustion, neutralisation* **a** $Pb(NO_3)_{2(aq)} + 2KCl_{(aq)} \rightarrow PbCl_{2(s)} + 2KNO_{3(aq)}$ **b** $C_6H_{12}O_{6(aq)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)}$ **c** $Mg_{(s)} + ZnCl_{2(aq)} \rightarrow MgCl_{2(aq)} + Zn_{(s)}$		3

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22	**a** **Identify** which substance in the following list could *not* be an acid. *HBr, CH_3COOH, K_2CO_3* **b** **Justify** your answer in part **a**.		3
23	**Predict** the approximate pH of the solutions described in **a** to **e** using the numbers shown below. *2, 4, 5, 7.5, 13* **a** coffee **b** oven cleaner **c** orange juice **d** rust remover **e** human blood		5
24	**Use** the numbers from the list below to **predict** the approximate pH of the solutions described in **a** to **c**. *1, 5, 7, 8, 13* **a** a concentrated solution of a strong acid **b** a neutral solution **c** a dilute solution of a weak base		3

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Science Focus 3 TEST

4: Sense and control

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 93 marks

Section A – Multiple choice (29 marks)

1	Which of the following is a sense? **A** sight **B** anger **C** moisture **D** temperature		1
2	The human eye focuses by: **A** changing the size of the pupil **B** changing the shape of the lens **C** moving the lens forwards or backwards within the eye **D** moving the head slightly closer or further from an object.		1
3	To reduce the amount of light entering the eye: **A** the pupil dilates **B** the pupil reduces in size **C** the lens thickens **D** the lens becomes slightly cloudy.		1
4	The section at the back surface of the eye lacking light-sensitive cells is called: **A** the fovea **B** the yellow spot **C** the blind spot **D** the black spot.		1
5	Binocular vision is: **A** the ability to see in three dimensions and judge distances **B** the magnified image you see when using binoculars **C** another name for perfect vision **D** when two people look at the same object.		1

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6	Colour blindness affects: **A** more women than men **B** men only **C** women only **D** more men than women.		1
7	The inability to focus properly on far-away objects is called: **A** accommodation **B** hyperopia **C** myopia **D** presbyteria.		1
8	How many main sections does the human ear contain? **A** one **B** two **C** three **D** four		1
9	The hammer, anvil and stirrup are: **A** three tiny bones in the ear **B** other names for the eardrum **C** attached directly to the cochlea **D** nerves that transmit sounds to the brain.		1
10	A bionic ear: **A** amplifies sounds before they reach the eardrum **B** is an artificial eardrum **C** replaces the cochlea with an electronic implant **D** produces electrical signals which stimulate auditory nerves.		1
11	When you smell a freshly baked cake, it's because: **A** vibrations caused by particles within the cake are passed into the air and in turn vibrate cells in your nose **B** your brain creates the illusion of smell based purely on what you see **C** tiny particles of cake have dissolved in the moist lining of your nose **D** cake particles carried by air currents dissolve on your tongue.		1

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12	How many different smells can a typical human nose detect? **A** 100 **B** 500 **C** 2000 **D** 10 000		1
13	Which of the following statements is correct? **A** Taste buds are on the sides of your tongue only. **B** Taste buds are on the sides of papillae (small bumps on the surface of the tongue). **C** Papillae are particles that dissolve samples of food so they may be detected by taste buds. **D** There are five different types of taste buds.		1
14	Which of the following is a type of skin cancer? **A** carcinoma **B** eczema **C** dermatitis **D** mole		1
15	When the level of carbon dioxide in your blood falls, the usual response of your body is to: **A** decrease breathing rate **B** increase breathing rate **C** produce the hormone oestrogen **D** produce the hormone insulin.		1
16	Which of the following is part of the peripheral nervous system? **A** the brain **B** spinal cord **C** heart muscle **D** taste buds on the tongue		1

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17	Interneurons transfer messages: **A** from receptors to the central nervous system **B** from the central nervous system to effectors **C** within the central nervous system **D** directly between sensory receptors.		1
18	Which of the following correctly matches receptor cells to the energy conversion they carry out? **A** Retina cells convert sound energy to electrical energy. **B** Cells in the taste buds convert chemical energy to electrical energy. **C** Thermoreceptor cells convert electrical energy to thermal energy. **D** Cochlea cells convert light energy to electrical energy.		1
19	Which of the following is *not* a hormone? **A** insulin **B** glucagon **C** dopamine **D** adrenalin		1
20	Which of the following is *not* a reflex action? **A** blinking **B** sneezing **C** eating **D** coughing		1
21	The part of the brain that gives you the sensation of hearing is the: **A** cerebrum **B** cerebellum **C** medulla **D** spinal cord.		1
22	Which of the following is usually largely controlled by the endocrine system? **A** breathing **B** heartbeat **C** tasting **D** water balance		1

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23	The conditions known as gigantism and dwarfism can result from abnormal levels of which hormone? **A** adrenalin **B** antidiuretic hormone (ADH) **C** human growth hormone (HGH) **D** oestrogen		1
24	Which hormone is known as the 'fight or flight' hormone? **A** adrenalin **B** antidiuretic hormone **C** human growth hormone **D** oestrogen		1
25	A fall in blood glucose levels is initially likely to result in the release of which hormone into the bloodstream? **A** insulin **B** thyroxin **C** progesterone **D** glucagon		1
26	Stroke occurs when: **A** the brain stops working **B** blood clots in your feet **C** you have a heart attack **D** the brain is deprived of blood.		1
27	Spina bidifa is a: **A** genetic hormone **B** growth hormone **C** genetic condition **D** disease.		1

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28	The disease diabetes mellitus is most often associated with which hormone? **A** insulin **B** growth hormone **C** thyroxin **D** testosterone		1
29	Which group of hormones is known to cause a plant to bend towards light? **A** adrenalin **B** antidiuretic hormone **C** auxin **D** oestrogen		1

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Section B – Written answers (64 marks)

1	**Identify** the part of the eye that is similar to that of a camera's: **a** diaphragm **b** film.		2
2	**Propose** an advantage of having an eye on each side of the head for animals such as rabbits and deer.		2
3	**Explain** how a sight defect may be corrected.		2

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4	Choose three parts of the eye and **summarise** their function.		6

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5	**Identify** the two types of light-sensitive cells in the eye, and **explain** the difference between them.		2
6	**Describe** any three parts of the ear.		6

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7	**Account** for the usefulness of the following glands: **a** sweat glands **b** sebaceous glands.		2
8	**Propose** two rules for avoiding skin cancer.		2
9	**Identify** two reasons why organisms need to respond to changes in their surroundings.		2
10	**a** **Define** 'homeostasis'. **b** **Identify** two substances whose levels are homeostatically controlled.		4

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11 **Define** the following terms and give an example of each.

a receptor

b effector

4

12 Complete the following table showing the receptors of various stimuli.

Stimulus	**Receptor and location**
light	
chemicals	
	semicircular canals in the ear
	cochlea cells in the inner ear

4

13 **Identify:**

a the meaning of 'the threshold' for a stimulus

b why it is useful to have thresholds.

3

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14	Suppose your body temperature increased after exercise. **Identify:** **a** one area where receptors that detected this increase would be located **b** the coordinating centre that would receive these messages **c** one organ or gland that might act as an effector **d** one response you might detect.		4
15	**Compare** and **contrast** neurons and most other animal cells. **State** two features that are similar and two features that are different.		4
16	**Describe** three ways in which the brain is protected from injury.		3
17	**Describe** the sequence of events that makes up the reflex action occurring when a bright light is shone in your eyes.		4

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18	**List** the most common causes of spinal cord injuries in Australia.		4
19	**Identify:** **a** one difference between pheromones and hormones **b** two examples of the use of pheromones.		4

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Science Focus 3 TEST

5: Reproduction

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 67 marks

Section A—Multiple choice (19 marks)

1	Which of the following methods of reproduction produces the most varied offspring? **A** budding **B** spore formation **C** fusion of gametes **D** fragmentation and regeneration		1
2	Hermaphrodites are organisms which: **A** produce both male and female gametes **B** reproduce asexually **C** produce pheromones to attract the opposite sex **D** can regenerate lost body parts.		1
3	Which of the following organisms reproduces using spores? **A** frog **B** fern **C** hydra **D** yeast		1
4	Female gametes are usually: **A** larger and more mobile than male gametes **B** smaller and more mobile than male gametes **C** larger and less mobile than male gametes **D** smaller and less mobile than male gametes.		1
5	Which of the following statements is correct? **A** Ovulation refers to the release of egg cells from the uterus. **B** Fertilisation refers to the production of female gametes. **C** Menstruation refers to the shedding of the lining of the ovaries. **D** Gestation refers to the time between fertilisation and birth.		1
6	The male part of a flower is made up of the: **A** stigma, style and ovary **B** petals and sepals **C** anther and filament **D** nectary and ovule.		1

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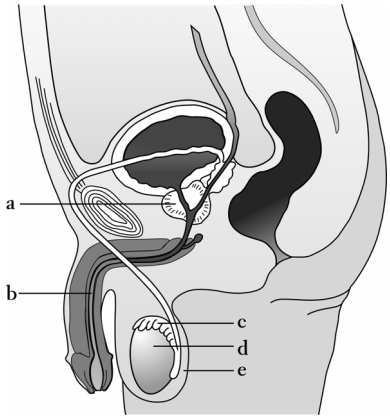
7	This question refers to the diagram below, which shows the human male reproductive system.  Which of the following correctly describes the function of the labelled structure? **A** Structure *c* stores sperm while they continue to develop. **B** Sperm are produced by structure *e*. **C** Structure *a* releases the major male hormone, testosterone. **D** Structure *d* is the only fluid-producing structure in the male reproductive system.		1
8	The urethra is a tube that carries: **A** sperm only **B** urine only **C** neither sperm nor urine **D** both sperm and urine (although not together).		1
9	Which of the following statements concerning gamete production in humans is correct? **A** Males have a store of gametes at birth; females produce gametes throughout their lives. **B** Females have a store of gametes at birth; males produce gametes throughout their lives. **C** Both males and females have a store of gametes at birth. **D** Both males and females produce gametes throughout their lives.		1

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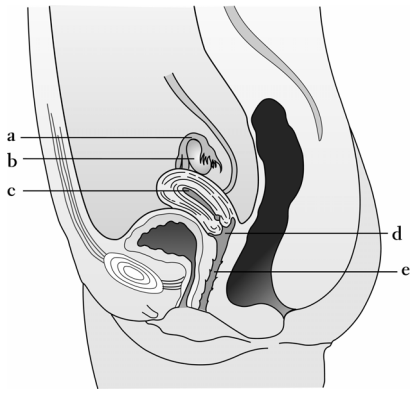
10	This question refers to the diagram below, which shows the human female reproductive system.  Which of the following correctly describes the function of the labelled structure? **A** Structure *d* is the site of implantation. **B** Structure *e* carries both urine and the menstrual flow. **C** Sperm are deposited in structure *c* during sexual intercourse. **D** Ova are released from structure *b*.		1
11	The site of fertilisation of the human egg cell is the: **A** ovary **B** fallopian tubes or oviduct **C** uterus **D** vagina.		1
12	The major female reproductive hormone is: **A** oestrogen **B** testosterone **C** adrenalin **D** insulin.		1
13	The major male reproductive hormone is: **A** sperm **B** testosterone **C** oestrogen **D** insulin.		1
14	The term menarche refers to the time in a human female's life when: **A** the first menstruation occurs **B** egg cells are no longer released from the ovaries **C** a zygote implants in the uterus **D** sexual intercourse first takes place.		1

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15	Which of the following shows the correct order of the stages of development of a human embryo? **A** zygote, foetus, blastocyst, morula **B** foetus, zygote, morula, blastocyst **C** zygote, blastocyst, morula, foetus **D** zygote, morula, blastocyst, foetus		1
16	Which of the following shows events in their correct order of occurrence? **A** copulation, ejaculation, fertilisation, implantation **B** ejaculation, fertilisation, copulation, implantation **C** implantation, ejaculation, fertilisation, copulation **D** copulation, ejaculation, implantation, fertilisation		1
17	Which of the following tests conducted during pregnancy does *not* involve the removal of fluid from either the mother or the foetus? **A** amniocentesis **B** ultrasound **C** testing for alpha-fetoprotein levels in the mother **D** chorionic villus sampling		1
18	Contraception refers to any process that prevents: **A** sexual intercourse from occurring **B** the transmission of STDs **C** pregnancy from occurring **D** gametes from being produced.		1
19	Which of the following processes does *not* normally occur during IVF? **A** Egg cells are retrieved from the ovary. **B** Eggs are placed in a salt solution at 45^{o}C. **C** Sperm are added to the eggs in the salt solution. **D** Fertilised eggs are transferred to the uterus after a short development time.		1

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5: Reproduction

Section B—Written answers (48 marks)

1	**a** **Explain** what is meant by asexual reproduction. **b** **List** two examples of asexual reproduction.		3
2	**Describe** two reasons why the offspring of asexually reproducing organisms may not all be identical.		2
3	Flowering plants may either be self-pollinated or cross-pollinated. **a** **Explain** what is meant by cross-pollination. **b** **Propose** two ways in which cross-pollination can be made to occur. **c** **Clarify** the advantage of cross-pollination.		4
4	A particular flower has its stigma placed higher than its anthers. **Predict** whether this flower could be self-pollinated. **Justify** your answer.		2

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5	**a** **State** in which structure the human testes are located. **b** **Outline** the advantage of the testes being in this structure, rather than deep in the body.		3
6	The ovaries of a 30-year-old woman have many scars on their surface. **Explain** what causes these scars?		2
7	**Describe** or **name** the events occurring at each of the following times in the human menstrual cycle. **a** days 1 to 5 **b** days 5 to 14 **c** day 14 **d** days 15 to 25		5
8	**a** **Describe** two ways in which twins can be produced. **b** **Identify** which method produces twins that are most alike.		5

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<table>
<tr><td>9</td><td>Describe three changes in females at puberty.</td><td></td><td>3</td></tr>
<tr><td>10</td><td>Describe three changes in males at puberty.</td><td></td><td>3</td></tr>
<tr><td>11</td><td>Complete the table below, which shows several methods of contraception.
<table>
<tr><th>Name of method</th><th>How it works</th></tr>
<tr><td></td><td>Rubber device that fits over the cervix.</td></tr>
<tr><td></td><td>Stops sperm entering the uterus and prevents implantation.</td></tr>
<tr><td>Contraceptive pills</td><td></td></tr>
</table></td><td></td><td>4</td></tr>
</table>

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12	**Describe** briefly each of the following embryonic stages. **a** morula **b** blastocyst		2
13	When referring to human reproduction, **define** what each of the following terms stands for. **a** STI **b** IUD **c** AIDS **d** IVF		4
14	**State** the length of time (in days) for each of the following: **a** the typical menstrual cycle in humans **b** human gestation **c** the number of days a human egg cell spends in the oviduct.		3

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15	Choose from the list below the approximate numbers for each of the quantities described in **a** to **c**. • *one* • *several hundred* • *several hundred thousand* • *several million* • *several hundred million* **a** The number of egg cells present in each human ovary at birth. **b** The number of sperm released by a human male at each ejaculation. **c** The number of sperm that reach the egg cell following copulation.		3

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Science Focus 3 TEST

6: Ecosystems

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 80 marks

Section A – Multiple choice (19 marks)

1	Consider the food chain shown below. *phytoplankton* → *zooplankton* → *small fish* → *large fish* → *sea eagles* Which of the following correctly describes the role of the zooplankton and large fish? **A** producer / third order consumer **B** first order consumer / fourth order consumer **C** first order consumer / third order consumer **D** second order consumer / fourth order consumer (*zooplankton* / *large fish*)		1
2	Which of the following organisms is *not* a consumer? **A** mushroom **B** blue-green algae **C** earthworm **D** elephant		1
3	Which of the following is recycled in an ecosystem? **A** matter only **B** matter and energy **C** energy only **D** neither matter nor energy		1

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4	Which type of reaction ultimately provides the energy for almost all life on Earth? **A** photosynthesis **B** respiration **C** nuclear fusion **D** decomposition		1
5	Consider the food chain shown below. *wheat* → *mouse* → *snake* → *kookaburra* If the wheat eaten by the mouse contains 1000 units of energy, approximately how many units of energy will the kookaburra receive? **A** 1 **B** 10 **C** 100 **D** 1000		1
6	A soil sample was analysed and four species of animal were found. The approximate numbers of each animal in the soil sample are shown below. • species W: 2 000 000 • species X: 2 • species Y: 2000 • species W: 20 000 Which species is likely to be the highest order consumer? **A** W **B** X **C** Y **D** Z		1
7	Transpiration is the name given to the loss of water by evaporation from: **A** moist soil **B** oceans **C** animals **D** plants.		1

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8	Approximately what percentage of all water on Earth is available as fresh water to organisms? **A** 1% **B** 10% **C** 50% **D** 98%		1
9	Nitrifying bacteria convert: **A** atmospheric nitrogen molecules to nitrate ions **B** atmospheric nitrogen molecules to amino acids **C** nitrate ions to atmospheric nitrogen molecules **D** ammonia to nitrate ions.		1
10	Which of the following is *not* responsible for nitrogen fixation? **A** the action of lightning on atmospheric nitrogen and oxygen **B** industrial production of nitrogen-based fertilisers **C** *Rhizobium* bacteria found in the root nodules of clover plants **D** denitrifying bacteria found in the soil		1
11	Which of the following processes removes carbon from the atmosphere? **A** combustion of natural gas **B** photosynthesis by green plants **C** anaerobic respiration by yeast **D** decomposition of fallen leaves by fungi		1
12	Which of the following is a renewable energy source? **A** coal **B** oil **C** geothermal **D** uranium		1
13	A photovoltaic cell is designed to convert: **A** solar energy to sound energy **B** solar energy to electrical energy **C** heat energy to electrical energy **D** heat energy to solar energy.		1

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14	Splitting one atom of uranium releases how much more energy than burning one molecule of natural gas? **A** 2.6 times **B** 26 times **C** 2.6 million times **D** 26 million times		1
15	Which of the following is *not* part of conservation? **A** things we do to prevent plants from becoming extinct **B** things we do to to keep habitats safe **C** cutting down trees **D** planting trees		1
16	What is an exotic species? **A** a species that is introduced **B** a species that is native **C** a species that is a pest **D** a species that is a pet		1
17	Which of the following energy sources is least used in Australia at present? **A** hydroelectricity **B** energy from wind **C** geothermal energy **D** solar energy		1
18	Salinisation is the process of: **A** planting trees in a desert **B** adding salt to the soil **C** losing good land to the desert **D** draining water.		1
19	Desertification is the process of: **A** planting trees in a desert **B** adding salt to the soil **C** losing good land to the desert **D** draining water.		1

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Section B – Written answers (61 marks)

1	**Examine** the food chain shown below. *mangroves → soldier crabs → spoonbills → sea eagles* For this food chain, **identify** the **a** producer **b** first order consumer **c** herbivore **d** third order consumer.		4
2	Draw a food chain, with four members only, by **interpreting** the following description. *The grasslands of Africa are home to many herbivores, including zebra, gnu and gazelles. These grazing animals are the prey of carnivores, including lions and cheetahs. Hyenas and vultures that scavenge for dead remains may, at times, eat any of the animals of the grassland.*		4
3	**Explain** why a food chain is unlikely to have more than six members.		3

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4	Consider the food chain shown below. *leaves* → *caterpillar* → *bird* → *cat* The total energy in the leaves is much greater than the energy received by the cat. **Describe** two different ways in which the energy of the leaves may be 'lost' and therefore not reach the cat.		2
5	**a** **Define** the Law of Conservation of Energy. **b** **Present** an example of how this law applies to energy in ecosystems.		3
6	**a** **Define** the Law of Conservation of Mass. **b** **Present** an example of how this law applies to matter in ecosystems.		3

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7	**Clarify** the meaning of each of the following, and give an example of each. **a** organic matter **b** an endothermic animal **c** an ectothermic animal		6
8	**Describe** two: **a** uses of nitrogen by plants **b** ways in which nitrogen may leave a plant **c** ways in which nitrogen may leave an animal.		6
9	The greenhouse effect is caused by an increase in the level of carbon dioxide in the atmosphere. **Describe** two ways in which this increase has occurred.		2

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<table>
<tr>
<td>10</td>
<td>Complete the following table, which shows the role of bacteria in the cycling of nitrogen.
<table>
<tr><th>Bacteria type</th><th>Role of bacteria</th></tr>
<tr><td>Nitrogen fixing</td><td>Convert nitrogen to</td></tr>
<tr><td></td><td>Convert ammonia to nitrates</td></tr>
<tr><td></td><td>Convert nitrates to</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>11</td>
<td>Explain what is meant by a non-renewable energy source. Include two examples in your answer.</td>
<td></td>
<td>3</td>
</tr>
<tr>
<td>12</td>
<td>Identify two ways in which solar energy may be used to generate electricity.</td>
<td></td>
<td>2</td>
</tr>
<tr>
<td>13</td>
<td>Explain two disadvantages of using geothermal energy to produce electricity.</td>
<td></td>
<td>2</td>
</tr>
<tr>
<td>14</td>
<td>Explain how a change in wind speed influences the amount of power supplied by a wind turbine generator.</td>
<td></td>
<td>2</td>
</tr>
</table>

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15	**Identify** the three main types of pollution.		3
16	**a** **Define** the term 'biomass'. **b** **Identify:** **i)** one direct use of biomass for energy production **ii)** one indirect use of biomass for energy production.		3
17	**Describe** two aspects of how Aboriginal people survived in balance with the environment.		4
18	**Explain** two negatives for the environment of white settlement in Australia.		2
19	**Identify** three introduced species that live in Australia.		3

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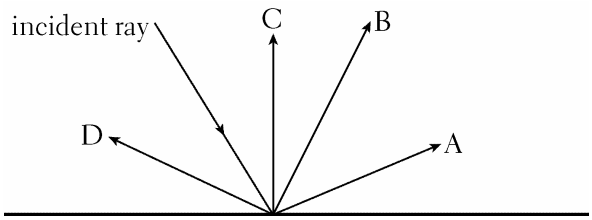
Science Focus 3 TEST

7: Light

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 76 marks

Section A—Multiple choice (21 marks)

1	Which ray below is the one that is reflected from a plane mirror?  		1
2	If light leaves a medium that has a lower refractive index than the new medium, which of the following will occur: **A** the light will slow down slightly and bend towards the normal **B** the light will slow down slightly and bend away from the normal **C** the light will speed up slightly and bend towards the normal **D** the light will speed up slightly and bend away from the normal.		1
3	When a light ray passing from glass to air strikes the boundary at the critical angle of incidence, the light ray: **A** is totally internally reflected **B** passes through to the air without any change in its direction **C** skims the surface of the glass **D** is reflected back along its original path.		1
4	When a pencil is viewed through water, it appears bent because light: **A** slows down as it passes from water to air **B** is reflected at the surface of the water **C** is refracted away from the normal as it passes from water to air **D** is refracted towards the normal as it passes from water to air.		1

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5	A fisherman under the water is attempting to spear a fish. The fisherman must aim: **A** above the apparent depth of the fish **B** directly at the fish as he sees it **C** below the apparent depth of the fish **D** to the left of the apparent position of the fish.		1
6	The image produced by a convex lens when the object is inside the focal length is enlarged: **A** real and upright **B** real and inverted **C** virtual and upright **D** virtual and inverted.		1
7	An object 4 mm high produces an image 16 mm high. The magnification is: **A** 0.25 **B** 4 **C** 12 **D** 20		1
8	The image produced on a screen by a slide projector is: **A** virtual, upright and enlarged **B** virtual, inverted and diminished **C** real, upright and enlarged **D** real, inverted and enlarged.		1
9	In an old-fashioned camera, photographic film records the images formed and is placed where the images would be. The images formed by the lenses in a camera must therefore be: **A** real **B** unreal **C** virtual **D** undecided, depending on the distance from the camera.		1
10	In a camera, the image formed is upside-down when recorded on the film. This suggests that the lenses in a camera are probably: **A** bi-concave **B** bi-plano **C** bi-convex **D** convex and concave.		1

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11	In the eye disorder known as long-sightedness, the image of an object forms behind the retina. Long-sightedness is corrected by bringing the image forward using: **A** convex lenses that bend the light less **B** convex lenses that bend the light more **C** concave lenses that bend the light less **D** concave lenses that bend the light more.		1
12	Blue skies and red sunsets are caused by the: **A** dispersion of light **B** refraction of light **C** absorption of light **D** scattering of light.		1
13	Optical fibres are used in: **A** microscopes **B** thermometers **C** endoscopes **D** decorative lights.		2
14	In a primary rainbow, what colour is at the top of the rainbow? **A** blue **B** green **C** orange **D** red		1
15	Mirages are basically caused because light is: **A** refracted differently at different temperatures **B** dispersed differently at different temperatures **C** reflected differently at different temperatures **D** scattered differently at different temperatures.		1
16	Which of the following colours of light is most strongly refracted? **A** red **B** yellow **C** green **D** violet		1
17	What colour will a green leaf placed in green light appear to be? **A** black **B** green **C** red **D** blue		1
18	What colour is produced when magenta and cyan pigments are mixed? **A** black **B** green **C** red **D** blue		1

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19	What colour is transmitted when white light goes through a red filter? **A** no colour (black) **B** white **C** red **D** green		1
20	Which colours are absorbed by blue paint? **A** blue only **B** red, orange and yellow **C** green, blue, indigo and violet **D** red only		1

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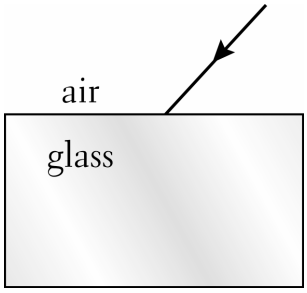
Section B—Written answers (55 marks)

1	**Explain** the difference between luminous and non-luminous objects.		2
2	**Define** the law of reflection.		1
3	**Explain** what is meant by the refraction of light?		2
4	**Explain** what is meant by the reflection of light.		1
5	**Recall** and **explain** the features of a mirage.		1
6	The diagram below shows a light ray travelling through air to the interface between the air and a glass block.  On the diagram, draw the normal, and a possible path for the ray in the glass.		2

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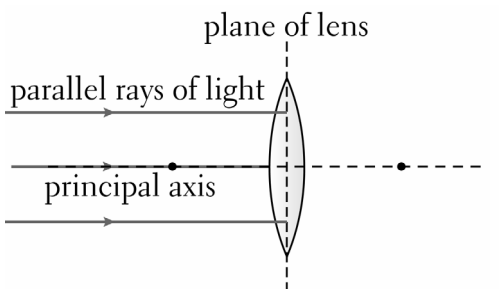
7	Light travelling from substance A to substance B is totally internally reflected. **a** **State** which substance (A or B) has the higher refractive index. **b** **Identify** at what angle the light ray must have struck the boundary of A and B for total internal reflection to occur.		2
8	Is light always refracted when it passes from one material to another? **Explain** your answer.		2
9	**List** three advantages of optical fibres over copper wire for transmitting data.		3
10	**Identify** which type of lens (convex or concave): **a** is fatter in the middle than at the ends **b** can produce a real image.		2
11	**Apply** the rules for drawing light rays to complete the diagram shown below by: **a** drawing in the refracted light rays **b** labelling the principal focus **c** labelling the focal length. 		4

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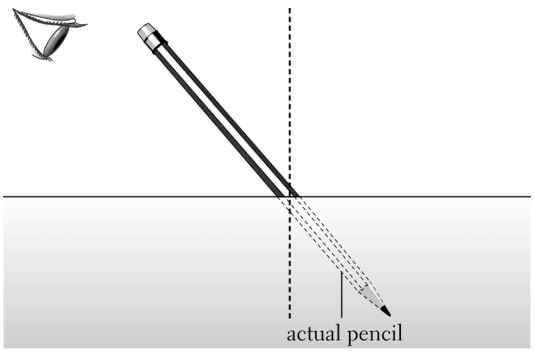
12	**Explain** why the sky appears blue during the day.		2
13	**Identify** what colour (if any) is transmitted when: **a** white light is shone on a blue filter **b** red light is shone on a blue filter **c** magenta light is shone on a red filter.		3
14	The diagram below shows a pencil placed in water. The pencil appears bent when viewed from the air.  Complete the diagram by drawing the position of the image of the pencil as seen by the viewer.		2
15	As the curvature of a lens increases, **state** whether: **a** the light is bent more or less **b** the focal length increases or decreases.		2

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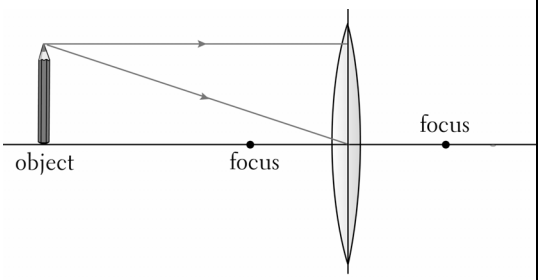
16	The diagram below shows a partly drawn ray-tracing diagram.  **a** Complete the ray-tracing diagram. **b** **Use** the correct terminology to **describe** the image.		5
17	The objective lens of a telescope produces an image just inside the focal length of the eyepiece lens. This image acts as the object for the eyepiece lens. **a** For a larger first image to be formed, **state** whether the objective lens should be thin or thick. **b** **Explain** why telescopes are long.		2

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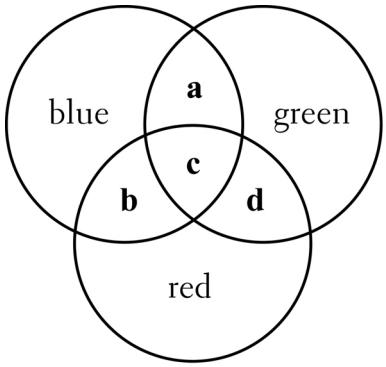
18	**a** **Recall** the primary colours of light. **b** **Explain** why they are called primary colours. **c** **Recall** the secondary colours of light. **d** **Explain** why they are called secondary colours. **e** Is black a colour? **Explain** your answer.		5
19	The diagram below shows combinations of light colours.  Light **Recall** the names of the colours marked **a** to **d**.		4
20	**Identify** the colour or colours of light that are: **a** absorbed by a blue shirt **b** reflected by a red shirt **c** combined to produce yellow light.		3

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21	**Recall** the name given to the: **a** spreading of light by particles in the sky **b** bending of light rays as they pass from one material to another **c** splitting of light into component colours by a prism.		3
22	Light from neon lamps is largely red and orange. **Propose** why some butchers place neon lamps above their window displays.		2

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Science Focus 3 TEST

8: The universe

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 65 marks

Section A—Multiple choice (21 marks)

1	Most of the stars you see at night are: **A** like our Sun, some much bigger **B** groups of stars called galaxies **C** planets **D** meteors.		1
2	Our nearest multiple star system is: **A** the Southern Cross **B** Ceres **C** Alpha Centauri **D** Orion.		1
3	A star gets its energy from: **A** a chemical reaction **B** a nuclear fission reaction **C** a nuclear fishing reaction **D** a nuclear fusion reaction.		1
4	Which of the following is the correct statement? **A** Nebulae may form stars, but stars do not form nebulae. **B** Nebulae may form stars and stars may explode to form nebulae. **C** A nebula is the name given to a star that is about to explode. **D** A nebula is the core of a star that remains after the star has exploded.		1
5	Our Sun will eventually become: **A** a white dwarf **B** a red dwarf **C** a neutron star **D** a black hole.		1
6	A pulsar is a: **A** radio wave that comes from a star **B** type of galaxy **C** rapidly rotating neutron star **D** galaxy whose centre is so bright it obscures the outer regions.		1

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7	Which of the following is a major space-based telescope? **A** Bubble **B** Hubble **C** Keck **D** Tech		1
8	The Milky Way is itself part of a larger group called: **A** the big dipper **B** the local group **C** the Andromeda super galaxy **D** the spiral arm.		1
9	As a fast-moving ambulance passes by, its siren goes from: **A** a high pitch to a low pitch **B** a low pitch to a high pitch **C** a high pitch to an even higher pitch **D** there is no change in pitch.		1
10	Which of the following statements concerning the colours of visible light is correct? **A** When light passes through a prism the colours are separated by reflection. **B** Red light has a lower energy than blue light. **C** Red and blue light have the same wavelength. **D** Red light has a higher frequency than blue light.		1
11	When a fast-moving star is moving away from the Earth, the absorption lines on the spectrum produced by the star will be shifted towards the: **A** blue end of the spectrum (light of longer wavelength) **B** red end of the spectrum (light of longer wavelength) **C** blue end of the spectrum (light of shorter wavelength) **D** red end of the spectrum (light of shorter wavelength).		1
12	Approximately how many years ago is the Big Bang thought to have occurred? **A** 13 thousand **B** 13 million **C** 13 billion **D** 13 trillion		1
13	A small fraction of a second after the Big Bang, the universe: **A** had a temperature of one billion degrees Celsius, and chemical elements had begun forming **B** was expanding slowly and stars were beginning to form **C** was cooling, and quarks were beginning to clump together to form neutrons **D** inflated suddenly and was filled with particles of matter and antimatter.		1

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14	Which theory of the universe includes the idea of a contracting universe? **A** open universe **B** closed universe **C** flat universe **D** accelerating universe		1
15	The open universe theory states that the universe will: **A** keep expanding forever but at a decreasing rate **B** eventually stop expanding and then contract into a smaller and smaller space **C** eventually stop expanding, but not contract again **D** expand at an ever-increasing rate.		1
16	Approximately how fast (in metres per second) does electromagnetic radiation such as radio waves travel? **A** 300 **B** 300 000 **C** 300 000 000 **D** 300 000 000 000		1
17	The orbital speed of the International Space Station is approximately: **A** 30 km/h **B** 400 km/h **C** 3000 km/h **D** 40 000 km/h.		1
18	The main oxygen source for the International Space Station is: **A** a generator that splits water into hydrogen and oxygen **B** photosynthesis by onboard plants **C** chemical decomposition of hydrogen peroxide to produce oxygen and water **D** oxygen tanks transported from Earth.		1
19	Electromagnetic radiation: **A** is laser waves **B** is radio waves **C** travels at the speed of light **D** is all of the above.		1
20	The term ‘dark energy’ is used to describe: **A** any light towards the high-energy end of the visible light spectrum **B** subatomic particles left over from the Big Bang **C** gas and dust orbiting stars that may condense to form planets **D** some mysterious cosmological force that overrides gravity.		1

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21	Passive sensors: **A** are the same as active sensors **B** make electromagnetic radiation **C** detect electromagnetic radiation **D** are space junk.		1

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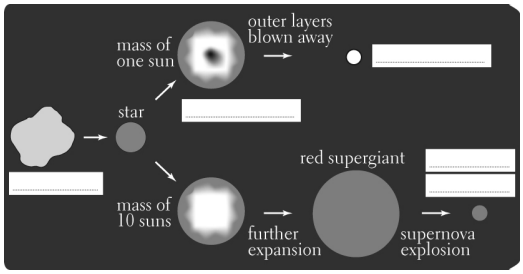
Section B—Written answers (44 marks)

1	Fill in the blanks in the diagram below showing the life of a star.  		4
2	**Describe** The Global Positioning System (GPS).		2
3	**List** the four types of satellites.		2
4	**Name** any three types of radiation that is emitted by the Sun.		3

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5	**a** If light cannot escape from black holes, **explain** how astronomers detect them. **b** Predict where the black hole that is closest to Earth is likely to be.		2
6	**Name** and sketch three different types of galaxy.		3
7	Complete these sentences about parts of the Milky Way galaxy. **a** An _____ cluster contains a few hundred stars within the spiral arm of the galaxy. **b** A _____ cluster may be seen with the naked eye as a fuzzy star, but actually contains up to a million stars.		2
8	**State** which sound wavelength (long or short) produces the higher pitched sound.		1

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9	Complete the following statement about a law by selecting the correct word in each part. Most galaxies are moving **a** *towards/away from* the Earth. The further away the galaxies are, the **b** *faster/slower* they are moving. This is known as **c** *Doppler's/Hubble's* law.		3
10	**Explain** why it is impractical to search for extraterrestrial life using current spacecraft.		1
11	**Describe** what would happen over time in a closed universe.		4
12	**Explain** why frequencies in the microwave region are preferred when broadcasting messages to try to make contact with extraterrestrial life.		1

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13	**a** **State** the height above the Earth's surface of a satellite in geostationary orbit. **b** **Explain** what it means if a satellite is in an asynchronous orbit. **c** **Clarify** what type of orbit observation satellites are usually placed in. **d** **Clarify** what type of orbit communication satellites are usually placed in.		4
14	**Define** microgravity.		1

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15	The ISS is a combined effort of several countries to construct a permanent space station. **a** **Recall** when the ISS started, and its expected completion date. **b** **Describe** one use of the ISS. **c** **State** the source of power for the ISS. **d** **Describe** two practical difficulties of daily life on the ISS.		6

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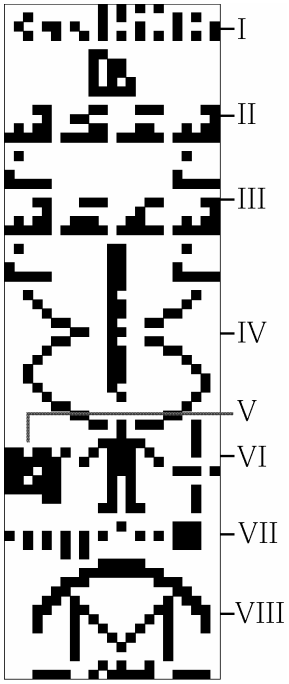
16	The diagram below shows a message beamed from the Arecibo radio telescope in 1974.  Match each description below to a symbol (I to VIII) on the message. **a** DNA double helix **b** solar system **c** numbers 1 to 10 in binary **d** atomic numbers of important elements **e** world population		5

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Science Focus 3 TEST

9: Earth's fragile crust

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 104 marks

Section A – Multiple choice (25 marks)

1	Which combination is thought to have made up the ancient supercontinent Laurasia? **A** Australia, Antarctica, South America, Africa and India **B** South America, North America and Europe **C** Europe, North America and most of Asia **D** North America, Africa and most of Asia		1
2	In 1915, Alfred Wegener proposed the idea of a supercontinent splitting to form the continents. His ideas were largely ignored because: **A** there was no evidence to support them **B** little was known about the shapes of the continents at that time **C** the distribution of reptilian fossils could not be explained by shifting continents **D** it was thought that the Earth was solid rock.		1
3	Studies of the magnetic stripes of rocks on the ocean floor indicate that the: **A** youngest rock is next to the ridges and the oldest is next to the trenches **B** youngest rock is next to the trenches and the oldest is next to the ridges **C** rocks are all approximately the same age **D** rocks are older than the rocks of the continents.		1
4	Which of the following boundaries between tectonic plates is also known as a conservative boundary? **A** spreading **B** collision **C** transform **D** constructive		1

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5	When tectonic plates collide, a variety of things can happen. Which of the following is most likely when an ocean plate collides with another ocean plate? **A** Both plates crumple and fold. **B** The less dense plate is forced under the more dense plate. **C** A rift valley forms. **D** The faster-moving plate is forced under the slower-moving plate.		1
6	When tectonic plates collide, a variety of things can happen. Which of the following is most likely when two continental plates collide? **A** Both plates crumple and fold. **B** The less dense plate is forced under the more dense plate. **C** A rift valley forms. **D** The faster-moving plate is forced under the slower-moving plate.		1
7	A 'mountain root' is most likely to form when: **A** two ocean plates collide **B** two continental plates collide **C** an ocean plate collides with a continental plate **D** two plates scrape along each other.		1
8	Which of the following statements concerning earthquakes is *incorrect*? **A** The focus is the point where an earthquake begins. **B** Earthquakes occur on a fault line at the edges of tectonic plates. **C** The epicentre is the point on the Earth's surface above the focus. **D** Seismic waves spread from the epicentre to the focus and beyond.		1
9	For which of the following waves is the vibration of the particles in the same direction as the movement of the wave? **A** sound **B** water **C** seismic Love (L) waves **D** seismic secondary (S) waves		1

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10	Which of the following is a property of secondary (S) body waves? S waves: **A** travel through both solid and molten rock **B** are the fastest-moving body waves **C** are transverse waves **D** hit the surface with an up-and-down or push-pull motion.		1
11	Both primary and secondary body waves: **A** are transverse waves **B** are refracted when they pass through rocks of different density **C** travel faster when they move through less dense rock **D** travel around the surface of the Earth.		1
12	A scientist who studies earthquakes is a: **A** sizemologist **B** seismologist **C** seismometer **D** physicist.		1
13	An earthquake of magnitude 5 on the Richter scale has: **A** one-quarter more energy than a magnitude 4 quake **B** ten times the energy of a magnitude 4 quake **C** twenty times the energy of a magnitude 4 quake **D** thirty times the energy of a magnitude 4 quake.		1
14	How many active volcanoes are there in the world? **A** 100 **B** 600 **C** 1500 **D** more than 2000		1
15	How many active volcanoes are there in Australia? **A** zero **B** one **C** six **D** eleven		1

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16	Which part of an active volcano does the lava move through? **A** magma **B** central vent **C** volcanic ash **D** all of the above		1
17	If the rock of a fault scarp is very hard and weathering is slow, which landscape feature is most likely to form? **A** cliff **B** gentle rise **C** volcano **D** syncline		1
18	When continental plates collide, rock may be folded to build mountain ranges. Upward folds in the rock are called: **A** horsts **B** graben **C** synclines **D** anticlines.		1
19	How do the plates at a transcurrent fault move? **A** One plate slides over another. **B** Two plates collide and crumple. **C** Both plates move sideways. **D** One plate moves upwards, the other downwards.		1

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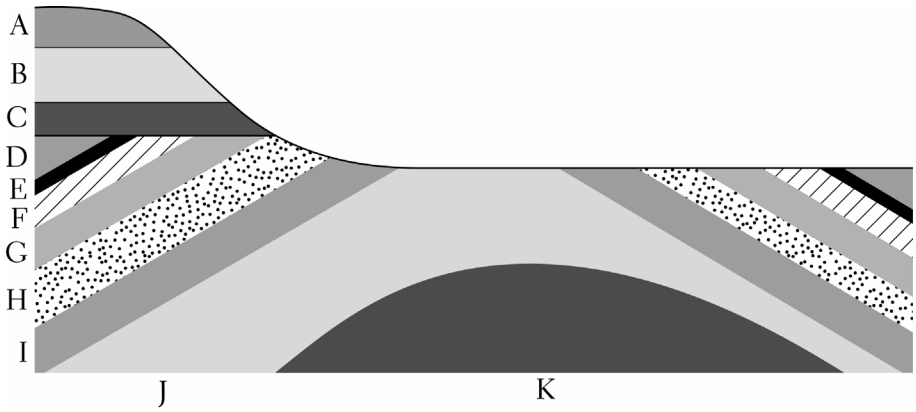
20	This question concerns the landform shown in the diagram below.  Which of the following statements is correct? **A** K represents a syncline. **B** C represents an unconformity. **C** Rock in E is older than rock in H. **D** J shows no signs of erosion.		1
21	Which type of volcano is the biggest? **A** shield cones **B** composite cones **C** cinder cones **D** batholiths		1
22	Which of the following statements concerning fossil fuels is *incorrect*? Fossil fuels: **A** form from decomposed plant and animal matter **B** take millions of years to form **C** are usually found near weaknesses in the Earth's crust **D** require low pressures and high temperatures to form.		1
23	A fossil is: **A** the preserved remains of an animal **B** evidence of past life found in a rock **C** a past life form buried in the sea bed **D** a skeleton embedded in rock.		1

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24	Radioactive dating of rocks works by: **A** measuring the amount of uranium in a rock that has turned into lead **B** finding the depth to which gamma rays penetrate within a rock **C** bombarding a rock with radiation and measuring the amount of rock that disintegrates **D** using X-rays to measure the space between atoms within rock.		1
25	Which of the following is a period of geological time? **A** Zirconic **B** Cretaceous **C** Rhombic **D** Dichotomous		1

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Section B – Written answers (79 marks)

1	**Describe** four pieces of evidence Alfred Wegener used to support his theory of the splitting of an ancient supercontinent to form the continents.		4

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2	**Recall** three important pieces of information obtained when the ocean floor was mapped during World War II.		3
3	The theory of plate tectonics may be explained using an analogy with pieces of toast floating on very thick, hot soup. By **applying** this analogy: **a** **state** what the soup represents **b** **state** what the pieces of toast represent. **c** If the soup is stirred the toast moves. **Explain** what is widely thought to cause the 'soup' to be stirred.		3
4	**Explain:** **a** why spreading boundaries are also called constructive boundaries **b** why collision boundaries are also called destructive boundaries.		2

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5	**Identify** which type of tectonic plate boundaries form, or formed, the: **a** earthquake activity around the San Andreas fault in California **b** East African rift valley **c** islands of Japan **d** Andes mountains and the Peru-Chile ocean trench which runs parallel to them.		4
6	**Apply** your knowledge of what happens when an oceanic plate collides with a continental plate to answer the following questions. **a** **Explain** why the oceanic plate is forced under the continental plate. **b** **Outline** a 'subduction zone'. **c** **Explain** why volcanoes often form on the continental plate after the collision.		5
7	**Clarify** the difference between the focus and the epicentre of an earthquake.		2
8	When an earthquake occurs, **identify** where the worst damage occurs.		1
9	**Explain** why no S waves are recorded on the Earth directly opposite the epicentre of the earthquake.		2

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10	The four types of seismic waves are: primary (P), secondary (S), Rayleigh (R) and Love (L). Use the letters P, S, R and L when answering the questions that follow. **Identify** the waves that: **a** travel through the body of the Earth **b** are recorded first by a seismometer **c** are recorded last by a seismometer **d** travel around the Earth's surface **e** are rolling waves, like surf at a beach.		7
11	**Identify** whether each of the following waves are transverse or longitudinal waves. **a** sound waves **b** P waves **c** S waves **d** waves that push and pull **e** waves that have an up-down motion like water waves		5
12	**Explain** how aftershocks can be just as dangerous as the original earthquake, even though they are usually much smaller.		2

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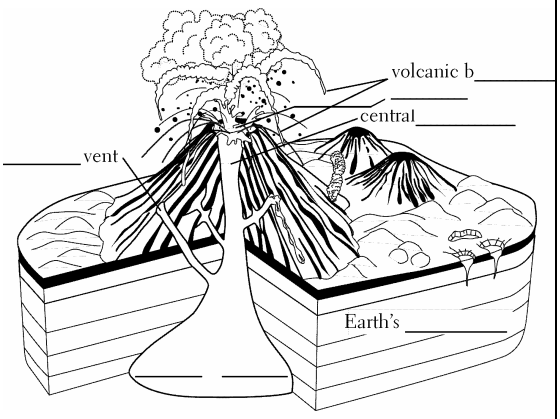
13	**Identify:** **a** an alternative scale to the Richter scale. **Explain:** **b** how the scale works **c** when it is used.		3
14	**Identify** the factors that affect how many lives are lost due to an earthquake.		2
15	Complete the labels on the volcano diagram below.  		6
16	**Explain** why many volcanic eruptions are not noticed.		2
17	**Identify** the problems that volcanic dust may cause.		2

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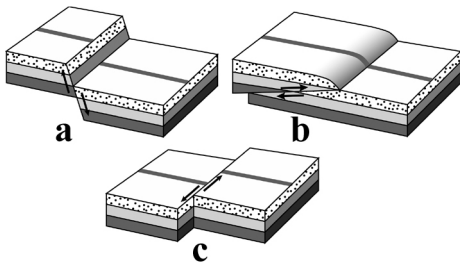
18	Three types of faults are shown in the diagram below. **Identify** each as either a transcurrent, normal or reverse fault.  		3
19	**Define** the term 'plastic behaviour'.		1
20	**Explain** how an unconformity is created. Include a diagram in your answer.		4
21	**Explain** how a volcano could form away from the edge of a tectonic plate.		2
22	**Explain** why igneous rock will not contain fossils of plant material.		2

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23	**Classify** the following steps of fossil formation in the correct order from start to finish. **A** Movement in the Earth's crust thrusts a layer of rock upwards. **B** Animal dies and falls to the sea floor. **C** Soft parts of body decay, leaving the shell. **D** Rock weathered. **E** Fossil exposed. **F** Shell covered by layers of sediment.		2
24	True or false? **a** A dinosaur footprint is a fossil. **b** An ammonite is a fossil animal. **c** More primitive fossils are likely to be found in layers of igneous rock. **d** Carbon may be used to date fossils.		4
25	Match one of the terms below to each of the following. *primitive life, no life, middle life, recent life* **a** Cenozoic **b** Azoic **c** Mesozoic **d** Archaeozoic		4
26	**a** **Explain** the connection between an era and a period. **b** **Recall** a period when dinosaurs existed.		2

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Science Focus 3 TEST ANSWERS

1: Forensics

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 78 marks

Section A—Multiple choice (17 marks)

1	The Bertillon system of identification involved: **A** taking fingerprints **B** making up a composite drawing **C** measuring and recording the dimensions of a series of bony body parts **D** retroactive interference.	C	1
2	Choose the best answer. Using photographs to identify people is difficult because: **A** they might be wearing different clothes **B** the photographs are unrealistic **C** drawings are better **D** the photographs themselves can distort a person's memory.	D	1
3	Who performs an autopsy? **A** the victim **B** a police officer **C** a forensic pathologist **D** a forensic scientist	C	1
4	The Identikit system of identification involved: **A** taking fingerprints **B** making up a composite drawing **C** measuring and recording the dimensions of a series of bony body parts **D** retroactive interference.	B	1
5	There are four types of fingerprints. Which of these is not one of them? **A** whirl **B** loop **C** composite **D** arch	A	1

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1: Forensics

6	Which of the following statements is true regarding iris and retina identification? **A** False eyes can be used to 'bluff' the iris and retina identification. **B** The chance of incorrect identification is very high. **C** Iris identification is less accurate than using fingerprints. **D** Scans are sometimes difficult to obtain from uncooperative people.	D	1
7	Which of the following contains your DNA and could be used to identify you? **A** your dandruff **B** your blood **C** your saliva **D** all of the above	D	1
8	Chromatography is: **A** a technique used to get inks to fluoresce **B** when oblique lighting is used to identify writing **C** a method used to counterfeit money **D** a technique that separates the colours in inks and dyes, allowing their identification.	D	1
9	Which of the following is incorrect? Good track impressions that can be used in forensics are created when you walk: **A** onto mud, clay or snow **B** out from somewhere wet onto a dry path **C** through something spilt (such as blood or grease) over a hard surface **D** through mud or dirt and then onto a clean surface.	B	1
10	The shape of blood drips on the ground can be used to determine: **A** the direction of movement of the victim **B** the time the crime was committed **C** the blood group of the victim **D** the DNA of the victim.	A	1
11	Latent fingerprints are usually left on non-porous materials. This means they would easily be found on: **A** newspaper **B** unpolished wood **C** polished wood **D** any animal without paws.	C	1

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1: Forensics

12	Forensic scientists usually: **A** help investigators collect evidence **B** make arrests **C** are pathologists **D** all of the above.	A	1
13	Evidence can be gathered from: **A** biological remains **B** computers **C** recording devices **D** all of the above.	D	1
14	Which of the following statements are true (there may be more than one)? **A** Blood sprays are not very useful in a crime investigation. **B** Wounds provide information about the type of attack, such as whether a blunt or a sharp object was used. **C** Gunshot wounds are more likely to kill near organs. **D** Gunshot wounds always kill.	B C	2
15	Where can fingerprints be *easily* found? **A** glass, plastic and mirrors **B** unpolished wood, mirrors and stone **C** in water or other liquids **D** all of the above	A	1
16	A cold case is: **A** a bag used to store your clothes when you go snow skiing **B** a box that is used to store fruit in a cold-store **C** an unsolved crime that has no more leads for police to investigate **D** a person who has the flu.	C	1

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1: Forensics

Section B—Written answers (61 marks)

1	**Define** biometric facial recognition.	Using a computer program to recognise a face by comparing the position of features such as eyes, chin, nose and ears with profiles stored on the computer system.	2
2	**State** whether the following statement is true, false or a bit of both. **Justify** your answer. *Every person's DNA is different.*	Except for identical twins, each person has different DNA. Hence, the statement is true for most people. However, technically, it is false, because it does not account for identical twins. It is therefore a bit of both.	3
3	Draw sketches of the following types of fingerprints, showing their main identifying characteristics. *loop*, *arch*	 **Loops** have ridges that enter and leave from the same side like a loop of string. Loops can begin and end on the left or the right side of the finger. **Arches** look like an arched bridge.	4
4	**Present** four reasons why iris identification is very reliable, providing almost 100% correct identification of an individual.	• Two scans are taken, one of each eye. • It cannot be forged with a false eye. A real iris moves constantly. • An iris has 266 identifiable features. • Your iris never changes, even as you age.	4

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5	**Describe** the methods of analysing writing and print? **Explain** each using one sentence.	Handwriting analysis: studying a person's handwriting to identify a document in question. Typewriter matching: different brands and models of typewriter have specific fonts and styles that can make it easy to identify a particular feature of the typewriter. Printer matching: this is more difficult, but still possible through oblique lighting which shows specific marks from the transport mechanism of the printer made onto the paper print-out. Ink analysis: chromatography or different lighting can be used to identify different inks.	8
6	A student has collapsed on the way home from school. **State** four ways that the police might use to identify him/her.	Any four of: • school student card • library card • L or P plate licence • credit card • school uniform—contact school and check photos • mobile phone address book.	4
7	Briefly **describe** two techniques that can be used to identify the ink used to write or print a document.	Many inks fluoresce under different coloured lights. Inks can be separated into their components using chromatography.	2

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8	**List** five methods that make it difficult to forge Australian banknotes.	Any five of : • notes made of special polymer film (plastic) • intaglio printing (raised printing) • microprinting (very small details, sentences, initials etc.) • water marks • metal bands • fluorescent inks • optically active devices • serial numbers.	5
9	The first $100 bills in Australia were printed on paper and were predominantly black, white and grey in colour. **Present** three reasons why these bills were easily forged when first introduced.	Few people knew what a $100 bill looked like and so they could easily be bluffed. The colours were easily reproduced using simple printing or photocopying. The bills were made of paper and not from polymer film (plastic).	3
10	**Explain** how you could tell whether a threatening document has been typed on an old-fashioned typewriter or printed from a computer printer.	Each brand and model of typewriter produces characteristic letter shapes and produces particular faults in the letters due to dirt and/or pits and cracks in the hammer. When lit obliquely, computer printouts will often show tracks of the feed mechanism.	2
11	Identity theft is an increasing crime. **Propose** what you think the term 'identity theft' means.	Identify theft is when someone uses someone else's identity (either their whole identity or some identifying card/document) for unauthorised purchases or to access a secure area.	2

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12	**Explain** why rumour and gossip have no value in court as evidence. Second, **explain** how rumours can be dangerous in court.	Although rumour and gossip might have some factual basis, they cannot be confirmed as fact. Evidence must be factual, in the form of verbal testimony of what a person has actually seen or heard or in the form of physical evidence such as DNA or fingerprints. Rumours can be dangerous especially if they imply a person is guilty when they are not or vise versa.	3
13	**List** an example of a fibre that is: **a** animal in origin **b** vegetable in origin **c** mineral or synthetic in origin.	Students may list any one example from each of parts **a–c** below. **a** fur, wool, hair, feathers, silk **b** cotton, string, linen, hessian **c** rock wool, glass fibres, polyester, nylon, rayon	3
14	**List** three features of shoes that can lead to the identification of a criminal.	• size of shoe • track impression: shape and pattern • track impression: wear and tear and characteristic cuts and marks	3
15	A decaying body has been found in the bush. It is covered in different types of insects, insect eggs, worms and maggots. **Explain** how you could use these to determine how long the body has been there.	The stages in the lifecycles of the different insects will give a good idea as to how long they have been breeding and growing on the body. From this, the time the body has been there can be determined.	2

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16	**Demonstrate** how information can be gained and used as evidence from each of a computer, a mobile phone or video footage.	Computer: binary code stored even when computer files deleted. Mobile phone: used to trace a subject or where they were during a crime. Video footage: security cameras can be set up in high-profile areas and, often, individual shops will have their own camera. These are generally CCTV and make individual frames which can be linked together to (possibly) show a crime being committed.	3
17	**Explain** four ways in which blood splatters and drips can tell a forensic scientist what happened at a crime scene.	Any four of: • movement (shape) • height (size) • fine sprays (gunshot) • splattering on ceiling (repeated striking) • fading (time of attack) • analysis (DNA and blood grouping).	8

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2: The periodic table

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 71 marks

Section A—Multiple choice (21 marks)

1	Hydrogen peroxide: **A** is water **B** has one more hydrogen atom than water **C** has one more oxygen atom than water **D** is none of the above.	C	1
2	The atomic number of an element is equal to the number of: **A** electrons and neutrons **B** protons **C** neutrons **D** protons, neutrons and electrons.	B	1
3	Which one of the following is *not* a property of non-metals? Non-metals are: **A** normally gases or liquids at room temperature **B** either poor electrical conductors or non-conductors **C** able to be hammered into sheets **D** dull, with little or no shine.	C	1
4	The atomic number of zinc (Zn) is 30. Therefore, the ion represented by the symbol $^{65}Zn^{2+}$ has: **A** 30 protons, 35 neutrons and 28 electrons **B** 65 protons, 30 neutrons and 32 electrons **C** 30 protons, 65 neutrons and 28 electrons **D** 30 protons, 35 neutrons and 32 electrons.	A	1
5	An atom of chlorine (Cl) has 17 protons, 18 neutrons and 17 electrons. Which of the following statements is *incorrect* about the atom of chlorine? **A** It has an atomic number of 17. **B** The atom would have no overall electrical charge. **C** It would be in Period VII. **D** It would be in Period 3.	C	1

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2: The periodic table

6	Use the periodic table to determine which of the following statements is *incorrect* about chlorine atoms. **A** They form the ion Cl^-. **B** Chlorine atoms are smaller than fluorine atoms. **C** They are 'related' to F, Br and I. **D** Chlorine belongs to the halogen family.	B	1
7	A compound forms when: **A** two or more elements chemically combine with each other **B** two or more elements are physically mixed together **C** a large number of identical atoms join together **D** a mixture is separated into its components.	A	1
8	The vertical columns in the periodic table are known as: **A** periods, and they contain elements with the same number of outer-shell electrons **B** periods, and they contain elements with the same number of used electron shells **C** groups, and they contain elements with the same number of outer-shell electrons **D** groups, and they contain elements with the same number of used electron shells.	C	1
9	An atom of calcium has 20 electrons. Its electron configuration would best be written as: **A** $^{42}_{20}Ca$ **B** 20 **C** 2,8,8,2 **D** 2,8,10.	C	1
10	The electron configuration of an element in Period 3, Group IV of the periodic table is: **A** 2,8,6 **B** 2,8,8,3 **C** 2,8,18,3 **D** 2,8,4.	D	1
11	In the flame colour experiment, different metallic salts gave out different colours when placed in a Bunsen burner flame. This was because: **A** the heat from the Bunsen burner caused the electrons to jump to another shell. Coloured light emerged when they jumped back **B** the heat from the Bunsen burner caused the electrons to make colours **C** the electrons moved very fast around the nucleus **D** the light came from the Bunsen burner.	A	1

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2: The periodic table

12	A particular element is shiny, malleable and a good conductor of electricity. Which section of the periodic table would this element *not* be found in? **A** transition elements **B** Group I **C** Group III **D** Group VII	D	1
13	Which of the following is *not* a physical property of an element? **A** melting point **B** colour **C** reactivity with oxygen **D** density	C	1
14	Which of the following lists contains only compounds? **A** water (H_2O), glucose ($C_6H_{12}O_6$), carbon dioxide (CO_2) **B** tungsten (W), phosphorus (P), chlorine (Cl_2) **C** Vegemite, limewater ($Ca(OH)_2$), plutonium (Pu) **D** salt (NaCl), air, iron (Fe)	A	1
15	Moving down Group VII in the periodic table, which property would be expected to decrease? **A** the reactivity **B** the number of outer-shell electrons in each atom **C** the size of atoms **D** the melting point	A	1
16	In 1869, Mendeleev devised a periodic table. Which of the following did Mendeleev *not* do in his periodic table? **A** Predict properties of undiscovered elements. **B** List elements with atomic numbers greater than 100. **C** Arrange elements in order of increasing atomic mass. **D** Place elements in vertical columns based on chemical properties.	B	1
17	Which of the following elements would be expected to conduct electricity? **A** helium (element number 2) **B** chlorine (element number 17) **C** scandium (element number 21) **D** krypton (element number 36)	C	1

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2: The periodic table

18	The mass of one electron is approximately equal to the mass of: **A** 1800 protons **B** 1/1800 of a proton **C** 1 proton **D** 1 neutron.	B	1
19	Semi-metals have properties similar to those of: **A** metals **B** gases **C** non-metals **D** both metals and non-metals.	D	1
20	The most likely formula for magnesium iodide is: **A** MgI **B** Mg_2I **C** Mg2I **D** MgI_2	D	1

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2: The periodic table

Section B—Written answers (50 marks)

1	In a neutral atom, **list** which two particles have: **a** equal but opposite charges **b** approximately the same mass.	**a** protons and electrons **b** protons and neutrons	2
2	Write the electron shell configuration for each of the following elements: **a** an atom of phosphorus (atomic number 15) **b** an atom of potassium (atomic number 19) **c** an atom with atomic number 7.	**a** 2,8,5 **b** 2,8,8,1 **c** 2,5	3
3	Write the chemical symbol for any element: **a** in the same group as fluorine (F) **b** in the same period as sodium (Na) **c** in the group known as the alkali metals.	Students may list any one symbol from each of parts **a–c** below. **a** Cl, Br, I, At **b** Mg, Al, Si, P, S, Cl, Ar **c** Li, Na, K, Rb, Cs, Fr	3
4	**List** three properties of non-metals.	Any three of: • poor conductors of electricity or they are insulators • low melting points • usually gases or liquids at room temperature • brittle and tend to crumble to powder • dull, with little or no shine.	3

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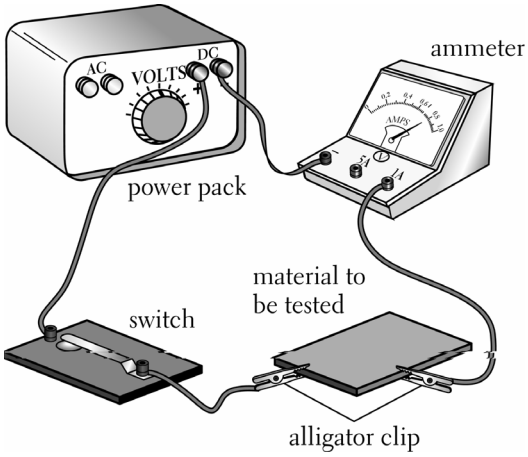
5	**List** three properties of metals.	Any three of: • good conductors of heat and electricity • lustrous • malleable and ductile • relatively high melting points • high density.	3
6	The apparatus shown below may be used to test a property of several materials.  **Identify:** **a** the property of materials being tested **b** which of the materials listed below would be expected to cause the ammeter needle to move when the switch is closed: *sulfur, lead, copper, phosphorus.* **Justify** your answer to part **b**.	**a** electrical conductivity **b** lead and copper Both are metals. Metals conduct electricity.	4

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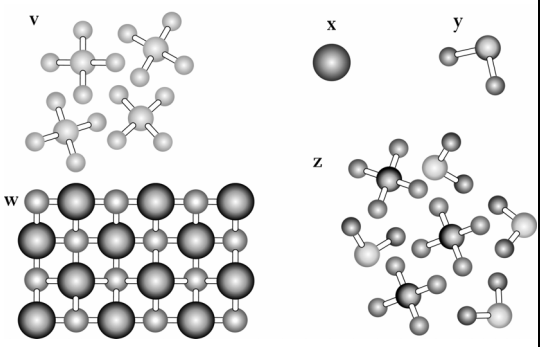
7 Five diagrams, labelled *v* to *z*, are shown below. Use the letters *v* to *z* to answer the questions that follow.



Identify the diagram that could represent:

a an atom of sulfur

b a water molecule (H_2O)

c a mixture of two compounds

d the potassium chloride lattice (KCl).

a x

b y

c z

d w

4

8 Complete the following table.

Atom and change	Name of ion formed	Symbol of ion formed
Magnesium (Mg) atom loses two electrons		
Nitrogen (N) atom gains three electrons		

Atom and change	Name of ion formed	Symbol of ion formed
Magnesium (Mg) atom loses two electrons	**magnesium**	**M g $^{2+}$**
Nitrogen (N) atom gains three electrons	**nitride**	**N $^{3-}$**

4

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<table>
<tr>
<td>9</td>
<td>Write the chemical formulas for:
a calcium oxide
b barium iodide
c beryllium nitride
d aluminium oxide.</td>
<td>a CaO
b BaI_2
c Be_3N_2
d Al_2O_3</td>
<td>4</td>
</tr>
<tr>
<td>10</td>
<td>The flame colours of metal salts are given in the table below.
<table>
<tr><th>Salt</th><th>Flame colour</th></tr>
<tr><td>strontium chloride</td><td>red</td></tr>
<tr><td>sodium chloride</td><td>orange</td></tr>
<tr><td>potassium chloride</td><td>purple</td></tr>
<tr><td>barium chloride</td><td>green</td></tr>
</table>
Produce evidence that shows that the colours came from the metals and not from the non-metal parts of the salts?</td>
<td>Different metals produce different colours.
If colour came from the non-metal part (Cl), then all the colours would be the same.</td>
<td>2</td>
</tr>
<tr>
<td>11</td>
<td>Identify the group of the periodic table described in each statement below:
a reactive metals whose atoms form ions with a charge of +2
b reactive non-metals whose atoms form ions with a charge of –1
c colourless gases whose atoms are very unreactive.</td>
<td>a II
b VII
c VIII</td>
<td>3</td>
</tr>
</table>

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12	Listed below are five atoms using symbols that are not the usual symbols of the elements. **Use** the letters *V* to *Z* to answer the following questions. $^{40}_{18}V$ $^{20}_{10}W$ $^{40}_{19}X$ $^{39}_{19}Y$ $^{40}_{20}Z$ **Identify:** **a** which two atoms have the same number of neutrons **b** which atom has the smallest mass number **c** which two atoms belong to the same element **d** which two symbols represent noble gases.	**a** *Y* and *Z* (20 neutrons) **b** *W* **c** *X* and *Y* (same atomic number) **d** *V* and *W*	4
13	Write the symbol of a: **a** Period 2 alkali metal **b** member of the lanthanide series **c** Period 5 transition element.	Students may write any one symbol from each of parts **a–c** below. **a** Li **b** Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu **c** Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd	3
14	**State** one way in which helium (He) is: **a** like all other gases in Group VIII **b** different from all other gases in Group VIII.	**a** It has a full outer shell and is therefore unreactive. **b** It has two outer-shell electrons, not eight.	2
15	**State** one way in which hydrogen (H) is: **a** similar to the Group VII elements **b** similar to the Group I elements.	**a** Both form ions with a –1 charge. **b** Both form ions with a +1 charge.	2

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16	**Identify** which element in each pair (atomic numbers are given in parentheses) is more reactive. **a** Li (3) and Cs (55) **b** F (9) and Cl (17) **c** Na (11) and Mg (12) **d** O (8) and Ne (10)	**a** Cs **b** F **c** Na **d** O	4

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Science Focus 3 TEST ANSWERS

3: Chemical change

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 80 marks

Section A—Multiple choice (18 marks)

1	Which of the following involves a chemical change? **A** A piece of chocolate becomes soft when warmed. **B** Dry ice (solid CO_2) is warmed to form gaseous CO_2. **C** Fireworks explode. **D** Water boils to become steam.	C	1
2	Most reactions need an input of energy. These reactions are called: **A** spontaneous **B** non-spontaneous **C** explosive **D** energy needing.	B	1
3	The equation for the overall reaction in photosynthesis is shown below. $energy + 6CO_{2(g)} + 6H_2O_{(l)} \rightarrow C_6H_{12}O_{6(aq)} + 6O_{2(g)}$ Photosynthesis is: **A** an endothermic reaction **B** an exothermic process **C** a combustion reaction **D** an acid-base reaction.	A	1
4	$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$ This is an example of a: **A** neutralisation reaction **B** displacement reaction **C** precipitation reaction **D** decomposition reaction.	D	1

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5	A precipitate is a: **A** clear and colourless solution **B** gas **C** solid lump **D** fine solid powder that forms when two solutions are mixed.	D	1
6	A precipitate: **A** may form when two clear solutions are mixed **B** is a soluble salt **C** forms because the ions in a solution repel each other **D** collects at the surface when two solutions are mixed.	A	1
7	A weak acid is an acid which: **A** easily donates its hydrogen ion to a base **B** does not easily donate its hydrogen ion to a base **C** has few solute particles dissolved in the solvent **D** is highly corrosive.	B	1
8	When solutions of sodium chloride (NaCl) and silver nitrate ($AgNO_3$) are mixed, solid silver chloride (AgCl) forms. Which ions remain dissolved in the solution? **A** Na^+ and Cl^- **B** Ag^+ and Cl^- **C** Na^+ and NO_3^- **D** Ag^+ and NO_3^-	C	1
9	An acid reacting with a metal hydroxide to produce salt and water is an example of a: **A** combustion reaction **B** displacement reaction **C** neutralisation reaction **D** decomposition reaction.	C	1
10	Reduction occurs when: **A** a metal atom forms a positively charged ion **B** a non-metal atom forms a negatively charged ion **C** two ions combine to form a precipitate **D** steam condenses to form liquid water.	B	1

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11	A combustion reaction: **A** always has carbon dioxide as a product **B** always has water as a product **C** always has oxygen as a reactant **D** is always endothermic.	C	1
12	Which of the following is a property of bases? **A** taste sour **B** turn blue litmus red **C** react with acids to produce a salt and hydrogen gas **D** feel soapy	D	1
13	Which test would show that a sample of gas is carbon dioxide? The gas: **A** produces a popping noise when a match is placed in it **B** causes a flame to flare up **C** is odourless and yellow **D** produces a precipitate when it is bubbled through a limewater solution.	D	1
14	When hydrochloric acid reacts with magnesium, the products are: **A** salt and hydrogen gas **B** salt and water **C** salt, carbon dioxide and water **D** salt and oxygen gas.	A	1
15	A change from pH 3 to pH 5 means that the hydrogen ion present in the solution: **A** decreases by a factor of 2 **B** increases by a factor of 2 **C** decreases by a factor of 100 **D** increases by a factor of 100.	C	1
16	What is the pH of pure or distilled water? **A** 0 **B** 5 **C** 7 **D** 10	C	1
17	A solution of pH 8 is: **A** slightly basic **B** slightly acidic **C** strongly basic **D** strongly acidic.	A	1

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18	100 mL of a solution with pH 10 is added slowly to 900 mL of cold water. What will be the pH of the diluted solution? **A** 8 **B** 9 **C** 11 **D** 12	B	1

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Section B—Written answers (62 marks)

1	Write a chemical equation for a reaction where the products are dissolved $CaCl_2$, gaseous CO_2 and liquid H_2O. The reactants are solid $CaCO_3$ and dissolved HCl.	$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$	2
2	**Identify** three signs that a chemical reaction has occurred.	Any three of: • there is a permanent colour change • a gas is given off • energy is released or absorbed • a precipitate forms in solution • one metal deposits on another.	3
3	**Propose** two reasons why some chemical reactions do not occur spontaneously.	The reaction is endothermic (the products have higher energy than the reactants). The products are more ordered than the reactants (spontaneous reactions tend to disorder).	2

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4 The table below shows the symbol or formula of several ions.

Name of ion	Symbol or formula
magnesium	Mg^{2+}
carbonate	CO_3^{2-}
hydroxide	OH^-
nitride	N^{3-}
ammonium	NH_4^+

Construct the chemical formula for:

a magnesium nitride

b ammonium hydroxide

c magnesium hydroxide.

a Mg_3N_2

b NH_4OH

c $Mg(OH)_2$

4

5 **Use** the information in the previous question to name these chemicals.

a $Al_2(SO_4)_3$

b $MgCO_3$

c $Ca(OH)_2$

a aluminium sulfate

b magnesium carbonate

c calcium hydroxide

3

6 **Identify** how many atoms of each element there are in the compounds below.

a $Zn(NO_3)_2$

b $(NH_4)_2SO_4$

a Zn = 1, N = 2, O = 6

b N = 2, H = 8, S = 1, O = 4

2

7 **Predict** the missing chemicals in the equations below.

a $HCl + NaOH \rightarrow NaCl + \ldots$

b $2HNO_3 + Zn \rightarrow Zn(NO_3)_2 + \ldots$

c $2HCl + \ldots \rightarrow CaCl_2 + H_2O$

a H_2O

b H_2

c CaO

3

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8	Complete the following word equation. acid + metal oxide →	acid + metal oxide → salt + water	1
9	**Name** the salt produced when: **a** hydrochloric acid (HCl) reacts with magnesium (Mg) **b** nitric acid (HNO_3) reacts with barium carbonate ($BaCO_3$)	**a** magnesium chloride ($MgCl_2$) **b** barium nitrate ($Ba(NO_3)_2$)	2
10	Two clear aqueous solutions, $BaCl_2$ and Na_2SO_4, are mixed. A white solid forms. **Identify:** **a** the solvent for this reaction **b** the name given to a solid that forms when the two solutions are mixed **c** the chemical formula of the solid that is formed in this reaction.	**a** water **b** precipitate **c** $BaSO_4$	3
11	Write a chemical equation for the reaction between solid zinc (Zn) and an aqueous solution of hydrochloric acid (HCl). One of the products of the reaction is a soluble salt, zinc chloride ($ZnCl_2$).	$Zn_{(s)} + 2HCl_{(aq)} \rightarrow H_{2(g)} + ZnCl_{2(aq)}$	2

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12 The table below shows the solubility of several ions.

Negative ion	Positive ion	Solubility of compounds
NO_3^-	all	soluble
SO_4^{2-}	Ba^{2+} Pb^{2+} others	insoluble soluble
Cl^-	Ag^+ Pb^{2+} others	insoluble soluble

Use this table to help you write equations for the following reactions. You should include states (aq, s, g) in your equations.

a Solutions of sodium sulfate (Na_2SO_4) and barium chloride ($BaCl_2$) are mixed.

b Solutions of lead(II) nitrate ($Pb(NO_3)_2$) and potassium chloride (KCl) are mixed.

a $Na_2SO_{4(aq)} + BaCl_{2(aq)} \rightarrow 2NaCl_{(aq)} + BaSO_{4(s)}$

b $Pb(NO_3)_{2(aq)} + 2KCl_{(aq)} \rightarrow PbCl_{2(s)} + 2KNO_{3(aq)}$

4

13 **Define** the terms 'thermolysis reaction' and 'decomposition temperature'.

Thermolysis reactions are special types of decomposition reactions where heat causes a reactant to break up into two or more products.

Decomposition temperature is the temperature at which a substance decomposes.

2

14 **Propose** whether water can decompose.

Yes, but only at temperatures over 2000ºC does it turn back into hydrogen and oxygen.

2

15 **Explain** what salts are made up of.

A crystal lattice of positive metal and negative non-metal ions.

1

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16	**Explain** why it is necessary to have solubility rules.	It is difficult to know whether or not a salt is soluble in water. The rules that scientists have developed make it easier for us to know whether or not they are soluble.	2
17	Write the chemical formula for: **a** dinitrogen pentoxide **b** phosphorus trichloride **c** sulfur hexafluoride.	**a** N_2O_5 **b** PCl_3 **c** SF_6	3
18	In a solution of pH 7, **state** the colour of: **a** red litmus **b** blue litmus.	**a** red **b** blue	2
19	You are provided with 10 mL of a solution of pH 4. **Describe** how you would prepare a solution of pH 5.	Slowly add the 10 mL of acid to 90 mL of cold water.	2
20	**List** three properties of acids.	• taste sour • turn blue litmus red • conduct electricity in aqueous solution	3
21	**Use** any of the terms in the list below to **name** each of the reactions shown. *decomposition, precipitation, displacement, combustion, neutralisation* **a** $Pb(NO_3)_{2(aq)} + 2KCl_{(aq)} \rightarrow PbCl_{2(s)} + 2KNO_{3(aq)}$ **b** $C_6H_{12}O_{6(aq)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)}$ **c** $Mg_{(s)} + ZnCl_{2(aq)} \rightarrow MgCl_{2(aq)} + Zn_{(s)}$	**a** precipitation **b** combustion **c** displacement	3

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22	**a** **Identify** which substance in the following list could *not* be an acid. *HBr, CH_3COOH, K_2CO_3* **b** **Justify** your answer in part **a**.	**a** K_2CO_3 **b** Acids donate hydrogen ions. This substance does not contain hydrogen, so cannot be an acid.	3
23	**Predict** the approximate pH of the solutions described in **a** to **e** using the numbers shown below. *2, 4, 5, 7.5, 13* **a** coffee **b** oven cleaner **c** orange juice **d** rust remover **e** human blood	**a** 5 **b** 13 **c** 4 **d** 2 **e** 7.5	5
24	**Use** the numbers from the list below to **predict** the approximate pH of the solutions described in **a** to **c**. *1, 5, 7, 8, 13* **a** a concentrated solution of a strong acid **b** a neutral solution **c** a dilute solution of a weak base	**a** 1 **b** 7 **c** 8	3

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Science Focus 3 TEST ANSWERS

4: Sense and control

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 93 marks

Section A – Multiple choice (29 marks)

1	Which of the following is a sense? **A** sight **B** anger **C** moisture **D** temperature	A	1
2	The human eye focuses by: **A** changing the size of the pupil **B** changing the shape of the lens **C** moving the lens forwards or backwards within the eye **D** moving the head slightly closer or further from an object.	B	1
3	To reduce the amount of light entering the eye: **A** the pupil dilates **B** the pupil reduces in size **C** the lens thickens **D** the lens becomes slightly cloudy.	B	1
4	The section at the back surface of the eye lacking light-sensitive cells is called: **A** the fovea **B** the yellow spot **C** the blind spot **D** the black spot.	C	1
5	Binocular vision is: **A** the ability to see in three dimensions and judge distances **B** the magnified image you see when using binoculars **C** another name for perfect vision **D** when two people look at the same object.	A	1

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6	Colour blindness affects: **A** more women than men **B** men only **C** women only **D** more men than women.	D	1
7	The inability to focus properly on far-away objects is called: **A** accommodation **B** hyperopia **C** myopia **D** presbyteria.	C	1
8	How many main sections does the human ear contain? **A** one **B** two **C** three **D** four	C	1
9	The hammer, anvil and stirrup are: **A** three tiny bones in the ear **B** other names for the eardrum **C** attached directly to the cochlea **D** nerves that transmit sounds to the brain.	A	1
10	A bionic ear: **A** amplifies sounds before they reach the eardrum **B** is an artificial eardrum **C** replaces the cochlea with an electronic implant **D** produces electrical signals which stimulate auditory nerves.	D	1
11	When you smell a freshly baked cake, it's because: **A** vibrations caused by particles within the cake are passed into the air and in turn vibrate cells in your nose **B** your brain creates the illusion of smell based purely on what you see **C** tiny particles of cake have dissolved in the moist lining of your nose **D** cake particles carried by air currents dissolve on your tongue.	C	1

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12	How many different smells can a typical human nose detect? **A** 100 **B** 500 **C** 2000 **D** 10 000	C	1
13	Which of the following statements is correct? **A** Taste buds are on the sides of your tongue only. **B** Taste buds are on the sides of papillae (small bumps on the surface of the tongue). **C** Papillae are particles that dissolve samples of food so they may be detected by taste buds. **D** There are five different types of taste buds.	B	1
14	Which of the following is a type of skin cancer? **A** carcinoma **B** eczema **C** dermatitis **D** mole	A	1
15	When the level of carbon dioxide in your blood falls, the usual response of your body is to: **A** decrease breathing rate **B** increase breathing rate **C** produce the hormone oestrogen **D** produce the hormone insulin.	A	1
16	Which of the following is part of the peripheral nervous system? **A** the brain **B** spinal cord **C** heart muscle **D** taste buds on the tongue	D	1

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4: Sense and control

17	Interneurons transfer messages: **A** from receptors to the central nervous system **B** from the central nervous system to effectors **C** within the central nervous system **D** directly between sensory receptors.	C	1
18	Which of the following correctly matches receptor cells to the energy conversion they carry out? **A** Retina cells convert sound energy to electrical energy. **B** Cells in the taste buds convert chemical energy to electrical energy. **C** Thermoreceptor cells convert electrical energy to thermal energy. **D** Cochlea cells convert light energy to electrical energy.	B	1
19	Which of the following is *not* a hormone? **A** insulin **B** glucagon **C** dopamine **D** adrenalin	C	1
20	Which of the following is *not* a reflex action? **A** blinking **B** sneezing **C** eating **D** coughing	C	1
21	The part of the brain that gives you the sensation of hearing is the: **A** cerebrum **B** cerebellum **C** medulla **D** spinal cord.	A	1
22	Which of the following is usually largely controlled by the endocrine system? **A** breathing **B** heartbeat **C** tasting **D** water balance	D	1

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23	The conditions known as gigantism and dwarfism can result from abnormal levels of which hormone? **A** adrenalin **B** antidiuretic hormone (ADH) **C** human growth hormone (HGH) **D** oestrogen	C	1
24	Which hormone is known as the 'fight or flight' hormone? **A** adrenalin **B** antidiuretic hormone **C** human growth hormone **D** oestrogen	A	1
25	A fall in blood glucose levels is initially likely to result in the release of which hormone into the bloodstream? **A** insulin **B** thyroxin **C** progesterone **D** glucagon	D	1
26	Stroke occurs when: **A** the brain stops working **B** blood clots in your feet **C** you have a heart attack **D** the brain is deprived of blood.	D	1
27	Spina bidifa is a: **A** genetic hormone **B** growth hormone **C** genetic condition **D** disease.	C	1

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28	The disease diabetes mellitus is most often associated with which hormone? **A** insulin **B** growth hormone **C** thyroxin **D** testosterone	A	1
29	Which group of hormones is known to cause a plant to bend towards light? **A** adrenalin **B** antidiuretic hormone **C** auxin **D** oestrogen	C	1

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4: Sense and control

Section B – Written answers (64 marks)

1	**Identify** the part of the eye that is similar to that of a camera's: **a** diaphragm **b** film.	**a** iris **b** retina	2
2	**Propose** an advantage of having an eye on each side of the head for animals such as rabbits and deer.	They can see much more on both sides at once and avoid predators.	2
3	**Explain** how a sight defect may be corrected.	Various answers are possible. Short-sightedness may be corrected using a concave lens (thinner in middle). Long-sightedness may be corrected using a convex lens (wider in middle). Laser surgery may be used to reshape the lens.	2

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4 Choose three parts of the eye and **summarise** their function.

Any three of the following.

Part	**Function**
conjunctiva	thin, clear layer covering front of eye
sclerotic layer	tough white of the eye
cornea	clear window in sclerotic layer
aqueous humour	fills front of eye
iris	controls amount of light entering eye
pupil	hole in the iris
lens	helps focus
suspensory ligaments	hold lens in position
ciliary muscles	change shape of lens
vitreous humour	jelly that fills the back of the eye
choroid	black layer that stops reflection
retina	layer of light-sensitive cells at the back of the eye
fovea	yellow spot, helps us to see colour
blind spot	where vessels and nerves join the eyeball
optic nerve	carries messages from eye to brain

6

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<table>
<tr>
<td>5</td>
<td>Identify the two types of light-sensitive cells in the eye, and explain the difference between them.</td>
<td>Rods help us to see light and dark.
Cones help us to see colour.</td>
<td>2</td>
</tr>
<tr>
<td>6</td>
<td>Describe any three parts of the ear.</td>
<td>Any three of the following.
<table>
<tr><th>Part</th><th>Description or function</th></tr>
<tr><td>pinna</td><td>fleshy ear flap, collects sound</td></tr>
<tr><td>eardrum</td><td>thin sheet of muscle which vibrates in response to sounds</td></tr>
<tr><td>ossicles</td><td>three bones (hammer, anvil and stirrup) which receive vibrations from eardrum</td></tr>
<tr><td>oval window</td><td>section at the boundary between the middle and the inner ear</td></tr>
<tr><td>cochlea</td><td>coiled, fluid-filled tube</td></tr>
<tr><td>semicircular canals</td><td>three perpendicular sections which help us balance</td></tr>
<tr><td>auditory nerves</td><td>pass messages to the brain which are interpreted as sound</td></tr>
</table>
</td>
<td>6</td>
</tr>
</table>

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7	**Account** for the usefulness of the following glands: **a** sweat glands **b** sebaceous glands.	**a** They produce sweat, which helps cool the skin by evaporation. **b** These produce oil, which keeps the skin soft and stops it from cracking.	2
8	**Propose** two rules for avoiding skin cancer.	Use sunscreen and reapply regularly if required. Wear a hat and shirt (and other clothing!) when in the sun.	2
9	**Identify** two reasons why organisms need to respond to changes in their surroundings.	Any two of: • to escape predators • to find food • to avoid extremes of temperature	2
10	**a** **Define** 'homeostasis'. **b** **Identify** two substances whose levels are homeostatically controlled.	**a** The maintenance of a constant internal environment despite changes in the surroundings. **b** Any two of: • glucose • water • carbon dioxide.	4

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11 **Define** the following terms and give an example of each.

a receptor

b effector

a A specialised area sensitive to changes in a particular condition; e.g. retina cells in the eye respond to changes in light.

b An organ or gland that causes a response to a stimulus; e.g. the pancreas releases insulin in response to increased blood glucose levels.

4

12 Complete the following table showing the receptors of various stimuli.

Stimulus	**Receptor and location**
light	
chemicals	
	semicircular canals in the ear
	cochlea cells in the inner ear

Stimulus	**Receptor and location**
light	*retina cells in the eye*
chemicals	*receptors on the tongue and in the nose*
gravity	semicircular canals in the ear
sound	cochlea cells in the inner ear

4

13 **Identify:**

a the meaning of 'the threshold' for a stimulus

b why it is useful to have thresholds.

a The minimum intensity that causes a response for that stimulus.

b To prevent a response occurring every time a stimulus is encountered, because the stimulus at very low levels may not be important.

3

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14	Suppose your body temperature increased after exercise. **Identify:** **a** one area where receptors that detected this increase would be located **b** the coordinating centre that would receive these messages **c** one organ or gland that might act as an effector **d** one response you might detect.	**a** the skin (thermoreceptors) **b** the brain **c** sweat glands in the skin **d** sweating	4
15	**Compare** and **contrast** neurons and most other animal cells. **State** two features that are similar and two features that are different.	Compare and contrast any two of: dendrites, axon, myelin sheath with any two of: nucleus, cytoplasm, cell membrane.	4
16	**Describe** three ways in which the brain is protected from injury.	• a bony case (the skull) • layers of connective tissue (the meninges) • a jacket of fluid (cerebrospinal fluid)	3
17	**Describe** the sequence of events that makes up the reflex action occurring when a bright light is shone in your eyes.	1 Receptors detect a change in light intensity. 2 An impulse is sent along a sensory neuron to the brain. 3 An impulse is sent along a motor neuron to iris muscles. 4 Iris muscles contract, causing the pupil to narrow.	4

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18	**List** the most common causes of spinal cord injuries in Australia.	• high-speed motor accidents • falls from ladders • sports-related injuries • water-related injuries	4
19	**Identify:** **a** one difference between pheromones and hormones **b** two examples of the use of pheromones.	**a** Hormones are released into the bloodstream. Pheromones are released into the environment. **b** • Many insects use pheromones to attract mates • Ants use pheromones to mark food trails	4

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Science Focus 3 TEST ANSWERS

5: Reproduction

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 67 marks

Section A—Multiple choice (19 marks)

1	Which of the following methods of reproduction produces the most varied offspring? **A** budding **B** spore formation **C** fusion of gametes **D** fragmentation and regeneration	C	1
2	Hermaphrodites are organisms which: **A** produce both male and female gametes **B** reproduce asexually **C** produce pheromones to attract the opposite sex **D** can regenerate lost body parts.	A	1
3	Which of the following organisms reproduces using spores? **A** frog **B** fern **C** hydra **D** yeast	B	1
4	Female gametes are usually: **A** larger and more mobile than male gametes **B** smaller and more mobile than male gametes **C** larger and less mobile than male gametes **D** smaller and less mobile than male gametes.	C	1
5	Which of the following statements is correct? **A** Ovulation refers to the release of egg cells from the uterus. **B** Fertilisation refers to the production of female gametes. **C** Menstruation refers to the shedding of the lining of the ovaries. **D** Gestation refers to the time between fertilisation and birth.	D	1
6	The male part of a flower is made up of the: **A** stigma, style and ovary **B** petals and sepals **C** anther and filament **D** nectary and ovule.	C	1

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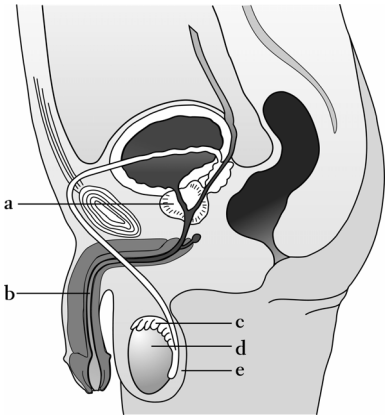
7	This question refers to the diagram below, which shows the human male reproductive system.  Which of the following correctly describes the function of the labelled structure? **A** Structure *c* stores sperm while they continue to develop. **B** Sperm are produced by structure *e*. **C** Structure *a* releases the major male hormone, testosterone. **D** Structure *d* is the only fluid-producing structure in the male reproductive system.	A	1
8	The urethra is a tube that carries: **A** sperm only **B** urine only **C** neither sperm nor urine **D** both sperm and urine (although not together).	D	1
9	Which of the following statements concerning gamete production in humans is correct? **A** Males have a store of gametes at birth; females produce gametes throughout their lives. **B** Females have a store of gametes at birth; males produce gametes throughout their lives. **C** Both males and females have a store of gametes at birth. **D** Both males and females produce gametes throughout their lives.	B	1

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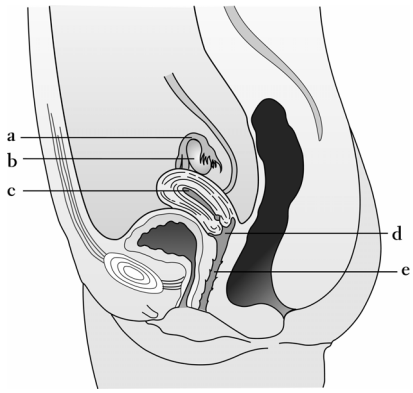
10	This question refers to the diagram below, which shows the human female reproductive system.  Which of the following correctly describes the function of the labelled structure? **A** Structure *d* is the site of implantation. **B** Structure *e* carries both urine and the menstrual flow. **C** Sperm are deposited in structure *c* during sexual intercourse. **D** Ova are released from structure *b*.	D	1
11	The site of fertilisation of the human egg cell is the: **A** ovary **B** fallopian tubes or oviduct **C** uterus **D** vagina.	B	1
12	The major female reproductive hormone is: **A** oestrogen **B** testosterone **C** adrenalin **D** insulin.	A	1
13	The major male reproductive hormone is: **A** sperm **B** testosterone **C** oestrogen **D** insulin.	B	1
14	The term menarche refers to the time in a human female's life when: **A** the first menstruation occurs **B** egg cells are no longer released from the ovaries **C** a zygote implants in the uterus **D** sexual intercourse first takes place.	A	1

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15	Which of the following shows the correct order of the stages of development of a human embryo? **A** zygote, foetus, blastocyst, morula **B** foetus, zygote, morula, blastocyst **C** zygote, blastocyst, morula, foetus **D** zygote, morula, blastocyst, foetus	D	1
16	Which of the following shows events in their correct order of occurrence? **A** copulation, ejaculation, fertilisation, implantation **B** ejaculation, fertilisation, copulation, implantation **C** implantation, ejaculation, fertilisation, copulation **D** copulation, ejaculation, implantation, fertilisation	A	1
17	Which of the following tests conducted during pregnancy does *not* involve the removal of fluid from either the mother or the foetus? **A** amniocentesis **B** ultrasound **C** testing for alpha-fetoprotein levels in the mother **D** chorionic villus sampling	B	1
18	Contraception refers to any process that prevents: **A** sexual intercourse from occurring **B** the transmission of STDs **C** pregnancy from occurring **D** gametes from being produced.	C	1
19	Which of the following processes does *not* normally occur during IVF? **A** Egg cells are retrieved from the ovary. **B** Eggs are placed in a salt solution at 45^{o}C. **C** Sperm are added to the eggs in the salt solution. **D** Fertilised eggs are transferred to the uterus after a short development time.	B	1

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5: Reproduction

Section B—Written answers (48 marks)

1	**a** **Explain** what is meant by asexual reproduction. **b** **List** two examples of asexual reproduction.	**a** production of a new organism without the fusion of gametes. **b** Any two of: • fission • budding • spores • regeneration • parthenogenesis.	3
2	**Describe** two reasons why the offspring of asexually reproducing organisms may not all be identical.	• Mutations may occur, resulting in new characteristics. • Different environmental conditions may result in different characteristics.	2
3	Flowering plants may either be self-pollinated or cross-pollinated. **a** **Explain** what is meant by cross-pollination. **b** **Propose** two ways in which cross-pollination can be made to occur. **c** **Clarify** the advantage of cross-pollination.	**a** Pollen is transferred from the anther of one flower to the stigma of another flower. **b** By wind or the action of insects. **c** Cross-pollination results in greater genetic variation in the offspring.	4
4	A particular flower has its stigma placed higher than its anthers. **Predict** whether this flower could be self-pollinated. **Justify** your answer.	No. The pollen cannot fall 'upwards' from the anther to the stigma.	2

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5	**a** **State** in which structure the human testes are located. **b** **Outline** the advantage of the testes being in this structure, rather than deep in the body.	**a** the scrotum **b** The scrotum allows the testes to be at a lower temperature than body temperature. This is better for sperm production.	3
6	The ovaries of a 30-year-old woman have many scars on their surface. **Explain** what causes these scars?	Each time an egg cell is released from the surface of the ovary, the follicle is left behind and leaves a scar.	2
7	**Describe** or **name** the events occurring at each of the following times in the human menstrual cycle. **a** days 1 to 5 **b** days 5 to 14 **c** day 14 **d** days 15 to 25	**a** menstruation (shedding of the uterine lining) **b** development of follicles in the ovary; repair of the uterine lining **c** ovulation (release of an egg cell) **d** thickening of the uterine lining	5
8	**a** **Describe** two ways in which twins can be produced. **b** **Identify** which method produces twins that are most alike.	**a** Any two of: • two separate eggs are fertilised (fraternal twins) • a single fertilised egg splits in two (identical twins) • an egg splits before fertilisation and each half is fertilised by a different sperm (half-identical twins) **b** Identical twins are formed when a single fertilised egg splits in two.	5

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<table>
<tr><td>9</td><td>Describe three changes in females at puberty.</td><td>Any three of:
growth spurt
breasts and buttocks develop
hips become wider
pubic and underarm hair develops
menstruation begins.</td><td>3</td></tr>
<tr><td>10</td><td>Describe three changes in males at puberty.</td><td>Any three of:
growth spurt
increased muscle mass
body and facial hair growth
lowering of voice
testes develop and sperm production begins.</td><td>3</td></tr>
<tr><td>11</td><td>Complete the table below, which shows several methods of contraception.<table><tr><th>Name of method</th><th>How it works</th></tr><tr><td></td><td>Rubber device that fits over the cervix.</td></tr><tr><td></td><td>Stops sperm entering the uterus and prevents implantation.</td></tr><tr><td>Contraceptive pills</td><td></td></tr></table></td><td><table><tr><th>Name of method</th><th>How it works</th></tr><tr><td>Cap or diaphragm</td><td>Rubber device that fits over the cervix.</td></tr><tr><td>IUD (intra-uterine device)</td><td>Stops sperm entering the uterus and prevents implantation.</td></tr><tr><td>Contraceptive pills</td><td>Consist of hormones that stop ovulation.</td></tr></table></td><td>4</td></tr>
</table>

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12	**Describe** briefly each of the following embryonic stages. **a** morula **b** blastocyst	**a** a clump of approximately 80 cells **b** a fluid-filled ball with more cells than the morula	2
13	When referring to human reproduction, **define** what each of the following terms stands for. **a** STI **b** IUD **c** AIDS **d** IVF	**a** sexually transmitted infection **b** intra-uterine device **c** acquired immune deficiency syndrome **d** in-vitro fertilisation	4
14	**State** the length of time (in days) for each of the following: **a** the typical menstrual cycle in humans **b** human gestation **c** the number of days a human egg cell spends in the oviduct.	**a** 28 days **b** 280 days **c** 7–8 days	3

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15	Choose from the list below the approximate numbers for each of the quantities described in **a** to **c**. • *one* • *several hundred* • *several hundred thousand* • *several million* • *several hundred million* **a** The number of egg cells present in each human ovary at birth. **b** The number of sperm released by a human male at each ejaculation. **c** The number of sperm that reach the egg cell following copulation.	**a** several hundred thousand (500 000) **b** several hundred million **c** several hundred	3

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Science Focus 3 TEST ANSWERS

6: Ecosystems

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 80 marks

Section A – Multiple choice (19 marks)

1	Consider the food chain shown below. *phytoplankton* → *zooplankton* → *small fish* → *large fish* → *sea eagles* Which of the following correctly describes the role of the zooplankton and large fish? **A** producer / third order consumer **B** first order consumer / fourth order consumer **C** first order consumer / third order consumer **D** second order consumer / fourth order consumer	C	1
2	Which of the following organisms is *not* a consumer? **A** mushroom **B** blue-green algae **C** earthworm **D** elephant	B	1
3	Which of the following is recycled in an ecosystem? **A** matter only **B** matter and energy **C** energy only **D** neither matter nor energy	A	1

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4	Which type of reaction ultimately provides the energy for almost all life on Earth? **A** photosynthesis **B** respiration **C** nuclear fusion **D** decomposition	C	1
5	Consider the food chain shown below. *wheat* → *mouse* → *snake* → *kookaburra* If the wheat eaten by the mouse contains 1000 units of energy, approximately how many units of energy will the kookaburra receive? **A** 1 **B** 10 **C** 100 **D** 1000	A	1
6	A soil sample was analysed and four species of animal were found. The approximate numbers of each animal in the soil sample are shown below. • species W: 2 000 000 • species X: 2 • species Y: 2000 • species W: 20 000 Which species is likely to be the highest order consumer? **A** W **B** X **C** Y **D** Z	B	1
7	Transpiration is the name given to the loss of water by evaporation from: **A** moist soil **B** oceans **C** animals **D** plants.	D	1

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8	Approximately what percentage of all water on Earth is available as fresh water to organisms? **A** 1% **B** 10% **C** 50% **D** 98%	A	1
9	Nitrifying bacteria convert: **A** atmospheric nitrogen molecules to nitrate ions **B** atmospheric nitrogen molecules to amino acids **C** nitrate ions to atmospheric nitrogen molecules **D** ammonia to nitrate ions.	D	1
10	Which of the following is *not* responsible for nitrogen fixation? **A** the action of lightning on atmospheric nitrogen and oxygen **B** industrial production of nitrogen-based fertilisers **C** *Rhizobium* bacteria found in the root nodules of clover plants **D** denitrifying bacteria found in the soil	D	1
11	Which of the following processes removes carbon from the atmosphere? **A** combustion of natural gas **B** photosynthesis by green plants **C** anaerobic respiration by yeast **D** decomposition of fallen leaves by fungi	B	1
12	Which of the following is a renewable energy source? **A** coal **B** oil **C** geothermal **D** uranium	C	1
13	A photovoltaic cell is designed to convert: **A** solar energy to sound energy **B** solar energy to electrical energy **C** heat energy to electrical energy **D** heat energy to solar energy.	B	1

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14	Splitting one atom of uranium releases how much more energy than burning one molecule of natural gas? **A** 2.6 times **B** 26 times **C** 2.6 million times **D** 26 million times	D	1
15	Which of the following is *not* part of conservation? **A** things we do to prevent plants from becoming extinct **B** things we do to to keep habitats safe **C** cutting down trees **D** planting trees	C	1
16	What is an exotic species? **A** a species that is introduced **B** a species that is native **C** a species that is a pest **D** a species that is a pet	A	1
17	Which of the following energy sources is least used in Australia at present? **A** hydroelectricity **B** energy from wind **C** geothermal energy **D** solar energy	C	1
18	Salinisation is the process of: **A** planting trees in a desert **B** adding salt to the soil **C** losing good land to the desert **D** draining water.	B	1
19	Desertification is the process of: **A** planting trees in a desert **B** adding salt to the soil **C** losing good land to the desert **D** draining water.	C	1

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Section B – Written answers (61 marks)

1	**Examine** the food chain shown below. *mangroves → soldier crabs → spoonbills → sea eagles* For this food chain, **identify** the **a** producer **b** first order consumer **c** herbivore **d** third order consumer.	**a** mangroves **b** soldier crabs **c** soldier crabs **d** sea eagles	4
2	Draw a food chain, with four members only, by **interpreting** the following description. *The grasslands of Africa are home to many herbivores, including zebra, gnu and gazelles. These grazing animals are the prey of carnivores, including lions and cheetahs. Hyenas and vultures that scavenge for dead remains may, at times, eat any of the animals of the grassland.*	Possible answers are: grass → zebra → cheetah → hyena grass → gnu → lion → vulture	4
3	**Explain** why a food chain is unlikely to have more than six members.	Energy is lost at each step in a food chain. Each organism receives only approximately 10% of the energy available from the previous organism in the chain. The last organism in a chain of six would receive too little energy (0.1%) to survive.	3

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4	Consider the food chain shown below. *leaves* → *caterpillar* → *bird* → *cat* The total energy in the leaves is much greater than the energy received by the cat. **Describe** two different ways in which the energy of the leaves may be 'lost' and therefore not reach the cat.	Any two of: • movement by the caterpillar and bird as it searches for food • sounds made by the bird • energy used by the bird in flying • heat used to maintain the body temperature of the bird.	2
5	**a** **Define** the Law of Conservation of Energy. **b** **Present** an example of how this law applies to energy in ecosystems.	**c** Energy is neither created nor destroyed, but can only be converted from one form to another. **d** When energy is passed along a food chain, approximately 10% is passed from one organism to the next. The remaining 90% is converted to other forms of energy such as energy of movement, heat, sound etc.	3
6	**a** **Define** the Law of Conservation of Mass. **b** **Present** an example of how this law applies to matter in ecosystems.	**c** Atoms are not created or destroyed in chemical reactions, just rearranged. **d** A carbon atom that is now part of your body was once in the atmosphere as a carbon dioxide molecule, and once part of another organism.	3

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7	**Clarify** the meaning of each of the following, and give an example of each. **a** organic matter **b** an endothermic animal **c** an ectothermic animal	**a** matter that contains the carbon atom and is derived from either plants or animals; e.g. chlorophyll in a plant leaf **b** an animal that maintains a constant body temperature; e.g. a cat **c** an animal whose body temperature depends on that of the external environment; e.g. a snake	6
8	**Describe** two*:* **a** uses of nitrogen by plants **b** ways in which nitrogen may leave a plant **c** ways in which nitrogen may leave an animal.	**a** making amino acids and nucleic acids **b** fallen leaves, eaten by an animal **c** waste products, such as urine, eaten by another animal	6
9	The greenhouse effect is caused by an increase in the level of carbon dioxide in the atmosphere. **Describe** two ways in which this increase has occurred.	• deforestation (removal of trees leading to less photosynthesis) • excessive combustion of fossil fuels	2

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<table>
<tr>
<td>10</td>
<td>Complete the following table, which shows the role of bacteria in the cycling of nitrogen.
<table>
<tr><th>Bacteria type</th><th>Role of bacteria</th></tr>
<tr><td>Nitrogen fixing</td><td>Convert nitrogen to</td></tr>
<tr><td></td><td>Convert ammonia to nitrates</td></tr>
<tr><td></td><td>Convert nitrates to</td></tr>
</table>
</td>
<td>
<table>
<tr><th>Bacteria type</th><th>Role of bacteria</th></tr>
<tr><td>Nitrogen fixing</td><td>Convert nitrogen to nitrates</td></tr>
<tr><td>Nitrifying</td><td>Convert ammonia to nitrates</td></tr>
<tr><td>Denitrifying</td><td>Convert nitrates to nitrogen</td></tr>
</table>
</td>
<td>4</td>
</tr>
<tr>
<td>11</td>
<td>Explain what is meant by a non-renewable energy source. Include two examples in your answer.</td>
<td>Energy sources that cannot be replaced. Examples include coal, oil, natural gas and uranium.</td>
<td>3</td>
</tr>
<tr>
<td>12</td>
<td>Identify two ways in which solar energy may be used to generate electricity.</td>
<td>• solar ponds
• solar (photovoltaic) cells</td>
<td>2</td>
</tr>
<tr>
<td>13</td>
<td>Explain two disadvantages of using geothermal energy to produce electricity.</td>
<td>Any two of:
• not all countries have access to this source of energy
• changes in underground pressures caused by this technique may result in earthquakes and rock cracking
• gaseous pollutants are produced.</td>
<td>2</td>
</tr>
<tr>
<td>14</td>
<td>Explain how a change in wind speed influences the amount of power supplied by a wind turbine generator.</td>
<td>The power supplied is proportional to the wind speed cubed. If the wind speed is doubled, the power is multiplied by a factor of eight.</td>
<td>2</td>
</tr>
</table>

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15	**Identify** the three main types of pollution.	• water pollution • air pollution • soil pollution	3
16	**a** **Define** the term 'biomass'. **b** **Identify:** **i)** one direct use of biomass for energy production **ii)** one indirect use of biomass for energy production.	**a** Organic material that has recently died and that can be used to generate energy. **b** **i)** burning of wood for heating **ii)** conversion of wastes from crops into fuels such as ethanol	3
17	**Describe** two aspects of how Aboriginal people survived in balance with the environment.	Answers could include the following: • bush tucker: eating food that didn't require cooking and that was native • firestick farming: burning off and then flushing out animals for hunting also encouraged regrowth.	4
18	**Explain** two negatives for the environment of white settlement in Australia.	Answers could include the following: • foreign animals • foreign plants • pollution • clearing of the land.	2
19	**Identify** three introduced species that live in Australia.	• rabbits • foxes • cane toads • cows • horses	3

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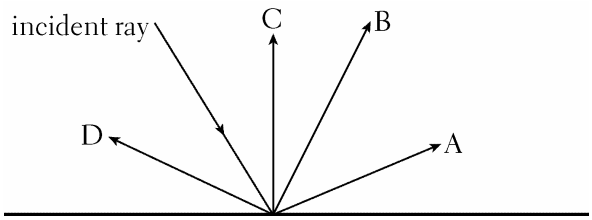
Science Focus 3 TEST ANSWERS

7: Light

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 76 marks

Section A—Multiple choice (21 marks)

1	Which ray below is the one that is reflected from a plane mirror?  	B	1
2	If light leaves a medium that has a lower refractive index than the new medium, which of the following will occur: **A** the light will slow down slightly and bend towards the normal **B** the light will slow down slightly and bend away from the normal **C** the light will speed up slightly and bend towards the normal **D** the light will speed up slightly and bend away from the normal.	A	1
3	When a light ray passing from glass to air strikes the boundary at the critical angle of incidence, the light ray: **A** is totally internally reflected **B** passes through to the air without any change in its direction **C** skims the surface of the glass **D** is reflected back along its original path.	C	1
4	When a pencil is viewed through water, it appears bent because light: **A** slows down as it passes from water to air **B** is reflected at the surface of the water **C** is refracted away from the normal as it passes from water to air **D** is refracted towards the normal as it passes from water to air.	C	1

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7: Light

5	A fisherman under the water is attempting to spear a fish. The fisherman must aim: **A** above the apparent depth of the fish **B** directly at the fish as he sees it **C** below the apparent depth of the fish **D** to the left of the apparent position of the fish.	B	1
6	The image produced by a convex lens when the object is inside the focal length is enlarged: **A** real and upright **B** real and inverted **C** virtual and upright **D** virtual and inverted.	C	1
7	An object 4 mm high produces an image 16 mm high. The magnification is: **A** 0.25 **B** 4 **C** 12 **D** 20	B	1
8	The image produced on a screen by a slide projector is: **A** virtual, upright and enlarged **B** virtual, inverted and diminished **C** real, upright and enlarged **D** real, inverted and enlarged.	D	1
9	In an old-fashioned camera, photographic film records the images formed and is placed where the images would be. The images formed by the lenses in a camera must therefore be: **A** real **B** unreal **C** virtual **D** undecided, depending on the distance from the camera.	A	1
10	In a camera, the image formed is upside-down when recorded on the film. This suggests that the lenses in a camera are probably: **A** bi-concave **B** bi-plano **C** bi-convex **D** convex and concave.	C	1

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11	In the eye disorder known as long-sightedness, the image of an object forms behind the retina. Long-sightedness is corrected by bringing the image forward using: **A** convex lenses that bend the light less **B** convex lenses that bend the light more **C** concave lenses that bend the light less **D** concave lenses that bend the light more.	B	1
12	Blue skies and red sunsets are caused by the: **A** dispersion of light **B** refraction of light **C** absorption of light **D** scattering of light.	D	1
13	Optical fibres are used in: **A** microscopes **B** thermometers **C** endoscopes **D** decorative lights.	C D	2
14	In a primary rainbow, what colour is at the top of the rainbow? **A** blue **B** green **C** orange **D** red	D	1
15	Mirages are basically caused because light is: **A** refracted differently at different temperatures **B** dispersed differently at different temperatures **C** reflected differently at different temperatures **D** scattered differently at different temperatures.	A	1
16	Which of the following colours of light is most strongly refracted? **A** red **B** yellow **C** green **D** violet	D	1
17	What colour will a green leaf placed in green light appear to be? **A** black **B** green **C** red **D** blue	B	1
18	What colour is produced when magenta and cyan pigments are mixed? **A** black **B** green **C** red **D** blue	D	1

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7: Light

19	What colour is transmitted when white light goes through a red filter? **A** no colour (black) **B** white **C** red **D** green	C	1
20	Which colours are absorbed by blue paint? **A** blue only **B** red, orange and yellow **C** green, blue, indigo and violet **D** red only	B	1

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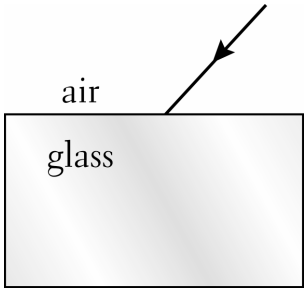
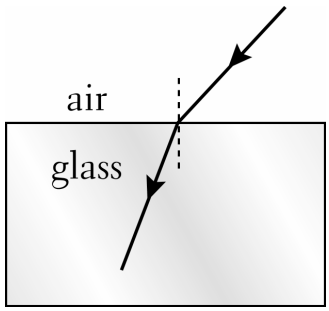
Section B—Written answers (55 marks)

1	**Explain** the difference between luminous and non-luminous objects.	Objects that are luminous make their own light to be seen (i.e. the Sun, glow worms), whereas objects that are non-luminous don't (i.e. the Moon, a tennis ball).	2
2	**Define** the law of reflection.	$i = r$ (angle of incidence = angle of reflection)	1
3	**Explain** what is meant by the refraction of light?	the bending of a light ray as it passes from one substance into another substance	2
4	**Explain** what is meant by the reflection of light.	where the light ray continues after bouncing off (reflecting) a surface	1
5	**Recall** and **explain** the features of a mirage.	A mirage is an optical illusion that reflects an image in a similar way to the reflection from the surface of a pool of water.	1
6	The diagram below shows a light ray travelling through air to the interface between the air and a glass block.  On the diagram, draw the normal, and a possible path for the ray in the glass.	 	2

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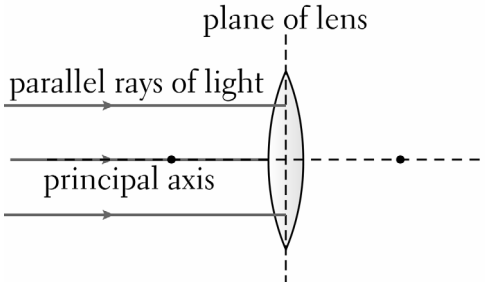
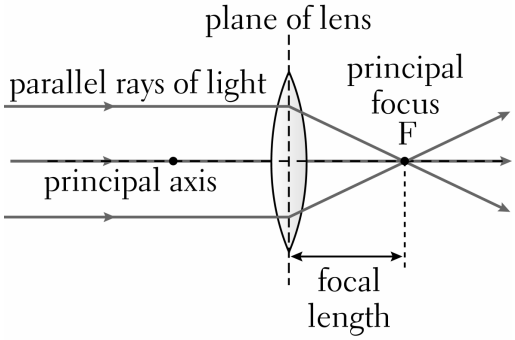
7	Light travelling from substance A to substance B is totally internally reflected. **a** **State** which substance (A or B) has the higher refractive index. **b** **Identify** at what angle the light ray must have struck the boundary of A and B for total internal reflection to occur.	**a** A **b** an angle greater than the critical angle	2
8	Is light always refracted when it passes from one material to another? **Explain** your answer.	No. When a light ray strikes perpendicular to a boundary, it is not bent.	2
9	**List** three advantages of optical fibres over copper wire for transmitting data.	Various answers are possible; e.g. thinner, cheaper, more durable, can carry more information.	3
10	**Identify** which type of lens (convex or concave): **a** is fatter in the middle than at the ends **b** can produce a real image.	**a** convex **b** convex	2
11	**Apply** the rules for drawing light rays to complete the diagram shown below by: **a** drawing in the refracted light rays **b** labelling the principal focus **c** labelling the focal length. 		4

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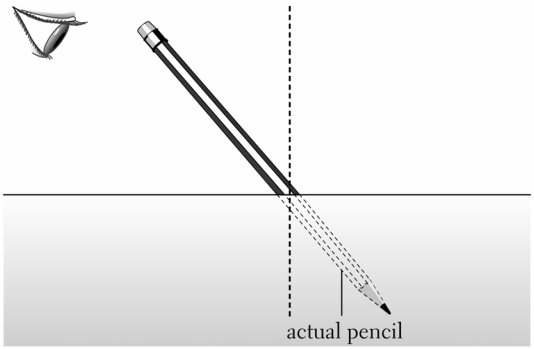
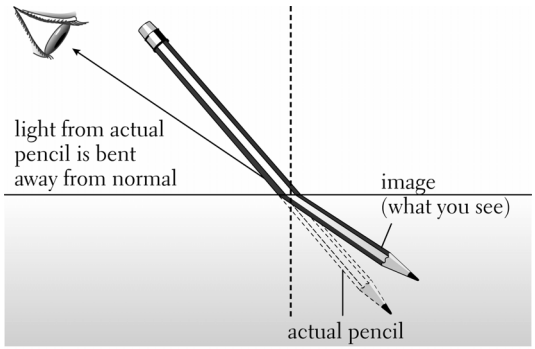
12	**Explain** why the sky appears blue during the day.	Blue light is scattered more by particles in the atmosphere than other colours. More blue light therefore reaches our eyes from particles in the sky.	2
13	**Identify** what colour (if any) is transmitted when: **a** white light is shone on a blue filter **b** red light is shone on a blue filter **c** magenta light is shone on a red filter.	**a** blue **b** no light **c** red	3
14	The diagram below shows a pencil placed in water. The pencil appears bent when viewed from the air.  Complete the diagram by drawing the position of the image of the pencil as seen by the viewer.	 	2
15	As the curvature of a lens increases, **state** whether: **a** the light is bent more or less **b** the focal length increases or decreases.	**a** more **b** decreases	2

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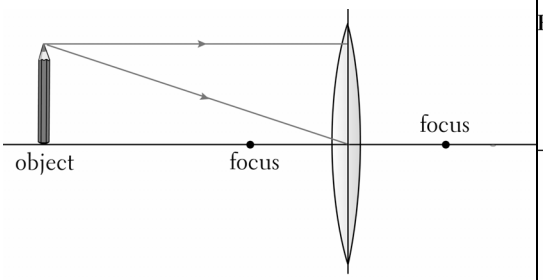
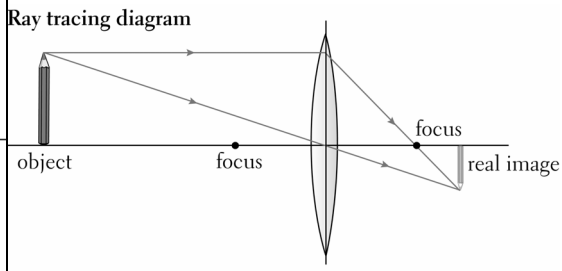
16	The diagram below shows a partly drawn ray-tracing diagram.  **a** Complete the ray-tracing diagram. **b** **Use** the correct terminology to **describe** the image.	**a**  **b** real, inverted and diminished	5
17	The objective lens of a telescope produces an image just inside the focal length of the eyepiece lens. This image acts as the object for the eyepiece lens. **a** For a larger first image to be formed, **state** whether the objective lens should be thin or thick. **b** **Explain** why telescopes are long.	**a** thin **b** Thin lenses have long focal lengths.	2

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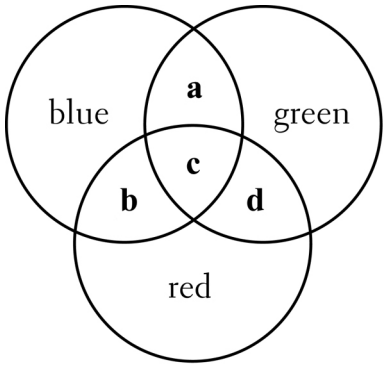
18	**a** **Recall** the primary colours of light. **b** **Explain** why they are called primary colours. **c** **Recall** the secondary colours of light. **d** **Explain** why they are called secondary colours. **e** Is black a colour? **Explain** your answer.	**a** red, green and blue **b** They can be combined in various proportions to form all the other colours. **c** cyan, magenta and yellow **d** They are produced when just two primary colours overlap. **e** No. Black is the absence of any colour.	5
19	The diagram below shows combinations of light colours.  Light **Recall** the names of the colours marked **a** to **d**.	**a** cyan **b** magenta **c** white **d** yellow	4
20	**Identify** the colour or colours of light that are: **a** absorbed by a blue shirt **b** reflected by a red shirt **c** combined to produce yellow light.	**a** all colours other than blue **b** red **c** red and green	3

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21	**Recall** the name given to the: **a** spreading of light by particles in the sky **b** bending of light rays as they pass from one material to another **c** splitting of light into component colours by a prism.	**a** scattering **b** refraction **c** dispersion	3
22	Light from neon lamps is largely red and orange. **Propose** why some butchers place neon lamps above their window displays.	The meat appears redder in the 'red' light than it does in white light.	2

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Science Focus 3 TEST ANSWERS

8: The universe

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right-hand column. Score: _____ / 65 marks

Section A—Multiple choice (21 marks)

1	Most of the stars you see at night are: **A** like our Sun, some much bigger **B** groups of stars called galaxies **C** planets **D** meteors.	A	1
2	Our nearest multiple star system is: **A** the Southern Cross **B** Ceres **C** Alpha Centauri **D** Orion.	C	1
3	A star gets its energy from: **A** a chemical reaction **B** a nuclear fission reaction **C** a nuclear fishing reaction **D** a nuclear fusion reaction.	D	1
4	Which of the following is the correct statement? **A** Nebulae may form stars, but stars do not form nebulae. **B** Nebulae may form stars and stars may explode to form nebulae. **C** A nebula is the name given to a star that is about to explode. **D** A nebula is the core of a star that remains after the star has exploded.	B	1
5	Our Sun will eventually become: **A** a white dwarf **B** a red dwarf **C** a neutron star **D** a black hole.	A	1
6	A pulsar is a: **A** radio wave that comes from a star **B** type of galaxy **C** rapidly rotating neutron star **D** galaxy whose centre is so bright it obscures the outer regions.	C	1

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8: The universe

7	Which of the following is a major space-based telescope? **A** Bubble **B** Hubble **C** Keck **D** Tech	B	1
8	The Milky Way is itself part of a larger group called: **A** the big dipper **B** the local group **C** the Andromeda super galaxy **D** the spiral arm.	B	1
9	As a fast-moving ambulance passes by, its siren goes from: **A** a high pitch to a low pitch **B** a low pitch to a high pitch **C** a high pitch to an even higher pitch **D** there is no change in pitch.	A	1
10	Which of the following statements concerning the colours of visible light is correct? **A** When light passes through a prism the colours are separated by reflection. **B** Red light has a lower energy than blue light. **C** Red and blue light have the same wavelength. **D** Red light has a higher frequency than blue light.	B	1
11	When a fast-moving star is moving away from the Earth, the absorption lines on the spectrum produced by the star will be shifted towards the: **A** blue end of the spectrum (light of longer wavelength) **B** red end of the spectrum (light of longer wavelength) **C** blue end of the spectrum (light of shorter wavelength) **D** red end of the spectrum (light of shorter wavelength).	B	1
12	Approximately how many years ago is the Big Bang thought to have occurred? **A** 13 thousand **B** 13 million **C** 13 billion **D** 13 trillion	C	1
13	A small fraction of a second after the Big Bang, the universe: **A** had a temperature of one billion degrees Celsius, and chemical elements had begun forming **B** was expanding slowly and stars were beginning to form **C** was cooling, and quarks were beginning to clump together to form neutrons **D** inflated suddenly and was filled with particles of matter and antimatter.	D	1

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8: The universe

14	Which theory of the universe includes the idea of a contracting universe? **A** open universe **B** closed universe **C** flat universe **D** accelerating universe	B	1
15	The open universe theory states that the universe will: **A** keep expanding forever but at a decreasing rate **B** eventually stop expanding and then contract into a smaller and smaller space **C** eventually stop expanding, but not contract again **D** expand at an ever-increasing rate.	A	1
16	Approximately how fast (in metres per second) does electromagnetic radiation such as radio waves travel? **A** 300 **B** 300 000 **C** 300 000 000 **D** 300 000 000 000	C	1
17	The orbital speed of the International Space Station is approximately: **A** 30 km/h **B** 400 km/h **C** 3000 km/h **D** 40 000 km/h.	C	1
18	The main oxygen source for the International Space Station is: **A** a generator that splits water into hydrogen and oxygen **B** photosynthesis by onboard plants **C** chemical decomposition of hydrogen peroxide to produce oxygen and water **D** oxygen tanks transported from Earth.	A	1
19	Electromagnetic radiation: **A** is laser waves **B** is radio waves **C** travels at the speed of light **D** is all of the above.	D	1
20	The term ‘dark energy’ is used to describe: **A** any light towards the high-energy end of the visible light spectrum **B** subatomic particles left over from the Big Bang **C** gas and dust orbiting stars that may condense to form planets **D** some mysterious cosmological force that overrides gravity.	D	1

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8: The universe

21	Passive sensors: **A** are the same as active sensors **B** make electromagnetic radiation **C** detect electromagnetic radiation **D** are space junk.	C	1

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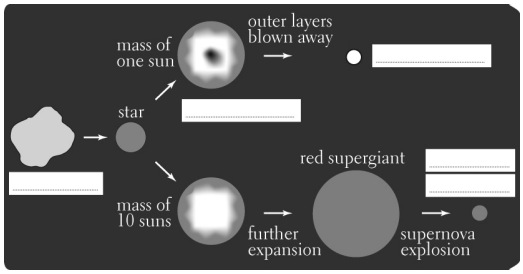
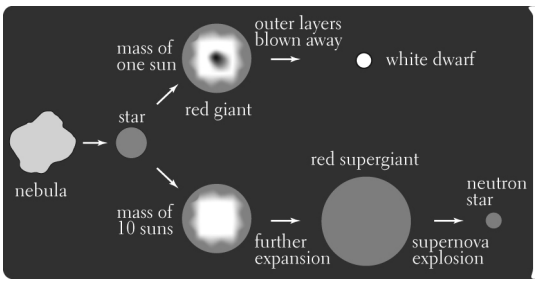
Section B—Written answers (44 marks)

1	Fill in the blanks in the diagram below showing the life of a star.  	 	4
2	**Describe** The Global Positioning System (GPS).	The GPS is a widely used and known satellite navigation system. There are 24 satellites over six orbits 20 000 km above the Earth.	2
3	**List** the four types of satellites.	• communication • navigation • weather • remote sensing	2
4	**Name** any three types of radiation that is emitted by the Sun.	Any three of: • IR • UV • visible light • X-rays • radio waves • gamma waves.	3

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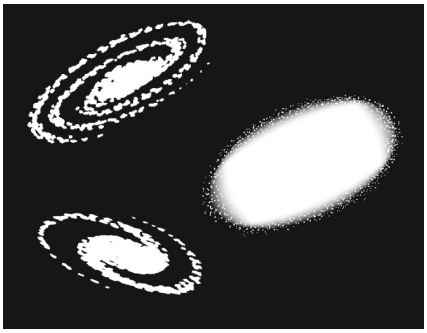
5	**a** If light cannot escape from black holes, **explain** how astronomers detect them. **b** Predict where the black hole that is closest to Earth is likely to be.	**a** by detecting X-rays emitted by black holes **b** at the centre of the Milky Way galaxy	2
6	**Name** and sketch three different types of galaxy.	 clockwise from top: spiral, elliptical and barred spiral	3
7	Complete these sentences about parts of the Milky Way galaxy. **a** An _____ cluster contains a few hundred stars within the spiral arm of the galaxy. **b** A _____ cluster may be seen with the naked eye as a fuzzy star, but actually contains up to a million stars.	**a** An *open* cluster contains a few hundred stars within the spiral arm of the galaxy. **b** A *globular* cluster may be seen with the naked eye as a fuzzy star, but actually contains up to a million stars.	2
8	**State** which sound wavelength (long or short) produces the higher pitched sound.	short	1

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9	Complete the following statement about a law by selecting the correct word in each part. Most galaxies are moving **a** *towards/away from* the Earth. The further away the galaxies are, the **b** *faster/slower* they are moving. This is known as **c** *Doppler's/Hubble's* law.	**a** away from **b** faster **c** Hubble's	3
10	**Explain** why it is impractical to search for extraterrestrial life using current spacecraft.	It would take 80 000 years to reach the nearest star.	1
11	**Describe** what would happen over time in a closed universe.	• The universe would stop expanding. • Gravitational attraction would pull all the matter closer together and the universe would contract into a smaller and smaller space. • Matter would become superheated and atoms would disintegrate. • A 'big crunch' would pack everything into a single black hole.	4
12	**Explain** why frequencies in the microwave region are preferred when broadcasting messages to try to make contact with extraterrestrial life.	There is less background interference when using these frequencies.	1

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13	**a** **State** the height above the Earth's surface of a satellite in geostationary orbit. **b** **Explain** what it means if a satellite is in an asynchronous orbit. **c** **Clarify** what type of orbit observation satellites are usually placed in. **d** **Clarify** what type of orbit communication satellites are usually placed in.	**a** 36 000 km **b** The satellite moves in the same rotational direction as the Earth, but more quickly. **c** asynchronous **d** geostationary	4
14	**Define** microgravity.	The apparent lack of gravity that produces a feeling of weightlessness.	1

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15	The ISS is a combined effort of several countries to construct a permanent space station. **a** **Recall** when the ISS started, and its expected completion date. **b** **Describe** one use of the ISS. **c** **State** the source of power for the ISS. **d** **Describe** two practical difficulties of daily life on the ISS.	**a** 1998; 2004 **b** For example, to study the effects of microgravity on crystal formation and on growth of tissue cultures. **c** solar panels **d** Answers could include the following. • Water is in limited supply so sponge baths are used rather than showers. • Toilets must have hand and foot restraints and must operate using a vacuum. • To exercise, astronauts must be strapped into a treadmill or an exercise bike. • For sleeping, astronauts must be strapped down to prevent them from floating away.	6

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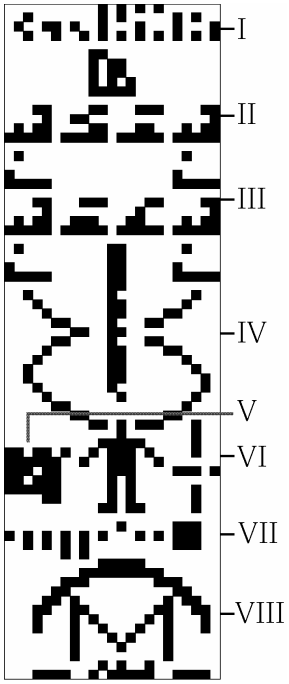
16	The diagram below shows a message beamed from the Arecibo radio telescope in 1974.  Match each description below to a symbol (I to VIII) on the message. **a** DNA double helix **b** solar system **c** numbers 1 to 10 in binary **d** atomic numbers of important elements **e** world population	**a** IV **b** VII **c** I **d** II **e** V	5

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Science Focus 3 TEST ANSWERS

9: Earth's fragile crust

Name: _____ Class: _____ Date: _____

Instructions: Write answers in the right hand column. Score: _____ / 104 marks

Section A – Multiple choice (25 marks)

1	Which combination is thought to have made up the ancient supercontinent Laurasia? **A** Australia, Antarctica, South America, Africa and India **B** South America, North America and Europe **C** Europe, North America and most of Asia **D** North America, Africa and most of Asia	C	1
2	In 1915, Alfred Wegener proposed the idea of a supercontinent splitting to form the continents. His ideas were largely ignored because: **A** there was no evidence to support them **B** little was known about the shapes of the continents at that time **C** the distribution of reptilian fossils could not be explained by shifting continents **D** it was thought that the Earth was solid rock.	D	1
3	Studies of the magnetic stripes of rocks on the ocean floor indicate that the: **A** youngest rock is next to the ridges and the oldest is next to the trenches **B** youngest rock is next to the trenches and the oldest is next to the ridges **C** rocks are all approximately the same age **D** rocks are older than the rocks of the continents.	A	1
4	Which of the following boundaries between tectonic plates is also known as a conservative boundary? **A** spreading **B** collision **C** transform **D** constructive	C	1

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5	When tectonic plates collide, a variety of things can happen. Which of the following is most likely when an ocean plate collides with another ocean plate? **A** Both plates crumple and fold. **B** The less dense plate is forced under the more dense plate. **C** A rift valley forms. **D** The faster-moving plate is forced under the slower-moving plate.	D	1
6	When tectonic plates collide, a variety of things can happen. Which of the following is most likely when two continental plates collide? **A** Both plates crumple and fold. **B** The less dense plate is forced under the more dense plate. **C** A rift valley forms. **D** The faster-moving plate is forced under the slower-moving plate.	A	1
7	A 'mountain root' is most likely to form when: **A** two ocean plates collide **B** two continental plates collide **C** an ocean plate collides with a continental plate **D** two plates scrape along each other.	B	1
8	Which of the following statements concerning earthquakes is *incorrect*? **A** The focus is the point where an earthquake begins. **B** Earthquakes occur on a fault line at the edges of tectonic plates. **C** The epicentre is the point on the Earth's surface above the focus. **D** Seismic waves spread from the epicentre to the focus and beyond.	D	1
9	For which of the following waves is the vibration of the particles in the same direction as the movement of the wave? **A** sound **B** water **C** seismic Love (L) waves **D** seismic secondary (S) waves	A	1

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10	Which of the following is a property of secondary (S) body waves? S waves: **A** travel through both solid and molten rock **B** are the fastest-moving body waves **C** are transverse waves **D** hit the surface with an up-and-down or push-pull motion.	C	1
11	Both primary and secondary body waves: **A** are transverse waves **B** are refracted when they pass through rocks of different density **C** travel faster when they move through less dense rock **D** travel around the surface of the Earth.	B	1
12	A scientist who studies earthquakes is a: **A** sizemologist **B** seismologist **C** seismometer **D** physicist.	B	1
13	An earthquake of magnitude 5 on the Richter scale has: **A** one-quarter more energy than a magnitude 4 quake **B** ten times the energy of a magnitude 4 quake **C** twenty times the energy of a magnitude 4 quake **D** thirty times the energy of a magnitude 4 quake.	D	1
14	How many active volcanoes are there in the world? **A** 100 **B** 600 **C** 1500 **D** more than 2000	C	1
15	How many active volcanoes are there in Australia? **A** zero **B** one **C** six **D** eleven	A	1

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16	Which part of an active volcano does the lava move through? **A** magma **B** central vent **C** volcanic ash **D** all of the above	B	1
17	If the rock of a fault scarp is very hard and weathering is slow, which landscape feature is most likely to form? **A** cliff **B** gentle rise **C** volcano **D** syncline	A	1
18	When continental plates collide, rock may be folded to build mountain ranges. Upward folds in the rock are called: **A** horsts **B** graben **C** synclines **D** anticlines.	D	1
19	How do the plates at a transcurrent fault move? **A** One plate slides over another. **B** Two plates collide and crumple. **C** Both plates move sideways. **D** One plate moves upwards, the other downwards.	C	1

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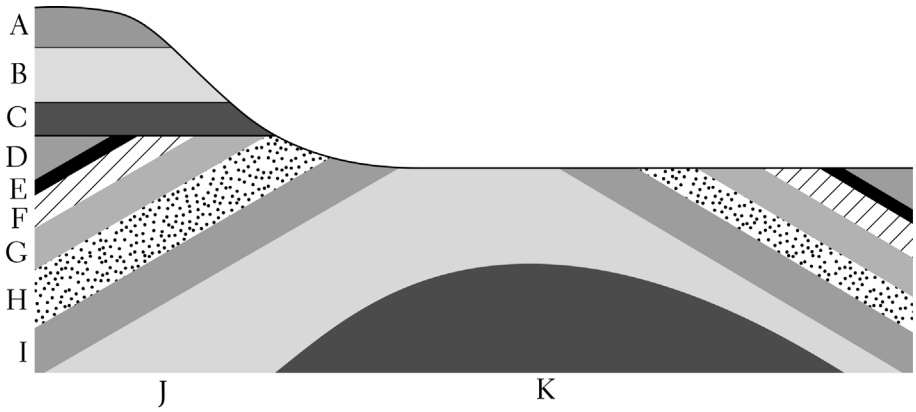
20	This question concerns the landform shown in the diagram below.  Which of the following statements is correct? **A** K represents a syncline. **B** C represents an unconformity. **C** Rock in E is older than rock in H. **D** J shows no signs of erosion.	B	1
21	Which type of volcano is the biggest? **A** shield cones **B** composite cones **C** cinder cones **D** batholiths	A	1
22	Which of the following statements concerning fossil fuels is *incorrect*? Fossil fuels: **A** form from decomposed plant and animal matter **B** take millions of years to form **C** are usually found near weaknesses in the Earth's crust **D** require low pressures and high temperatures to form.	D	1
23	A fossil is: **A** the preserved remains of an animal **B** evidence of past life found in a rock **C** a past life form buried in the sea bed **D** a skeleton embedded in rock.	B	1

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9: Earth's fragile crust

24	Radioactive dating of rocks works by: **A** measuring the amount of uranium in a rock that has turned into lead **B** finding the depth to which gamma rays penetrate within a rock **C** bombarding a rock with radiation and measuring the amount of rock that disintegrates **D** using X-rays to measure the space between atoms within rock.	A	1
25	Which of the following is a period of geological time? **A** Zirconic **B** Cretaceous **C** Rhombic **D** Dichotomous	B	1

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Section B – Written answers (79 marks)

1	**Describe** four pieces of evidence Alfred Wegener used to support his theory of the splitting of an ancient supercontinent to form the continents.	Any four of the following. • Fossils of ferns and reptiles found across the southern continents suggested a common ancestry, followed by movement as the land mass split and the pieces drifted apart. • Rock and mountain formations on one continent are similar to those on another continent; e.g. those in the USA are similar to those in Europe. • Evidence of glaciers is found in places now too warm for their formation. • Decomposed plant material (e.g. coal) is found in places now too cold for plants to grow there. • 'Mini-magnets' in ancient rocks only align properly when the continents are joined together.	4

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2	**Recall** three important pieces of information obtained when the ocean floor was mapped during World War II.	Any three of: • volcanic ranges run down the centre of most oceans • rocks of the ocean floor vary in age from new to 200 million years old, and are younger than continental rocks • deep ocean trenches exist • continental rock is less dense than ocean floor rock • magnetic 'stripes' on ocean rocks indicate that the oldest rock is next to the trenches, and the youngest is next to the ridges.	3
3	The theory of plate tectonics may be explained using an analogy with pieces of toast floating on very thick, hot soup. By **applying** this analogy: **a** **state** what the soup represents **b** **state** what the pieces of toast represent. **c** If the soup is stirred the toast moves. **Explain** what is widely thought to cause the 'soup' to be stirred.	**a** the asthenosphere **b** slabs of the lithosphere **c** convection currents within the mantle	3
4	**Explain:** **a** why spreading boundaries are also called constructive boundaries **b** why collision boundaries are also called destructive boundaries.	**a** New rock is being made on the ocean floor. **b** Rock is melted and returned for recycling.	2

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5	**Identify** which type of tectonic plate boundaries form, or formed, the: **a** earthquake activity around the San Andreas fault in California **b** East African rift valley **c** islands of Japan **d** Andes mountains and the Peru-Chile ocean trench which runs parallel to them.	**a** transform (or scraping) **b** spreading (or constructive) **c** collision (or destructive) **d** collision (or destructive)	4
6	**Apply** your knowledge of what happens when an oceanic plate collides with a continental plate to answer the following questions. **a** **Explain** why the oceanic plate is forced under the continental plate. **b** **Outline** a 'subduction zone'. **c** **Explain** why volcanoes often form on the continental plate after the collision.	**d** The oceanic plate is denser than the continental plate. **e** A subduction zone is the angled dive (at 20–60° to the surface) that the oceanic plate makes when it collides with the continental plate. **f** Steam forms when water enters the fault line between the plates. This steam is carried down into the subduction zone. The resulting gas-filled molten rock forces its way to the surface as a volcano.	5
7	**Clarify** the difference between the focus and the epicentre of an earthquake.	The focus is where an earthquake occurs in the Earth's lithosphere, and the focus is the point directly above this on the surface.	2
8	When an earthquake occurs, **identify** where the worst damage occurs.	the epicentre	1
9	**Explain** why no S waves are recorded on the Earth directly opposite the epicentre of the earthquake.	S waves cannot travel through molten rock. The liquid outer core blocks the path of the S waves through the Earth.	2

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10	The four types of seismic waves are: primary (P), secondary (S), Rayleigh (R) and Love (L). Use the letters P, S, R and L when answering the questions that follow. **Identify** the waves that: **a** travel through the body of the Earth **b** are recorded first by a seismometer **c** are recorded last by a seismometer **d** travel around the Earth's surface **e** are rolling waves, like surf at a beach.	**a** P and S **b** P **c** L **d** R and L **e** R	7
11	**Identify** whether each of the following waves are transverse or longitudinal waves. **a** sound waves **b** P waves **c** S waves **d** waves that push and pull **e** waves that have an up-down motion like water waves	**a** longitudinal **b** longitudinal **c** transverse **d** longitudinal **e** transverse	5
12	**Explain** how aftershocks can be just as dangerous as the original earthquake, even though they are usually much smaller.	Buildings may have become unstable due to the original earthquake and may be toppled by even a weak aftershock.	2

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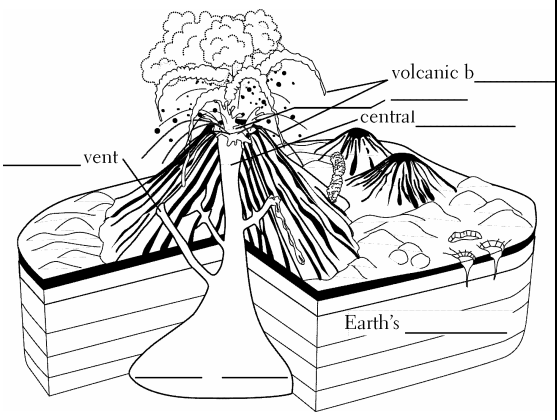
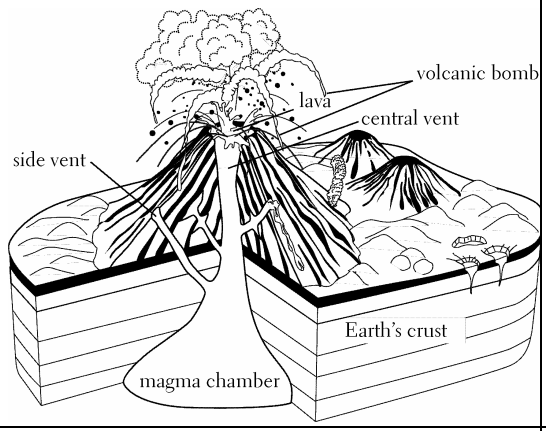
13	**Identify:** **a** an alternative scale to the Richter scale. **Explain:** **b** how the scale works **c** when it is used.	**a** the modified Mercalli scale **b** It consists of various descriptions of the effect of a quake matched to numbers from I to XII. **c** It is used when no instruments are available to measure a quake.	3
14	**Identify** the factors that affect how many lives are lost due to an earthquake.	population density, local geography, quality of buildings, the available medical and emergency services	2
15	Complete the labels on the volcano diagram below.  	 	6
16	**Explain** why many volcanic eruptions are not noticed.	They may be small, in remote areas or under water.	2
17	**Identify** the problems that volcanic dust may cause.	Dust can blanket the Earth, block the Sun, or may even bury/crush a town. Rain may mix with dust to form lahar and create a river of mud.	2

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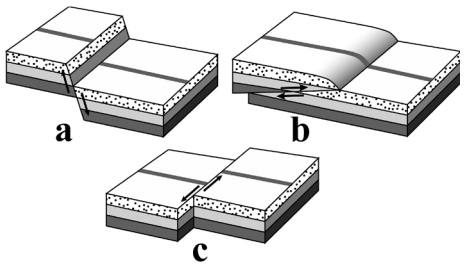
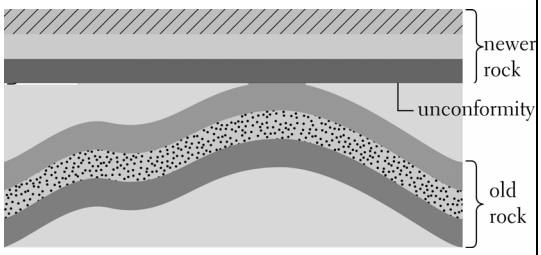
18	Three types of faults are shown in the diagram below. **Identify** each as either a transcurrent, normal or reverse fault.  	**a** normal **b** reverse **c** transcurrent	3
19	**Define** the term 'plastic behaviour'.	able to change shape without breaking	1
20	**Explain** how an unconformity is created. Include a diagram in your answer.	Erosion wears away the rock at the top of a fold. New sediments are laid on top of these eroded old rocks, forming an unconformity.  	4
21	**Explain** how a volcano could form away from the edge of a tectonic plate.	At hot spots or plumes the magma is under so much pressure that it forces its way through the plate.	2
22	**Explain** why igneous rock will not contain fossils of plant material.	Igneous rock forms when magma cools below the surface. The intense heat would destroy plant material.	2

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23	**Classify** the following steps of fossil formation in the correct order from start to finish. **A** Movement in the Earth's crust thrusts a layer of rock upwards. **B** Animal dies and falls to the sea floor. **C** Soft parts of body decay, leaving the shell. **D** Rock weathered. **E** Fossil exposed. **F** Shell covered by layers of sediment.	**B → C → F → A → D→ E**	2
24	True or false? **a** A dinosaur footprint is a fossil. **b** An ammonite is a fossil animal. **c** More primitive fossils are likely to be found in layers of igneous rock. **d** Carbon may be used to date fossils.	**a** True **b** True **c** False **d** True	4
25	Match one of the terms below to each of the following. *primitive life, no life, middle life, recent life* **a** Cenozoic **b** Azoic **c** Mesozoic **d** Archaeozoic	**a** recent life **b** no life **c** middle life **d** primitive life	4
26	**a** **Explain** the connection between an era and a period. **b** **Recall** a period when dinosaurs existed.	**a** An era contains several periods on the geological time scale. **b** Any one of: Cretaceous, Jurassic, Triassic.	2

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Unit 1.2 Prac 2 Chromatography catches a criminal

Description of procedure/requirements

Chemicals required: **Class sets**: Ethanol or methylated spirits

Equipment: A 'ransom' note written with an unknown brand of felt pen, several different black felt pens including the one that wrote the note, the name of each 'suspect' attached, gas jars, scissors, paper

Procedure: Refer to Prac 2 Unit 1.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Ethanol (C_2H_5OH) or methylated spirits	3	Highly flammable	**R11 –** Highly flammable.	**S7** – Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Other hazards and safety considerations

Methylated spirits and ethanol are highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or gas heaters. Methylated spirits or ethanol may be substituted with salt solution (0.1 M) with water-based pens.

Decant flammable liquids into class sets in a fume hood. In case of ethanol spill, eliminate all ignition sources; absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of

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Unit 1.2 Prac 2 Chromatography catches a criminal

chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult the **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide a container for flammable liquid (ethanol) collection (preferably within an operating fume hood). Recycle the ethanol or discard it via a chemical waste disposal company. Allow ethanol to evaporate from the paper in a fume hood. Discard the paper via the school waste system.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
To be a valid assessment, the MSDS expiry dates should be added and this risk assessment should be signed and dated.
This assessment is valid for 5 years from the earliest MSDS expiry date.

Disclaimer: This Risk Assessment Sheet is provided to offer guidance only. It must not be construed to waive or modify any legal obligation of the school to ensure the safety of students when conducting the experiment or activity. It is the responsibility of the school to have the content of this sheet checked against the Material Safety Data Sheets (MSDS) provided by the manufacturer of chemicals used in the school's laboratories. This sheet must not be used in the school's laboratories until it has been checked against the school's MSDS, signed and dated. To the maximum extent permitted by law, the Publisher disclaims all responsibility for actions taken or not taken in relation to this Risk Assessment Sheet.

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Unit 1.3 Prac 1 Comparing fingerprints

Description of procedure/requirements

Chemicals required: **Class sets**: Graphite powder
Equipment: Clean microscope slides, small soft bristle brushes or puff-brushes, sticky tape, paper
Procedure: Refer to Prac 1 Unit 1.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Graphite (C)	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Avoid inhalation of graphite. For small graphite spill, sweep up with a dustpan and brush or a clean non-sparking tool and place into a loosely covered plastic container for later disposal. Wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Small quantities of graphite can be discarded via the school waste system. No waste generated by technical activity. No prac waste generated.

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Unit 1.3 Prac 1 Comparing fingerprints

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
To be a valid assessment, the MSDS expiry dates should be added and this risk assessment should be signed and dated.
This assessment is valid for 5 years from the earliest MSDS expiry date.

Disclaimer: This Risk Assessment Sheet is provided to offer guidance only. It must not be construed to waive or modify any legal obligation of the school to ensure the safety of students when conducting the experiment or activity. It is the responsibility of the school to have the content of this sheet checked against the Material Safety Data Sheets (MSDS) provided by the manufacturer of chemicals used in the school's laboratories. This sheet must not be used in the school's laboratories until it has been checked against the school's MSDS, signed and dated. To the maximum extent permitted by law, the Publisher disclaims all responsibility for actions taken or not taken in relation to this Risk Assessment Sheet.

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Unit 2.1 Prac 1 Making elements

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M sulfuric acid (H_2SO_4)
Equipment: Large beaker, 9–12 V power supply or battery, alligator clips, 2 metal electrodes, safety glasses
Procedure: Refer to Prac 1 Unit 2.1

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Concentrated sulfuric acid (H_2SO_4) (98%)	8	Very corrosive	**R35 –** Causes severe burns.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S30 –** Never add water to this product. **S45 –** In case of accident or if you feel unwell seek medical advice immediately.	
Nickel (Ni) electrodes	None allocated	Harmful, irritant, listed as a carcinogen, category 3 in List of Designated Hazardous Substances – NOHSC.	**R40(3) –** Possible risk of irreversible effects. **R43 –** May cause sensitisation by skin contact.	**S22 –** Do not breathe dust. **S36 –** Wear suitable protective clothing.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 2.1 Prac 1 Making elements

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

0.1 M Sulfuric acid (H_2SO_4) (0.98%)	Two metal electrodes – Platinum (Pt)	Carbon electrodes (C)	**PRODUCT** Hydrogen gas (H_2) (very small volume)
PRODUCT Oxygen gas (O_2)			

Other hazards and safety considerations

Ensure that power supplies are in a safe working order. Minimise the quantity and volume of acid provided in the class sets. Prepare solutions according to laboratory precautions. Use a fume cupboard to prepare acid solutions. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. For acid spills on skin, irrigate with large quantities of water: washing should be continued for at least ten to fifteen minutes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Neutralise sulfuric acid solutions with a weak solution of sodium carbonate (soda ash) and pour down the sink with plenty of water. Provide a waste container for used sulfuric acid solutions. Washed electrodes can be stored for future use.

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Unit 2.1 Prac 1 Making elements

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Notes

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Unit 2.1 Prac 2 Making a compound

Description of procedure/requirements

Chemicals required: Class sets: Limewater, marble chips, 2 M hydrochloric acid
Equipment: Two test tubes, test-tube rack, drinking straw, 1-hole rubber stopper with glass tubing, safety glasses
Procedure: Refer to Prac 2 Unit 2.1

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	
Concentrated hydrochloric acid (HCl) 32%	8 	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	

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Unit 2.1 Prac 2 Making a compound

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

2 M Hydrochloric acid (HCl) (7%)	Marble chips ($CaCO_3$)	**PRODUCT** Calcium chloride ($CaCl_2$)(aqueous)	**PRODUCT** Carbon dioxide (CO_2) gas

Other hazards and safety considerations

Use a fume cupboard to make up hydrochloric acid solution. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result.** Risks minimised by wearing safety glasses, lab coat and gloves. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Strain unconsumed marble chips to wash with distilled water for future use. Filtrate may be washed down the drain using plenty of water.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

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Unit 2.1 Prac 2 Making a compound

Notes
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Unit 2.2 Prac 1 Investigating a physical property

Description of procedure/requirements

Chemicals required: Class sets: Cubes or cylinders of aluminium, brass, lead, wood, ice; a collection of small items such as pebbles, candles, chunks of concrete or cement, copper wire

Equipment: Electronic balance, ruler, beakers, measuring cylinders

Procedure: Refer to Prac 1 Unit 2.2

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61(1) –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 2.2 Prac 1 Investigating a physical property

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Aluminium (Al)	Brass	Copper wire (Cu)	

Other hazards and safety considerations

Lead may be absorbed through the skin on prolonged exposure, which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac. In case of prolonged exposure or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

There is no waste generated in this practical. Samples can be stored for future use.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.2 Prac 2 Comparing elements

Description of procedure/requirements

Chemicals required: Class sets: Sulfur, aluminium, carbon, silicon, tin, zinc, lead, magnesium, calcium, iron, steel wool
Equipment: Three or four test tubes and rack, power pack about 2 V or battery, wires with alligator clips, light globe, safety glasses
Procedure: Refer to Prac 2 Unit 2.2

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R33 –** Danger of cumulative effects. **R62 –** Possible risk of impaired fertility. **R61(1) –** May cause harm to the unborn child. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R20/22 –** Harmful by inhalation and if swallowed.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Calcium granules (Ca)	4.3	Irritant, highly flammable	**R15 –** Contact with water liberates extremely flammable gases. **R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S8 –** Keep container dry. **S24/25 –** Avoid contact with skin and eyes. **S43(1) –** In case of fire, use alcohol-resistant foam/dry powder/CO_2. Never use water.	
Carbon (C)	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves	

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Unit 2.2 Prac 2 Comparing elements

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
				and eye/face protection.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	
Sulfur powder (S)	4.1	Highly flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition; no smoking.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Aluminium (Al)	Silicon (Si)	Zinc (Zn)	Iron (Fe)
Steel wool	Tin pieces (Sn)		

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Lead may be absorbed through the skin on prolonged exposure which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac. Consider setting up this prac at stations around the room to which students rotate. Power supplies should be checked for damaged leads and malfunction prior to use. Instruct students on the safe use of power supplies. Construct the circuit with the power off. Do not touch the circuit while the power is on. Use only the recommended voltage. Do not allow power supplies/electrical equipment to come into contact with liquids. Use a sulfur roll to test conductivity and sulfur powder for the solubility test. A carbon electrode makes an ideal carbon sample as it minimises the irritant nature of carbon powder. Provide only a small quantity of calcium and use forceps to handle it (calcium granules are a safer option). If alligator clips will not 'clip' to the samples, provide electrodes (nails work well) that can be touched onto the sample to

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Unit 2.2 Prac 2 Comparing elements

complete the circuit. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Collect prac waste from classroom. Place sulfur, lead and tin waste into separate hazardous waste containers for later disposal via a chemical waste company.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 2.3 Prac 1 Firework colours

Description of procedure/requirements

Chemicals required: Class sets: Barium chloride, copper chloride, potassium chloride, sodium chloride and strontium chloride (suggested concentration 0.1 M) in which to soak icy-pole sticks

Equipment: Bunsen burner, bench mat and matches, tongs, safety glasses, wooden icy-pole sticks soaked overnight in distilled water and solutions of barium chloride, copper chloride, potassium chloride, sodium chloride and strontium chloride; spectroscope (optional)

Procedure: Refer to Prac 1 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Barium chloride hydrated ($BaCl_2.2H_2O$) solid	6.1	Toxic	**R20 –** Harmful by inhalation. **R25 –** Toxic if swallowed.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S37/39 –** Wear suitable gloves and eye/face protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Copper(II) chloride ($CuCl_2$) solid	8	Harmful, irritant, dangerous for the environment	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin. **R52/53 –** Harmful to aquatic organisms; may cause long-term adverse effects in the aquatic environment.	**S22 –** Do not breathe dust. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheet.	
Strontium chloride hydrated ($SrCl_3.6H_2O$) solid	None allocated	Harmful	**R22 –** Harmful if swallowed.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S37/39 –** Wear suitable gloves and eye/face protection.	

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Unit 2.3 Prac 1 Firework colours

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

0.1 M Barium chloride ($BaCl_2$) solution (2.4%)	Potassium chloride (KCl) solid	0.1 M Potassium chloride (KCl) (0.75%)	Sodium chloride (NaCl) solid
0.1 M Sodium chloride (NaCl) (0.58%)	0.1 M Strontium chloride ($SrCl_2$) (2.7%)	0.1 M Copper(II) chloride ($CuCl_2$) (1.7%)	

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Solid spills should be swept up with a dust pan, avoiding inhalation and contact with skin and eyes. Dispose of copper chloride and barium chloride spill into separate hazardous waste containers for disposal via a chemical waste company. Other solid spills can be disposed of in the normal rubbish. In tall, labelled jars, soak the icy-pole sticks in the solutions overnight, then decant off most of the solution. After the prac, you can rinse and discard the remaining icy-pole sticks, return the solution to the jar, replace the lid and store the chemical solutions for the next time. Provide a beaker of water in which to place used icy-pole sticks (this extinguishes them and rinses off chemical residues). Be aware of the sensitivity of your fire alarm system. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

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Unit 2.3 Prac 1 Firework colours

Disposal of wastes

Solutions may be stored away for future use. Prac waste for disposal: separate used icy-pole sticks from the water (sieve). Discard icy-pole sticks via the school waste system. Discard the dousing/rinsing water to the sewer with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.3 Prac 3 Metal crystals

Description of procedure/requirements

Chemicals required: Class sets: 1 cm × 4 cm strip clean zinc sheet, one 0.3 g sample of silver nitrate, 0.5 g of agar powder, 40 mL distilled water
Equipment: Sterilised Petri dish, 250 mL beaker, Bunsen burner, tripod, gauze mat, matches, bench mat, stirring rod, stereo microscope (optional), safety glasses, gloves
Procedure: Refer to Prac 3 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous to the environment	**R34 –** Causes burns. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Lead(II) nitrate ($Pb(NO_3)_2$) solid	5.1/6.1	Toxic, dangerous to the environment	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 2.3 Prac 3 Metal crystals

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			impaired fertility.		
Copper(II) sulfate ($CuSO_4$) solid	9	Harmful, irritant dangerous for the environment	**R22** – Harmful if swallowed. **R36/38** – Irritating to eyes and skin. **R50** – Very toxic to aquatic organisms. **R53** – May cause long-term adverse effects in the aquatic environment.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S36/37/39** – Wear suitable protective clothing, gloves and eye/face protection. **S60** – This material and its container must be disposed of as hazardous waste. **S61** – Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Tin(II) chloride ($SnCl_2.2H_2O$) solid	None allocated	Harmful	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin. **R43 –** May cause sensitisation by skin contact.	**S24** – Avoid contact with skin. **S26** – In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S37** – Wear suitable gloves. **S46** – If swallowed, seek medical advice immediately and show the label.	
PRODUCT Lead (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61(1) –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 2.3 Prac 3 Metal crystals

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Agar	Zinc strips (Zn)	**PRODUCT** Silver (Ag)	**PRODUCT** Tin (Sn)
PRODUCT Copper (Cu)	**PRODUCT** Zinc nitrate solution in agar (very low concentration)	**PRODUCT** Zinc sulfate solution in agar (very low concentration)	**PRODUCT** Zinc chloride solution in agar (very low concentration)

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Significant risks to students are minimised if the chemicals in this practical are pre-weighed. Exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. For silver nitrate and lead nitrate spills, sweep up and place into separate hazardous waste containers for disposal via a chemical waste company. Other spills may be swept up and disposed of in the normal rubbish. Wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of chemical in eye, an eye wash or safety shower should be used. In case of inhalation, ingestion or felling unwell, consult your local **Poisons Information Centre 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

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Unit 2.3 Prac 3 Metal crystals

Disposal of wastes

Collect prac waste from classroom. Place the metal strips containing silver residues into a hazardous waste container for disposal via a chemical waste company. Petri dishes and other waste may be disposed of in the normal rubbish.

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Unit 2.3 Prac 4 More crystals

Description of procedure/requirements

Chemicals required: **Class sets**: 10 to 15 cm length of copper wire and/or strip of copper foil, silver nitrate solution (suggested concentration 0.1 M)
Equipment: 100 mL conical flask, cork or rubber stopper, silver nitrate solution
Procedure: Refer to Prac 4 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous to the environment	**R34 –** Causes burns. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous Goods class)

0.1 M Silver nitrate ($AgNO_3$) (1.7%)	Sodium thiosulfate 4% (silver nitrate spill clean-up)	Copper (Cu)	**PRODUCT** Silver (Ag)

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Unit 2.3 Prac 4 More crystals

PRODUCT Copper nitrate solution (weak concentration)			

Other hazards and safety considerations

Prepare solutions according to recipes and standard laboratory safety procedures. Suggested safe concentration for silver nitrate solution is 0.1 M. Consider providing the silver nitrate solution in stoppered flasks (this will minimise student contact with the silver nitrate and the risk of spills.) Silver nitrate stains: spills can be decolourised with sodium thiosulfate 4%.

Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Collect prac waste for disposal. After separating the silver by filtration it can be stored for later disposal via a chemical waste company. The unreacted copper wire may be placed in the normal waste whereas the filtrate can be placed in the copper waste bottle.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.3 Prac 6 Using metals to make non-metals

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

2 M Hydrochloric acid (HCl) (7%)	Copper (Cu) solid	Iron (Fe) solid	**PRODUCT** Magnesium chloride ($MgCl_2$)(aqueous)
PRODUCT Ferrous chloride ($FeCl_2$) weak solution	**PRODUCT** Hydrogen (H_2) gas (very small volume)		

Other hazards and safety considerations

Use a fume cupboard to make up hydrochloric acid solution. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result.** Risks minimised by wearing safety glasses, lab coat and gloves. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide a container for prac waste. Separate liquid waste from metal waste. Iron nails and copper wire may be disposed of via normal waste. Store any unconsumed magnesium metal for disposal by a chemical waste company (magnesium is not suitable to be discarded via the school waste system). Liquid waste may be poured down the sink with plenty of water.

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Unit 2.3 Prac 4 More crystals

Notes

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Unit 2.3 Prac 6 Using metals to make non-metals

Description of procedure/requirements

Chemicals required: **Class sets**: Magnesium, iron and copper, 1–2 M hydrochloric acid in a dropping bottle
Equipment: Test tubes and rack, matches, safety glasses
Procedure: Refer to Prac 6 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand never use water.	

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Unit 2.3 Prac 6 Using metals to make non-metals

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Unit 2.4 Prac 1 The alkaline earths

Description of procedure/requirements

Chemicals required: Class sets: 5 cm strip of magnesium, small sample of calcium, phenolphthalein indicator

Equipment: Crucible and a lid, clay triangle, 2 test tubes and rack, Bunsen burner, tripod, bench mat, matches, tongs, safety glasses, distilled water, steel wool or emery paper

Procedure: Refer to Prac 1 Unit 2.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium granules (Ca)	4.3	Irritant, highly flammable	**R15 –** Contact with water liberates extremely flammable gases. **R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S8 –** Keep container dry. **S24/25 –** Avoid contact with skin and eyes. **S43(1) –** In case of fire, use alcohol-resistant foam/dry powder/CO_2. Never use water.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	
Phenolphthalein indicator	3	Highly flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition – No smoking. **S23 –** Do not breathe vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S33 –** Take precautionary measures against static discharges.	

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Unit 2.4 Prac 1 The alkaline earths

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Phenolphthalein powder	**PRODUCT** Mixture of phenolphthalein indicator and magnesium hydroxide solutions (weak concentration)	**PRODUCT** Mixture of phenolphthalein indicator and calcium hydroxide solutions (weak concentration)	

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Risks minimised by wearing safety glasses, lab coat and gloves. For small calcium spill, collect spill using a dry utensil and place into a dry hazardous waste container for later disposal. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Use a sieve to separate the metals from the liquid. Magnesium waste should be discarded via a chemical waste disposal company. The weak alkaline liquid can be poured down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 2.4 Prac 2 Group IV

Description of procedure/requirements

Chemicals required: Class sets: Charcoal, graphite, silicon, lead
Equipment: Power pack or battery, leads with alligator clips, light
Procedure: Refer to Prac 2 Unit 2.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R62 –** Possible risk of impaired fertility. **R61(1) –** May cause harm to the unborn child.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Graphite (C)	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/3739 –** Wear suitable protective clothing, gloves and eye/face protection.	

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Unit 2.4 Prac 2 Group IV

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Charcoal powder (C)	Silicon (Si)		

Other hazards and safety considerations

Ensure all electrical appliances are in safe working order. Prepare chemicals according to laboratory precautions. Lead may be absorbed through the skin on prolonged exposure, which may result in symptoms of poisoning. Risks minimised by wearing safety glasses, lab coat and gloves. In case of prolonged exposure or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

There is no waste generated in this practical. Samples can be stored for future use.

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Unit 2.4 Prac 2 Group IV

Notes
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Unit 3.1 Prac 2 Signs of chemical change

Description of procedure/requirements

Chemicals required: Class sets: Copper carbonate, magnesium, solid zinc, dilute nitric acid, dilute sodium hydroxide, dilute sodium sulfate, dilute copper sulfate, dilute barium nitrate (0.1 M is a suitable concentration for all the solutions except for nitric acid where a concentration of 1 M will produce the desired amount of gas)

Equipment: Splint, matches, Bunsen burner, test-tube holder/tongs, test-tube rack, 5 test tubes (1 with stopper), thermometer, dropper or Pasteur pipette, lab coat, safety glasses

Procedure: Refer to Prac 2 Unit 3.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper carbonate ($CuCO_3$) solid	None allocated	Harmful, irritant	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
Concentrated nitric acid (HNO_3) 70%	8	Very corrosive, oxidising	**R35 –** Causes severe burns.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
1 M nitric acid (HNO_3) (6.3%)	8	Corrosive	**R35 –** Causes severe burns.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

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Unit 3.1 Prac 2 Signs of chemical change

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium hydroxide (NaOH)	8	Very corrosive	**R35 –** Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Barium nitrate ($Ba(NO_3)_2$) solid	5.1/6.1	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S17 –** Keep away from combustible material. **S28 –** After contact with skin, wash immediately with plenty of soap-suds.	
0.1 M barium nitrate ($Ba(NO_3)_2$)	None allocated	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S28 –** After contact with skin, wash immediately with plenty of soap-suds.	
Copper(II) sulfate ($CuSO_45H_2O$) solid	9	Harmful, irritant, dangerous to the environment	**R22 –** Harmful if swallowed. **R36/38 –** Irritating to eyes and skin. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
PRODUCT Copper(II) oxide (CuO) solid	None allocated	Harmful	**R22 –** Harmful if swallowed.	**S22 –** Do not breathe dust.	

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Unit 3.1 Prac 2 Signs of chemical change

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	Sodium sulfate (Na_2SO_4) solid	0.1 M Sodium sulfate (Na_2SO_4) (1.4%)	Zinc (Zn) solid
0.1 M Copper(II) sulfate (2.4%)	**PRODUCT** Carbon dioxide gas (CO_2)	**PRODUCT** Magnesium nitrate ($Mg(NO_3)_2$) (weak solution)	**PRODUCT** Hydrogen gas (H_2) (very small volume)
PRODUCT Sodium nitrate ($NaNO_3$) (weak solution)	**PRODUCT** Barium sulfate ($BaSO_4$) solid	**PRODUCT** Zinc sulfate ($ZnSO_4$) (very diluted solution)	**PRODUCT** Copper (Cu)

Other hazards and safety considerations

Refer to Heating RA. Prepare solutions according to laboratory precautions. Use fume cupboard to make up acid solutions. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid**, as uncontrolled boiling may result. In case of nitric acid spill, cover spill with dry earth and cover with plastic to prevent spreading. Collect spill, then dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Concentrated spills should be neutralised with solid citric acid, then diluted with water and small concentrated spills can be neutralised with a weak concentration of HCl solution, then wiped up using paper towel, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 3.1 Prac 2 Signs of chemical change

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide copper and barium waste containers for waste from parts a, e and d reactions. Dilute the nitric acid waste with water and neutralise with sodium carbonate or calcium hydroxide. Neutralised liquid may be poured down the sink. Dilute the sodium hydroxide waste with water, then neutralise with a weak concentration of hydrochloric acid solution and pour down the sink using plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.2 Prac 1 Burning magnesium ribbon (teacher demonstration)

Description of procedure/requirements

Chemicals required: 15–20 cm of magnesium ribbon, steel wool for cleaning magnesium
Equipment: Tongs, lighter, safety glasses, gloves, lab coat
Procedure: Refer to Prac 1 Unit 3.2 (teacher demonstration)

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium ribbon (Mg)	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S43 –** In case of fire, use sand; never use water. **S7/8 –** Keep container tightly closed and dry.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous Goods class)

Steel wool	**PRODUCT** Magnesium oxide (MgO) solid			

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Unit 3.2 Prac 1 Burning magnesium ribbon (teacher demonstration)

Other hazards and safety considerations

Refer to Heating RA. Provide a cleaned magnesium ribbon. Risks minimised by wearing safety glasses, lab coat and gloves. Avoid inhalation when disposing of magnesium oxide. In case of inhalation, consult your local **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Magnesium oxide may be disposed of in the normal waste.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.2 Prac 2 Stopping combustion

Description of procedure/requirements

Chemicals required: **Class sets**: Sodium bicarbonate, 0.1 M hydrochloric acid, limewater
Equipment: 250 mL conical flask, rubber stopper with a flexible plastic tube passing through it, large test tube with rubber stopper, wooden splint, matches
Procedure: Refer to Prac 2 Unit 3.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Concentrated hydrochloric acid (HCl) 32%	**8**	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	

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Unit 3.2 Prac 2 Stopping combustion

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.37%)	**PRODUCT** Sodium chloride (NaCl) solution	**PRODUCT** Water (H_2O)	**PRODUCT** Carbon dioxide (CO_2) gas
Sodium bicarbonate solid			

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Use a fume cupboard to make up hydrochloric acid solution. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result.** Risks minimised by wearing safety glasses, lab coat and gloves. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

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Unit 3.2 Prac 2 Stopping combustion

Disposal of wastes

Test samples may be disposed of down the sink with plenty of water. Any remaining dilute hydrochloric acid can be stored for future use.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

Description of procedure/requirements

Chemicals required: Class sets: Copper carbonate, limewater
Equipment: Bunsen burner, matches, retort stand and clamp, 2 large test tubes, stopper for one of the test tubes and a delivery tube, lab coat, safety glasses
Procedure: Refer to Prac 3 Unit 3.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper carbonate ($CuCO_3$) solid	None allocated	Harmful, irritant	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	
PRODUCT Copper(II) oxide (CuO) solid	None allocated	Harmful	**R22 –** Harmful if swallowed	**S22 –** Do not breathe dust.	

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) with NO allocated Dangerous goods class)

PRODUCT Carbon dioxide gas (CO_2)			

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. For small solid spills, sweep up immediately, avoiding inhalation. Place spill into a hazardous waste container for disposal via a chemical waste company. In case of limewater spill, wipe up with paper towel, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide copper waste container. Collect the copper waste to be disposed of via a chemical waste company.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

Notes
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Unit 3.3 Prac 1 Precipitation reactions

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M solutions of silver nitrate ($AgNO_3$), sodium chloride (NaCl), sodium hydroxide (NaOH), barium nitrate ($Ba(NO_3)_2$) and copper sulfate ($CuSO_4.5H_2O$)

Equipment: Test tubes, test tube rack, Pasteur pipettes, lab coat, safety glasses, gloves

Procedure: Refer to Prac 1 Unit 3.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous to the environment	**R34 –** Causes burns. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S26 -** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45** - In case of accident or if you feel unwell, seek medical advice immediately. **S60** – This material and its container must be disposed of as hazardous waste. **S61**- Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Sodium hydroxide (NaOH)	8	Very corrosive	**R35** – Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Barium nitrate ($Ba(NO_3)_2$) solid	5.1/6.1	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S17 –** Keep away from combustible material. **S28 –** After contact with skin, wash immediately with plenty of soap suds.	

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Unit 3.3 Prac 1 Precipitation reactions

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
0.1 M barium nitrate ($Ba(NO_3)_2$)	None allocated	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S28 –** After contact with skin, wash immediately with plenty of soap suds.	
Copper(II) sulfate ($CuSO_4.5H_2O$) solid	9	Harmful, irritant, dangerous to the environment	**R22 –** Harmful if swallowed. **R36/38 –** Irritating to eyes and skin. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Silver nitrate ($AgNO_3$) (1.7%)	Sodium chloride (NaCl) Solid	0.1 M Sodium chloride (NaCl) (0.6%)	0.1 M Sodium hydroxide (NaOH) (0.4%)
0.1 M Copper(II) sulfate (2.4%)	**PRODUCT** Silver chloride (AgCl) solid	**PRODUCT** Silver oxide (Ag_2O) solid	**PRODUCT** Sodium nitrate ($NaNO_3$) solution
PRODUCT Barium sulfate ($BaSO_4$) solid	**PRODUCT** Copper nitrate ($Cu(NO_3)_2$) solution		

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Unit 3.3 Prac 1 Precipitation reactions

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Dispense the solutions in dropping bottles to minimise spillage. Exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. For silver and barium nitrate spills, sweep up, avoiding inhalation and place into separate hazardous waste containers for disposal via a chemical waste company. Other spills may be swept up and disposed of in the normal rubbish. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide containers for silver and barium wastes solutions. Place heavy metal waste, barium and silver into separate hazardous waste containers for disposal via a chemical waste company. The dilute solutions of chemicals can be stored for future use.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.3 Prac 2 Precipitation of unknowns

Description of procedure/requirements

Chemicals required: Class sets: Unknown 0.1 M solutions labelled A, B, C, D, E. These are (not in order) sodium iodide, sodium chloride, sodium sulfate, sodium carbonate and sodium nitrate; 0.1 M solutions of silver, lead, calcium and barium nitrate

Equipment: 20 semi-micro test tubes, Pasteur pipettes, lab coat, safety glasses, gloves

Procedure: Refer to Prac 2 Unit 3.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous for the environment	**R34 –** Causes burns. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45** – In case of accident or if you feel unwell, seek medical advice immediately. **S60** – This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Lead(II) nitrate ($Pb(NO_3)_2$) solid	5.1/6.1	Toxic, dangerous for the environment	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R62 – Possible risk of impaired fertility.		
0.1 M lead nitrate ($Pb(NO_3)_2$) (3.3%)	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R61 –** May cause harm to the unborn child.	**S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Calcium nitrate ($Ca(NO_3)_2$) solid	5.1	Irritant, oxidising	**R8 –** Contact with combustible material may cause fire. **R36/38 –** Irritating to eyes and skin.	**S17 –** Keep away from combustible material. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
Barium nitrate ($Ba(NO_3)_2$) solid	5.1 Subrisk 6.1	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S17 –** Keep away from combustible material. **S28 –** After contact with skin, wash immediately with plenty of soap-suds.	
0.1 M barium nitrate ($Ba(NO_3)_2$)	None Allocated	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S28** After contact with skin, wash immediately with plenty of soap-suds.	
Sodium nitrate ($NaNO_3$) solid	5.1	Harmful, irritant, oxidising	**R8 –** Contact with combustible material may cause fire. **R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S17 –** Keep away from combustible material. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium carbonate anhydrous (Na_2CO_3) solid	None allocated	Irritant	**R36 –** Irritating to eyes.	**S22 –** Do not breathe dust. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.	
Sodium iodide (NaI) solid	None allocated	Irritant	**R36/38 –** Irritating to eyes and skin. (If diluted with a large amount of water, this chemical released directly or indirectly into the environment is not expected to have a significant impact.)	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.	
PRODUCT Lead chloride ($PbCl_2$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
PRODUCT Lead sulfate ($PbSO_4$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R53 – May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S61 –** Avoid release to the environment. Refer to special instructions/Materials Safety Data Sheets.	
PRODUCT Lead iodide (PbI_2) solid	6.1	Toxic, dangerous for the environment	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
PRODUCT Lead carbonate ($PbCO_3$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R61 – May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.		
PRODUCT Silver sulfate (Ag_2SO_4) solid	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
PRODUCT Barium carbonate ($BaCO_3$) solid	6.1	Harmful	**R22 –** Harmful if swallowed.	**S24/25 –** Avoid contact with skin and eyes.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Sodium chloride (NaCl) solid	0.1 M Sodium chloride (NaCl) (0.6%)	Sodium sulfate (Na_2SO_4) solid	0.1 M Sodium sulfate (Na_2SO_4) (1.4%)
0.1 M Sodium iodide (NaI) (1.5%)	0.1 M Sodium nitrate ($NaNO_3$) (0.85%)	0.1 M Sodium carbonate (Na_2CO_3) (1%)	0.1 M Silver nitrate solution ($AgNO_3$) (1.7%)
0.1 M Calcium nitrate ($Ca(NO_3)_2$)	**PRODUCT** Silver chloride (AgCl) solid	**PRODUCT** Silver iodide (AgI) solid	**PRODUCT** Silver carbonate (Ag_2CO_3) solid

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Unit 3.3 Prac 2 Precipitation of unknowns

PRODUCT	**PRODUCT**	**PRODUCT**	
Barium sulfate ($BaSO_4$) solid	Calcium carbonate ($CaCO_3$) solid	Calcium sulfate ($CaSO_4$) solid	

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. For silver, lead and barium nitrate spills, sweep up, avoiding inhalation and place into separate hazardous waste containers for disposal via a chemical waste company. Other spills may be swept up and disposed of in the normal rubbish. Wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of chemical in eye, an eye wash or safety shower should be used. In case of inhalation, ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide waste containers for barium, lead, silver waste. Dispose of barium, lead, silver waste via a chemical waste company. Other waste may be poured down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 1 Common indicators

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M solutions of sodium hydroxide and hydrochloric acid, distilled water, liquid red and blue litmus, universal indicator, methyl orange, methyl red, bromothymol blue, phenolphthalein

Equipment: Three test tubes, test-tube rack, 3 × 100 mL beakers, lab coat, safety glasses

Procedure: Refer to Prac 1 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium hydroxide (NaOH) solid	8	Very corrosive	**R35** – Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	
Methyl orange solid	None allocated	Toxic	**R25 –** Toxic if swallowed.	**S37 –** Wear suitable gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

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Unit 3.4 Prac 1 Common indicators

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Methyl red solid	None allocated	Harmful	**R22 –** Harmful if swallowed.	**S46 –** If swallowed, seek medical advice immediately and show container or label.	
Ethanol (C_2H_5OH)	3	Highly flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S7 –** Keep container tightly closed.	
Phenolphthalein indicator	3	Highly flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking. **S23 –** Do not breathe vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S33 –** Take precautionary measures against static discharges.	
Methyl red solution	3	Flammable	**R10 –** Flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S23(2) –** Do not breathe vapour. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S33 –** Take precautionary measures against static discharges. **S36/37 –** Wear suitable protective clothing and gloves.	

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Unit 3.4 Prac 1 Common indicators

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Bromothymol blue solution (0.04%)	3	None allocated	**R10 –** Flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S23/25 –** Avoid contact with skin and eyes.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)	Red and blue litmus indicator	Methyl orange indicator (0.04%)
Phenolphthalein powder	Bromothymol blue solid		

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use a fume cupboard to make up sodium hydroxide solution and hydrochloric acid solution. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Dilute sodium hydroxide spill with water, then neutralise with a weak concentration of hydrochloric acid solution and pour down the sink using plenty of water. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. In case of ethanol spill, eliminate all ignition sources; absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

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Unit 3.4 Prac 1 Common indicators

Disposal of wastes

Provide waste containers. Collect all liquid waste. Wear PPE. Check the pH and adjust as necessary to between a pH of 6 and 9. Discard the liquid waste down the sink with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 2 Universal indicator

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M hydrochloric acid, 1.0 M sodium hydroxide, distilled water, universal indicator
Equipment: Pasteur pipette, 10 mL measuring cylinder, waterproof felt-tip pen, 14 large test tubes, test-tube rack, lab coat, safety glasses
Procedure: Refer to Prac 2 Unit 3.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium hydroxide (NaOH) solid	8	Very corrosive	**R35** – Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
1 M sodium hydroxide (NaOH) (4%)	8	Corrosive	**R34** – Causes burns.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37 –** Wear suitable protective clothing and gloves. **S45** – In case of accident or if you feel unwell, seek medical advice immediately.	
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	

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Unit 3.4 Prac 2 Universal indicator

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S7 –** Keep container tightly closed.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.35%)			

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use a fume cupboard to make up sodium hydroxide solution and hydrochloric acid solution. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Dilute sodium hydroxide spill with water, then neutralise with a weak concentration of hydrochloric acid solution and pour down the sink using plenty of water. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. In case of ethanol spill, eliminate all ignition sources; absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 3.4 Prac 2 Universal indicator

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide waste containers. Collect all liquid waste. Wear PPE. Check the pH and adjust as necessary to between a pH of 6 and 9. Discard the liquid waste to down the sink with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.4 Prac 3 Natural indicators

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M solutions of sodium hydroxide and hydrochloric acid, distilled water

Equipment: Pink or red flower petals, beetroot juice, tea bag, 3 × 100 mL beakers, filter paper cut into strips, Bunsen burner, heatproof mat, tripod, gauze, matches, 3 test tubes, test-tube rack, Pasteur pipettes, lab coat, safety glasses

Procedure: Refer to Prac 3 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium hydroxide (NaOH) solid	8	Very corrosive	**R35** – Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)	Beetroot juice	Flower petals

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Unit 3.4 Prac 3 Natural indicators

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use a fume cupboard to make up sodium hydroxide solution and hydrochloric acid solution. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Dilute sodium hydroxide spill with water, then neutralise with a weak concentration of hydrochloric acid solution and pour down the sink using plenty of water. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Mix hydrochloric acid solution and sodium hydroxide solution together to neutralise and pour down the sink with plenty of water. Flower petals, beetroot juice and filter paper may be disposed of in the normal rubbish.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 4 Testing household solutions

Description of procedure/requirements

Chemicals required: Class sets: Universal indicator, non-hazardous household solutions including orange juice, soft drink, fresh and sour milk, vinegar (solids may be used if dissolved in water first), distilled water, blue and red litmus paper

Equipment: Two test tubes, test-tube rack, Pasteur pipettes, 2 watch glasses, safety glasses, lab coat

Procedure: Refer to Prac 4 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Non-hazardous household solutions including orange juice, soft drink, fresh and sour milk, vinegar. Solids may be used if dissolved in water first.			

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Unit 3.4 Prac 4 Testing household solutions

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Waste may be poured down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 5 Acids and hydroxides

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M sodium hydroxide, 0.1 M hydrochloric acid
Equipment: 250 mL beaker, 50 mL measuring cylinder, digital pH meter or universal indicator, Pasteur pipette, lab coat, safety glasses
Procedure: Refer to Prac 5 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium hydroxide (NaOH) solid	8	Very corrosive	**R35** – Causes severe burns.	**S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing and gloves. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62** – If swallowed, do not induce vomiting; seek medical advice immediately and show this container or label.	

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Unit 3.4 Prac 5 Acids and hydroxides

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S7 –** Keep container tightly closed.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)		

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use fume cupboard to make up sodium hydroxide solution and hydrochloric acid solution. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Dilute sodium hydroxide spill with water, then neutralise with a weak concentration of hydrochloric acid solution and pour down the sink using plenty of water. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. Small spills should be wiped up immediately using paper towel and water, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

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Unit 3.4 Prac 5 Acids and hydroxides

Disposal of wastes

Provide waste containers. Collect all liquid waste. Wear PPE. Check the pH and adjust as necessary to between a pH of 6 and 9. Discard the liquid waste down the sink with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.4 Prac 6 Acids and carbonates

Description of procedure/requirements

Chemicals required: Class sets: Limewater, solid sodium hydrogen carbonate, lithium carbonate, sodium carbonate, ammonium carbonate, 0.1 M solutions of nitric acid and hydrochloric acid

Equipment: Test tubes, test-tube rack, rubber stopper, 100 mL beaker, matches, spatula, safety glasses, lab coat

Procedure: Refer to Prac 6 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S37/39 –** Wear suitable gloves and eye/face protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment.	
Concentrated hydrochloric acid (HCl) 32%	8 	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62 –** If swallowed, do not induce vomiting; seek medical advice immediately and show container or label.	

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Unit 3.4 Prac 6 Acids and carbonates

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Concentrated nitric acid (HNO_3) 70%	8	Very corrosive, oxidising	**R35 –** Causes severe burns.	**S23** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	
Sodium carbonate anhydrous (Na_2CO_3) solid	None allocated	Irritant	**R36 –** Irritating to eyes.	**S22 –** Do not breathe dust. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.35%)	0.1 M Nitric acid (HNO_3) (0.63%)	Sodium hydrogen carbonate ($NaHCO_3$) solid	Ammonium carbonate ($(NH_4)_2CO_3$) solid
Lithium carbonate (Li_2CO_3) solid	**PRODUCT** Carbon dioxide (CO_2) gas	**PRODUCT** Sodium nitrate ($NaNO_3$) (very diluted solution)	**PRODUCT** Lithium nitrate ($LiNO_3$) (very diluted solution)
PRODUCT Ammonium nitrate (NH_4NO_3) (very diluted solution)	**PRODUCT** Sodium chloride (NaCl) (very diluted solution)	**PRODUCT** Lithium chloride (LiCl) (very diluted solution)	**PRODUCT** Ammonium chloride (NH_4Cl) (very diluted solution)

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use a fume cupboard to make up acid solutions. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result.** In case of concentrated acid spill, cover spill with dry earth and cover with plastic to prevent spreading. Collect spill, then dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. When preparing sodium hydroxide solution, add pellets to water in small amounts to avoid boiling and splattering. Concentrated spills should

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Unit 3.4 Prac 6 Acids and carbonates

be neutralised with solid citric acid, then diluted with water. Small concentrated spills can be neutralised with a weak concentration of HCl solution, then wiped up using paper towel, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

The salt and nitrate solutions produced in this experiment are of weak concentration and may be disposed of down the sink using plenty of water. Any remaining acid solutions can be stored away for future use or neutralised with sodium carbonate or calcium hydroxide and poured down the sink.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 3.4 Prac 7 Acids and metals

Description of procedure/requirements

Chemicals required: Class sets: Small pieces of aluminium, magnesium, zinc, iron, tin, 0.1 M solutions of hydrochloric acid, sulfuric acid and ethanoic (acetic) acid
Equipment: Test tubes with stoppers, test-tube rack, matches, 100 mL beaker, lab coat, safety glasses
Procedure: Refer to Prac 7 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Concentrated hydrochloric acid (HCl) 32%	8	Corrosive	**R34 –** Causes burns. **R37 –** Irritating to respiratory system.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S27 –** Take off immediately all contaminated clothing. **S37/39 –** Wear suitable gloves and eye protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S62 –** If swallowed, do not induce vomiting; seek medical advice immediately and show container or label.	
Concentrated sulfuric acid (H_2SO_4) 98%	8	Very corrosive	**R35 –** Causes severe burns.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S30 –** Never add water to this product. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

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Unit 3.4 Prac 7 Acids and metals

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Ethanoic acid glacial (acetic acid) CH_3COOH 99.8%	8/3	Very corrosive	**R10 –** Flammable. **R35 –** Causes severe burns.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.35%)	0.1 M Ethanoic (acetic) acid (0.3%)	0.1 M Sulfuric acid (0.98%)	Aluminium (Al) solid
Zinc (Zn) solid	Iron (Fe) solid	Tin pieces (Sn)	**PRODUCT** Hydrogen gas (H_2) (very small volume)
PRODUCT Aluminium chloride ($AlCl_3$) (very diluted solution)	**PRODUCT** Magnesium chloride ($MgCl_2$) (very diluted solution)	**PRODUCT** Zinc chloride ($ZnCl_2$) (very diluted solution)	**PRODUCT** Iron(II) chloride ($FeCl_2$) (very diluted solution)
PRODUCT Tin(II) chloride ($SnCl_2$) (very diluted solution)	**PRODUCT** Magnesium sulfate ($MgSO_4$) (very diluted solution)	**PRODUCT** Iron(II) sulfate ($FeSO_4$) (very diluted solution)	**PRODUCT** Magnesium ethanoate ($(CH_3COO)_2Mg$) (very diluted solution)

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Unit 3.4 Prac 7 Acids and metals

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use a fume cupboard to prepare acid solutions. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. For acid spills on the skin, irrigate with large quantities of water: washing should be continued for at least ten to fifteen minutes. Seek immediate medical advice. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide waste containers for each acid. Allow the reactions to be completed. Neutralise acid waste and discard to the sewer with plenty of water. Magnesium is not suitable for disposal via the school waste system. Discard via a chemical waste company. Zinc, tin and iron can be disposed of in the normal rubbish.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 4.1 Prac 3 Eye dissection

Description of procedure/requirements

Chemicals required Class sets: Bleach solution 0.5%

Equipment: Frozen cow or bullock eye, sharp fine scissors, paper towel, dissection board, newspaper

Procedure: Refer to Prac 3 Unit 4.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Bleach 5%	Bleach solution 0.5%		

Other hazards and safety considerations

Frozen eyes are recommended as preserved eyes have an overpowering, unpleasant smell. Prepare solutions according to laboratory precautions. 5% bleach solution is NOT classified as hazardous by NOHSC criteria; however, care should be taken to not breathe in vapours. Animal tissue should be handled with care and should be treated as a potential source of infection. Cuts, sores or broken skin should be covered. Gloves may be worn. Wash any animal blood/tissue off skin immediately using antibacterial hand wash. Cut away from yourself and keep fingers out of cutting path. To disinfect all work areas, use a commercial bleach with 5% available chlorine, as it is an effective broad-spectrum germicide and can be diluted up to 1:10 (0.5%) for safe use in the

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Unit 4.1 Prac 3 Eye dissection

classroom without losing its effectiveness. Safety goggles, lab coat and gloves should also be worn to avoid contact of animal tissue with skin and eyes. If feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Sterilise used dissecting equipment in disinfectant or diluted bleach. Dispose of waste daily, or freeze and discard when all dissections are completed. Dispose of waste as normal food waste.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 4.3 Prac 1 Taste regions

Description of procedure/requirements

Chemicals required: **Class sets**: Sugar, salt, vinegar, lemon juice, coffee

Equipment: Clean cotton buds, samples of the following solutions in new plastic cups: sugar, salt, vinegar or lemon juice, coffee, blindfold

Procedure: Refer to Prac 1 Unit 4.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Sugar	Salt	Vinegar	Lemon juice
Coffee			

Other hazards and safety considerations

Tasting should occur outside the lab environment, under controlled hygienic conditions. Provide fresh stock solutions for each class. Disinfect surfaces, line with clean butcher's paper if in doubt. Use new disposable containers; e.g. disposable cups. Do not use beakers or other lab glassware. Contamination risks can be minimised by providing a new cup per person, and the teacher dispensing the stock solutions. Allow one use only of cotton buds and ensure immediate disposal into a lined waste bin. Wash hands prior to handling food.

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Unit 4.3 Prac 1 Taste regions

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Waste can be disposed of in a sealed bag via the general school waste system.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 4.5 Prac 1 Brain dissection

Description of procedure/requirements

Chemicals required: **Class sets**: Bleach solution 0.5%
Equipment: Partly frozen lamb's brain, dissection board, scalpel, dissecting scissors, newspaper
Procedure: Refer to Prac 1 Unit 4.5

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Bleach 5%	Bleach solution 0.5%		

Other hazards and safety considerations

Partially frozen brains provide students with a firmer object to cut—they retain integrity of the structures for a period of time. Prepare solutions according to laboratory precautions. 5% bleach solution is NOT classified as hazardous by NOHSC criteria; however, care should be taken to not breathe in vapours. Animal tissue should be handled with care and should be treated as a potential source of infection. Cuts, sores or broken skin should be covered. Gloves may be worn. Students instructed to wash any animal blood/tissue off skin immediately using antibacterial hand wash. Students instructed to cut away from themselves and to keep fingers out of cutting path. To disinfect all work areas, use a commercial bleach with 5% available chlorine as it is an effective broad-spectrum germicide and can be diluted up to 1:10 (0.5%) for safe use in the classroom without losing its effectiveness. It is NOT classified as hazardous by NOHSC criteria. However, care should be taken to not breathe in vapours. Safety goggles, lab coat and gloves should be worn to avoid contact with skin and eyes. If feeling unwell consult **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 4.5 Prac 1 Brain dissection

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Sterilise used dissecting equipment in disinfectant or diluted bleach. Dispose of waste daily, or freeze and discard when all dissections are completed. Dispose of waste as normal food waste.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 6.2 Prac 1 Testing for water

Description of procedure/requirements

Chemicals required: **Class sets**: Anhydrous copper(II) sulfate, cobalt chloride paper, various liquids (tap water, methylated spirits, salt water, sucrose solution, acetic acid (2 M), unknown liquids X,Y and Z

Equipment: Watch glass, dropper, paper towel

Procedure: Refer to Prac 1 Unit 6.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper(II) sulfate, anhydrous ($CuSO_4$) solid	9	Harmful, irritant	**R22** – Harmful if swallowed. **R36/38** – Irritating to eyes and skin. **R50** – Very toxic to aquatic organisms. **R53** – May cause long-term adverse effects in the aquatic environment.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S60** – This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Methylated spirits (C_2H_5OH)	3	Highly flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	
Ethanoic acid glacial (acetic acid) CH_3COOH 99.8%	8/3	Very corrosive	**R10 –** Flammable. **R35 –** Causes severe burns.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

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Unit 6.2 Prac 1 Testing for water

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
2 M acetic (ethanoic) acid (12%)	None allocated	Irritant	**R36/38** – Irritating to eyes and skin.	**S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	
Cobalt chloride paper	6.1	Toxic	**R22** – Harmful if swallowed. **R42/43** – May cause sensitisation by inhalation and skin contact. **R49(2)** – May cause cancer by inhalation. **R60 –** May impair fertility. **R68** – Possible risk of irreversible effects.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S45** – In case of accident or if you feel unwell, seek medical advice immediately. **S53** – Avoid exposure: obtain special instructions before use. **S60** –This material and its container must be disposed of as hazardous waste.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Tap water	Salt	Salt water	Sucrose
Sucrose solution	**PRODUCT** Copper(II) sulfate ($CuSO_4$) (very dilute solution)		

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Unit 6.2 Prac 1 Testing for water

Other hazards and safety considerations

Prepare solutions according to laboratory precautions. Use fume cupboard to prepare acid solutions. When diluting acids, acids should always be slowly added to the water in small amounts. **Never use hot water and never add water to the acid, as uncontrolled boiling may result**. In case of concentrated acid spill, absorb spill with earth, sand or other non-combustible material; dilute small amounts with water and neutralise with sodium carbonate or calcium hydroxide. Filter or strain to separate sand from liquid. Dispose of sand in the normal waste and the neutralised liquid may be poured down the sink. For acid spills on the skin, irrigate with large quantities of water: washing should be continued for at least ten to fifteen minutes. Seek immediate medical advice.

Methylated spirits is highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or gas heaters. Decant flammable liquids into class sets in a fume hood and dispense in dropping bottles. In case of methylated spirits spill, eliminate all ignition sources, absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. Consider choosing non-hazardous solution for the unknown liquids. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult the **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Provide container for cobalt chloride paper waste. Wipe the used watch glasses with paper towel. Allow ethanol to evaporate from the paper towel in a fume hood. Discard the paper via the school's waste system.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 6.2 Prac 1 Testing for water

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Unit 6.2 Prac 2 Measuring the boiling point of water

Description of procedure/requirements

Chemicals required: Class sets: Distilled water, salt

Equipment: Bunsen burner and mat, tripod, 4 × 250 mL beakers, measuring cylinder, tablespoon, thermometer able to record temperatures higher than 100ºC or a temperature probe

Procedure: Refer to Prac 2 Unit 6.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Salt			

Other hazards and safety considerations

Prepare chemicals as per laboratory precautions. Instruct students to be cautious when heating water as the beaker can easily be knocked over, splashing nearby persons. Use a spirit thermometer rather than a mercury thermometer. There are no hazardous substances used or produced in this experiment. Risks minimised by wearing heatproof gloves, safety glasses and lab coat. Spills should be wiped up immediately using paper towel, avoiding contact with skin and eyes.

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Unit 6.2 Prac 2 Measuring the boiling point of water

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Heatproof mat, heatproof gloves and tongs
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Test samples may be washed down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 7.1 Prac 2 Observing light

Description of procedure/requirements

Chemicals required: **Class sets**: Milk, filtered water

Equipment: 20 L rectangular fish tank or clear Perspex box, HeNe laser pointer or other collimated light source, dropper, retort stand, clamp, boss head clamp

Procedure: Refer to Prac 2 unit 7.1

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Milk			

Other hazards and safety considerations

Only Class 1 and Class 2 laser pointers are permitted in schools. Check all electrical equipment for damaged leads prior to use. Never operate the laser at eye-level as the beam, if deflected into the eye, could do serious damage. Before operation of a laser pointer or collimated light source, the teacher in charge of the demonstration or experiment should warn students of the potential hazards and instruct them never to look into the direct beam or the reflected beams. The laser beam should be turned off when it is not in use. Minimise the risks of lasers by the use of Class 1 laser pointers, by performing this practical as a teacher demonstration and by wearing safety glasses. For further information refer to http://www.nhmrc.gov.au

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes			
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

No hazardous waste is generated in this practical. The water and milk waste may be disposed of down the drain.

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Unit 7.1 Prac 2 Observing light

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Unit 7.3 Prac 4 Measuring light absorption

Description of procedure/requirements

Chemicals required: coloured paint: black, white red, blue and green

Equipment: 5 metallic cans of equal size with plastic lids; e.g. coffee cans, 5 thermometers, timer, light source; e.g. the Sun or a flood lamp

Procedure: Refer to Prac 4 Unit 7.3

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Acrylic paint (water based)			

Other hazards and safety considerations

Short-term exposure to the Sun can result in sunburn and injuries to the eye. To minimise risk, avoid the UV peak time to carry out this practical activity and take precautions for sunburn, such as wearing hats and sun glasses. Allocate one colour can per group to minimise sun exposure. If using a flood lamp, instruct students not to touch the lamp as it can become very hot. Instruct students to wash any paint off skin immediately. In case of paint in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back			Wipe bench/wash hands on completion of prac	

Disposal of wastes

Any unused paint may be stored for future use.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

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Unit 7.3 Prac 4 Measuring light absorption

Notes

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Unit 9.1 Prac 2 Convection currents

Description of procedure/requirements

Chemicals required: **Class sets**: Potassium permanganate ($KMnO_4$)
Equipment: Large (500 mL or 1000 mL) beaker, tweezers, hot plate or Bunsen burner, bench mat, tripod, gauze mat, plastic bag, ice
Procedure: Refer to Prac 2 Unit 9.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Potassium permanganate ($KMnO_4$) solid	5.1	Harmful, oxidising, dangerous for the environment	**R8 –** Contact with combustible material may cause fire. **R22 –** Harmful if swallowed. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment.	**S17 –** Keep away from combustible material. **S24/25 –** Avoid contact with skin and eyes. **37/39 –** Wear suitable gloves and eye/face protection. **S61 –** Avoid release to the environment. Refer to special instructions/MSDS.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous sood	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

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Unit 9.1 Prac 2 Convection currents

Other hazards and safety considerations

Prepare chemicals according to laboratory precautions. Potassium permanganate solution will readily stain skin and surfaces. Stains can be removed from surfaces using a solution of hydrogen peroxide. Care should be taken when handling hydrogen peroxide as it is corrosive. Stains to skin will generally fade within 48 hours. For small, solid potassium permanganate spills, collect with a non-sparking tool and place into a hazardous waste container. For small solution spills, dilute with water and wipe up using paper towel. Risks are minimised by wearing safety glasses, lab coat and gloves and dispensing minimal quantities. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Collect potassium permanganate waste from classroom. Reduce the potassium permanganate solution with a weak reducing agent such as sodium thiosulfate and neutralise with dilute hydrochloric acid. The minimal quantities used could be poured down the drain with plenty of water or disposed of via a chemical waste company.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 9.1 Prac 2 Convection currents

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Unit 9.4 Prac 1 Volcanic clouds

Description of procedure/requirements

Chemicals required: Class sets: Food dye
Equipment: Large beaker or pneumatic trough or transparent jar, 100 mL flask, rubber stopper with hole, electric kettle (optional)
Procedure: Refer to Prac 1 Unit 9.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant gubstance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Food dye			

Other hazards and safety considerations

Suggestion: Heating the water in an electric kettle at the front bench, and allowing the students to collect it from the front, is the safest way to do this experiment (instead of all students heating water). There are no hazardous substances used or produced in this experiment. Risks minimised by wearing heatproof gloves, safety glasses and lab coat. Spills should be wiped up immediately using paper towel, avoiding contact with skin and eyes.

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Unit 9.4 Prac 1 Volcanic clouds

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Heatproof mat, heatproof gloves and tongs
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Test samples may be washed down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 1.2 Prac 2 Chromatography catches a criminal

Description of procedure/requirements

Chemicals required: **Class sets**: Ethanol or methylated spirits
Equipment: A 'ransom' note written with an unknown brand of felt pen, several different black felt pens including the one that wrote the note, the name of each 'suspect' attached, gas jars, scissors, paper
Procedure: Refer to Prac 2 Unit 1.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Ethanol (C_2H_5OH) or methylated spirits	**3**	Highly flammable	**R11** – Highly flammable.	**S7** – Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

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Unit 1.2 Prac 2 Chromatography catches a criminal

Other hazards and safety considerations

Methylated spirits and ethanol are highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or heaters. Methylated spirits or ethanol may be substituted with salt solution (0.1 M) with water base pens.

In case of ethanol spill, eliminate all ignition sources; absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. Risks minimised by wearing safety glasses, lab coat and gloves. Students instructed to wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave prac waste for the laboratory technician to collect.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 1.3 Prac 1 Comparing fingerprints

Description of procedure/requirements

Chemicals required: **Class sets**: Graphite powder
Equipment: Clean microscope slides, small soft bristle brushes or puff-brushes, sticky tape, paper
Procedure: Refer to Prac 1 Unit 1.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Graphite (C)	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

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Unit 1.3 Prac 1 Comparing fingerprints

Other hazards and safety considerations

Instruct students to exercise care when handling microscope slides as they may shatter. Handle broken slides carefully and avoid sharp areas. Avoid inhalation of graphite. For small graphite spill, sweep up with a dustpan and brush or a clean non-sparking tool and place into a loosely covered plastic container for later disposal. Students instructed to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult your local **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

There is no waste generated.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

Notes
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Unit 2.1 Prac 1 Making elements

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M sulfuric acid (H_2SO_4)
Equipment: Large beaker, 9–12 V power supply or battery, alligator clips, 2 metal electrodes, safety glasses
Procedure: Refer to Prac 1 Unit 2.1

HAZARDOUS as classified by ASCC

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Nickel (Ni) electrodes	None allocated	Harmful, irritant, listed as a carcinogen, category 3 in List of Designated Hazardous Substances – NOHSC.	**R40(3) –** Possible risk of irreversible effects. **R43 –** May cause sensitisation by skin contact.	**S22 –** Do not breathe dust. **S36 –** Wear suitable protective clothing.	

NOT hazardous as classified by ASCC

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous Goods class)

0.1 M Sulfuric acid (H_2SO_4) (0.98%)	Two metal electrodes - Platinum (Pt)	Carbon electrodes (C)	**PRODUCT** Hydrogen gas (H_2) (very small volume)

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Unit 2.1 Prac 1 Making elements

PRODUCT Oxygen gas (O_2)			

Other hazards and safety considerations

Instruct students on safe use of power supplies. Construct the circuit with the power off. Do not touch the circuit while the power is on. Use only the recommended voltage. Do not allow power supplies/electrical equipment to come into contact with liquids. Risks minimised by providing the minimum quantity and volume of acid. Risks also minimised by wearing safety glasses, lab coat and gloves. For small diluted acid spill, dilute spill with water and wipe up with paper towel, avoiding contact with skin and eyes. Instruct students to wash any chemical off skin immediately. For acid spills on skin, irrigate with large quantities of water: washing should be continued for at least ten to fifteen minutes. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes	Yes	Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Students place the waste solutions of sulfuric acid in the provided container to be disposed of by the lab technician. Leave the electrodes to be collected by the lab tech to be stored for future use.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 2.1 Prac 1 Making elements

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Unit 2.1 Prac 2 Making a compound

Description of procedure/requirements

Chemicals required: Class sets: Limewater, marble chips, 2 M hydrochloric acid
Equipment: Two test tubes, test-tube rack, drinking straw, 1-hole rubber stopper with glass tubing, safety glasses
Procedure: Refer to Prac 2 Unit 2.1

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

2 M Hydrochloric acid (HCl) (7%)	Marble chips	**PRODUCT** Calcium chloride ($CaCl_2$)(aqueous)	**PRODUCT** Carbon dioxide (CO_2) gas

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Unit 2.1 Prac 2 Making a compound

Other hazards and safety considerations

Students must wear safety glasses and use a new straw. Instruct students to blow gently to avoid spray coming into contact with eyes. A long piece of tubing with a disposable straw end will increase the distance of the limewater from students' faces. For diluted hydrochloric acid spill, dilute spill with water and wipe up using paper towel. Students instructed to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult your local **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

If the marble chips have been completely consumed in the reaction, then the product, calcium chloride, may be disposed of down the sink using plenty of water. If there are unconsumed marble chips remaining in the product, leave for the laboratory technician to filter and recycle.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.2 Prac1 Investigating a physical property

Description of procedure/requirements

Chemicals required: Class sets: Cubes or cylinders of aluminium, brass, lead, wood, ice; a collection of small items such as pebbles, candles, chunks of concrete or cement, copper wire

Equipment: Electronic balance, ruler, beakers, measuring cylinders

Procedure: Refer to Prac 1 Unit 2.2

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61(1) –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 2.2 Prac1 Investigating a physical property

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Other hazards and safety considerations

Lead may be absorbed through the skin on prolonged exposure, which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac. In case of prolonged exposure or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

There is no waste generated in this practical. Samples can be stored for future use.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.2 Prac1 Investigating a physical property

Notes
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Unit 2.2 Prac 2 Comparing elements

Description of procedure/requirements

Chemicals required: Class sets: Sulfur, aluminium, carbon, silicon, tin, zinc, lead, magnesium, calcium, iron, steel wool
Equipment: Three or four test tubes and rack, power pack about 2 V or battery, wires with alligator clips, light globe, safety glasses
Procedure: Refer to Prac 2 Unit 2.2

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R33 –** Danger of cumulative effects. **R62 –** Possible risk of impaired fertility. **R61(1) –** May cause harm to the unborn child. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R20/22 –** Harmful by inhalation and if swallowed.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Calcium (Ca) solid	4.3	Irritant, highly flammable	**R15 –** Contact with water liberates extremely flammable gases. **R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S8 –** Keep container dry. **S24/25 –** Avoid contact with skin and eyes. **S43(1) –** In case of fire, use sand – never use water.	

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Unit 2.2 Prac 2 Comparing elements

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Carbon (C)	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	
Sulfur powder (S)	4.1	Highly flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition; no smoking.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Aluminium (Al)	Silicon (Si)	Zinc (Zn)	Iron (Fe)
Steel wool	Tin pieces (Sn)		

Other hazards and safety considerations

Lead may be absorbed through the skin on prolonged exposure which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac. Consider setting up this prac at stations around the room to which students rotate. Power supplies should be checked for damaged leads and malfunction prior to use. Use only the recommended voltage. Do not allow power supplies/electrical equipment to come into contact

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Unit 2.2 Prac 2 Comparing elements

with liquids. Use sulfuric roll to test conductivity and sulfur powder for the solubility test. A carbon electrode makes an ideal carbon sample as it minimises the irritant nature of carbon powder. Provide only a small quantity of calcium and use forceps to handle it (calcium granules are a safer option). If alligator clips will not 'clip' to the samples, provide electrodes (nails work well) that can be touched onto the sample to complete the circuit. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave prac waste for the laboratory technician to collect. Sulfur, lead and tin waste will be placed into separate hazardous waste containers for later disposal via a chemical waste company.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.3 Prac 1 Firework colours

Description of procedure/requirements

Chemicals required: Class sets: Barium chloride, copper chloride, potassium chloride, sodium chloride and strontium chloride (suggested concentration 0.1 M) in which to soak icy-pole sticks
Equipment: Bunsen burner, bench mat and matches, tongs, safety glasses, wooden icy-pole sticks soaked overnight in distilled water and solutions of barium chloride, copper chloride, potassium chloride, sodium chloride and strontium chloride; spectroscope (optional)
Procedure: Refer to Prac 1 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

0.1 M Barium chloride ($BaCl_2$) solution (2.4%)	0.1 M Copper(II) chloride ($CuCl_2$) (1.7%)	0.1 M Potassium chloride (KCl) (0.75%)	0.1 M Sodium chloride (NaCl) (0.58%)
0.1 M Strontium chloride ($SrCl_2$) (2.7%)			

Other hazards and safety considerations

Instruct students not to 'burn' the icy-pole sticks. Do not allow chemical-soaked icy-pole sticks to come into contact with skin—use tongs to handle them. Do not breathe in any vapours or fumes. When using the spectroscope, do not aim the spectroscope, or look directly, at the Sun. Spills should be wiped up using paper towel and water, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical

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Unit 2.3 Prac 1 Firework colours

assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place used icy-pole sticks into a beaker of water and set aside for the laboratory technician to collect and dispose of.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 2.3 Prac 3 Metal crystals

Description of procedure/requirements

Chemicals required: Class sets: 1 cm × 4 cm strip clean zinc sheet, one 0.3 g sample of silver nitrate, 0.5 g of agar powder, 40 mL distilled water
Equipment: Sterilised Petri dish, 250 mL beaker, Bunsen burner, tripod, gauze mat, matches, bench mat, stirring rod, stereo microscope (optional), safety glasses, gloves
Procedure: Refer to Prac 3 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous to the environment	**R34 –** Causes burns. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Lead(II) nitrate ($Pb(NO_3)_2$) solid	5.1/6.1	Toxic, dangerous to the environment	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause long-term adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 2.3 Prac 3 Metal crystals

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper(II) sulfate ($CuSO_4$) solid	9	Harmful, irritant dangerous for the environment	**R22** – Harmful if swallowed. **R36/38** – Irritating to eyes and skin. **R50** – Very toxic to aquatic organisms. **R53** – May cause long-term adverse effects in the aquatic environment.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S36/37/39** – Wear suitable protective clothing, gloves and eye/face protection. **S60** – This material and its container must be disposed of as hazardous waste. **S61** – Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Tin(II) chloride ($SnCl_2.2H_2O$) solid	None allocated	Harmful	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin. **R43 –** May cause sensitisation by skin contact.	**S24** – Avoid contact with skin. **S26** – In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S37** – Wear suitable gloves. **S46** – If swallowed, seek medical advice immediately and show the label.	
PRODUCT Lead (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61(1) –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S20/21 –** When using, do not eat, drink or smoke. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately. **S53 –** Avoid exposure; obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 2.3 Prac 3 Metal crystals

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Agar	Zinc strips (Zn)	**PRODUCT** Silver (Ag)	**PRODUCT** Tin (Sn)
PRODUCT Copper (Cu)	**PRODUCT** Zinc nitrate solution in agar (very low concentration)	**PRODUCT** Zinc sulfate solution in agar (very low concentration)	**PRODUCT** Zinc chloride solution in agar (very low concentration)

Other hazards and safety considerations

Students to exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. Significant risks to the students are minimised if the chemicals in this practical are pre-weighed. Risks also minimised by wearing safety glasses, lab coat and gloves. For silver nitrate and lead nitrate spills, sweep up and place into separate hazardous waste containers for disposal via a chemical waste company. Other spills may be swept up and disposed of in the normal rubbish. Instruct students to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of inhalation or ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave Petri dishes for the laboratory technician to collect and dispose of.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

Notes

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Unit 2.3 Prac 3 Metal crystals

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Unit 2.3 Prac 4 More crystals

Description of procedure/requirements

Chemicals required: **Class sets**: 10 to 15 cm length of copper wire and/or strip of copper foil, silver nitrate solution (suggested concentration 0.1 M)
Equipment: 100 mL conical flask, cork or rubber stopper, silver nitrate solution
Procedure: Refer to Prac 4 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous Goods class)

0.1 M Silver nitrate ($AgNO_3$) (1.7%)	Sodium thiosulfate 4% (silver nitrate spill clean-up)	Copper (Cu)	**PRODUCT** Silver (Ag)
PRODUCT Copper nitrate solution (weak concentration)			

Other hazards and safety considerations

Suggested safe concentration for silver nitrate solution is 0.1 M. Consider providing the silver nitrate solution in stoppered flasks. (This will minimise student

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Unit 2.3 Prac 4 More crystals

contact with the silver nitrate and the risk of spills.) Silver nitrate stains: spills can be decolourised with sodium thiosulfate 4%.

Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place the waste solutions and the copper wires in the containers provided to be collected by the lab tech.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 2.3 Prac 6 Using metals to make non-metals

Description of procedure/requirements

Chemicals required: **Class sets**: Magnesium, iron and copper, 1–2 M hydrochloric acid in a dropping bottle
Equipment: Test tubes and rack, matches, safety glasses
Procedure: Refer to Prac 6 Unit 2.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S43 –** In case of fire, use sand, never use water. **S7/8 –** Keep container tightly closed and dry.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

2 M Hydrochloric acid (HCl) (7%)	Copper (Cu) solid	Iron (Fe) solid	**PRODUCT** Magnesium chloride ($MgCl_2$) (aqueous)
PRODUCT Ferrous chloride ($FeCl_2$) weak solution	**PRODUCT** Hydrogen (H_2) gas (very small volume)		

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Unit 2.3 Prac 6 Using metals to make non-metals

Other hazards and safety considerations

Use stoppers rather than fingers to collect the gas. Reducing the concentration to 1 M should still provide effective results. For diluted hydrochloric acid spill, dilute spill with water and wipe up using paper towel. Students instructed to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult your local **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place liquid and metal waste in the container provided.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 2.4 Prac 1 The alkaline earths

Description of procedure/requirements

Chemicals required: Class sets: 5 cm strip of magnesium, small sample of calcium, phenolphthalein indicator
Equipment: Crucible and a lid, clay triangle, 2 test tubes and rack, Bunsen burner, tripod, bench mat, matches, tongs, safety glasses, distilled water, steel wool or emery paper
Procedure: Refer to Prac 1 Unit 2.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium granules (Ca)	4.3	Irritant, highly flammable	**R15 –** Contact with water liberates extremely flammable gases. **R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S8 –** Keep container dry. **S24/25 –** Avoid contact with skin and eyes. **S43(1) –** In case of fire, use alcohol-resistant foam/dry powder/CO_2. Never use water.	

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	
Phenolphthalein	3	Highly	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed.	

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Unit 2.4 Prac 1 The alkaline earths

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
indicator		flammable		**S16 –** Keep away from sources of ignition–no smoking. **S23 –** Do not breathe vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S33 –** Take precautionary measures against static discharges.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

PRODUCT Mixture of phenolphthalein indicator and calcium hydroxide solutions (weak concentration)	**PRODUCT** Mixture of phenolphthalein indicator and magnesium hydroxide solutions (weak concentration)		

Other hazards and safety considerations

Beware of hot objects. Students perform a pop test detecting the presence of hydrogen. Students may need to lightly stopper the test tube to collect enough gas. Instruct students to gently hold the stopper in place, otherwise pressure build-up may eject it. Refer teacher to Risk Assessment: Heating. For small calcium spill, collect spill using a dry utensil and place into a dry hazardous waste container for later disposal. Instruct students to wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave prac waste for the laboratory technician to collect.

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Unit 2.4 Prac 1 The alkaline earths

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Unit 2.4 Prac 2 Group IV

Description of procedure/requirements

Chemicals required: Class sets: Charcoal, graphite, silicon, lead
Equipment: Power pack or battery, leads with alligator clips, light
Procedure: Refer to Prac 2 Unit 2.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Lead pieces (Pb)	None allocated	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment. **R61(1) –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S20/21** – When using, do not eat, drink or smoke. **S36/37/39** – Wear suitable protective clothing, gloves and eye/face protection. **S45** – In case of accident or if you feel unwell, seek medical advice immediately. **S53** – Avoid exposure; obtain special instructions before use. **S60** – This material and its container must be disposed of as hazardous waste. **S61** – Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Graphite (C)	None allocated	Irritant	**R36/37/38** – Irritating to eyes, respiratory system and skin.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S36/3739** – Wear suitable protective clothing, gloves and eye/face protection.	

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Unit 2.4 Prac 2 Group IV

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Charcoal powder (C)	Silicon (Si)		

Other hazards and safety considerations

Lead may be absorbed through the skin on prolonged exposure, which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac. In case of prolonged exposure or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

There is no waste generated in this practical. Samples can be stored for future use.

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Unit 2.4 Prac 2 Group IV

Notes
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Unit 3.1 Prac 2 Signs of chemical change

Description of procedure/requirements

Chemicals required: Class sets: Copper carbonate, magnesium, solid zinc, dilute nitric acid, dilute sodium hydroxide, dilute sodium sulfate, dilute copper sulfate, dilute barium nitrate (0.1 M is a suitable concentration for all the solutions except for nitric acid where a concentration of 1 M will produce the desired amount of gas)
Equipment: Splint, matches, Bunsen burner, test-tube holder/tongs, test-tube rack, 5 test tubes (1 with stopper), thermometer, dropper or Pasteur pipette, lab coat, safety glasses
Procedure: Refer to Prac 2 Unit 3.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper carbonate ($CuCO_3$) solid	None allocated	Harmful, Irritant	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
1 M nitric acid (HNO_3) (6.3%)	8	Corrosive	**R35 –** Causes severe burns.	**S23 –** Do not breathe gas/fumes/vapour/spray. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S45 –** In case of accident or if you feel unwell, seek medical advice immediately.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 3.1 Prac 2 Signs of chemical change

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M sodium hydroxide (NaOH) (0.4%)	0.1 M sodium sulfate (Na_2SO_4) (1.4%)	Zinc (Zn) solid	0.1 M copper(II) sulfate (2.4%)
PRODUCT Carbon dioxide gas (CO_2)	**PRODUCT** Magnesium nitrate ($Mg(NO_3)_2$) (weak solution)	**PRODUCT** Hydrogen gas (H_2) (very small volume)	**PRODUCT** Sodium nitrate ($NaNO_3$) (weak solution)
PRODUCT Barium sulfate ($BaSO_4$) solid	**PRODUCT** Zinc sulfate ($ZnSO_4$) (very diluted solution)		

Other hazards and safety considerations

Refer to Heating RA. Beware of hot objects. Small diluted nitric acid spills can be neutralised with a weak sodium carbonate solution, then wiped up using paper towel, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Students instructed to wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place copper and barium waste in the containers provided. Leave nitric acid and sodium hydroxide waste for the laboratory technician to collect and neutralise before disposing of down the sink.

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Unit 3.1 Prac 2 Signs of chemical change

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Unit 3.2 Prac 2 Stopping combustion

Description of procedure/requirements

Chemicals required: **Class sets**: Sodium bicarbonate, 0.1 M hydrochloric acid, limewater
Equipment: 250 mL conical flask, rubber stopper with a flexible plastic tube passing through it, large test tube with rubber stopper, wooden splint, matches
Procedure: Refer to Prac 2 Unit 3.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.37%)	**PRODUCT** Sodium chloride (NaCl) solution	**PRODUCT** Water (H_2O)	**PRODUCT** Carbon dioxide (CO_2) gas
Sodium bicarbonate solid			

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Unit 3.2 Prac 2 Stopping combustion

Other hazards and safety considerations

Refer to Heating RA. Small diluted acid and limewater spills can be diluted with water, then wiped up using paper towel, avoiding contact with skin and eyes. For small solid spills, sweep up immediately and place into the normal waste, avoiding inhalation. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Test samples may be disposed of down the sink with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

Description of procedure/requirements

Chemicals required: Class sets: Copper carbonate, limewater
Equipment: Bunsen burner, matches, retort stand and clamp, 2 large test tubes, stopper for one of the test tubes and a delivery tube, lab coat, safety glasses
Procedure: Refer to Prac 3 Unit 3.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper carbonate ($CuCO_3$) solid	None allocated	Harmful, Irritant	**R22 –** Harmful if swallowed. **R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S38 –** If insufficient ventilation, wear suitable respiratory equipment. **S37/39 –** Wear suitable gloves and eye/face protection.	
PRODUCT Copper(II) oxide (CuO) solid	None allocated	Harmful	**R22 –** Harmful if swallowed	**S22 –** Do not breathe dust.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

PRODUCT Carbon dioxide gas (CO_2)			

Other hazards and safety considerations

Refer to Heating RA. Beware of hot objects. For small solid spills, sweep up immediately avoiding inhalation. Place spill into a hazardous waste container for disposal via a chemical waste company. In case of limewater spill, wipe up with paper towel, avoiding contact with skin and eyes. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place copper waste in the container provided.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

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Unit 3.2 Prac 3 Decomposition of a metal carbonate

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Unit 3.3 Prac 1 Precipitation reactions

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M solutions of silver nitrate ($AgNO_3$), sodium chloride (NaCl), sodium hydroxide (NaOH), barium nitrate ($Ba(NO_3)_2$) and copper sulfate ($CuSO_4.5H_2O$)

Equipment: Test tubes, test tube rack, Pasteur pipettes, lab coat, safety glasses, gloves

Procedure: Refer to Prac 1 Unit 3.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
0.1 M barium nitrate ($Ba(NO_3)_2$)	None allocated	Harmful	**R20/22 –** Harmful by inhalation and if swallowed.	**S28 –** After contact with skin, wash immediately with plenty of soap suds.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous Goods class)

0.1 M Silver nitrate ($AgNO_3$) (1.7%)	0.1 M Sodium chloride (NaCl) (0.6%)	0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Copper(II) sulfate (2.4%)
PRODUCT Silver chloride (AgCl) solid	**PRODUCT** Silver oxide (Ag_2O) solid	**PRODUCT** Sodium nitrate ($NaNO_3$) solution	**PRODUCT** Barium sulfate ($BaSO_4$) solid

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Unit 3.3 Prac 1 Precipitation reactions

PRODUCT Copper nitrate ($Cu(NO_3)_2$) solution (weak concentration)			

Other hazards and safety considerations

Students to exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. Risks minimised by wearing safety glasses, lab coat and gloves. Small spills may be wiped up with paper towel and disposed of in the normal rubbish. Instruct students to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of inhalation or ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance..

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place barium and silver wastes into the containers provided.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.3 Prac 1 Precipitation reactions

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Unit 3.3 Prac 2 Precipitation of unknowns

Description of procedure/requirements

Chemicals required: Class sets: Unknown 0.1 M solutions labelled A, B, C, D, E. These are (not in order) sodium iodide, sodium chloride, sodium sulfate, sodium carbonate and sodium nitrate; 0.1 M solutions of silver, lead, calcium and barium nitrate
Equipment: 20 semi-micro test tubes, Pasteur pipettes, lab coat, safety glasses, gloves
Procedure: Refer to Prac 2 Unit 3.3

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
0.1 M lead nitrate ($Pb(NO_3)_2$) 3.3%	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R61 –** May cause harm to the unborn child.	**S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
0.1 M barium nitrate ($Ba(NO_3)_2$)	None Allocated	Harmful	**R20/22** Harmful by inhalation and if swallowed.	**S28** After contact with skin, wash immediately with plenty of soap suds.	
PRODUCT Lead chloride ($PbCl_2$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R62 – Possible risk of impaired fertility.		
PRODUCT Lead sulfate ($PbSO_4$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
PRODUCT Lead iodide (PbI_2) solid	6.1	Toxic, dangerous for the environment	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	

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Unit 3.3 Prac 2 Precipitation of unknowns

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R62 – Possible risk of impaired fertility.		
PRODUCT Lead carbonate ($PbCO_3$) solid	6.1	Toxic	**R20/22 –** Harmful by inhalation and if swallowed. **R33 –** Danger of cumulative effects. **R50 –** Very toxic to aquatic organisms. **R53 –** May cause adverse effects in the aquatic environment. **R61 –** May cause harm to the unborn child. **R62 –** Possible risk of impaired fertility.	**S45 –** In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible). **S53 –** Avoid exposure. Obtain special instructions before use. **S60 –** This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
PRODUCT Silver sulfate (Ag_2SO_4) solid	None allocated	Irritant	**R36/37/38 –** Irritating to eyes, respiratory system and skin.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36 –** Wear suitable protective clothing.	
PRODUCT Barium carbonate ($BaCO_3$) solid	6.1	Harmful	**R22 –** Harmful if swallowed.	**S24/25 –** Avoid contact with skin and eyes.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 3.3 Prac 2 Precipitation of unknowns

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium chloride (NaCl) (0.6%)	0.1 M Sodium sulfate (Na_2SO_4) (1.4%)	0.1 M Sodium iodide (NaI) (1.5%)	0.1 M Sodium nitrate ($NaNO_3$) (0.85%)
0.1 M Sodium carbonate (Na_2CO_3) (1%)	0.1 M Silver nitrate ($AgNO_3$) solution (1.7%)	0.1 M Calcium nitrate ($Ca(NO_3)_2$)	**PRODUCT** Silver chloride (AgCl) solid
PRODUCT Silver iodide (AgI) solid	**PRODUCT** Silver carbonate (Ag_2CO_3) solid	**PRODUCT** Barium sulfate ($BaSO_4$) solid	**PRODUCT** Calcium carbonate ($CaCO_3$) solid
PRODUCT Calcium sulfate ($CaSO_4$) solid			

Other hazards and safety considerations

Students to exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. Risks minimised by wearing safety glasses, lab coat and gloves. Small spills may be wiped up with paper towel and disposed of in the normal rubbish. Instruct students to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of inhalation or ingestion or feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place barium, lead and silver waste into the container provided. Other waste may be poured down the drain with plenty of water.

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Unit 3.3 Prac 2 Precipitation of unknowns

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Unit 3.4 Prac 1 Common indicators

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M solutions of sodium hydroxide and hydrochloric acid, distilled water, liquid red and blue litmus, universal indicator, methyl orange, methyl red, bromothymol blue, phenolphthalein
Equipment: Three test tubes, test-tube rack, 3 × 100 mL beakers, lab coat, safety glasses
Procedure: Refer to Prac 1 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	
Phenolphthalein indicator	3	Highly flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking. **S23 –** Do not breathe vapour/spray. **S24/25 –** Avoid contact with skin and eyes. **S33 –** Take precautionary measures against static discharges.	
Methyl red solution	3	Flammable	**R10 –** Flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S23(2) –** Do not breathe vapour.	

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Unit 3.4 Prac 1 Common indicators

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
				S26 – In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S33 –** Take precautionary measures against static discharges. **S36/37 –** Wear suitable protective clothing and gloves.	
Bromothymol blue solution (0.04%)	3 	None allocated	**R10 –** Flammable.	**S16 –** Keep away from sources of ignition–No smoking. **S24/25 –** Avoid contact with skin and eyes.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)	Red and blue litmus indicator	Methyl orange indicator (0.04%)

Other hazards and safety considerations

Phenolphthalein, methyl red, bromothymol blue and universal indicators are flammable. Do not use near a flame or sources of ignition. For small hydrochloric acid, sodium hydroxide and indicator spills, dilute spill with water and wipe up using paper towel. Students instructed to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult your local **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back			Wipe bench/wash hands on completion of prac	

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Unit 3.4 Prac 1 Common indicators

Disposal of wastes

Leave waste for the laboratory technician to collect. The laboratory technician will neutralise waste before disposing of it down the sink.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 2 Universal indicator

Description of procedure/requirements

Chemicals required: **Class sets**: 0.1 M hydrochloric acid, 1.0 M sodium hydroxide, distilled water, universal indicator
Equipment: Pasteur pipette, 10 mL measuring cylinder, waterproof felt-tip pen, 14 large test tubes, test-tube rack, lab coat, safety glasses
Procedure: Refer to Prac 2 Unit 3.4

HAZARDOUS as classified by ASCC

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
1 M sodium hydroxide (NaOH) (4%)	8	Corrosive	**R34** – Causes burns.	**S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S36/37 –** Wear suitable protective clothing and gloves. **S45** – In case of accident or if you feel unwell, seek medical advice immediately.	

NOT hazardous as classified by ASCC

Reactant Substance	Dangerous Good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S7 –** Keep container tightly closed.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

0.1 M hydrochloric acid (HCl) (0.35%)			

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Unit 3.4 Prac 2 Universal indicator

Other hazards and safety considerations

Universal indicator is flammable; do not use near a flame or near sources of ignition. For small hydrochloric acid, sodium hydroxide and indicator spills, dilute spill with water and wipe up using paper towel. Instruct students to wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave waste for the laboratory technician to collect. The laboratory technician will neutralise waste before disposing of it down the sink.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.4 Prac 3 Natural indicators

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M solutions of sodium hydroxide and hydrochloric acid, distilled water
Equipment: Pink or red flower petals, beetroot juice, tea bag, 3 × 100 mL beakers, filter paper cut into strips, Bunsen burner, heatproof mat, tripod, gauze, matches, 3 test tubes, test-tube rack, Pasteur pipettes, lab coat, safety glasses
Procedure: Refer to Prac 3 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)	Beetroot juice	Flower petals

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Unit 3.4 Prac 3 Natural indicators

Other hazards and safety considerations

Refer to Heating RA. Beware of hot objects. For diluted hydrochloric acid and sodium hydroxide spill, dilute spill with water and wipe up using paper towel. Students instructed to wash any chemical off skin immediately. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Mix hydrochloric acid solution and sodium hydroxide solution together to neutralise and pour down the sink with plenty of water. Flower petals, beetroot juice and filter paper may be disposed of in the normal rubbish.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 3.4 Prac 4 Testing household solutions

Description of procedure/requirements

Chemicals required: Class sets: Universal indicator, non-hazardous household solutions including orange juice, soft drink, fresh and sour milk, vinegar (solids may be used if dissolved in water first), distilled water, blue and red litmus paper
Equipment: Two test tubes, test-tube rack, Pasteur pipettes, 2 watch glasses, safety glasses, lab coat
Procedure: Refer to Prac 4 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Non-hazardous household solutions including orange juice, soft drink, fresh and sour milk, vinegar. Solids may be used if dissolved in water first.			

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Unit 3.4 Prac 4 Testing household solutions

Other hazards and safety considerations

Universal indicator is flammable. Do not use near a flame or sources of ignition. For small spills, dilute spill with water and wipe up using paper towel. Instruct students to wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Waste may be poured down the drain with plenty of water.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

Notes
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Unit 3.4 Prac 5 Acids and hydroxides

Description of procedure/requirements

Chemicals required: Class sets: 0.1 M sodium hydroxide, 0.1 M hydrochloric acid
Equipment: 250 mL beaker, 50 mL measuring cylinder, digital pH meter or universal indicator, Pasteur pipette, lab coat, safety glasses
Procedure: Refer to Prac 5 Unit 3.4

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Universal indicator	3	Flammable	**R11 –** Highly flammable.	**S16 –** Keep away from sources of ignition. No smoking. **S7 –** Keep container tightly closed.	

NOT hazardous as classified by ASCC (with NO allocated Dangerous Goods class)

0.1 M Sodium hydroxide (NaOH) (0.4%)	0.1 M Hydrochloric acid (HCl) (0.35%)				

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Unit 3.4 Prac 5 Acids and hydroxides

Other hazards and safety considerations

Universal indicator is flammable. Do not use near a flame or sources of ignition. For small hydrochloric acid, sodium hydroxide and indicator spills, dilute spill with water and wipe up using paper towel. Instruct students to wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave waste for the laboratory technician to collect. The laboratory technician will neutralise waste before disposing of it down the sink.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 3.4 Prac 6 Acids and carbonates

Description of procedure/requirements

Chemicals required: Class sets: Limewater, solid sodium hydrogen carbonate, lithium carbonate, sodium carbonate, ammonium carbonate, 0.1 M solutions of nitric acid and hydrochloric acid
Equipment: Test tubes, test-tube rack, rubber stopper, 100 mL beaker, matches, spatula, safety glasses, lab coat
Procedure: Refer to Prac 6 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Sodium carbonate anhydrous (Na_2CO_3) solid	None allocated	Irritant	**R36 –** Irritating to eyes.	**S22 –** Do not breathe dust. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.	
Calcium hydroxide $Ca(OH)_2$ solid Limewater – saturated solution	None allocated	Irritant	**R38 –** Irritating to skin. **R41 –** Risk of serious damage to eyes.	**S22 –** Do not breathe dust. **S24/25 –** Avoid contact with skin and eyes. **S26 –** In case of contact with eyes, rinse immediately with plenty of water and seek medical advice. **S37/39 –** Wear suitable gloves and eye/face protection. **S38 –** If insufficient ventilation, wear suitable respiratory equipment.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

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Unit 3.4 Prac 6 Acids and carbonates

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.35%)	0.1 M Nitric acid (HNO_3) (0.63%)	Sodium hydrogen carbonate ($NaHCO_3$) solid	Ammonium carbonate ($(NH_4)_2CO_3$) solid
Lithium carbonate (Li_2CO_3) solid	**PRODUCT** Carbon dioxide (CO_2) gas	**PRODUCT** Sodium nitrate ($NaNO_3$) (very diluted solution)	**PRODUCT** Lithium nitrate ($LiNO_3$) (very diluted solution)
PRODUCT Ammonium nitrate (NH_4NO_3) (very diluted solution)	**PRODUCT** Sodium chloride (NaCl) (very diluted solution)	**PRODUCT** Lithium chloride (LiCl) (very diluted solution)	**PRODUCT** Ammonium chloride (NH_4Cl) (very diluted solution)

Other hazards and safety considerations

Beware of hot objects. Small diluted acid and limewater spills can be diluted with water, then wiped up using paper towel, avoiding contact with skin and eyes. For small solid spills, sweep up immediately and place into the normal waste, avoiding inhalation. Risks minimised by wearing safety glasses, lab coat and gloves. Students instructed to wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

The salt and nitrate solutions produced in this experiment are of weak concentration and may be disposed of down the sink using plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 3.4 Prac 6 Acids and carbonates

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Unit 3.4 Prac 7 Acids and metals

Description of procedure/requirements

Chemicals required: Class sets: Small pieces of aluminium, magnesium, zinc, iron, tin, 0.1 M solutions of hydrochloric acid, sulfuric acid and ethanoic (acetic) acid
Equipment: Test tubes with stoppers, test-tube rack, matches, 100 mL beaker, lab coat, safety glasses
Procedure: Refer to Prac 7 Unit 3.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Magnesium (Mg) ribbon	4.1	Highly flammable	**R11 –** Highly flammable. **R15 –** Contact with water liberates extremely flammable gases.	**S7/8 –** Keep container tightly closed and dry. **S43 –** In case of fire, use sand, never use water.	

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

0.1 M Hydrochloric acid (HCl) (0.35%)	0.1 M Ethanoic (acetic) acid (0.3%)	0.1 M Sulfuric acid (0.98%)	Aluminium (Al) solid
Zinc (Zn) solid	Iron (Fe) solid	Tin pieces (Sn)	**PRODUCT** Hydrogen gas (H_2) (very small volume)

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Unit 3.4 Prac 7 Acids and metals

PRODUCT Aluminium chloride ($AlCl_3$) (very diluted solution)	**PRODUCT** Magnesium chloride ($MgCl_2$) (very diluted solution)	**PRODUCT** Zinc chloride ($ZnCl_2$) (very diluted solution)	**PRODUCT** Iron(II) chloride ($FeCl_2$) (very diluted solution)
PRODUCT Tin(II) chloride ($SnCl_2$) (very diluted solution)	**PRODUCT** Magnesium sulfate ($MgSO_4$) (very diluted solution)	**PRODUCT** Iron(II) sulfate ($FeSO_4$) (very diluted solution)	**PRODUCT** Magnesium ethanoate ($(CH_3COO)_2Mg$) (very diluted solution)

Other hazards and safety considerations

For small diluted acid spills, dilute spill with water and wipe up with paper towel. Students instructed to wash any chemical off skin immediately. Risks minimised by wearing safety glasses, lab coat and gloves. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion or inhalation, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place acid and metal wastes in each of the waste containers provided.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

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Unit 3.4 Prac 7 Acids and metals

Notes
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Unit 4.1. Prac 3 Eye dissection

Description of procedure/requirements

Chemicals required Class sets: Bleach solution 0.5%

Equipment: Frozen cow or bullock eye, sharp fine scissors, paper towel, dissection board, newspaper

Procedure: Refer to Prac 3 Unit 4.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Bleach 0. 5%			

Other hazards and safety considerations

Frozen eyes are recommended as preserved eyes have an overpowering, unpleasant smell. Prepare solutions according to laboratory precautions. 5% bleach solution is NOT classified as hazardous by NOHSC criteria; however, care should be taken to not breathe in vapours. Animal tissue should be handled with care and should be treated as a potential source of infection. Cuts, sores or broken skin should be covered. Gloves may be worn. Wash any animal blood/tissue off skin immediately using antibacterial hand wash. Cut away from yourself and keep fingers out of cutting path. To disinfect all work areas, use a commercial bleach with 5% available chlorine, as it is an effective broad-spectrum germicide and can be diluted up to 1:10 (0.5%) for safe use in the classroom without losing its effectiveness. Safety goggles, lab coat and gloves should also be worn to avoid contact of animal tissue with skin and eyes. If feeling unwell, consult **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 4.1. Prac 3 Eye dissection

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Sterilise used dissecting equipment in disinfectant or diluted bleach. Dispose of waste as normal food waste.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 4.3 Prac 1 Taste regions

Description of procedure/requirements

Chemicals required: **Class sets**: Sugar, salt, vinegar, lemon juice, coffee

Equipment: Clean cotton buds, samples of the following solutions in new plastic cups: sugar, salt, vinegar or lemon juice, coffee, blindfold

Procedure: Refer to Prac 1 Unit 4.3

HAZARDOUS as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (with NO allocated Dangerous Goods class)

Sugar	Salt	Vinegar	Lemon juice
Coffee			

Other hazards and safety considerations

Tasting should occur outside the lab environment, under controlled hygienic conditions. Provide fresh stock solutions for each class. Disinfect surfaces, line with clean butcher's paper if in doubt. Use new disposable containers; e.g. disposable cups. Do not use beakers or other lab glassware. Contamination risks can be minimised by providing a new cup per person, teacher dispensing the stock solutions and also wearing face masks. Allow one use only of cotton buds and ensure immediate disposal into a lined waste bin. Wash hands prior to handling food.

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Unit 4.3 Prac 1 Taste regions

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Waste can be disposed of in a sealed bag via the general school waste system.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 4.5 Prac 1 Brain dissection

Description of procedure/requirements

Chemicals required: **Class sets**: Bleach solution 0.5%
Equipment: Partly frozen lamb's brain, dissection board, scalpel, dissecting scissors, newspaper
Procedure: Refer to Prac 1 Unit 4.5

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Bleach solution 0.5%			

Other hazards and safety considerations

Partially frozen brains provide students with a firmer object to cut—they retain integrity of the structures for a period of time. Animal tissue should be handled with care and should be treated as a potential source of infection. Cuts, sores or broken skin should be covered. Gloves may be worn. Students instructed to wash any animal blood/tissue off skin immediately using antibacterial hand wash. Students instructed to cut away from themselves and to keep fingers out of cutting path. To disinfect all work areas, use a commercial bleach with 5% available chlorine as it is an effective broad-spectrum germicide and can be diluted up to 1:10 (0.5%) for safe use in the classroom without losing its effectiveness. It is NOT classified as hazardous by NOHSC criteria. However, care should be taken to not breathe in vapours. Safety goggles, lab coat and gloves should be worn to avoid contact with skin and eyes. If feeling unwell consult **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 4.5 Prac 1 Brain dissection

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Place used utensils and dissected waste in receptacles provided.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes
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Unit 6.2 Prac 1 Testing for water

Description of procedure/requirements

Chemicals required: **Class sets**: Anhydrous copper(II) sulfate, cobalt chloride paper, various liquids (tap water, methylated spirits, salt water, sucrose solution, acetic acid (2 M), unknown liquids X,Y and Z
Equipment: Watch glass, dropper, paper towel
Procedure: Refer to Prac 1 Unit 6.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Copper(II) sulfate, anhydrous ($CuSO_4$) solid	9	Harmful, irritant	**R22** – Harmful if swallowed. **R36/38** – Irritating to eyes and skin. **R50** – Very toxic to aquatic organisms. **R53** – May cause long-term adverse effects in the aquatic environment.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection. **S60** – This material and its container must be disposed of as hazardous waste. **S61 –** Avoid release to the environment. Refer to special instructions/Material Safety Data Sheets.	
Methylated spirits (C_2H_5OH)	3	Highly flammable	**R11 –** Highly flammable.	**S7 –** Keep container tightly closed. **S16 –** Keep away from sources of ignition. No smoking.	
2 M acetic (ethanoic) acid (12%)	None allocated	Irritant	**R36/38** – Irritating to eyes and skin.	**S36/37/39 –** Wear suitable protective clothing, gloves and eye/face protection.	
Cobalt chloride paper	6.1	Toxic	**R22** – Harmful if swallowed. **R42/43** – May cause sensitisation by inhalation and skin contact. **R49(2)** – May cause cancer by inhalation.	**S22** – Do not breathe dust. **S24/25** – Avoid contact with skin and eyes. **S45** – In case of accident or if you feel unwell, seek medical advice immediately. **S53** – Avoid exposure: Obtain special instructions before use. **S60** –This material and its container must be disposed of as hazardous waste.	

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Unit 6.2 Prac 1 Testing for water

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
			R60 – May impair fertility. **R68** – Possible risk of irreversible effects.		

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous Goods class)

Tap water	Salt	Salt water	Sucrose
Sucrose solution	**PRODUCT** Copper(II) sulfate ($CuSO_4$) (very dilute solution)		

Other hazards and safety considerations

Methylated spirits is highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or gas heaters. In case of methylated spirits spill, eliminate all ignition sources, absorb spill with earth, sand or other non-combustible material. Use clean, non-sparking tools to collect material and place it in loosely covered metal or plastic containers for later disposal. In case of dilute acid spill, dilute spill with water, then wipe up using paper towel, avoiding contact with skin and eyes. Risks minimised by wearing safety glasses, lab coat and gloves. Wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult the **Poisons Information Centre PH 131126** and seek medical assistance.

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Unit 6.2 Prac 1 Testing for water

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Paper towel
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Wipe the used watch glasses with paper towel. Allow ethanol to evaporate from the paper towel in a fume hood. Discard the paper via the school waste system. Placed the used cobalt chloride paper in container provided.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 6.2 Prac 2 Measuring the boiling point of water

Description of procedure/requirements

Chemicals required: Class sets: Distilled water, salt

Equipment: Bunsen burner and mat, tripod, 4 × 250 mL beakers, measuring cylinder, tablespoon, thermometer able to record temperatures higher than 100ºC or a temperature probe

Procedure: Refer to Prac 2 Unit 6.2

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Salt			

Other hazards and safety considerations

Refer to heating RA. Beware of hot objects. Instruct students to be cautious when heating water as the beaker can easily be knocked over, splashing persons nearby. Use a spirit thermometer rather than a mercury thermometer. There are no hazardous substances used or produced in this experiment. Risks minimised by wearing heatproof gloves, safety glasses and lab coat. Spills should be wiped up immediately using paper towel, avoiding contact with skin and eyes.

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Unit 6.2 Prac 2 Measuring the boiling point of water

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Heatproof mat, heatproof gloves and tongs
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Test samples may be washed down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Unit 7.1 Prac 2 Observing light

Description of procedure/requirements

Chemicals required: **Class sets**: Milk, filtered water

Equipment: 20 L rectangular fish tank or clear Perspex box, HeNe laser pointer or other collimated light source, dropper, retort stand, clamp, boss head clamp

Procedure: Refer to Prac 2 unit 7.1

NOT hazardous as classified by ASCC (with NO allocated Dangerous goods class)

Milk			

Other hazards and safety considerations

Only Class 1 and Class 2 laser pointers are permitted in schools. Check all electrical equipment for damaged leads prior to use. Never operate the laser at eye-level as the beam, if deflected into the eye, could do serious damage. Before operation of a laser pointer or collimated light source, the teacher in charge of the demonstration or experiment should warn students of the potential hazards and instruct them never to look into the direct beam or the reflected beams. The laser beam should be turned off when it is not in use. Minimise the risks of lasers by the use of Class 1 laser pointers, by performing this practical as a teacher demonstration and by wearing safety glasses. For further information refer to http://www.nhmrc.gov.au

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes			
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

No hazardous waste is generated in this practical. The water and milk waste may be disposed of down the drain.

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Unit 7.1 Prac 2 Observing light

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Unit 7.3 Prac 4 Measuring light absorption

Description of procedure/requirements

Chemicals required: coloured paint: black, white red, blue and green

Equipment: 5 metallic cans of equal size with plastic lids; e.g. coffee cans, 5 thermometers, timer, light source; e.g. the Sun or a flood lamp

Procedure: Refer to Prac 4 Unit 7.3

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Acrylic paint (water based)			

Other hazards and safety considerations

Short-term exposure to the Sun can result in sunburn and injuries to the eye. To minimise risk, avoid the UV peak time to carry out this practical activity and take precautions for sunburn, such as wearing hats and sun glasses. Allocate one colour can per group to minimise sun exposure. If using a flood lamp, instruct students not to touch the lamp as it can become very hot. Instruct students to wash any paint off skin immediately. In case of paint in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Any unused paint may be stored for future use.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

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Unit 7.3 Prac 4 Measuring light absorption

Notes

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Unit 9.1 Prac 2 Convection currents

Description of procedure/requirements

Chemicals required: **Class sets**: Potassium permanganate ($KMnO_4$)
Equipment: Large (500 mL or 1000 mL) beaker, tweezers, hot plate or Bunsen burner, bench mat, tripod, gauze mat, plastic bag, ice
Procedure: Refer to Prac 2 Unit 9.1

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Potassium permanganate ($KMnO_4$) solid	5.1	Harmful, oxidising, dangerous for the environment	**R8 –** Contact with combustible material may cause fire. **R22 –** Harmful if swallowed. **R50/53 –** Very toxic to aquatic organisms; may cause long-term adverse effects in the aquatic environment.	**S17 –** Keep away from combustible material. **S24/25 –** Avoid contact with skin and eyes. **37/39 –** Wear suitable gloves and eye/face protection. **S61 –** Avoid release to the environment. Refer to special instructions/MSDS.	

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

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Unit 9.1 Prac 2 Convection currents

Other hazards and safety considerations

Beware of hot objects. Refer to **Heating Risk Assessment**. Students to exercise care when handling potassium permanganate as it can readily stain skin and surfaces. Teacher or laboratory technician can remove stains from surfaces using a solution of hydrogen peroxide. Care should be taken when handling hydrogen peroxide as it is corrosive. Stains to skin will generally fade within 48 hours. For small, solid potassium permanganate spills, collect with a non-sparking tool and place into a hazardous waste container. For small solution spills, dilute with water and wipe up using paper towel. Risks are minimised by wearing safety glasses, lab coat and gloves and dispensing minimal quantities. Instruct students to wash any chemical off skin immediately. In case of inhalation, move to fresh air. In case of chemical in eye, an eye wash or safety shower should be used. In case of ingestion, consult **Poisons Information Centre PH 131126** and seek medical assistance.

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Leave potassium permanganate waste in the beakers for the laboratory technician to collect. The laboratory technician will reduce the potassium permanganate solution with a weak reducing agent such as sodium thiosulfate and neutralise with dilute hydrochloric acid. The waste can then be poured down the drain with plenty of water or disposed of via a chemical waste company.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

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Unit 9.1 Prac 2 Convection currents

Notes

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Unit 9.4 Prac 1 Volcanic clouds

Description of procedure/requirements

Chemicals required: Class sets: Food dye
Equipment: Large beaker or pneumatic trough or transparent jar, 100 mL flask, rubber stopper with hole, electric kettle (optional)
Procedure: Refer to Prac 1 Unit 9.4

HAZARDOUS as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC)

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue

NOT hazardous as classified by ASCC (NOHSC) (with NO allocated Dangerous goods class)

Food dye			

Other hazards and safety considerations

Suggestion: Heating the water in an electric kettle at the front bench, and allowing the students to collect it from the front, is the safest way to do this experiment (instead of all students heating water). There are no hazardous substances used or produced in this experiment. Risks minimised by wearing heatproof gloves, safety glasses and lab coat. Spills should be wiped up immediately using paper towel, avoiding contact with skin and eyes.

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Unit 9.4 Prac 1 Volcanic clouds

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	Yes		Heatproof mat, heatproof gloves and tongs
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Test samples may be washed down the drain with plenty of water.

Assessor/ s: _____ Date: _____ Earliest MSDS expiry date: _____

Notes

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Heating

Apply to all pracs requiring the use of Bunsen burners and/or matches

Description of procedure/requirements

Lighting of, and use of, Bunsen burners and matches.

HAZARDOUS substance as classified by ASSC (NOHSC) criteria

Reactant substance	Dangerous good	Hazard category	Risk phrases	Safety phrases	MSDS issue
Gas – mains supply (methane) gas	2 Flammable gas		**R12 –** Extremely flammable. **R20 –** Harmful by inhalation. **R33 –** Danger of cumulative effects.	**S44 –** If you feel unwell, contact a doctor or **Poisons Information Centre immediately (PH 131126)** (show the label if possible).	
Matches	4.1		Extremely flammable.	Wear suitable protective clothing and glasses. Store away from naked flames, sparks and ignition sources.	

Other hazards and safety considerations

Ensure the Bunsen burner is on a heatproof mat and the area is clear of flammable items.
Always use the safety flame when the Bunsen burner is lit but not in use.
When heating test tubes, ensure they point away from people and equipment.
Use tongs or heatproof gloves to handle hot items. Place hot items on a heatproof mat to cool.
Beware of the scald risk associated with hot liquids. Where possible, allow to cool prior to moving.
In the case of a gas tap being inadvertently left on, remove ignition sources, ventilate the area and remove personnel as required.
Allow all equipment to cool prior to packing away.
Staff must be trained in location and use of fire-fighting equipment.
Staff must know the location of safety cut-off switches, especially for the gas supply.

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Heating

Protective measures

Lab coat	Glasses	Gloves	Fume cupboard	Other
Yes	Yes	—	—	
Tie long hair back Wipe bench/wash hands on completion of prac				

Disposal of wastes

Allow matches to cool on an inflammable surface prior to discarding in the bin.

Assessor/ s: _____ **Date: _____** **Earliest MSDS expiry date: _____**

Notes
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1: Forensics

The Safety Notes in *Science Focus 3* are designed to provide a quick reference guide for teachers to assess the requirements and Health and Safety considerations for each practical.

Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
1.1.1	Make your own Identikit	—	*Equipment*: access to digital camera with computer, printer, A4 paper, scissors or Stanley knife, sticky tape or glue, camera tripod (optional)	Instruct students on the safe use of equipment. Make sure you have permission to take photos of other students (get parental permission if needed). Stanley knives are extremely sharp and will cause serious injury if not used carefully.	**No safety equipment required.**
1.1.2	Fingerprints	—	*Equipment:* blank paper, ink pad	Use non-toxic ink. Ink will stain clothes and it may irritate the skin of some students.	Lab coat. Wash hands. Wipe benches.
1.1.3	Collecting teeth impressions	—	'jelly' lollies such as frogs or snakes (one per student) *Equipment*: grey-lead pencil, string, ruler, access to a mirror, tracing paper (optional)	Instruct students to wash hands before touching lollies. Instruct students not to consume lollies as some students may be diabetic or allergic to the dyes and flavourings in the lollies.	Wash hands.
1.1.4	Forensic anthropometry	—	*Equipment*: string, metre ruler or tape measure, pencil and calculator, access to a model human skeleton	Only take the measurements of willing participants.	**No safety equipment required.**
1.2.1	Testing handwriting	—	*Equipment*: 2 pieces of paper, hard-tipped pen (not felt pen), grey-lead pencil, lamp or torch	Burn risk: Do not touch the lamp, as it can become very hot.	**No safety equipment required.**

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1: Forensics

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
1.2.2	Chromatography catches a criminal	ethanol or methylated spirits	*Equipment*: a 'ransom' note written with an unknown brand of black felt pen, several different black felt pens including the one that wrote the note, the name of each 'suspect' attached, gas jars, scissors, paper	Methylated spirits is highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or gas heaters. Methylated spirits or ethanol may be substituted with salt solution (0.1 M) with water-based pens.	**Refer to Risk Assessment:** **RA 1.2.2: Chromatography catches a criminal.** Lab coat/safety glasses. Wash hands.
1.2.3	Examining money	—	*Equipment*: access to banknote (doesn't need to be Australian), magnifying glass or stereo microscope, access to UV light (optional)	The lens of a magnifying glass can easily shatter. When transporting the stereo microscope, instruct students to hold the microscope with two hands, one supporting the stage and the other the neck. Students should take care not to bump a person using a microscope as serious eye injury can occur.	**No safety equipment required.**
1.2.4	Forging a photograph	—	*Equipment*: computer, digital camera, a classmate, image manipulation software (Adobe Photoshop, Microsoft Photodraw etc.)	Make sure you have permission to take photos of other students (get parental permission if needed).	**No safety equipment required.**

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1: Forensics

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
1.3.1	Comparing fingerprints	graphite powder	*Equipment*: clean microscope slides, small soft bristle brushes or puff-brushes, sticky tape, paper	The glass microscope slide may shatter. Handle broken slides carefully and avoid sharp areas. Discard used slides into a glass or sharps bin. Avoid inhalation of graphite. Wash hands thoroughly after prac.	**Refer to Risk Assessment: RA 1.3.1: Comparing fingerprints.** Lab coat/safety glasses. Wash hands. Wipe benches.
1.3.2	Fibre analysis	—	*Equipment*: sticky tape, hairbrush, white paper, monocular and stereo microscopes, two microscope slides, pencil, samples of fibres (natural, animal, vegetable and synthetic)	Instruct students on the use and care of microscopes, especially in transporting them. Warn students that incorrect focusing techniques may result in broken cover slips/slides. Handle broken slides carefully and avoid sharp areas. Discard used slides into a glass or sharps bin. Some fibres (e.g. animal fur) may cause allergic reactions. Do not use glass fibres (used to make fibreglass) as they pierce the skin easily. Do not use asbestos fibres as they cause asbestosis (lung disease).	**No safety equipment required.**

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1: Forensics

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
1.3.3	Collecting foot impressions	—	plaster of Paris, salt, water *Equipment*: a 5-cm-wide cardboard strip, paddle-pop sticks, plastic disposable cups, stapler, shoes	If you wish to perform this prac in the lab, provide tubs of soil for the impressions. Soil presents a potential pathogen risk. Do not use soil that has had recent manure or organic content additions. Soil can be discarded onto the school gardens or via the school waste system. Plaster of Paris poses a mechanical irritant risk on contact with eyes. Do not inhale dust. Using washable trays or newspaper under areas where the plaster is used may assist in the clean-up. Ensure work areas are wiped clean at the end of the prac. Mixing the plaster in a flexible/disposable container (e.g. ice-cream container) will allow solid residues to be easily removed and the container will be reusable. Allow excess plaster to set and discard it via the school waste system. Do not flush it down the sink. Providing a container of water (small bucket) to wash the finished plaster shoe prints will minimise soil in the lab pipes. The water can be discarded on the garden.	Lab coat. Wash hands. Wipe benches.

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1: Forensics

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
1.3.4	Investigating stride length	—	*Equipment*: may need rulers or measuring tapes	Devise and test a method for determining the distance between footprints when walking. Instruct students to treat all the equipment and activities with care and respect.	**No safety equipment required.**
1.3.5	Blood drips	natural gas, matches	starch, red ink (e.g. for a stamp pad), brown food colouring *Equipment*: 100 mL hot water, butcher's paper, metre ruler, electronic balance, 250 mL beaker, dropper bottle, spatula, small funnel, heating apparatus	Shattering of glass beakers and funnels. Heating water: fire/scald risk. Instruct students to be cautious when heating water as boiling water may scald. Ink may cause skin irritation in some students. Non-toxic red food dye can be substituted for red ink. Use a soluble starch such as potato starch. Ensure the 'dripping' of blood is a controlled exercise and that the surrounding area is well protected from stray droplets (newspaper is ideal).	**Refer to Risk Assessment: RA: Heating.** Lab coat. Wash hands. Wipe benches.

*** = No Safety Notes required**

DG = Dangerous good

HS = Hazardous substance as classified by ASCC criteria

NOT HS = NOT a hazardous substance as classified by ASCC criteria

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2: The periodic table

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.1.1	Making elements	nickel (Ni) electrodes	0.1 M sulfuric acid (H_2SO_4) 2 metal electrodes: platinum (Pt) **PRODUCT** hydrogen gas (H_2) (very small volume) **PRODUCT** oxygen gas (O_2) *Equipment:* large beaker, 9–12 V power supply or battery, alligator clips, safety glasses	Shattering of glassware. Suggestion: use nickel or platinum electrodes which are hardly oxidised. Carbon electrodes are cheaper and safer to use, although they oxidise. Minimise risk by using a weak concentration of solution. Instruct students on safe use of power supplies. Construct the circuit with the power off. Do not touch the circuit while the power is on. Use only the recommended voltage. Make sure the electrodes do not contact each other. Do not allow power supplies/electrical equipment to come into contact with liquids.	**Refer to Risk Assessment: RA 2.1.1: Making elements.** Lab coat, safety glasses and gloves. Paper towel. Tie long hair back.

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2: The periodic table

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.1.2	Making a compound	limewater (saturated solution of calcium hydroxide)	marble chips ($CaCO_3$), 2 M hydrochloric acid **PRODUCT** carbon dioxide (CO_2) gas **PRODUCT** calcium chloride ($CaCl_2$)(aqueous) *Equipment:* 2 test tubes, test-tube rack, drinking straw, 1-hole rubber stopper with glass tubing, safety glasses	Students must wear safety glasses and use a new straw. Instruct students to blow gently to avoid spray coming into contact with their eyes. A long piece of tubing with a disposable straw end will increase the distance of the limewater from students' faces.	**Refer to Risk Assessment: RA 2.1.2: Making a compound.** Lab coat/safety glasses. Gloves could be provided. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.1.3	Compound in soft drinks	—	a selection of 375 mL cans of soft drink, including 'lite' drinks *Equipment*: access to an electronic balance (accuracy 0.01 g) able to measure to 400 g. If scales are not able to read to 400 g, choose small 'mixer' cans of tonic water and dry ginger instead, 500 mL measuring cylinder, large beaker (over 400 mL), stirring rod	Soft drinks will release more CO_2 at room temperature than at a refrigerated temperature. To assist students in obtaining measurable results ensure that each soft drink tested is at room temperature. Handle glassware with care. Ensure liquid is contained and floors do not become a slip hazard.	Lab coat. Wash hands. Wipe benches.
2.2.1	Investigating a physical property	cube or cylinder of lead	cubes or cylinders of aluminium, brass, wood, ice, a collection of small items such as pebbles, candles, chunks of concrete, copper wire *Equipment:* electronic balance, rulers, beakers, measuring cylinders	Lead may be absorbed through the skin on prolonged exposure which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac.	**Refer to Risk Assessment: RA 2.2.1: Investigating a physical property.** Lab coat/safety glasses. Gloves. Wash hands. Wipe benches.

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2.2.2	Comparing elements	lead, magnesium ribbon, calcium granules	sulfur, aluminium, carbon, silicon, tin, zinc, iron, steel wool *Equipment*: 3 or 4 test tubes and rack, power pack about 2 V or battery, wires with alligator clips, light globe, safety glasses	Lead may be absorbed through the skin on prolonged exposure which may result in symptoms of poisoning. Consider omitting this element or have the lead set up as a demonstration station. Students are advised to wear gloves and wash hands thoroughly on completion of prac. Power supplies should be checked for damaged leads and malfunction prior to use. Instruct students on the safe use of power supplies. Construct the circuit with the power off. Do not touch the circuit while the power is on. Use only the recommended voltage. Do not allow power supplies/electrical equipment to come into contact with liquids.	**Refer to Risk Assessment: RA 2.2.2: Comparing elements.** Lab coat/safety glasses. Gloves. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.3.1	Firework colours	natural gas, matches, 0.1 M of barium chloride, strontium chloride solutions	0.1 M copper chloride, potassium chloride, sodium chloride *Equipment*: Bunsen burner, bench mat, tongs, safety glasses, wooden icy-pole sticks soaked overnight in distilled water, spectroscope (optional)	Suggested concentration for solutions is 0.1 M. Gas/matches/flame: Heating: burning of hazardous and non-hazardous chemicals. Beware of flames: fire risk. Remove flammable items from vicinity. Be aware of the sensitivity of your fire alarm system. Instruct students not to 'burn' the icy-pole sticks. Extinguish icy-pole sticks by immersing them in a beaker of water. Do not allow chemical-soaked icy-pole sticks to come into contact with skin; use tongs. When using the spectroscope ensure you are a safe distance from the Bunsen burner flame. Do not aim the spectroscope, or look directly, at the Sun.	**Refer to Risk Assessment:** **RA: Heating.** **RA: 2.3.1: Firework colours.** Lab coat/safety glasses. Heatproof mat. Tongs. Tie back long hair. Wash hands. Wipe benches.
2.3.2	Ions get together!	—	*Equipment:* photocopy of Worksheet 2.3 'Ions'	Instruct students to cut away from themselves and to keep fingers out of cutting path.	No safety equipment required.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.3.3	Metal crystals	natural gas, matches, solid of silver nitrate $AgNO_3$, lead(II) nitrate $Pb(NO_3)_2$, copper(II) sulfate $CuSO_4$, tin(II) chloride $SnCl_2.2H_2O$ **PRODUCT** lead (Pb)	1 cm × 4 cm strip clean zinc sheet, agar powder, 40 mL distilled water **PRODUCT** silver (Ag) **PRODUCT** tin (Sn) **PRODUCT** copper (Cu) **PRODUCT** zinc nitrate solution in agar (very low concentration) **PRODUCT** zinc sulfate solution in agar (very low concentration) **PRODUCT** zinc chloride solution in agar (very low concentration) *Equipment*: sterilised Petri dish, 250 mL beaker, Bunsen burner, tripod, gauze mat, bench mat, stirring rod,	Gas/matches/flame: Heating: preparing agar. Beware of scald/burn risk. Consider performing this prac as a demonstration or provide prepared Petri dishes for students to observe. Silver nitrate stains: stained skin and spills can be decolourised with 4% sodium thiosulfate. **Note**: 0.3g of lead nitrate, copper(II) sulfate or tin(II) chloride can be used instead of silver nitrate although the crystals are not nearly as well formed or as large. Beware of hot objects. To prevent burns, turn off any burner not in use or turn it to the visible yellow flame.	**Refer to Risk Assessment: RA: Heating. RA 2.3.3: Metal crystals.** Lab coat/safety glasses. Gloves. Heatproof mat. Heatproof gloves. Tie back long hair. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
			stereo microscope (optional), safety glasses, gloves		
2.3.4	More crystals		0.1 M silver nitrate ($AgNO_3$) (1.7%), copper wire/foil, 4% sodium thiosulfate (spill clean-up) **PRODUCT** silver (Ag) **PRODUCT** copper nitrate solution (weak concentration) *Equipment*: 100 mL conical flask, cork or rubber stopper, silver nitrate solution, 10–15 cm length of copper wire and/or strip of copper foil, safety glasses, gloves	Suggestion for the concentration of silver nitrate solution is 0.1 M solution. HS: student designs the experiment using Figure 2.3.9. Assess the safety of the experiments prior to them being performed. Silver nitrate stains: spills can be decolourised with 4% sodium thiosulfate. Shattering of glassware.	**Refer to Risk Assessment: RA 2.3.4: More metal crystals.** Lab coat/safety glasses. Gloves could be provided. Wash hands. Wipe benches. Other safety requirements as assigned by the teacher.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.3.5	Changing the properties of metals	natural gas, matches	*Equipment*: four steel hairpins, steel wool, Bunsen burner, bench mat, 250 mL beaker filled with water, wooden peg, safety glasses, pliers (optional)	Gas/match/flame: Heating: metal. Beware of hot items and burn risk. Use tongs to hold hot items and place them on a heatproof mat to cool. Shattering of glassware.	**Refer to Risk Assessment: RA: Heating.** Lab coat/safety glasses. Heatproof mat. Tongs. Tie back long hair.
2.3.6	Using metals to make non-metals	matches, magnesium ribbon	iron, copper, 2 M hydrochloric acid in a dropping bottle **PRODUCT** magnesium chloride ($MgCl_2$) (aqueous) **PRODUCT** ferrous chloride ($FeCl_2$) (weak solution) **PRODUCT** hydrogen (H_2) gas (very small volume) *Equipment:* test tubes and rack, safety glasses	Matches/flame: igniting hydrogen (pop test). Use stoppers rather than fingers to collect the gas. Reducing the concentration to 1 M should still provide effective results. Shattering of glassware.	**Refer to Risk Assessment: RA: Heating.** **RA 2.3.6: Using metals to make non-metals.** Lab coat/safety glasses. Gloves could be provided. Tie back long hair. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
2.4.1	The alkaline earths	natural gas, matches, 5 cm strip of magnesium, a small sample of calcium, phenolphthalein indicator	distilled water **PRODUCT** mixture of phenolphthalein indicator and magnesium hydroxide solutions (weak concentration) **PRODUCT** mixture of phenolphthalein indicator and calcium hydroxide solutions (weak concentration) *Equipment*: crucible, lid, clay triangle, 2 test tubes and rack, Bunsen burner, tripod, bench mat, tongs, safety glasses, steel wool or emery paper	Gas/matches/flame: Heating: liquids in a test tube. Beware of hot items and burn risk. Point test tubes away from people when heating. Use tongs to hold hot test tubes and place them in a test tube rack to cool. Shattering of glassware.	**Refer to Risk Assessment: RA: Heating** **RA 2.4.1: The alkaline earths.** Lab coat/safety glasses. Heatproof mat. Tongs. Tie back long hair. Wash hands. Wipe benches.
2.4.2	Group IV	lead, graphite	charcoal, silicon *Equipment*: power pack or battery, leads with alligator clips, light	Lead may be absorbed through the skin on prolonged exposure, which may result in symptoms of poisoning. Students are advised to wear gloves and wash hands thoroughly on completion of prac.	**Refer to Risk Assessment: RA 2.4.2: Group IV.** Lab coat. Gloves. Wash hands. Wipe benches.

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*** = No Safety Notes required**

DG = Dangerous good

HS = Hazardous substance as classified by ASCC criteria

NOT HS = NOT a hazardous substance as classified by ASCC criteria

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3: Chemical change

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.1.1	Chemistry in the kitchen		*Equipment*: ingredients and kitchen utensils to make your favourite cookie or muffin recipe or pre-made cake-mix, access to an oven	Beware of hot items and burn risk. Use heatproof gloves.	Heatproof gloves. Tie back long hair.

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3: Chemical change

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.1.2	Signs of chemical change	natural gas, matches, solid copper carbonate, magnesium, dilute nitric acid, 0.1 M barium nitrate	solid zinc, 0.1 M sodium sulfate, 0.1 M sodium hydroxide, 0.1 M copper sulfate **PRODUCT** carbon dioxide gas (CO_2) **PRODUCT** magnesium nitrate (weak solution) ($Mg(NO_3)_2$) **PRODUCT** hydrogen gas (H_2) (very small volume) **PRODUCT** sodium nitrate (weak solution) ($NaNO_3$) **PRODUCT** barium sulfate ($BaSO_4$) **PRODUCT** zinc sulfate ($ZnSO_4$) (very diluted solution) *Equipment*: splint, Bunsen burner, test-tube holder/tongs, test-tube rack, 5 test tubes (1 with stopper), thermometer, lab coat, safety glasses	Note: 0.1 M is a suitable concentration for all the solutions except for nitric acid where a concentration of 1 M will produce a desired amount of gas. Gas/matches/flame: Heating: solids in a test tube. Beware of hot items and burn risk. Use tongs to hold hot test tubes and place them in a test-tube rack to cool. Use spirit thermometers rather than mercury thermometers.	**Refer to Risk Assessment RA: Heating.** **RA 3.1.2: Signs of chemical change.** Lab coat/safety glasses. Gloves. Heatproof mat. Tongs. Tie back long hair. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.1.3	Light sticks: chemiluminescence	—	*Equipment*: 2 × 250 mL beakers, ice, hot tap water, 2 light sticks (from scuba diving store)	Refer to the light stick safety instructions for proper use and disposal. The chemicals are enclosed in a sealed container and do not pose a risk to students when used according to instructions.	Lab coat.
3.2.1	Burning magnesium ribbon (teacher demonstration)	15–20 cm of magnesium ribbon, natural gas, matches	**PRODUCT** magnesium oxide (MgO) solid *Equipment:* tongs, lighter, safety glasses, gloves, lab coat	Gas/match/flame: Heating Do not look directly at the magnesium as it is very bright and may result in eye damage. Beware of flames and hot items.	**Refer to Risk Assessment:** **RA: Heating.** **RA: 3.2.1 Burning magnesium ribbon (teacher demonstration).** Lab coat/safety glasses. Gloves. Heatproof mat. Tongs. Tie back long hair.

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3: Chemical change

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.2.2	Stopping combustion	matches, limewater $Ca(OH)_2$)	sodium bicarbonate ($NaHCO_3$), 0.1 M hydrochloric acid **PRODUCT** sodium chloride (NaCl) solution **PRODUCT** carbon dioxide (CO_2) gas **PRODUCT** water (H_2O) *Equipment:* 250 mL conical flask, rubber stopper with a flexible plastic tube passing through it, large test tube with rubber stopper, wooden splint, matches	Beware of flames. Extinguish the wooden splint in a safe way.	**Refer to Risk Assessment:** **RA: Heating.** **RA: 3.2.2 Stopping combustion.** Lab coat/safety glasses. Gloves. Tie back long hair.
3.2.3	Decomposition of a metal carbonate	copper carbonate ($CuCO_3$), limewater ($Ca(OH)_2$), natural gas, matches **PRODUCT** copper(II) oxide (CuO) solid	**PRODUCT** carbon dioxide (CO_2) gas *Equipment:* two large test tubes, stopper for the test tube with a delivery tube, retort stand and clamp, Bunsen burner	Beware of hot objects. Shattering of glassware. Hint: instruct the students to lift the delivery tube from the limewater as soon as the heating is stopped to avoid any liquid sucking back.	**Refer to Risk Assessment:** **RA: Heating.** **RA: 3.2.3 Decomposition of a metal carbonate.** Lab coat/safety glasses. Gloves. Tie back long hair.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.3.1	Precipitation reactions	0.1 M barium nitrate ($Ba(NO_3)_2$)	0.1 M solutions of silver nitrate ($AgNO_3$) (1.7%), sodium chloride (NaCl) (0.6%), sodium hydroxide (NaOH) (0.4%), copper sulfate ($CuSO_4.5H_2O$) (2.4%) **PRODUCT** silver chloride (AgCl) **PRODUCT** silver oxide (Ag_2O) **PRODUCT** sodium nitrate solution ($NaNO_3$) **PRODUCT** barium sulfate ($BaSO_4$) **PRODUCT** copper nitrate solution ($Cu(NO_3)_2$) (weak concentration) *Equipment:* test tubes, test-tube rack, Pasteur pipettes, lab coat, safety glasses, gloves	Students to exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. Hint: dispense the solutions in dropping bottles to minimise spillage.	**Refer to Risk Assessment:** **RA: 3.3.1 Precipitation reactions.** Lab coat/safety glasses. Gloves. Tie back long hair.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.3.2	Precipitation of unknowns	0.1 M solutions of lead nitrate and barium nitrate **PRODUCT** lead chloride ($PbCl_2$) solid **PRODUCT** lead sulfate ($PbSO_4$) solid **PRODUCT** lead iodide (PbI_2) solid **PRODUCT** lead carbonate ($PbCO_3$) solid **PRODUCT** silver sulfate (Ag_2SO_4) solid **PRODUCT** barium carbonate ($BaCO_3$) solid	0.1 M solutions of sodium iodide, sodium chloride, sodium sulfate, sodium carbonate, sodium nitrate, silver nitrate, calcium nitrate **PRODUCT** silver chloride (AgCl) solid **PRODUCT** silver iodide (AgI) solid **PRODUCT** silver carbonate **(**Ag_2CO_3) solid **PRODUCT** barium sulfate ($BaSO_4$) solid **PRODUCT** calcium carbonate ($CaCO_3$) solid **PRODUCT** calcium sulfate ($CaSO_4$) solid *Equipment*: 20 semi-micro test tubes, Pasteur pipettes, lab coat, safety glasses, gloves	Students to exercise care when handling silver nitrate as it can readily burn or stain skin and surfaces. Silver nitrate stains can be decolourised with sodium thiosulfate 4%. Hint: slightly more concentrated solutions are needed to form silver sulfate, calcium sulfate and lead chloride precipitates.	**Refer to Risk Assessment: RA 3.3.2: Precipitation of unknowns.** Lab coat/safety glasses. Gloves. Tie back long hair. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.4.1	Common indicators	universal indicator, methyl red solution, phenolphthalein indicator, bromothymol blue solution	0.1 M sodium hydroxide, 0.1 M hydrochloric acid, methyl orange solution, liquid red and blue litmus, distilled water *Equipment*: 3 test tubes, test-tube rack, 3 × 100 mL beakers, lab coat, safety glasses	Phenolphthalein indicator is flammable. Do not use near a flame or source of ignition. Indicators may stain the skin. Gloves should be worn and hands washed thoroughly on completion of prac.	**Refer to Risk Assessment: RA 3.4.1: Common indicators.** Lab coat/safety glasses. Gloves. Wash hands. Wipe benches.
3.4.2	Universal indicator	1.0 M sodium hydroxide, universal indicator	0.1 M hydrochloric acid, distilled water *Equipment*: Pasteur pipette, 10 mL measuring cylinder, waterproof felt-tip pen, 14 large test tubes, test-tube rack, lab coat, safety glasses	Universal indicator is flammable. Do not use near a flame or sources of ignition. Instruct students to exercise care when handling 1 M sodium hydroxide as it is corrosive. Gloves should be worn and hands washed thoroughly on completion of prac.	**Refer to Risk Assessment: RA 3.4.2: Universal indicator.** Lab coat/safety glasses. Gloves. Wash hands. Wipe benches.

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3: Chemical change

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.4.3	Natural indicators	natural gas, matches	0.1 M sodium hydroxide, 0.1 M hydrochloric acid, distilled water, pink or red flower petals, beetroot juice, tea bag *Equipment*: 3 × 100 mL beakers, filter paper cut into strips, Bunsen burner, heatproof mat, tripod, gauze, 3 test tubes, test-tube rack, Pasteur pipettes, lab coat, safety glasses	Gas/matches/flame: boiling liquids. Beware of hot items/burn risk and flame/fire risk. Use tongs or heatproof gloves to handle hot items. Place them on a heatproof surface to cool. Use plants that are not allergenic or toxic.	**Refer to Risk Assessment: RA: Heating.** **RA3.4.3: Natural indicators.** Lab coat/safety glasses. Heatproof mat. Heatproof gloves or tongs. Tie back long hair. Wash hands. Wipe benches.
3.4.4	Testing household solutions	universal indicator	non-hazardous household solutions including orange juice, soft drink, fresh and sour milk, vinegar (solids may be used if dissolved in water first), distilled water, blue and red litmus paper *Equipment*: test tubes, test-tube rack, Pasteur pipettes, 2 watch-glasses, safety glasses, lab coat	Assess the hazards of the household chemicals selected and provide appropriate safety precautions if hazardous substances are chosen. Universal indicator is a flammable liquid and should not be stored or used near ignition sources. Indicators stain: gloves can be provided.	**Refer to Risk Assessment: RA 3.4.4: Testing household solutions.** Lab coat/safety glasses. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.4.5	Acids and hydroxides	universal indicator	0.1 M sodium hydroxide, 0.1 M hydrochloric acid *Equipment*: 250 mL beaker, 50 mL measuring cylinder, digital pH meter, Pasteur pipette, lab coat, safety glasses	Instruct students to handle the digital pH meters with care. Universal indicator is a flammable liquid and should not be stored or used near ignition sources. Indicators stain: gloves can be provided.	**Refer to Risk Assessment: RA 3.4.5: Acids and hydroxides.** Lab coat/safety glasses. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
3.4.6	Acids and carbonates	matches, limewater ($Ca(OH)_2$), solid sodium carbonate	solid samples of sodium hydrogen carbonate, lithium carbonate, ammonium carbonate, 0.1 M hydrochloric acid, 0.1 M nitric acid (HNO_3) (0.63%) **PRODUCT** carbon dioxide (CO_2) gas **PRODUCT** sodium nitrate ($NaNO_3$) (very diluted solution) **PRODUCT** lithium nitrate ($LiNO_3$) (very diluted solution) **PRODUCT** lithium nitrate ($LiNO_3$) **PRODUCT** sodium chloride (NaCl) (very diluted solution) **PRODUCT** lithium chloride (LiCl) (very diluted solution) **PRODUCT** ammonium chloride (NH_4Cl) (very diluted solution)	Matches/flame: testing for carbon dioxide. Minimise the quantity and volume of acid provided in the class sets. Students produce, and test for, carbon dioxide (extinguish a match). The quantity of gas produced is not hazardous in a well-ventilated area.	**Refer to Risk Assessment:** **RA: Heating.** **RA 3.4.6: Acids and carbonates.** Lab coat/safety glasses. Gloves. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
			Equipment: test tubes, test-tube rack, rubber stopper, 100 mL beaker, spatula, lab coat, safety glasses		
3.4.7	Acids and metals	matches, magnesium	aluminium, zinc, iron, tin, 0.1 M hydrochloric acid, 0.1 M sulfuric acid, 0.1 M acetic acid **PRODUCT** hydrogen gas (H_2) (very small volume) **PRODUCT** aluminium chloride ($AlCl_3$) (very diluted solution) **PRODUCT** magnesium chloride ($MgCl_2$) (very diluted solution) **PRODUCT** zinc chloride ($ZnCl_2$) (very diluted solution) **PRODUCT** iron(II) chloride ($FeCl_2$) (very diluted solution) **PRODUCT** tin(II) chloride ($SnCl_2$) (very diluted solution)	Avoid splashes of acid to eyes and skin. Wear safety glasses. Handle hot objects and glassware with care.	**Refer to Risk Assessment:** **RA: Heating.** **RA 3.4.7: Acids and metals.** Lab coat/safety glasses. Tie back long hair. Wash hands. Wipe benches.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
			PRODUCT magnesium sulfate ($MgSO_4$) (very diluted solution) **PRODUCT** iron(II) sulfate ($FeSO_4$) (very diluted solution) **PRODUCT** magnesium ethanoate ($(CH_3COO)_2Mg$) (very diluted solution) *Equipment:* test tubes with stoppers, test-tube rack, 100 mL beaker, lab coat, safety glasses		

*** = No Safety Notes required**

DG = Dangerous good

HS = Hazardous substance as classified by ASCC criteria

NOT HS = NOT a hazardous substance as classified by ASCC criteria

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4: Sense and control

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
4.1.1	Eye tests		*Equipment*: eye charts	No safety notes required.	No safety equipment required.
4.1.2	Persistence of vision		*Equipment*: wooden rod (e.g. bamboo skewer), stiff white cardboard, pencils, sticky tape, scissors	If using skewers, cut off pointed end with strong scissors to minimise risk of stab injury.	No safety equipment required.
4.1.3	Eye dissection	No	0.5 % bleach solution *Equipment*: a cow's eye, sharp fine scissors, paper towel, dissection board, newspaper	Frozen eyes are recommended as preserved eyes have an overpowering, unpleasant smell. Animal tissue should be handled with care and should be treated as a potential source of infection. Cuts, sores or broken skin should be covered. Protect bench area with newspaper. Instruct students to wash any animal blood/tissue off skin immediately using antibacterial hand wash. Instruct students to cut away from themselves and to keep fingers out of cutting path.	**Refer to Risk Assessment: RA: 4.1.3.** Lab coat/safety glasses. Gloves may be worn. Wash hands after prac. Tie back long hair. Disinfect bench and equipment.
4.1.4	Lens and retina		*Equipment*: convex lens, sheet of paper	Instruct students to avoid focusing towards sunlight.	No safety equipment required.
4.2.1	Hearing tests 1		blindfold, tape measure	Being blindfolded can be a frightening experience for some people. Encourage students to ask their partner's permission before blindfolding them.	No safety equipment required.

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4: Sense and control

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
4.2.2	Modelling hearing difficulties		*Equipment*: a grey-lead pencil, a book that you are allowed to write in with a pencil, eraser	No safety notes required.	No safety equipment required.
4.2.3	Measuring decibels		*Equipment*: a sound level meter	Instruct students to keep a safe distance from traffic.	No safety equipment required.
4.3.1	Taste regions		sugar, salt, vinegar or lemon juice, coffee *Equipment*: clean cotton buds, samples of the solutions in new plastic cups, blindfold	Provide fresh stock solutions for each class. Disinfect surfaces, line with clean butcher's paper if in doubt. Use new disposable containers; e.g. disposable cups. Do not use beakers or other lab glassware. Contamination risks can be minimised by providing a new cup per person and the teacher dispensing the stock solutions. Allow one use only of cotton buds and ensure immediate disposal into a lined waste bin. Waste can be disposed of in a sealed bag via the general school waste system.	**Refer to Risk Assessment: RA: 4.3.1.** Lab coat/safety glasses. Gloves may be worn. Wash hands after prac. Tie back long hair. Wash hands prior to handling food.
4.3.2	Taste trickery		*Equipment*: blindfold, apple, potato, pear	Being blindfolded can be a frightening experience for some people. Encourage students to ask their partner's permission before blindfolding them.	No safety equipment required.

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4.3.3	Skin receptors		*Equipment*: toothpicks, tape, ruler, blindfold	Instruct students to take care when touching their partner's skin with toothpicks. Being blindfolded can be a frightening experience for some people. Encourage students to ask their partner's permission before blindfolding them.	No safety equipment required.
4.4.1	Sweet and salty	—	2–3 mL each of solutions of sugar of varying concentrations (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%), 2–3 mL each of solutions of salt (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%) *Equipment*: 12 new small paper cups, waste jar for rinsings, bottled water for rinsing mouth	The solutions are to be tasted. Use food salt and sugar. Prepare the solutions hygienically into appropriate containers, not into lab glassware. Handling of food/eating/tasting should occur outside the lab environment, under controlled hygienic conditions. Consider a room swap for this prac. Students must have individual cups, bottled rinsing water and waste jars. Students should discard waste via the sewer with plenty of water and wash or discard the jars.	Wash hands. Wipe benches.
4.4.2	Sound threshold	—	*Equipment*: an old-fashioned ticking clock, metre ruler or tape measure	If electrical equipment is to be used, check that it is in safe working order before use.	No safety equipment required.

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4.5.1	Brain dissection	—	5% sodium hypochlorite, 0.5% sodium hypochlorite, partly frozen lamb's brain *Equipment*: dissection board, scalpel, dissecting scissors, newspaper	Cuts, sores or broken skin should be covered. Gloves may be worn. Warning: scalpels are extremely sharp. Instruct students on the safe use of scalpels. When cutting, always draw the scalpel AWAY from your hands and body. Protect bench area with newspaper. Sterilise used dissecting equipment in disinfectant or diluted bleach. Consider washable plastic dissection boards. Wipe down benches. Dispose of waste daily or freeze and dispose of when all dissections are completed. Dispose of waste as normal food waste. Preserving brains: fresh brains deteriorate rapidly. Providing frozen brains overcomes the availability problem. Partially frozen brains provide students with a firmer object to cut—they retain integrity of the structures for a period of time. Another method of preserving brains: immerse fresh brains in a concentrated salt solution at 4ºC. Change the solution, increasing the concentration a number of times over 24–48 hours.	**Refer to Risk Assessment: RA: 4.5.1: Brain dissection.** Lab coat/safety glasses. Gloves may be worn. Wash hands. Disinfect bench and equipment.

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4.5.2	A model brain	—	food dye, unpeeled orange, assorted lollies (e.g. 1 banana, 3 jubes, 1 marshmallow, 3 snakes of different colours, 1 spearmint leaf), 2 sultanas *Equipment*: toothpicks, cotton buds, plastic knife, newspaper	Handling/eating of food should occur outside the lab environment, under controlled hygienic conditions. Consider a room swap for this prac. If food is to be eaten, ensure it is handled hygienically. Check students for diabetes, allergies or reactions. Provide the food dye diluted to a minimum effective concentration.	
4.5.3	Brain wars: the Stroop effect	—	*Equipment*: Figure 4.5.17, stopwatch	No safety notes required.	No safety equipment required.
4.5.4	Response time		*Equipment*: examples include a bell, blindfold, a ball, tape measure or metre ruler	Being blindfolded can be a frightening experience for some people. Encourage students to ask their partner's permission before blindfolding them.	No safety equipment required.
4.5.5	What's that smell?		*Equipment*: access to 'secret smell' containers (film canisters with a small ball of cotton wool with a drop or two of a scent capped to stop smells escaping)	Check students for allergies or reactions.	No safety equipment required.

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4.5.6	Chunking		*Equipment*: three lists of words (A, B and C) on an A4 sheet, access to clock or timer, paper, pen	No safety notes required.	No safety equipment required.
4.6.1	Plant hormones	matches	melted paraffin wax, water *Equipment:* 6 shoots of the plant *Tradescantia* (commonly known as wandering jew), 6 test tubes, a darkened area and a well-lit area to place plants	Matches/flame/burning candle: fire risk. Hot wax: burn risk. Para film may provide a suitable alternative to paraffin wax to seal the test tube. This will replace the use of flames and prevent dripping wax.	**Refer to Risk Assessment: RA: Heating.** Heatproof mat. Tie back long hair.
4.6.2	Plants and gravity			All experiments designed by students should be assessed by the teacher for safety issues before carrying them out.	

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
5.1.1	Asexual reproduction in plants	–	an onion and a potato with eyes *Equipment*: knife Note: many other things could be substituted for the onion and the potato; e.g. strawberry runners, Chinese willow stem cuttings or orchid bulbs	Protect benches and provide a washable cutting board. Instruct students on the safe use of the knife. Choose non-toxic/non-allergenic plants. Onion juice may irritate the skin or eyes. Instruct students not to eat any plant, fruit or vegetable.	Lab coat. Wash hands. Wipe benches.
5.1.2	Examination of spores	–	fern leaf with visible spores *Equipment*: stereo microscope, tweezers, filter paper	Instruct students on the care and use of the microscope.	No safety equipment required.
5.1.3	Flower dissection	–	large flower *Equipment*: dissecting instruments, hand lens	Instruct students on the safe use of the dissecting equipment. Protect benches and provide a dissecting board or washable cutting board. Select low-allergenic flowers. Ask students about allergies and asthma. The sap of some plants may (e.g. agapanthus) may cause skin irritation in some people. Wash hands thoroughly after prac.	Lab coat. Wash hands. Wipe benches.

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5.2.1	Model reproductive systems	–	*Equipment*: scrap A4 paper, scissors, coloured felt-tip pens or pencils, sticky tape, cotton thread	Instruct students to take care when using scissors and to keep fingers out of cutting path.	No safety equipment required.
5.3.1	Survey of twins	–	*Equipment*: paper, pen, Excel spreadsheet or calculator and protractor	No safety notes required.	No safety equipment required.
5.4.1	Methods of contraception	–	*Equipment*: access to a selection of contraceptive devices (condom, a blister pack of pills, IUD, diaphragm)	Ensure that the contraceptive pill pack is tamperproof. Education kits containing tamperproof pills can be acquired through the 'Family Planning Association' in your state or territory.	No safety equipment required.

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6: Ecosystems

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
6.1.1	Biomass		*Equipment* beaker, fresh plant matter, access to a drying oven, electronic balance, paper bag	Beware of hot objects.	No safety equipment required.
6.1.2	Food pyramid		*Equipment* ruler, calculator, strips of paper	No safety notes required.	No safety equipment required.
6.2.1	Testing for water	anhydrous copper(II) sulfate, acetic (ethanoic) acid (2 M), cobalt chloride paper, methylated spirits	distilled water, various liquids (tap water, salt water, sucrose solution), unknown liquids X,Y and Z *Equipment* watch-glass, dropper, paper towel	Methylated spirits is highly flammable and should be kept away from all sources of heat such as matches, Bunsen burner flames or gas heaters. Gloves should be worn when handling cobalt chloride paper. Hint: consider choosing a non-hazardous solution for the unknown liquids and using a less concentrated solution of acetic acid.	**Refer to Risk Assessment: RA 6.2.1: Testing for water.** Lab coat/safety glasses. Gloves. Tie back long hair.

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6: Ecosystems

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
6.2.2	Measuring the boiling point of water	matches	distilled water, salt, *Equipment* Bunsen burner and mat, tripod, 4 × 250 mL beakers, measuring cylinder, tablespoon, thermometer able to record temperatures higher than 100ºC or a temperature probe	Beware of hot objects. Heating water: fire/scald risk. Use heatproof gloves to handle hot-water beakers. Instruct students to be cautious when heating water as the beaker can easily be knocked over, splashing persons nearby. Use a spirit thermometer rather than a mercury thermometer.	**Refer to Risk Assessment:** **RA: Heating.** **RA 6.2.2: Measuring the boiling point of water** Lab coat/safety glasses. Gloves. Tie back long hair. Heatproof gloves
6.3.1	Water pollution		cooking oil, liquid soap, vinegar, soil, sand, grass clippings, alfalfa sprouts *Equipment* jar with lid, toilet paper, strips of plastic, teaspoon, wire gauze, blotting paper, cotton wool, filter funnel, filter paper, 3 beakers labelled A,B and C, Petri dish	Check students' records for allergies. Some grasses are known to cause allergic reactions in some people.	Lab coat/safety glasses. Gloves.
6.3.2	Climate in a beaker	matches	ice cubes *Equipment* large beaker or glass bowl, aluminium foil, paper tapers, thermometer	Beware of hot objects.	**Refer to Risk assessment:** **RA: Heating.**

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
6.3.3	Simulating global warming		*Equipment* two small juice cartons (poppers), sticky tape, 2 thermometers, scissors, at least 4 microscope slides	Use a spirit thermometer rather than a mercury thermometer. Dispose of used glass slides in the glass bin.	No safety equipment required.

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7: Light

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
7.1.1	Measuring angles of refraction	–	*Equipment:* light box and single-slit slide, 12 V power source, sheet of paper, ruler, polar graph paper, (or protractor or Mathomat), semicircular slab of Perspex	Instruct students on the correct use and care of light boxes and power supplies. Ensure the voltage selected is suitable for the globe tolerance in your light boxes. A darkened room will assist this exercise.	No safety equipment required.
7.1.2	Observing light	–	filtered water, milk *Equipment:* 20 L rectangular fish tank or clear Perspex box, HeNe laser pointer or other collimated light source, dropper, retort stand, clamp, boss head clamp	Only Class 1 and Class 2 laser pointers are permitted in schools. Never operate the laser at eye-level as the beam, if deflected into the eye, could do serious damage. Before operation of a laser pointer or collimated light source, the teacher in charge of the demonstration or experiment should warn students of the potential hazards and instruct them never to look into the direct beam or the reflected beams. The laser beam should be turned off when it is not in use. Risks can be minimised by the use of Class 1 laser pointers, by performing this practical as a teacher demonstration and by wearing safety glasses. For further information refer to http://www.nhmrc.gov.au	**Refer to Risk Assessment: RA: 7.1.2.** Lab coat. Safety glasses. Tie back long hair.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
7.1.3	Refraction illusions	–	water *Equipment:* beaker, ruler, opaque bowl (e.g. icecream container), coin or small heavy object	Ensure water is contained and floors do not become a slip hazard.	No safety equipment required.
7.1.4	Apparent depth	–	*Equipment:* white paper with several parallel lines drawn on it, rectangular glass/Perspex block, a pin in a cork	Check glass blocks for chips, cracks and sharp edges that may cut skin.	No safety equipment required.
7.2.1	Water lenses	–	*Equipment:* eye dropper or pipette, a printed A4 sheet of glossy paper with fonts of various sizes, small beaker of water, wire loop as shown in Fig 7.2.21	Ensure water is contained and floors do not become a slip hazard.	No safety equipment required.
7.2.2	Lenses and light box	–	*Equipment*: a light box, multiple-slit slide, 12 V power supply, light box lenses set, sheet of paper	Instruct students on the correct use and care of light boxes and power supplies. Ensure the voltage selected is suitable for the globe tolerance in your light boxes. A darkened room will assist this exercise.	No safety equipment required.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
7.2.3	Images in a convex lens	–	wax candle, matches *Equipment*: a convex lens, a concave lens, white card or screen, plasticine or lens holder, metre ruler, candle or small globe with power supply	Matches/flame: burning a candle as a light source. Keep the flame away from flammable material, especially clothing or hair. Minimise fire risk: use a globe as a light source if possible. A darkened room will assist this exercise.	**Refer to Risk Assessment: RA: Heating.** Lab coat. Heatproof mat. Tie back long hair.
7.2.4	Simulating imperfect vision	–	*Equipment:* a convex lens, white card or screen, extra piece of card, plasticine or lens holder, metre ruler, candle or small globe with power supply	Matches/flame: burning a candle as a light source. Keep the flame away from flammable material, especially clothing or hair. Minimise fire risk: use a globe as a light source if possible. A darkened room will assist this exercise.	**Refer to Risk Assessment: RA: Heating.** Lab coat. Heatproof mat. Tie back long hair.
7.2.5	Telescopes and microscopes	–	*Equipment:* 2 convex lenses, one thin (e.g. focal length 25 cm), one thick (e.g. focal length 5 cm), cardboard, scissors, tracing paper or other translucent material (e.g. thin plastic from a shopping bag), lamp, small object to view	Beware of hot wax and the possible fire risk of the candle. Use a globe to minimise risk. Keep the flame away from flammable material, especially clothing or hair.	**Refer to Risk Assessment: RA: Heating.** Lab coat. Heatproof mat. Tie back long hair.

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
7.2.6	Exploring curved surfaces	–	*Equipment:* metal soup spoon	Instruct students to take care when placing objects close to the eye.	No safety equipment required.
7.2.7	Forming images with curved mirrors	–	wax candle, matches *Equipment:* convex mirror, concave mirror, white card or screen, plasticine or lens holder, metre ruler	Matches/flame: burning a candle as a light source Keep the flame away from flammable material, especially clothing or hair. Minimise fire risk: use a globe as a light source if possible. A darkened room will assist this exercise.	**Refer to Risk Assessment: RA: Heating.** Lab coat. Heatproof mat. Tie back long hair.
7.3.1	Dispersion: splitting white light	–	*Equipment:* triangular glass or Perspex prism, light box and power supply, slide with single slit, white paper	Instruct students on correct use and care of light boxes and power supplies. Ensure the voltage selected is suitable for the globe tolerance in your light boxes. A darkened room will assist this exercise.	No safety equipment required.
7.3.2	Mixing coloured light	–	*Equipment:* light box and power supply, variety of coloured slides, white paper	Instruct students on correct use and care of light boxes and power supplies and also not to leave filters in the light box for a long period of time. Ensure the voltage selected is suitable for the globe tolerance in your light boxes. A darkened room will assist this exercise.	No safety equipment required.

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7.3.3	Seeing things in a different light	–	*Equipment:* light box and power supply, variety of coloured slides, variety of coloured cards (Note: the slides and cards should be labelled with their colour.)	Instruct students on the correct use and care of light boxes and power supplies and also not to leave filters in the light box for a long period of time. Ensure the voltage selected is suitable for the globe tolerance in your light boxes. A darkened room will assist this exercise.	No safety equipment required.
7.3.4	Measuring light absorption	–	*Equipment:* 5 metallic cans of equal size with plastic lids; e.g. coffee cans, 5 thermometers, coloured paint (black, white, red, blue and green), timer, light source; e.g. the Sun or a flood lamp	Burn risk: do not touch the lamp, as it can become very hot. Instruct students to handle the glass thermometers with care and to notify the teacher immediately if one breaks. Minimise the risks of students handling paint by providing painted cans. Use water-based paint such as acrylic. Short-term exposure to the Sun can result in sunburn and injuries to the eye. Minimise the risks by avoiding the UV peak time to carry out the practical activity and taking precautions for sunburn. Hint: allocate one colour of can per group to minimise the time of Sun exposure.	**Refer to Risk Assessment: RA: 7.3.4.** Lab coat. Gloves. Tie back long hair.

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7.3.5	Mixing pigments	–	a set of water-colour or acrylic paints *Equipment:* paint brushes, blank paper	Hint: provide small plastic containers per group to mix the colours. Paints are chemicals, so avoid contact with your skin.	Lab coat. Gloves. Tie back long hair.

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8: The universe

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
8.1.1	Using a spectroscope		*Equipment:* spectroscope	Do not aim the spectroscope, or look directly, at the Sun. It can damage your eyes and cause permanent blindness.	No safety equipment required.
8.1.2	A balloon universe		*Equipment:* balloon, pen for marking dots on the balloon	Ensure that each student has one new balloon each and does not share in the task of blowing up the balloon.	No safety equipment required.

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9: Earth's fragile crust

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
9.1.1	The planet Splatter		*Equipment:* A4 photocopy of the map in Fig 9.1.10 or photocopy of Worksheet 9.1	Instruct students to cut away from themselves and to keep fingers out of cutting path.	No safety equipment required.
9.1.2	Convection currents	potassium permanganate ($KMnO_4$)	*Equipment:* large (500 mL or 1000 mL) beaker, tweezers, hot plate or Bunsen burner, bench mat, tripod, gauze mat, plastic bag, ice	Potassium permanganate solution will readily stain skin and surfaces. Beware of hot objects. If possible, use a hot plate instead of a Bunsen burner.	**Refer to Risk Assessments:** **RA: Heating** **RA 9.1.3: Convection currents.** Lab coat/safety glasses. Tie back long hair. Wash hands. Wipe benches.
9.1.3	Future Earth		*Equipment:* map of tectonic plates, A4 sheet of paper, 6 overhead transparencies, 1 sheet of tracing paper, overhead transparency pen or marker, sticky tape, photocopy of Worksheet 9.2	No safety notes required.	No safety equipment required.

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9.2.1	Plates that separate		*Equipment:* A3 sheet of paper, coloured pencils or highlighters, scissors, sticky tape, pegs	No safety notes required.	No safety equipment required.
9.2.2	Colliding plates		*Equipment:* a stack of about 30 A4 pages (recycle: use scrap paper), textbook	No safety notes required.	No safety equipment required.
9.3.1	Slinky springs	—	*Equipment:* slinky spring, dense smaller diameter slinky, sheet of paper or cardboard, sticky tape, string	Ensure that the slinkies on the floor do not create a trip hazard. To minimise tangles in the slinky, instruct students on the appropriate use of this equipment.	No safety equipment required.
9.3.2	Building a seismometer		*Equipment:* retort stand, bosshead, clamp, springs, heavy weight, adhesive tape, felt pen, sheet of paper	Beware of physical injury from heavy weight.	No safety equipment required.
9.3.3	Locating the epicentre	—	*Equipment:* compass, ruler, photocopy of Workbook 9.11	No safety notes required	No safety equipment required.

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9: Earth's fragile crust

The Safety Notes in *Science Focus 3* are designed to provide a quick reference guide for teachers to assess the requirements and Health and Safety considerations for each practical.

Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
9.4.1	Volcanic clouds		hot and cold water *Equipment:* large beaker or pneumatic trough or transparent jar, 100 mL flask, rubber stopper with hole, food dye	Beware of hot water.	**Refer to Risk assessment: RA 9.4.1: Volcanic clouds.** Heatproof mat, heatproof gloves and tongs. Safety glasses, lab coat. Tie back long hair.
9.4.2	Build a 3D model of a volcano		*Equipment:* cardboard, scissors, blades, glue, chicken wire, newspaper, clay, plasticine, string, wood, different colours of paint, paint brushes, ruler	Protect the bench with a mat when cutting the plasticine with the blade. Provide instructions on the appropriate use of the blade. Using plasticine on washable trays or mats may assist in the clean up.	Lab coat.
9.5.1	Faults and folds	—	*Equipment:* plasticine in four colours, a rolling pin, fine wire or a hacksaw blade	Protect the bench with a mat when cutting the plasticine with the hacksaw. Provide instructions on the appropriate use of the hacksaw. Using plasticine on washable trays or mats may assist in the clean-up.	No safety equipment required.

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9: Earth's fragile crust

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
9.5.2	Shaping volcanoes	matches	materials such as fine sand, coarse sand, flour, fine blue metal screenings, candles (old stumps will do) *Equipment:* funnel, retort stand, bosshead and clamp, 1 sheet of graph paper, ruler with millimetre markings, protractor, access to a scientific calculator	Beware of hot objects, especially hot wax. Using washable trays under the graph paper will contain the fine solids and assist in the clean-up.	**Refer to Risk Assessment: RA: Heating.** Heatproof mat. Tie back long hair.
9.6.1	Inspecting fossils	—	*Equipment:* access to a fossils kit, a range of skulls and/or skeletons, grey-lead pencil, eraser	No safety notes required.	No safety equipment required.
9.6.2	Dinosaur fossils		*Equipment:* clay or modelling clay (to make mould), a pin, a probe or blunt pencil, tracing paper or photocopy of the skeleton on page 329, rolling pin or piece of dowel, cardboard or shoe box lid, plaster mix, water	Plaster of Paris poses a mechanical irritant risk on contact with eyes. Do not inhale dust.	Lab coat and gloves.

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9: Earth's fragile crust

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Prac	Name	HS/DG	Not HS	Other hazard/precautions/hints	Recommended safety requirements
9.6.3	'Radioactive' cubes		*Equipment:* 50 or more small wooden cubes, each with one face marked (e.g. with a dot), a cup, graph paper	No safety notes required.	No safety equipment required.

*** = No Safety Notes required**

DG = Dangerous good

HS = Hazardous substance as classified by ASCC criteria

NOT HS = NOT a hazardous substance as classified by ASCC criteria

Disclaimer: The Safety Notes and related Risk Assessment Sheets are provided to offer guidance only. They must not be construed to waive or modify any legal obligation of the school to ensure the safety of students when conducting the experiment or activity. It is the responsibility of the school to have the content of the Risk Assessment Sheets checked against the Material Safety Data Sheets (MSDS) provided by the manufacturer of chemicals used in the school's laboratories. Risk Assessment Sheets must not be used in the school's laboratories until they have been checked against the school's MSDS, signed and dated. To the maximum extent permitted by law, the publisher disclaims all responsibility for actions taken or not taken in relation to the Safety Notes and Risk Assessment Sheets.

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Technician's Checklist: Chemicals Used

(Hazardous classification according to ASCC (NOHSC); check information against your suppliers' current MSDS) (d) = Teacher Demonstration

HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
No	Acrylic paint	None allocated	–		7.3.4, 7.3.5
No	Agar	None allocated	–		2.3.3
No	Aluminium – cubes or cylinders (Al)	None allocated	–		2.2.1
No	Aluminium pieces (Al)	None allocated	–		2.2.2, 3.4.7
No	Ammonium carbonate (NH_4CO_3) solid	None allocated	–		3.4.6
Yes	Barium chloride-hydrated ($BaCl_2.2H_2O$) solid	6.1	Toxic		2.3.1
No	Barium chloride-hydrated ($BaCl_2.2H_2O$) 0.1 M (2.4%)	None allocated	–		2.3.1
Yes	Barium nitrate ($Ba(NO_3)_2$) solid	5.1 Sub. Risk 6.1	Harmful		3.1.2, 3.3.1, 3.3.2
Yes	Barium nitrate ($Ba(NO_3)_2$) 0.1 M	None allocated	Harmful		3.1.2, 3.3.1, 3.3.2
No	Bleach solution (sodium hypochlorite) 5%	None allocated	–		4.1.3, 4.5.1
No	Bleach solution (sodium hypochlorite) 0.5%	None allocated	–		4.1.3, 4.5.1
No	Brass – cubes or cylinders	None allocated	–		2.2.1
No	Bromothymol blue solid	None allocated	–		3.4.1
Yes	Bromothymol blue solution (0.04%)	3	Flammable		3.4.1
Yes	Calcium hydroxide ($Ca(OH)_2$)	None allocated	Irritant		2.1.2, 3.2.2, 3.2.3, 3.4.6
Yes	Calcium metal/granules (Ca)	4.3	Irritant, highly flammable		2.2.2, 2.4.1
Yes	Calcium nitrate ($Ca(NO_3)_2$) solid	5.1	Irritant, oxidising		3.3.2

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
No	Calcium nitrate ($Ca(NO_3)_2$) 0.1 M	None allocated	–		3.3.2
No	Carbon (C) electrodes	None allocated	–		2.1.1
Yes	Carbon (C) powder	None allocated	Irritant		2.2.2
No	Charcoal powder (C) activated	None allocated	–		2.4.2
Yes	Cobalt(II) chloride paper ($CoCl_2.6H_2O$)	6.1	Toxic		6.2.1
Yes	Copper carbonate ($CuCO_3$)	None allocated	Harmful, irritant		3.1.2, 3.2.3
Yes	Copper(II) chloride ($CuCl_2$) solid	8	Harmful, irritant, dangerous for the environment		2.3.1
No	Copper(II) chloride ($CuCl_2$) 0.1 M (1.7%)	None allocated	–		2.3.1
No	Copper piece (Cu)	None allocated	–		2.3.6
No	Copper wire/ foil (Cu)	None allocated	–		2.2.1, 2.3.4
Yes	Copper(II) sulfate anhydrous ($CuSO_4$) solid	9	Harmful, irritant		6.2.1
Yes	Copper(II) sulfate ($CuSO_4.5H_2O$) solid	9	Harmful, irritant, dangerous for the environment		2.3.3, 3.1.2, 3.3.1
No	Copper(II) sulfate ($CuSO_4.5H_2O$) 0.1 M (2.5%)	None allocated	–		3.1.2, 3.3.1
No	Disinfectant	None allocated	–		4.5.1
Yes	Ethanoic acid – glacial (acetic acid CH_3COOH) (99.8%) Conc>=90%: C; R35 >=25%Conc<90%: C; R34 >=10%Conc<25%: Xi; R36/38	8/3	Very corrosive		3.4.7, 6.2.1

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
Yes	Ethanoic acid – (acetic acid) 2 M (12%)	None allocated	Irritant		6.2.1
No	Ethanoic acid – (acetic acid) 0.1 M (0.6%)	None allocated	–		3.4.7
Yes	Ethanol (C_2H_5OH) or methylated spirits	3	Highly flammable		1.2.2, 2.4.1, 3.4.1, 6.2.1
No	Food colouring: brown	None allocated	–		1.3.5, 4.5.2
Yes	Graphite powder	None allocated	Irritant		1.3.1, 2.4.2
No	Household substances including orange juice, soft drink, fresh and sour milk, vinegar	None allocated	–		3.4.4
Yes	Hydrochloric acid (HCl) 32% – concentrated Conc>=25%: C; R34; R37 >=10%Conc<25%: Xi; R36/37/38	8	Corrosive		2.1.2, 2.3.6, 3.2.2, 3.4.1, 3.4.2, 3.4.3, 3.4.5, 3.4.6, 3.4.7
No	Hydrochloric acid (HCl) (7%) 2.0 M	None allocated	–		2.1.2, 2.3.6
No	Hydrochloric acid (HCl) (0.37%) 0.1 M	None allocated	–		3.2.2, 3.4.1, 3.4.2, 3.4.3, 3.4.5, 3.4.6, 3.4.7
No	Iron (Fe) nails	None allocated	–		2.2.2, 2.3.6, 3.4.7
Yes	Lead – cubes or cylinders (Pb)	None allocated	Toxic		2.2.1
Yes	Lead piece (Pb)	None allocated	Toxic		2.2.2, 2.4.2

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
Yes	Lead(II) nitrate ($Pb(NO_3)_2$) solid	5.1/6.1	Toxic, dangerous to the environment		2.3.3, 3.3.2
Yes	Lead(II) nitrate ($Pb(NO_3)_2$) 0.1 M (3.3%)	6.1	Toxic		3.3.2
Yes	Limewater (saturated $Ca(OH)_2$)	None allocated	Irritant		2.1.2, 3.2.2, 3.2.3, 3.4.6
No	Lithium carbonate (Li_2CO_3)	None allocated	–		3.4.6
No	Litmus indicator, red and blue	None allocated	–		3.4.1
No	Litmus paper, red and blue	None allocated	–		3.4.4
No	Magnesium strips/ribbon (Mg)	4.1	Highly flammable		2.2.2, 2.3.6, 2.4.1, 3.2.1, 3.4.7
No	Marble chips – calcium carbonate ($CaCO_3$)	None allocated	–		2.1.2
Yes	Methyl orange indicator – solid	None allocated	Toxic		3.4.1
No	Methyl orange indicator (0.04%)	None allocated	–		3.4.1
Yes	Methyl red solid	None allocated	Harmful		3.4.1
Yes	Methyl red solution	3	Flammable		3.4.1
Yes	Nickel (Ni) electrodes (solid)	None allocated	Harmful, irritant, listed as a carcinogen, category 3 in List of Designated Hazardous Substances – ASCC (NOHSC)		2.1.1

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
Yes	Nitric acid (HNO_3) (70%) – concentrated	8	Very corrosive, oxidising		3.1.2, 3.4.6
Yes	Nitric acid (HNO_3) 1 M (6.3%)	8	Corrosive		3.1.2
No	Nitric acid (HNO_3) 0.1 M (0.63%)	None allocated	–		3.4.6
No	Paraffin wax	None allocated	–		4.6.1
No	Phenolphthalein ($C_{20}H_{14}O_4$) solid	None allocated	–		2.4.1, 3.4.1
No	Phenolphthalein indicator (0.1% in ethanol)	3	Flammable		2.4.1, 3.4.1
No	Plaster of Paris	None allocated	–		1.3.3
No	Platinum (Pt)	None allocated	–		2.1.1
No	Potassium chloride (KCl) solid	None allocated	–		2.3.1
No	Potassium chloride (KCl) 0.1 M (0.75%)	None allocated	–		2.3.1
Yes	Potassium permanganate ($KMNO_4$) solid	5.1	Harmful, oxidising		9.1.2
No	Red ink (e.g. for a stamp pad)	None allocated	–		1.3.5
No	Silicon	None allocated	–		2.2.2, 2.4.2
Yes	Silver nitrate ($AgNO_3$) solid	5.1	Corrosive, dangerous to the environment		2.3.3, 2.3.4, 3.3.1, 3.3.2
No	Silver nitrate ($AgNO_3$) 0.1 M (1.7%)	None allocated	–		2.3.4, 3.3.1, 3.3.2
No	Salt (sodium chloride)	None allocated	–		1.3.3, 4.3.1, 4.4.1, 6.2.1, 6.2.2

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
No	Salt (sodium chloride) solution (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, and 0.5%)	None allocated	–		4.4.1
Yes	Sodium carbonate anhydrous (Na_2CO_3) solid	None allocated	Irritant		3.3.2, 3.4.6
No	Sodium chloride (NaCl) solid (salt)	None allocated	–		1.3.3, 2.3.1, 3.3.1, 3.3.2
No	Sodium chloride (NaCl) 0.1 M (0.6%)	None allocated	–		2.3.1, 3.3.1, 3.3.2
No	Sodium hydrogen carbonate (sodium bicarbonate) ($NaHCO_3$) solid	None allocated	–		3.2.2, 3.4.6
Yes	Sodium hydroxide (NaOH) solid Conc>=5%: C; R35 >=2%Conc<5%: C; R34 >=0.5%Conc<2%: Xi; R36/38	8	Very corrosive		3.1.2, 3.3.1, 3.4.1, 3.4.2, 3.4.3, 3.4.5
Yes	Sodium hydroxide (NaOH) (4%) 1.0 M	8	Corrosive		3.4.2
No	Sodium hydroxide (NaOH)(0.4%) 0.1 M	None allocated	–		3.1.2, 3.3.1, 3.4.1, 3.4.3, 3.4.5
Yes	Sodium iodide (NaI)	None allocated	Irritant		3.3.2
No	Sodium iodide (NaI) 0.1 M (1.5%)	None allocated	–		3.3.2
Yes	Sodium nitrate ($NaNO_3$) solid	5.1	Harmful, irritant, oxidising		3.3.2
No	Sodium nitrate ($NaNO_3$) 0.1 M	None allocated	–		3.3.2
No	Sodium sulfate (Na_2SO_4) solid	None allocated	–		3.1.2, 3.3.2
No	Sodium sulfate (Na_2SO_4) 0.1 M	None allocated	–		3.1.2, 3.3 2

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
No	Sodium thiosulfate ($Na_2S_2O_3$)	None allocated	–		2.3.3, 2.3.4
No	Sodium thiosulfate ($Na_2S_2O_3$) 4%	None allocated	–		2.3.3, 2.3.4
No	Starch	None allocated	–		1.3.5
No	Steel wool	None allocated	–		2.2.2, 2.3.5
Yes	Strontium chloride hydrated ($SrCl_3.6H_2O$) solid	None allocated	Harmful		2.3.1
No	Strontium chloride ($SrCl_2$) 0.1 M (2.7%)	None allocated	–		2.3.1
No	Sugar	None allocated	–		4.3.1, 4.4.1, 6.2.1
No	Sugar of varying concentrations (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%)	None allocated	–		4.4.1
No	Sulfur powder (S)	4.1	Highly flammable		2.2.2
Yes	Sulfuric acid (H_2SO_4) (98%) concentrated (Conc>=15%: C; R35 >=5%Conc<15%: Xi; R36/38)	8	Very corrosive		2.1.1, 3.4.7
No	Sulfuric acid (H_2SO_4) (0.98%) 0.1 M	None allocated	–		2.1.1, 3.4.7
No	Sultana		–		4.5.2
Yes	Tin piece (Sn)	None allocated	–		2.2.2, 3.4.7
Yes	Tin(II) chloride ($SnCl_2.2H_2O$) solid	None allocated	Harmful		2.3.3
No	Universal indicator	3	Flammable		3.4.1, 3.4.2, 3.4.4, 3.4.5

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HS	Substances used	DG	Hazard category	MSDS issue	Req'd for prac
No	Vinegar (white) (4% acetic acid)	None allocated	–		4.3.1, 6.3.1
No	Wood: cubes or cylinders	None allocated	–		2.2.1
No	Zinc pieces (Zn)	None allocated	–		2.2.2, 3.1.2, 3.4.7
No	Zinc strips (Zn)	None allocated	–		2.3.3

Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Barium chloride, hydrated 0.1 M	$BaC1_2.2H_2O$	244.28	Dissolve 24.43 g of barium chloride in distilled water and make up to 1 litre with distilled water.
Barium nitrate 0.1 M	$Ba(NO_3)_2$	261.35	Dissolve 26.14 g of barium nitrate in distilled water and make up to 1 litre with distilled water.
Bleach solution (sodium hypochlorite) 0.5%			Measure 10 mL of household bleach 5% (White King), add distilled water and make up to 100 mL with distilled water.
Bromothymol blue solution (0.04%)			Dissolve 0.4 g of bromothymol blue in 200 mL of ethanol. Make up to 1 litre with distilled water. Filter the solution. Yellow – pH 6.0; Blue – pH 7.6.

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Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Calcium nitrate 0.1 M	$Ca(NO_3)_2$	236.15	Dissolve 23.62 g of calcium nitrate in distilled water and make up to 1 litre with distilled water.
Copper(II) sulfate 0.1 M	$CuSO_4\ 5H_2O$	249.68	Dissolve 24.97 g of copper(II) sulfate in distilled water and make up to 1 litre with distilled water.
Ethanoic acid – glacial acetic acid 2.0 M	CH_3COOH	60.05	Supplied as 99–100% In a fume hood, measure 114 mL of ethanoic acid-glacial; add to two-thirds of distilled water and make up to 1 litre with distilled water.
Ethanoic acid – glacial acetic acid 0.1 M	CH_3COOH	60.05	Supplied as 99–100% In a fume hood, measure 5.7 mL of ethanoic acid-glacial; add to two-thirds of distilled water and make up to 1 litre with distilled water.
Ethanol or methylated spirits	C_2H_5OH	46.08	Supplied as 99–100% or 95% Decant in a fume hood.
Hydrochloric acid (7%) 2.0 M	HCl	36.46	The concentrated acid is available in two concentrated solutions, 32% and 36%. The 32% is less unpleasant to handle and is suitable for school purposes. Supplied as 32%. In a fume hood, measure 195 mL of hydrochloric acid; add to two-thirds of distilled water and make up to 1 litre with distilled water.

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Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Hydrochloric acid (0.37%) 0.1 M	HCl	36.46	The concentrated acid is available in two concentrated solutions, 32% and 36%. The 32% is less unpleasant to handle and is suitable for school purposes. Supplied as 32%. In a fume hood, measure 10 mL of hydrochloric acid; add to two-thirds of distilled water and make up to 1 litre with distilled water.
Lead(II) nitrate 0.1 M	$Pb(NO_3)_2$	331.20	Dissolve 33.12 g of lead(II) nitrate in distilled water and make up to 1 litre with distilled water.
Limewater (saturated)	$Ca(OH)_2$	74.09	To make 1 litre, shake that quantity of distilled water with 2.5g of calcium hydroxide and allow settling before using. Decant supernatant to class sets for use.
Methyl orange indicator (0.04%)	$C_{14}H_{14}N_3NaO_3S$	327.34	Dissolve 0.4 g of methyl orange in 200 mL of distilled water and make up to 1 litre with distilled water. pH range: red <2.8 yellow >4.6
Methyl red indicator	$C_{15}H_{15}N_3O_2$	269.30	Dissolve 0.2 g of methyl red in 300 mL of ethanol and make up to 1 litre with distilled water.
Nitric acid (70%) concentrated	HNO_3	63.01	Supplied as 16 M (70%).
Nitric acid 1 M (6.3%)	HNO_3	63.01	Supplied as 16 M (70%). In a fume hood, measure 62 mL of nitric acid; add to two-thirds of distilled water and make up to 1 litre with distilled water. Store in dark bottles.

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Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Nitric acid 0.1 M (0.63%)	HNO_3	63.01	Supplied as 70%. In a fume hood, measure 6 mL of nitric acid; add to two-thirds of distilled water and make up to 1 litre with distilled water. Store in dark bottles.
Phenolphthalein solution (0.1%) in ethanol	$C_{20}H_{14}O_4$	318.31	Dissolve 1 g of phenolphthalein in 600 cm^3 of ethanol and make up to 1 litre with distilled water.
Potassium chloride 0.1 M	KCl	74.56	Dissolve 7.50 g of potassium chloride in distilled water and make up to 1 litre with distilled water.
Silver nitrate 0.1 M	$AgNO_3$	169.87	Dissolve 16.98 g of silver nitrate in distilled water and make up to 1 litre with distilled water.
Salt (sodium chloride) solutions (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%)	NaCl	58.44	0.5% solution: Dissolve 0.5 g of salt in distilled water and make up to 100 mL with distilled water. 0.1% solution: Measure 20 mL of 0.5% solution and add 80 mL of distilled water. 0.05% solution: Measure 10 mL of 0.5% solution and add 90 mL of distilled water. 0.01% solution: Measure 10 mL of 0.1% solution and add 90 mL of distilled water. 0.005% solution: Measure 10 mL of 0.05% solution and add 90 mL of distilled water. 0.001% solution: Measure 10 mL of 0.01% solution and add 90 mL of distilled water.
Sodium chloride 0.1 M (0.6%)	NaCl	58.44	Dissolve 5.85 g of sodium chloride in distilled water and make up to 1 litre with distilled water.
Sodium hydroxide 1.0 M (4%)	NaOH	40.00	Dissolve 40.00 g of sodium hydroxide in distilled water and make up to 1 litre with distilled water.

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Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Sodium hydroxide 0.1 M (0.4%)	NaOH	40.00	Dissolve 4.00 g of sodium hydroxide in distilled water and make up to 1 litre with distilled water.
Sodium iodide (NaI) 0.1 M (1.5%)	NaI	149.9	Dissolve 14.90 g of sodium iodide in distilled water and make up to 1 litre with distilled water.
Sodium nitrate 0.1 M (0.85%)	$NaNO_3$	85.00	Dissolve 8.50 g of sodium nitrate in distilled water and make up to 1 litre with distilled water.
Sodium sulfate 0.1 M (1.4%	Na_2SO_4	142.04	Dissolve 14.20 g of sodium sulfate in distilled water and make up to 1 litre with distilled water.
Sodium thiosulfate 4%	$Na_2S_2O_3$	248.17	Dissolve 4 g of sodium thiosulfate in distilled water and make up to 100 mL with distilled water.
Strontium chloride 0.1 M (2.7%)	$SrCl_2$	267	Dissolve 26.70 g of strontium chloride in distilled water and make up to 1 litre with distilled water.
Sugar solutions (0.001%, 0.005%, 0.01%, 0.05%, 0.1%, 0.5%)			0.5% solution: Dissolve 0.5 g of sugar in distilled water and make up to 100 mL with distilled water. 0.1% solution: Measure 20 mL of 0.5% solution and add 80 mL of distilled water. 0.05% solution: Measure 10 mL of 0.5% solution and add 90 mL of distilled water. 0.01% solution: Measure 10 mL of 0.1% solution and add 90 mL of distilled water. 0.005% solution: Measure 10 mL of 0.05% solution and add 90 mL of distilled water. 0.001% solution: Measure 10 mL of 0.01% solution and add 90 mL of distilled water.

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Recipes

Chemical	Formula	Molecular weight	Method (review associated prac risk assessment, and safety precautions prior to preparing any solution)
Sulfuric acid 0.1 M (0.98%)	H_2SO_4	98.08	Supplied as 18 M (98%). In a fume hood, measure 5 mL of sulfuric acid; add to two-thirds of distilled water and make up to 1 litre with distilled water.

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